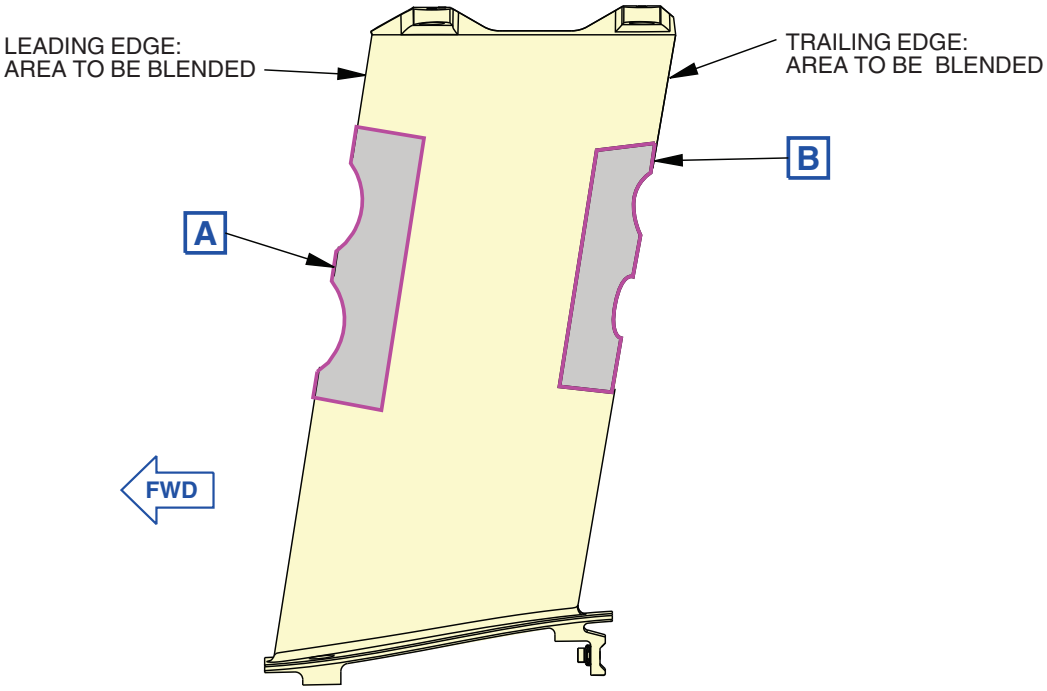
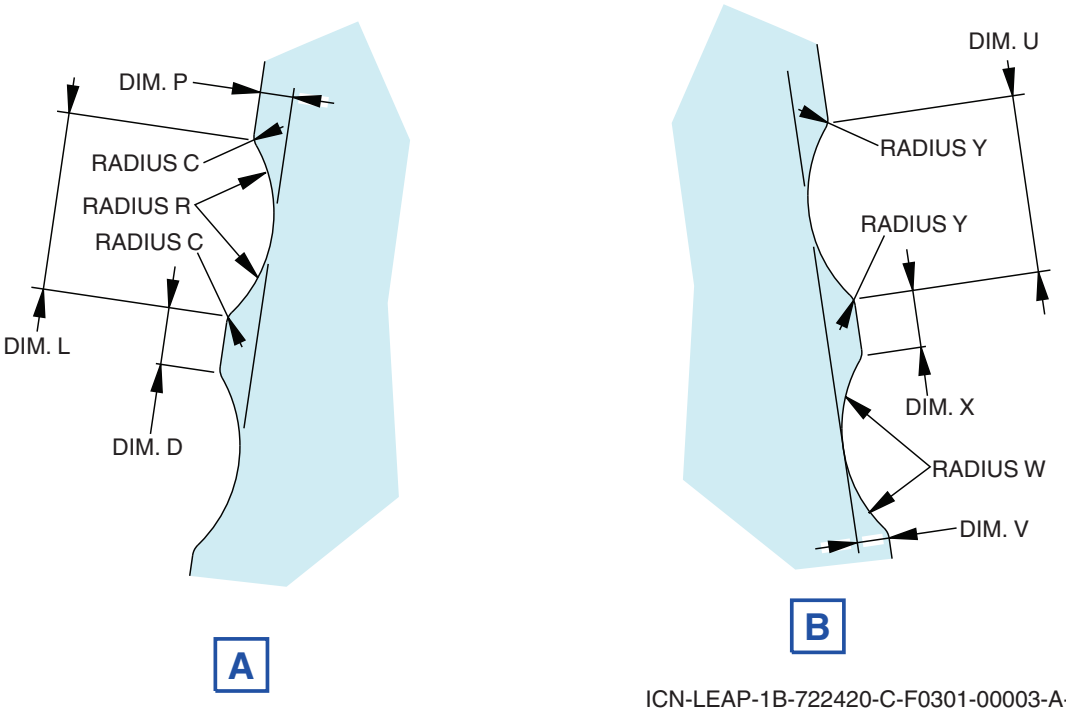


TYPICAL VIEW OF BLENDING ON LEADING EDGE AND TRAILING EDGE FOR DIMENSION LIMITS



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2800666 S0000640793_V2

Aluminum Outlet Guide Vane Blending Dimension
Figure 802/72-21-03-990-805-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-21-03-320-803-G00

3. Titanium Outlet Guide Vane Leading Edge Blending Repair

(Figure 803 or Figure 804)

A. General

(1) This procedure gives instructions to blend the leading edge of the titanium Outlet Guide Vane (OGV).

B. References

Reference	Title
70-11-06-230-802-G00	Fluorescent Penetrant Inspection (P/B 201)
70-11-06-230-803-G00	Fluorescent Penetrant Inspection (General) (P/B 201)
70-21-23-100-801-G00	Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701)
70-43-24-100-802-G00	Swab Etching Procedure and Cleaning (P/B 201)
70-43-42-300-802-G00	Blending Procedures - Hand Blending (P/B 201)
70-43-42-300-803-G00	Blending Procedures - Power Blending (P/B 201)
72-21-03-000-801-G00	Outlet Guide Vane Removal (P/B 401)
72-21-03-400-801-G00	Outlet Guide Vane Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
G51485 [S1093]	Etchant - Titanium Alloy	

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Repair

SUBTASK 72-21-03-010-009-G00

(1) Remove the OGV that you will blend repair (TASK 72-21-03-000-801-G00).

(a) The following table lists the part numbers that are applicable to this repair.

Table 803/72-21-03-993-801-G00 OGV Part Numbers

Part Number	Designation	CSN
364-062-201-0	Vane, Outlet Guide	01-380, 72–24–00
364-062-301-0	Vane, Outlet Guide	01-390, 72–24–00

F. Outlet Guide Vane Blending Repair

WARNING USE PROTECTIVE GLOVES TO PREVENT INJURY FROM SHARP EDGES AND ROUGH SURFACES. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

WHEN YOU REMOVE DAMAGE FROM INITIAL TITANIUM PARTS, PREVENT SOURCES OF IGNITION. SMALL PARTICLES AND SHAVINGS OF TITANIUM ARE A FIRE HAZARD AND CAN BE A SOURCE OF IGNITION. DO NOT USE FOAM, WATER, HALON, CARBON TETRACHLORIDE, OR CARBON DIOXIDE TO EXTINGUISH A FIRE WHILE YOU WORK WITH TITANIUM. SMALL PARTICLES OF TITANIUM ARE FLAMMABLE, AND AN EXPLOSION CAN OCCUR. USE DRY TALC, CALCIUM CARBONATE, SAND OR GRAPHITE. WATER CAN CAUSE A STEAM EXPLOSION. TITANIUM DUST IS FLAMMABLE AND IN CONCENTRATION CAN CAUSE AN EXPLOSION AND INJURY TO PERSONNEL.

WARNING

DO NOT USE CHLORINATED HYDROCARBONS AND HALOGEN MATERIALS. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

CAUTION

DO NOT USE TOOLS MADE FROM OR WITH CADMIUM, BISMUTH, TIN, ZINC, GOLD OR SILVER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION

DO NOT CAUSE HEAT THAT CAUSES A THERMAL TENSION DURING THE OPERATION OF POWER TOOLS. IF YOU DO NOT OBEY, YOU CAN CAUSE DAMAGE TO THE PART.

SUBTASK 72-21-03-210-009-G00

(1) Count the number of OGV and OGV struts blended on the engine.

NOTE: All OGV wheels included in the count: OGV struts, OGV aluminum and OGV titanium.

SUBTASK 72-21-03-210-010-G00

(2) Make sure that the quantity of blended OGV and OGV struts is less than five.

NOTE: If the quantity of blended OGV and OGV struts is equal to five, you cannot repair this part on the engine with this procedure.

SUBTASK 72-21-03-110-006-G00

(3) If necessary, remove the grease from the part (TASK 70-21-23-100-801-G00).

SUBTASK 72-21-03-350-002-G00

WARNING

METAL PARTICLES FROM MACHINING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT.

CAUTION

REMOVE MINIMUM MATERIAL WHEN YOU BLEND TO THE DEPTH OF THE OGV DAMAGE. DAMAGE TO THE ENGINE CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

DO NOT DO MORE THAN TWO REPAIRS FOR EACH LEADING EDGE OF THE OGV STRUTS. IF YOU DO, ENGINE FAILURE CAN OCCUR.

(4) Blend the area to repair (TASK 70-43-42-300-803-G00 or TASK 70-43-42-300-802-G00).

(a) Make sure that you do not blend the leading edge fillet.

SUBTASK 72-21-03-350-003-G00

(5) Obey these limits when blending:

(a) The depth DIM. P is less than or equal to 0.315 in. (8.0 mm).

(b) The length DIM. L is less than or equal to 1.771 in. (45.0 mm).

(c) The RADIUS R is more than 4 times DIM. P.

(d) The RADIUS C is more than 0.04 in. (1.0 mm).

(e) If there are two areas blended, make sure that the distance DIM. D is more than or equal to:

1) 10 times DIM. P, if DIM. P is less than or equal to 0.087 in. (2.2 mm).

2) 0.866 in. (22.0 mm), if DIM. P is more than 0.087 in. (2.2 mm).

SUBTASK 72-21-03-211-005-G00

(6) After you do the repair, make sure that the leading edge contour and thickness are similar to the shape of the initial leading edge contour.

SUBTASK 72-21-03-211-006-G00

(7) Make sure that after you do the blending the roughness of the part is less than 63 microinches (1.6 micrometers).

SUBTASK 72-21-03-160-009-G00

WARNING

USE EYE PROTECTION WHEN YOU USE COMPRESSED AIR TO CLEAN, COOL, OR DRY PARTS OR TOOLS. PARTICLES CAN CAUSE AN INJURY TO YOUR EYES. DO NOT USE MORE THAN 30 PSI (207 KPA). DO NOT POINT COMPRESSED AIR AT YOURSELF OR OTHER PERSONS.

(8) Blow the blended areas with dry and oil free shop air.

SUBTASK 72-21-03-110-007-G00

(9) Clean the blended area (TASK 70-21-23-100-801-G00).

SUBTASK 72-21-03-230-003-G00

(10) Locally swab all the blended areas to be etched with titanium alloy etchant, G51485 [S1093] (TASK 70-43-24-100-802-G00).

SUBTASK 72-21-03-230-004-G00

(11) Do a fluorescent-penetrant inspection of the blended areas (TASK 70-11-06-230-803-G00 or TASK 70-11-06-230-802-G00).

(a) Make sure that there are no cracks.

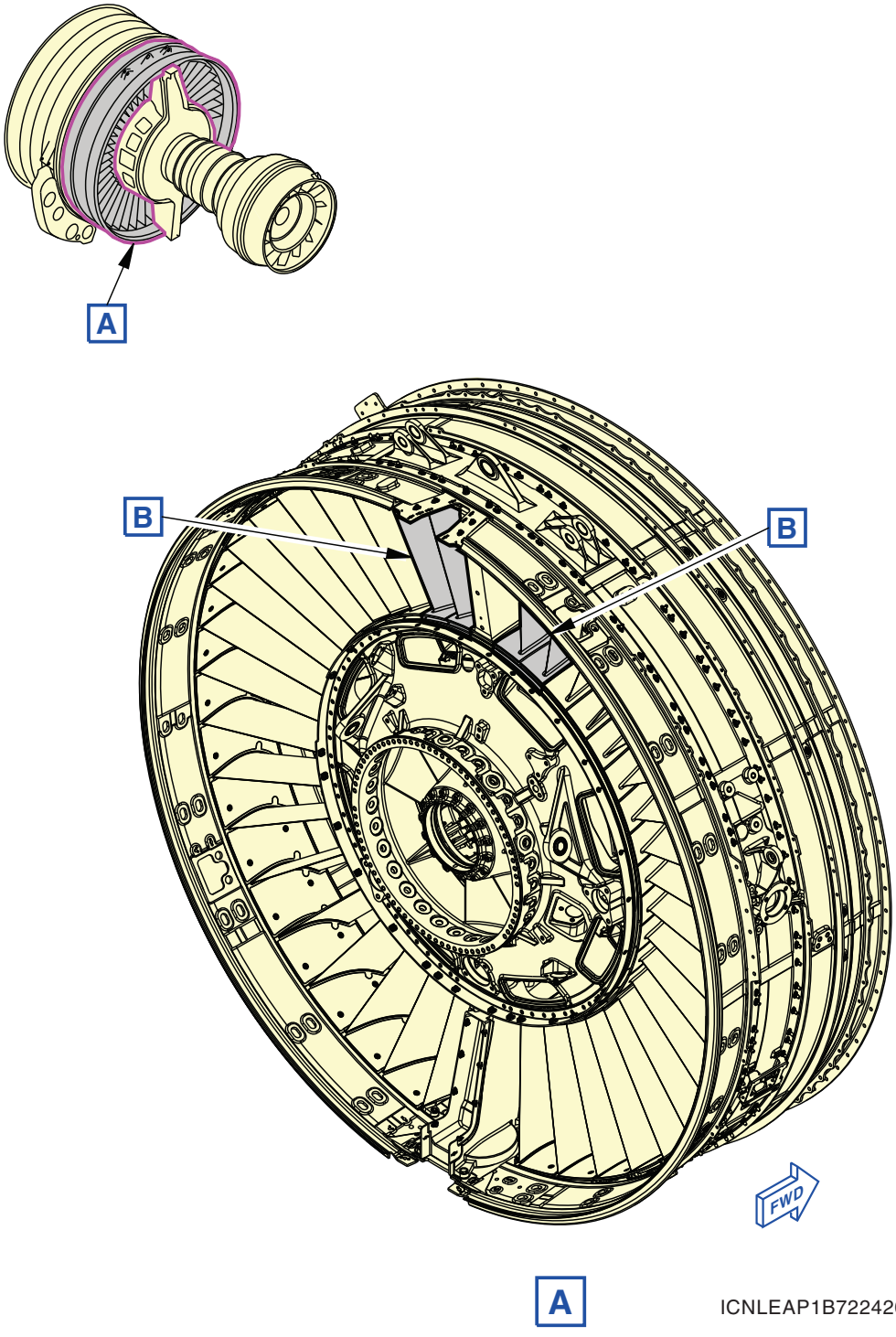
G. Put the Airplane Back to its Usual Condition

SUBTASK 72-21-03-410-013-G00

(1) Install the OGV (TASK 72-21-03-400-801-G00).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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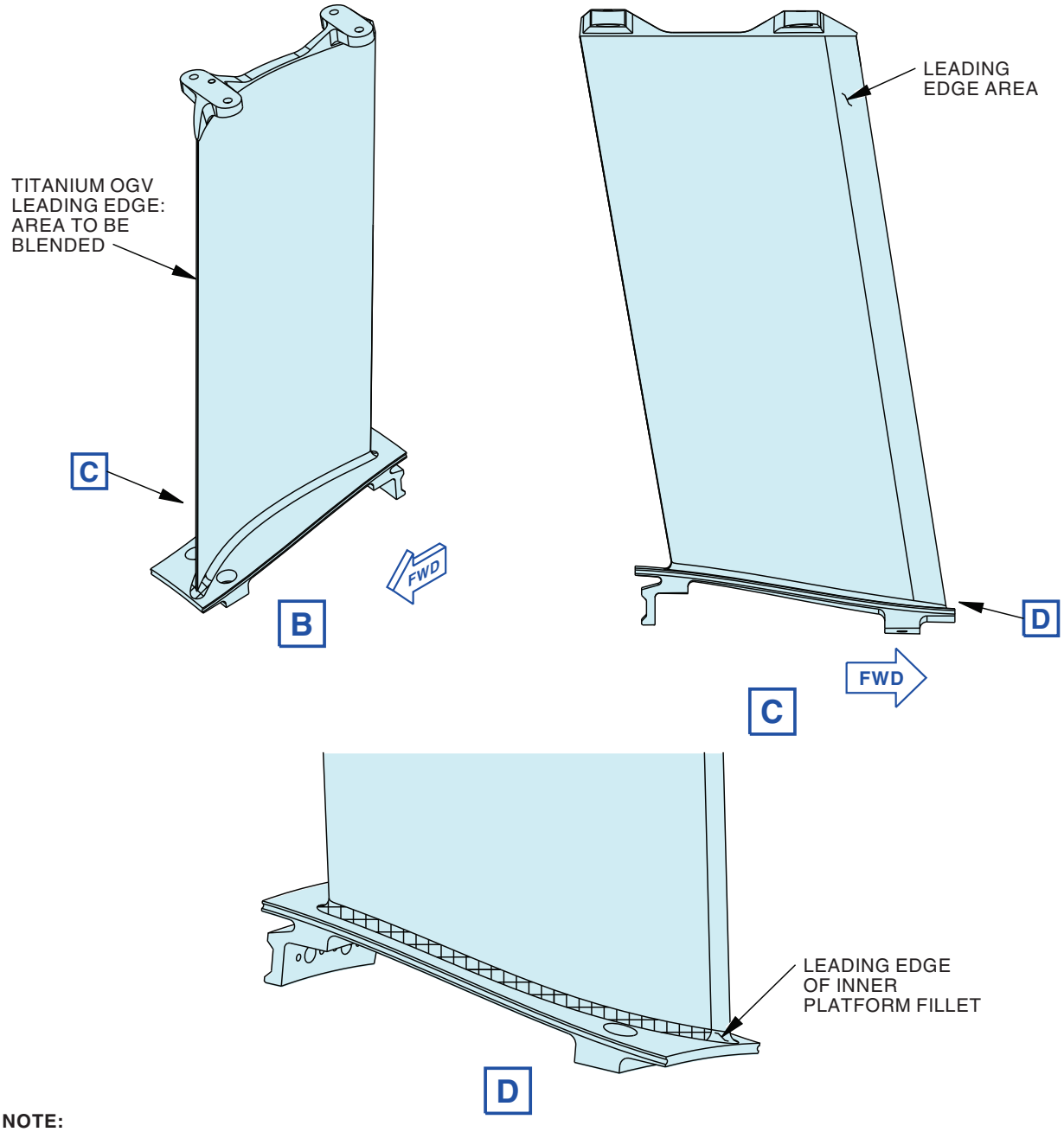
Titanium Outlet Guide Vane Leading Edge Blend Area
Figure 803/72-21-03-990-817-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

INNER PLATFORM FILLET AREA

LEADING EDGE OF INNER PLATFORM FILLET AREA

BLENDING IS NOT PERMITTED ON LEADING EDGE OF INNER PLATFORM FILLET AREA

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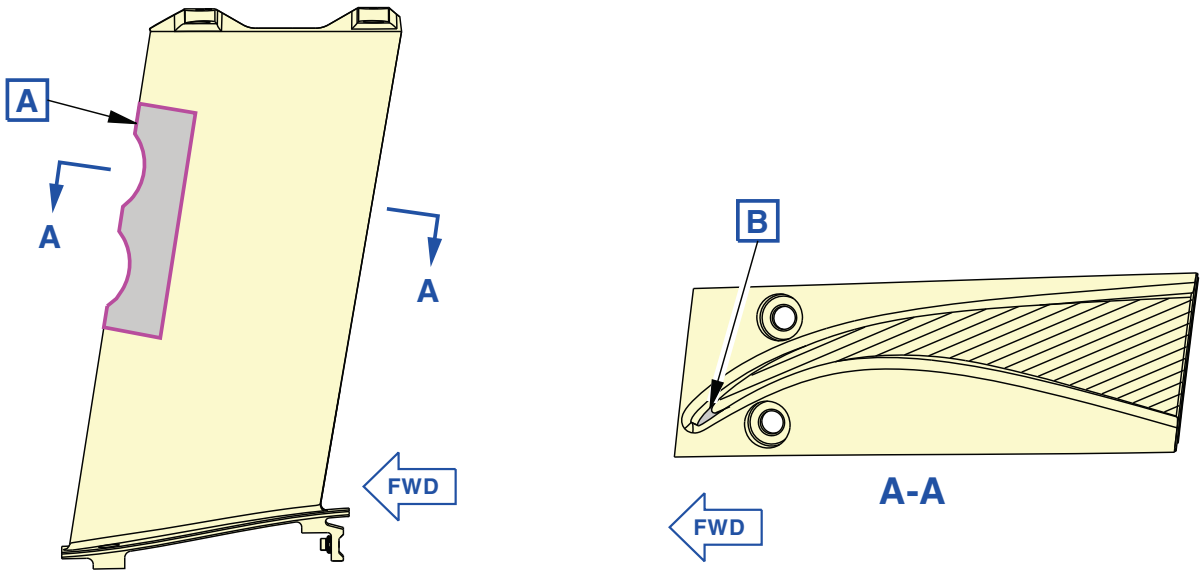
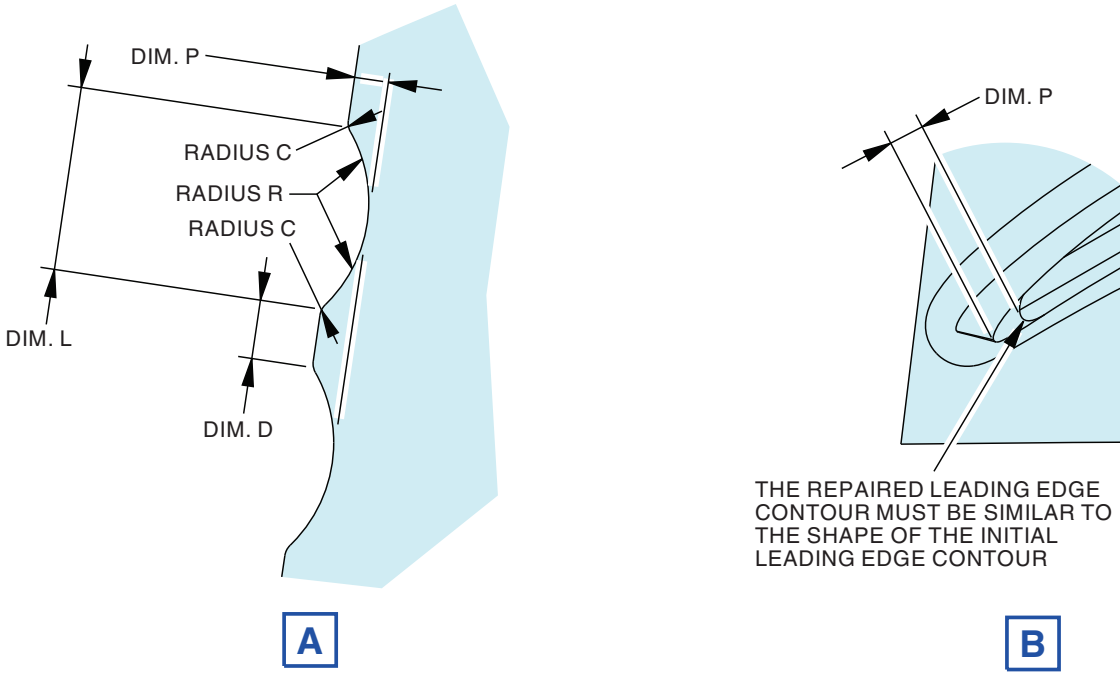
Titanium Outlet Guide Vane Leading Edge Blend Area
Figure 803/72-21-03-990-817-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TYPICAL VIEW OF BLENDINGS ON LEADING EDGE FOR DIMENSION LIMITS



NOTE:

ONLY TWO BLENDINGS ARE PERMITTED FOR EACH LEADING EDGE

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Titanium Outlet Guide Vane Blending Dimension
Figure 804/72-21-03-990-818-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-21-03-320-802-G00

4. Outlet Guide Vane Struts 1 and 5 Leading Edge Blending Repair

(Figure 805 and Figure 806)

A. General

(1) This task gives instructions to blend the leading edge of the Outlet Guide Vane (OGV) struts 1 and 5.

B. References

Reference	Title
70-11-06-230-802-G00	Fluorescent Penetrant Inspection (P/B 201)
70-11-06-230-803-G00	Fluorescent Penetrant Inspection (General) (P/B 201)
70-43-24-100-802-G00	Swab Etching Procedure and Cleaning (P/B 201)
70-43-42-300-802-G00	Blending Procedures - Hand Blending (P/B 201)
70-43-42-300-803-G00	Blending Procedures - Power Blending (P/B 201)
72-21-03-000-801-G00	Outlet Guide Vane Removal (P/B 401)
72-21-03-400-801-G00	Outlet Guide Vane Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
B50333 [CP2687]	Solvent - Degreasing, Diestone DLS	
G01043	Cloth - Lint-free	
G51485 [S1093]	Etchant - Titanium Alloy	

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Repair

SUBTASK 72-21-03-970-002-G00

(1) Count the number of blended leading edges in the OGV wheel (OGV strut, aluminum OGV and titanium OGV included) and record it.

SUBTASK 72-21-03-210-007-G00

(2) Make sure that the quantity of blended leading edges is less than five.

NOTE: If the quantity of blended leading edges is equal to five, you cannot repair this part on the engine with this procedure.

SUBTASK 72-21-03-410-011-G00

(3) Remove the OGV 1 or OGV 5 that you will repair (TASK 72-21-03-000-801-G00).

SUBTASK 72-21-03-160-007-G00

WARNING

ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

(4) Clean the part you removed with lint-free cloth, G01043, moisten with isopropyl alcohol, B00676 [CP1041], or Diestone DLS degreasing solvent, B50333 [CP2687], cleaning solvent.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Outlet Guide Vane Struts 1 and 5 Leading Edge Blending Repair

SUBTASK 72-21-03-320-002-G00

(1) Do the blend repair of the OGV as follows (TASK 70-43-42-300-803-G00 or TASK 70-43-42-300-802-G00):

CAUTION

DO NOT DO MORE THAN TWO REPAIRS FOR EACH LEADING EDGE OF THE OGV STRUTS. IF YOU DO, ENGINE FAILURE CAN OCCUR.

(a) Make sure that there is no more than one repair already on the leading edge of the OGV.

CAUTION

REMOVE MINIMUM MATERIAL WHEN YOU BLEND TO THE DEPTH OF THE OGV DAMAGE. DAMAGE TO THE ENGINE CAN OCCUR.

(b) Remove a minimum of material while blending to more than the depth of deviation.

SUBTASK 72-21-03-300-002-G00

(2) Make sure that you do not blend repair on the leading edge fillet.

SUBTASK 72-21-03-211-002-G00

(3) Obey these limits when blending:

(a) The depth DIM. P is less than or equal to 0.314 in. (8 mm).

(b) The length DIM. L is less than or equal to 1.771 in. (45 mm).

(c) The RADIUS R is more than 4 times DIM. P.

(d) The RADIUS C is more than 0.04 in. (1 mm).

(e) If there are two areas blended, make sure that the distance DIM. D between the two blending is more than or equal to:

1) 10 times DIM. P, if DIM. P is less than or equal to 0.087 in. (2.2 mm).

2) 0.866 in. (22 mm), if DIM. P is more than 0.087 in. (2.2 mm).

SUBTASK 72-21-03-211-003-G00

(4) After you do the repair, make sure that the leading edge contour and thickness are similar to the shape of the initial leading edge contour.

SUBTASK 72-21-03-211-004-G00

(5) Make sure that after you do the blending the roughness of the part is less than 63 microinches (1.6 micrometers).

SUBTASK 72-21-03-160-008-G00

WARNING

BEFORE YOU USE COMPRESSED AIR, PUT ON GOGGLES FOR EYE PROTECTION. DO NOT POINT THE NOZZLE AT OTHER PERSONNEL. IF YOU DO NOT OBEY THESE PRECAUTIONS, INJURIES TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

PUT ON THE PARTICULATE RESPIRATOR, STD-6435, SAFETY GOGGLES, STD-3904, NEOPRENE OR NITRILE GLOVE, STD-6517, AND HOODED COVERALLS, STD-6439 BEFORE YOU TOUCH THE FILTER. THE FILTER REMOVES SMALL PARTICLES (SMOKE, DUST, LINT, FIBERS, POLLEN) AND INFECTIOUS MATERIALS (BACTERIA, VIRUSES, MOLD SPORES, FUNGI) FROM THE AIR WHICH CAN CAUSE ILLNESSES.

(6) Blow the blended areas with dry and oil free shop air.

SUBTASK 72-21-03-110-005-G00

WARNING

ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

(7) Clean the part you removed with lint-free cloth, G01043, moisten with isopropyl alcohol, B00676 [CP1041], or Diestone DLS degreasing solvent, B50333 [CP2687], cleaning solvent.

SUBTASK 72-21-03-230-002-G00

(8) Do an Fluorescent Penetrant Inspection (FPI) as follows:

(a) Locally swab all the blended areas to be etched with titanium alloy etchant, G51485 [S1093] (TASK 70-43-24-100-802-G00).

(b) Do an FPI of the blended areas (TASK 70-11-06-230-803-G00 and TASK 70-11-06-230-802-G00).

1) Do an FPI sensitivity level 3.

(c) Cracks that you find are not permitted.

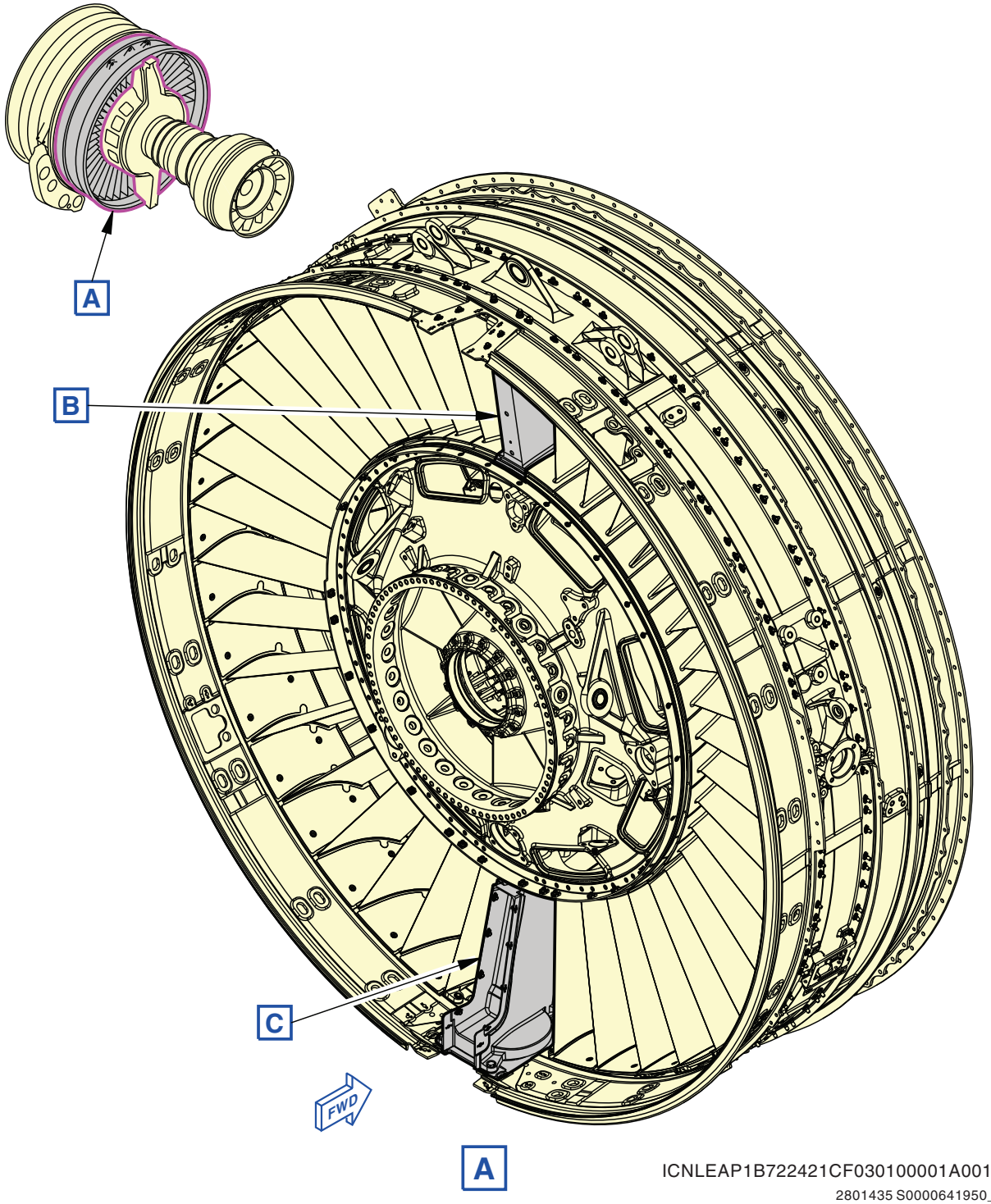
G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-03-410-012-G00

(1) Install the OGV (TASK 72-21-03-400-801-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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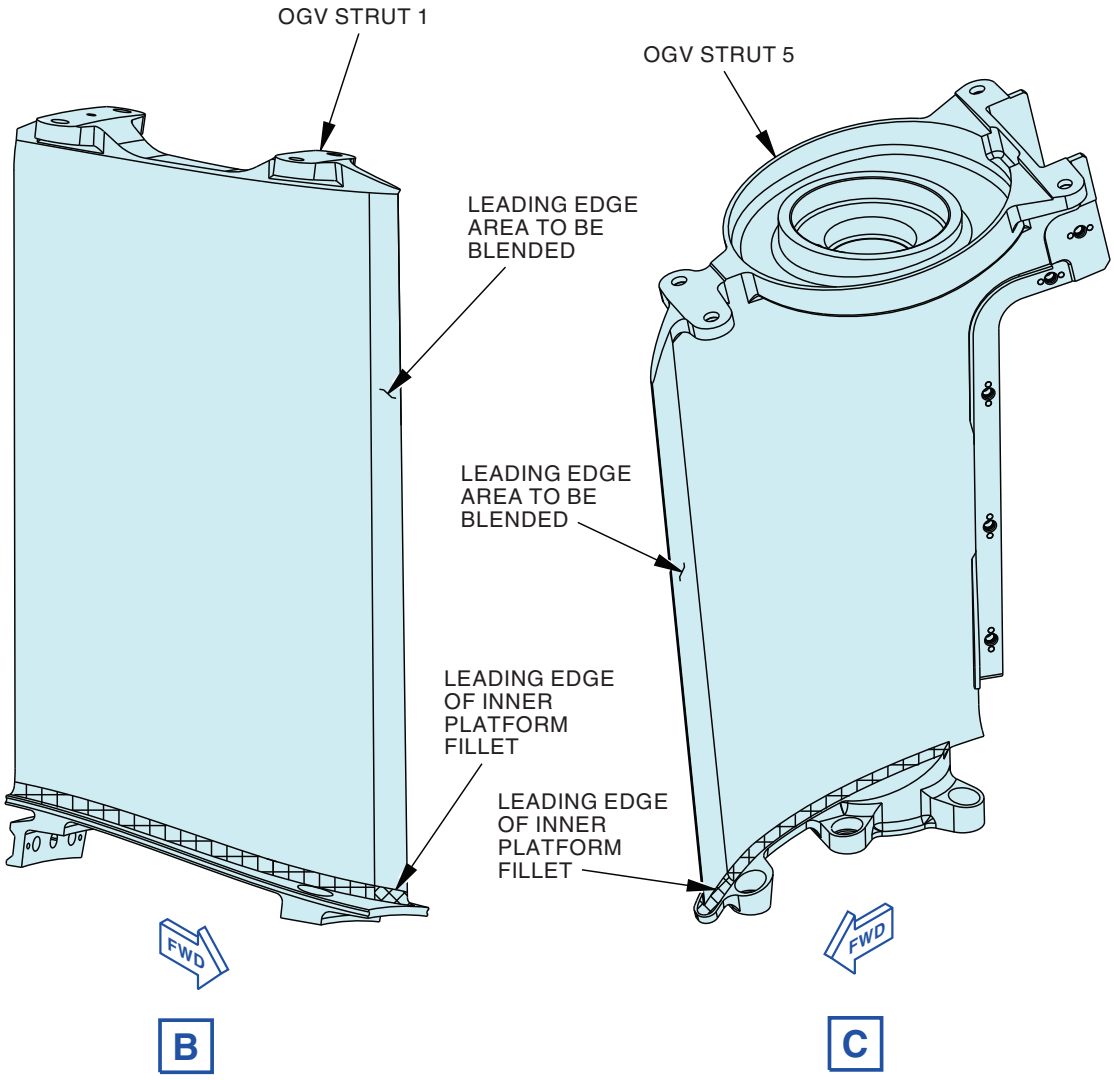
OGV Struts 1 and 5 Blending Repair Area
Figure 805/72-21-03-990-815-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

INNER PLATFORM FILLET AREA

LEADING EDGE OF INNER PLATFORM FILLET AREA

BLENDING IS NOT PERMITTED ON LEADING EDGE OF INNER PLATFORM FILLET AREA

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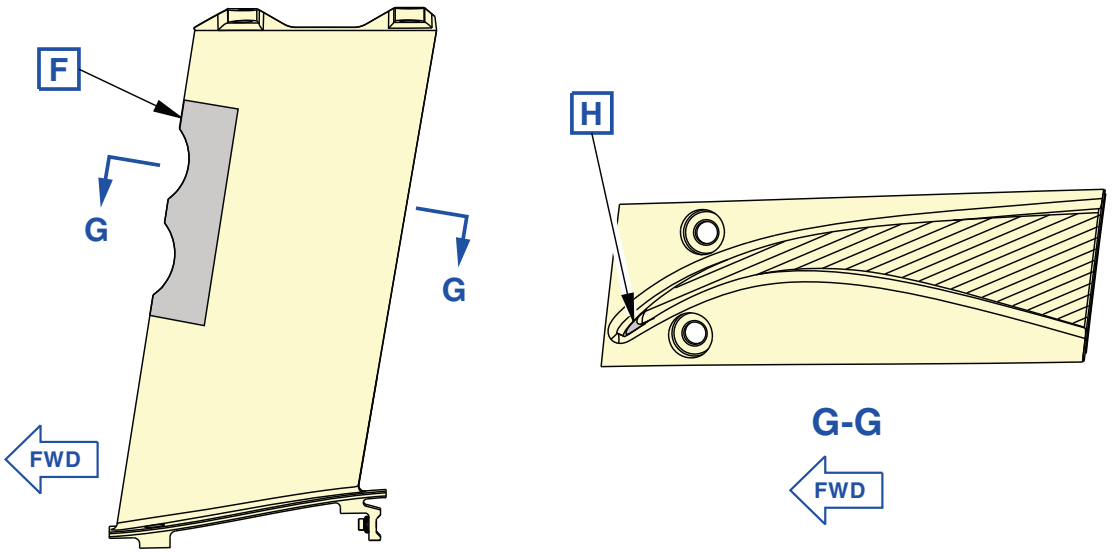
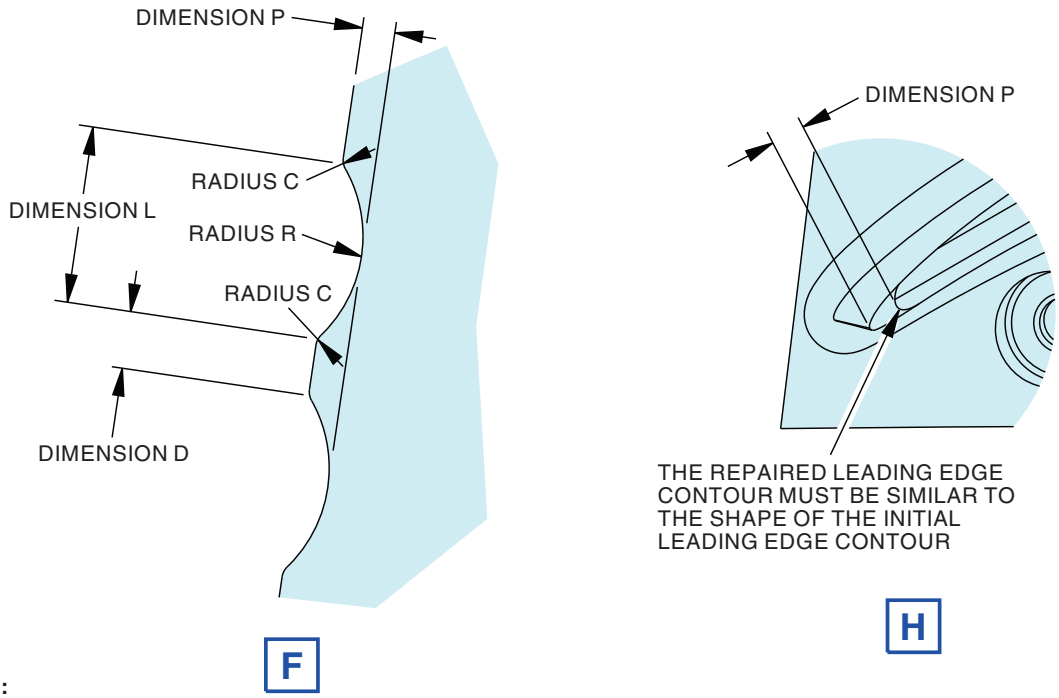
OGV Struts 1 and 5 Blending Repair Area
Figure 805/72-21-03-990-815-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TYPICAL VIEW OF BLENDINGS ON LEADING EDGE FOR DIMENSION LIMITS



NOTE:

ONLY TWO BLENDINGS ARE PERMITTED FOR EACH LEADING EDGE

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OGV Struts Blending Dimensions
Figure 806/72-21-03-990-816-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-21-03-370-801-G00

5. Outlet Guide Vanes Coating Local Repair

(Figure 807, Figure 808, Figure 809, Figure 810, and Figure 811)

A. General

(1) This task gives instructions to do local repair of primer and anti-erosion paint or local repair of primer only on the Outlet Guide Vane (OGV).

(2) The table below gives the list of the part numbers that are applicable to this repair.

Table 804/72-21-03-993-806-G00 OGV Part Numbers/Configurations

Part Number	Designation
364-068-111-0	Vane, Outlet Guide
364-068-211-0	Vane, Outlet Guide
364-068-311-0	Vane, Outlet Guide
364-068-411-0	Vane, Outlet Guide
364-068-511-0	Vane, Outlet Guide
364-068-611-0	Vane, Outlet Guide
364-068-711-0	Vane, Outlet Guide
364-068-811-0	Vane, Outlet Guide
364-068-911-0	Vane, Outlet Guide

B. References

Reference	Title
70-21-01-100-801-G00	Dry Abrasive Blast Cleaning with Plastic Media (P/B 701)
70-21-01-100-802-G00	Dry Abrasive Blast Cleaning with Fruit Stones (P/B 701)
70-21-23-100-801-G00	Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701)
70-43-07-300-801-G00	Specifications Chemical Touchup for Aluminum Standard Repair (P/B 201)
70-43-31-300-801-G00	Blind Rivet Assemblies Repair (P/B 201)
70-43-42-300-802-G00	Blending Procedures - Hand Blending (P/B 201)
70-43-42-300-803-G00	Blending Procedures - Power Blending (P/B 201)
70-64-29-300-801-G00	Polyurethane Anti-Erosion Paint - LBYH 203 Application (P/B 201)
70-64-30-300-801-G00	Epoxy-Amine Anti-Corrosion Paint - SP350 Application (P/B 201)
72-21-03-000-801-G00	Outlet Guide Vane Removal (P/B 401)
72-21-03-400-801-G00	Outlet Guide Vane Installation (P/B 401)
72-23-00-211-804-G00	Fan Frame Module Inspection (P/B 601)

C. Tools/Equipment

Reference	Description
STD-564	Lens - Magnifying, 20x
STD-26434	Lens - Magnifying, 12x

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
B50333 [CP2687]	Solvent - Degreasing, Diestone DLS	
C50360 [CP2948]	Paint - Finishing, Top Coat, Black - LBYH 203	
C50479 [S1316]	Primer - Epoxy, SP350	
G01043	Cloth - Lint-free	
G51486 [CP2109]	Tape - Plastic, Colored - 3M No. 471	
G51742 [CP2441]	Tape - Vinyl - 3M No. 472	
G52019 [CP2185]	Tape - Masking - 3M 202	
G52020	Paper - Abrasive, Silicon Carbide, 120 Grit	
G52021	Paper - Abrasive, Silicon Carbide, 320 Grit	
G52022	Paper - Abrasive, Aluminum Oxide, 120 Grit	
G52023	Paper - Abrasive, Aluminum Oxide, 320 Grit	
G52024 [CP9006]	Film - Preservation, VCI (Volatile Corrosion Inhibitor)	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Prepare for the Repair

SUBTASK 72-21-03-010-006-G00

(1) Remove the OGV that you will repair, do this task: Outlet Guide Vane Removal, TASK 72-21-03-000-801-G00.

G. Outlet Guide Vanes (OGV) Coating Local Repair

SUBTASK 72-21-03-020-006-G00

(1) If it is necessary, remove and discard the rivets and nut plate or gang channel attached to the aft conical flange on OGV (TASK 70-43-31-300-801-G00) (Figure 808).

SUBTASK 72-21-03-120-002-G00

(2) It is recommended to blend the finishing paint over 1.18 in. (30 mm) maximum all around the area to repair to decrease the thickness of the topcoat (TASK 70-43-42-300-802-G00) (Figure 810).

(a) Make sure that you do not remove the base material.

SUBTASK 72-21-03-160-004-G00

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THIS PRODUCT. IF NOT, IT CAN CAUSE INJURY.

WARNING ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING PUT ON GLOVES FOR PROTECTION BEFORE YOU TOUCH THE SHARP EDGES. THE SHARP EDGES CAN CUT YOU.

(3) Clean the area to repair with a lint-free cloth, G01043, moist with alcohol, B00676 [CP1041], or degreasing solvent, B50333 [CP2687] (TASK 70-21-23-100-801-G00).

SUBTASK 72-21-03-950-003-G00

(4) If it is necessary, apply masking 3M 472 tape, G51742 [CP2441], or 3M 202 tape, G52019 [CP2185], or 3M 471 tape, G51486 [CP2109], around the area to repair (Figure 808, Figure 809, and Figure 810).

SUBTASK 72-21-03-120-003-G00

CAUTION REMOVE MINIMUM MATERIAL WHEN YOU BLEND TO THE DEPTH OF THE OGV DAMAGE. DAMAGE TO THE ENGINE CAN OCCUR.

(5) If it is necessary, remove the anti-erosion paint on the area to repair.

(a) Make sure that you do not remove the base material.

WARNING DO NOT BREATHE THE PARTICLES FROM BLENDING OR LET THE PARTICLES TOUCH YOU. THE PARTICLES FROM BLENDING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PROTECTION EQUIPMENT. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR.

(b) Alternative procedures available. Blend the area (TASK 70-43-42-300-802-G00).

1) Use 120 grit abrasive paper, G52020, or 120 grit abrasive paper, G52022.

2) If you will remove only the finishing paint, finish removing the end of anti-erosion paint with 320 grit abrasive paper, G52021, or 320 grit abrasive paper, G52023.

WARNING DO NOT BREATHE THE PARTICLES FROM BLENDING OR LET THE PARTICLES TOUCH YOU. THE PARTICLES FROM BLENDING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PROTECTION EQUIPMENT. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR.

(c) Alternative procedure. Blend the area (TASK 70-43-42-300-803-G00).

1) If you will remove only the finishing paint, finish removing the end of anti-erosion paint with 320 grit abrasive paper, G52021, or 320 grit abrasive paper, G52023.

WARNING PUT ON A FACE SHIELD, GLOVES, PROTECTIVE CLOTHING, AND PROTECTIVE SHOES. INJURIES TO PERSONNEL CAN OCCUR.

(d) Alternative procedure. Do a dry abrasive blast of the area (TASK 70-21-01-100-801-G00 and TASK 70-21-01-100-802-G00).

NOTE: It is permitted to increase the pressure from the 25 psi (172.4 kPa) - 40 psi (276 kPa) range to a 58 psi (399.9 kPa) - 72.5 psi (499.87 kPa) range if you use plastic media to remove paint.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Do a visual inspection of the blasted part area.
   a) If it is necessary, use a 12x magnifying lens, STD-26434.
   b) In blasted areas, make sure that the part is uniformly clean and there are no abrasive blast residues, no trace of protuberances, cracks, beads, folding, burrs, blows, or pollutions and aspect is uniform.
   c) In non-blasted areas, make sure that there is no sandblast traces.

(e) Make sure that you remove all the finishing paint from the area to repair.

(f) Make sure that it is possible to measure the thickness of paint on the repaired area (Figure 810).
   1) If it is necessary, increase the surface of the repaired area.

SUBTASK 72-21-03-220-002-G00

(6) If there is remaining primer on the area to repair, it is permitted to not remove the primer.
   (a) Measure the thickness and visually inspect the primer.
      1) If it is necessary, use a 20x magnifying lens, STD-564.
   (b) If the thickness of the primer is 0.00099 in. (0.025 mm) - 0.00137 in. (0.035 mm) and if on painted surfaces there are no missing primer, occlusions of foreign material, cracks, impacts, pitting, orange peel, scratches, run-outs, craters, blisters, and aspect is uniform and smooth, it is not necessary to remove and apply the primer in the steps below.

SUBTASK 72-21-03-120-004-G00

CAUTION

REMOVE MINIMUM MATERIAL WHEN YOU BLEND TO THE DEPTH OF THE OGV DAMAGE. DAMAGE TO THE ENGINE CAN OCCUR.

(7) If it is necessary, remove the primer on the area to repair.

NOTE: It is permitted to let a thin layer of primer on the repaired area to avoid affecting the base material.

(a) Make sure that you do not remove the base material.

WARNING

DO NOT BREATHE THE PARTICLES FROM BLENDING OR LET THE PARTICLES TOUCH YOU. THE PARTICLES FROM BLENDING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PROTECTION EQUIPMENT. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR.

(b) Alternative procedure available. Blend the area (TASK 70-43-42-300-802-G00).
   1) Use 320 grit abrasive paper, G52021, or 320 grit abrasive paper, G52023.

WARNING

PUT ON A FACE SHIELD, GLOVES, PROTECTIVE CLOTHING, AND PROTECTIVE SHOES. INJURIES TO PERSONNEL CAN OCCUR.

(c) Alternative procedure. Do a dry abrasive blast of the area (TASK 70-21-01-100-801-G00 and TASK 70-21-01-100-802-G00).

NOTE: It is permitted to increase the pressure from the 25 psi (172.4 kPa) - 40 psi (276 kPa) range to a 58 psi (399.9 kPa) - 72.5 psi (499.87 kPa) range if you use plastic media to remove paint.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Do a visual inspection of the blasted part area (Figure 810).
   a) If it is necessary, use a 12x magnifying lens, STD-26434.
   b) In blasted areas, make sure that the part is uniformly clean and there are no abrasive blast residues, no trace of protuberances, cracks, beads, folding, burrs, blows, or pollutions and aspect is uniform.
   c) In non-blasted areas, make sure that there is no sandblast traces.

SUBTASK 72-21-03-120-005-G00

(8) If you did not do a finish hand blending with 320 grit abrasive paper, G52021, or 320 grit abrasive paper, G52023, do these steps:

WARNING DO NOT BREATHE THE PARTICLES FROM BLENDING OR LET THE PARTICLES TOUCH YOU. THE PARTICLES FROM BLENDING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PROTECTION EQUIPMENT. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR.

(a) Rub down lightly the area to paint (TASK 70-43-42-300-802-G00).
   1) Make sure that you do not remove the base material.
   2) Use 320 grit abrasive paper, G52021, or 320 grit abrasive paper, G52023.

(b) Measure the thickness and visually inspect the primer.
   1) If it is necessary, use a 20x magnifying lens, STD-564.

(c) If the thickness of the primer is 0.00099 in. (0.025 mm) - 0.00137 in. (0.035 mm), it is not necessary to apply the primer in the steps below.

SUBTASK 72-21-03-950-004-G00

(9) If applied, remove the masking.
   (a) No deterioration of the paint permitted on adjacent areas.

SUBTASK 72-21-03-100-001-G00

WARNING USE EYE PROTECTION WHEN YOU USE COMPRESSED AIR TO CLEAN, COOL, OR DRY PARTS OR TOOLS. PARTICLES CAN CAUSE AN INJURY TO YOUR EYES. DO NOT USE MORE THAN 30 PSIG (200 KPA). DO NOT POINT COMPRESSED AIR AT YOURSELF OR OTHER PERSONS.

(10) Blow the area to repair with dry and oil free shop air.

SUBTASK 72-21-03-110-009-G00

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THIS PRODUCT. IF NOT, IT CAN CAUSE INJURY.

WARNING ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

PUT ON GLOVES FOR PROTECTION BEFORE YOU TOUCH THE SHARP EDGES. THE SHARP EDGES CAN CUT YOU.

(11) Clean the area to paint with a lint-free cloth, G01043, moist with alcohol, B00676 [CP1041], or degreasing solvent, B50333 [CP2687].

SUBTASK 72-21-03-210-012-G00

(12) Examine the area to paint.

(a) Alteration, embrittlement, segregations, folds, inclusions, cracks, linear indications (length ≥ 4 times the width), facing indications, and corrosion are not permitted.

SUBTASK 72-21-03-530-001-G00

(13) If you will apply primer or oxide film coating more than 6 hours after blending, pack completely the part.

(a) Before packing the part, wait minimum of 30 minutes after the last cleaning.

(b) Use VCI preservation film, G52024 [CP9006].

(c) Use the products applicable on aluminum alloys.

(d) The package must be airtight.

SUBTASK 72-21-03-120-006-G00

(14) If you apply primer or oxide film coating more than 6 hours after the previous blending do these steps:

(a) If it is necessary, apply masking with 3M 472 tape, G51742 [CP2441], or 3M 202 tape, G52019 [CP2185], or 3M 471 tape, G51486 [CP2109], around the area to repair (Figure 808, Figure 809, and Figure 810).

(b) Rub down lightly the area to paint (TASK 70-43-42-300-802-G00).

1) Make sure that you do not remove the base material.

2) Use 320 grit abrasive paper, G52021, or 320 grit abrasive paper, G52023.

(c) If applied, remove the masking.

1) No deterioration of the paint permitted on adjacent areas.

WARNING

USE EYE PROTECTION WHEN YOU USE COMPRESSED AIR TO CLEAN, COOL, OR DRY PARTS OR TOOLS. PARTICLES CAN CAUSE AN INJURY TO YOUR EYES. DO NOT USE MORE THAN 30 PSIG (200 KPA). DO NOT POINT COMPRESSED AIR AT YOURSELF OR OTHER PERSONS.

(d) Blow the area to repair with dry and oil free shop air.

WARNING

REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THIS PRODUCT. IF NOT, IT CAN CAUSE INJURY.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

WARNING

PUT ON GLOVES FOR PROTECTION BEFORE YOU TOUCH THE SHARP EDGES. THE SHARP EDGES CAN CUT YOU.

(e) Clean the area to paint with a lint-free cloth, G01043, moist with alcohol, B00676 [CP1041], or degreasing solvent, B50333 [CP2687].

(f) Examine the area to paint.

1) Alteration, embrittlement, segregations, folds, inclusions, cracks, linear indications (length ≥ 4 times the width), facing indications, and corrosion are not permitted.

SUBTASK 72-21-03-370-010-G00

CAUTION

MAKE SURE THAT YOU DO NOT APPLY PAINT IN AREAS THAT ARE NOT PERMITTED. IF YOU DO, DAMAGE CAN OCCUR.

(15) Apply a thickness of SP350 primer, C50479 [S1316] (TASK 70-64-30-300-801-G00) (Figure 808 and Figure 810).

NOTE: It is not mandatory to process a test piece provided that polymerization, paint thickness and paint adhesion criterias are respected.

NOTE: It is recommended to process a test piece.

(a) It is permitted to stir manually when you prepare SP350 primer, C50479 [S1316], before application but you must obtain a homogeneous product.

NOTE: It is recommended to use an automatic equipment when you prepare the primer.

(b) Apply masking with 3M 472 tape, G51742 [CP2441], or 3M 202 tape, G52019 [CP2185], or 3M 471 tape, G51486 [CP2109], around the area to repair.

(c) Remove the masking before curing.

1) No deterioration of the paint permitted on adjacent areas.

(d) For recoverable paint (if the paint is used as primer and covered by LBYH 203 paint, C50360 [CP2948]), if you choose to cure the part with the cycle of temperature 167°F (75°C) - 185°F (85°C), cure the part for 30 minutes - 1 hour 15 minutes (Figure 809).

NOTE: Better results are achieved if you use the cycle of temperature 167°F (75°C) - 185°F (85°C) when you apply primer.

NOTE: It is recommended to use an oven for curing the paint.

NOTE: It is permitted to apply primer overspray on the finishing paint during primer application without removing it.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING DO NOT BREATHE THE PARTICLES FROM BLENDING OR LET THE PARTICLES TOUCH YOU. THE PARTICLES FROM BLENDING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PROTECTION EQUIPMENT. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR.

(e) If the primer thickness is too high and not between 0.00099 in. (0.025 mm) - 0.00137 in. (0.035 mm), blend the area to have the correct thickness (TASK 70-43-42-300-802-G00).

1) Use 320 grit abrasive paper, G52021, or 320 grit abrasive paper, G52023.

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THIS PRODUCT. IF NOT, IT CAN CAUSE INJURY.

WARNING ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

WARNING PUT ON GLOVES FOR PROTECTION BEFORE YOU TOUCH THE SHARP EDGES. THE SHARP EDGES CAN CUT YOU.

(f) If it is necessary, clean the repaired area with a lint-free cloth, G01043, moist with alcohol, B00676 [CP1041], or degreasing solvent, B50333 [CP2687].

(g) If it is necessary, do a touch-up of the repaired area, do the repair again from the beginning.

(h) Do a visual inspection of the repaired area.

1) Primer overspray on finishing paint is permitted.

2) The appearance of the repaired area must be homogeneous, mat.

3) Shades of primer color are permitted.

SUBTASK 72-21-03-370-011-G00

CAUTION MAKE SURE THAT YOU DO NOT APPLY PAINT IN AREAS THAT ARE NOT PERMITTED. IF YOU DO, DAMAGE CAN OCCUR.

(16) If it is necessary, apply LBYH 203 paint, C50360 [CP2948], on the area to paint (TASK 70-64-29-300-801-G00) (Figure 810).

NOTE: It is not mandatory to process a test piece provided that polymerization, paint thickness and paint adhesion criterias are respected.

NOTE: It is recommended to process a test piece.

(a) Make sure that you follow the limits of the area to paint (TASK 72-23-00-211-804-G00).

(b) It is permitted to stir manually when you prepare LBYH 203 paint, C50360 [CP2948], before application but you must obtain a homogeneous product.

NOTE: It is recommended to use an automatic equipment when you prepare the finishing paint.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Apply masking with 3M 472 tape, G51742 [CP2441], or 3M 202 tape, G52019 [CP2185], or 3M 471 tape, G51486 [CP2109], around the area to repair.

(d) Apply a thickness of LBYH 203 paint, C50360 [CP2948], between 0.00197 in. (0.050 mm) and 0.00275 in. (0.070 mm).

(e) Make sure that you do a smooth transition between repaired area and non-repaired area.

NOTE: It is recommended to have the thickness of the topcoat decreased over 1.18 in. (30 mm) maximum all around the area to do the smooth transition.

WARNING

DO NOT BREATHE THE PARTICLES FROM BLENDING OR LET THE PARTICLES TOUCH YOU. THE PARTICLES FROM BLENDING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PROTECTION EQUIPMENT. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR.

1) If it is necessary, use the same product specified in the repair for blending steps and 120 grit abrasive paper, G52020, or 120 grit abrasive paper, G52022 (TASK 70-43-42-300-802-G00).

(f) After application and before curing, let dry in clean air for 30 minutes minimum.

(g) Remove the masking before curing.

1) No deterioration of the paint permitted on adjacent areas.

(h) For local touch-up of anti-erosion paint, it is permitted to cure the paint with the following alternative cycles:

NOTE: Better results are achieved if you use the cycle of temperature 167°F (75°C) - 185°F (85°C) when you apply anti-erosion paint.

NOTE: It is recommended to use an oven for curing the paint.

- 24 hours minimum at ambient temperature
- 1 hour minimum at 104°F (40°C) - 185°F (85°C).

(i) Make sure that the finishing paint thickness on repaired area and adjacent areas is correct.

WARNING

DO NOT BREATHE THE PARTICLES FROM BLENDING OR LET THE PARTICLES TOUCH YOU. THE PARTICLES FROM BLENDING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PROTECTION EQUIPMENT. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR.

1) If the finishing paint thickness is too high, blend the area to have the correct thickness (TASK 70-43-42-300-802-G00).

a) Use 320 grit abrasive paper, G52021, or 320 grit abrasive paper, G52023.

WARNING

REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THIS PRODUCT. IF NOT, IT CAN CAUSE INJURY.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

WARNING PUT ON GLOVES FOR PROTECTION BEFORE YOU TOUCH THE SHARP EDGES. THE SHARP EDGES CAN CUT YOU.

(j) If it is necessary, clean the repaired area with a lint-free cloth, G01043, moist with alcohol, B00676 [CP1041], or degreasing solvent, B50333 [CP2687].

(k) If it is necessary, do a touch-up of the repaired area, do the repair again from the beginning.

(l) Examine the repaired area (TASK 72-23-00-211-804-G00).

1) In the area exposed to the air flow, the transition between the repaired area and adjacent areas must be smooth (Figure 811).

2) Shades of black are permitted on anti-erosion paint.

3) Primer must not be visible after application of finishing paint.

SUBTASK 72-21-03-110-010-G00

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THIS PRODUCT. IF NOT, IT CAN CAUSE INJURY.

WARNING ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

WARNING PUT ON GLOVES FOR PROTECTION BEFORE YOU TOUCH THE SHARP EDGES. THE SHARP EDGES CAN CUT YOU.

(17) If you blended anti-erosion paint on the repaired area, clean the repaired area with a lint-free cloth, G01043, moist with alcohol, B00676 [CP1041], or degreasing solvent, B50333 [CP2687].

SUBTASK 72-21-03-300-003-G00

CAUTION DO THE TOUCH UP OF THE OXIDE LAYER IN FOUR HOURS OR LESS AFTER THE COATING REMOVAL. IF YOU DO NOT, DAMAGE CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

DO NOT APPLY THE OXIDE LAYER ON THE PRIMER AND PAINT THAT STAY. DAMAGE TO THE PART CAN OCCUR.

(18) If it is not possible to paint the areas with primer, do a local touch-up of oxide film coating only on the counterbores and bolt holes (TASK 70-43-07-300-801-G00) (Figure 808).

NOTE: It is also permitted to apply touch up of the oxide film coating before application of the primer and anti-erosion paint.

SUBTASK 72-21-03-420-005-G00

(19) If removed, install the new rivets and nut plate or gang channel attached to the aft conical flange on the OGV (TASK 72-21-03-960-801-G00).

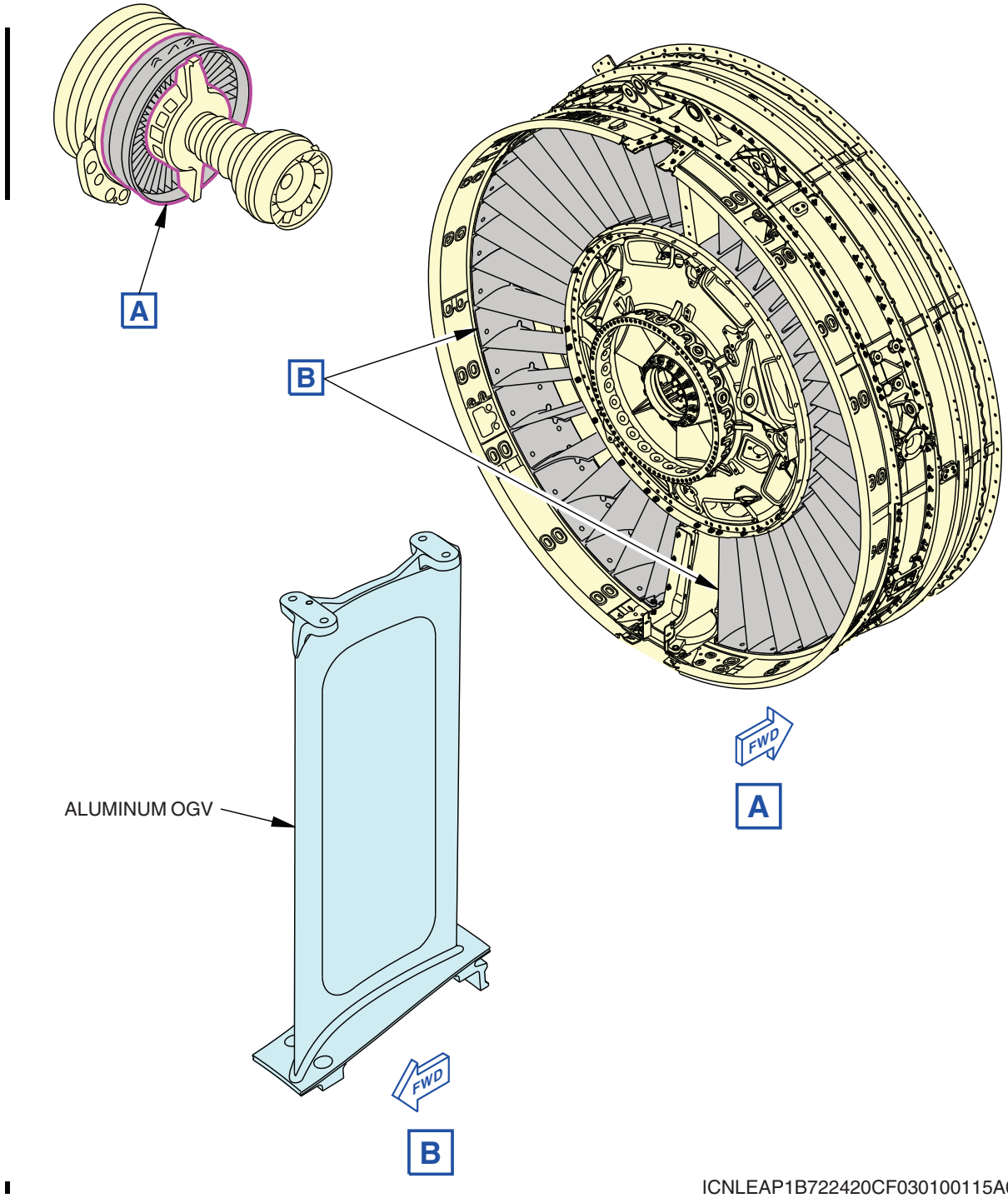
H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-03-410-008-G00

(1) Do this task: Outlet Guide Vane Installation, TASK 72-21-03-400-801-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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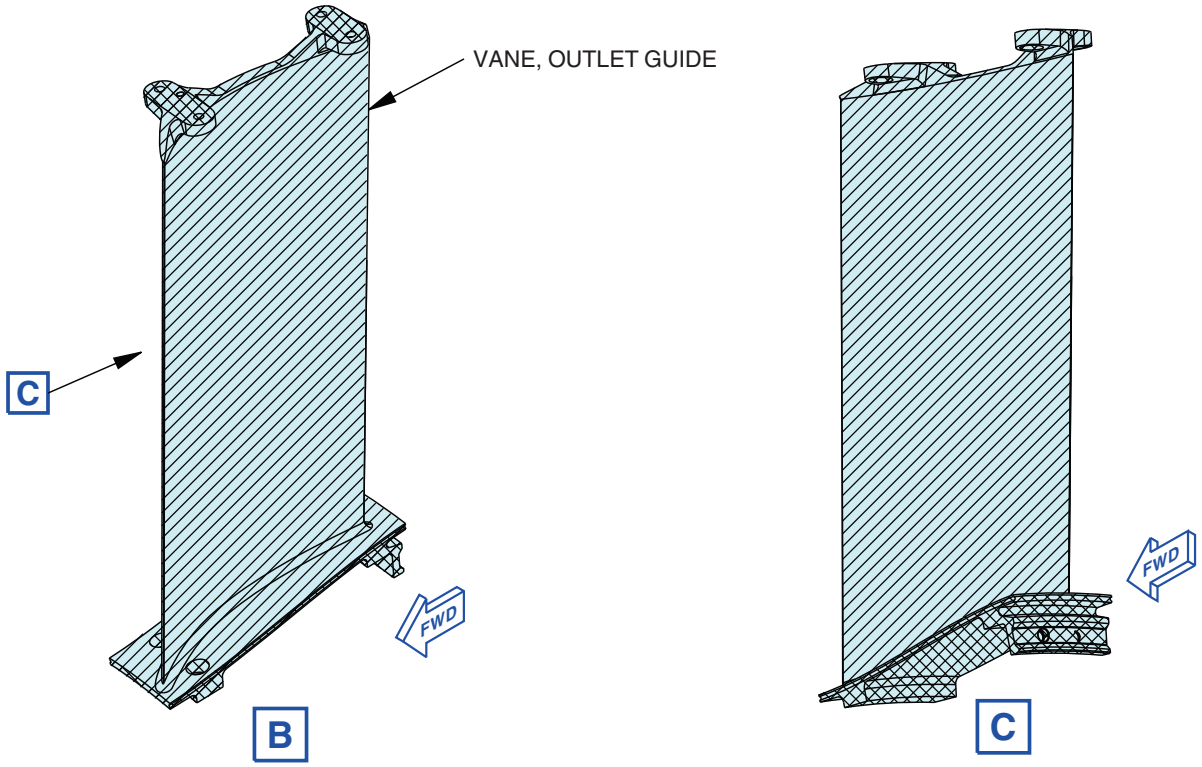
2806516 S0000641166_V2

Outlet Guide Vane
Figure 807/72-21-03-990-821-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

PRIMER ANTI CORROSION OR PRIMER ANTI CORROSION AND OXIDE FILM COATING (ONLY ON COUNTERBORE AND BOLT HOLES)

ANTI-EROSION PAINT

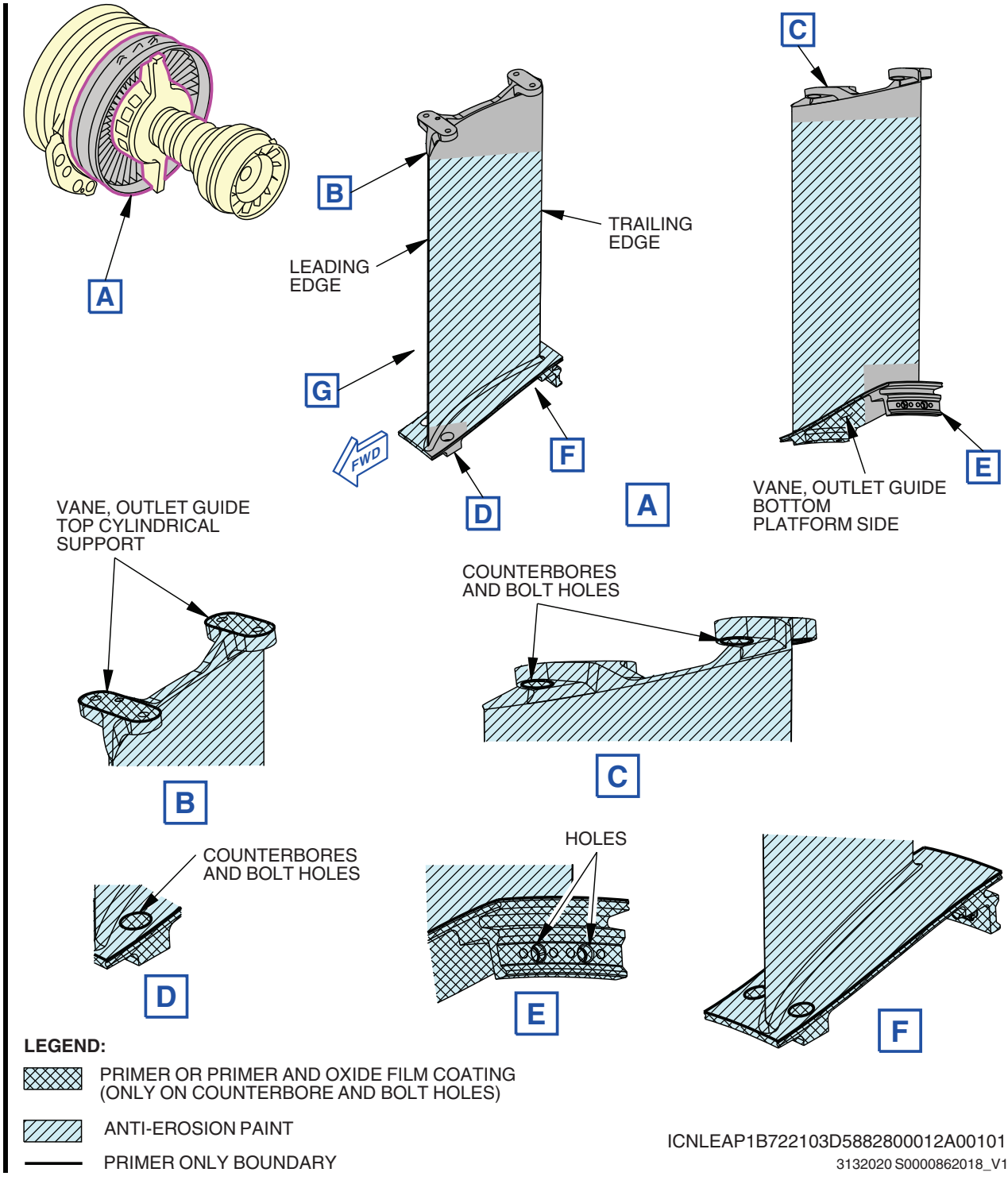
NOTE:

MATERIAL OF THE OGV: 2219

ICNLEAP1B722103D5882800011A00101
3132010 S0000862017_V1

Outlet Guide Vane
Figure 807/72-21-03-990-821-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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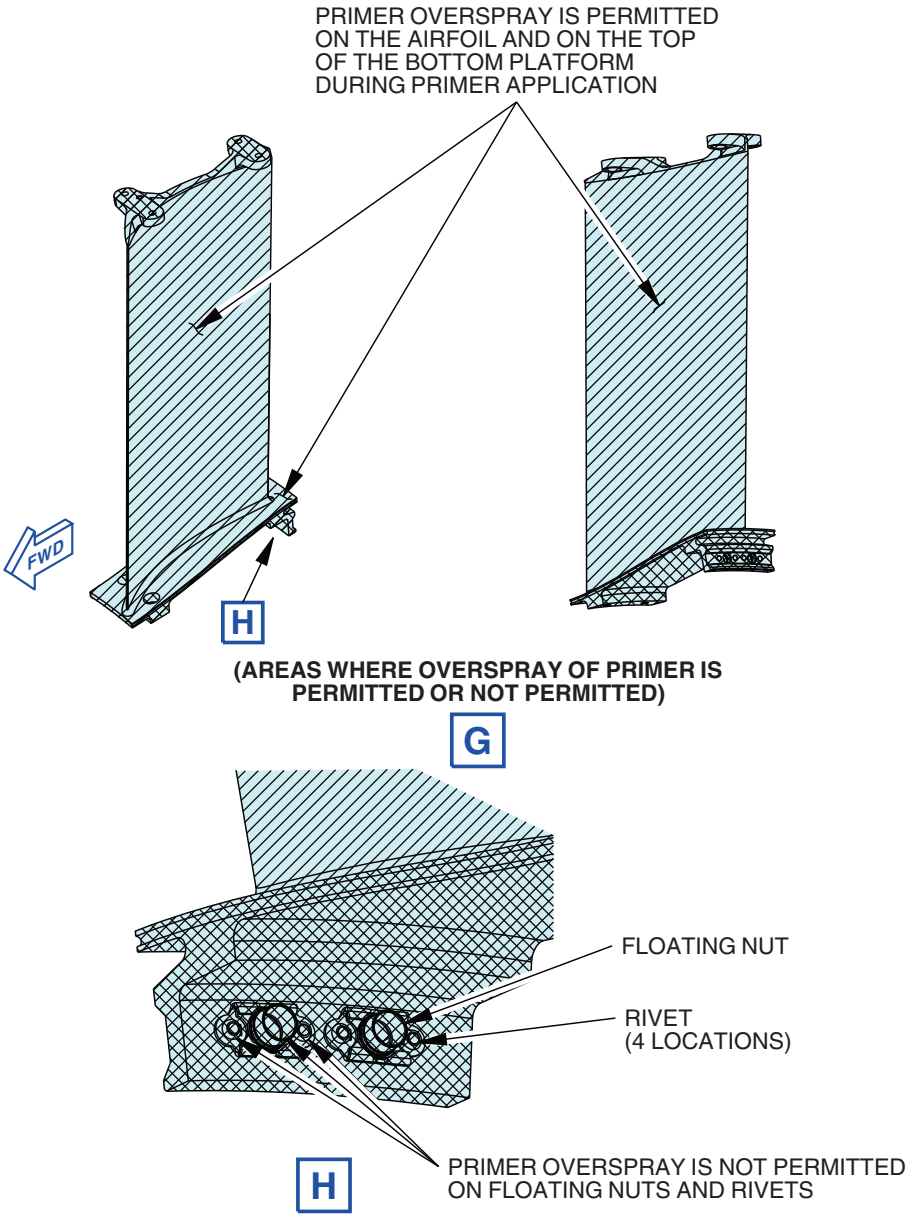
3132020 S0000862018_V1

Painted Areas with Primer or Touched-Up by Oxide Film Coating
Figure 808/72-21-03-990-822-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

PRIMER OR PRIMER AND OXIDE FILM COATING
(ONLY ON COUNTERBORE AND BOLT HOLES)

ANTI-EROSION PAINT

NOTE:

PRIMER OVERSPRAY IS PERMITTED:
-ON ANTI-EROSION PAINT (AIRFOIL AND TOP OF
THE BOTTOM PLATFORM INCLUDED)

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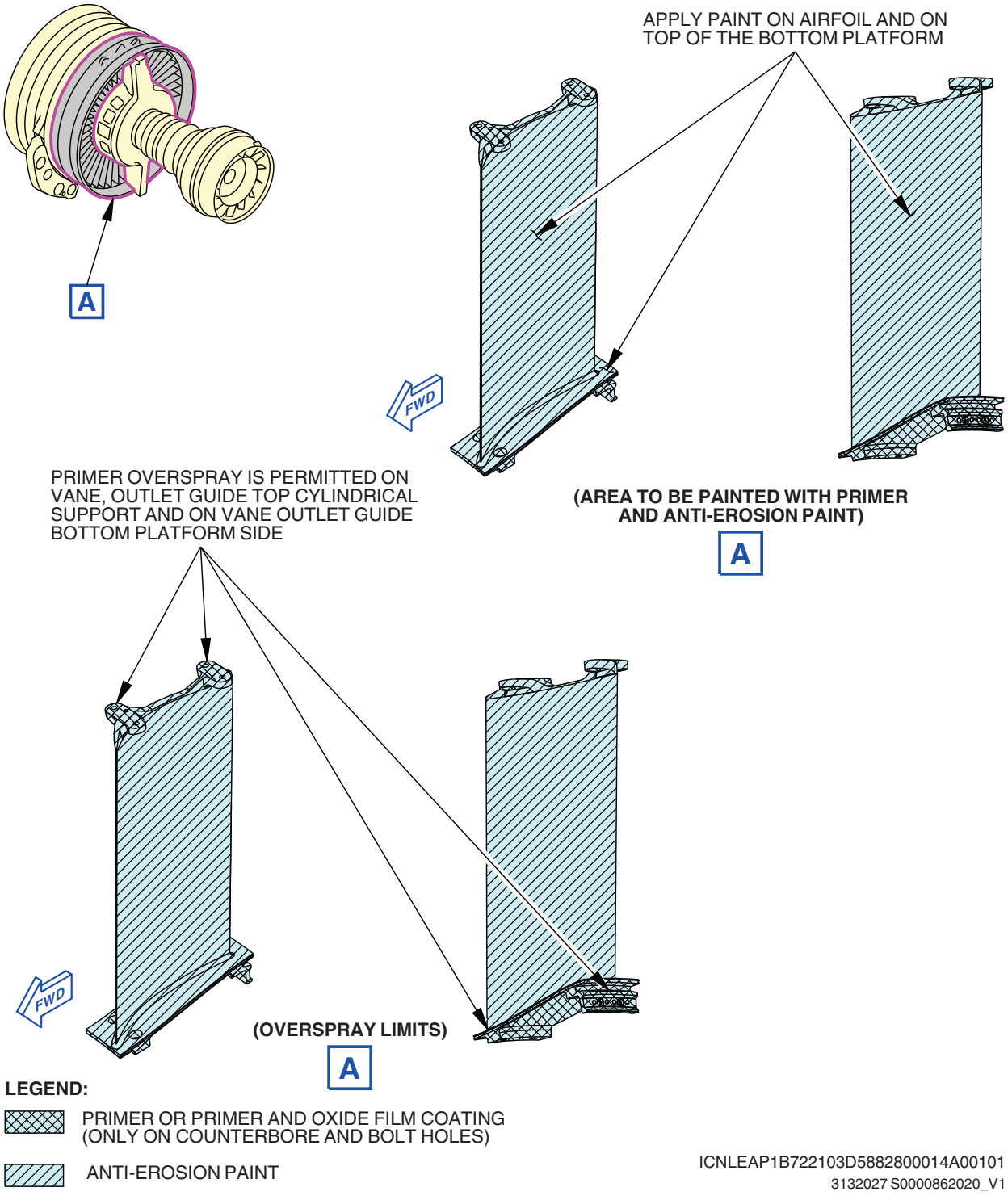
3132026 S0000862019_V1

Painted Areas with Primer or Touched-Up by Oxide Film Coating
Figure 808/72-21-03-990-822-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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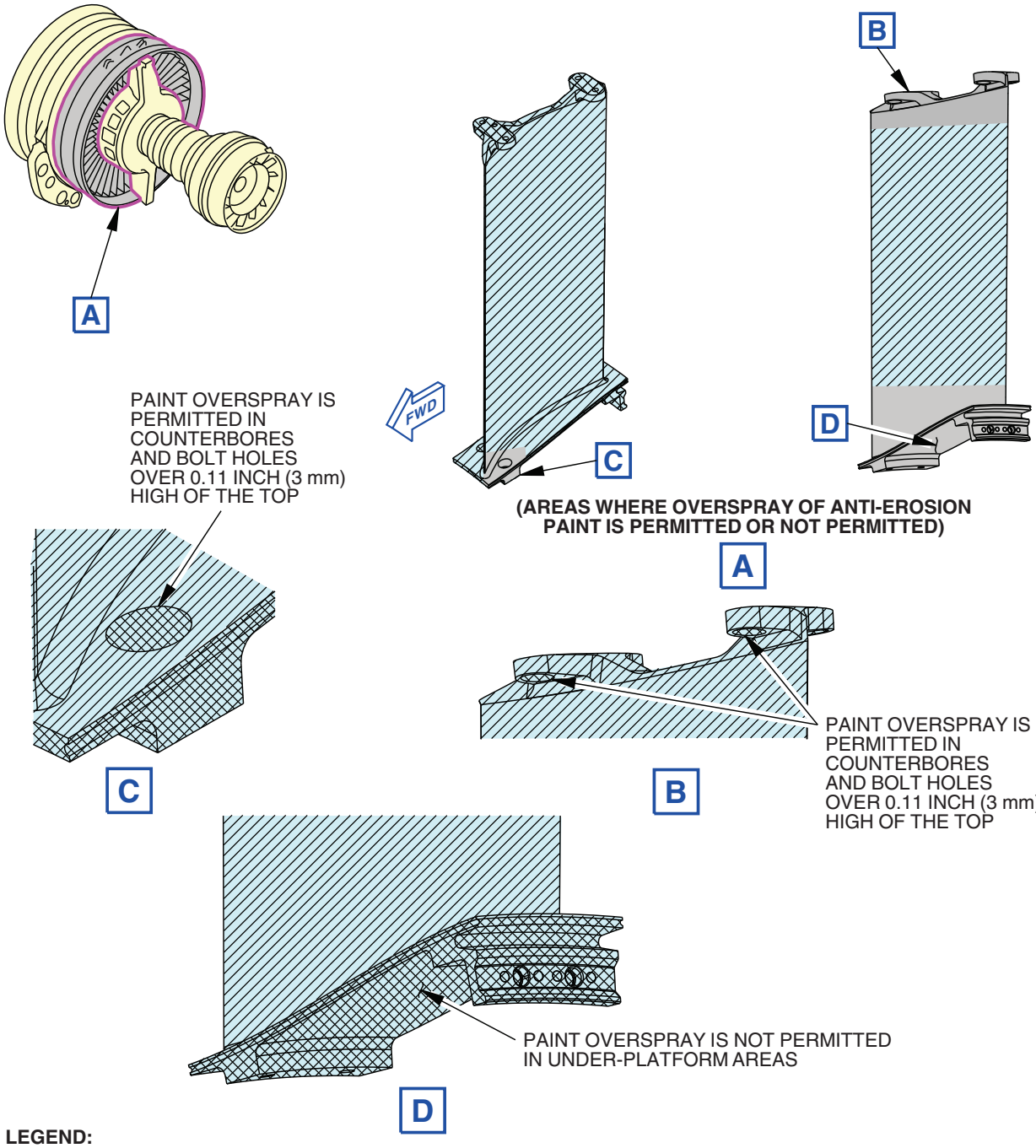
3132027 S0000862020_V1

Painted Areas with Primer or Touched-Up by Oxide Film Coating
Figure 809/72-21-03-990-823-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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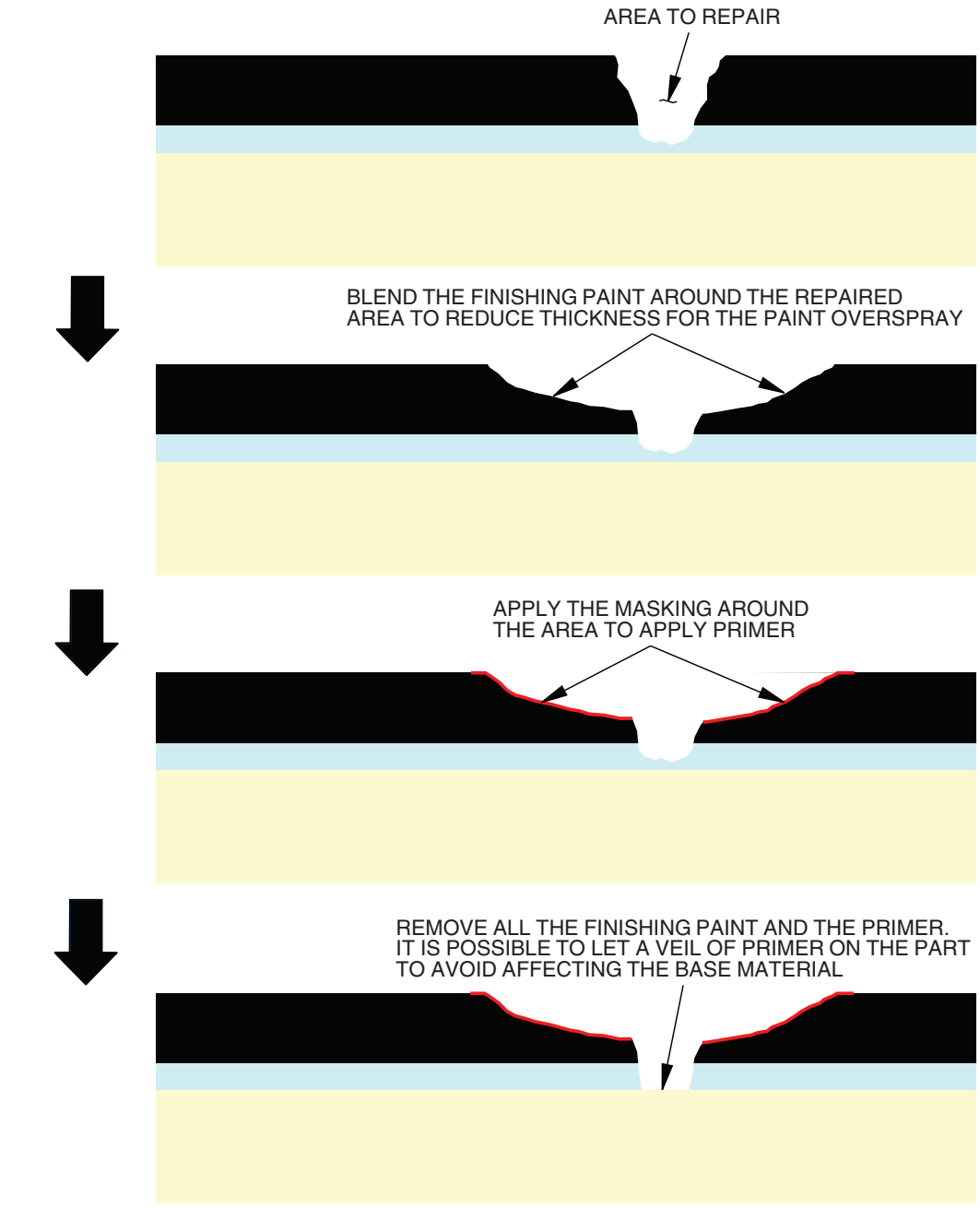
3132029 S0000862021_V1

Painted Areas with Primer or Touched-Up by Oxide Film Coating
Figure 809/72-21-03-990-823-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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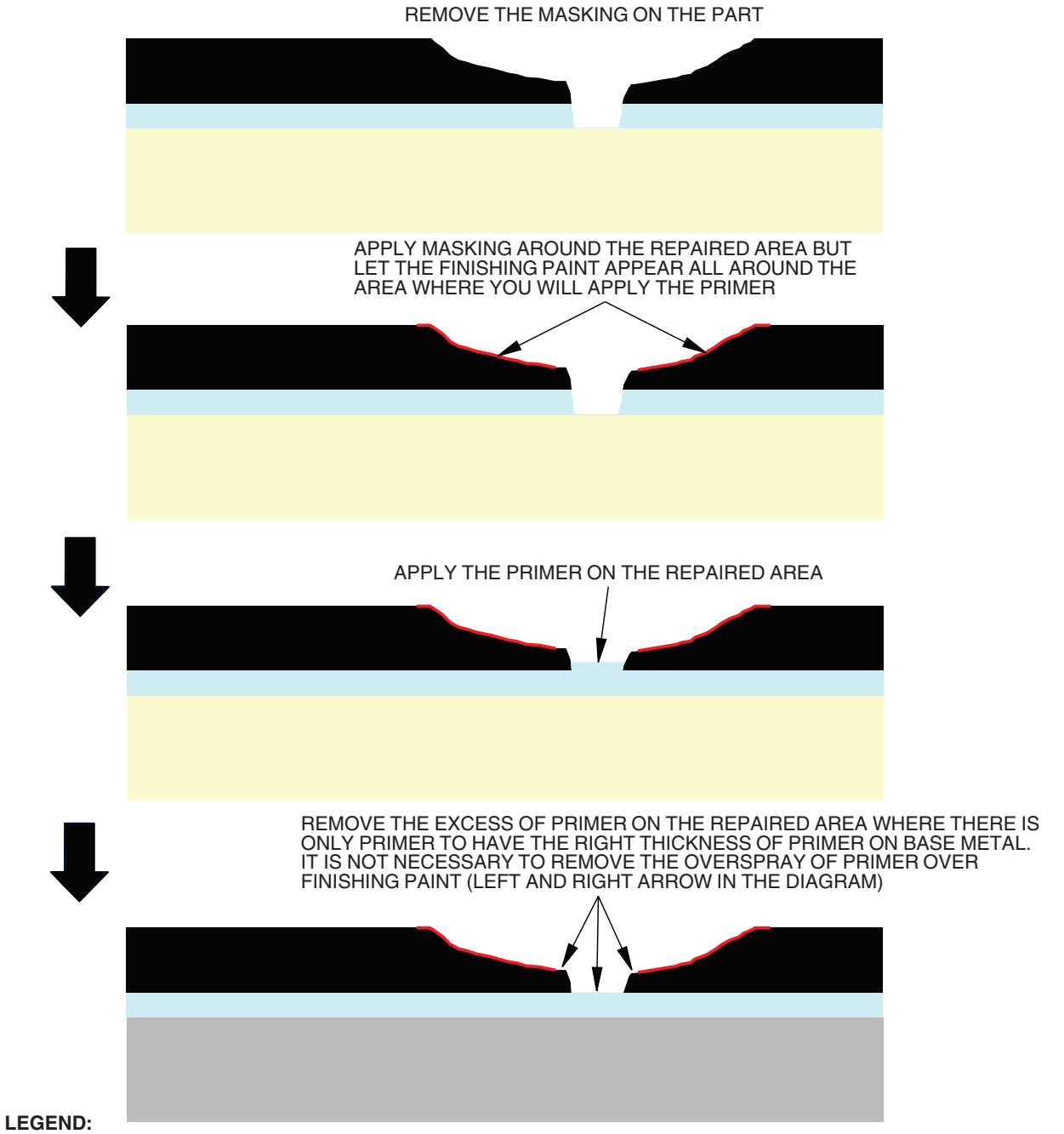
3132039 S0000862022_V1

Illustrated Steps of the Repair
Figure 810/72-21-03-990-824-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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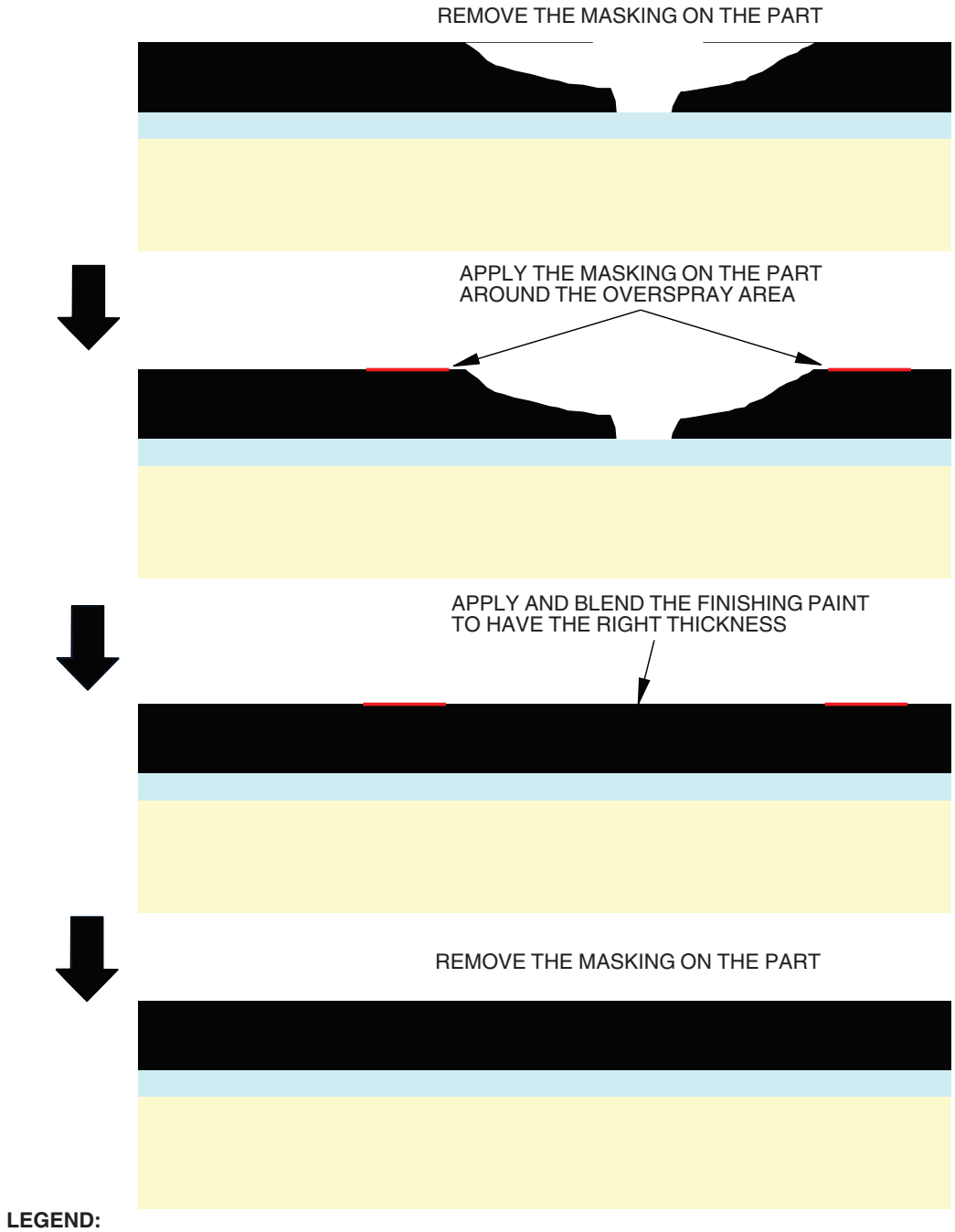
3132040 S0000862023_V1

Illustrated Steps of the Repair
Figure 810/72-21-03-990-824-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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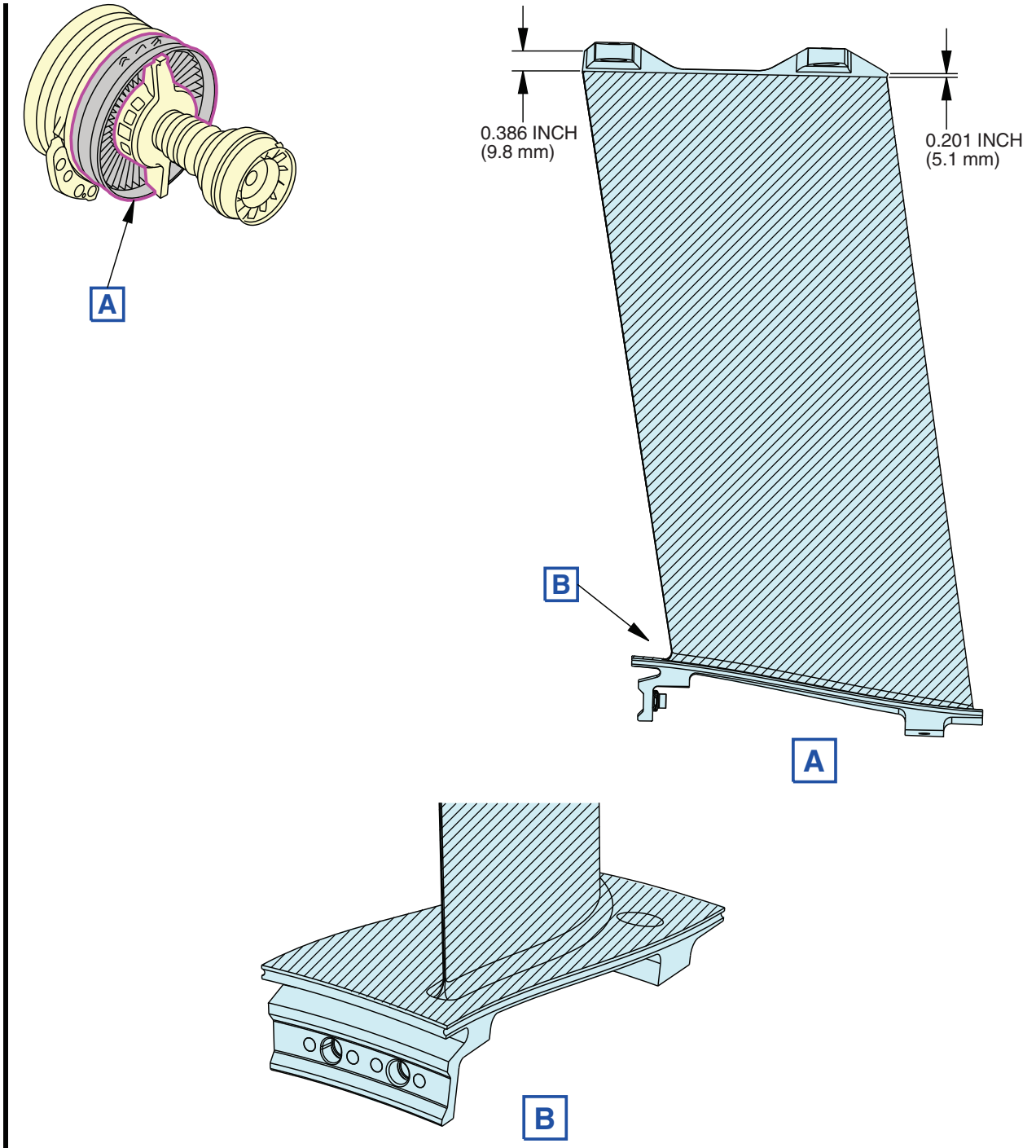
3132044 S0000862024_V1

Illustrated Steps of the Repair
Figure 810/72-21-03-990-824-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Area of the OGV Exposed to the Air Flow
Figure 811/72-21-03-990-825-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-21-03-370-802-G00

6. Outlet Guide Vanes Paint Primer Local Repair

(Figure 812 or Figure 813 or Figure 814 or Figure 815)

A. General

(1) This task gives the instructions how to repair the primer (anti-corrosion paint) of the Outlet Guide Vanes (OGV) with it removed.

B. References

Reference	Title
72-21-03-000-801-G00	Outlet Guide Vane Removal (P/B 401)
72-21-03-400-801-G00	Outlet Guide Vane Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
A50488 [CP2724]	Primer - Epoxy, Polyamide - PCEH 100	
B00676 [CP1041]	Alcohol - Isopropyl	
B50333 [CP2687]	Solvent - Degreasing, Diestone DLS	
C50361 [CP2286]	Primer - Epoxy, Adhesive - BR-127	
C50362 [CP2200 and CP2201]	Paint - Finishing, Anti-Erosion - PRIAM 32005 (Part A and B)	
G01043	Cloth - Lint-free	
G50143 [CP2208]	Tape - Glass Cloth with Silicone Adhesive, High Temperature, Abrasive Resistant - 3M 361	
G51168 [C10-189]	Tape - Adhesive, 3M #250	
G51412 [CP2721]	Cloth - Abrasive, Silicone Carbide - CS 333X	

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Repair

SUBTASK 72-21-03-010-007-G00

(1) Remove the OGV that you will repair (TASK 72-21-03-000-801-G00).

NOTE: The table below gives the list of the part numbers that are applicable to this repair and the configurations which corresponds.

OGV Part Numbers/Configurations

Part Number	Service Bulletin	CSN	Configuration*[1]
364-052-011-0	-	(01-290, 72-24-00) (01-660, 72-24-00)	1
364-052-611-0	-	(01-350, 72-24-00)	1
364-052-711-0	-	(01-360, 72-24-00)	1
364-052-811-0	-	(01-370, 72-24-00)	1
364-068-111-0	-	(01-290, 72-24-00) (01-660, 72-24-00)	2

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OGV Part Numbers/Configurations (Continued)

Part Number	Service Bulletin	CSN	Configuration*[1]
364-068-211-0	-	(01-300, 72-24-00)	2
364-068-311-0	-	(01-310, 72-24-00) (01-670, 72-24-00)	2
364-068-411-0	-	(01-320, 72-24-00)	2
364-068-511-0	-	(01-330, 72-24-00) (01-380, 72-24-00)	2
364-068-611-0	-	(01-340, 72-24-00) (01-690, 72-24-00)	2
364-068-711-0	-	(01-350, 72-24-00)	2
364-068-811-0	-	(01-360, 72-24-00)	2
364-068-911-0	-	(01-370, 72-24-00)	2

*[1] Configuration 1: Aluminum welded Outlet Guide Vanes
Configuration 2: Aluminum bonded Outlet Guide Vanes

SUBTASK 72-21-03-940-004-G00

(2) Make sure that you have the equipment and conditions for the preparation and application of the different coatings:

(a) There is sufficient airflow in the area.

(b) The area is free of dust.

(c) Protective safety masks are worn.

(d) There is compressed air for the spray guns that is dry, clean and free of oil.

(e) Mechanical stirrer to mix the paint are available.

(f) There are heating facilities (e.g. ovens, stoves, infrared equipment, etc.) capable of curing per instructions in the repair.

SUBTASK 72-21-03-940-005-G00

(3) Make sure that you follow these conditions for the preparation and application of the different coatings:

(a) The temperature is between 59°F (15°C) and 95°F (35°C) with recording system.

(b) The humidity is between 30% and 80% with recording system.

SUBTASK 72-21-03-110-002-G00

WARNING

ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

WARNING

PUT ON GLOVES FOR PROTECTION BEFORE YOU TOUCH THE SHARP EDGES. THE SHARP EDGES CAN CUT YOU.

(4) Clean the part you removed with a lint-free cloth, G01043, moisten with isopropyl alcohol, B00676 [CP1041], or Diestone DLS degreasing solvent, B50333 [CP2687], cleaning solvent.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-21-03-210-005-G00

(5) Compare the color of the primer on the OGV foot with the OGV Color Chart Check, find and record the primer to apply.

SUBTASK 72-21-03-940-006-G00

(6) Make sure that you use the primer before its pot life expires as follows:

NOTE: Cure the product between 64°F (18°C) and 95°F (35°C) for 24 hours before you use it.

(a) For Priam PCEH 100 primer, A50488 [CP2724], the pot life is 3 hours in the required conditions.

(b) For PRIAM 32005 paint, C50362 [CP2200 and CP2201], the pot life is 3 hours in the required conditions.

(c) For BR-127 primer, C50361 [CP2286], the pot life is 5 days at a maximum 90°F (32°C).

SUBTASK 72-21-03-370-008-G00

(7) Mix primer SP350 in these proportions:

(a) 3.53 oz (100 g) base of the SP350 primer

(b) 0.39 oz (11 g) hardening agent of the SP350 primer

(c) A maximum of 0.35 oz (10 g) thinner.

NOTE: The pot life of the SP350 primer is three hours between 59°F (15°C) and 86°F (30°C) and one hour between 86°F (30°C) and 95°F (35°C).

NOTE: After you mix the primer, close the pot to have a better shelf life of the mixture.

F. Outlet Guide Vanes Paint Primer Local Repair

SUBTASK 72-21-03-940-007-G00

(1) Make sure that you paint the primer within the specified limits of the area.

NOTE: Overspray is permitted in the counter-bores and bolt holes. Overspray is not permitted on floating nuts and rivets.

SUBTASK 72-21-03-110-003-G00

WARNING

ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

WARNING

PUT ON GLOVES FOR PROTECTION BEFORE YOU TOUCH THE SHARP EDGES. THE SHARP EDGES CAN CUT YOU.

(2) Clean the area that you will paint with a lint-free cloth, G01043, moisten with isopropyl alcohol, B00676 [CP1041], or Diestone DLS degreasing solvent, B50333 [CP2687], cleaning solvent (SPM LEAP-00-70-21-11-01A-251A-D).

SUBTASK 72-21-03-950-001-G00

(3) Mask the area that you will not paint with 3M 361 tape, G50143 [CP2208], or 3M tape #250, G51168 [C10-189], as follows:

(a) Remove minimal coating more than the missing paint to have remaining paint quality.

(b) Make sure that you apply the Priam PCEH 100 primer, A50488 [CP2724], or BR-127 primer, C50361 [CP2286], or SP350 (Primer Color Brown) as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) For Priam PCEH 100 primer, A50488 [CP2724], or BR-127 primer, C50361 [CP2286], make sure that the surface is less than or equal 1 in^2 (7 cm^2).

NOTE: There is no limit for primer SP350.

SUBTASK 72-21-03-150-001-G00

CAUTION

DO THE TOUCH UP OF THE OXIDE LAYER IN FOUR HOURS OR LESS AFTER THE COATING REMOVAL. IF YOU DO NOT, DAMAGE CAN OCCUR.

(4) Locally remove the finishing paint, the primer and the anodizing with one of these procedures:

NOTE: Make sure that you do not remove the parent material.

(a) Locally remove the finishing paint, the primer and the anodizing with fruit stones (SPM LEAP-00-70-22-15-01A-252A-D).

(b) Locally remove the finishing paint, the primer and the anodizing with organic compound (SPM LEAP-00-70-22-18-01A-252A-D).

(c) Rub down lightly with strips of abrasive cloth, G51412 [CP2721], grade 400 minimum (SPM LEAP-00-70-41-11-01A-655B-D).

SUBTASK 72-21-03-950-002-G00

(5) If the surface to repair is larger than the area to paint, do these steps:

(a) Remove the masking.

(b) Make the remaining primer rough with strips of abrasive cloth, G51412 [CP2721], CS 333 X, grade 400 minimum (SPM LEAP-00-70-22-15-01A-252A-D).

NOTE: This will prepare the area for painting.

(c) Make sure that the overspray is not larger than 1.18 in. (30 mm) as specified in DIM. S.

SUBTASK 72-21-03-160-006-G00

(6) Clean the area that you will paint as follows:

(a) Remove the masking tape.

WARNING

USE EYE PROTECTION WHEN YOU USE COMPRESSED AIR TO CLEAN, COOL, OR DRY PARTS OR TOOLS. PARTICLES CAN CAUSE AN INJURY TO YOUR EYES. DO NOT USE MORE THAN 30 PSIG (200 KPA). DO NOT POINT COMPRESSED AIR AT YOURSELF OR OTHER PERSONS.

(b) Blow the surface with compressed air that is dry, clean and free of oil.

WARNING

ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

(c) Clean the part with a lint-free cloth, G01043 moisten with isopropyl alcohol, B00676 [CP1041] or Diestone DLS degreasing solvent, B50333 [CP2687] (SPM LEAP-00-70-21-11-01A-251A-D).

SUBTASK 72-21-03-210-006-G00

(7) Examine the area to where you will do the touch-up for these conditions that are not permitted:

(a) Dents, nicks, scratches

(b) Erosion

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Corrosion

(d) Cracks

SUBTASK 72-21-03-940-008-G00

(8) Prepare a test specimen in aluminum 2219 as follows:

(a) Process a test specimen (strip of sheet metal), along with the part to paint, for quality assurance purposes.

(b) Identify the correct primer with color and prepare the test specimen as follows:

1) Compare the color of the primer on the OGV foot with the OGV Color Chart Check, find and record the good primer to apply.

2) For Priam PCEH 100 primer, A50488 [CP2724], or BR-127 primer, C50361 [CP2286], rub down lightly the test specimen with strips of abrasive cloth, G51412 [CP2721], CS 333 X, grade 400 minimum (SPM LEAP-00-70-41-11-01A-655B-D).

3) For primer SP350 (Primer Color Brown), do the dry abrasive blast with (CP2404) Corundum White 180 Micrometers Only (SPM LEAP-00-70-51-12-01A-651A-D).

WARNING ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

(c) Clean the test specimen with a lint-free cloth, G01043, moisten with isopropyl alcohol, B00676 [CP1041], or Diestone DLS degreasing solvent, B50333 [CP2687].

(d) Measure and record the thickness of the test specimen before you apply the primer to determine the required thickness.

SUBTASK 72-21-03-370-009-G00

CAUTION MANUFACTURER'S INSTRUCTIONS, SHELF LIFE AND POT LIFE TIMES MUST BE OBEYED.

(9) Apply the primer on the test specimen and on the part as follows:

WARNING ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

(a) Clean the area you will paint with a lint-free cloth, G01043, moisten with isopropyl alcohol, B00676 [CP1041], or Diestone DLS degreasing solvent, B50333 [CP2687].

(b) Make sure that the shelf life of the mixture is not expired.

(c) Apply and do a check for correct application of the selected primer per the color chart check as follows:

1) For Priam PCEH 100 primer, A50488 [CP2724], (configuration 1), do these steps (SPM LEAP-00-70-62-14-01A-257A-D):

CAUTION DO THE TOUCH UP WITH A PEN IN LESS THAN 72 HOURS. DAMAGE TO THE PART CAN OCCUR IF YOU DO THE TOUCH UP AFTER 72 HOURS.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION DO NOT APPLY THE OXIDE LAYER ON THE PRIMER AND PAINT THAT STAY. DAMAGE TO THE PART CAN OCCUR.

a) Do the spot touch-up of the oxide film coating of the area to paint with a pen (SPM LEAP-00-70-62-14-01A-644A-D).

NOTE: Let dry in clean air.

b) If it is necessary, apply new masking on the areas that you will not paint as follows:

<1> Remove the masking.

WARNING ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

<2> Clean the area you will paint with a lint-free cloth, G01043, moisten with isopropyl alcohol, B00676 [CP1041], or Diestone DLS degreasing solvent, B50333 [CP2687].

<3> Apply 3M 361 tape, G50143 [CP2208], or 3M tape #250, G51168 [C10-189], new masking on the areas not to paint.

c) Apply Priam PCEH 100 primer, A50488 [CP2724], to a thickness between 0.00079 in. (0.020 mm) and 0.00118 in. (0.030 mm) on the area that you will paint and on the test specimen at the same time.

<1> If it is applicable, do these steps:

<a> Remove the masking.

WARNING ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

<b> Clean the area that you will paint with a lint-free cloth, G01043, moisten with isopropyl alcohol, B00676 [CP1041], or Diestone DLS degreasing solvent, B50333 [CP2687].

<c> Apply the primer with overspray..

<2> Make sure that the overspray area is not larger than 1.18 in. (30 mm) as specified in DIM. S.

<3> Make sure that the thickness of the primer decreases to 0 in (0 mm) in the transition paint area.

d) Let dry in clean air and cure the painted area as follows:

<1> Make sure that the part stays at a temperature between 167°F (75°C) and 185°F (85°C) during curing.

NOTE: Use of an infrared system with feed-back control for curing is permitted.

e) Examine the repaired area as follows (SPM LEAP-00-70-62-14-01A-257A-D):

<1> Measure the thickness of the primer on the test specimen.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

<2> Do a visual examination of the repaired area.

<3> Make sure that after polymerization of the primer the roughness of the part is less than 63 microinches (1.6 micrometers).

<4> Do a local peeling-off test for paint stripping with Q1163 adhesive tape.

<5> If the thickness of the primer is in the limits given before, remove all masking tape.

2) For BR-127 primer, C50361 [CP2286], (configuration 2), do these steps:

CAUTION DO THE TOUCH UP WITH A PEN IN LESS THAN 72 HOURS. DAMAGE TO THE PART CAN OCCUR IF YOU DO THE TOUCH UP AFTER 72 HOURS.

CAUTION DO THE TOUCH UP OF THE OXIDE LAYER IN FOUR HOURS OR LESS AFTER THE COATING REMOVAL. IF YOU DO NOT, DAMAGE CAN OCCUR.

a) Do the spot touch-up of the oxide film coating of the area to paint with a pen (SPM LEAP-00-70-62-14-01A-644A-D).

NOTE: Let dry in clean air.

b) If it is necessary, apply new masking on the areas you will not paint as follows:

<1> Remove the masking.

WARNING ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

<2> Clean the area you will paint with a lint-free cloth, G01043, moisten with isopropyl alcohol, B00676 [CP1041], or Diestone DLS degreasing solvent, B50333 [CP2687].

<3> Apply 3M 361 tape, G50143 [CP2208], or 3M tape #250, G51168 [C10-189], new masking on the areas not to paint.

c) Apply BR-127 primer, C50361 [CP2286] to a thickness between 0.00020 in. (0.005 mm) and 0.00118 in. (0.030 mm) on the area that you will paint and on the test specimen at the same time.

<1> If it is applicable, do these steps:

<a> Remove the masking.

WARNING ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

<b> Clean the area you will paint with a lint-free cloth, G01043 moisten with isopropyl alcohol, B00676 [CP1041], or Diestone DLS degreasing solvent, B50333 [CP2687].

<c> Apply the primer with overspray..

<2> Make sure that the overspray area is not larger than 1.18 in. (30 mm) as specified in DIM. S.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

<3> Make sure that the thickness of the primer decreases to 0 in (0 mm) in the transition paint area.

d) Let dry in clean air for a minimum of 30 minutes to a maximum of 45 minutes.

e) Cure the painted area at a temperature between 167°F (75°C) and 185°F (85°C) for one hour.

NOTE: Make sure that the part stays between 167°F (75°C) and 185°F (85°C) during curing.

NOTE: Use of an infrared system with feed-back control for curing is permitted.

f) Examine the repaired area as follows (SPM LEAP-00-70-62-14-01A-257A-D):

<1> Measure the thickness of the primer on the test specimen.

<2> Primer missing in the repaired area is not permitted.

<3> Paint step is not permitted.

<4> The color and appearance of the repaired area must be homogenous, mat.

<5> The repaired area must be smooth and free of run, pitting, pinhole, bump, crater or blister.

<6> The repaired area must not have inclusions of foreign object or other defect that affects the paint quality.

<7> The repaired area must not have isolated pitting or craters.

<8> Make sure that after polymerization of the primer the roughness of the part is less than 59 micro-inches (1.5 micro-meters)

<9> Do a local peeling-off test for paint stripping with Q1163 adhesive tape.

<a> Paint tearing is not permitted.

<10> If the thickness of the primer is in the limits as before, remove the masking.

3) For primer SP350 (Primer Color Brown), do these steps:

a) Mix 3.53 oz (100 g) of the base SP350 primer with 0.39 oz (11 g) SP350 primer hardening agent and a maximum of 0.35 oz (10 g) thinner.

NOTE: The pot life of the SP350 primer is three hours between 59°F (15°C) and 86°F (30°C) and one hour between 86°F (30°C) and 95°F (35°C).

NOTE: After you mix the primer, close the pot to have a better shelf life of the mixture.

b) Do the abrasive blast on the area to paint with (CP2404) Corundum White 180 micrometers only (SPM LEAP-00-70-51-12-01A-651A-D).

c) If it is necessary, apply new masking on the areas that you will not paint as follows:

<1> Remove the masking.

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WARNING ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

<2> Clean the area you will paint with a lint-free cloth, G01043 moisten with isopropyl alcohol, B00676 [CP1041], or Diestone DLS degreasing solvent, B50333 [CP2687].

<3> Apply 3M 361 tape, G50143 [CP2208], or 3M tape #250, G51168 [C10-189], new masking on the areas not to paint.

d) Apply a thickness of the primer SP350 between 0.00099 in. (0.025 mm) and 0.00137 in. (0.035 mm) on the area to paint and on the test specimen at the same time and as follows:

<1> If it is applicable, do these steps:

<a> Remove the masking.

WARNING ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

<b> Clean the area that you will paint with a lint-free cloth, G01043, moisten with isopropyl alcohol, B00676 [CP1041], or Diestone DLS degreasing solvent, B50333 [CP2687].

<c> Apply the primer with overspray.

<2> Make sure that the overspray area is not larger than 1.18 in. (30 mm) as specified in DIM. S.

<3> Make sure that the thickness of the primer decreases to 0 in (0 mm) in the transition paint area.

e) Let dry in clean air for 15 minutes.

f) Cure the painted area at a temperature between 167°F (75°C) and 185°F (85°C) for one hour and thirty minutes.

NOTE: Make sure that the part stays between 167°F (75°C) and 185°F (85°C) during curing.

NOTE: Use of an infrared system with feed-back control for curing is permitted.

g) Examine the repaired area as follows:

<1> Measure the thickness of the primer on the test specimen.

<2> Primer missing in the repaired area is not permitted.

<3> Paint step is not permitted.

<4> The color and appearance of the repaired area must be homogenous, mat.

<5> The repaired area must be smooth and free of run, pitting, pinhole, bump, crater or blister.

<6> The repaired area must not have inclusions of foreign object or other defect that affects the paint quality.

<7> The repaired area must not have isolated pitting or craters.

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<8> Make sure that after polymerization of the primer the roughness of the part is less than 59 microinches (1.5 micrometers)

<9> Do a local peeling-off test for paint stripping with Q1163 adhesive tape.

<a> Paint tearing is not permitted.

<10> If the thickness of the primer is in the limits as before, remove the masking.

SUBTASK 72-21-03-110-004-G00

WARNING

ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

(10) Clean the painted area with a lint-free cloth, G01043, moisten with isopropyl alcohol, B00676 [CP1041], or Diestone DLS degreasing solvent, B50333 [CP2687] (SPM LEAP-00-70-21-11-01A-251A-D).

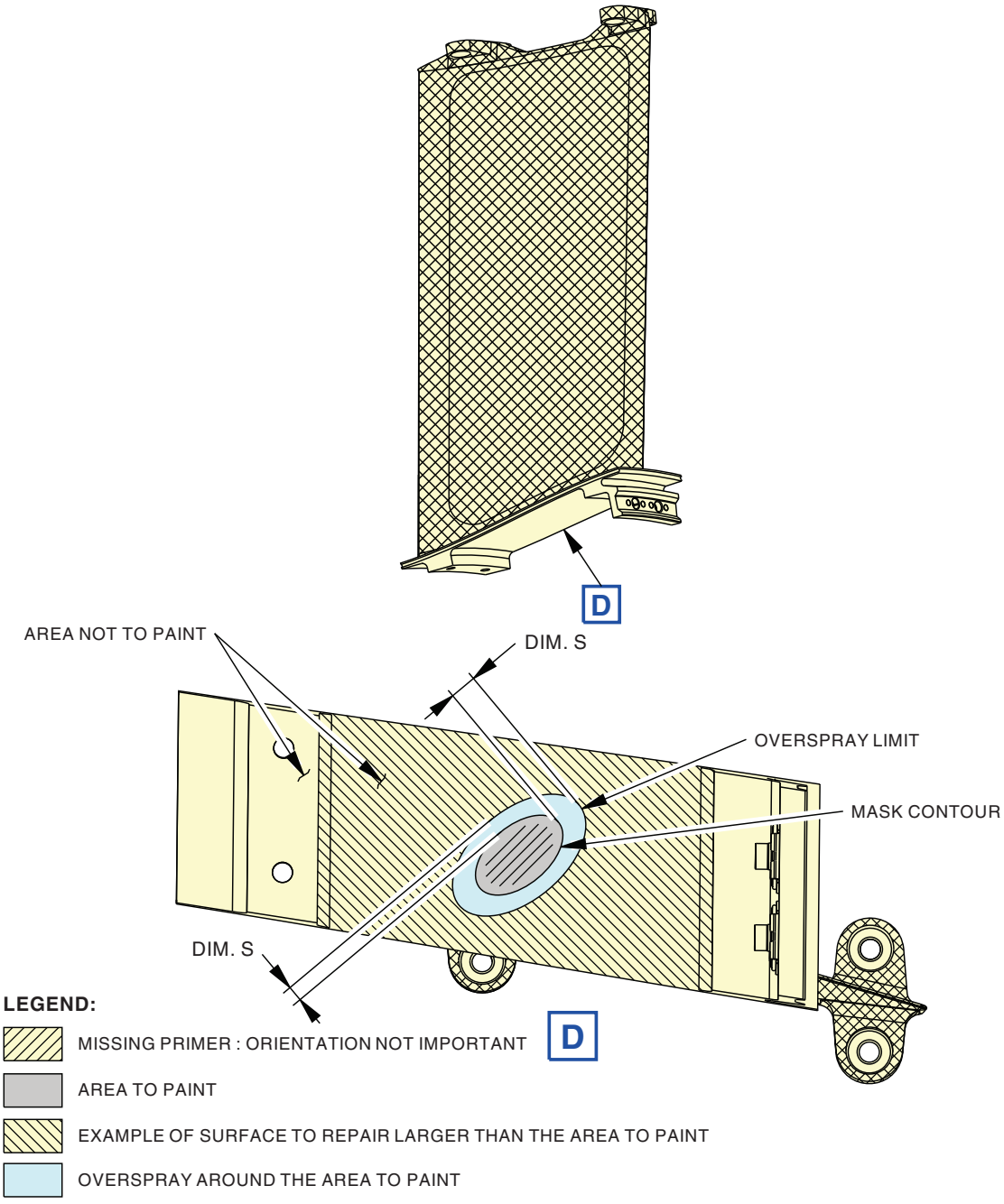
G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-03-410-009-G00

(1) Install the OGV (TASK 72-21-03-400-801-G00).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



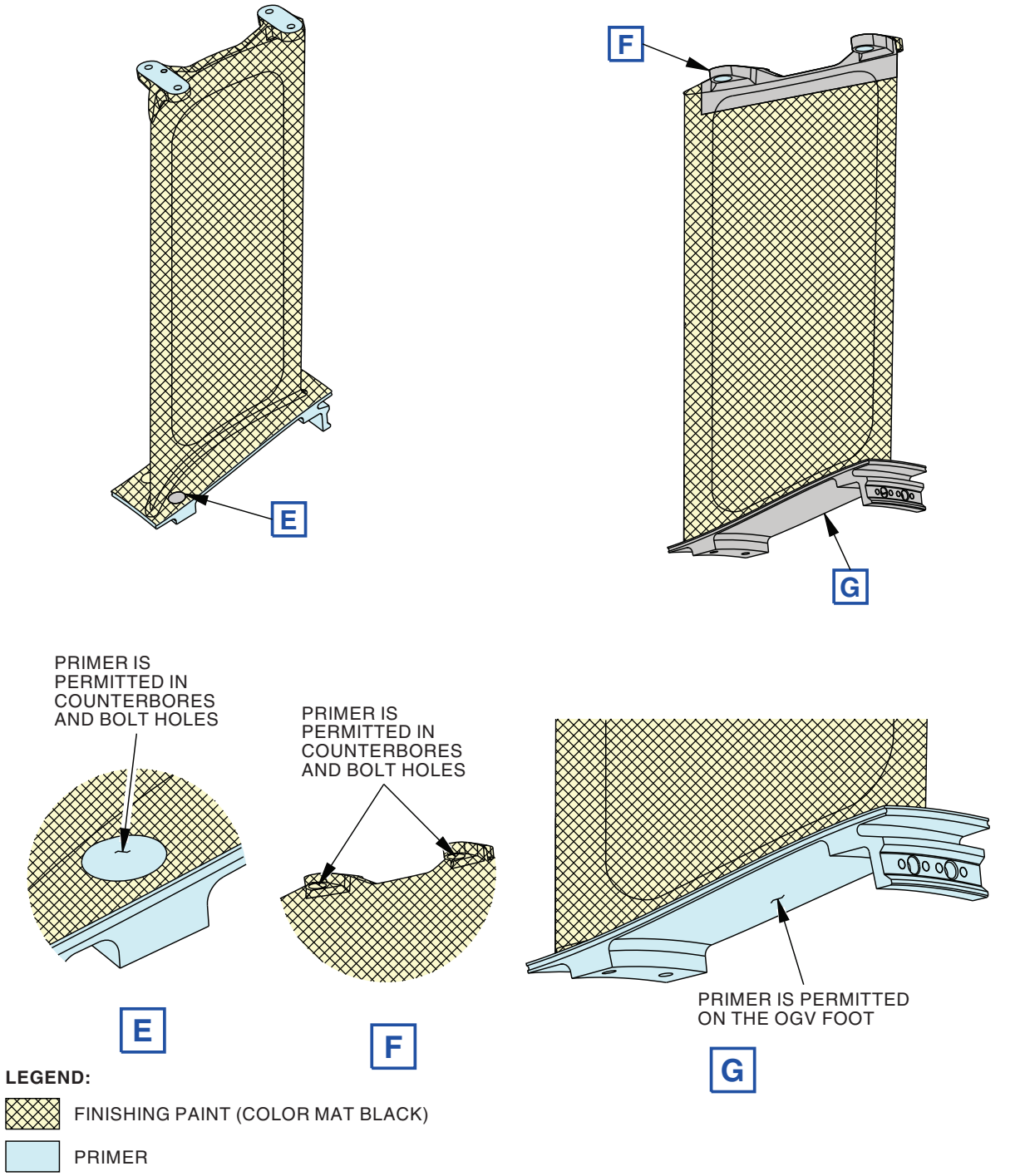
NOTE:

PAINT WITH OVERSPRAY IF THE SURFACE TO REPAIR IS SUFFICIENT

2805086 S0000641706_V1

OGV Mask Contour and Permitted Overspray Area
Figure 812/72-21-03-990-811-G00

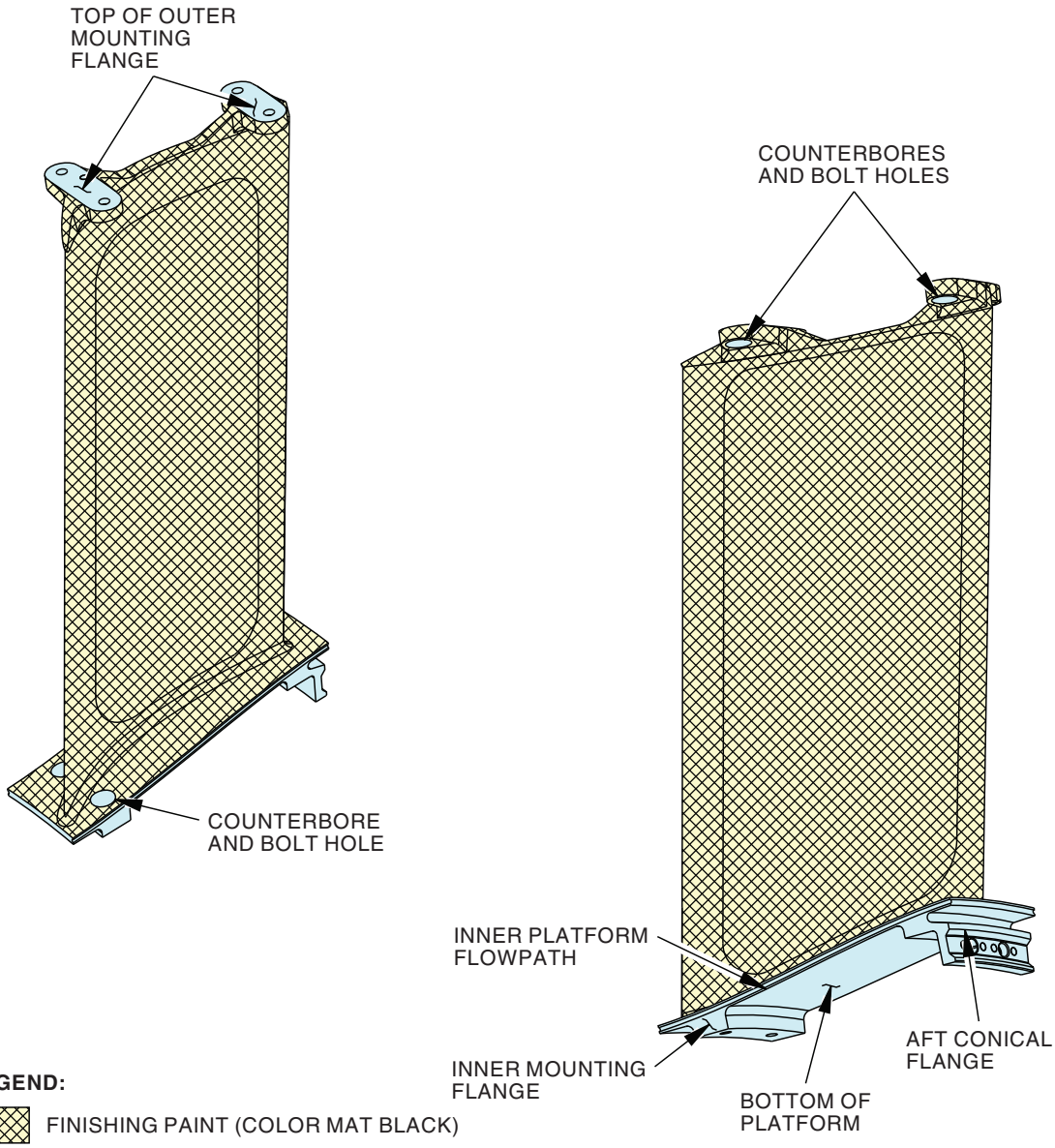
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2805113 S0000641716_V1

OGV Painted Areas
Figure 813/72-21-03-990-812-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



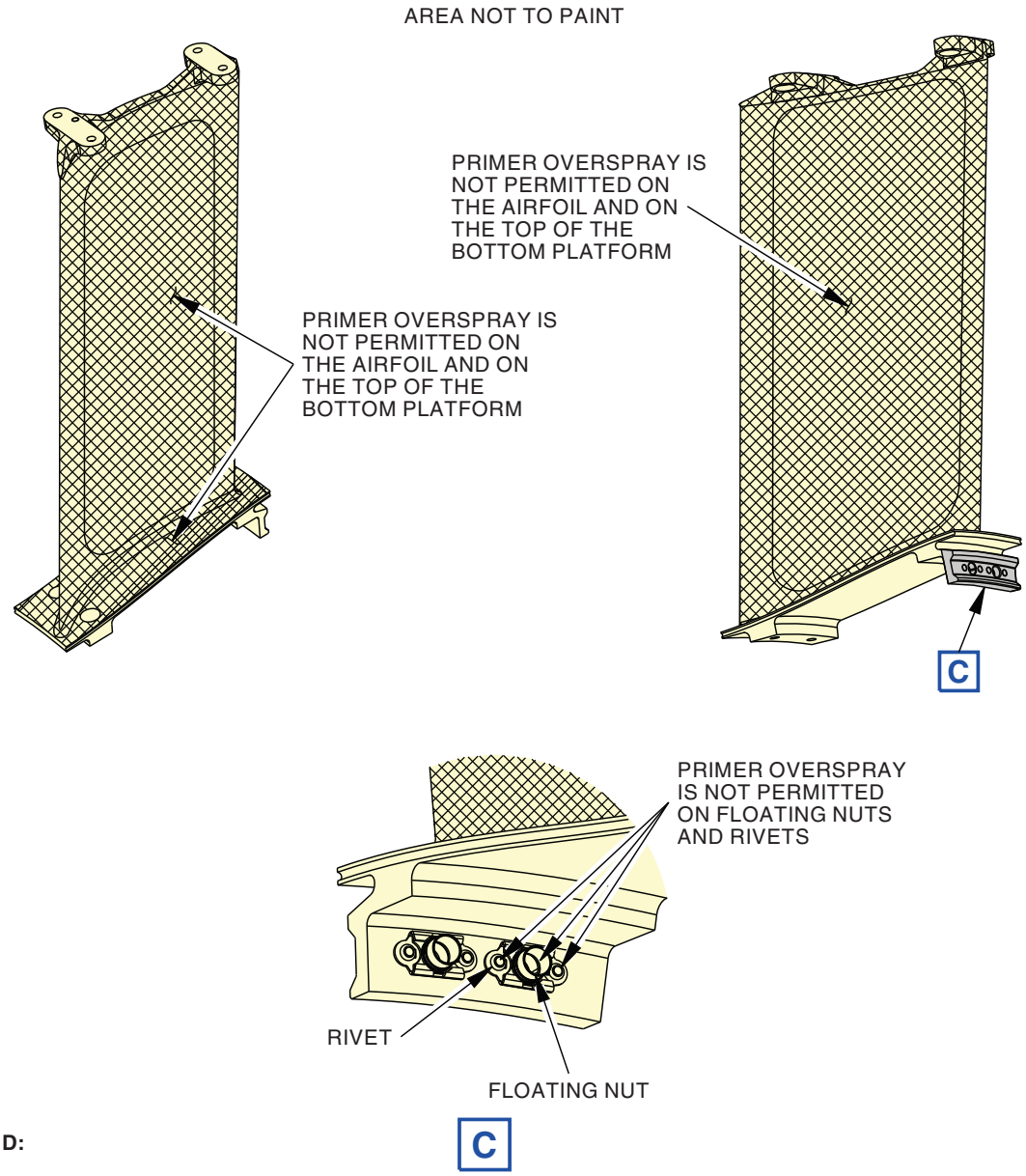
NOTE:

PRIMER IS PERMITTED :
- ON THE TOP OF OUTER MOUNTING FLANGE
- ON THE OGV FOOT (AFT CONICAL FLANGE, BOTTOM OF PLATFORM, INNER PLATFORM FLOWPATH AND INNER MOUNTING FLANGE INCLUDED)
- IN COUNTERBORES AND BOLT HOLES

2805114 S0000641717_V1

OGV Painted Areas
Figure 813/72-21-03-990-812-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

FINISHING PAINT (COLOR MAT BLACK)

PRIMER

NOTE:

PRIMER OVERSPRAY IS NOT PERMITTED :
ON FINISHING PAINT MAT BLACK (AIRFOIL AND TOP OF THE BOTTOM PLATFORM INCLUDED)
ON RIVETS AND FLOATING NUTS

2805128 S0000641718_V1

OGV Not Permitted Overspray Areas
Figure 814/72-21-03-990-813-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2807033 S0000641173_V1

Outlet Guide Vane Color Check Chart
Figure 815/72-21-03-990-814-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-21-03-960-801-G00

7. Outlet Guide Vane Two-Lug Nut Plate/Assembly Gang Channel Nuts Replacement

(Figure 816, Figure 817)

A. General

(1) This task gives the instructions to replace the rivets, two-lug nut plates, and assembly gang channel nuts on the Outlet Guide Vane (OGV).

B. References

Reference	Title
70-43-31-300-801-G00	Blind Rivet Assemblies Repair (P/B 201)
72-21-03-000-801-G00	Outlet Guide Vane Removal (P/B 401)
72-21-03-400-801-G00	Outlet Guide Vane Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
B50333 [CP2687]	Solvent - Degreasing, Diestone DLS	
D50321 [CP2773]	Compound - Jointing, Corrosion Inhibiting, Chromate Free, Mastinox CA1010	

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Replacement

SUBTASK 72-21-03-010-010-G00

(1) Remove the applicable OGV (TASK 72-21-03-000-801-G00).

(a) The table below gives a list of the part numbers that are applicable and the configuration that corresponds.

Table 805/72-21-03-993-802-G00

Part Number	Designation	Configuration
364-052-011-0	Vane, Outlet Guide	1
364-052-611-0	Vane, Outlet Guide	1
364-052-711-0	Vane, Outlet Guide	2
364-052-811-0	Vane, Outlet Guide	2
364-068-111-0	Vane, Outlet Guide	3
364-068-211-0	Vane, Outlet Guide	3
364-068-311-0	Vane, Outlet Guide	3
364-068-411-0	Vane, Outlet Guide	3
364-068-511-0	Vane, Outlet Guide	3
364-068-611-0	Vane, Outlet Guide	3
364-068-711-0	Vane, Outlet Guide	3
364-068-811-0	Vane, Outlet Guide	4

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 805/72-21-03-993-802-G00 (Continued)

Part Number	Designation	Configuration
364-068-911-0	Vane, Outlet Guide	4
364-062-201-0	Vane, Outlet Guide	1
364-062-301-0	Vane, Outlet Guide	1

F. Outlet Guide Vane Two-Lug Nut Plate/Assembly Gang Channel Nuts Replacement

SUBTASK 72-21-03-020-005-G00

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

WARNING USE CARE WHEN YOU REWORK TITANIUM. SMALL PARTICLES AND FINE SHAVINGS OF TITANIUM ARE HIGHLY FLAMMABLE. TITANIUM DUST IS HIGHLY FLAMMABLE AND CAN CAUSE AN EXPLOSION. WATER WITH HOT TITANIUM CAN CAUSE A STEAM EXPLOSION. EXTINGUISH ALL FIRES OF TITANIUM WITH FULLY DRY TALC, CALCIUM CARBONATE, SAND, OR GRAPHITE. APPLY THE POWDER TO A DEPTH OF 1/2 INCH OR MORE OVER THE BURNING METAL. DO NOT USE FOAM, WATER, CARBON TETRACHLORIDE, OR CARBON DIOXIDE. IF YOU DO NOT OBEY, YOU CAN CAUSE INJURY TO PERSONS.

WARNING METAL PARTICLES FROM MACHINING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT.

WARNING USE PROTECTIVE GLOVES TO PREVENT INJURY FROM SHARP EDGES AND ROUGH SURFACES. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

CAUTION DO NOT USE TOOLS MADE FROM OR WITH CADMIUM, BISMUTH, TIN, ZINC, GOLD OR SILVER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION KEEP CHLORINATED HYDROCARBONS AND HALOGEN MATERIALS AWAY FROM TITANIUM PARTS. IF YOU DO NOT OBEY, CHLORINATED HYDROCARBONS AND HALOGEN MATERIALS CAN CAUSE THE DAMAGE TO THE TITANIUM PARTS.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

DO NOT USE TOOLS ON TITANIUM THAT WERE USED ON OTHER MATERIALS. IT WILL CAUSE CONTAMINATION OF THE TITANIUM. KEEP THE BLENDING ABRASIVE TOOL USED FOR THE TITANIUM ISOLATED FROM TOOLS USED FOR OTHER METALS. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Remove and discard the two-lug nut plate or the assembly gang channel nuts and their rivets (TASK 70-43-31-300-801-G00).

NOTE: A 0.144 in. (3.66 mm) drill bit can be used to drill the rivet head.

SUBTASK 72-21-03-110-008-G00

(2) Clean the area where the new parts will be installed with a clean cloth moistened with degreasing solvent, B50333 [CP2687], or alcohol, B00676 [CP1041].

SUBTASK 72-21-03-210-011-G00

(3) Do a visual inspection of the chamfers and the holes of the rivets and the adjacent area.

(a) Primer missing is not permitted.

(b) Deviation to a correct condition is not permitted.

SUBTASK 72-21-03-420-004-G00

CAUTION

MAKE SURE THAT THE TITANIUM SURFACE IS CLEAN. DO NOT LET SALT FALL ON THE SURFACE. DO NOT TOUCH THE SURFACE. USE CLEAN WHITE COTTON GLOVES. SALT AND FINGERPRINTS CAN CAUSE CORROSION ON THE SURFACE.

(4) Install and make sure of correct installation of the two-lug nut plate or the assembly gang channel nuts (TASK 70-43-31-300-801-G00).

(a) The table below gives a list of the part numbers that are applicable and the configuration that corresponds.

Table 806/72-21-03-993-803-G00

Part Number	Designation	Configuration
649-313-025-0	Nut plate, two lug	1 and 3
649-785-309-0	Assembly gang channel nuts	2 and 4
NAS1721C4L5P	Rivet	1 to 4

(b) For configurations 3 and 4, do the following steps (Table 805):

1) Before installation, apply jointing compound, D50321 [CP2773], on the AREAS A of the two-lug nut plate, assembly gang channel nuts, and rivet body (Figure 817).

NOTE: A brush is not recommended to apply the compound.

a) Do not apply more jointing compound, D50321 [CP2773], than necessary.

2) After installation, use a clean dry cloth to remove the unwanted jointing compound, D50321 [CP2773], around the rivets, two-lug nut plate, assembly gang channel nuts, and from the OGV surfaces (Figure 816).

a) Solvent product is not permitted for cleaning the anticorrosion compound.

NOTE: Solvent might cancel the expected efficiency of the compound.

b) Chunks of anticorrosion compound are not permitted.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

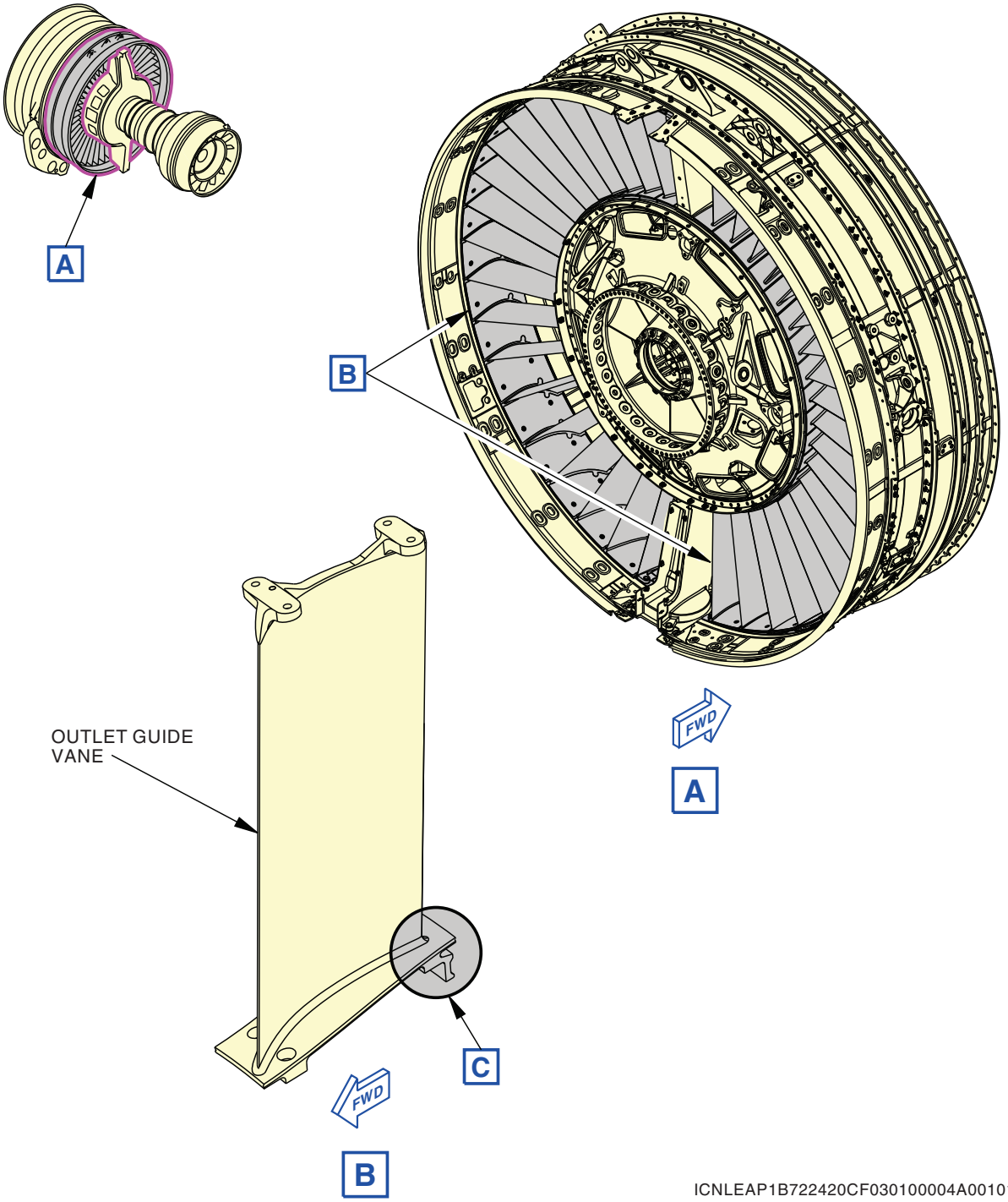
G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-03-410-014-G00

(1) Install the OGV that was removed (TASK 72-21-03-400-801-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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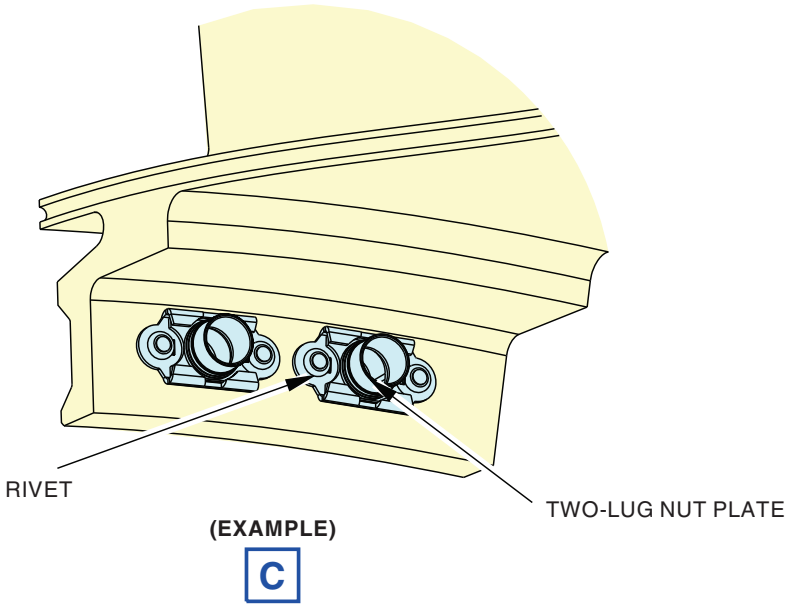
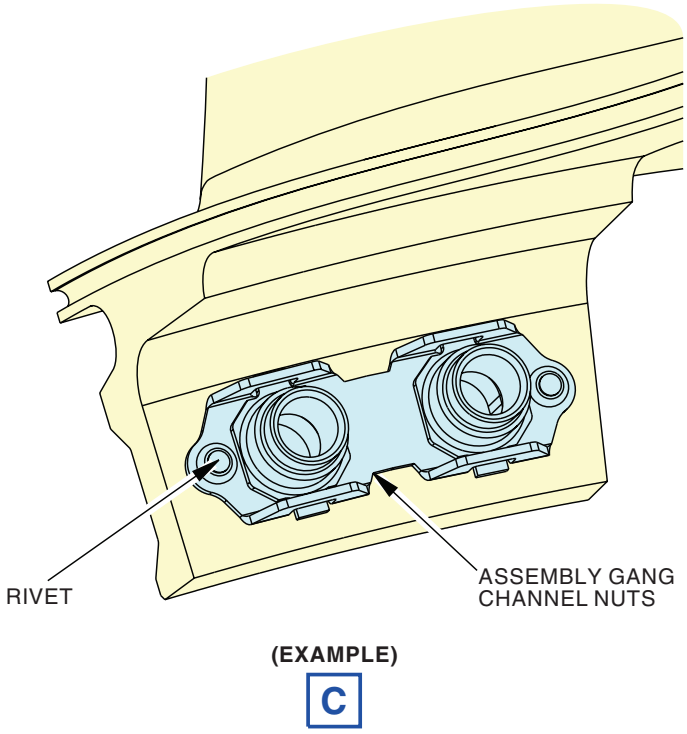
Outlet Guide Vane Two-Lug Nut Plate/Assembly Gang Channel Nuts Replacement
Figure 816/72-21-03-990-819-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





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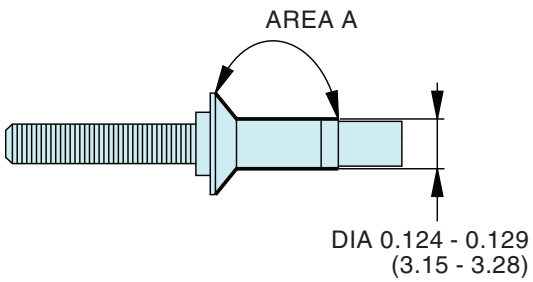
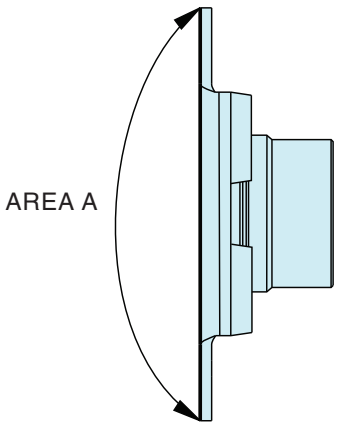
Outlet Guide Vane Two-Lug Nut Plate/Assembly Gang Channel Nuts Replacement
Figure 816/72-21-03-990-819-G00 (Sheet 2 of 2)

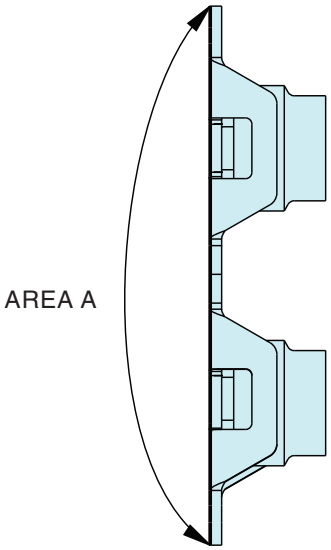
ECCN 9E991 BOEING PROPRIETARY - See title page for details



RIVET ONLY



TWO-LUG NUT PLATE ONLY



ASSEMBLY GANG CHANNEL NUTS ONLY

NOTE:
ALL DIMENSIONS ARE IN INCHES
(MILLIMETERS ARE IN PARENTHESES).

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Outlet Guide Vane Anti Corrosion Paste Application
Figure 817/72-21-03-990-820-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LOW PRESSURE COMPRESSOR STAGE 1 INLET GUIDE VANE - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Low pressure compressor stage 1 inlet guide vane removal

(2) Low pressure compressor stage 1 inlet guide vane installation.

TASK 72-21-04-000-801-G00

2. Low Pressure Compressor Stage 1 Inlet Guide Vane Removal

(Figure 401)

A. General

(1) This task provides the instructions for the removal of the low pressure compressor stage 1 inlet guide vane.

(2) The low pressure compressor stage 1 inlet guide vane is referred to as the LPC stage 1 IGV.

B. References

Reference	Title
72-21-11-000-801-G00	Low Pressure Compressor (LPC) Rotating Ring Spacer Removal (P/B 401)
72-22-02-000-803-G00	Fan Blades (Set) Removal (P/B 401)
72-22-05-000-801-G00	Flowpath Fairing Removal (P/B 401)
72-22-06-000-801-G00	Splitter Fairing Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-15155	Screw Set - Low Pressure Compressor Vane Remove Part #: 956A6880G02 Supplier: 58828 Opt Part #: 956A6880G01 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Removal

SUBTASK 72-21-04-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-21-04-860-003-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

F. Low Pressure Compressor Stage 1 Inlet Guide Vane Removal

SUBTASK 72-21-04-010-001-G00

(1) Remove the LPC stage 1 IGV [1] as follows:

(a) Do this task: Fan Blades (Set) Removal, TASK 72-22-02-000-803-G00.

(b) Do this task: Flowpath Fairing Removal, TASK 72-22-05-000-801-G00.

(c) Do this task: Splitter Fairing Removal, TASK 72-22-06-000-801-G00.

(d) Do this task: Low Pressure Compressor (LPC) Rotating Ring Spacer Removal, TASK 72-21-11-000-801-G00.

(e) Remove the 20 nuts [3] and the 20 bolts [4] that attach the hose [2] to the LPC stage 1 IGV [1].

(f) Remove the three nuts [6] and the three bolts [7] that attach the LPC stage 1 IGV [1] to the LPC stage 2 vane [5].

(g) Install the three screw sets, SPL-15155 in the threaded holes of the LPC stage 2 vane [5] forward flange.

(h) Tighten the three screw sets, SPL-15155 in a criss cross pattern to disengage the LPC stage 1 IGV [1].

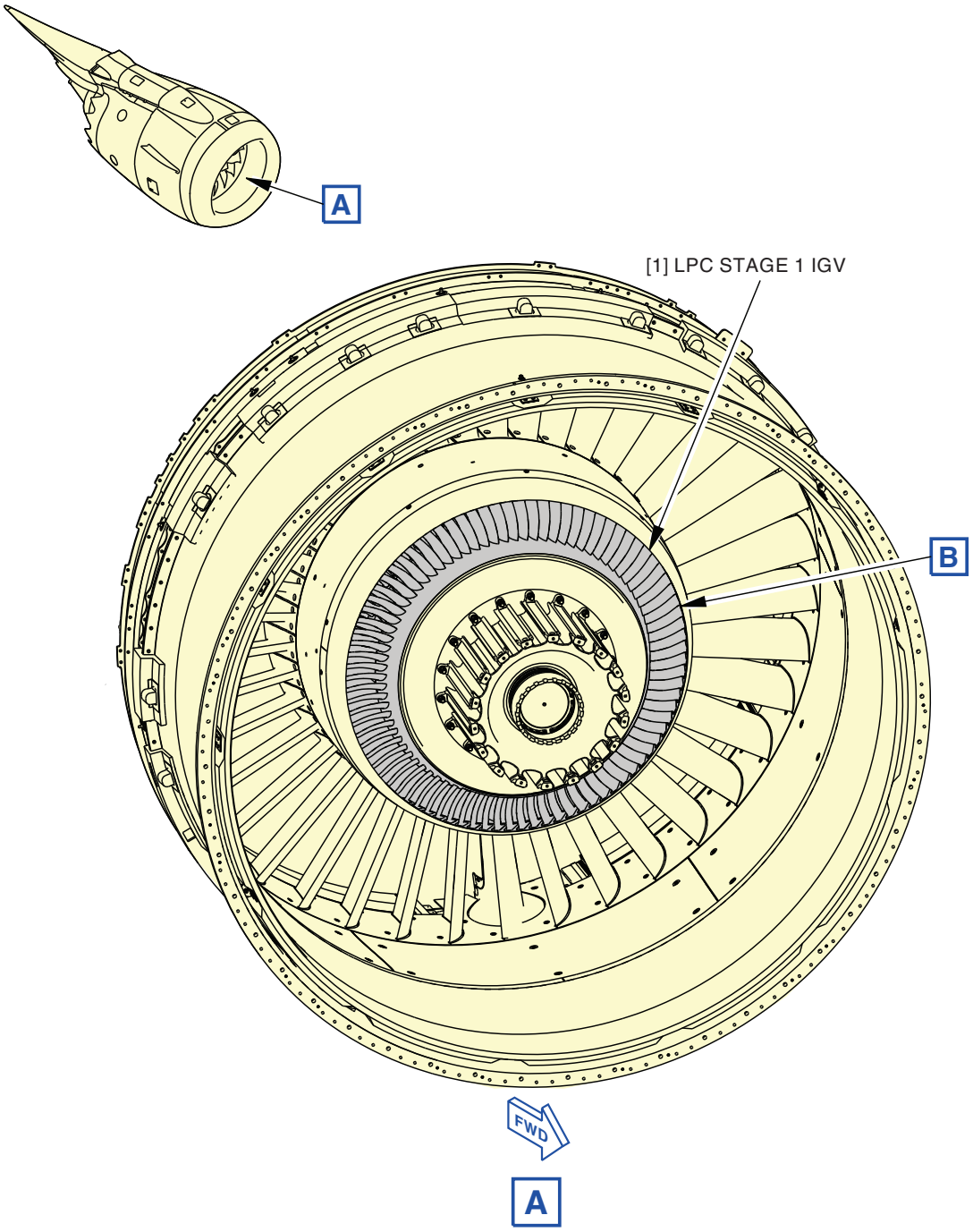
(i) Remove the three screw sets, SPL-15155 from the threaded holes of the LPC stage 2 vane [5] forward flange.

(j) Carefully move the LPC stage 1 IGV [1] forward to disengage from the LPC stage 2 vane [5].

(k) Remove the LPC stage 1 IGV [1] from the engine.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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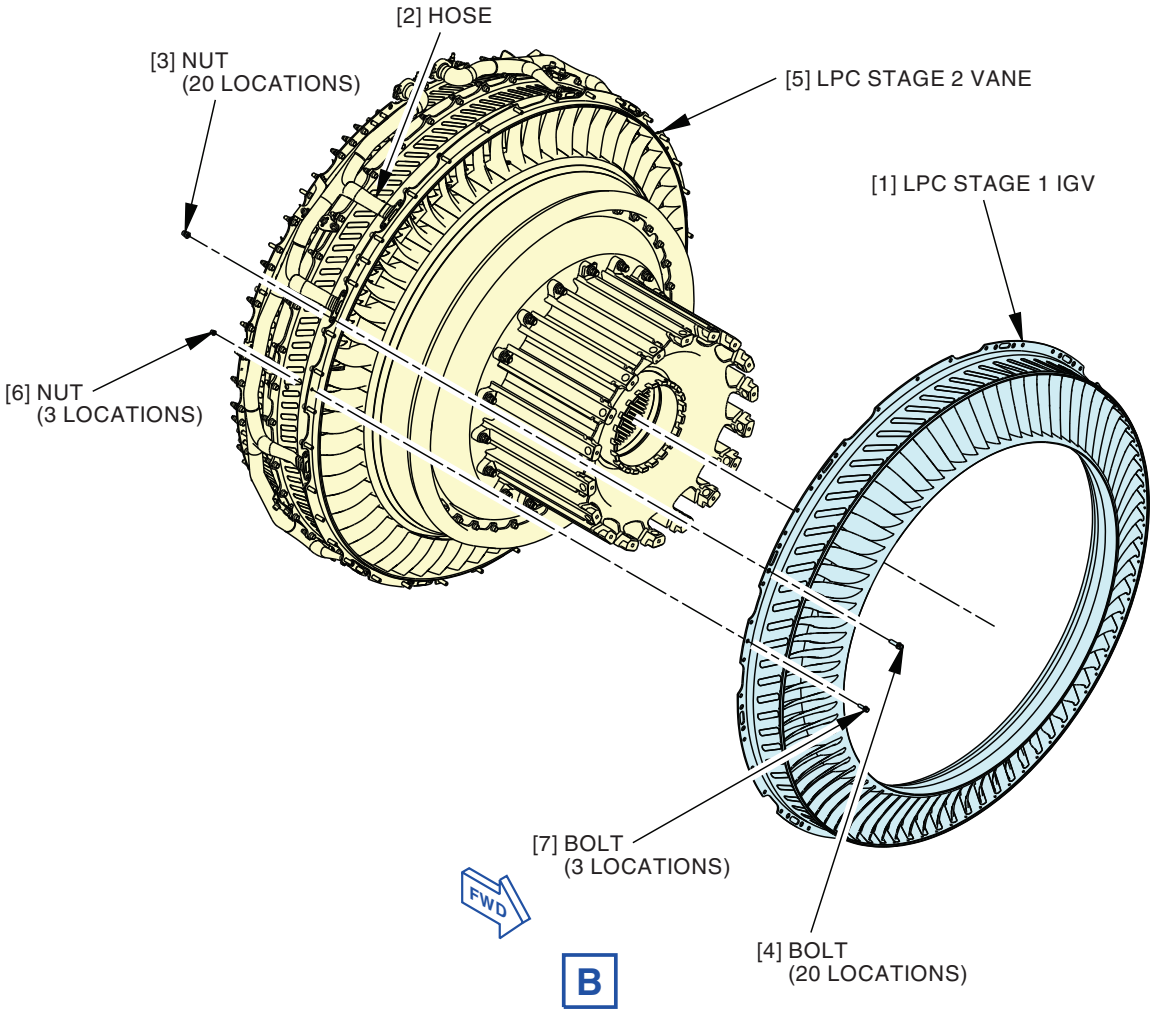
Low Pressure Compressor Stage 1 Inlet Guide Vane Installation
Figure 401/72-21-04-990-802-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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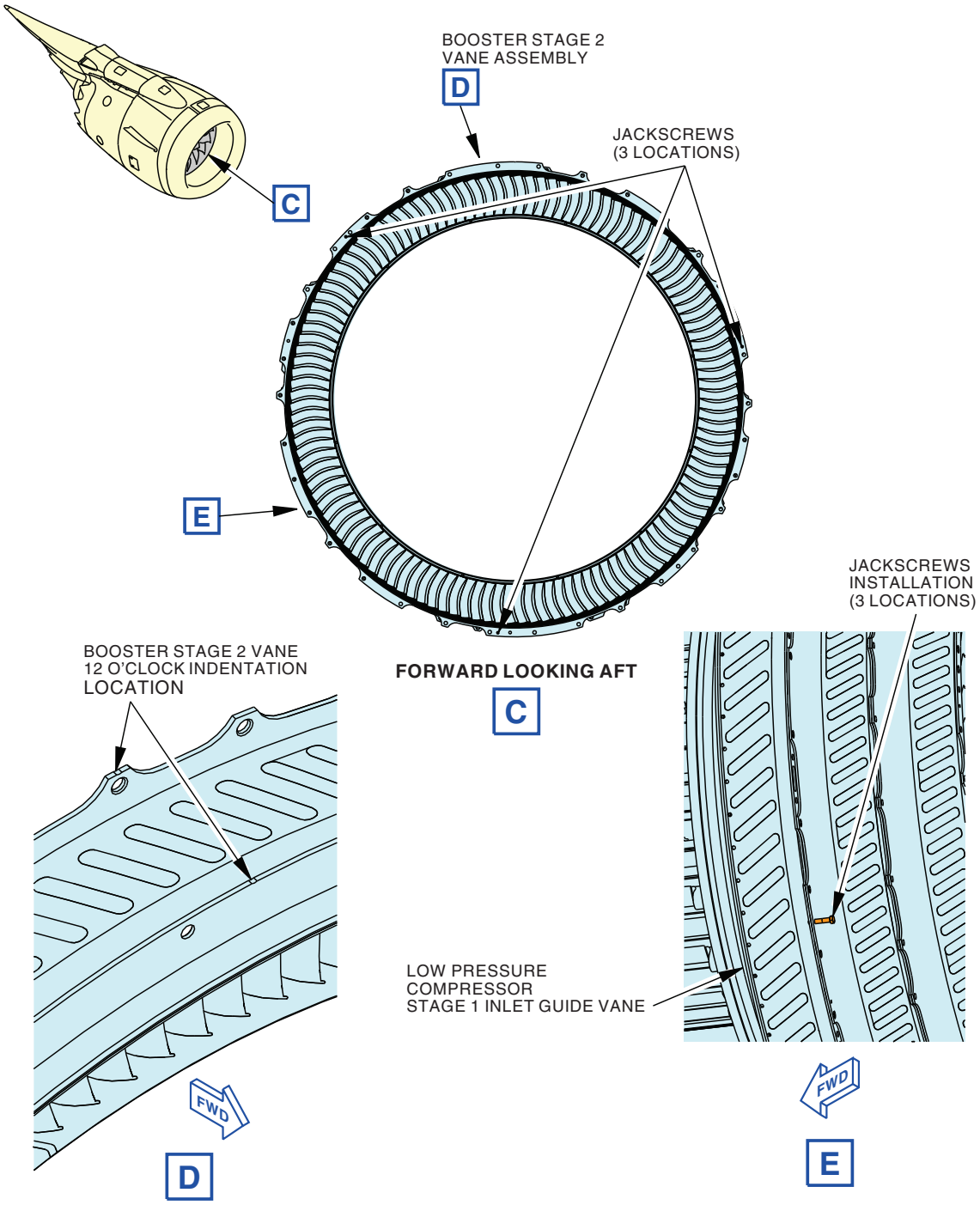
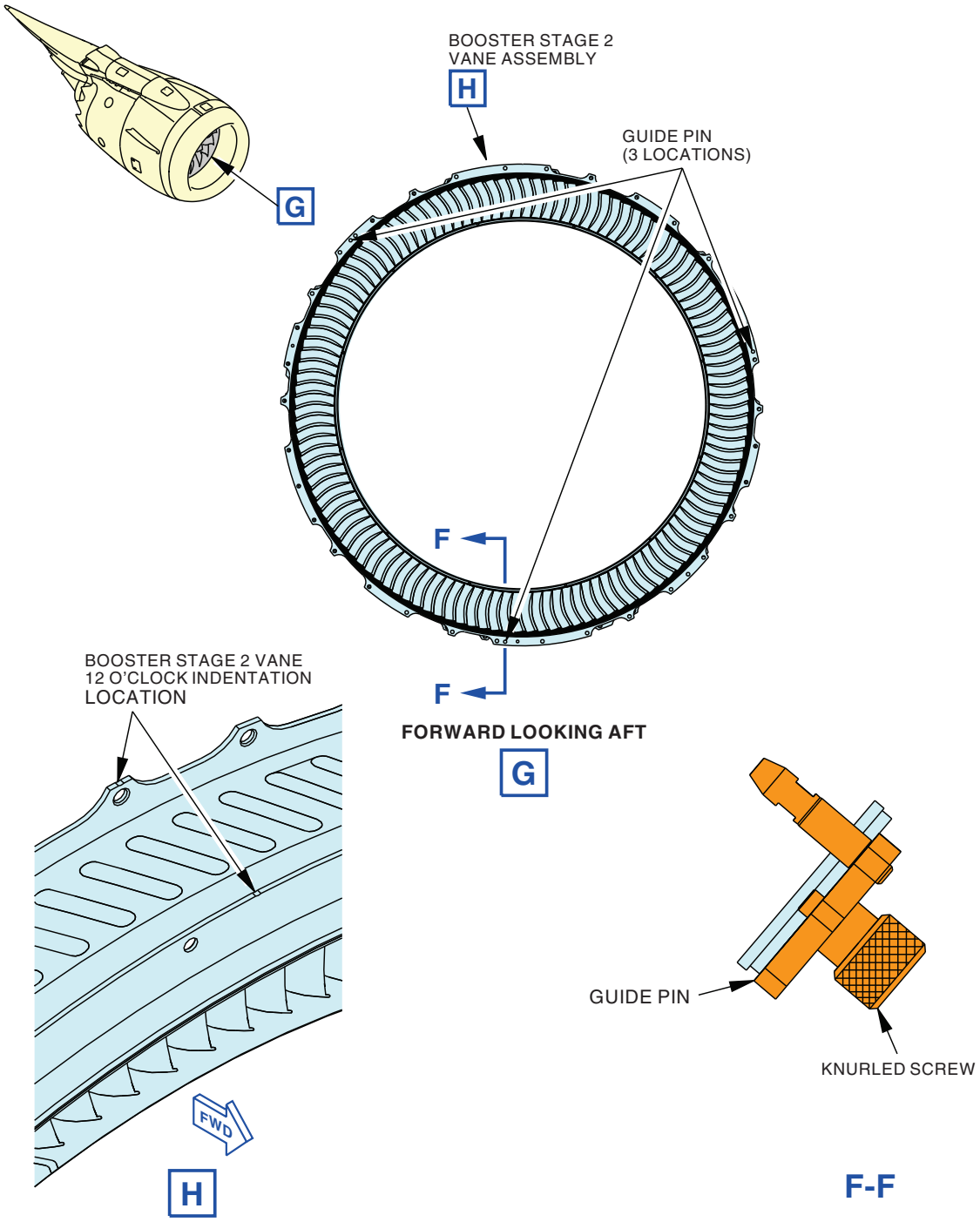
Low Pressure Compressor Stage 1 Inlet Guide Vane Installation
Figure 401/72-21-04-990-802-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Low Pressure Compressor Stage 1 Inlet Guide Vane Installation
Figure 401/72-21-04-990-802-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Low Pressure Compressor Stage 1 Inlet Guide Vane Installation
Figure 401/72-21-04-990-802-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-21-04-400-801-G00

3. Low Pressure Compressor Stage 1 Inlet Guide Vane Installation

(Figure 401)

A. General

(1) This task provides the instructions for the installation of the low pressure compressor stage 1 inlet guide vane.

(2) The low pressure compressor stage 1 inlet guide vane is referred to as the LPC stage 1 IGV.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
72-21-11-400-801-G00	Low Pressure Compressor (LPC) Rotating Ring Spacer Installation (P/B 401)
72-22-02-400-803-G00	Fan Blades (Set) Installation (P/B 401)
72-22-05-400-801-G00	Flowpath Fairing Installation (P/B 401)
72-22-06-400-801-G00	Splitter Fairing Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-15154	Pin - Guide, Low Pressure Compressor Vane Part #: 956A8864G01 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00601 [CP2101]	Vaseline - Graphite Mineral	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	LPC stage 1 IGV	72-22-04-01-390	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Low Pressure Compressor Stage 1 Inlet Guide Vane Installation

SUBTASK 72-21-04-410-001-G00

(1) Install the LPC stage 1 IGV [1] as follows:

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Lubricate the threads of the 20 bolts [4], and the three bolts [7] with the graphite mineral vaseline, D00601 [CP2101].

(c) Identify the three extraction holes of the LPC stage 2 vane [5].

(d) Install the three guide pins, SPL-15154 in the holes of the LPC stage 2 vane [5] and safety with the knurled screws in the extraction holes.

(e) Install the LPC stage 1 IGV [1] in the correct position on the LPC stage 2 vane [5].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Install the three bolts [7] and the three nuts [6] that attach the LPC stage 1 IGV [1] to the LPC stage 2 vane [5].

(g) Tighten the bolts [7] to 46.55 in-lb (5.259 N·m) - 51.45 in-lb (5.813 N·m).

(h) Loosen the knurled screws of the guide pins, SPL-15154 and remove the three guide pins from the holes of the LPC stage 2 vane [5].

(i) Install the 20 bolts [4] and the 20 nuts [3] that attach the hose [2] to the LPC stage 1 IGV [1].

(j) Tighten the bolts [4] to 109.25 in-lb (12.34 N·m) - 120.75 in-lb (13.64 N·m).

SUBTASK 72-21-04-160-001-G00

(2) Clean the work area and clear the tools and other items.

SUBTASK 72-21-04-410-002-G00

(3) Do this task: Splitter Fairing Installation, TASK 72-22-06-400-801-G00.

SUBTASK 72-21-04-410-003-G00

(4) Do this task: Flowpath Fairing Installation, TASK 72-22-05-400-801-G00.

SUBTASK 72-21-04-410-005-G00

(5) Do this task: Low Pressure Compressor (LPC) Rotating Ring Spacer Installation, TASK 72-21-11-400-801-G00.

SUBTASK 72-21-04-410-004-G00

(6) Do this task: Fan Blades (Set) Installation, TASK 72-22-02-400-803-G00.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-04-860-002-G00

(1) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 72-21-04-860-004-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

I. Low Pressure Compressor Stage 1 Inlet Guide Vane Installation Test

SUBTASK 72-21-04-710-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN CASE FORWARD ACOUSTIC SHROUD - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Fan case forward acoustic shroud removal.

(2) Fan case forward acoustic shroud installation.

TASK 72-21-05-000-801-G00

2. Fan Case Forward Acoustic Shroud Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the fan case forward acoustic shroud.

B. References

Reference	Title
71-11-01-000-801-G00	Inlet Cowl Removal (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Removal

SUBTASK 72-21-05-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-21-05-860-003-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-21-05-010-001-G00

(3) Remove the inlet cowl [2] (TASK 71-11-01-000-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Fan Case Forward Acoustic Shroud Removal

SUBTASK 72-21-05-010-002-G00

WARNING THE FAN CASE FORWARD ACOUSTIC SHROUD IS HEAVY. IT IS NECESSARY FOR TWO PERSONS TO LIFT THE FAN CASE FORWARD ACOUSTIC SHROUD. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Remove the fan case forward acoustic shroud [1] as follows:

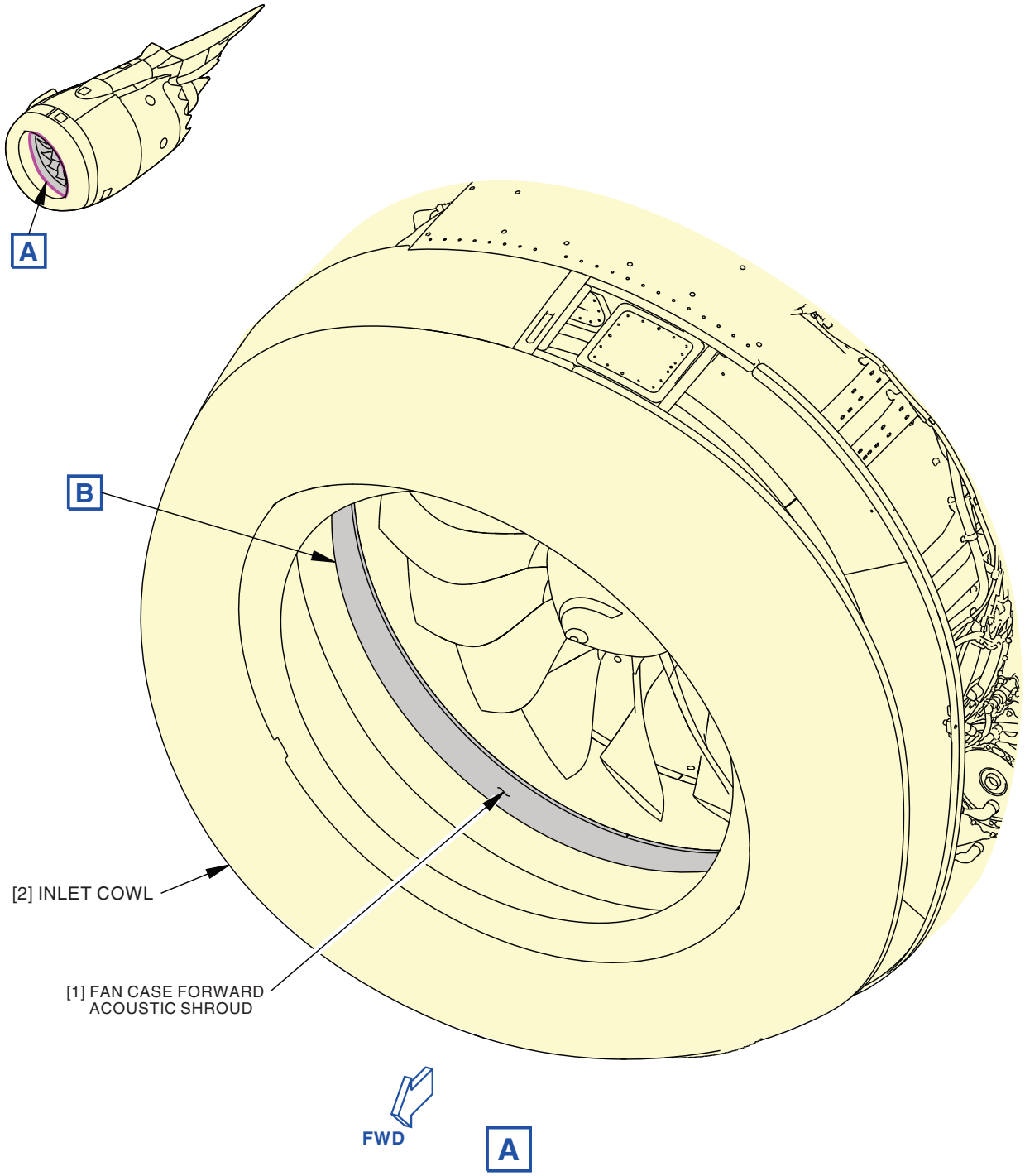
(a) Remove the 13 bolts [4] that attach the fan case forward acoustic shroud [1] to the fan case [3].

(b) Carefully move the fan case forward acoustic shroud [1] forward to disengage from the fan case [3].

(c) Remove the fan case forward acoustic shroud [1] from the engine.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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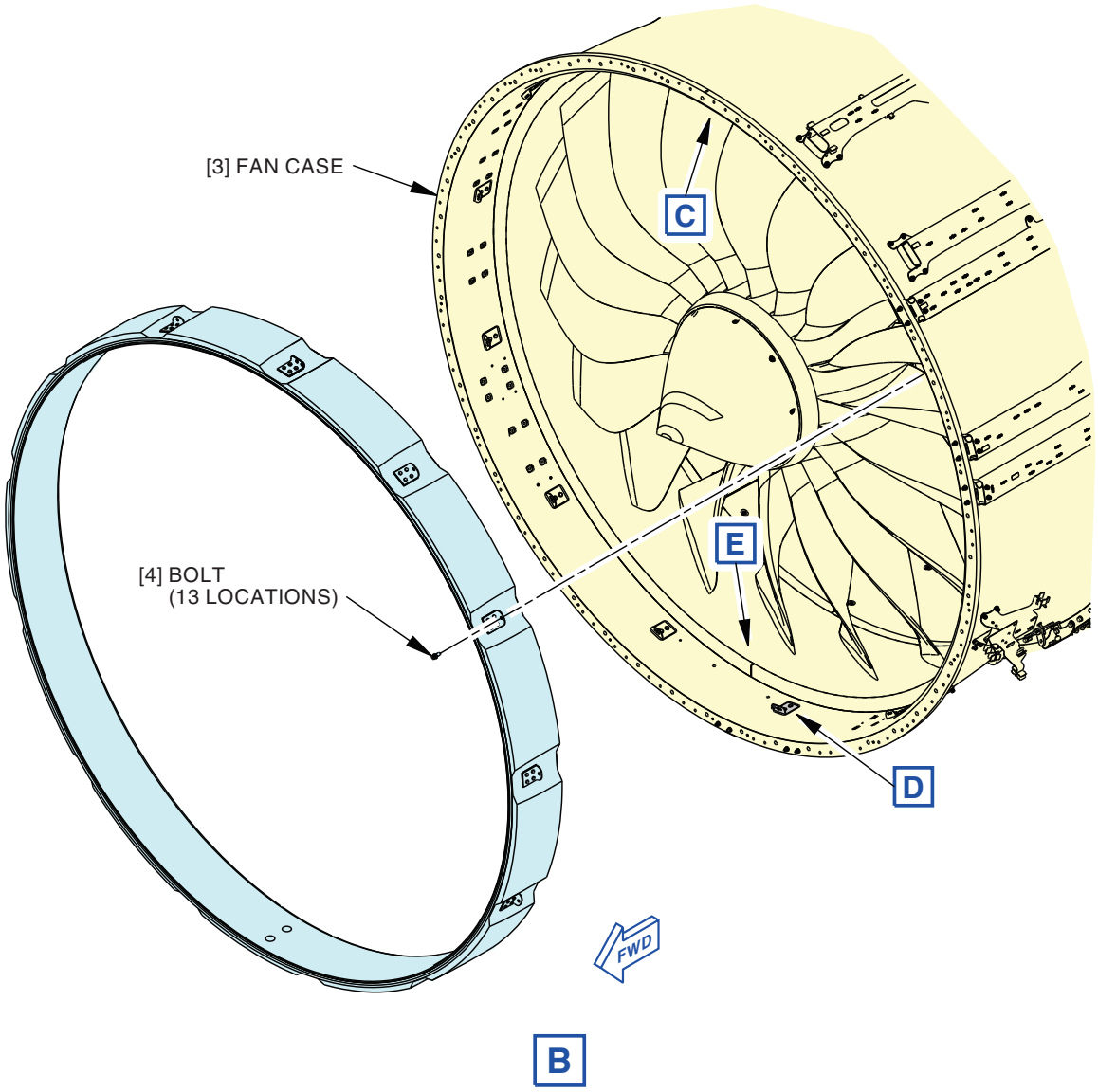
Fan Case Forward Acoustic Shroud Installation
Figure 401/72-21-05-990-801-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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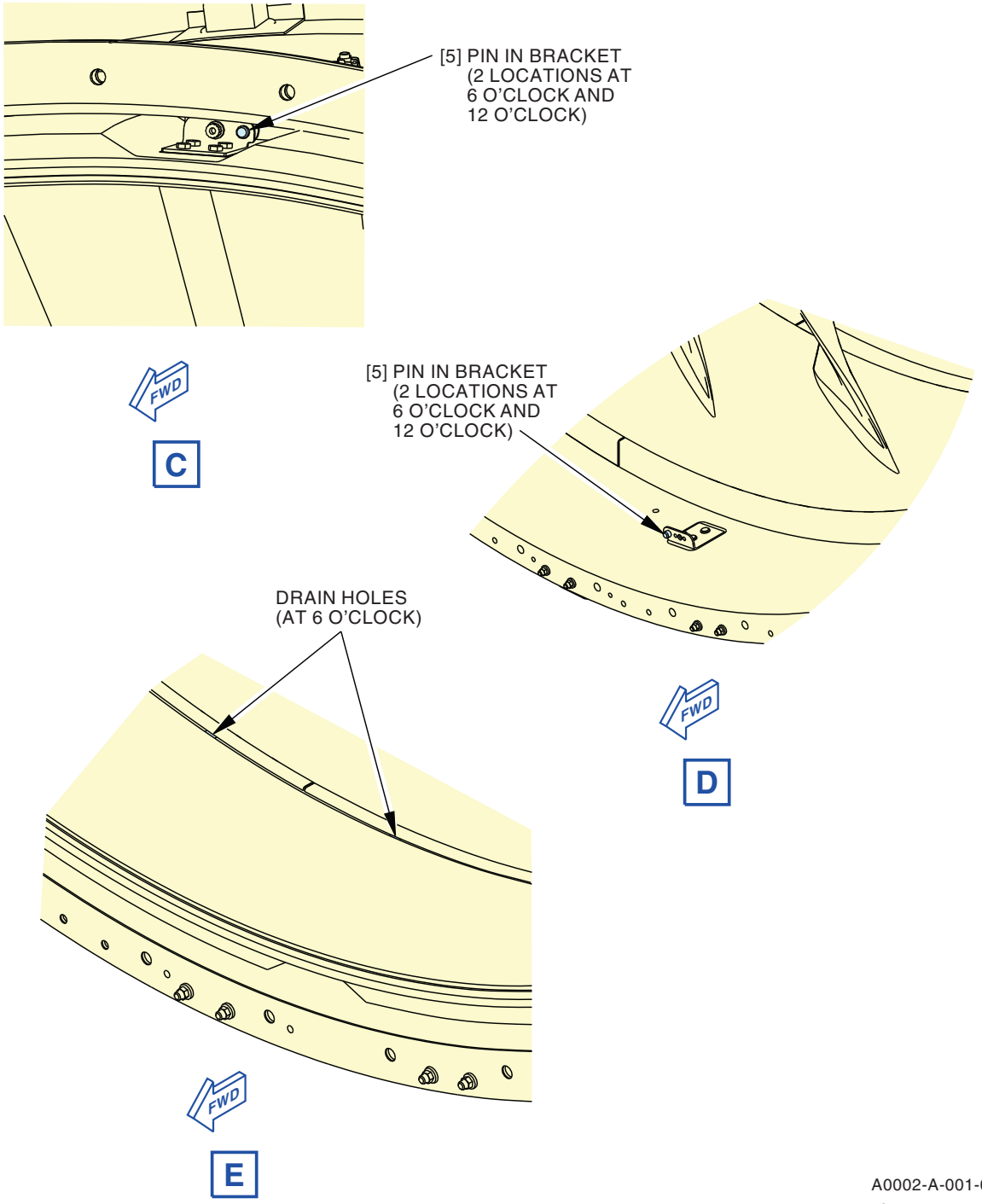
Fan Case Forward Acoustic Shroud Installation
Figure 401/72-21-05-990-801-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Fan Case Forward Acoustic Shroud Installation
Figure 401/72-21-05-990-801-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-21-05-400-801-G00

3. Fan Case Forward Acoustic Shroud Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the fan case forward acoustic shroud.

(2) You must do the tests that are given in the power plant test reference table after you install the fan case forward acoustic shroud.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-01-400-801-G00	Inlet Cowl Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00601 [CP2101]	Vaseline - Graphite Mineral	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Fan case forward acoustic shroud	72-21-05-01-090	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Fan Case Forward Acoustic Shroud Installation

SUBTASK 72-21-05-410-001-G00

WARNING THE FAN CASE FORWARD ACOUSTIC SHROUD IS HEAVY. IT IS NECESSARY FOR TWO PERSONS TO LIFT THE FAN CASE FORWARD ACOUSTIC SHROUD. INJURY TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Install the fan case forward acoustic shroud [1] as follows:

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Install the fan case forward acoustic shroud [1] in the correct position with the pin in brackets [5] on the fan case [3].

(c) Make sure that the drain holes are at the bottom of the fan case forward acoustic shroud [1].

(d) Lubricate the threads of the 13 bolts [4] with graphite mineral vaseline, D00601 [CP2101].

(e) Install the 13 bolts [4] that attach the fan case forward acoustic shroud [1] to the fan case [3].

(f) Tighten the 13 bolts [4] to 115 in-lb (13 N·m) - 168 in-lb (19 N·m).

(g) Make sure that the work area is clean of all tools and other items.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-05-410-002-G00

(1) Install the inlet cowl [2] (TASK 71-11-01-400-801-G00).

SUBTASK 72-21-05-860-004-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-21-05-860-002-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

I. Fan Case Forward Acoustic Shroud Installation Test

SUBTASK 72-21-05-710-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN CASE FORWARD ACOUSTIC SHROUD - REPAIR

1. General

A. This procedure has these tasks:

(1) A repair of the acoustic shroud by filling up

(2) A replacement of the acoustic shroud seal.

TASK 72-21-05-300-801-G00

2. Acoustic Shroud Repair by Filling Up

(Figure 801)

A. General

(1) This task gives instructions for the repair of the acoustic shroud by material filling.

B. References

Reference	Title
70-21-23-100-801-G00	Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701)
70-43-42-300-802-G00	Blending Procedures - Hand Blending (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-10679	Saw - Hole, Bearing Removal Part #: A32091-1920 Supplier: 81205
STD-585	Mat - Protective, Closed Cell Neoprene, 30-50 duro, Weather and Oil Resistant, 3/8 Inch Min Thickness, Minimum 42x60 Inches with Warning Streamers
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
G51480 [CP2772]	Cloth - Abrasive, Alumina	
G51894 [CP2923]	Compound - Epoxy, Void Filling and Edge Sealing, Scotch-Weld EC-3524 B/A Black	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Acoustic shroud	72-21-05-01-090	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Prepare for the Repair

SUBTASK 72-21-05-010-003-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-21-05-860-005-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-21-05-480-001-G00

CAUTION PUT THE PROTECTIVE MAT IN THE INLET COWL. THIS WILL PREVENT DAMAGE TO THE INLET COWL OR ENGINE BY TOOLS, PARTS, OR UNWANTED MATERIALS THAT FALL ON THE INLET COWL SURFACE.

(3) Put the protective mat, STD-585, in the lower half of the inlet cowl to prevent damage to the inlet cowl and engine.

H. Acoustic Shroud Repair by Filling Up

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THIS PRODUCT. IF NOT, IT CAN CAUSE INJURY.

WARNING DO NOT BREATHE THE PARTICLES FROM THE REPAIR OF COMPOSITE MATERIALS OR LET THE PARTICLES TOUCH YOU. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR. THE PARTICLES CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT.

SUBTASK 72-21-05-140-001-G00

(1) Find all repair areas at the acoustic shroud [1].

(a) The diameter of the repaired area must not be more than 2.0 in. (50.8 mm).

(b) In case of multiple areas to repair the total cumulated repaired surface must not be more than 15.7 in^2 (10,129.0 mm^2).

(c) The distance between two repaired areas is more than five times the length of the longest repaired diameter.

SUBTASK 72-21-05-140-002-G00

(2) Remove the indications at the acoustic shroud [1] as follows (Figure 801):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Cut the area around the indication to 0.25 in. (6.4 mm) larger than the indication in a circular shape using a hole saw, SPL-10679.

NOTE: A change of sound occurs during the hole-sawing when the carbon skin layer is cut and you reach the honeycomb material.

(b) Remove all the indications of the front sheet carbon skin layers and/or honeycomb such as loose particles, cracks and delamination.

NOTE: For better results, remove all of the honeycomb material from the repaired diameter.

(c) Hand blend the area to be filled to obtain a clean surface for material fill-up and bonding (TASK 70-43-42-300-802-G00).

WARNING

WHEN YOU USE THE COMPRESSED AIR TO CLEAN, TO COOL OR TO DRY, DO NOT EXCEED 30 PSIG (207 KPA). MAKE SURE YOU WEAR EYE PROTECTION AND DO NOT DIRECT COMPRESSED AIR AT YOURSELF AND OTHERS. INJURY TO PERSONNEL CAN OCCUR.

(d) Clean the area with clean de-oiled compressed air and alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

SUBTASK 72-21-05-330-001-G00

(3) Fill with epoxy compound, G51894 [CP2923], as follows:

CAUTION

YOU MUST WEAR GLOVES THAT ARE FREE OF TALC OR SILICONE TO PREVENT CONTAMINATION OF ABRADABLE COMPOUND. IF NOT, YOU CAN CAUSE DAMAGE.

(a) Prepare the epoxy compound, G51894 [CP2923], as follows:

1) Base Scotch-Weld EC-3524 Part B black: 100 parts by weight.

2) Accelerator Scotch-Weld EC-3524 Part A black: 94 parts by weight.

(b) Stir the above mixture manually until you get an homogenous material with a dark grey color.

1) Mix both parts slowly to prevent air bubbles in the mixture.

(c) If it is necessary, increase the temperature of the above mixture to 77°F (25°C) to make the application of product to the part easier.

NOTE: The most appropriate work temperature is 68°F (20°C) - 77°F (25°C).

SUBTASK 72-21-05-330-002-G00

(4) Fill-up the area to repair with the abradable material.

(a) If it is necessary, increase the temperature of the area to repair or the working environment to 68°F (20°C) - 77°F (25°C) to make the application of product to the part easier.

(b) Apply a very thin layer of abradable material on the area to repair.

1) Apply sufficient pressure with your thumb to obtain better results.

(c) Fill the remaining thickness with abradable material.

1) Fill with a 10% increase of thickness compared to the flush thickness.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-21-05-330-003-G00

CAUTION DO NOT USE HEAT FOR THE ACOUSTIC SHROUD ADJACENT AREAS AND ADJACENT FAN BLADES TO THE REPAIR AREAS. A TEMPERATURE MORE THAN THE POLYMERIZATION TEMPERATURE WILL OCCUR AND CAN CAUSE DAMAGE TO THE PARTS.

CAUTION DO NOT USE A LOCAL HEAT GUN. IF YOU DO, IT CAN CAUSE DAMAGE TO THE MATERIAL.

(5) Do the polymerization of the abradable material as follows:

(a) Alternative procedure available. Polymerize at room temperature during 24 h.

NOTE: Room temperature must be between 59°F (15°C) - 77°F (25°C). For better results, use 77°F (25°C).

(b) Alternative procedure. Polymerize at 122°F (50°C) - 140°F (60°C) during 5 h.

(c) Alternative procedure. Polymerize at 158°F (70°C) - 194°F (90°C) during 2 h.

SUBTASK 72-21-05-720-001-G00

(6) Make sure that the hardness is between 40 - 60 Shore D.

(a) Sand the abradable surface flush to the adjacent carbon plies surface that is in good condition with abrasive cloth, G51480 [CP2772] (TASK 70-43-42-300-802-G00).

(b) Do an inspection of the repaired area.

1) Do a visual inspection of the liner for bubbles or missed areas.

a) Air bubbles are not permitted.

<1> If it is necessary, do the repair again.

b) Lack of abradable is not permitted.

<1> If it is necessary, do the repair again.

2) Do a check of liner adherence with a wooden mallet.

a) A dull sound shows that the abradable material is not correctly bonded.

<1> If it is necessary, do the repair again.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-05-860-006-G00

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-21-05-410-004-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

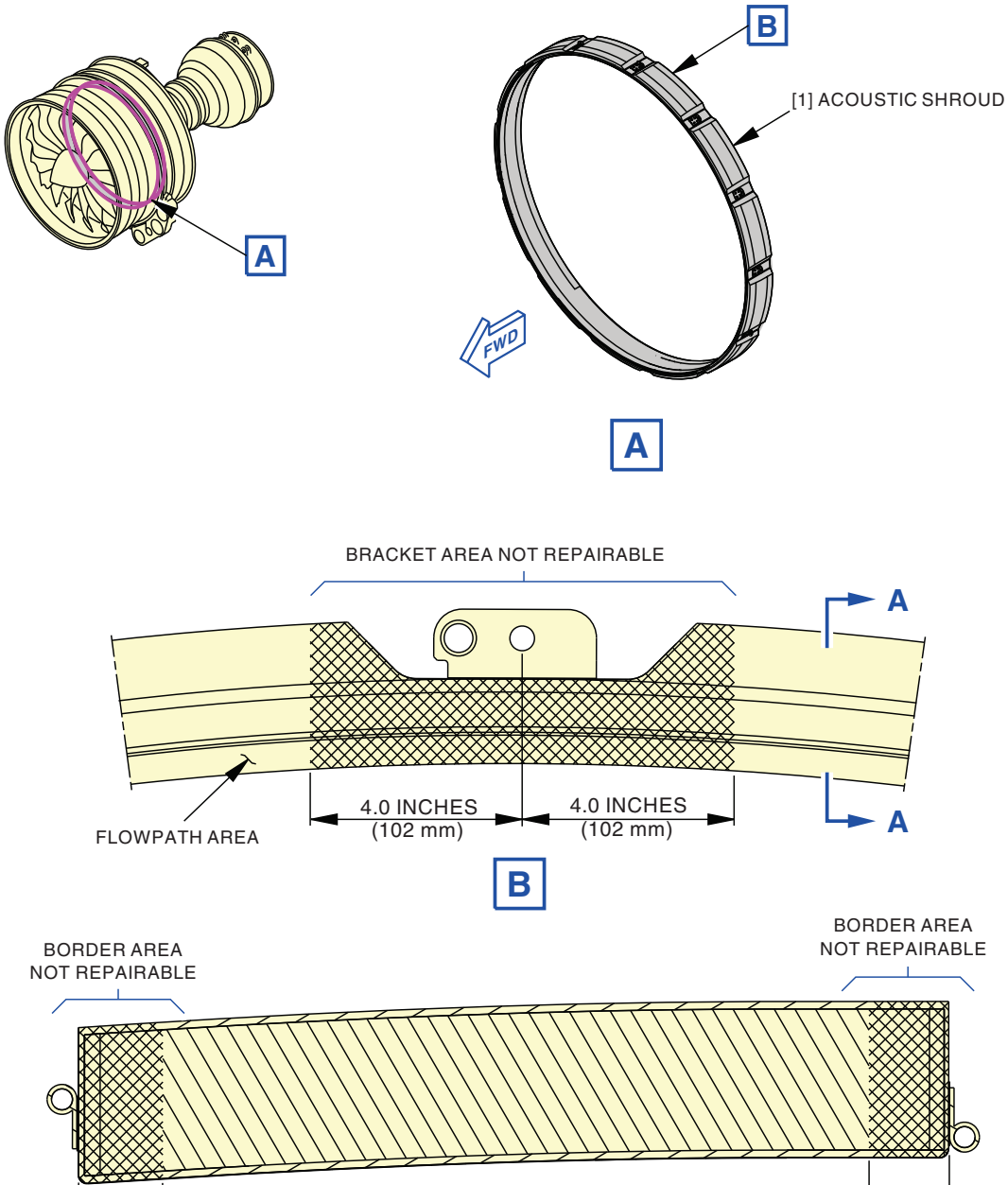
ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

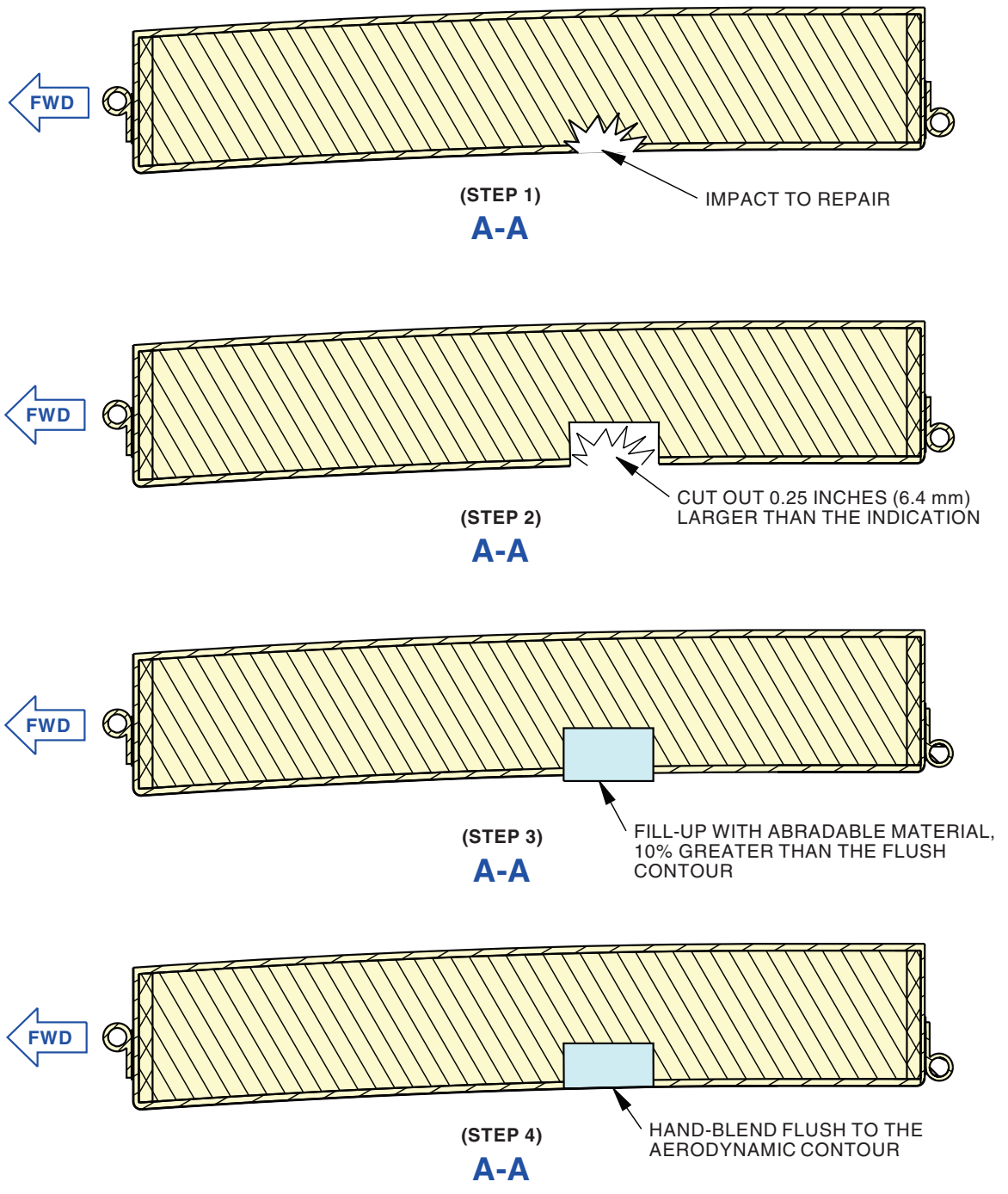
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Acoustic Shroud Repair
Figure 801/72-21-05-990-802-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Acoustic Shroud Repair
Figure 801/72-21-05-990-802-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-21-05-300-802-G00

3. Acoustic Shroud Seal Replacement

(Figure 802)

A. General

(1) This task gives instructions for the local or full re-bond of the elastomer acoustic shroud.

(2) The following table gives a list of the part numbers that are applicable to this repair.

Table 801/72-21-05-993-801-G00

Part number	Designation
364-062-101-0	SHROUD, ACOUSTIC

(3) The following table gives a dimensional information.

Table 802/72-21-05-993-802-G00

Description	Minimum In-Process Dimension	Maximum In-Process Dimension	Minimum Finish Dimension	Maximum Finish Dimension	Reference Dimension
Dim. A	-	-	0.217 in. (5.5 mm)	0.255 in. (6.5 mm)	-
Dim. B	-	-	0.512 in. (13.0 mm)	0.590 in. (15.0 mm)	-
Radius C	-	-	-	-	171.3°arc (2.990 rad)
Radius D	-	-	-	-	176.7°arc (3.084 rad)
Dim. E	-	-	0.36 in. (9 mm)	0.43 in. (11 mm)	-

(4) The following conditions must be provided during primer/adhesive application and bonding/polymerization process:

- Temperature above 59°F (15°C)
- Humidity between 40% and 70%.

(5) Acoustic shroud seal is referred to as seal in this task.

B. References

Reference	Title
72-21-05-000-801-G00	Fan Case Forward Acoustic Shroud Removal (P/B 401)
72-21-05-400-801-G00	Fan Case Forward Acoustic Shroud Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
A50236 [CP2244]	Primer - Liquid For Preparation of Surfaces Prior to Elastomer Application	
A50615 [CP2344]	Adhesive - Silicone Elastomer, Dowsil 3145 RTV	
B00676 [CP1041]	Alcohol - Isopropyl	
G51480 [CP2772]	Cloth - Abrasive, Alumina	

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
21	Seal	72-21-05-01-097	PTF ALL

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Prepare for the Repair

SUBTASK 72-21-05-010-005-G00

(1) Do this task: Fan Case Forward Acoustic Shroud Removal, TASK 72-21-05-000-801-G00.

G. Acoustic Shroud Seal Replacement

SUBTASK 72-21-05-140-003-G00

CAUTION

DO NOT TOUCH THE CARBON FIBER PLIES WHEN YOU REMOVE THE SILICONE SEAL AT THE FAN-CASE ACOUSTIC SHROUD. IF YOU DO NOT OBEY, DAMAGE TO PART CAN OCCUR.

(1) Remove the unbonded or worn elastomer seal [21] and remaining silicone (primer and adhesive).

SUBTASK 72-21-05-350-001-G00

(2) If it is necessary, prepare the new elastomer silicon seal [21] (Views E and F, Figure 802).

NOTE: The seal can be made of multiple parts.

SUBTASK 72-21-05-320-001-G00

WARNING

USE PROTECTIVE GLOVES TO PREVENT INJURY FROM SHARP EDGES AND ROUGH SURFACES. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

(3) Rough the seal [21] and acoustic shroud [1] surfaces to be bonded with grade 80/120 abrasive cloth, G51480 [CP2772], in criss-cross pattern.

SUBTASK 72-21-05-110-001-G00

(4) De-grease the surfaces to bond with a cloth moist in alcohol, B00676 [CP1041].

SUBTASK 72-21-05-100-001-G00

WARNING

WHEN YOU USE THE COMPRESSED AIR TO CLEAN, TO COOL OR TO DRY, DO NOT EXCEED 30 PSIG (207 KPA). MAKE SURE YOU WEAR EYE PROTECTION AND DO NOT DIRECT COMPRESSED AIR AT YOURSELF AND OTHERS. INJURY TO PERSONNEL CAN OCCUR.

(5) Blow the surfaces to bond with dry and de-oiled air.

SUBTASK 72-21-05-300-001-G00

(6) Apply primer 1200 OS, A50236 [CP2244] on the acoustic shroud [1] and on the seal [21] with a brush or spray gun.

(a) Let the primer to dry for 30 minutes.

SUBTASK 72-21-05-300-002-G00

(7) Apply Dowsil 3145 RTV adhesive, A50615 [CP2344], on the primer 1200 OS, A50236 [CP2244] on the acoustic shroud [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) The thickness of the adhesive must be approximately 0.019 in. (0.5 mm).

SUBTASK 72-21-05-300-003-G00

(8) Bond the elastomer acoustic seal [21] on the Dowsil 3145 RTV adhesive, A50615 [CP2344] (Views C, D, E, and F, Figure 802), (Table 802).

(a) Let the adhesive to polymerize for 24 hours at ambient temperature.

SUBTASK 72-21-05-210-001-G00

(9) Make sure that the elastomer acoustic seal [21] is bonded.

(a) The seal [21] must be uniform and free from indications such as bubbles and blisters.

(b) The seal [21] must be elastic and tack-free.

SUBTASK 72-21-05-220-001-G00

(10) Do a dimensional inspection (Table 802).

(a) For the front elastomer acoustic seal [21] (View E, Figure 802):

1) Make sure that the sum of all the gaps (J_x) between the seal sectors is less than 0.020 in. (0.5 mm).

(b) For the rear elastomer acoustic seal [21] (View F, Figure 802):

1) Make sure that the gaps between each seal sector is not more than 0.04 in. (1 mm).

2) Make sure that the sum of all the gaps (J_x) between the seal sectors is less than 0.19 in. (5 mm).

SUBTASK 72-21-05-300-004-G00

(11) If it is necessary, cut and remove completely the rear seal [21] in the 2 specific areas (View G, Figure 802), (Table 802).

NOTE: These holes are for the drain at the bottom of the fan case.

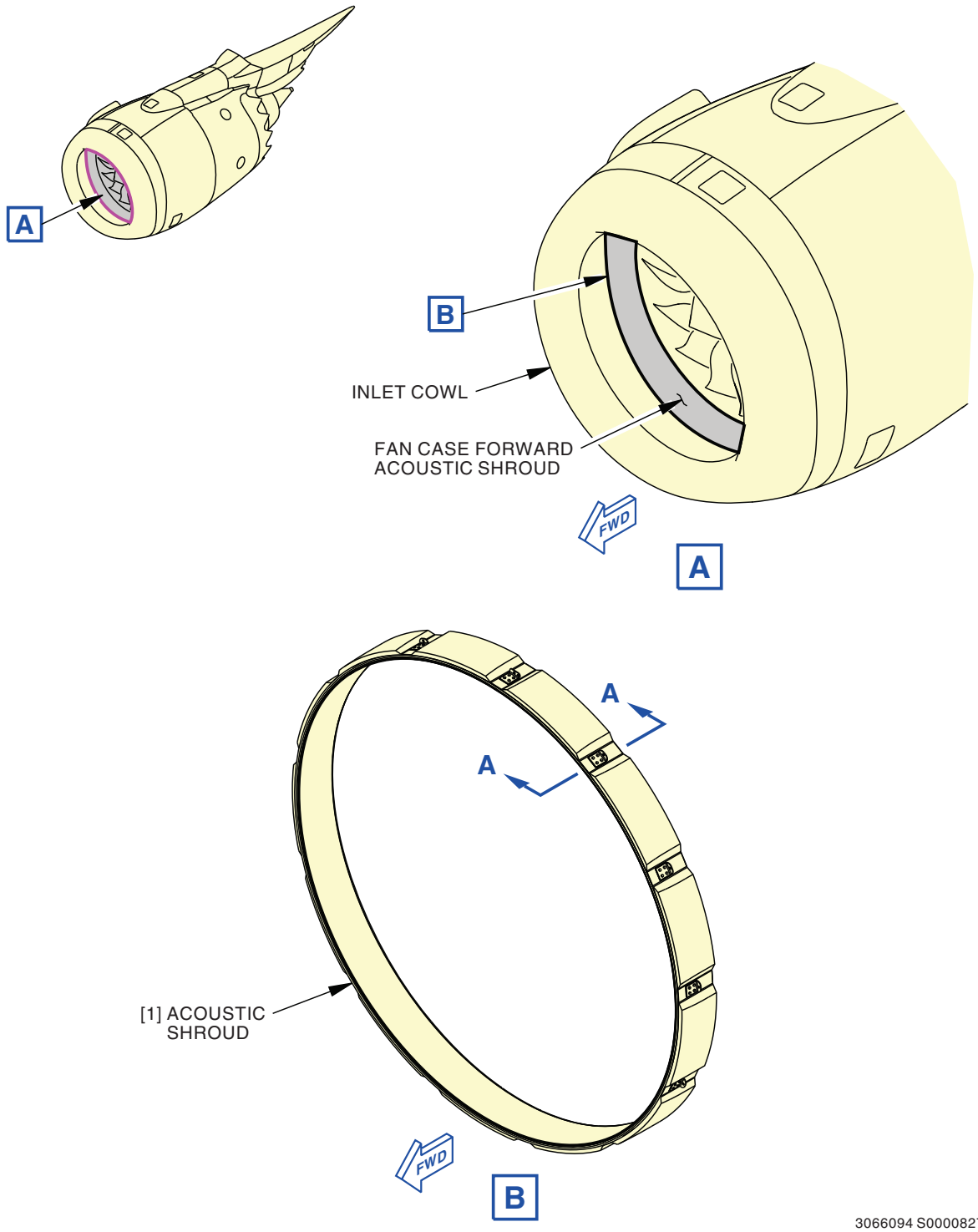
H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-05-410-005-G00

(1) Do this task: Fan Case Forward Acoustic Shroud Installation, TASK 72-21-05-400-801-G00.

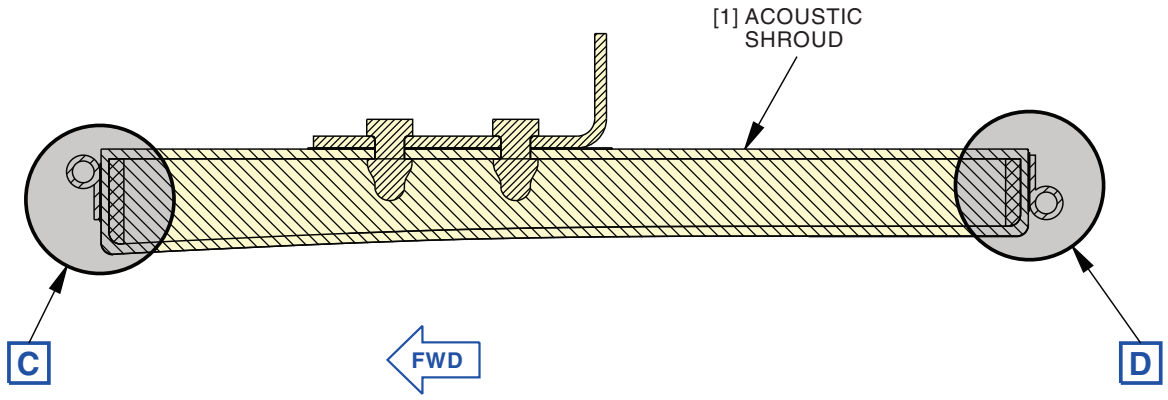
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

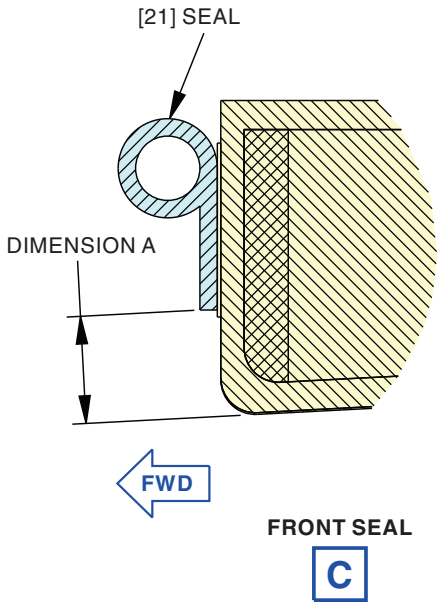


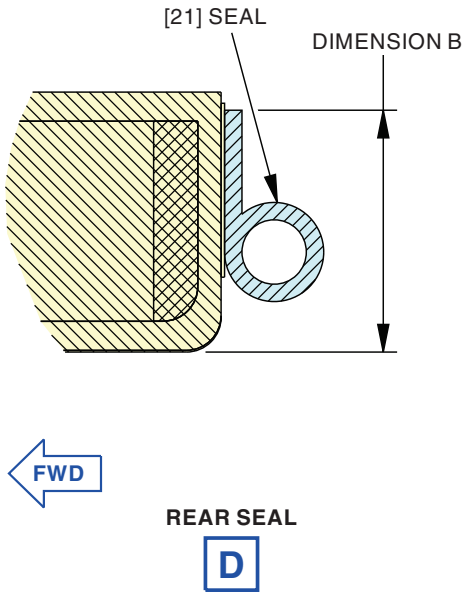
Acoustic Shroud Seal Replacement
Figure 802/72-21-05-990-803-G00 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



A-A





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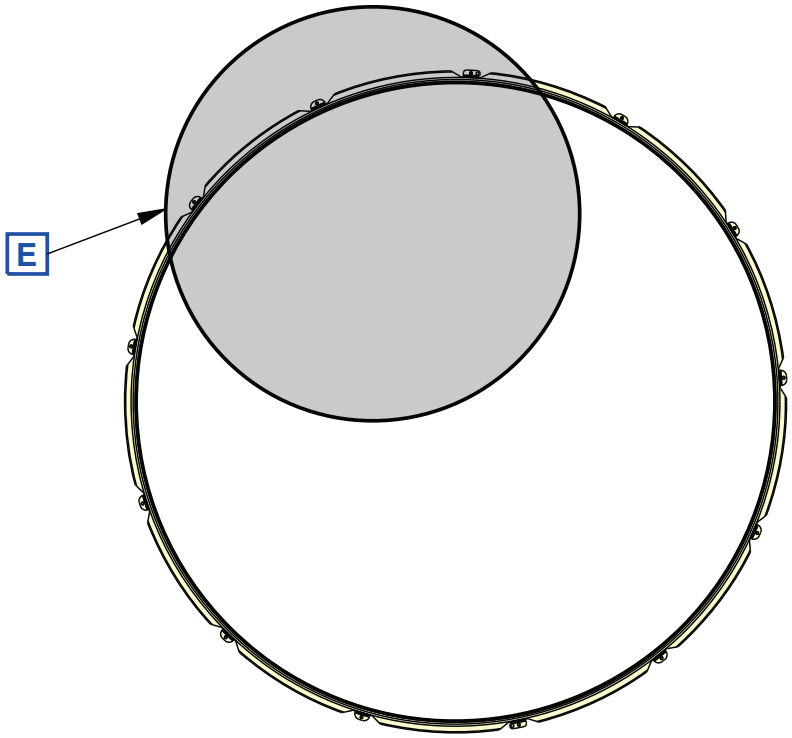
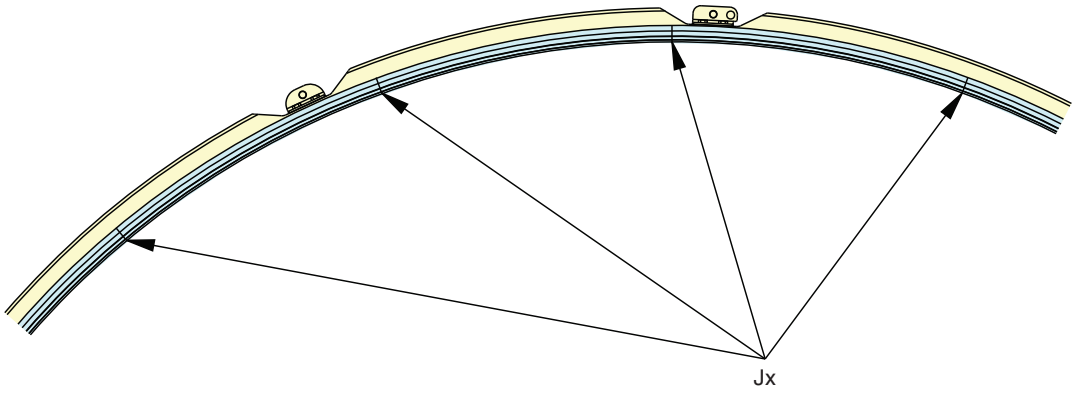
Acoustic Shroud Seal Replacement
Figure 802/72-21-05-990-803-G00 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(FRONT VIEW)

B



J1 + J2 + ... + Jn < 0.020 (0.5)

FRONT SEAL

E

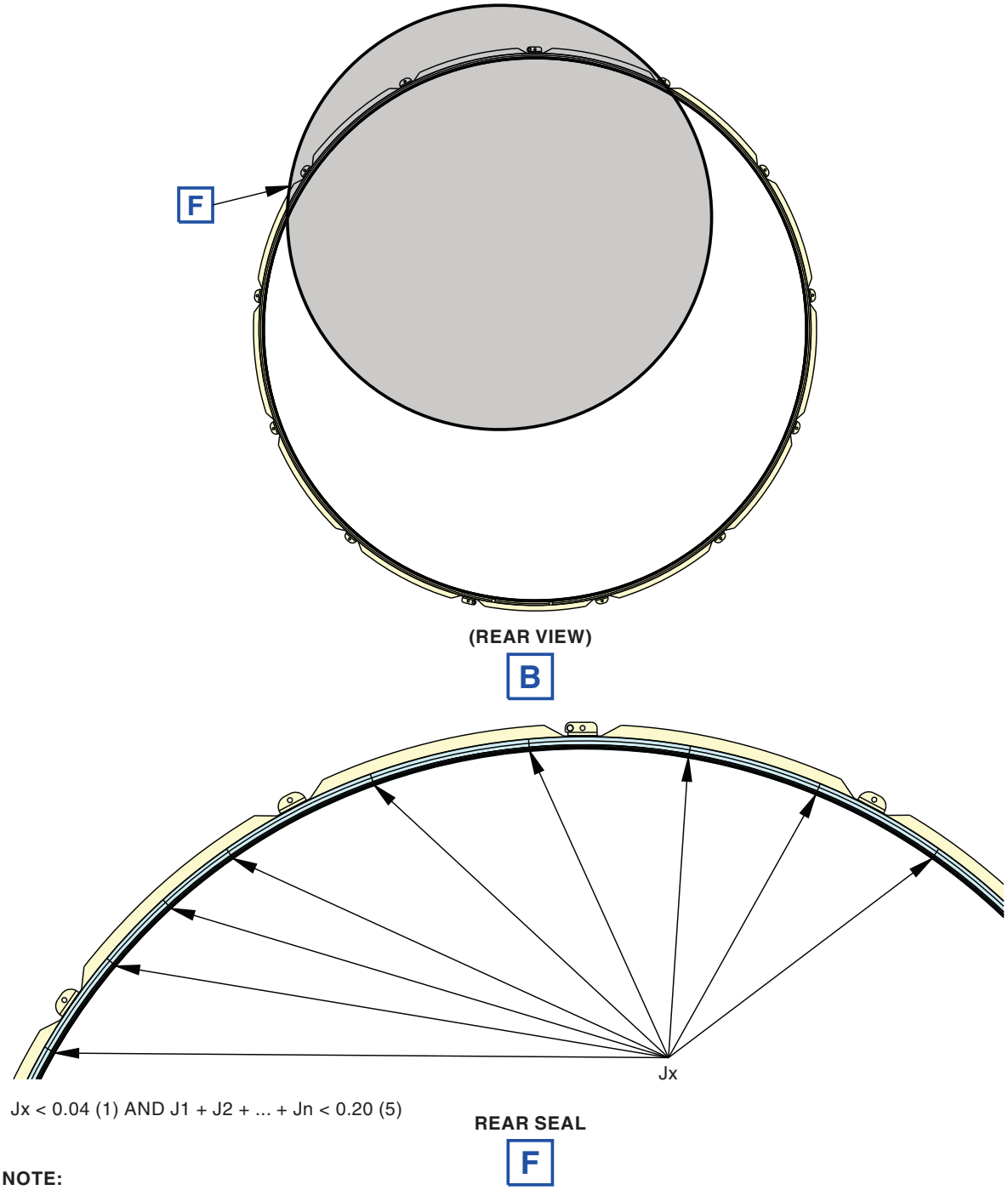
NOTE:

ALL DIMENSIONS ARE IN INCHES
(MILLIMETERS ARE IN PARENTHESES).

3066166 S0000827857_V1

Acoustic Shroud Seal Replacement
Figure 802/72-21-05-990-803-G00 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



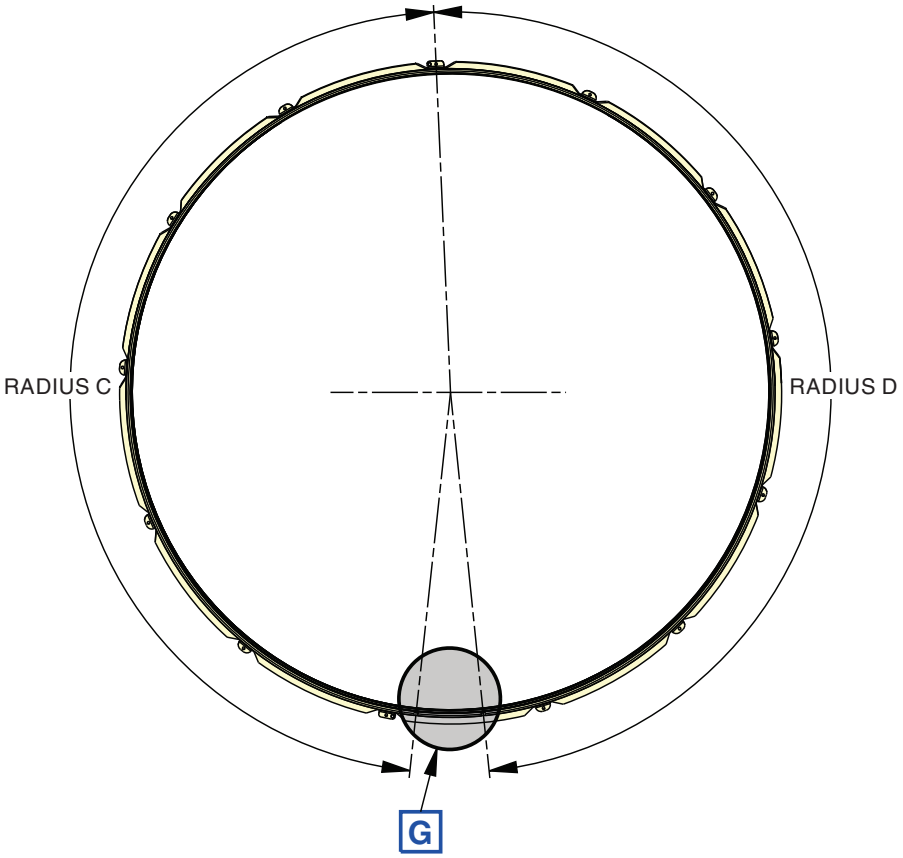
NOTE:

ALL DIMENSIONS ARE IN INCHES
(MILLIMETERS ARE IN PARENTHESES).

3066167 S0000827858_V1

Acoustic Shroud Seal Replacement
Figure 802/72-21-05-990-803-G00 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Acoustic Shroud Seal Replacement
Figure 802/72-21-05-990-803-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AFT FAN CASE HEAT SHIELDS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Aft Fan Case Heat Shield No. 1 Removal
(2) Aft Fan Case Heat Shield No. 1 Installation
(3) Aft Fan Case Heat Shield No. 2 Removal
(4) Aft Fan Case Heat Shield No. 2 Installation
(5) Aft Fan Case Heat Shield No. 3 Removal
(6) Aft Fan Case Heat Shield No. 3 Installation
(7) Aft Fan Case Heat Shield No. 4 Removal
(8) Aft Fan Case Heat Shield No. 4 Installation
(9) Aft Fan Case Heat Shield No. 5 Removal
(10) Aft Fan Case Heat Shield No. 5 Installation
(11) Aft Fan Case Heat Shield No. 6 Removal
(12) Aft Fan Case Heat Shield No. 6 Installation
(13) Aft Fan Case Heat Shield No. 7 Removal
(14) Aft Fan Case Heat Shield No. 7 Installation
(15) Aft Fan Case Heat Shield No. 8 Removal
(16) Aft Fan Case Heat Shield No. 8 Installation.

TASK 72-21-07-000-801-G00

2. Aft Fan Case Heat Shield No. 1 Removal

(Figure 401)

A. General

(1) This task provides the instructions on how to remove the aft fan case heat shield No. 1.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
73-21-04-000-801-G00	Fuel Metering Unit Removal (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Removal

SUBTASK 72-21-07-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-21-07-860-002-G00

(2) For the left engine, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 72-21-07-860-003-G00

(3) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-21-07-940-001-G00

WARNING BE CAREFUL WHEN YOU WORK ON THE ENGINE COMPONENTS IMMEDIATELY AFTER THE ENGINE IS SHUTDOWN. THE ENGINE COMPONENTS CAN STAY HOT FOR UP TO ONE HOUR AND CAN CAUSE INJURY.

(4) Make sure that the engine temperature is cool enough to work on.

SUBTASK 72-21-07-010-012-G00

(5) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. Aft Fan Case Heat Shield No. 1 Removal

SUBTASK 72-21-07-020-001-G00

(1) Remove the aft fan case heat shield No. 1 [1] as follows:

(a) Do this task: Fuel Metering Unit Removal, TASK 73-21-04-000-801-G00.

(b) Remove the three caps [2] from the heat shield.

(c) Remove the three nuts [3] from the three fan case inserts [5].

(d) Remove and discard the three washers [4].

(e) Remove the two bolts [6] that attach the aft fan case heat shield No. 1 [1] to the fan frame shroud [7].

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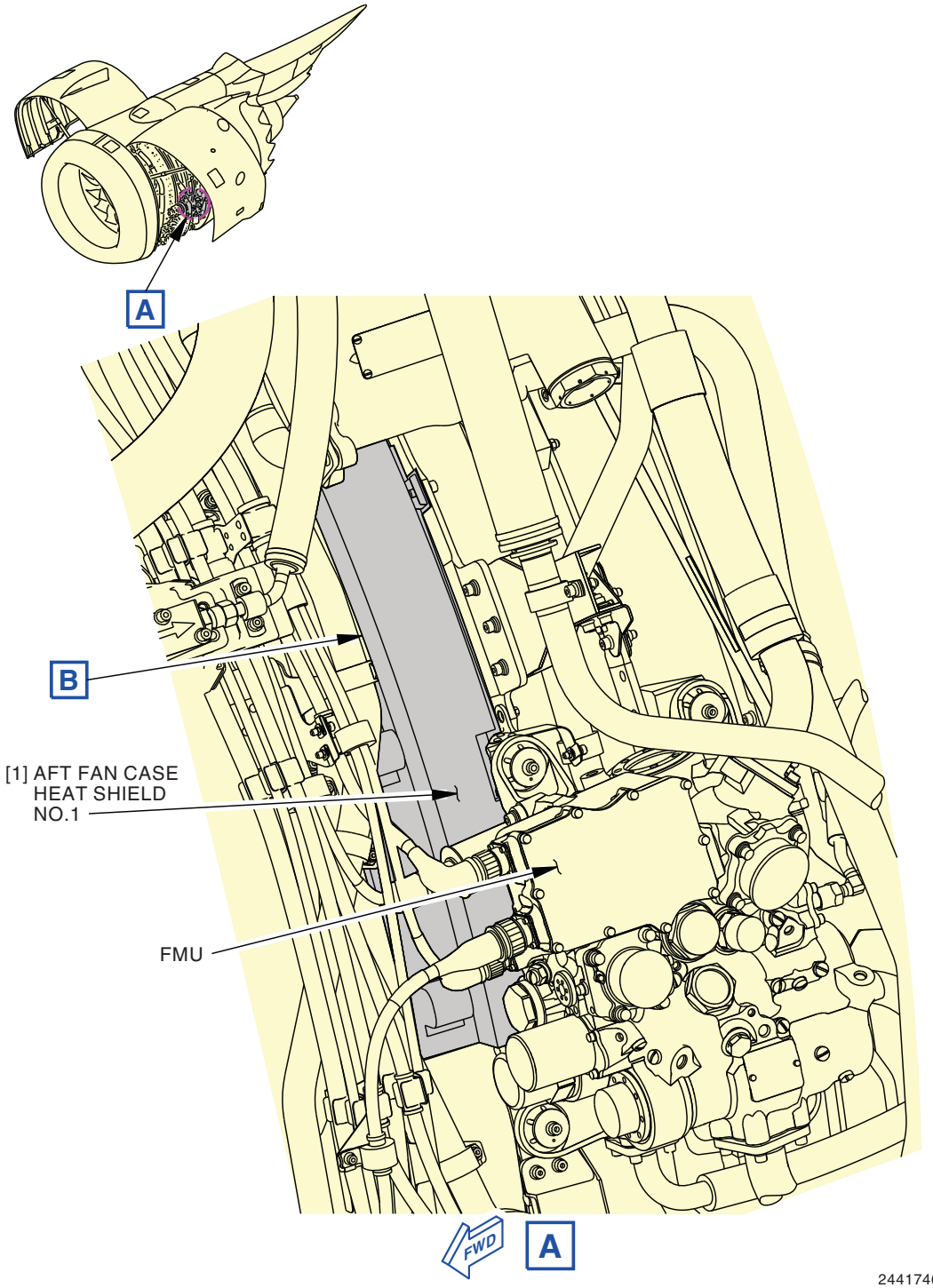
(f) Remove the aft fan case heat shield No. 1 [1] from the fan frame shroud [7].

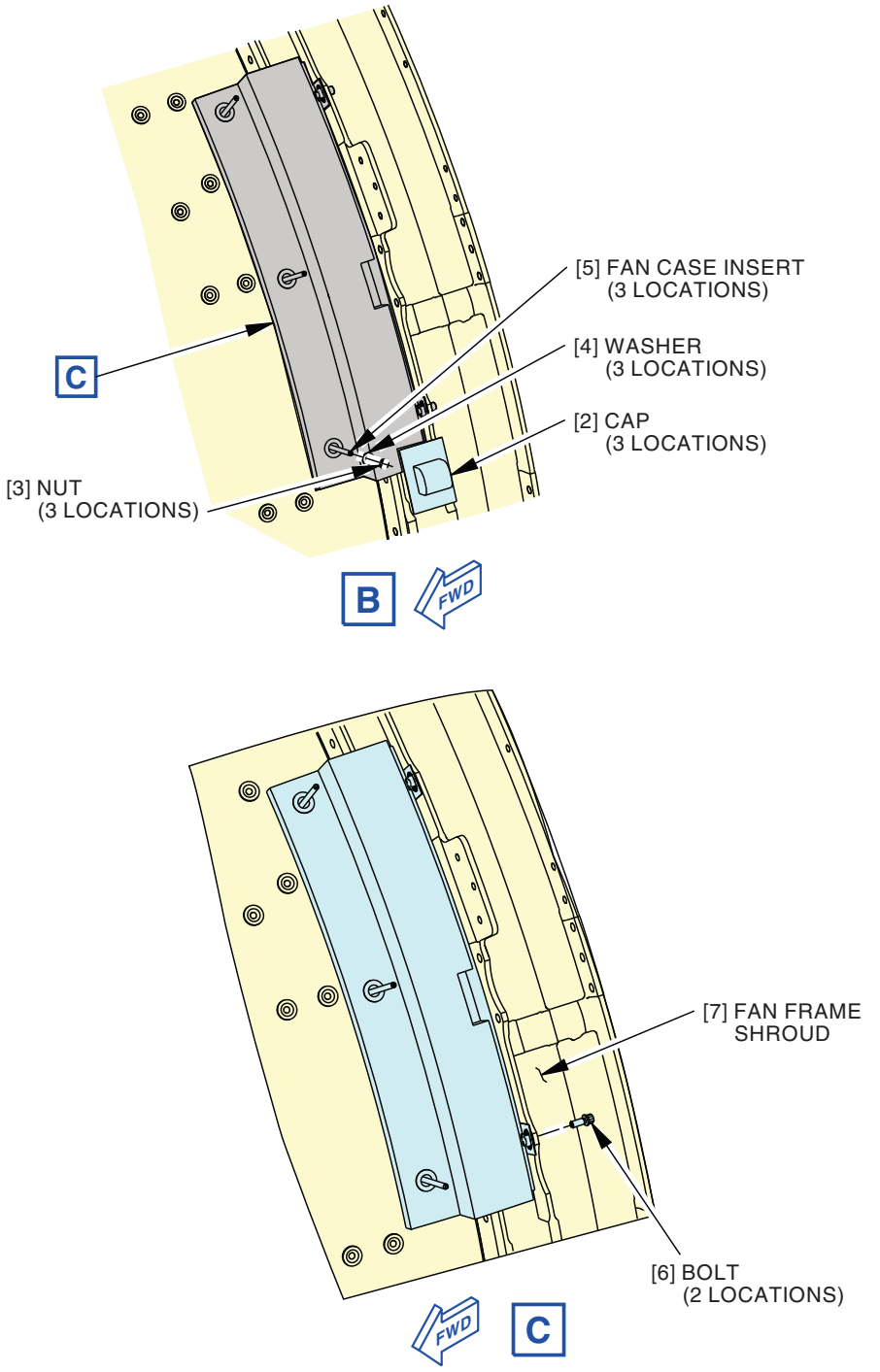
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aft Fan Case Heat Shield No. 1 Installation
Figure 401/72-21-07-990-801-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Aft Fan Case Heat Shield No. 1 Installation
Figure 401/72-21-07-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-21-07-400-801-G00

3. Aft Fan Case Heat Shield No. 1 Installation

(Figure 401)

A. General

(1) This task provides the instructions on how to install the aft fan case heat shield No. 1.

B. References

Reference	Title
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-21-04-400-801-G00	Fuel Metering Unit Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Aft fan case heat shield No. 1	72-21-07-01-250	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Aft Fan Case Heat Shield Installation

SUBTASK 72-21-07-420-001-G00

(1) Install the aft fan case heat shield No. 1 [1] as follows:

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Install the aft fan case heat shield No. 1 [1] in the correct position on the engine.

(c) Lubricate the threads of the three nuts [3] with graphite mineral vaseline, D00601 [CP2101].

(d) Lubricate the threads of the two bolts [6] with engine oil, D00599 [CP2442].

(e) Install the two bolts [6] to attach the aft fan case heat shield No. 1 [1] on the fan frame shroud [7].

(f) Tighten the bolts [6] to 67.5 in-lb (7.6 N·m) - 74.5 in-lb (8.4 N·m).

(g) Install the three new washers [4] and the three nuts [3] on the fan case inserts [5].

(h) Tighten the nuts [3] to 30.1 in-lb (3.4 N·m) - 31.9 in-lb (3.6 N·m).

(i) Install the three caps [2] on the heat shield.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(j) Do this task: Fuel Metering Unit Installation, TASK 73-21-04-400-801-G00.

(k) Clean the work area and remove all tools used to do the installation.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-07-410-001-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-21-07-860-004-G00

(2) For the left engine, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 72-21-07-860-005-G00

(3) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-21-07-860-006-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

TASK 72-21-07-000-802-G00

4. Aft Fan Case Heat Shield No. 2 Removal

(Figure 402)

A. General

(1) This task provides the instructions on how to remove the aft fan case heat shield No. 2.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
72-63-01-000-801-G00	Accessory Gearbox Assembly Removal (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 72-21-07-860-007-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-21-07-860-008-G00

(2) For the left engine, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 72-21-07-860-009-G00

(3) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-21-07-860-010-G00

WARNING BE CAREFUL WHEN YOU WORK ON THE ENGINE COMPONENTS IMMEDIATELY AFTER THE ENGINE IS SHUTDOWN. THE ENGINE COMPONENTS CAN STAY HOT FOR UP TO ONE HOUR AND CAN CAUSE INJURY.

(4) Make sure that the engine temperature is cool enough to work on.

SUBTASK 72-21-07-010-003-G00

(5) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Aft Fan Case Heat Shield No. 2 Removal

SUBTASK 72-21-07-020-009-G00

(1) Remove the aft fan case heat shield No. 2 [21] as follows:

(a) Remove the accessory gearbox assembly [22] (TASK 72-63-01-000-801-G00).

(b) Remove the two caps [23] from the heat shield.

(c) Remove the two nuts [24] from the two fan case inserts [26].

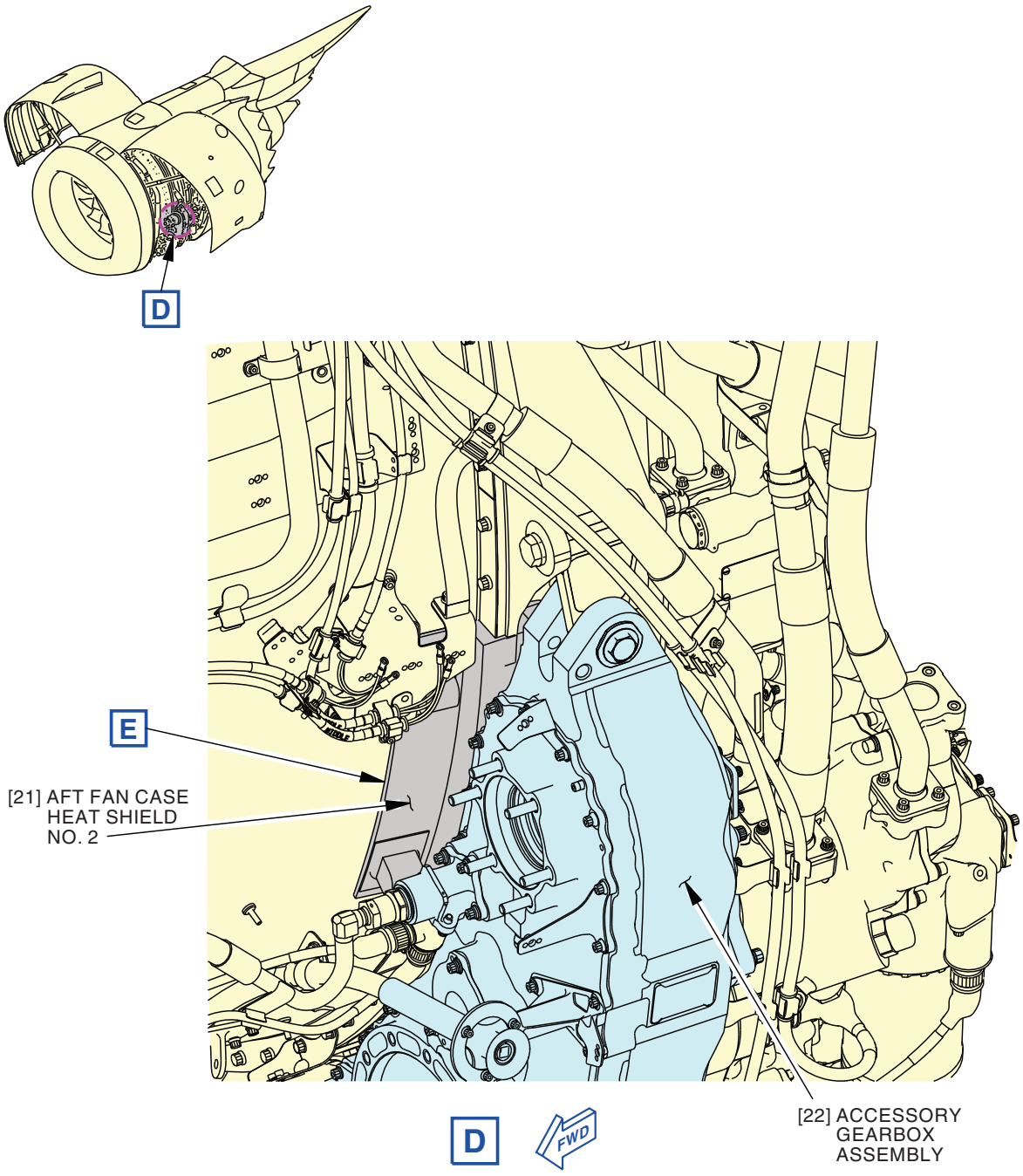
(d) Remove and discard the washers [25].

(e) Remove the two bolts [27] that attach the aft fan case heat shield No. 2 [21] to the fan frame shroud [7].

(f) Remove the aft fan case heat shield No. 2 [21] from the fan frame shroud [7].

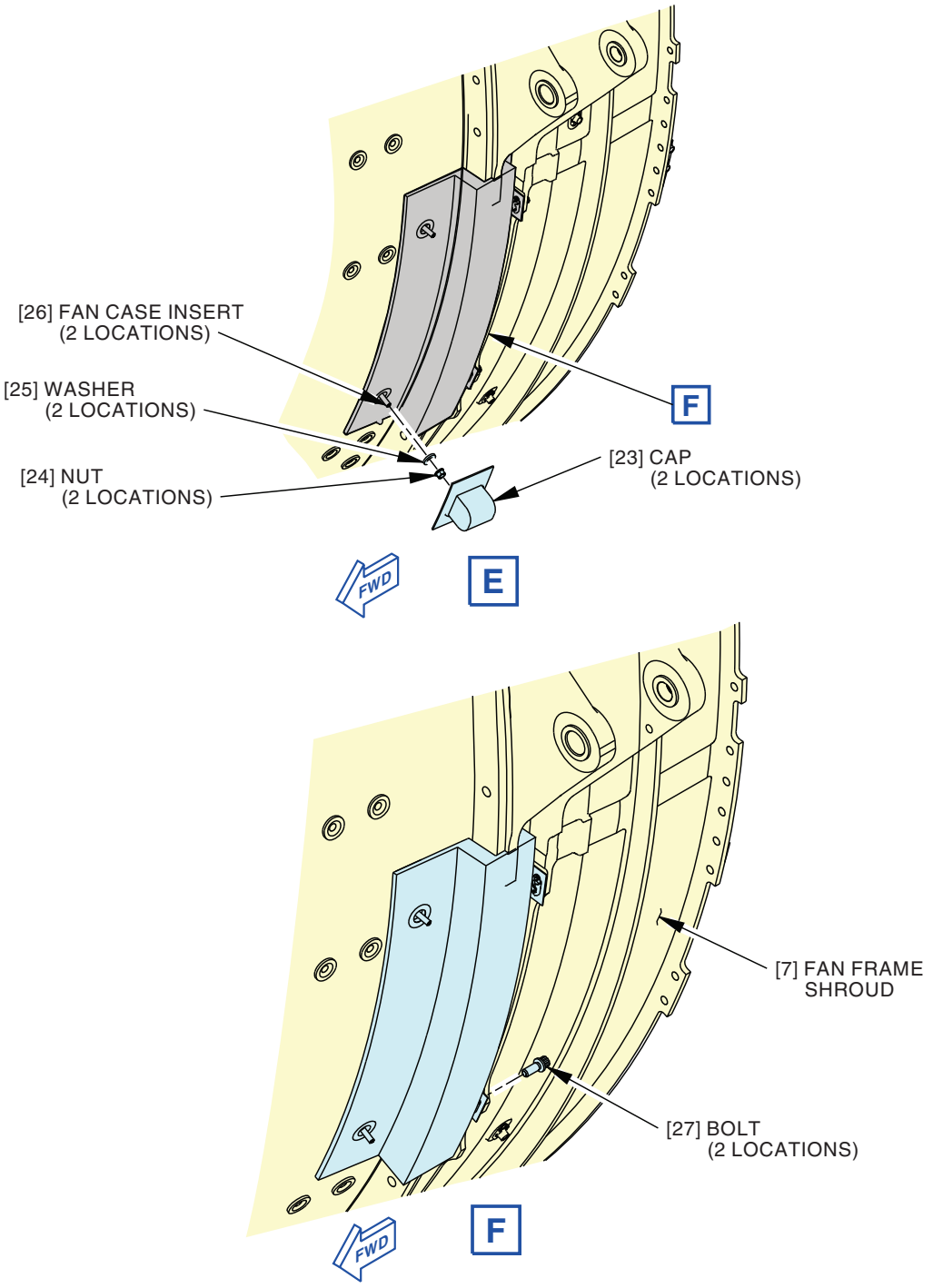
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aft Fan Case Heat Shield No. 2 Installation
Figure 402/72-21-07-990-809-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Aft Fan Case Heat Shield No. 2 Installation
Figure 402/72-21-07-990-809-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-21-07-400-802-G00

5. Aft Fan Case Heat Shield No. 2 Installation

(Figure 402)

A. General

(1) This task provides the instructions on how to install the aft fan case heat shield No. 2.

B. References

Reference	Title
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-63-01-400-801-G00	Accessory Gearbox Assembly Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
21	Aft fan case heat shield No. 2	72-21-07-01-260	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Aft Fan Case Heat Shield No. 2 Installation

SUBTASK 72-21-07-420-009-G00

(1) Install the aft fan case heat shield No. 2 [21] as follows:

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Install the aft fan case heat shield No. 2 [21] into the correct position on the engine.

(c) Lubricate the threads of the two nuts [24] with graphite mineral vaseline, D00601 [CP2101].

(d) Lubricate the threads of the two bolts [27] with engine oil, D00599 [CP2442].

(e) Install the two bolts [27] to attach the aft fan case heat shield No. 2 [21] on the fan frame shroud [7].

(f) Tighten the bolts [27] to 67.5 in-lb (7.6 N·m) - 74.5 in-lb (8.4 N·m).

(g) Install the two new washers [25] and the two nuts [24] on the two fan case insert [26].

(h) Tighten the nuts [24] to 30.1 in-lb (3.4 N·m) - 31.9 in-lb (3.6 N·m).

(i) Install the two caps [23] on the heat shield.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(j) Install the accessory gearbox assembly [22] (TASK 72-63-01-400-801-G00).

(k) Clean the work area and remove all tools used to do the installation.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-07-410-003-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-21-07-860-011-G00

(2) For the left engine, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 72-21-07-860-012-G00

(3) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-21-07-860-013-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

TASK 72-21-07-000-803-G00

6. Aft Fan Case Heat Shield No. 3 Removal

(Figure 403)

A. General

(1) This task provides the instructions on how to remove the aft fan case heat shield No. 3.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 72-21-07-860-014-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-21-07-860-015-G00

(2) For the left engine, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 72-21-07-860-016-G00

(3) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-21-07-700-001-G00

WARNING BE CAREFUL WHEN YOU WORK ON THE ENGINE COMPONENTS IMMEDIATELY AFTER THE ENGINE IS SHUTDOWN. THE ENGINE COMPONENTS CAN STAY HOT FOR UP TO ONE HOUR AND CAN CAUSE INJURY.

(4) Make sure that the engine temperature is cool enough to work on.

SUBTASK 72-21-07-010-004-G00

(5) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. Aft Fan Case Heat Shield No. 3 Removal

SUBTASK 72-21-07-020-010-G00

(1) Remove the aft fan case heat shield No. 3 [41] as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the cap [42] from the heat shield.

(b) Remove the nut [43] from the fan case insert [45].

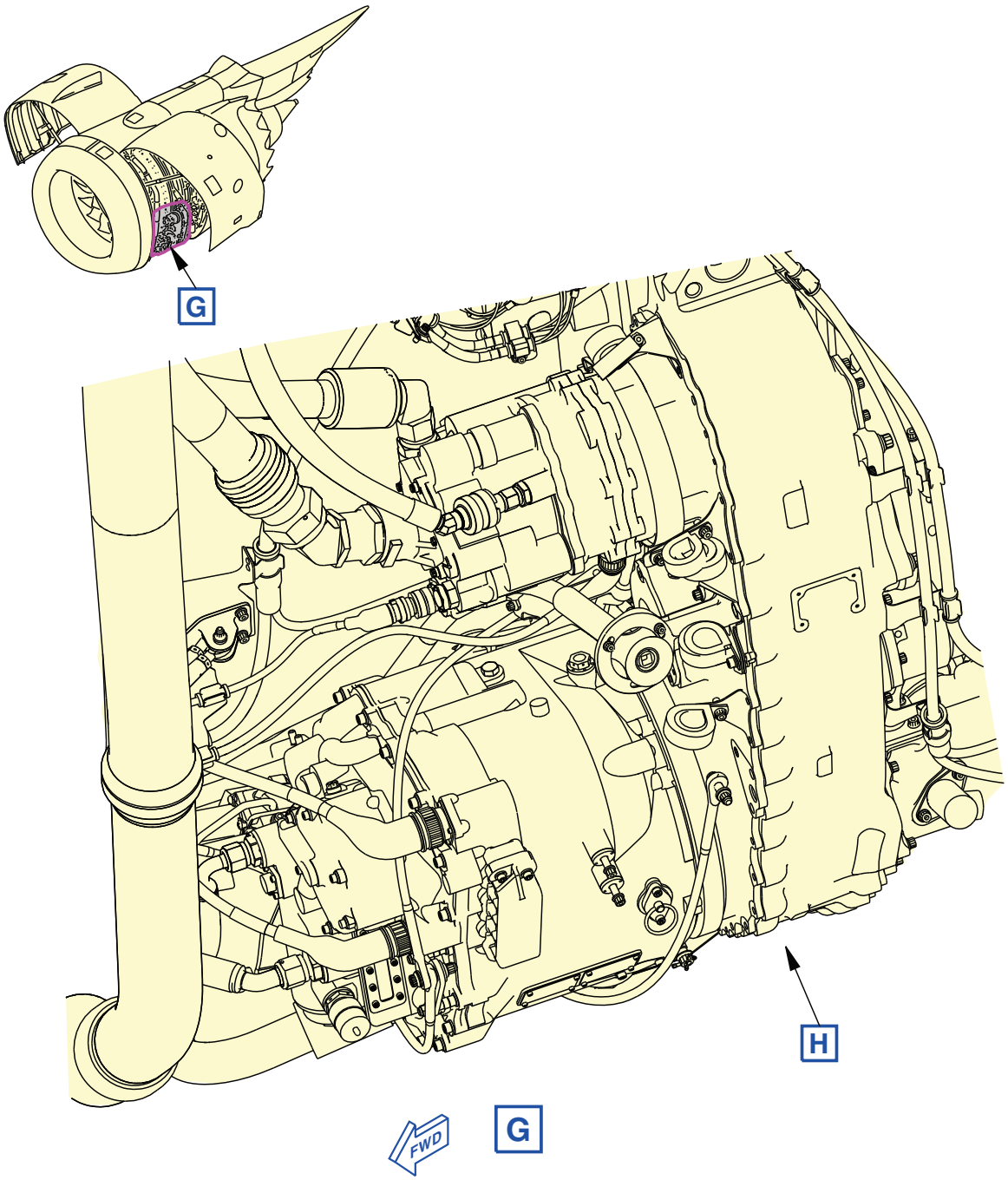
(c) Remove and discard the washer [44].

(d) Remove the bolt [46] that attaches the aft fan case heat shield No. 3 [41] to the fan frame shroud [7].

(e) Remove the aft fan case heat shield No. 3 [41] from the fan frame shroud [7].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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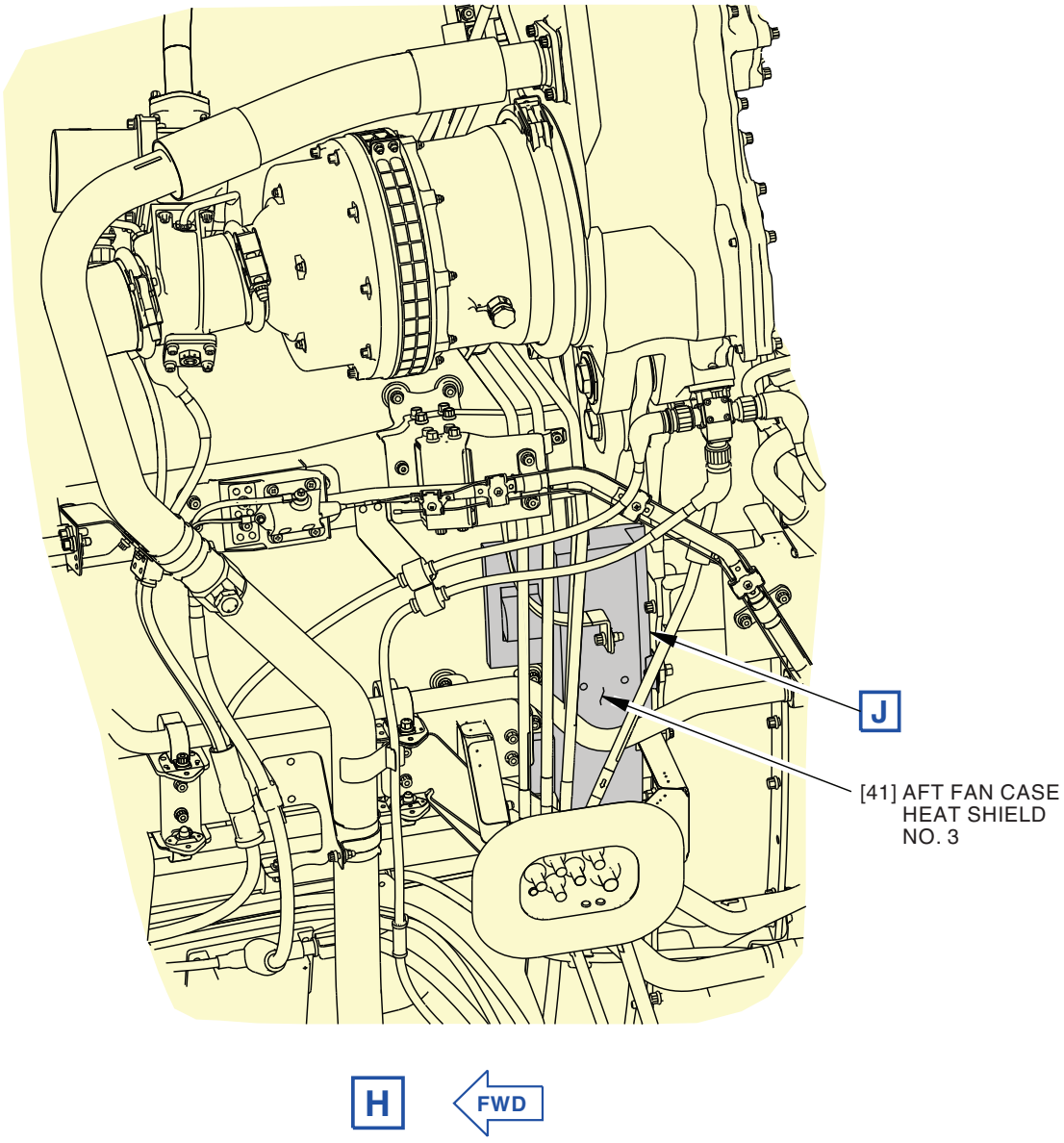
Aft Fan Case Heat Shield No. 3 Installation
Figure 403/72-21-07-990-810-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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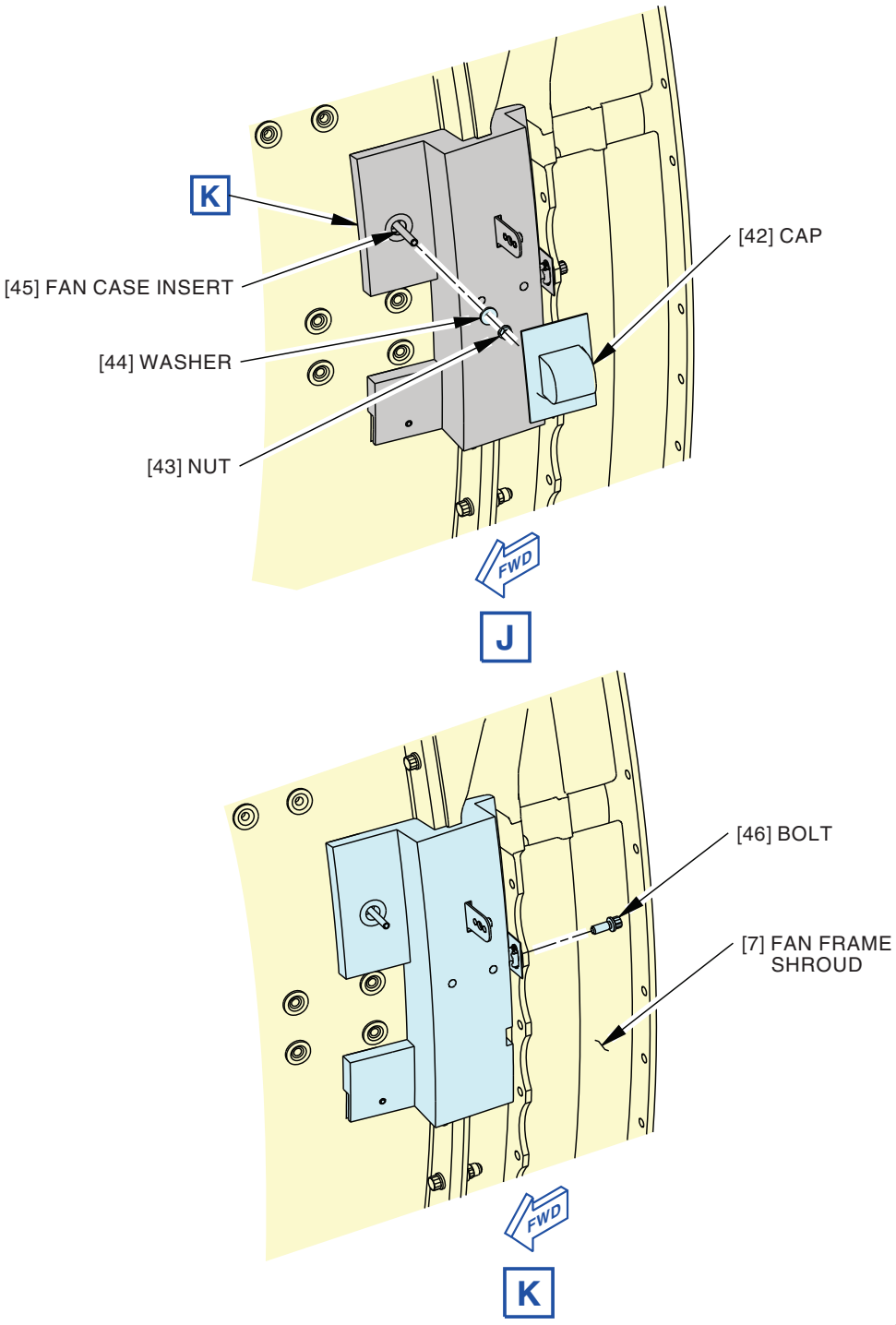
Aft Fan Case Heat Shield No. 3 Installation
Figure 403/72-21-07-990-810-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aft Fan Case Heat Shield No. 3 Installation
Figure 403/72-21-07-990-810-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-21-07-400-803-G00

7. Aft Fan Case Heat Shield No. 3 Installation

(Figure 403)

A. General

(1) This task provides the instructions on how to install the aft fan case heat shield No. 3.

B. References

Reference	Title
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
41	Aft fan case heat shield No. 3	72-21-07-01-270	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Aft Fan Case Heat Shield No. 3 Installation

SUBTASK 72-21-07-420-010-G00

(1) Install the aft fan case heat shield No. 3 [41] as follows:

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Install the aft fan case heat shield No. 3 [41] in the correct position on the engine.

(c) Lubricate the threads of the nut [43] with graphite mineral vaseline, D00601 [CP2101].

(d) Lubricate the threads of the bolt [46] with engine oil, D00599 [CP2442].

(e) Install the bolts [46] to attach the aft fan case heat shield No. 3 [41] on the fan frame shroud [7].

(f) Tighten the bolt [46] to 67.5 in-lb (7.6 N·m) - 74.5 in-lb (8.4 N·m).

(g) Install the new washer [44] and the nut [43] on the two fan case insert [26].

(h) Tighten the nut [43] to 30.1 in-lb (3.4 N·m) - 31.9 in-lb (3.6 N·m).

(i) Install the cap [42] on the heat shield.

(j) Clean the work area and remove all tools used to do the installation.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-07-410-004-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-21-07-860-017-G00

(2) For the left engine, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 72-21-07-860-018-G00

(3) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-21-07-860-019-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

TASK 72-21-07-000-804-G00

8. Aft Fan Case Heat Shield No. 4 Removal

(Figure 404)

A. General

(1) This task gives the instructions to remove the aft fan case heat shield No. 4.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 72-21-07-860-020-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-21-07-860-021-G00

(2) For the left engine, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 72-21-07-860-022-G00

(3) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-21-07-700-002-G00

WARNING BE CAREFUL WHEN YOU WORK ON THE ENGINE COMPONENTS IMMEDIATELY AFTER THE ENGINE IS SHUTDOWN. THE ENGINE COMPONENTS CAN STAY HOT FOR UP TO ONE HOUR AND CAN CAUSE INJURY.

(4) Make sure that the engine temperature is cool enough to work on.

SUBTASK 72-21-07-010-005-G00

(5) On the applicable engine, open the right fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

G. Aft Fan Case Heat Shield No. 4 Removal

SUBTASK 72-21-07-020-011-G00

(1) Remove the aft fan case heat shield No. 4 [61] as follows:

(a) Remove the eleven clips [62] from the harnesses adjacent to the aft fan case heat shield No. 4 [61].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Remove the bolt [64] from the clamp [65] that attaches the blower duct [63] to the bracket [66].

(c) Remove the bolt [67] from the clamp [68] that attaches blower duct [63] to the bracket [69].

(d) Remove the clamp [70] and the seal [71] to disconnect the blower duct [63].

(e) Remove and discard the seal [71].

(f) Remove the clamp [72] and the seal [73] to remove the blower duct [63].

(g) Remove and discard the seal [73].

(h) Remove the two caps [74] from the heat shield.

(i) Remove the two nuts [75] from the two fan case insert [77].

(j) Remove and discard the two washers [76].

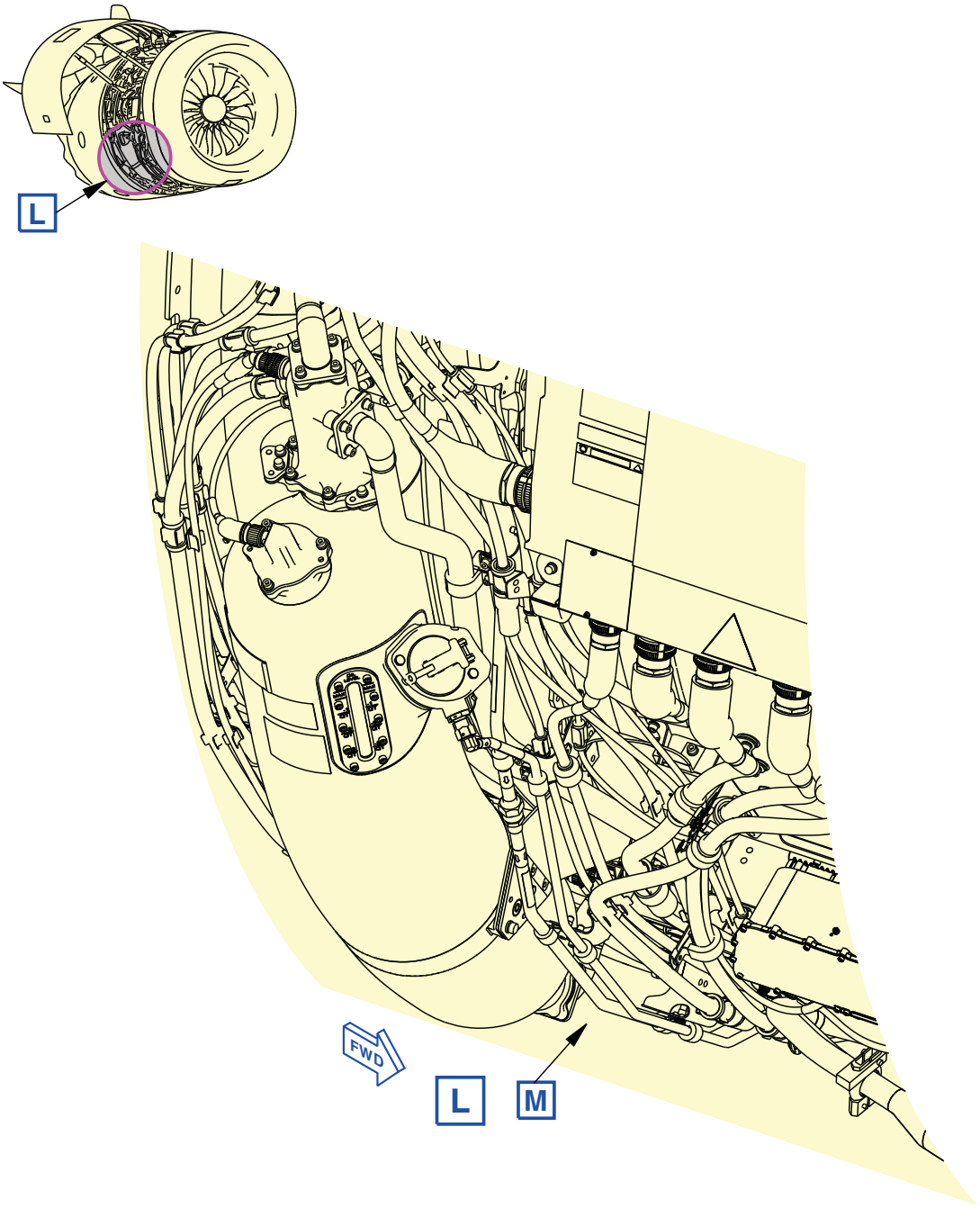
(k) Remove the two bolts [78] that attach the aft fan case heat shield No. 4 [61] to the fan frame shroud [7].

(l) Remove the aft fan case heat shield No. 4 [61] from the fan frame shroud [7].

(m) Install the protective covers on the openings where the blower duct [63] is connected.

——— END OF TASK ———

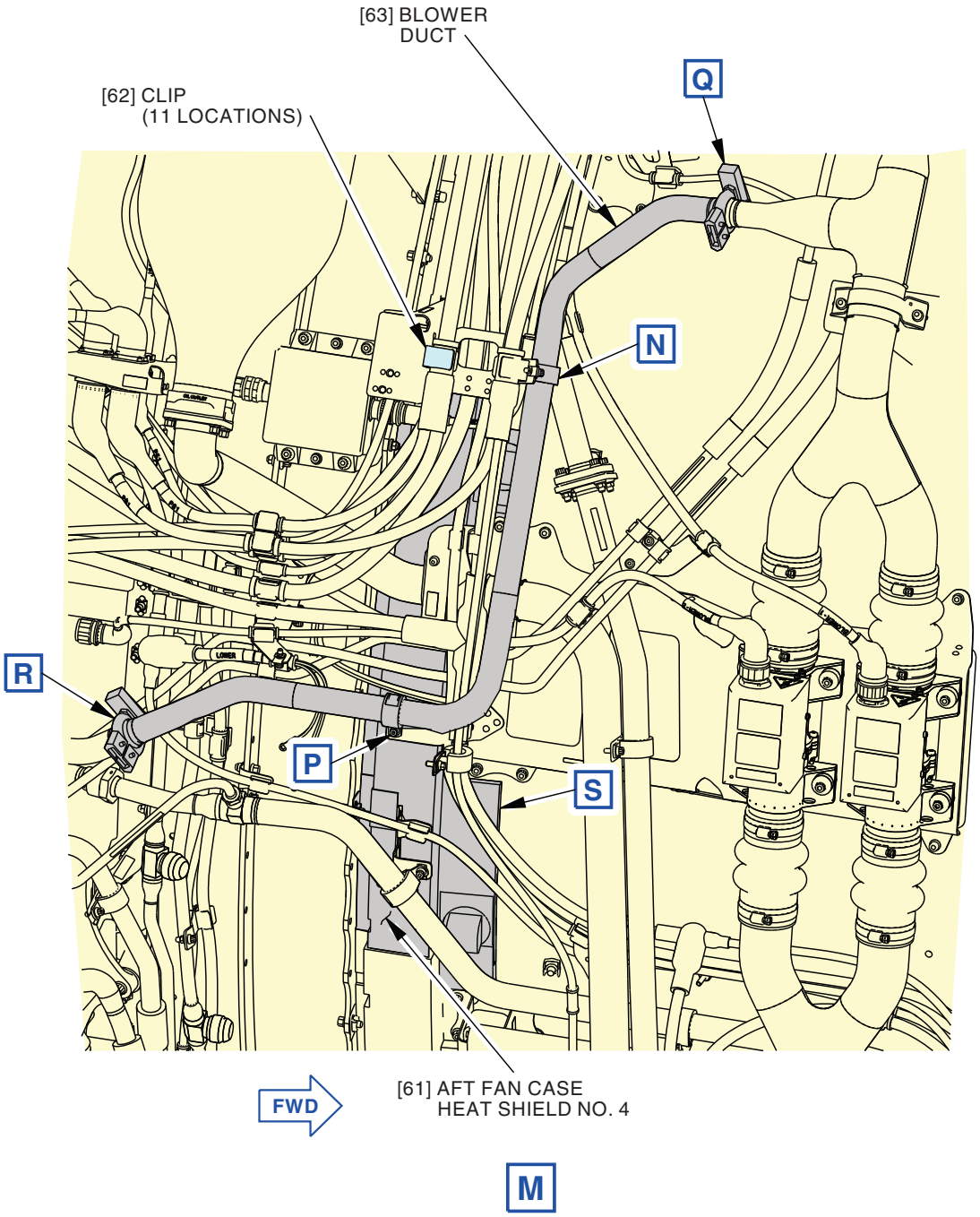
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2441919 S0000566770_V1

Aft Fan Case Heat Shield No. 4 Installation
Figure 404/72-21-07-990-811-G00 (Sheet 1 of 6)

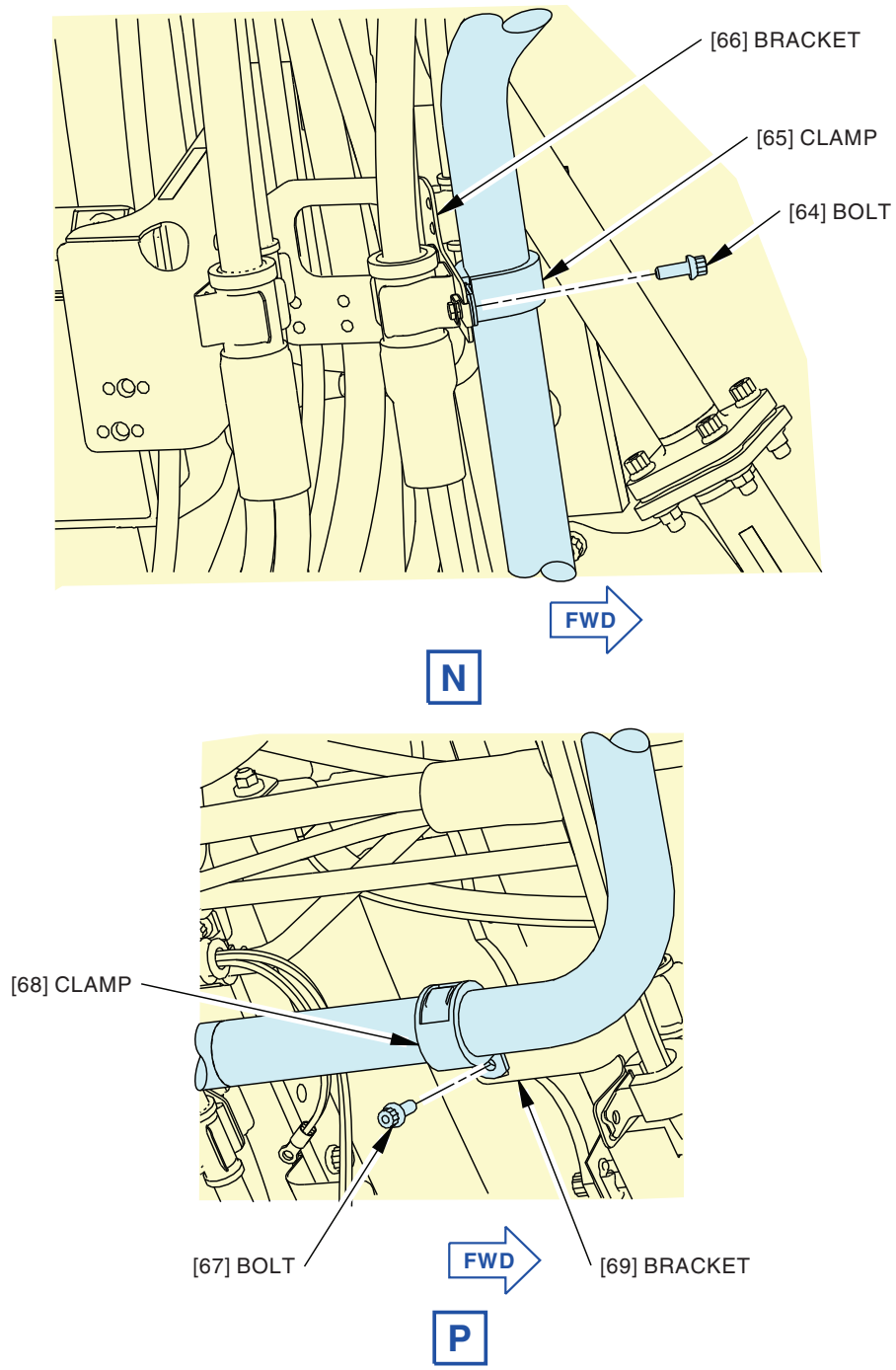
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2441929 S0000566771_V2

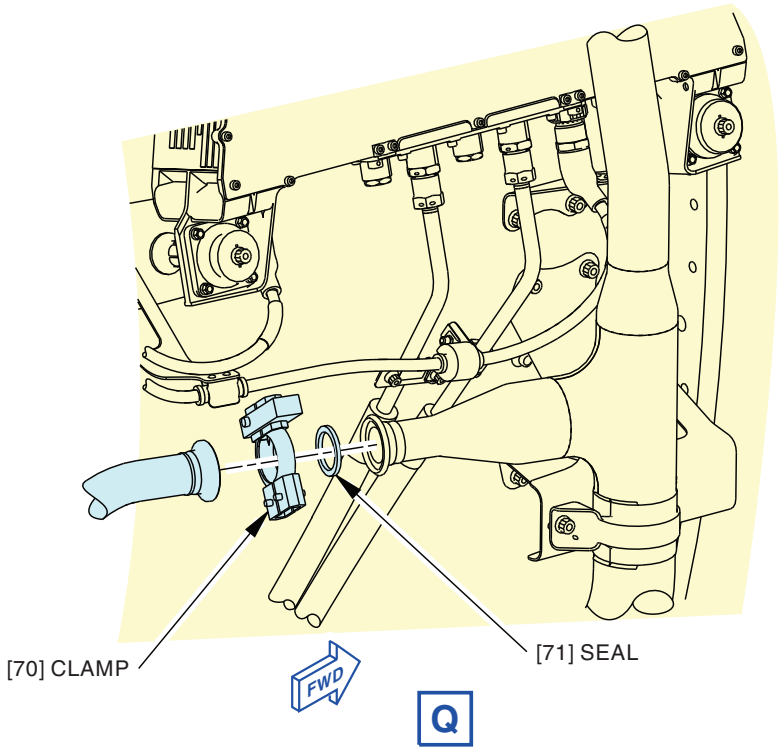
Aft Fan Case Heat Shield No. 4 Installation
Figure 404/72-21-07-990-811-G00 (Sheet 2 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



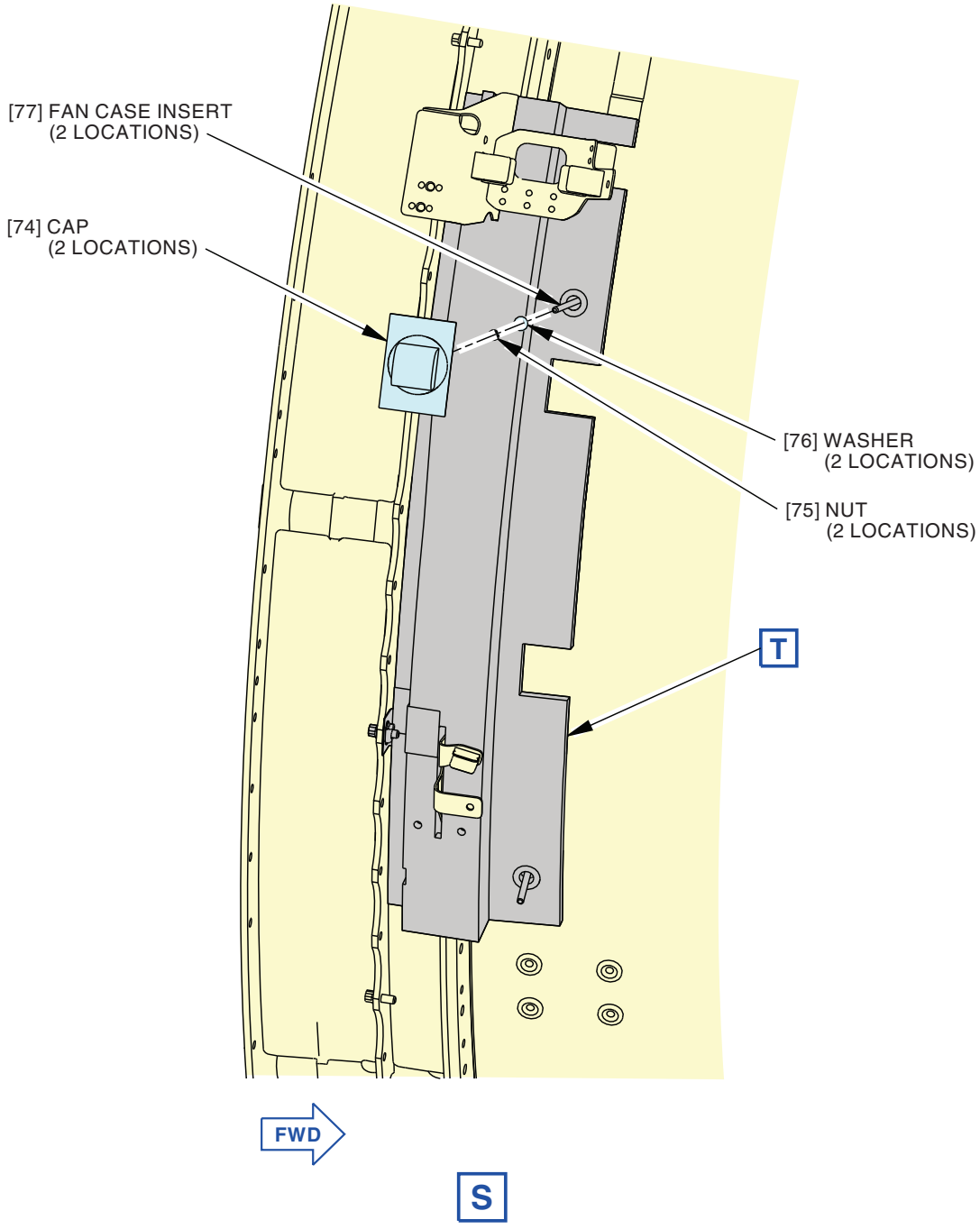
Aft Fan Case Heat Shield No. 4 Installation
Figure 404/72-21-07-990-811-G00 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aft Fan Case Heat Shield No. 4 Installation
Figure 404/72-21-07-990-811-G00 (Sheet 4 of 6)

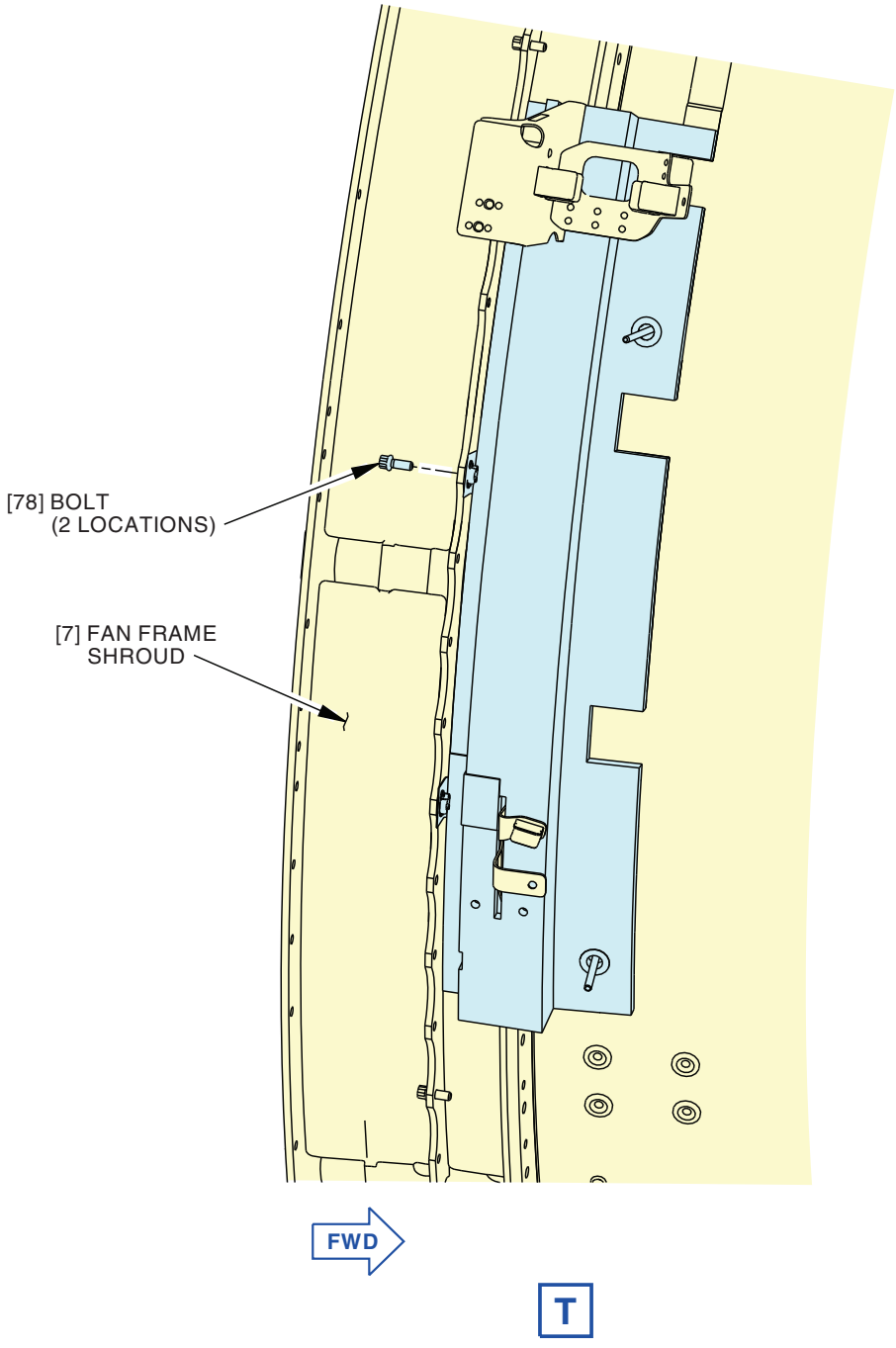
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2441980 S0000566774_V2

Aft Fan Case Heat Shield No. 4 Installation
Figure 404/72-21-07-990-811-G00 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2441996 S0000566775_V2

Aft Fan Case Heat Shield No. 4 Installation
Figure 404/72-21-07-990-811-G00 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-21-07-400-804-G00

9. Aft Fan Case Heat Shield No. 4 Installation

(Figure 404)

A. General

(1) This task gives the instructions to install the aft fan case heat shield No. 4.

B. References

Reference	Title
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
61	Aft fan case heat shield No. 4	72-21-07-01-280	PTF ALL
71	Seal	72-00-00-14-670	PTF ALL
73	Seal	72-00-00-14-670	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

H. Aft Fan Case Heat Shield No. 4 Installation

SUBTASK 72-21-07-420-011-G00

(1) Install the aft fan case heat shield No. 4 [61] as follows:

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Remove the protective covers from the openings for the blower duct [63].

(c) Install the aft fan case heat shield No. 4 [61] in the correct position on the engine.

(d) Lubricate the new seal [71], the new seal [73], the two nuts [75], and the threads of the two bolts [78] with engine oil, D00599 [CP2442].

(e) Lubricate the threads of the two nuts [75] with graphite mineral vaseline, D00601 [CP2101].

(f) Install the bolts [78] to attach the aft fan case heat shield No. 4 [61] on the fan frame shroud [7].

(g) Tighten the bolts [78] to 67.5 in-lb (7.6 N·m) - 74.5 in-lb (8.4 N·m).

(h) Install the two new washers [76] and the nuts [75] on the two fan case inserts [77].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(i) Tighten the nuts [75] to 30.1 in-lb (3.4 N·m) - 31.9 in-lb (3.6 N·m).

(j) Install the two caps [74] on the heat shield.

(k) Install the new seal [73] and the clamp [72] to connect the blower duct [63].

(l) Install the new seal [71] and the clamp [70] to connect the blower duct [63].

(m) Install the bolt [67] on the clamp [68] that attaches the blower duct [63] to the bracket [69].

(n) Tighten the bolt [67] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

(o) Install the bolt [64] on the clamp [65] that attaches the blower duct [63] to the bracket [69].

(p) Tighten the bolt [64] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

(q) Install the 11 clips [62] to all the harnesses adjacent to the aft fan case heat shield No. 4 [61].

(r) Clean the work area and clear the tools and other items.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-07-410-005-G00

(1) On the applicable engine, close the right fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

SUBTASK 72-21-07-860-023-G00

(2) For the left engine, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 72-21-07-860-024-G00

(3) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-21-07-860-025-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-21-07-000-805-G00

10. Aft Fan Case Heat Shield No. 5 Removal

(Figure 405)

A. General

(1) This task gives the instructions to remove the aft fan case heat shield No. 5.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-193	Container
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 72-21-07-860-026-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-21-07-860-027-G00

(2) For the left engine, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 72-21-07-860-028-G00

(3) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-21-07-700-003-G00

WARNING BE CAREFUL WHEN YOU WORK ON THE ENGINE COMPONENTS IMMEDIATELY AFTER THE ENGINE IS SHUTDOWN. THE ENGINE COMPONENTS CAN STAY HOT FOR UP TO ONE HOUR AND CAN CAUSE INJURY.

(4) Make sure that the engine temperature is cool enough to work on.

SUBTASK 72-21-07-010-006-G00

(5) On the applicable engine, open the right fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

G. Aft Fan Case Heat Shield No. 5 Removal

SUBTASK 72-21-07-020-012-G00

(1) Remove the aft fan case heat shield No. 5 [91] as follows:

WARNING BE CAREFUL WHEN YOU DO WORK NEAR THE ENGINE OIL SYSTEM COMPONENTS AFTER THE ENGINE IS SHUTDOWN. THE OIL TEMPERATURE STAYS HIGH FOR A LONG TIME AFTER THE ENGINE STOPS. THE HIGH TEMPERATURE CAN CAUSE INJURIES TO PERSONS.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET OIL STAY ON YOUR SKIN. THE SYNTHETIC OIL CONTAINS ADDITIVES THAT CAN BE POISONOUS IF THEY ARE ABSORBED THROUGH THE SKIN. CLEAN AWAY ALL OIL THAT GETS ON THE SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(a) Put a 1.32 gal (5 liters) container, STD-193 below the engine.

(b) Remove the nine clips [92] from all the harnesses adjacent to the aft fan case heat shield No. 5 [91].

(c) Remove the bolt [94] from the clamp [95] that attaches the drain tube [93] to the bracket [96].

(d) Remove the bolt [98] from the clamp [99] that attaches the drain tube [97] to the bracket [96].

(e) Remove the bolt [101] from the clamp [102] that attaches the HJ3B engine wiring harness [100] to the bracket [96].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Remove the bolt [103] from the clamp [104] that attaches the drain tube [93] to the bracket [105].

(g) Remove the bolt [106] from the clamp [107] that attaches the drain tube [97] to the bracket [105].

(h) Remove the four bolts [109] that attach the oil tube [108] to the air oil separator [111].

(i) Disconnect the oil tube [108] from the air oil separator [111].

(j) Remove and discard the gasket seal [110].

(k) Remove the bolt [112] from the clamp [113] that attaches the oil tube [108] to the bracket [114].

(l) Remove the bolt [115] from the clamp [116] that attaches the oil tube [108] to the bracket [117].

(m) Remove the four nuts [118] and the four bolts [119] that connect the oil tube [108] to the oil tube [121].

(n) Disconnect the oil tube [108] from the oil tube [121].

(o) Remove and discard the gasket C seal [120].

(p) Remove the two caps [122] from the heat shield.

(q) Remove the two nuts [123] from the two fan case inserts [125].

(r) Remove and discard the two washers [124].

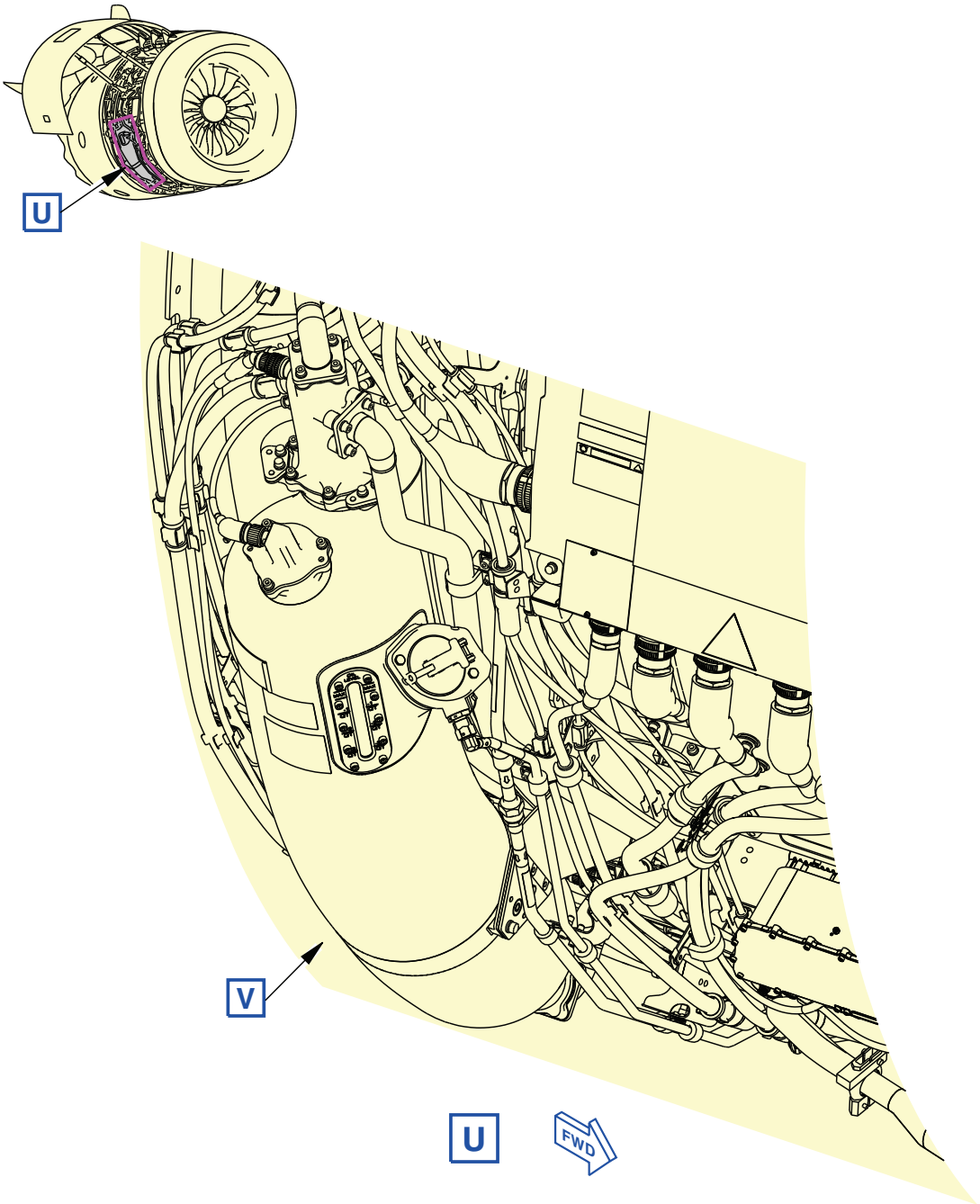
(s) Remove the three bolts [126] that attach the aft fan case heat shield No. 5 [91] to the fan frame shroud [7].

(t) Remove the aft fan case heat shield No. 5 [91] from the fan frame shroud [7].

(u) Install the protective covers on the oil tube [108], the oil tube [121], and the openings on the air oil separator [111].

——— END OF TASK ———

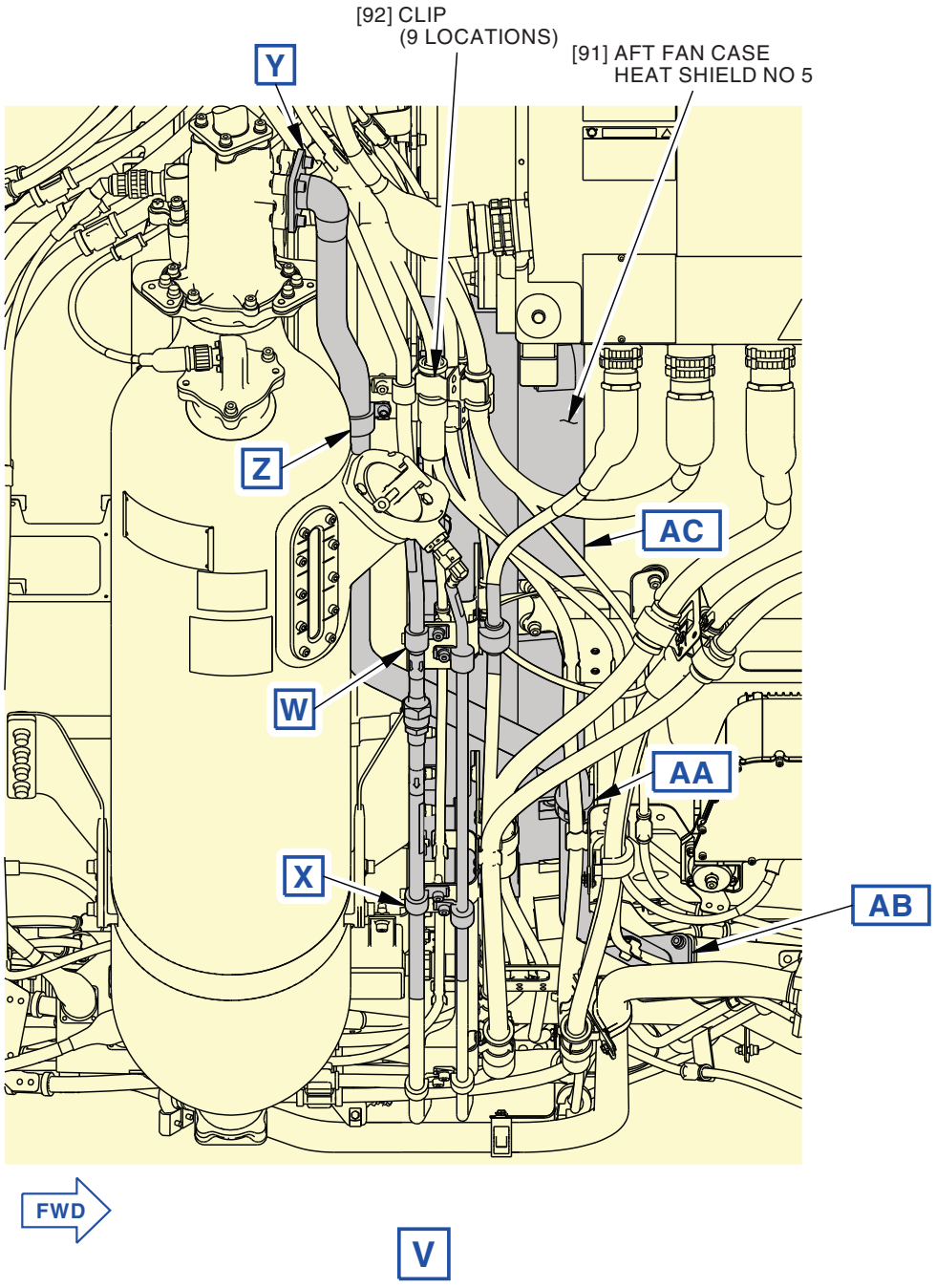
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2442009 S0000566776_V1

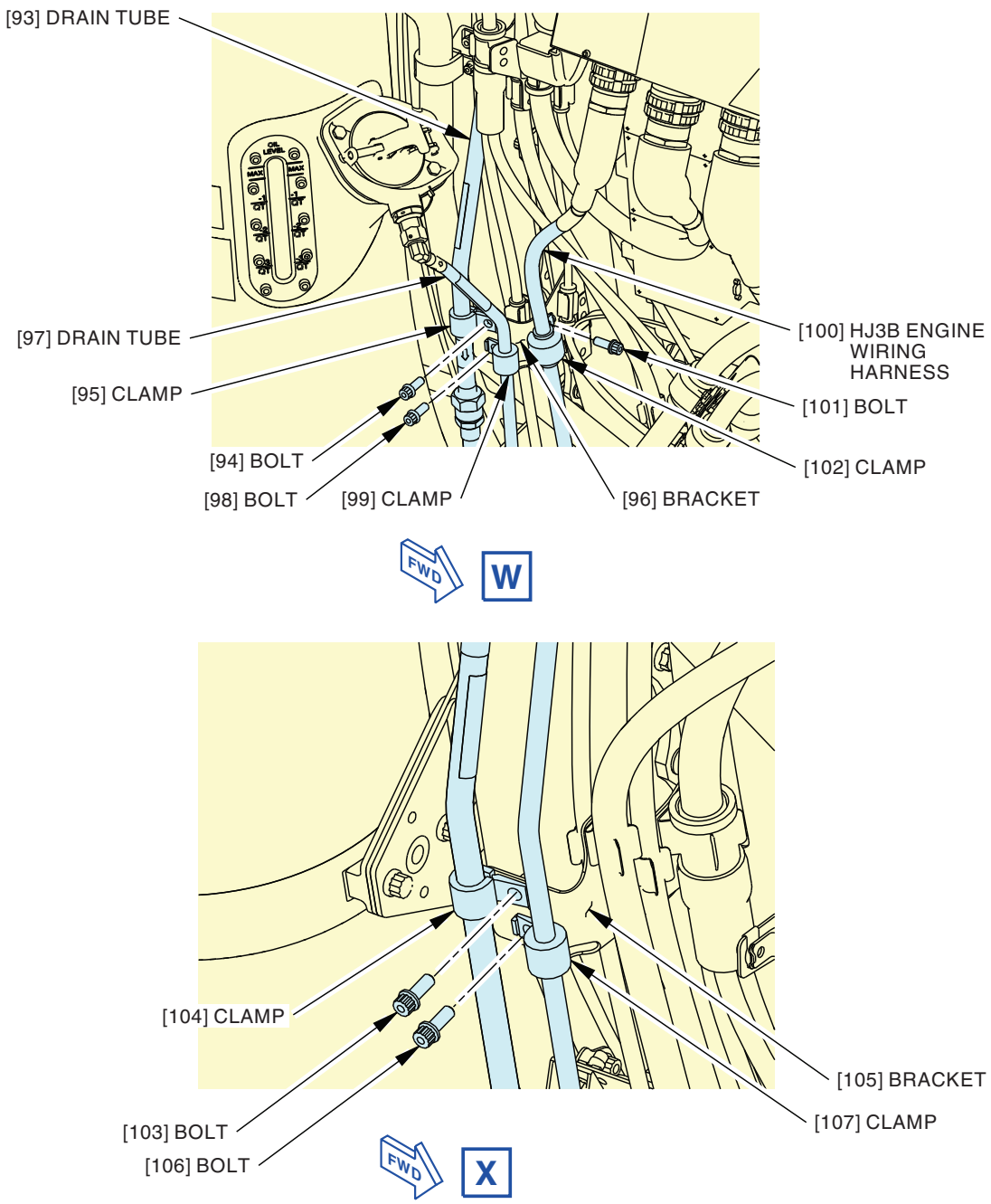
Aft Fan Case Heat Shield No. 5 Installation
Figure 405/72-21-07-990-812-G00 (Sheet 1 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aft Fan Case Heat Shield No. 5 Installation
Figure 405/72-21-07-990-812-G00 (Sheet 2 of 7)

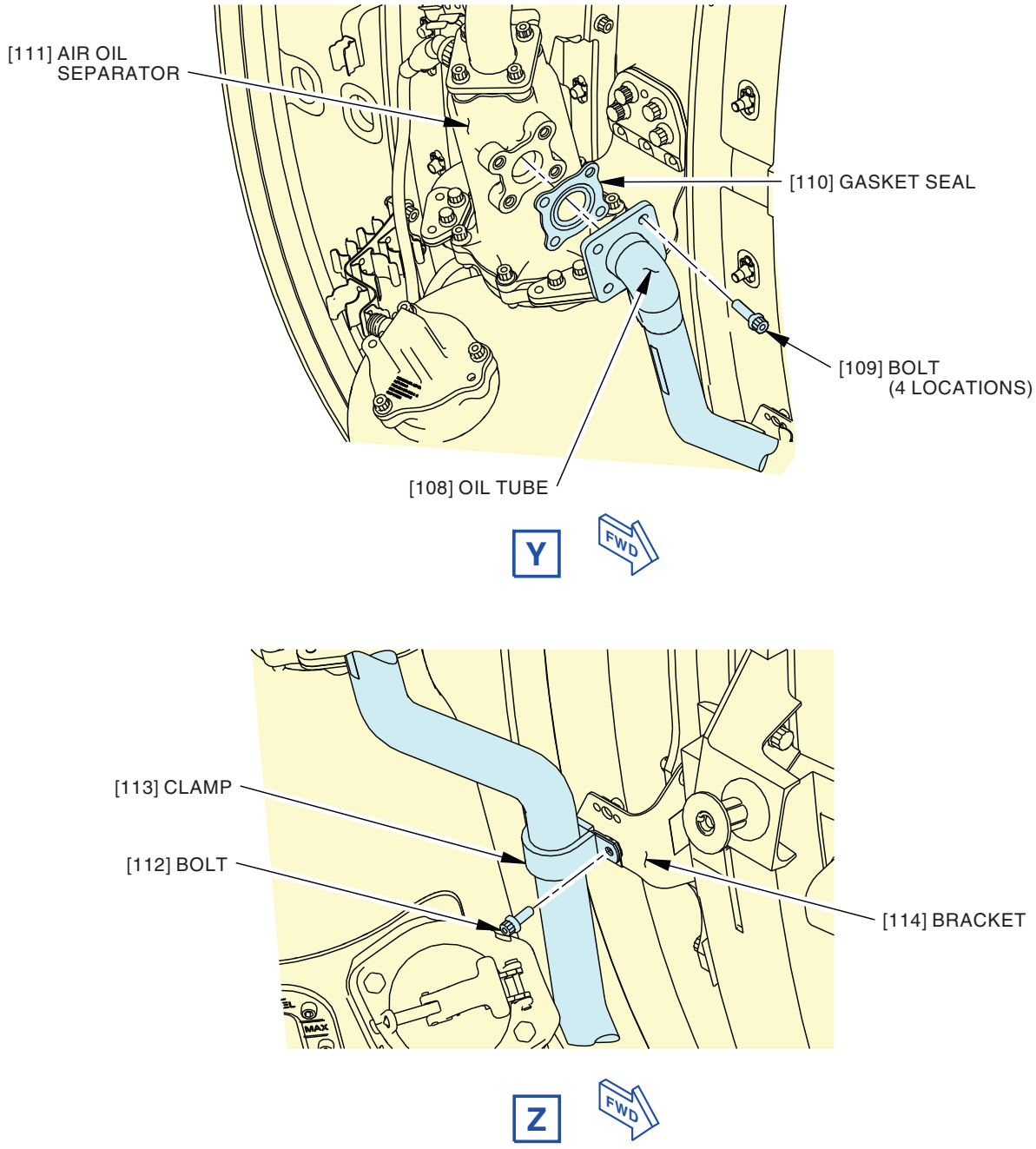
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2442024 S0000566778_V2

Aft Fan Case Heat Shield No. 5 Installation
Figure 405/72-21-07-990-812-G00 (Sheet 3 of 7)

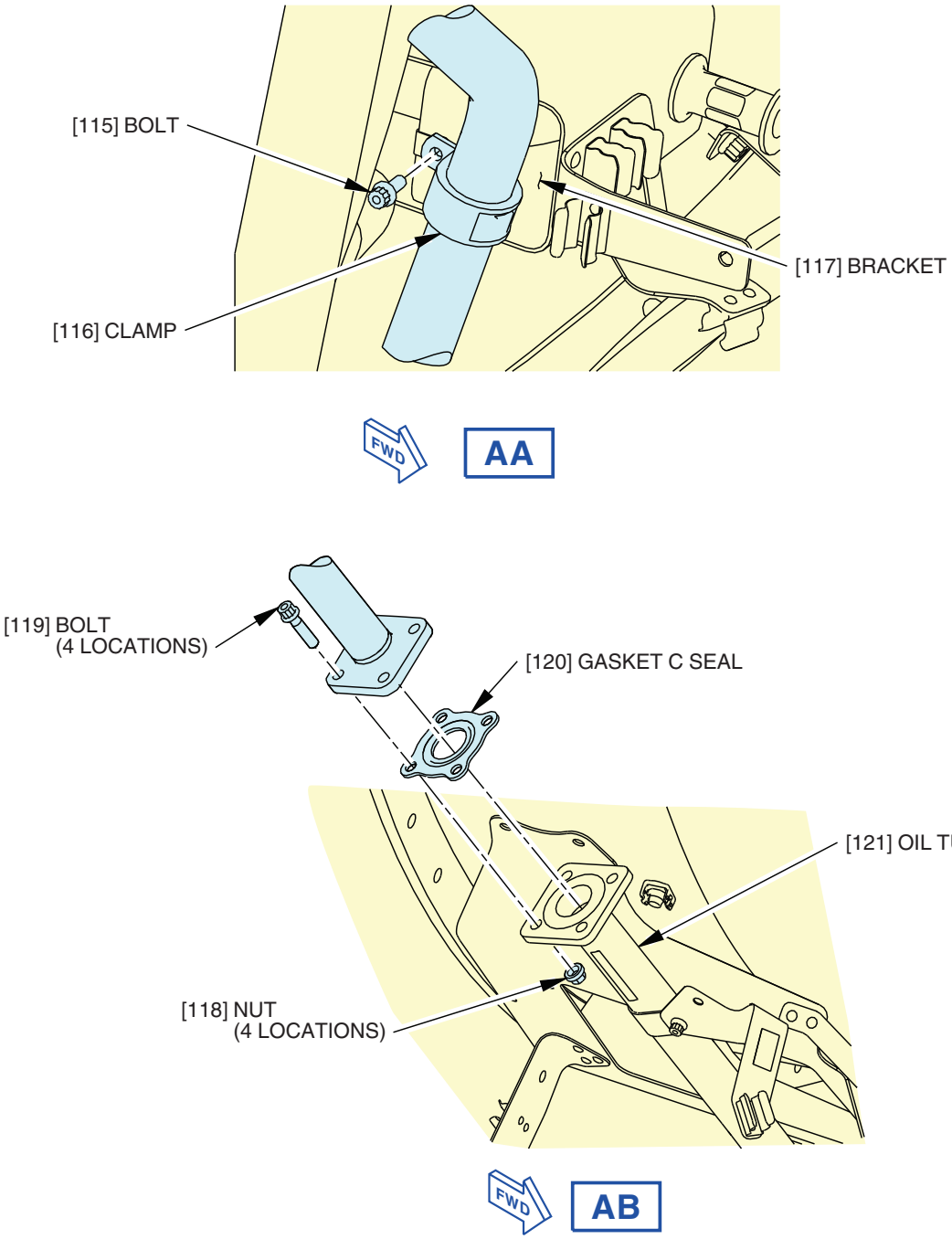
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2442030 S0000566779_V2

Aft Fan Case Heat Shield No. 5 Installation
Figure 405/72-21-07-990-812-G00 (Sheet 4 of 7)

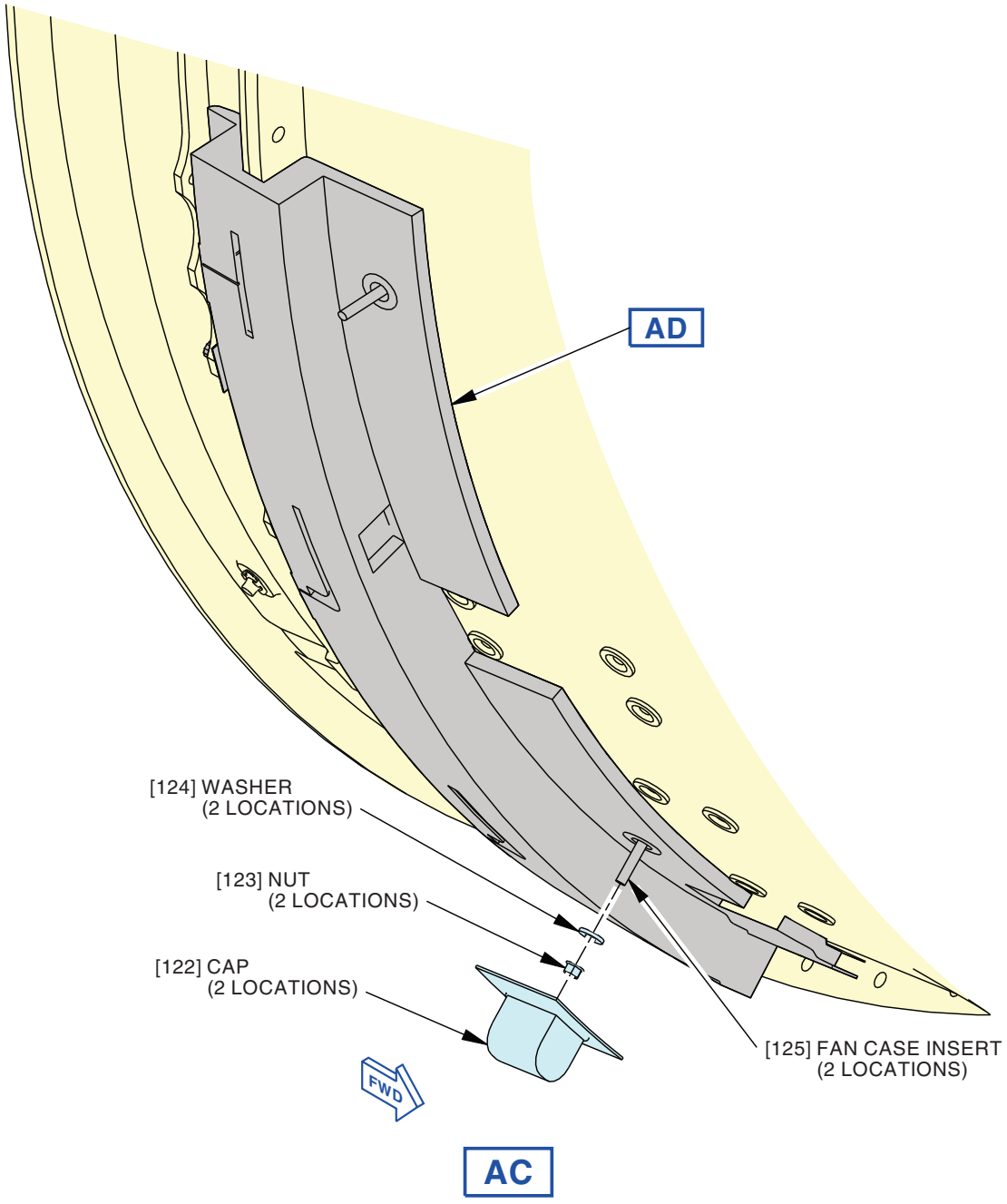
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2442033 S0000566780_V3

Aft Fan Case Heat Shield No. 5 Installation
Figure 405/72-21-07-990-812-G00 (Sheet 5 of 7)

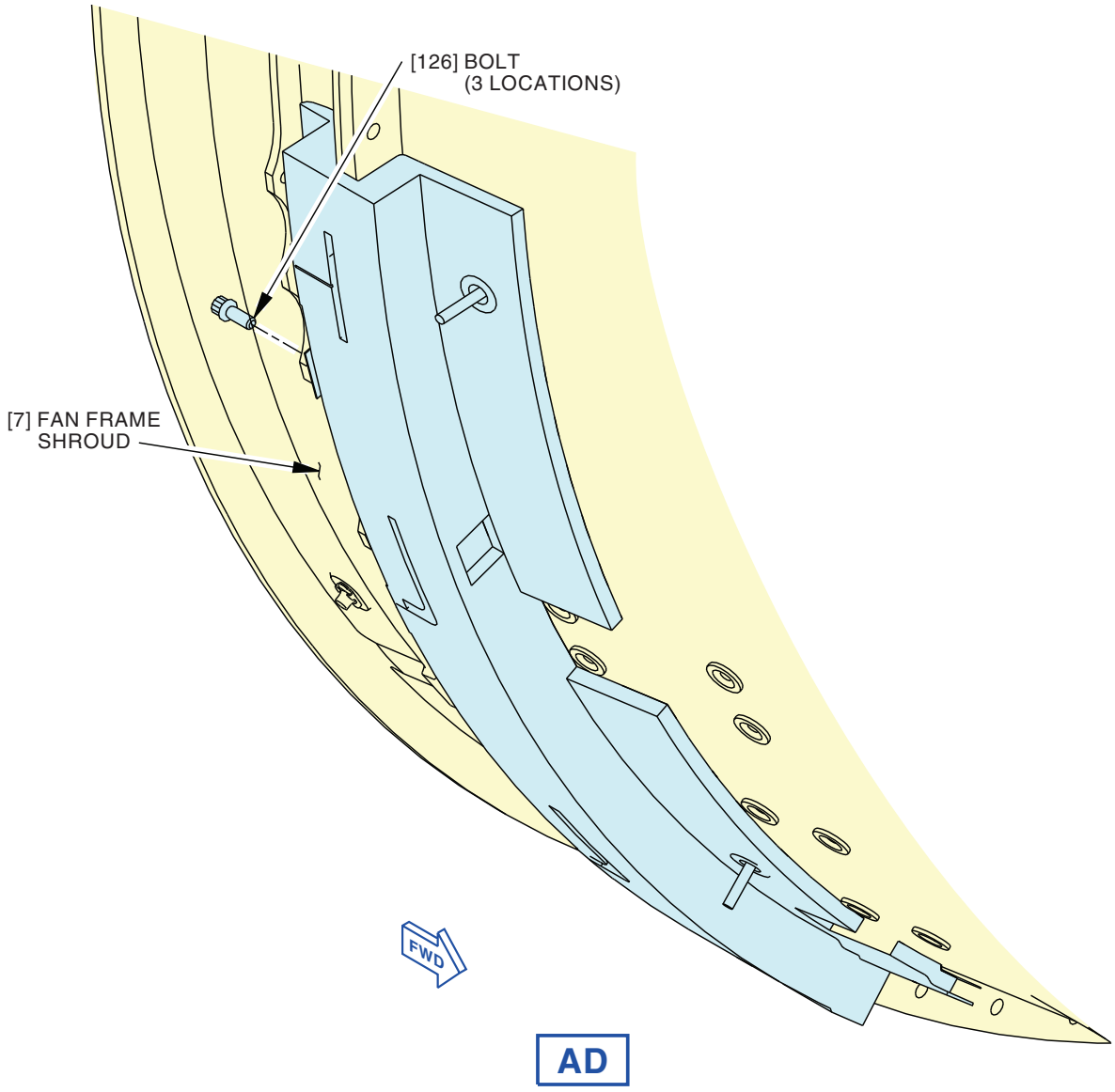
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2442034 S0000566781_V3

Aft Fan Case Heat Shield No. 5 Installation
Figure 405/72-21-07-990-812-G00 (Sheet 6 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2442154 S0000566782_V2

Aft Fan Case Heat Shield No. 5 Installation
Figure 405/72-21-07-990-812-G00 (Sheet 7 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-21-07-400-805-G00

11. Aft Fan Case Heat Shield No. 5 Installation

(Figure 405)

A. General

(1) This task gives the instructions to install the aft fan case heat shield No. 5.

B. References

Reference	Title
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-193	Container
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236
G51335 [CP2674]	Lint-Free Cloth	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
91	Aft fan case heat shield No. 5	72-21-07-01-290	PTF ALL
110	Gasket seal	72-21-00-17-790	PTF ALL
120	Gasket C seal	72-00-00-13-520	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

H. Aft Fan Case Heat Shield No. 5 Installation

SUBTASK 72-21-07-420-012-G00

(1) Install the aft fan case heat shield No. 5 [91] as follows:

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Remove the protective covers from the oil tube [108], the oil tube [121], and the openings on the air oil separator [111].

(c) Install the aft fan case heat shield No. 5 [91] in the correct position on the engine.

(d) Lubricate the heads and the pressure faces of the three bolts [126] with engine oil, D00599 [CP2442].

(e) Lubricate each side of the elastomer part of a new gasket seal [110] with vaseline, D00672 [CP5070], or engine oil, D00599 [CP2442].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Lubricate the threads of the two nuts [123] and the threads and pressure faces of the four bolts [109] with graphite mineral vaseline, D00601 [CP2101].

(g) Install the three bolts [126] to attach the aft fan case heat shield No. 5 [91] on the fan frame shroud [7].

(h) Tighten the bolts [126] to 67.5 in-lb (7.6 N·m) - 74.5 in-lb (8.4 N·m).

(i) Install the two new washers [124] and the two nuts [123] on the two fan case inserts [77].

(j) Tighten the nuts [123] to 30.1 in-lb (3.4 N·m) - 31.9 in-lb (3.6 N·m).

(k) Install the two caps [122] on the heat shield.

(l) Install the oil tube [108] as follows:

1) Install a new gasket C seal [120] and the oil tube [108] on the oil tube [121].

2) Install the four bolts [119] and the four nuts [118] to connect the oil tube [108] to the oil tube [121].

3) Tighten the nuts [118] to 84.1 in-lb (9.5 N·m) - 92.9 in-lb (10.5 N·m) in a criss cross pattern.

4) Install the bolt [115] to the clamp [116] that attaches the oil tube [108] to the bracket [117].

5) Tighten the bolt [115] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

6) Install the bolt [112] to the clamp [113] that attaches the oil tube [108] to the bracket [114].

7) Tighten the bolt [112] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

8) Install the new gasket seal [110] and the oil tube [108] to the air oil separator [111].

9) Attach the oil tube [108] to the air oil separator [111] with the four bolts [109].

10) Tighten the bolts [109] to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m) in a criss cross pattern.

(m) Install the bolt [106] to the clamp [107] that attaches the drain tube [97] to the bracket [105].

(n) Tighten the bolt [106] to 78.0 in-lb (8.8 N·m) - 82.0 in-lb (9.3 N·m).

(o) Install the bolt [103] to the clamp [104] that attaches the drain tube [93] to the bracket [105].

(p) Tighten the bolt [103] to 78.0 in-lb (8.8 N·m) - 82.0 in-lb (9.3 N·m).

(q) Install the bolt [101] to the clamp [102] that attaches the HJ3B engine wiring harness [100] to the bracket [96].

(r) Tighten the bolt [101] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

(s) Install the bolt [98] to the clamp [99] that attaches the drain tube [97] to the bracket [96].

(t) Tighten the bolt [98] to 78.0 in-lb (8.8 N·m) - 82.0 in-lb (9.3 N·m).

(u) Install the bolt [94] to the clamp [95] that attaches the drain tube [93] to the bracket [96].

(v) Tighten the bolt [94] to 78.0 in-lb (8.8 N·m) - 82.0 in-lb (9.3 N·m).

(w) Install the nine clips [92] to all harnesses adjacent to the aft fan case heat shield No. 5 [91].

(x) Remove the unwanted vaseline, D00672 [CP5070], graphite mineral vaseline, D00601 [CP2101], and unwanted engine oil, D00599 [CP2442], with a lint-free cloth, G51335 [CP2674].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(y) Remove the 1.32 gal (5.00 l) container, STD-193 from below the engine.

(z) Clean the work area and clear the tools and other items.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-07-410-006-G00

(1) On the applicable engine, close the right fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

SUBTASK 72-21-07-860-029-G00

(2) For the left engine, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 72-21-07-860-030-G00

(3) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-21-07-860-031-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

TASK 72-21-07-000-806-G00

12. Aft Fan Case Heat Shield No. 6 Removal

(Figure 406)

A. General

(1) This task provides the instructions on how to remove the aft fan case heat shield No. 6.

B. References

Reference	Title
73-21-05-000-802-G00	Electronic Engine Control (EEC) Channel B Removal (P/B 401)

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Prepare for the Removal

SUBTASK 72-21-07-860-033-G00

(1) For the left engine, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 72-21-07-860-034-G00

(2) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-21-07-010-010-G00

(3) Do this task: Electronic Engine Control (EEC) Channel B Removal, TASK 73-21-05-000-802-G00.

SUBTASK 72-21-07-700-004-G00

WARNING BE CAREFUL WHEN YOU WORK ON THE ENGINE COMPONENTS IMMEDIATELY AFTER THE ENGINE IS SHUTDOWN. THE ENGINE COMPONENTS CAN STAY HOT FOR UP TO ONE HOUR AND CAN CAUSE INJURY.

(4) Make sure that the engine temperature is cool enough to work on.

E. Aft Fan Case Heat Shield No. 6 Removal

SUBTASK 72-21-07-020-013-G00

(1) Remove the aft fan case heat shield No. 6 [141] as follows:

(a) Remove the cap [142] from the heat shield.

(b) Remove the nut [143] from the fan case insert [145].

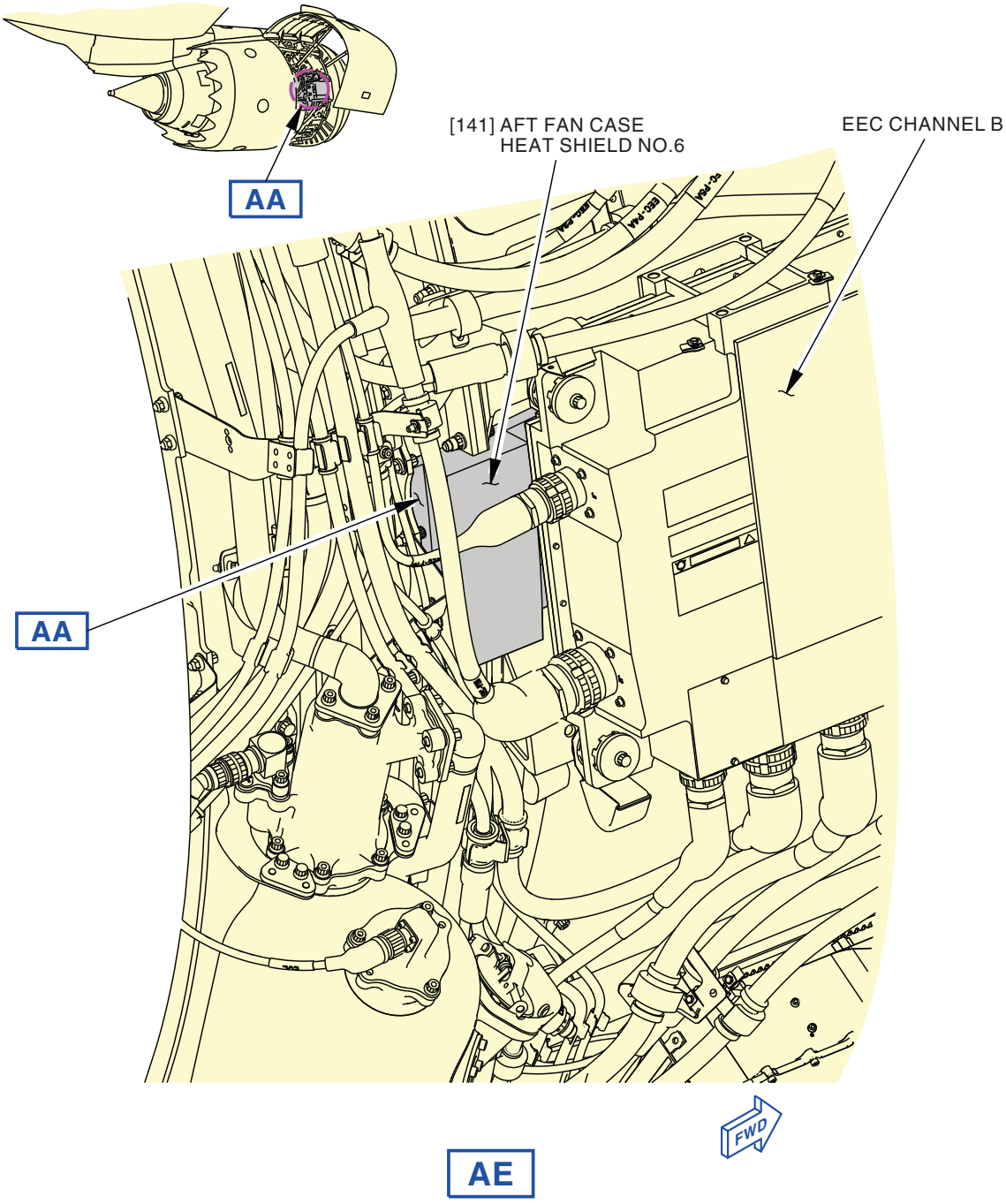
(c) Remove and discard the washer [144].

(d) Remove the bolt [146] that attaches the aft fan case heat shield No. 6 [141] to the fan frame shroud [7].

(e) Remove the aft fan case heat shield No. 6 [141] from the fan frame shroud [7].

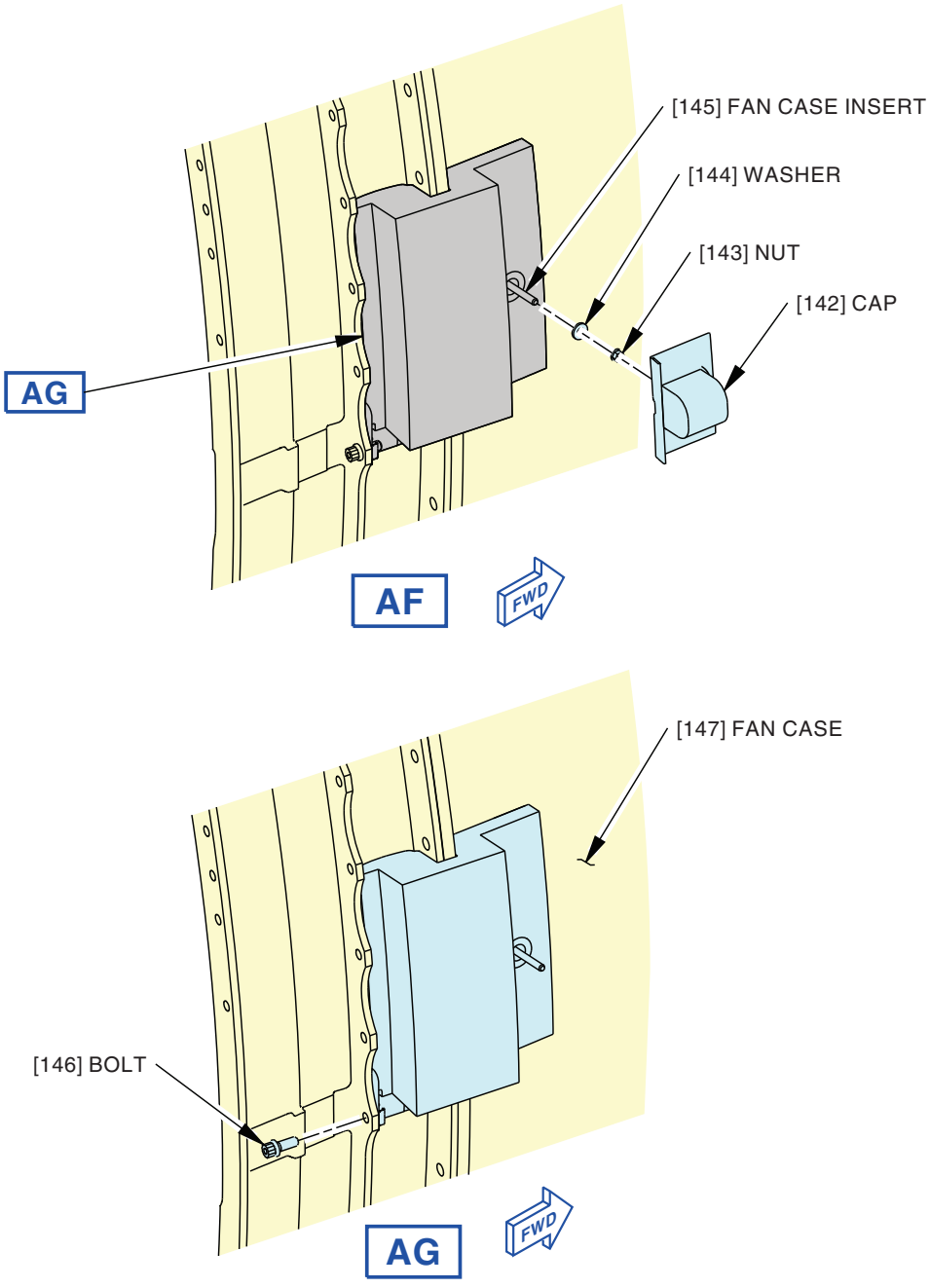
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aft Fan Case Heat Shield No. 6 Installation
Figure 406/72-21-07-990-813-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2442180 S0000566785_V2

Aft Fan Case Heat Shield No. 6 Installation
Figure 406/72-21-07-990-813-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-21-07-400-806-G00

13. Aft Fan Case Heat Shield No. 6 Installation

(Figure 406)

A. General

(1) This task provides the instructions on how to install the aft fan case heat shield No. 6.

B. References

Reference	Title
73-21-05-400-802-G00	Electronic Engine Control (EEC) Channel B Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
141	Aft fan case heat shield No. 6	72-21-07-01-300	PTF ALL

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Aft Fan Case Heat Shield No. 6 Installation

SUBTASK 72-21-07-420-013-G00

(1) Install the aft fan case heat shield No. 6 [141] as follows:

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Install the aft fan case heat shield No. 6 [141] in the correct position on the engine.

(c) Lubricate the threads of the nut [143] with graphite mineral vaseline, D00601 [CP2101].

(d) Lubricate the threads of the bolt [146] with engine oil, D00599 [CP2442].

(e) Install the bolt [146] to attach the aft fan case heat shield No. 6 [141] on the fan frame shroud [7].

(f) Tighten the bolt [146] to 67.5 in-lb (7.6 N·m) - 74.5 in-lb (8.4 N·m).

(g) Install the new washer [144] and the nut [143] on the fan case insert [145].

(h) Tighten the nut [143] to 30.1 in-lb (3.4 N·m) - 31.9 in-lb (3.6 N·m).

(i) Install the cap [142] on the heat shield.

(j) Clean the work area and remove all tools used to do the installation.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-07-410-010-G00

(1) Do this task: Electronic Engine Control (EEC) Channel B Installation, TASK 73-21-05-400-802-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-21-07-860-035-G00

(2) For the left engine, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 72-21-07-860-036-G00

(3) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

——— END OF TASK ———

TASK 72-21-07-000-807-G00

14. Aft Fan Case Heat Shield No. 7 Removal

(Figure 407)

A. General

(1) This task provides the instructions on how to remove the aft fan case heat shield No. 7.

B. References

Reference	Title
73-21-05-000-801-G00	Electronic Engine Control (EEC) Channel A Removal (P/B 401)

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Prepare for the Removal

SUBTASK 72-21-07-860-039-G00

(1) For the left engine, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 72-21-07-860-040-G00

(2) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-21-07-700-005-G00

WARNING BE CAREFUL WHEN YOU WORK ON THE ENGINE COMPONENTS IMMEDIATELY AFTER THE ENGINE IS SHUTDOWN. THE ENGINE COMPONENTS CAN STAY HOT FOR UP TO ONE HOUR AND CAN CAUSE INJURY.

(3) Make sure that the engine temperature is cool enough to work on.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-21-07-010-011-G00

(4) Do this task: Electronic Engine Control (EEC) Channel A Removal, TASK 73-21-05-000-801-G00.

E. Aft Fan Case Heat Shield No. 7 Removal

SUBTASK 72-21-07-020-014-G00

(1) Remove the aft fan case heat shield No. 7 [161] as follows:

(a) Remove the cap [162] from the heat shield.

(b) Remove the nut [163] from the fan case insert [165].

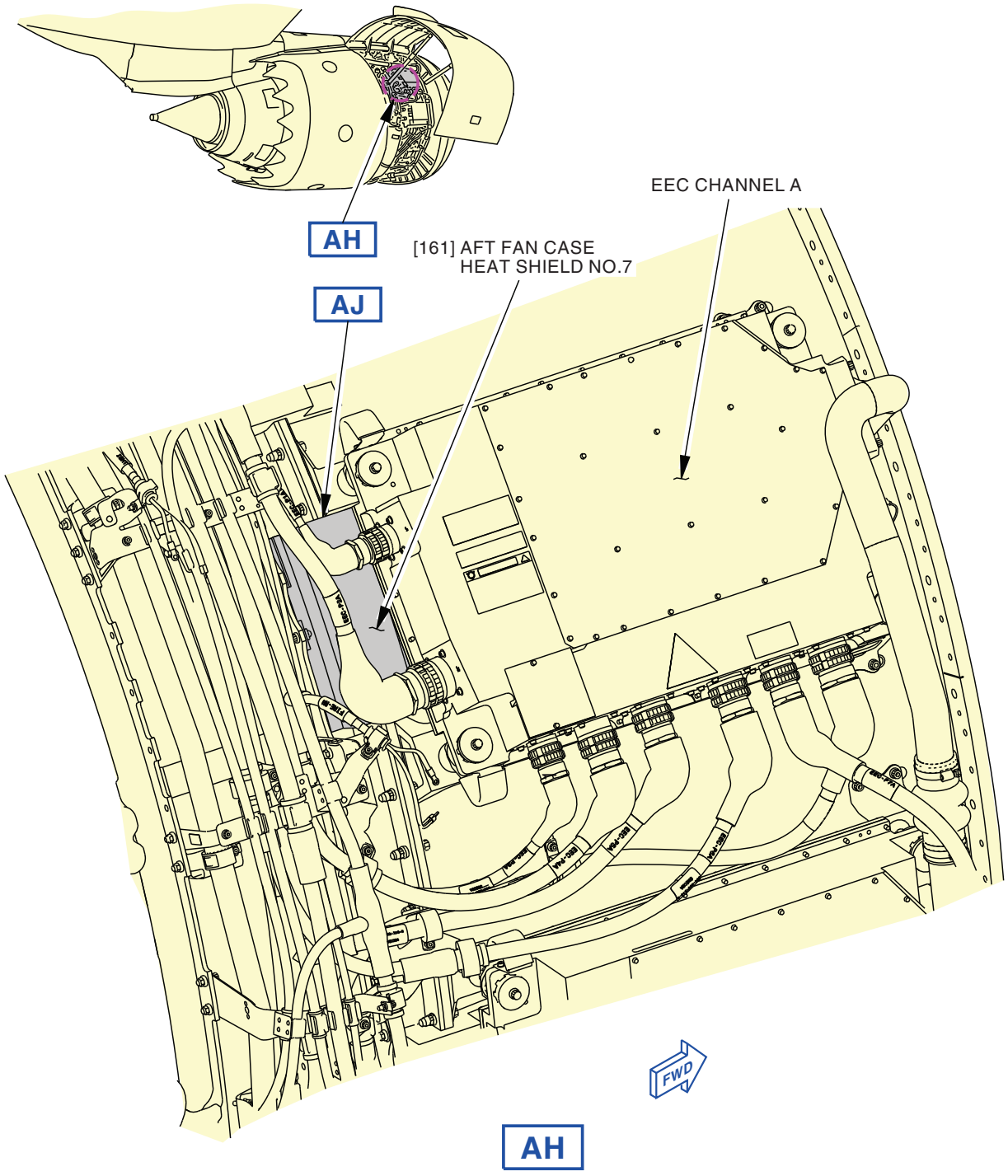
(c) Remove and discard the washer [164].

(d) Remove the bolt [166] that attaches the aft fan case heat shield No. 7 [161] to the fan case [167].

(e) Remove the aft fan case heat shield No. 7 [161] from the fan case [167].

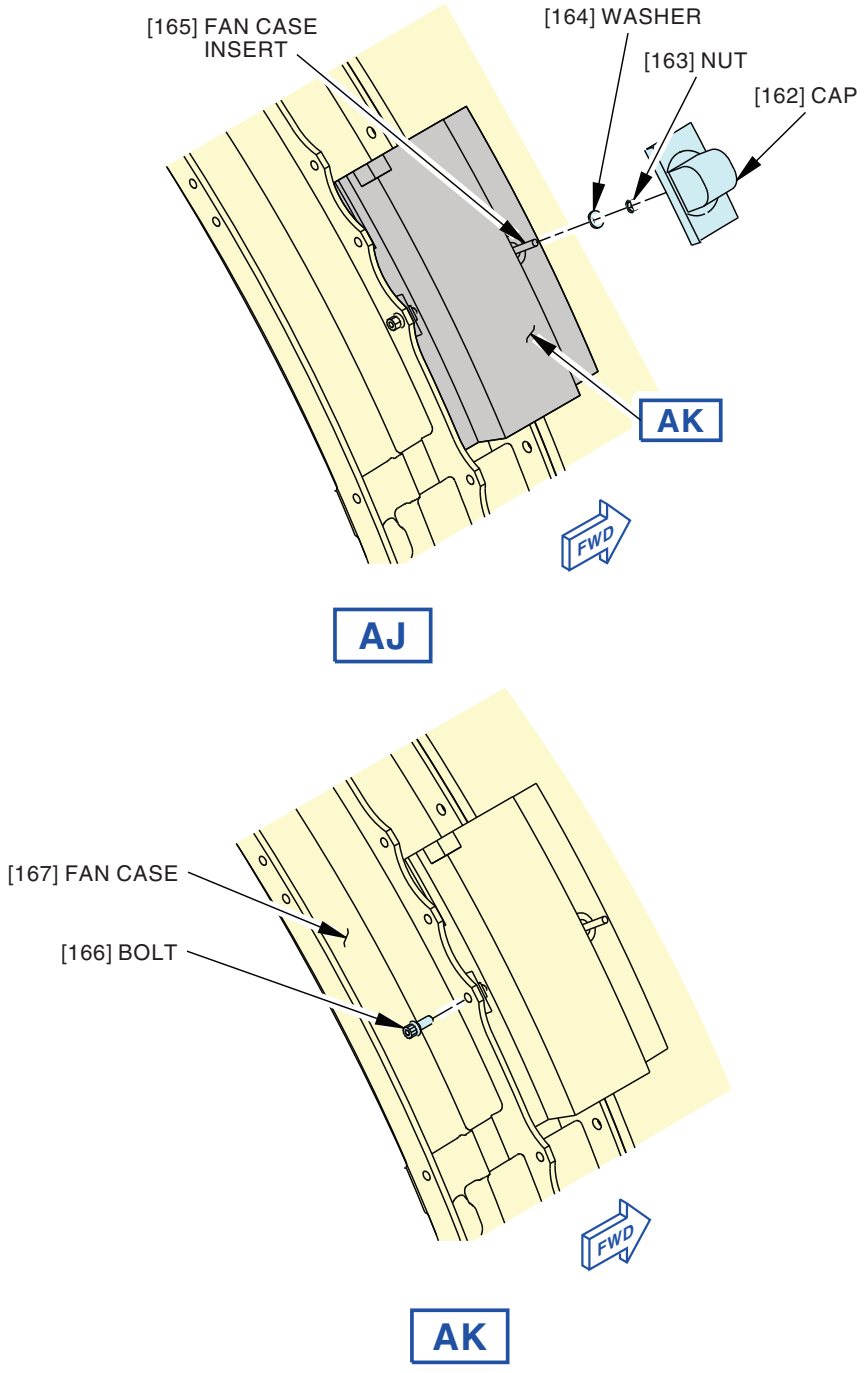
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aft Fan Case Heat Shield No. 7 Installation
Figure 407/72-21-07-990-814-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Aft Fan Case Heat Shield No. 7 Installation
Figure 407/72-21-07-990-814-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-21-07-400-807-G00

15. Aft Fan Case Heat Shield No. 7 Installation

(Figure 407)

A. General

(1) This task provides the instructions on how to install the aft fan case heat shield No. 7.

B. References

Reference	Title
73-21-05-400-801-G00	Electronic Engine Control (EEC) Channel A Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
161	Aft fan case heat shield No. 7	72-21-07-01-310	PTF ALL

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Aft Fan Case Heat Shield No. 7 Installation

SUBTASK 72-21-07-420-014-G00

(1) Install the aft fan case heat shield No. 7 [161] as follows:

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Install the aft fan case heat shield No. 7 [161] in correct position on the engine.

(c) Lubricate the threads of the nut [163] with graphite mineral vaseline, D00601 [CP2101].

(d) Lubricate the threads of the bolt [166] with engine oil, D00599 [CP2442].

(e) Install the bolt [166] to attach the aft fan case heat shield No. 7 [161] on the fan frame shroud [7].

(f) Tighten the bolt [166] to 67.5 in-lb (7.6 N·m) - 74.5 in-lb (8.4 N·m).

(g) Install the new washer [164] and the nut [163] on the fan case insert [165].

(h) Tighten the nut [163] to 30.1 in-lb (3.4 N·m) - 31.9 in-lb (3.6 N·m).

(i) Install the cap [162] on the heat shield.

(j) Clean the work area and remove all tools used to do the installation.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-07-410-011-G00

(1) Do this task: Electronic Engine Control (EEC) Channel A Installation, TASK 73-21-05-400-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-21-07-860-041-G00

(2) For the left engine, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 72-21-07-860-042-G00

(3) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

——— END OF TASK ———

TASK 72-21-07-000-808-G00

16. Aft Fan Case Heat Shield No. 8 Removal

(Figure 408)

A. General

(1) This task provides the instructions on how to remove the aft fan case heat shield No. 8.

(2) It is necessary to remove the engine from the pylon to remove the aft fan case heat shield No. 8.

B. References

Reference	Title
71-00-02-000-801-G00	Power Plant - Removal (P/B 401)

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Prepare for the Removal

SUBTASK 72-21-07-010-009-G00

(1) Do this task: Power Plant - Removal, TASK 71-00-02-000-801-G00.

E. Aft Fan Case Heat Shield No. 8 Removal

SUBTASK 72-21-07-020-015-G00

(1) Remove the aft fan case heat shield No. 8 [181] as follows:

(a) Remove the bolt [183] from the clamp [184] that attaches the W200 engine wiring harness [182] to the bracket [194].

(b) Remove the bolt [186] from the clamp [187] that attaches the HJ2A engine wiring harness [185] to the bracket [194].

(c) Remove the bolt [189] from the clamp [190] that attaches the HJ2A engine wiring harness [188] to the bracket [194].

(d) Remove the bolt [192] from the clamp [193] that attaches the HJ4A engine wiring harness [191] to the bracket [194].

(e) Remove the five caps [195] from the heat shield.

(f) Remove the five nuts [196] from the five fan case inserts [198].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

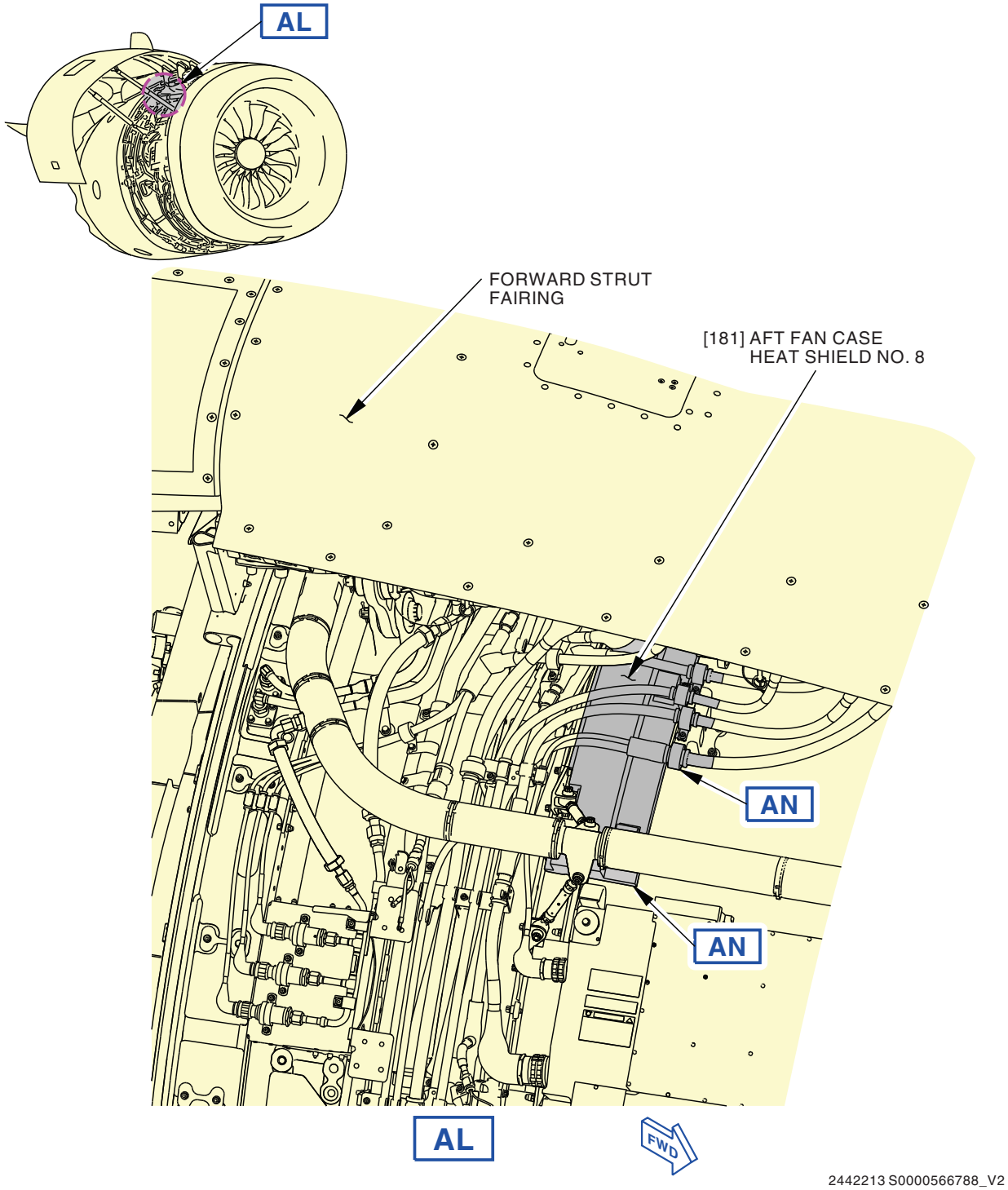
(g) Remove and discard the five washers [197].

(h) Remove the four bolts [199] that attach the aft fan case heat shield No. 8 [181] to the fan case [200].

(i) Remove the aft fan case heat shield No. 8 [181] from the fan case [200].

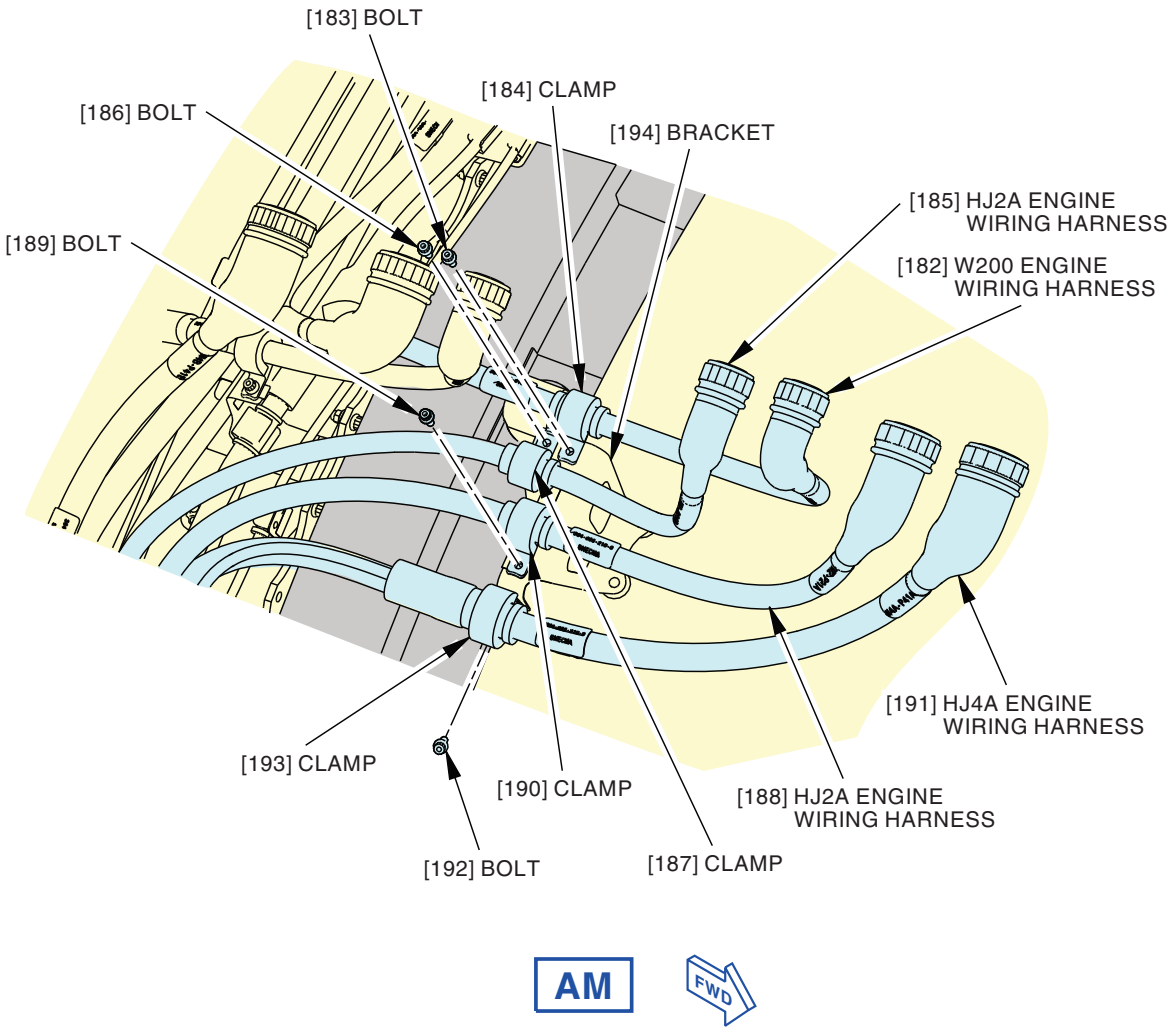
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aft Fan Case Heat Shield No. 8 Installation
Figure 408/72-21-07-990-815-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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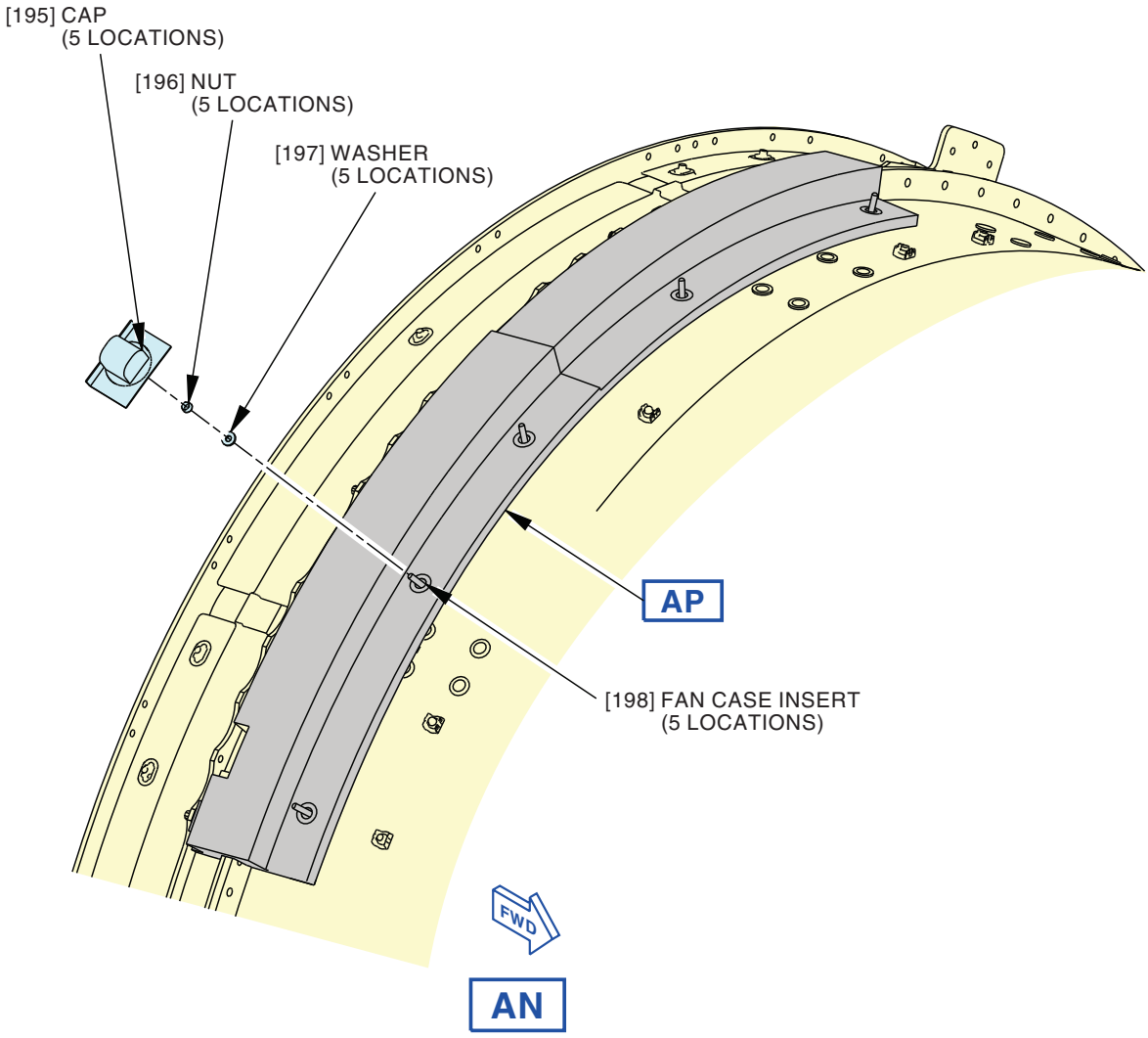
Aft Fan Case Heat Shield No. 8 Installation
Figure 408/72-21-07-990-815-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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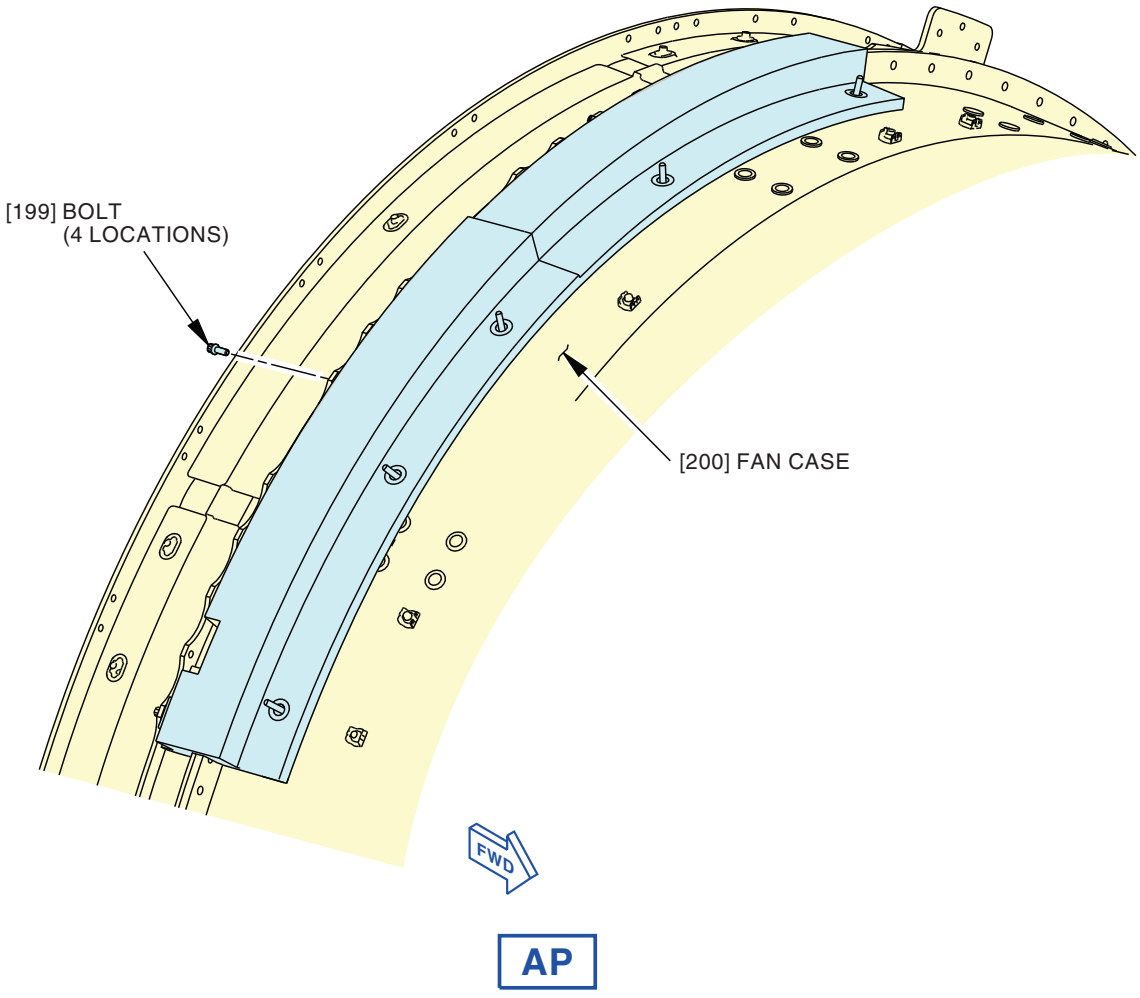
Aft Fan Case Heat Shield No. 8 Installation
Figure 408/72-21-07-990-815-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Aft Fan Case Heat Shield No. 8 Installation
Figure 408/72-21-07-990-815-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-21-07-400-808-G00

17. Aft Fan Case Heat Shield No. 8 Installation

(Figure 408)

A. General

(1) This task provides the instructions on how to install the aft fan case heat shield No. 8.

B. References

Reference	Title
71-00-02-400-801-G00	Power Plant - Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
181	Aft fan case heat shield No. 8	72-21-07-01-240	PTF ALL

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Aft Fan Case Heat Shield No. 8 Installation

SUBTASK 72-21-07-420-015-G00

(1) Install the aft fan case heat shield No. 8 [181] as follows:

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Install the aft fan case heat shield No. 8 [181] in the correct position on the engine.

(c) Lubricate the threads of the five bolts [199] with engine oil, D00599 [CP2442].

(d) Lubricate the threads of the five nuts [196] with graphite mineral vaseline, D00601 [CP2101].

(e) Install the five bolt [199] to attach the aft fan case heat shield No. 8 [181] on the fan frame shroud [7].

(f) Tighten the bolts [199] to 67.5 in-lb (7.6 N·m) - 74.5 in-lb (8.4 N·m).

(g) Install the five new washers [197] and the five nuts [196] to the five fan case inserts [198].

(h) Tighten the nuts [196] to 30.1 in-lb (3.4 N·m) - 31.9 in-lb (3.6 N·m).

(i) Install the five caps [195] on the heat shield.

(j) Install the bolt [192] on the clamp [193] that attaches the HJ4A engine wiring harness [191] to the bracket [194].

(k) Tighten the bolt [192] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

(l) Install the bolt [189] on the clamp [190] that attaches the HJ2A engine wiring harness [188] to the bracket [194].

(m) Tighten the bolt [189] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

(n) Install the bolt [186] on the clamp [187] that attaches the HJ2A engine wiring harness [185] to the bracket [194].

(o) Tighten the bolt [186] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(p) Install the bolt [183] on the clamp [184] that attaches the W200 engine wiring harness [182] to the bracket [194].

(q) Tighten the bolt [183] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

(r) Clean the work area and remove all tools used to do the installation.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-07-410-009-G00

(1) Do this task: Power Plant - Installation, TASK 71-00-02-400-801-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AFT ACOUSTICAL PANEL - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Aft acoustical panel removal

(2) Aft acoustical panel installation.

TASK 72-21-08-000-801-G00

2. Aft Acoustical Panel Removal

(Figure 401 or Figure 402)

A. General

(1) This task gives the instructions to remove the aft acoustical panel.

B. References

Reference	Title
72-21-09-000-801-G00	Outlet Guide Vane Platform Removal (P/B 401)
72-22-02-000-803-G00	Fan Blades (Set) Removal (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A50472 [CP2734]	Product - Silicone Elastomer Sealing	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Prepare for the Removal

SUBTASK 72-21-08-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-21-08-860-002-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-21-08-010-001-G00

(3) Do this task: Fan Blades (Set) Removal, TASK 72-22-02-000-803-G00.

G. Aft Acoustical Panel Removal

SUBTASK 72-21-08-010-003-G00

(1) Remove one OGV platform adjacent to the first acoustical panel which will be removed (TASK 72-21-09-000-801-G00).

NOTE: You can remove one OGV platform for each acoustical panel if necessary.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181

SUBTASK 72-21-08-010-002-G00

(2) Remove the aft acoustical panel [1]:

NOTE: This subtask provides the instructions to remove the aft acoustical panel [1]. The remaining aft acoustical panels removal is similar to the aft acoustical panel [1] removal.

(a) Remove the six bolts [2] that attach the aft acoustical panel [1] to the fan case [3].

WARNING USE EYE PROTECTION WHEN YOU CUT MATERIAL OR PARTS. THE PARTICLES CAN CAUSE DAMAGE TO YOUR EYES.

WARNING BE CAREFUL WHEN YOU USE SHARP OR POINTED TOOLS. FAILURE TO DO SO CAN CAUSE INJURY TO PERSONS.

CAUTION DO NOT USE A SHARP TOOL TO REMOVE THE ACOUSTIC PANELS. DAMAGE TO THE ACOUSTIC PANELS OR THE FAN BOOSTER CAN OCCUR.

(b) Carefully cut and remove all the RTV 103 adhesive, A50472 [CP2734] from the aft acoustical panel [1].

(c) Carefully move the aft acoustical panel [1] forward to disengage from the fan case [3].

(d) Remove the aft acoustical panel [1] from the engine.

(e) Do the steps above again to remove the remaining five aft acoustical panels.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181

SUBTASK 72-21-08-010-004-G01

(3) Remove the aft acoustical panel [5]:

(a) Remove the five bolts [2] and the bolt [4] that attach the aft acoustical panel [5] to the fan case [3].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181 (Continued)

WARNING USE EYE PROTECTION WHEN YOU CUT MATERIAL OR PARTS. THE PARTICLES CAN CAUSE DAMAGE TO YOUR EYES.

WARNING BE CAREFUL WHEN YOU USE SHARP OR POINTED TOOLS. FAILURE TO DO SO CAN CAUSE INJURY TO PERSONS.

CAUTION DO NOT USE A SHARP TOOL TO REMOVE THE ACOUSTIC PANELS. DAMAGE TO THE ACOUSTIC PANELS OR THE FAN BOOSTER CAN OCCUR.

(b) Carefully cut and remove all the RTV 103 adhesive, A50472 [CP2734] from the aft acoustical panel [5].

(c) Carefully move the aft acoustical panel [5] forward to disengage from the fan case [3].

(d) Remove the aft acoustical panel [5] forward from the engine.

SUBTASK 72-21-08-010-005-G01

(4) Remove the aft acoustical panel [1]:

NOTE: This subtask provides the instructions to remove the aft acoustical panel [1]. The remaining aft acoustical panels removal is similar to the aft acoustical panel [1] removal.

(a) Remove the six bolts [2] that attach the aft acoustical panel [1] to the fan case [3].

WARNING USE EYE PROTECTION WHEN YOU CUT MATERIAL OR PARTS. THE PARTICLES CAN CAUSE DAMAGE TO YOUR EYES.

WARNING BE CAREFUL WHEN YOU USE SHARP OR POINTED TOOLS. FAILURE TO DO SO CAN CAUSE INJURY TO PERSONS.

CAUTION DO NOT USE A SHARP TOOL TO REMOVE THE ACOUSTIC PANELS. DAMAGE TO THE ACOUSTIC PANELS OR THE FAN BOOSTER CAN OCCUR.

(b) Carefully cut and remove all the RTV 103 adhesive, A50472 [CP2734] from the aft acoustical panel [1].

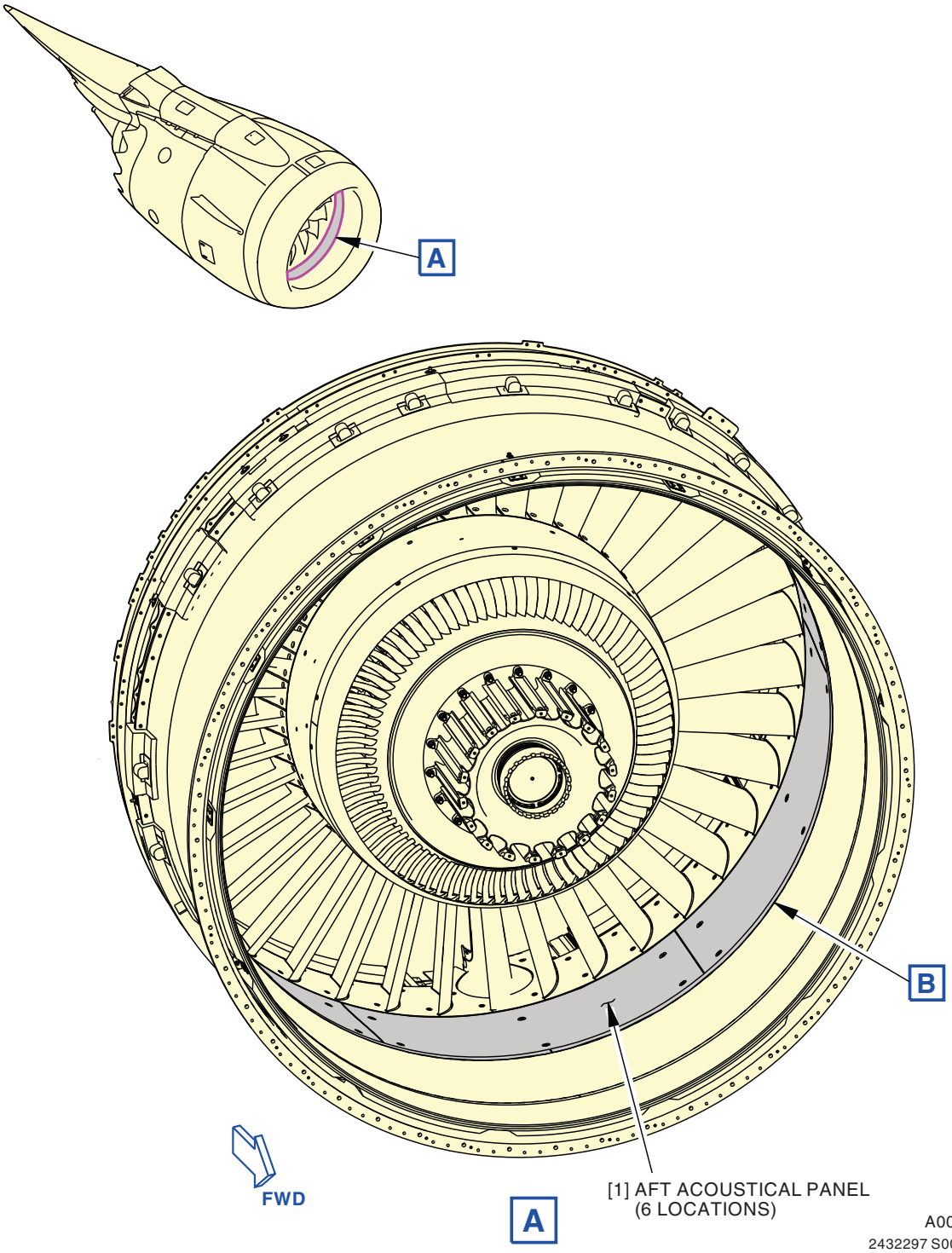
(c) Carefully move the aft acoustical panel [1] forward to disengage from the fan case [3].

(d) Remove the aft acoustical panel [1] from the engine.

(e) Do the steps above again to remove the remaining four aft acoustical panels.

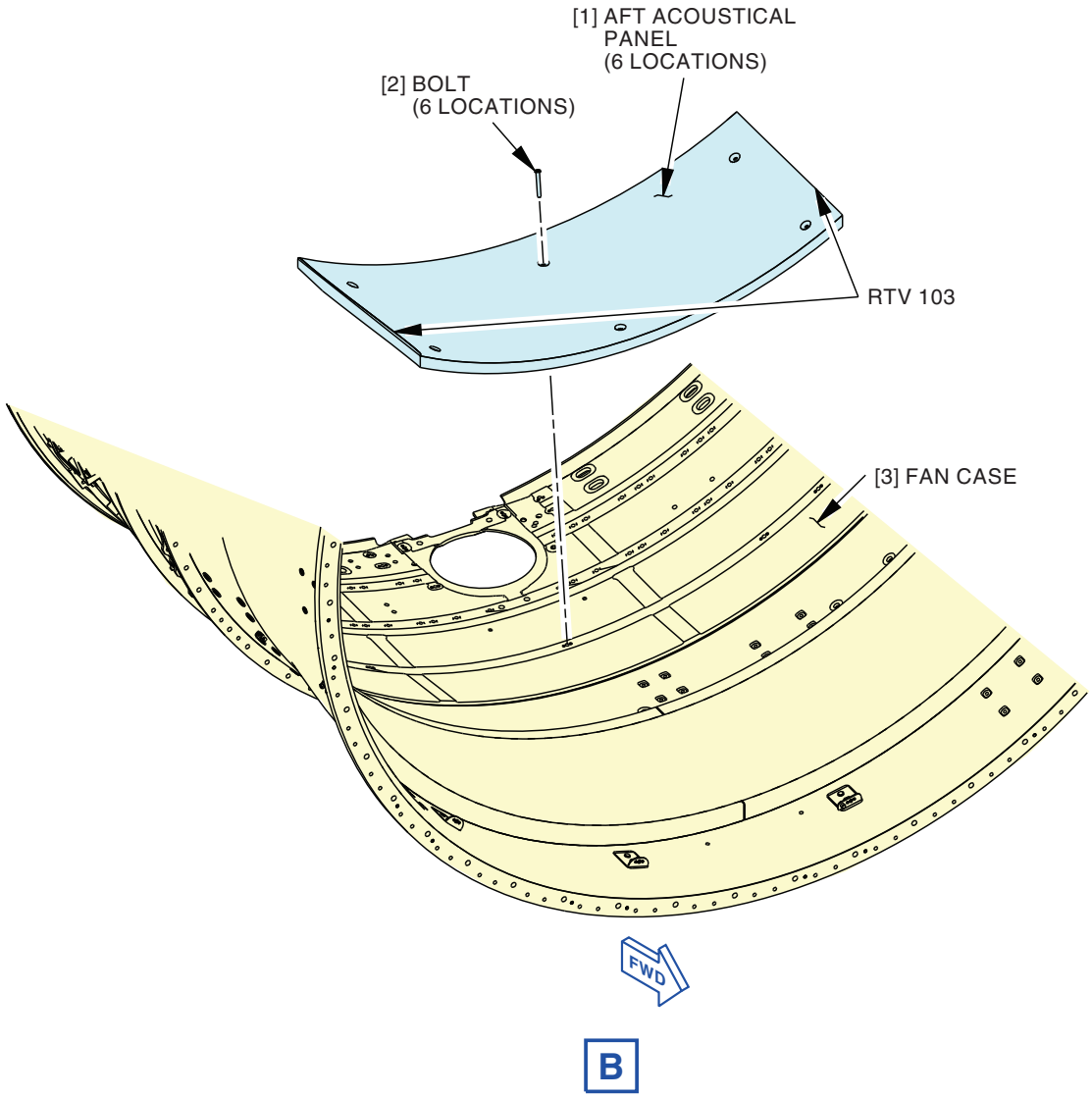
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aft Acoustical Panel Installation
Figure 401/72-21-08-990-801-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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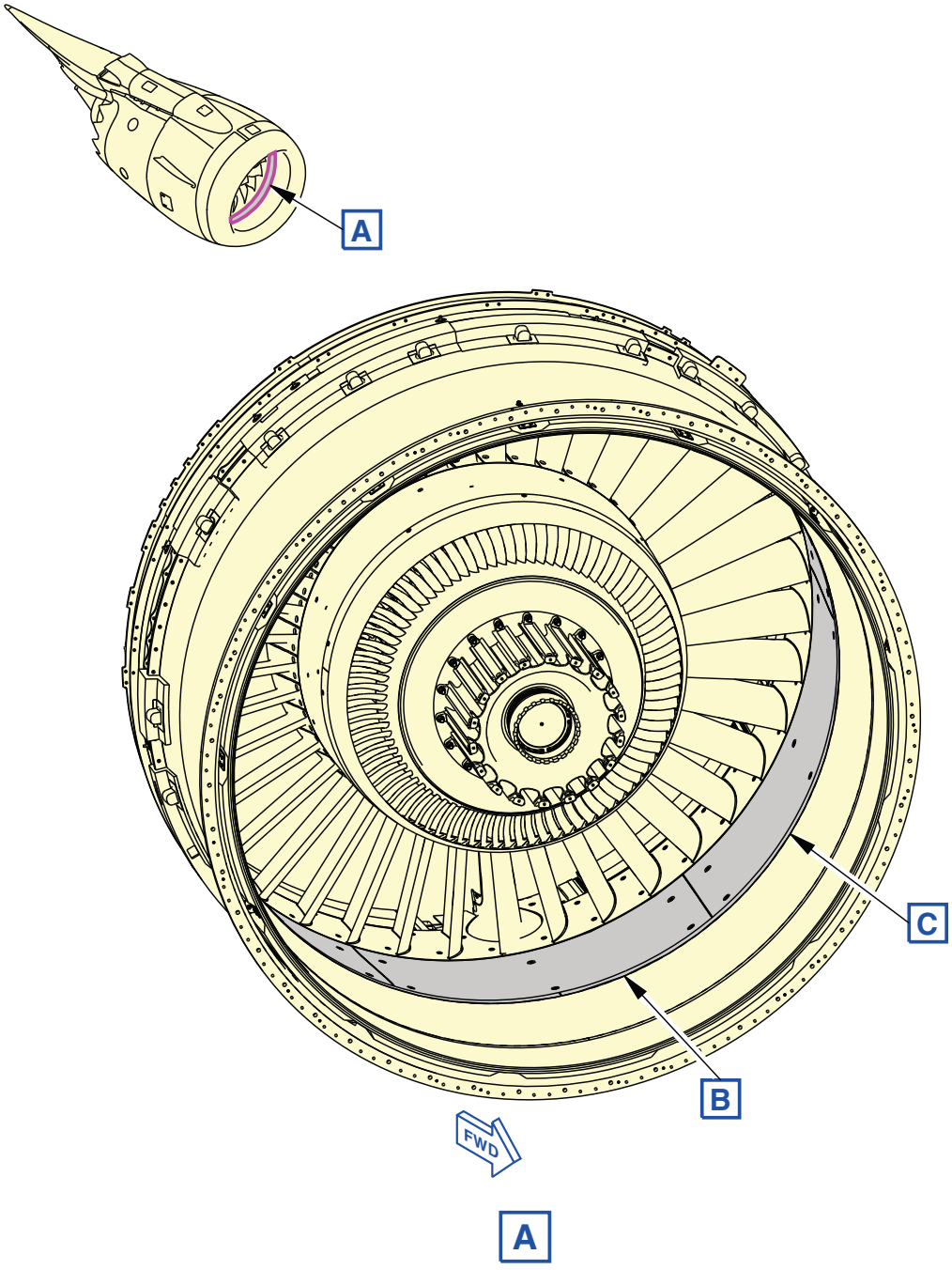
Aft Acoustical Panel Installation
Figure 401/72-21-08-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Aft Acoustical Panel Installation
Figure 402/72-21-08-990-802-G00 (Sheet 1 of 3)

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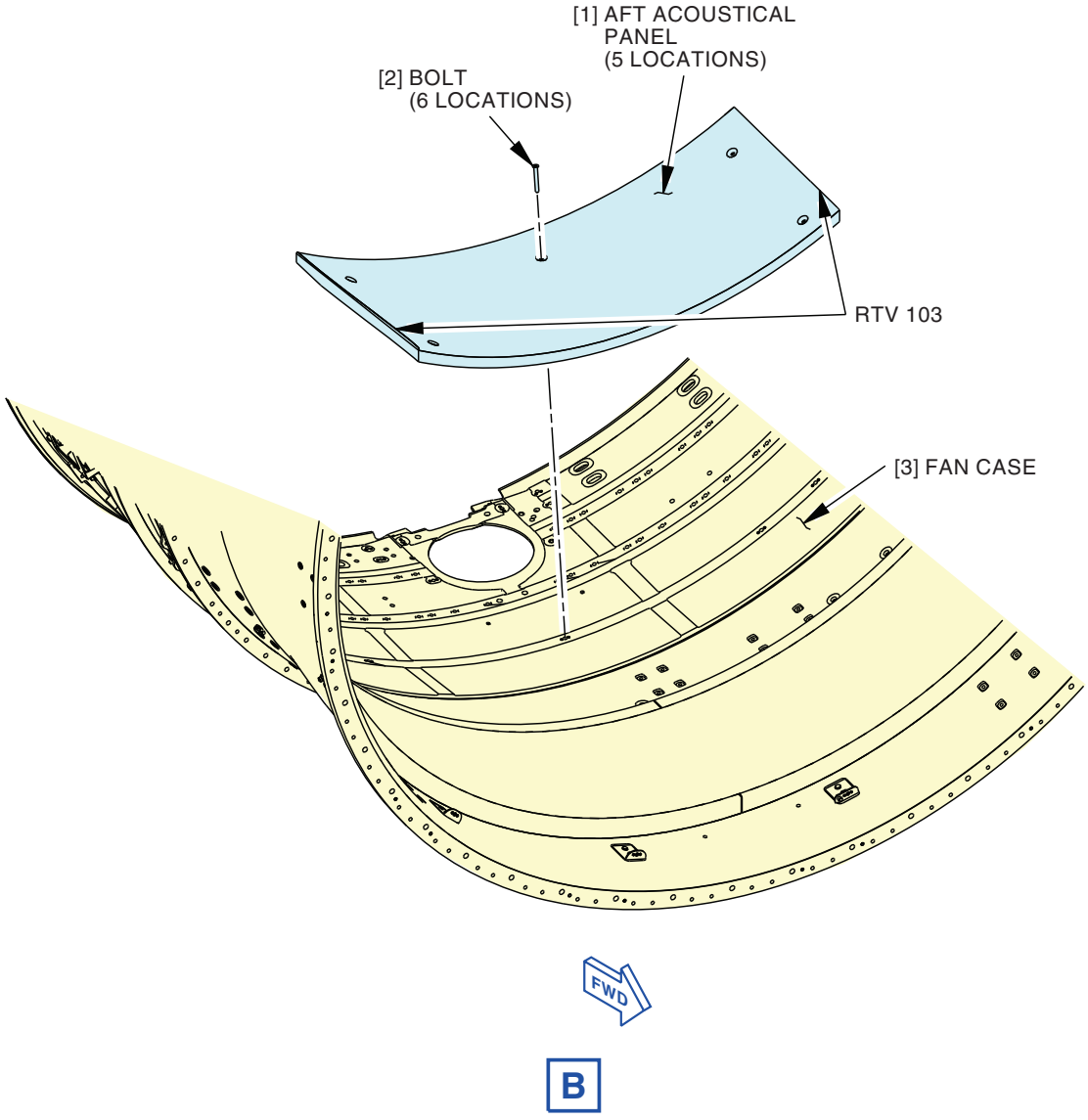

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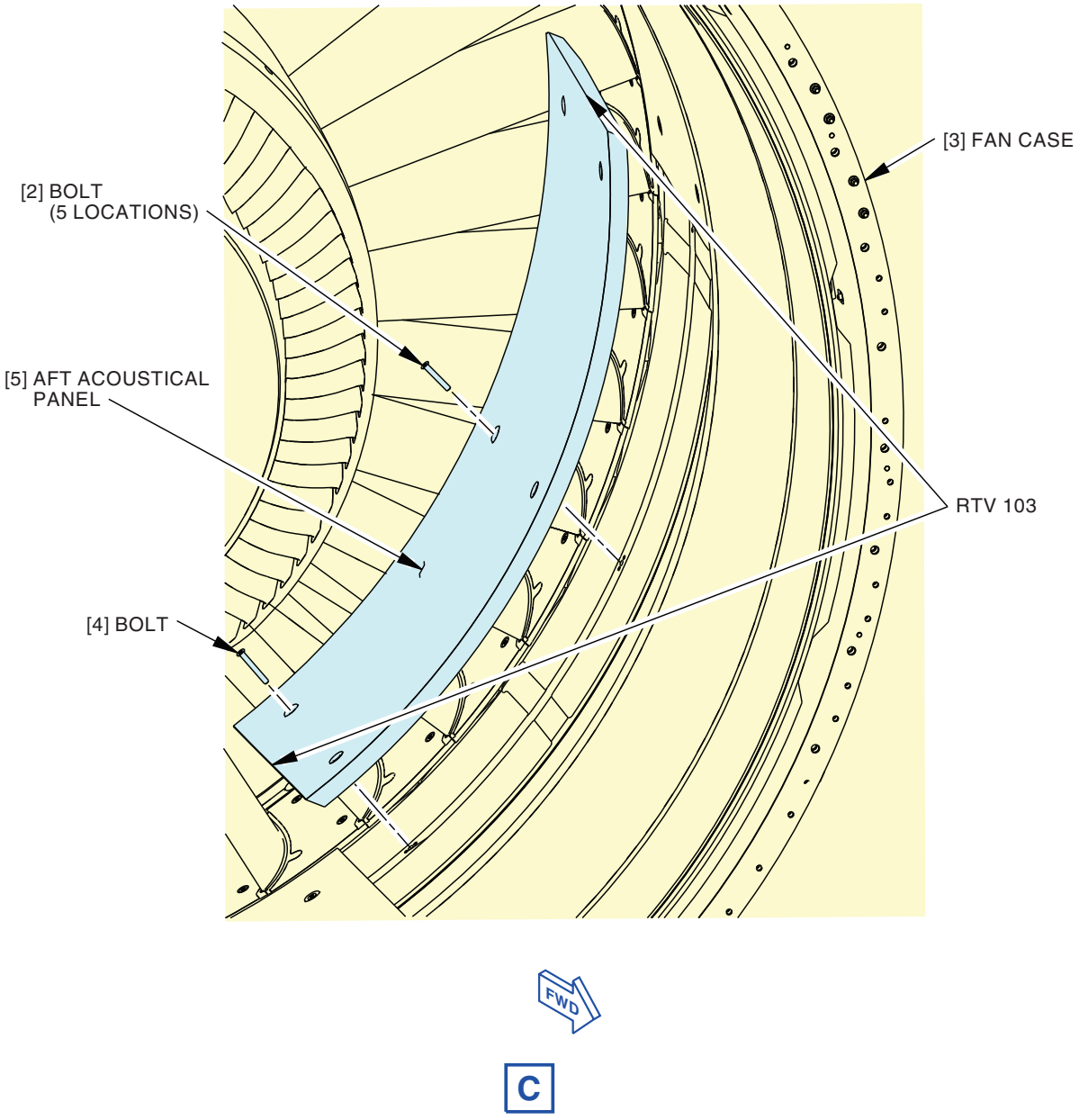
Aft Acoustical Panel Installation
Figure 402/72-21-08-990-802-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Aft Acoustical Panel Installation
Figure 402/72-21-08-990-802-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-21-08-400-801-G00

3. Aft Acoustical Panel Installation

(Figure 401 or Figure 402)

A. General

(1) This task gives the instructions to install the aft acoustical panel.

NOTE: The total acoustic panel blockage for the engine installation must be recalculated when components are changed per SRM SUBJECT 54-00-01.

B. References

Reference	Title
72-21-09-400-801-G00	Outlet Guide Vane Platform Installation (P/B 401)
72-22-02-400-803-G00	Fan Blades (Set) Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A50472 [CP2734]	Product - Silicone Elastomer Sealing	
D00601 [CP2101]	Vaseline - Graphite Mineral	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Aft acoustical panel	72-21-08-02-430	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Aft Acoustical Panel Installation

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181

SUBTASK 72-21-08-410-001-G00

(1) Install the aft acoustical panel [1]:

NOTE: This subtask provides the instructions to install the aft acoustical panel. The remaining aft acoustical panels installation are similar as the aft acoustical panel installation.

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Install the aft acoustical panel [1] in a correct position on the fan case [3].

(c) Lubricate the threads and under the head of the six bolts [2] with the graphite mineral vaseline, D00601 [CP2101].

(d) Install the six bolts [2] that attach the aft acoustical panel [1] to the fan case [3].

(e) Tighten the bolts [2] to 109.25 in-lb (12.34 N·m) - 120.75 in-lb (13.64 N·m).

(f) Install the Outlet Guide Vane (OGV) platform that was removed (TASK 72-21-09-400-801-G00).

(g) Do the steps above again to install the remaining five aft acoustical panels.

(h) Apply the RTV 103 adhesive, A50472 [CP2734] between each aft acoustical panel [1].

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181 (Continued)

(i) Clean the work area and clear the tools and other items.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181

SUBTASK 72-21-08-410-004-G00

(2) Install the aft acoustical panel [5]:

NOTE: This subtask provides the instructions to install the aft acoustical panel located at 8 o'clock position.

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Install the aft acoustical panel [5] in a correct position on the fan case [3].

(c) Lubricate the threads and under the head of the five bolts [2] and of the bolt [4] with the graphite mineral vaseline, D00601 [CP2101].

(d) Tighten the bolts [2] and the bolt [4] to 109.25 in-lb (12.34 N·m) - 120.75 in-lb (13.64 N·m).

(e) Install the OGV platform that was removed (TASK 72-21-09-400-801-G00).

(f) Apply the RTV 103 adhesive, A50472 [CP2734] between each aft acoustical panel [5] and the adjacent aft acoustical panels.

(g) Clean the work area and clear the tools and other items.

SUBTASK 72-21-08-410-005-G00

(3) Install the five remaining aft acoustical panels [1]:

NOTE: This subtask provides the instructions to install the five remaining aft acoustical panels.

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Install one of the five remaining aft acoustical panels [1] in a correct position on the fan case [3].

(c) Lubricate the threads and under the head of the six bolts [2] with the graphite mineral vaseline, D00601 [CP2101].

(d) Install the six bolts [2] that attach the aft acoustical panel [1] to the fan case [3].

(e) Tighten the six bolts [2] to 109.25 in-lb (12.34 N·m) - 120.75 in-lb (13.64 N·m).

(f) Install the OGV platform that was removed (TASK 72-21-09-400-801-G00).

(g) Do the steps above again to install the remaining four aft acoustical panels.

(h) Apply the RTV 103 adhesive, A50472 [CP2734] between each aft acoustical panel [1].

(i) Clean the work area and clear the tools and other items.

PTF ALL

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-08-410-002-G00

(1) Do this task: Fan Blades (Set) Installation, TASK 72-22-02-400-803-G00.

SUBTASK 72-21-08-860-003-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-21-08-860-004-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AFT ACOUSTICAL PANEL - REPAIR

1. General

A. This procedure has this task:

(1) A repair of the acoustic panel by filling up.

TASK 72-21-08-300-801-G00

2. Acoustic Panel Impact Repair by Filling Up

(Figure 801, Figure 802, and Figure 803)

A. General

(1) This task gives instructions for the repair of the acoustical panel [1] inner face by material filling.

(2) Dimensional information.

Table 801/72-21-08-993-802-G00

Description	Minimum In-Process Dimension	Maximum In-Process Dimension	Minimum Finish Dimension	Maximum Finish Dimension	Reference Dimension
DIM. C	-	-	2.17 in. (55.12 mm)	-	-
DIA. D	-	-	-	2.36 in. (59.94 mm)	-
DIM. E	-	-	1.97 in. (50.04 mm)	-	-

B. References

Reference	Title
70-21-23-100-801-G00	Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701)
70-43-42-300-802-G00	Blending Procedures - Hand Blending (P/B 201)
70-43-42-300-803-G00	Blending Procedures - Power Blending (P/B 201)
72-21-08-000-801-G00	Aft Acoustical Panel Removal (P/B 401)
72-21-08-400-801-G00	Aft Acoustical Panel Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-10679	Saw - Hole, Bearing Removal Part #: A32091-1920 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
G51328 [CP2674]	Cloth - Lint-Free	
G51412 [CP2721]	Cloth - Abrasive, Silicone Carbide - CS 333X	
G51480 [CP2772]	Cloth - Abrasive, Alumina	

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(Continued)

Reference	Description	Specification
G51697 [CP2929]	Tape - Flashbreaker 1	
G51826 [CP2508]	Tape - Flashbreaker No. 5	
G51894 [CP2923]	Compound - Epoxy, Void Filling and Edge Sealing, Scotch-Weld EC-3524 B/A Black	
G51919	Abrasive - Paper, Aluminum Oxide, Grade 75 or finer	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Prepare for the Repair

SUBTASK 72-21-08-020-001-G00

(1) Remove the acoustical panel [1], do this task: Aft Acoustical Panel Removal, TASK 72-21-08-000-801-G00.

G. Acoustic Panel Impact Repair by Filling Up

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THIS PRODUCT. IF NOT, IT CAN CAUSE INJURY.

WARNING ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

SUBTASK 72-21-08-100-001-G00

(1) If it is necessary, clean the damaged area with a lint-free cloth, G51328 [CP2674], moist with alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

SUBTASK 72-21-08-350-001-G00

(2) Obey these limits of the repair (Figure 802):

(a) It is permitted to repair several indications in a same repaired area, but the total diameter (DIA. D) of one repaired area must not exceed 2.36 in. (59.94 mm).

(b) The distance (DIM. E) between the repaired area and the edge of the panel must be 1.97 in. (50.04 mm) minimum.

(c) The distance (DIM.E) between the repaired area and the fixation areas of the panels must be 1.97 in. (50.04 mm) minimum.

(d) Maximum cumulated repaired surface is not more than 13.14 in^2 (8477.40 mm^2).

(e) The distance (DIM. C) between two repaired areas must be 2.17 in. (55.12 mm) minimum.

(f) It is possible to apply the repair on a maximum of 3 isolated areas.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-21-08-010-006-G00

WARNING DO NOT BREATHE THE PARTICLES FROM THE REPAIR OF COMPOSITE MATERIALS OR LET THE PARTICLES TOUCH. USE LOCAL MECHANICAL EXHAUST AIRFLOW OR AN APPROVED RESPIRATOR. USE PROTECTION EQUIPMENT. IF YOU DO NOT OBEY, THE PARTICLES CAN CAUSE DAMAGE, INJURY, OR IRRITATION.

(3) Remove the indications at the acoustical panel [1] as follows (Figure 802 and Figure 803):

(a) Use the hole saw, SPL-10679, to cut the diameter 0.252 in. (6.4 mm) larger than the maximal diameter of the indication.

1) Make sure that the cut diameter is not more than 2.36 in. (60 mm).

NOTE: A change of sound occurs during the hole-sawing when the carbon skin layer is cut and you reach the honeycomb material.

(b) Apply Flashbreaker 5 tape, G51826 [CP2508], or Flashbreaker 1 tape, G51697 [CP2929], around the area to repair.

SUBTASK 72-21-08-010-007-G00

(4) Remove all the indications of the front sheet carbon skin layers and honeycomb such as loose particles, cracks and delamination.

(a) Do not damage the carbon plies on the external face of the panel.

NOTE: For better results, remove all of the honeycomb material from the repaired diameter.

SUBTASK 72-21-08-320-001-G00

(5) Hand blend the area to be filled to obtain a clean surface for material fill-up and bonding (TASK 70-43-42-300-802-G00 or TASK 70-43-42-300-803-G00).

SUBTASK 72-21-08-010-008-G00

(6) Remove the masking tape.

SUBTASK 72-21-08-100-002-G00

WARNING ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

(7) Clean the damaged area with a lint-free cloth, G51328 [CP2674], moist with alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

SUBTASK 72-21-08-380-001-G00

(8) Apply Flashbreaker 5 tape, G51826 [CP2508], or Flashbreaker 1 tape, G51697 [CP2929], around the area to repair.

SUBTASK 72-21-08-940-001-G00

CAUTION YOU MUST WEAR GLOVES THAT ARE FREE OF TALC OR SILICONE TO PREVENT CONTAMINATION OF ABRADABLE COMPOUND. IF NOT, YOU CAN CAUSE DAMAGE.

(9) Prepare epoxy compound, G51894 [CP2923], as follows:

(a) Base Scotch-WeldTM EC-3524 Part B black: 100 parts by weight.

(b) Accelerator Scotch-WeldTM EC-3524 Part A: 94 parts by weight.

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(c) Slowly mix the base and accelerator by hand to get a homogeneous material with a dark grey color.

(d) If it is necessary, heat the above mixture to 77 ±9°F (25 ±5°C) to make the application of product to the part easier.

NOTE: The most applicable work temperature is between 68°F (20°C) and 77°F (25°C).

SUBTASK 72-21-08-380-002-G00

(10) Apply the abradable material as follows (Figure 803):

(a) If it is necessary, heat the area to repair or the working environment between 68°F (20°C) and 77°F (25°C) to make the application of product to the part easier.

(b) Make sure that the application surface is clean and not disbonded.

(c) Apply a very thin layer of abradable material on the area to repair.

1) Apply sufficient pressure with your thumb to obtain better results.

(d) Fill the remaining thickness with abradable material.

NOTE: Fill with an increase of thickness compared to the flush profile (profile with the masking tape) to flush the abradable with the masking tape and to counteract the shrinkage.

NOTE: Avoid overflowing the surface of the hole when apply the abradable paste.

SUBTASK 72-21-08-380-003-G00

CAUTION

DO NOT APPLY HEAT TO THE ACOUSTIC SHROUD ADJACENT AREAS AND ADJACENT FAN BLADES TO THE REPAIR AREAS. A TEMPERATURE MORE THAN THE POLYMERISATION TEMPERATURE CAN CAUSE DAMAGE TO THE PARTS.

CAUTION

DO NOT USE A HEAT GUN ON THE ABRADABLE MATERIAL. IF YOU DO NOT OBEY, DAMAGE TO THE ABRADABLE MATERIAL CAN OCCUR.

(11) Do one of these steps to polymerize the abradable material:

(a) Polymerize at room temperature for 24 hours.

NOTE: Room temperature must be between 59°F (15°C) and 86°F (30°C). Use 86°F (30°C) to obtain better results.

(b) Polymerize at 122°F (50°C) - 140°F (60°C) for 5 hours.

(c) Polymerize at 158°F (70°C) - 194°F (90°C) for 2 hours.

SUBTASK 72-21-08-380-005-G00

(12) Sand the abradable surface flush to the adjacent masked area with an abrasive cloth, G51480 [CP2772], or abrasive cloth, G51412 [CP2721] (grade 75 or finer), or abrasive paper, G51919 (TASK 70-43-42-300-802-G00).

SUBTASK 72-21-08-010-009-G00

(13) Remove the masking tape.

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SUBTASK 72-21-08-100-003-G00

WARNING ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

(14) Clean the repaired area with a lint-free cloth, G51328 [CP2674], moist with alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

SUBTASK 72-21-08-200-001-G00

(15) Do a hardness check of the abradable material.

(a) Make sure that the hardness is between 40 and 60 Shore D.

SUBTASK 72-21-08-220-002-G00

(16) Do an inspection of the repaired area.

(a) Do a visual inspection of the liner for bubbles or missing abradable material.

(b) Air bubbles are not permitted.

1) If it is necessary, do the local touch-up of the repaired area below.

(c) Lack of abradable is not permitted.

1) If it is necessary, do the repair again.

NOTE: Up to two adjacent rows of holes filled with abradable material around the repaired area are permitted.

SUBTASK 72-21-08-380-004-G00

(17) If it is necessary, do a local touch-up of the repaired area as follows:

(a) Apply the Flashbreaker 5 tape, G51826 [CP2508], or Flashbreaker 1 tape, G51697 [CP2929], around the area to repair.

(b) If it is necessary, slightly sand the surface of the abradable coating with abrasive cloth, G51480 [CP2772], or abrasive cloth, G51412 [CP2721] (grade 75 or finer), or abrasive paper, G51919, to remove all the indications of the surface and/or unbonded abradable material from the liner.

1) Round the sharp edges (TASK 70-43-42-300-802-G00).

WARNING ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

(c) Clean the repaired area with a lint-free cloth, G51328 [CP2674], moist with alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

(d) Do the repair again.

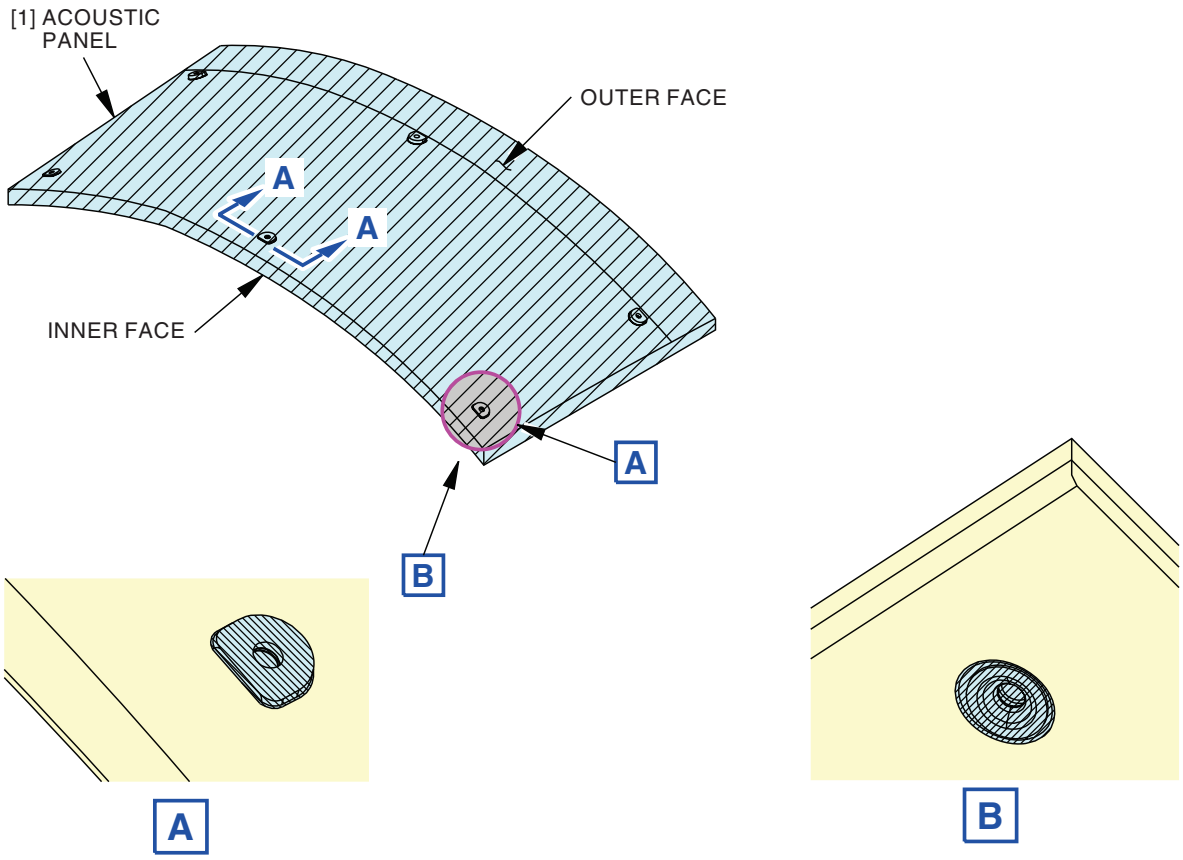
H. Put the Airplane Back to Its Usual Condition

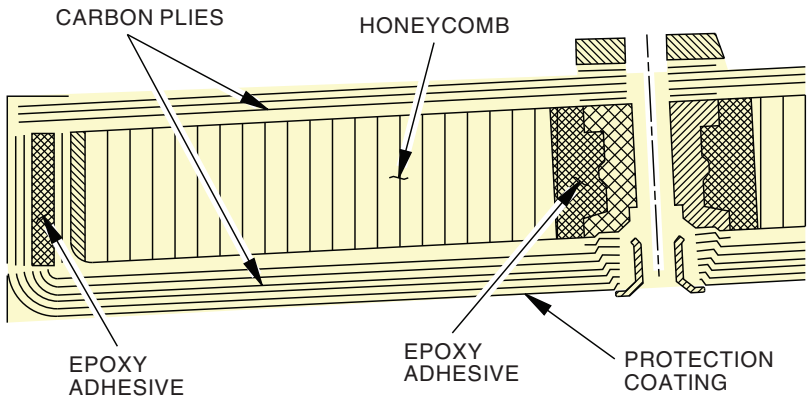
SUBTASK 72-21-08-420-001-G00

(1) Install the acoustical panel [1], do this task: Aft Acoustical Panel Installation, TASK 72-21-08-400-801-G00.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details





A-A

MATERIAL:

- COMPOUND MATERIAL
- POLYETHERIMIDE
- AISI 321 (X6CRNITI18)

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Acoustic Panel - Materials
Figure 801/72-21-08-990-803-G00

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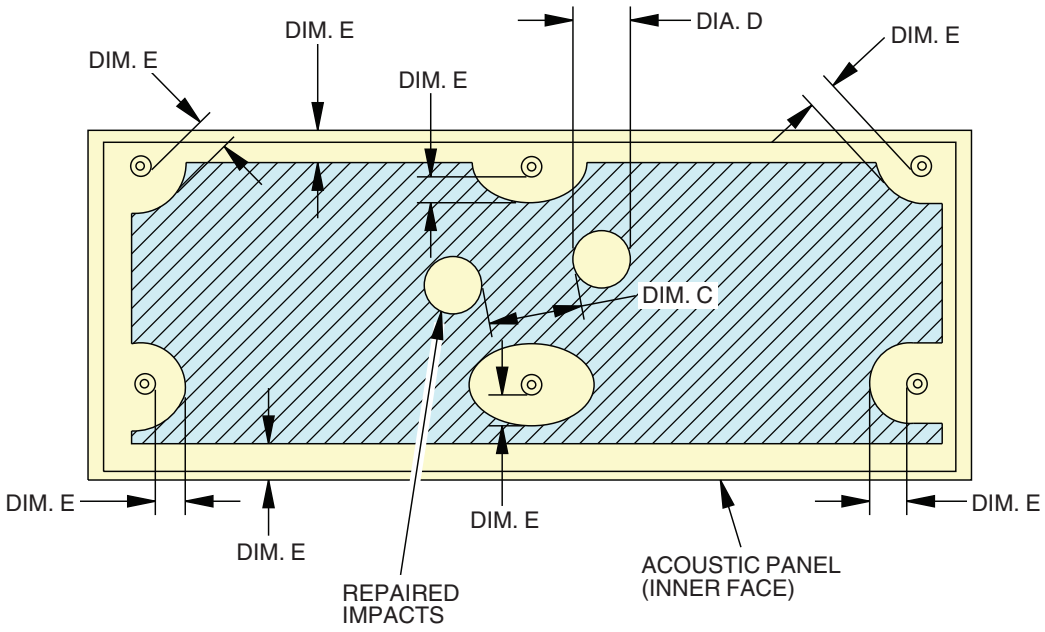

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Acoustic panel - Repairable Areas
Figure 802/72-21-08-990-804-G00

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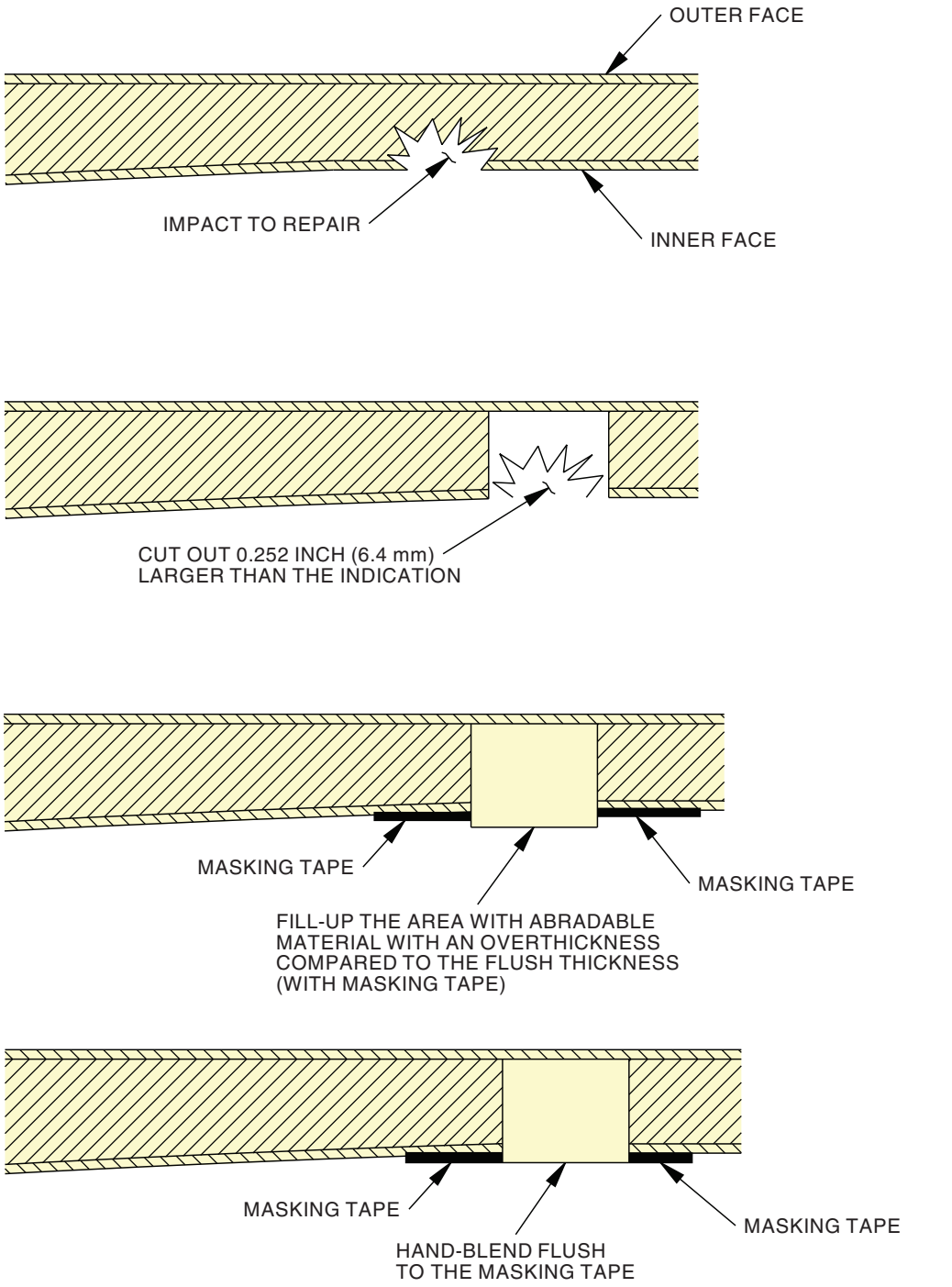


Acoustic Panel - Main Steps of the Repair
Figure 803/72-21-08-990-805-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OUTLET GUIDE VANE PLATFORM - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Outlet guide vane platform removal

(2) Outlet guide vane platform installation.

TASK 72-21-09-000-801-G00

2. Outlet Guide Vane Platform Removal

(Figure 401)

A. General

(1) This task provides the instructions on how to remove the Outlet Guide Vane (OGV) platform.

(2) The OGV platform is referred to as the OGV platform.

B. References

Reference	Title
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Removal

SUBTASK 72-21-09-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-21-09-010-005-G00

(2) Do this task: Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00.

SUBTASK 72-21-09-860-002-G00

(3) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

F. Outlet Guide Vane Platform Removal

SUBTASK 72-21-09-010-001-G00

(1) Remove the OGV platform [1], OGV platform [4], and OGV platform [5]:

(a) Remove the OGV platform [1]:

1) Remove the two bolts [3] that attach the OGV platform [1] to the fan case.

2) Move carefully the OGV platform [1] forward to disengage from the recess of the OGV [2].

3) Remove the OGV platform [1] from the engine.

4) Do the above steps to remove the remaining 37 OGV platforms from the engine.

(b) Remove the OGV platform [4], approximately at the 6:00 o'clock, as follows:

1) Remove the two bolts [6] that attach the OGV platform [4] to the fan case.

2) Move carefully the OGV platform [4] forward to disengage from the recess of the OGV [2].

3) Remove the OGV platform [4] from the engine.

(c) Remove the OGV platform [5], approximately at the 6:00 o'clock, as follows:

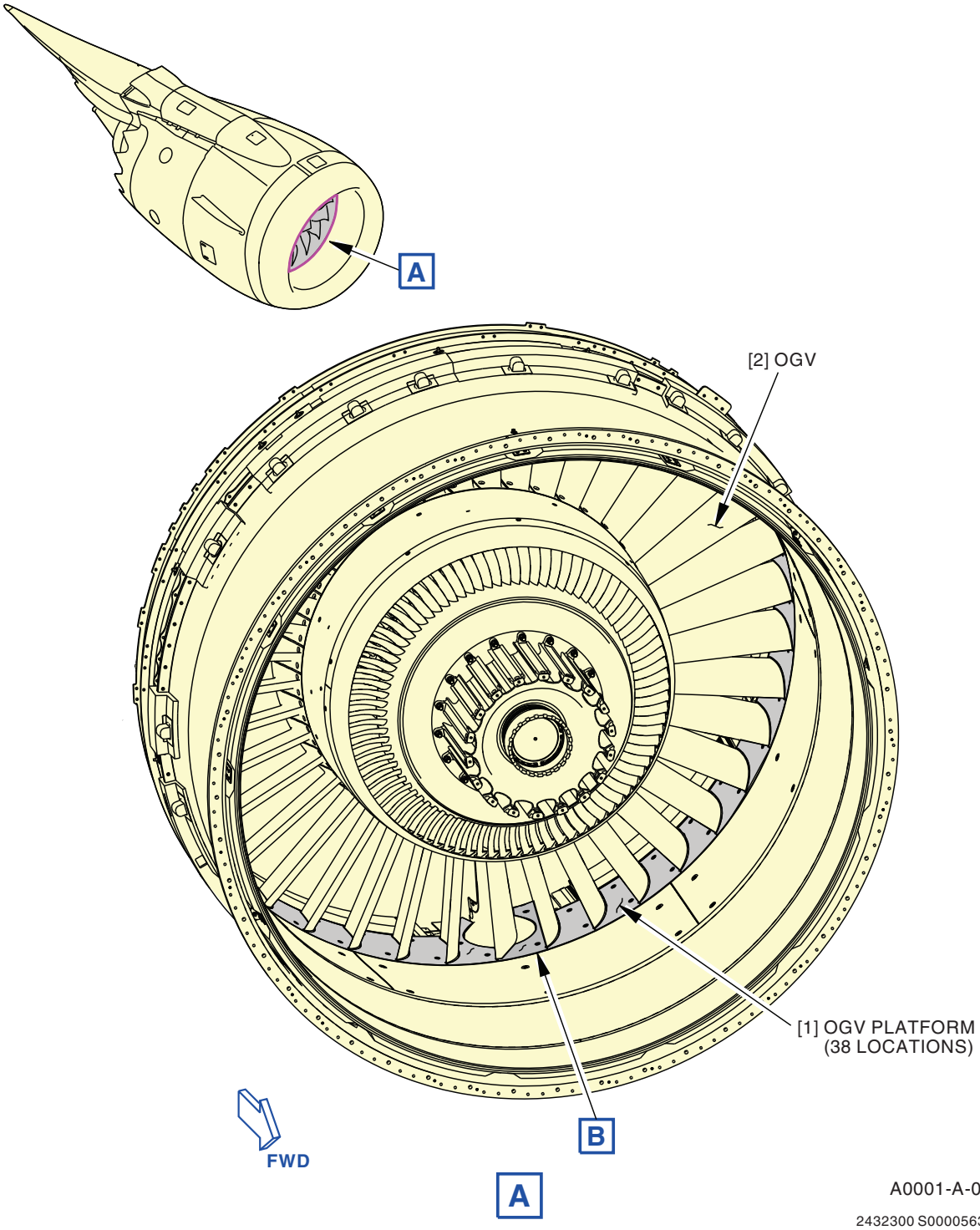
1) Remove the three bolts [7] that attach the OGV platform [5] to the fan case.

2) Move carefully the OGV platform [5] forward to disengage from the recess of the OGV [2].

3) Remove the OGV platform [5] from the engine.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Outlet Guide Vane Platform Installation
Figure 401/72-21-09-990-801-G00 (Sheet 1 of 3)

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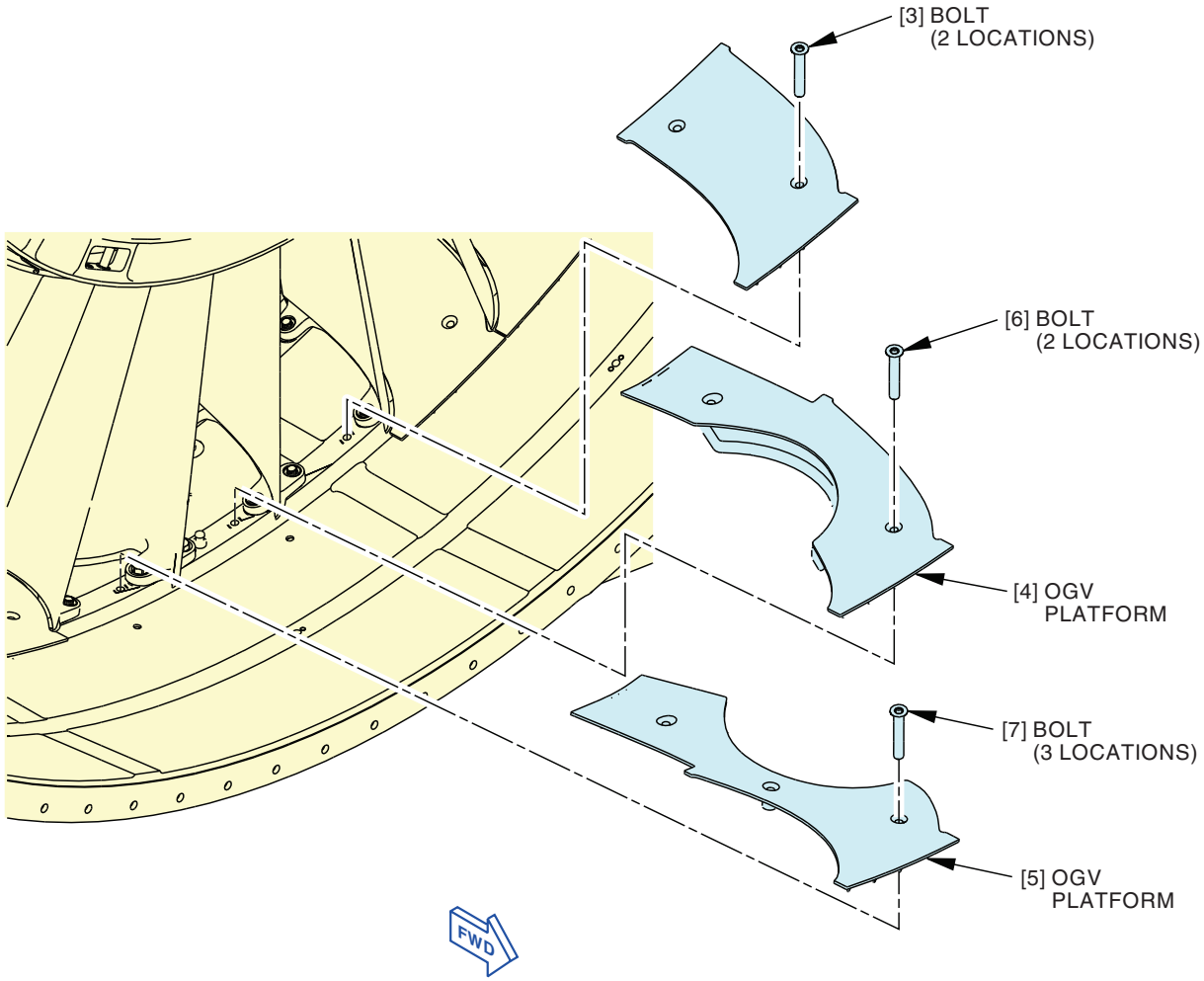

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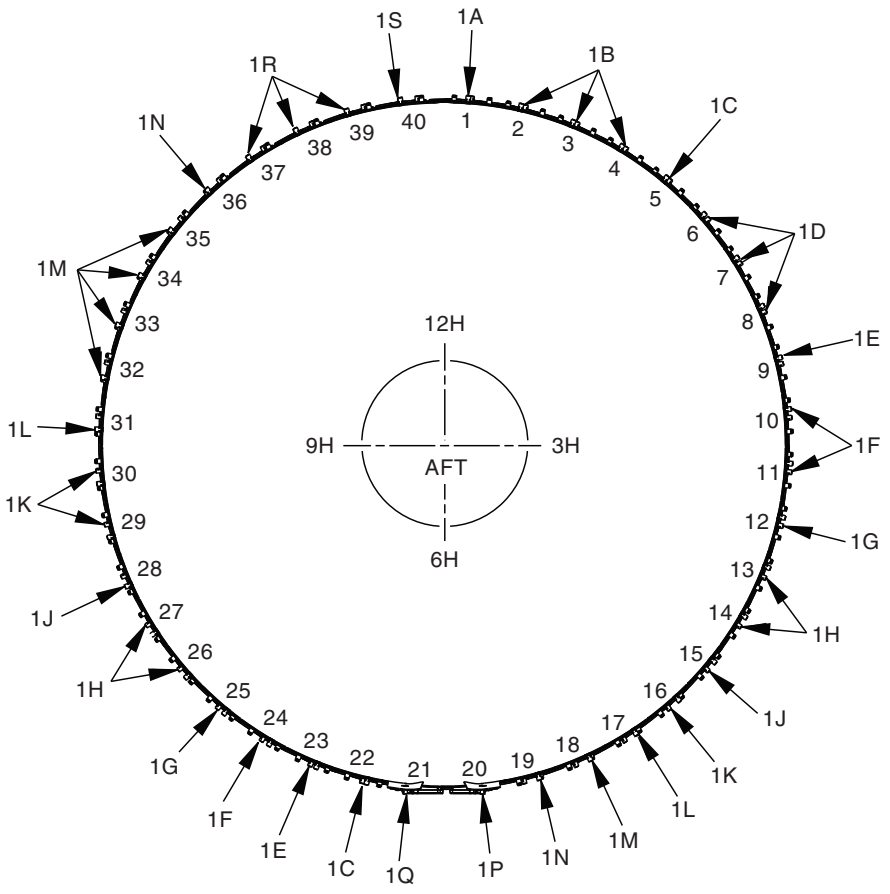
Outlet Guide Vane Platform Installation
Figure 401/72-21-09-990-801-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



REP.	QTY	REF.	REP.	QTY	REF.
1A	1		1K	3	
1B	3		1L	2	
1C	2		1M	5	
1D	3		1N	2	
1E	2		1P	1	
1F	3		1Q	1	
1G	2		1R	3	
1H	4		1S	1	
1J	2				

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Outlet Guide Vane Platform Installation
Figure 401/72-21-09-990-801-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-21-09-400-801-G00

3. Outlet Guide Vane Platform Installation

(Figure 401)

A. General

(1) This task provides the instructions on how to install the Outlet Guide Vane (OGV) platform.

(2) The OGV platform is referred to as the OGV platform.

B. References

Reference	Title
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A50447	Compound - Jointing [CP2773], Corrosion Inhibiting, Chromate Free, Mastinox CA1010	
D00601 [CP2101]	Vaseline - Graphite Mineral	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Outlet Guide Vane Platform Installation

SUBTASK 72-21-09-410-001-G00

(1) Install the OGV platform [1], OGV platform [4], and OGV platform [5]:

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Install the OGV platform [5] at the approximately 6:00 o'clock position:

1) Move carefully the OGV platform [5] rearward to engage in the recess of the OGV [2].

2) Install the OGV platform [5] in the correct position on the engine.

3) Lubricate the threads of the three bolts [7] with mineral grease graphite mineral vaseline, D00601 [CP2101].

4) Install the three bolts [7] that attach the OGV platform [5] to the fan case.

5) Tighten the bolts [7] to 58.9 in-lb (6.65 N·m) - 65.1 in-lb (7.36 N·m).

(c) Install the OGV platform [4] at the approximately 6:00 o'clock position:

1) Carefully move the OGV platform [4] rearward to engage in the recess of the OGV [2].

2) Install the OGV platform [4] in a correct position on the engine.

3) Lubricate the threads of the two bolts [6] with mineral grease graphite mineral vaseline, D00601 [CP2101].

4) Install the two bolts [6] that attach the OGV platform [4] to the fan case.

5) Tighten the bolts [6] to 58.9 in-lb (6.65 N·m) - 65.1 in-lb (7.36 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Install the OGV platform [1]:

1) Carefully move the OGV platform [1] rearward to engage in the recess of the OGV [2].
2) Install the OGV platform [1] in the correct position on the engine.
3) Lubricate the threads of the two bolts [3] with jointing compound, A50447.
4) Install the two bolts [3] that attach the OGV platform [1] to the fan case.
5) Tighten the bolts [3] to 76 in-lb (8.55 N·m) - 84 in-lb (9.45 N·m).
6) Do the above steps again to install the remaining 37 OGV platforms on the engine

(e) Clean the work area and clear the tools and other items.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-09-860-003-G00

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-21-09-410-005-G00

(2) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

SUBTASK 72-21-09-860-004-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OUTER OUTLET GUIDE VANE PLATFORM - DAMPER REPAIRS

1. General

A. This procedure has this task:

(1) Outer Outlet Guide Vane Platform - Damper Replacement.

TASK 72-21-09-960-802-G00

2. Outer Outlet Guide Vane Platform - Damper Replacement

(Figure 801)

A. General

(1) This task gives instructions to replace the dampers on the Outer Outlet Guide Vane Platform (OOGVP).

B. References

Reference	Title
72-21-09-000-801-G00	Outlet Guide Vane Platform Removal (P/B 401)
72-21-09-400-801-G00	Outlet Guide Vane Platform Installation (P/B 401)
72-23-00-211-804-G00	Fan Frame Module Inspection (P/B 601)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A50012 [CP2242]	Adhesive - Silicone, Thixotropic Paste, One-Part RTV - RTV 732	MIL-A-46106 Type I
B00676 [CP1041]	Alcohol - Isopropyl	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Prepare to the OOGVP Damper Replacement

SUBTASK 72-21-09-860-007-G00

(1) Do these step to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) On the P5 overhead panel, make sure that the applicable ENGINE START switch is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) On the P10 control stand, make sure that the applicable START LEVER switch is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-21-09-860-005-G00

(2) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

G. OOGVP Damper Replacement

SUBTASK 72-21-09-010-004-G00

(1) If it is necessary, remove the applicable OOGVP (TASK 72-21-09-000-801-G00).

SUBTASK 72-21-09-020-002-G00

(2) Remove and discard the damper [1].

SUBTASK 72-21-09-210-002-G00

(3) Do a visual inspection of the hole and adjacent area where the damper [1] will be installed.

NOTE: Deviation to a correct condition is not permitted.

(a) If you find signs of glue in the hole, the part is not repairable with this procedure.

SUBTASK 72-21-09-110-002-G00

WARNING

ISOPROPYL ALCOHOL IS POISONOUS AND FLAMMABLE. USE PROTECTION EQUIPMENT. MAKE SURE THAT YOU USE ISOPROPYL ALCOHOL IN AN AREA WITH AIRFLOW. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

(4) Clean the hole where you will bond a new damper [1], the adjacent area and new damper [1] with a clean cloth soaked in the alcohol, B00676 [CP1041].

SUBTASK 72-21-09-420-002-G00

(5) Bond the new damper [1] in the non-through hole as follows:

WARNING

REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

(a) Apply thin coat of RTV 732 adhesive, A50012 [CP2242], on the body of the damper [1].

(b) Push and turn the damper [1] in the OOGVP until the stop.

WARNING

ISOPROPYL ALCOHOL IS POISONOUS AND FLAMMABLE. USE PROTECTION EQUIPMENT. MAKE SURE THAT YOU USE ISOPROPYL ALCOHOL IN AN AREA WITH AIRFLOW. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

(c) Remove the unwanted adhesive from around the damper [1] with a clean cloth soaked in the alcohol, B00676 [CP1041].

(d) Keep the damper [1] in contact and let it dry 24 hours.

SUBTASK 72-21-09-211-002-G00

(6) Do a visual inspection of the damper [1] (TASK 72-23-00-211-804-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-21-09-410-004-G00

(7) If it is necessary, install the applicable OOGVP (TASK 72-21-09-400-801-G00).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-09-860-006-G00

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-21-09-860-008-G00

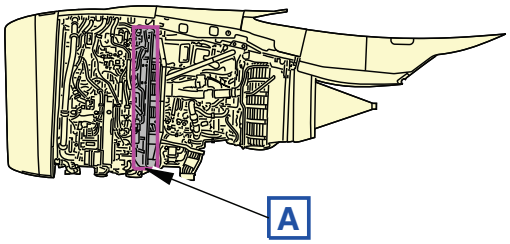
(2) Do these steps to remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch and START LEVER switch:

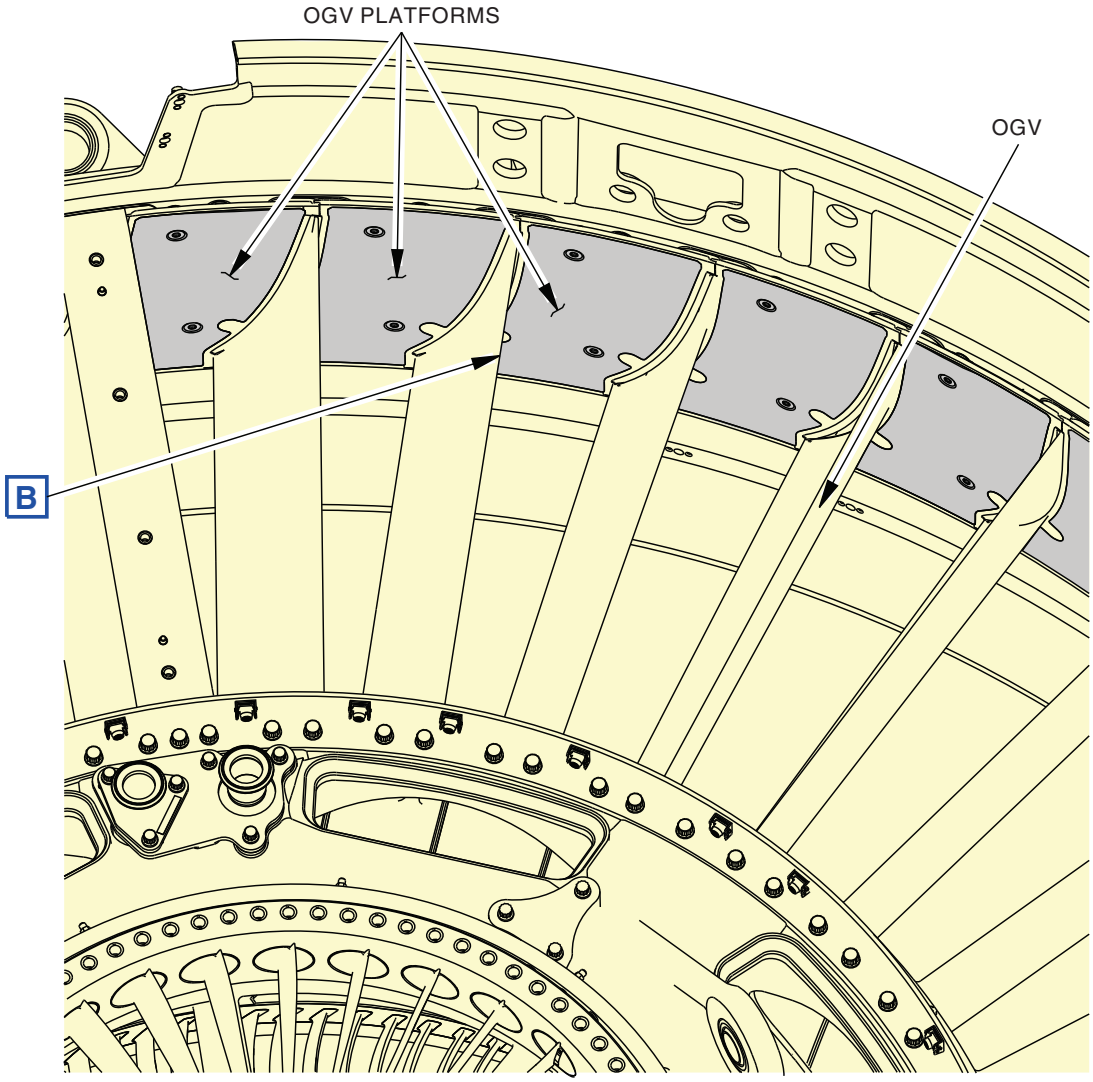
(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

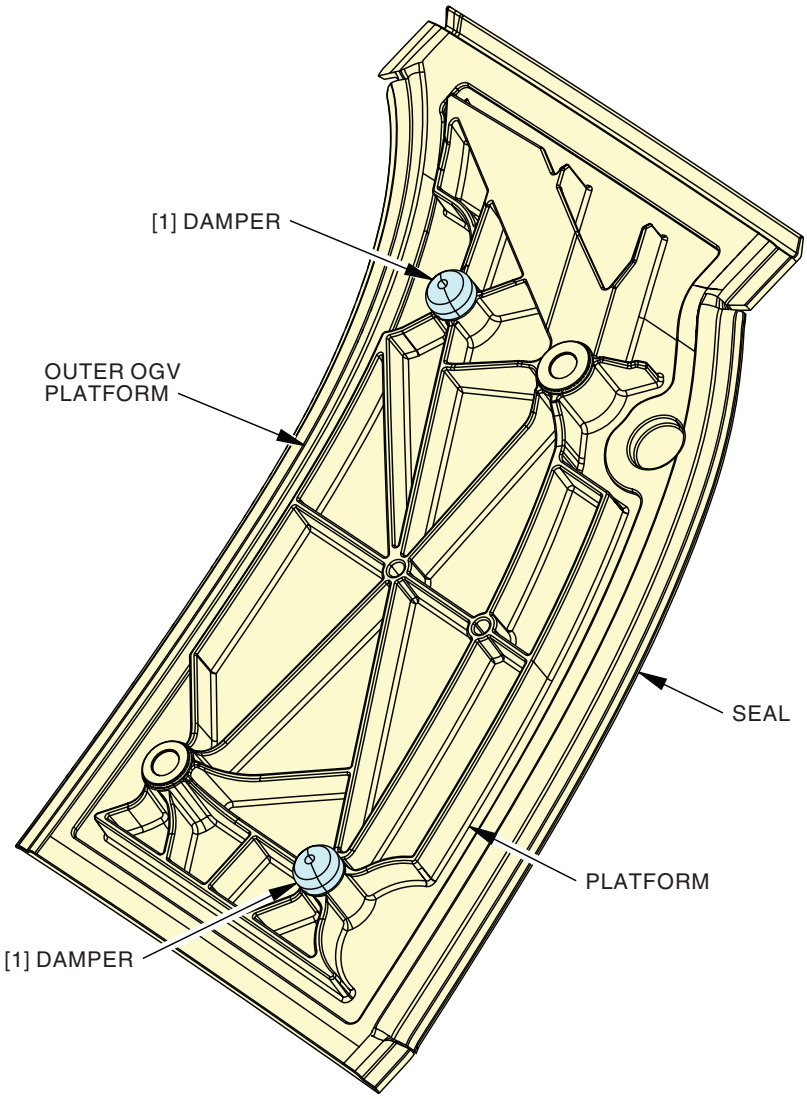
ECCN 9E991 BOEING PROPRIETARY - See title page for details





OOGVP Dampers Replacement
Figure 801/72-21-09-990-803-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TYPICAL VIEW OF OUTER OGV PLATFORMS

B

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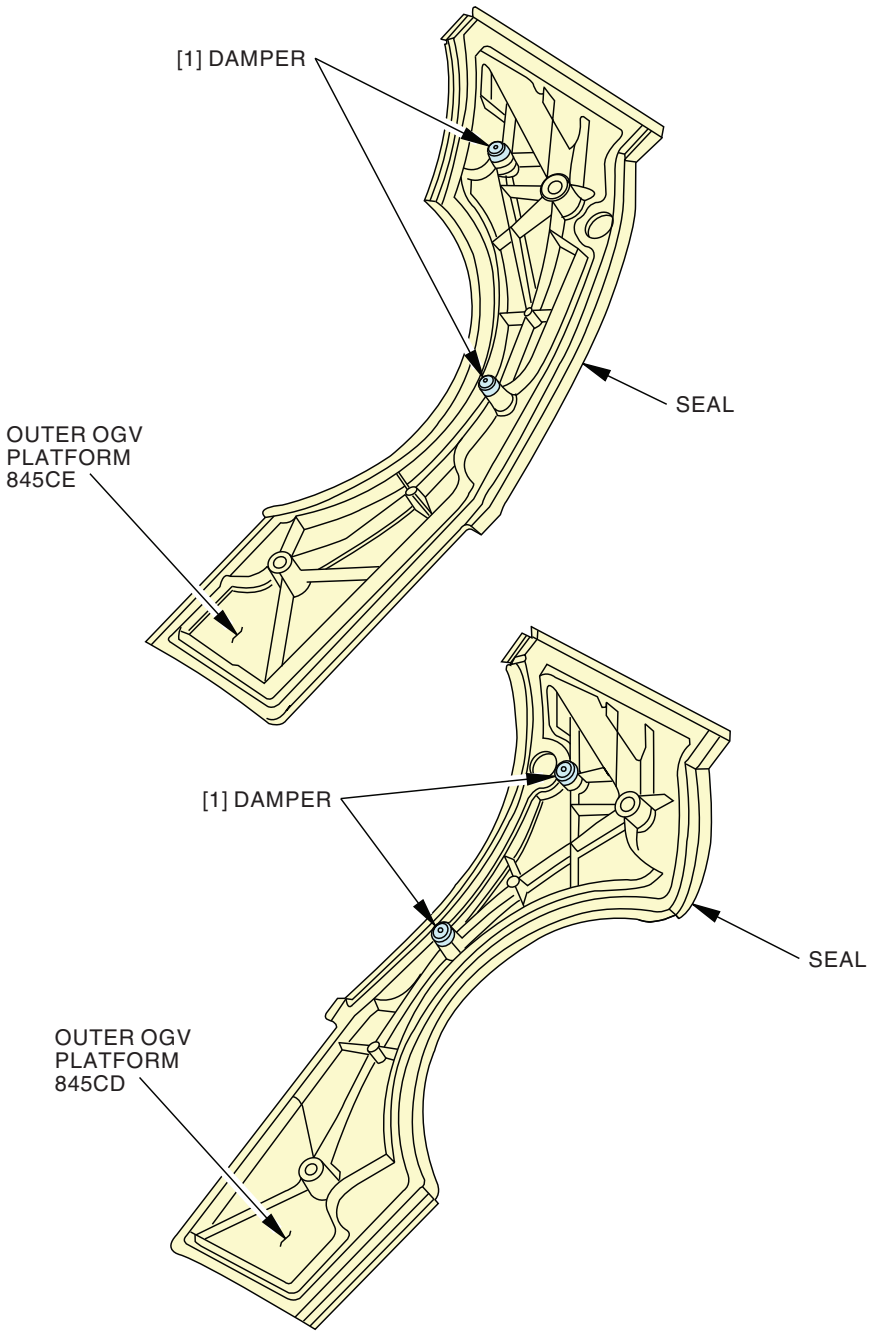
OOGVP Dampers Replacement
Figure 801/72-21-09-990-803-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



OOGVP Dampers Replacement
Figure 801/72-21-09-990-803-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN DISK SHIELD - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Fan disk shield removal

(2) Fan disk shield installation.

TASK 72-21-10-000-801-G00

2. Fan Disk Shield Removal

(Figure 401)

A. General

(1) This task provides the instructions on how to remove the fan disk shield.

B. References

Reference	Title
72-22-02-000-803-G00	Fan Blades (Set) Removal (P/B 401)

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Prepare for the Removal

SUBTASK 72-21-10-020-001-G00

(1) Do this task: Fan Blades (Set) Removal, TASK 72-22-02-000-803-G00.

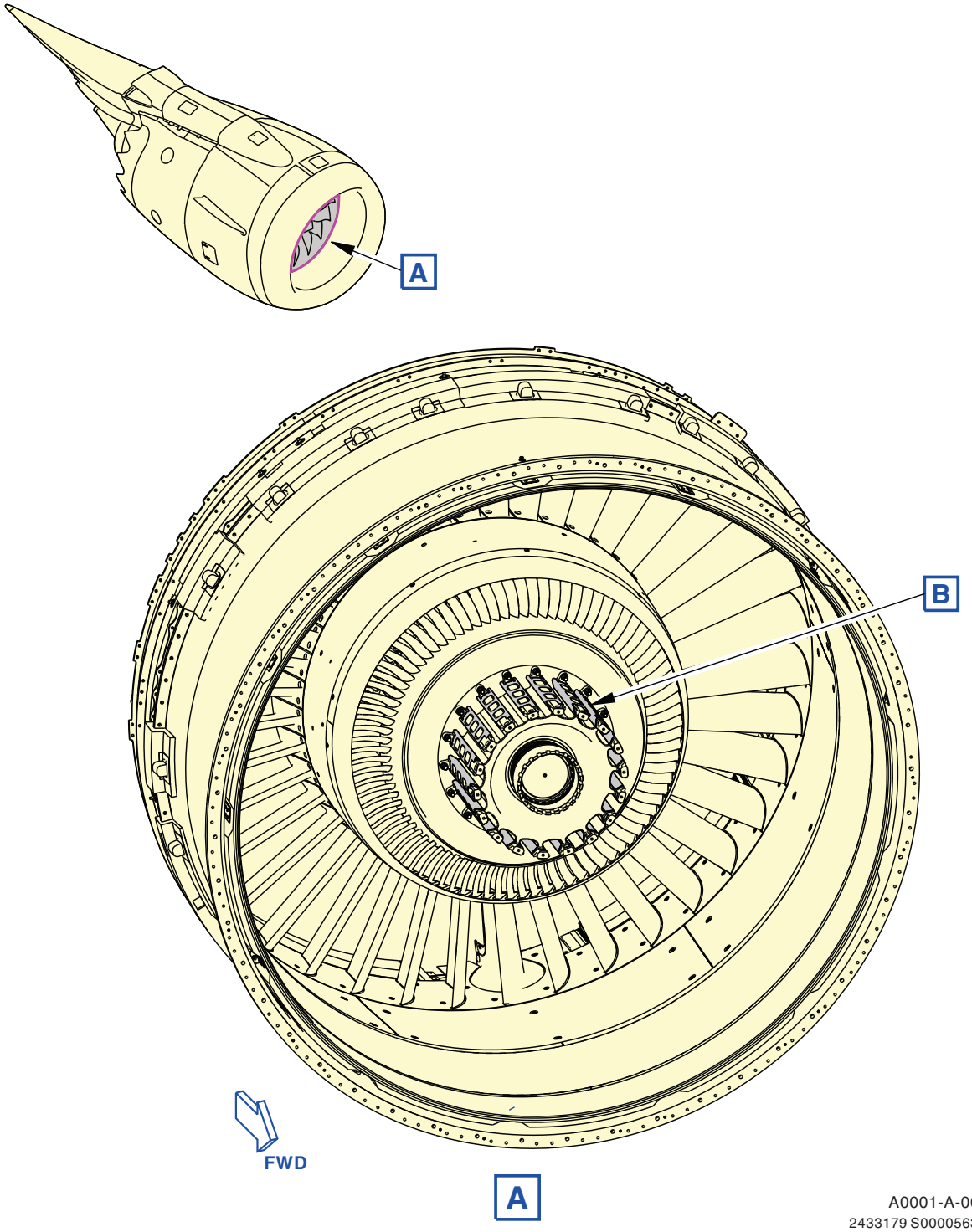
E. Fan Disk Shield Removal

SUBTASK 72-21-10-020-002-G00

(1) Remove the eighteen fan disk shields [2] from the fan disk [1].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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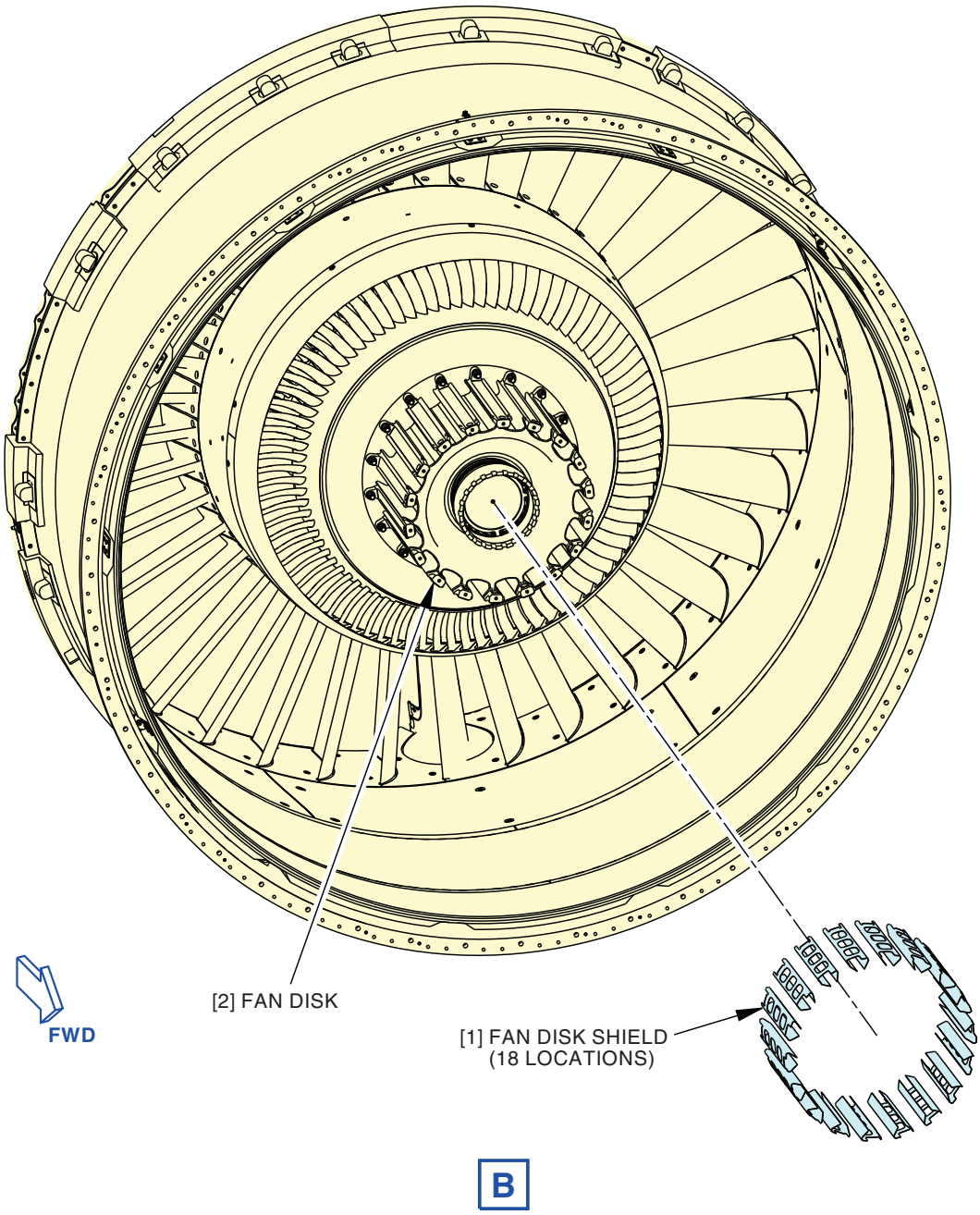
Fan Disk Shield Installation
Figure 401/72-21-10-990-801-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Fan Disk Shield Installation
Figure 401/72-21-10-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-21-10-400-801-G00

3. Fan Disk Shield Installation

(Figure 401)

A. General

(1) This task provides the instructions on how to install the fan disk shield.

B. References

Reference	Title
72-22-02-400-803-G00	Fan Blades (Set) Installation (P/B 401)

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Fan Disk Shield Installation

SUBTASK 72-21-10-420-001-G00

(1) Install the eighteen fan disk shields [2] as follows:

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Install the eighteen fan disk shields [2] in the correct positions on the fan disk [1].

E. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-10-420-002-G00

(1) Do this task: Fan Blades (Set) Installation, TASK 72-22-02-400-803-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LOW PRESSURE COMPRESSOR ROTATING RING SPACER - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Low Pressure Compressor (LPC) rotating ring spacer removal

(2) Low Pressure Compressor (LPC) rotating ring spacer installation.

TASK 72-21-11-000-801-G00

2. Low Pressure Compressor (LPC) Rotating Ring Spacer Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the Low Pressure Compressor (LPC) rotating ring spacer.

B. References

Reference	Title
72-22-02-000-803-G00	Fan Blades (Set) Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-15154	Pin - Guide, Low Pressure Compressor Vane Part #: 956A8864G01 Supplier: 58828
SPL-15155	Screw Set - Low Pressure Compressor Vane Remove Part #: 956A6880G02 Supplier: 58828 Opt Part #: 956A6880G01 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Removal

SUBTASK 72-21-11-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-21-11-860-002-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

F. Low Pressure Compressor (LPC) Rotating Ring Spacer Removal

SUBTASK 72-21-11-020-001-G00

(1) Remove the LPC rotating ring spacer [1] as follows:

(a) Remove the fan blades (TASK 72-22-02-000-803-G00).

(b) Remove three equally spaced bolts [2] that attach the LPC rotating ring spacer [1] to the LPC spool [3].

(c) Make sure that there are 11 bolts installed between each of the three guide pins, SPL-15154.

(d) Install the three guide pins, SPL-15154, through the three bolt holes of the LPC rotating ring spacer [1] to the LPC spool [3].

(e) Remove the remaining 33 bolts [2] that attach the LPC rotating ring spacer [1] to the LPC spool [3].

(f) Install the six screw sets, SPL-15155, in the six extraction holes [4] of the LPC rotating ring spacer [1].

(g) Tighten the six screw sets, SPL-15155, in a criss-cross pattern to remove the LPC rotating ring spacer [1] from the LPC spool [3].

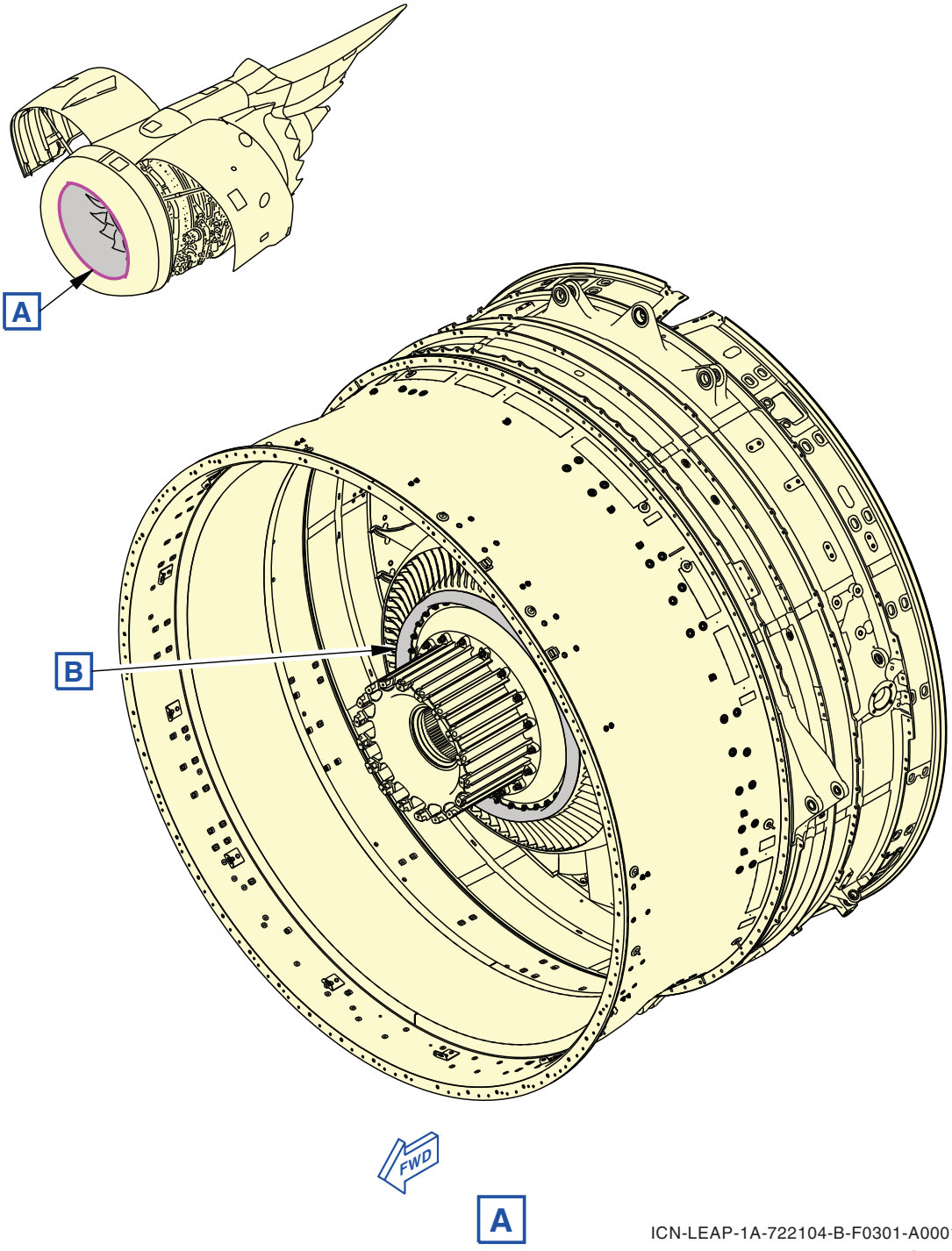
(h) Remove the six screw sets, SPL-15155, from the LPC rotating ring spacer [1].

(i) Remove the LPC rotating ring spacer [1] from the LPC spool [3].

(j) Remove the guide pin, SPL-15154, of the LPC spool [3].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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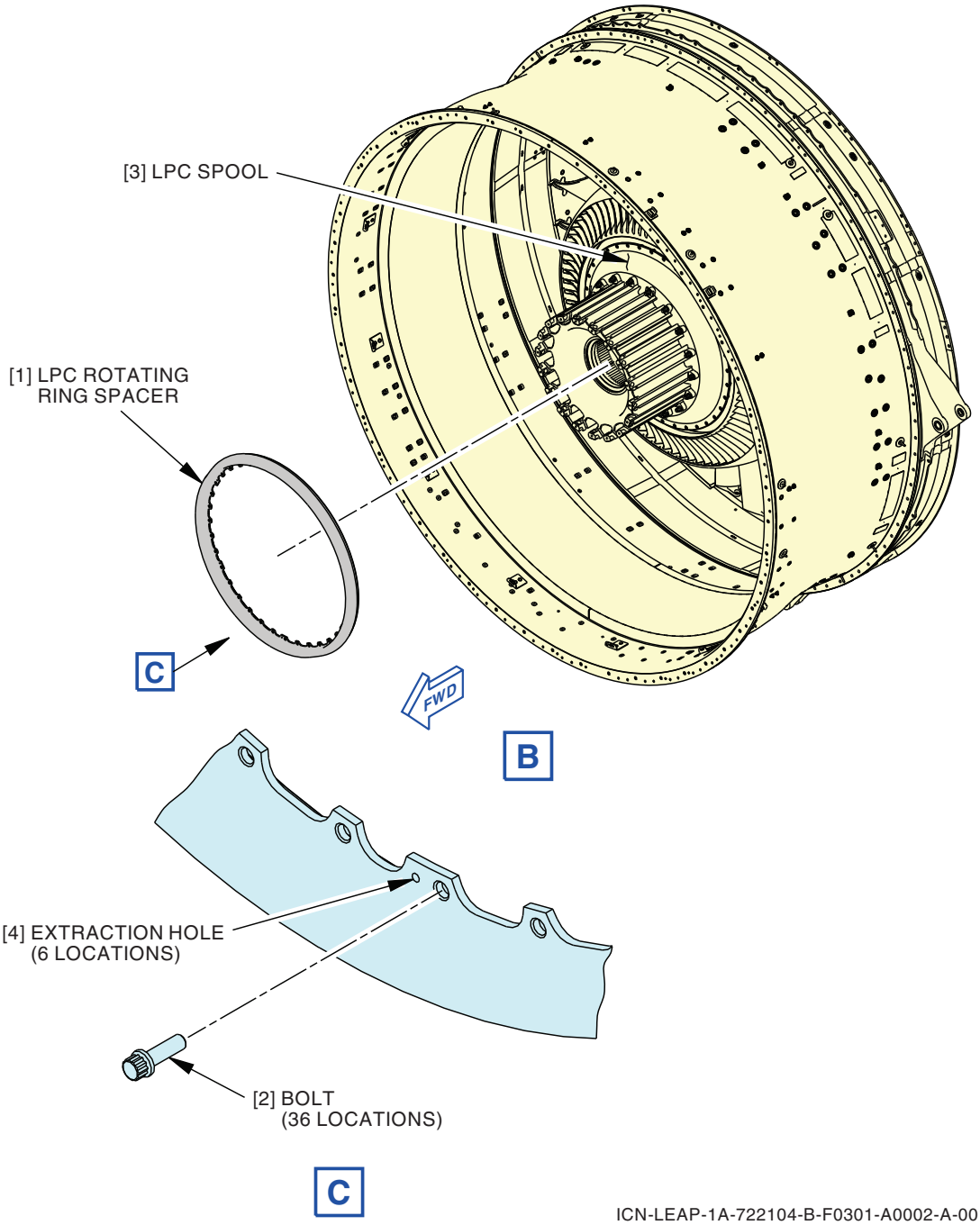
Low Pressure Compressor (LPC) Rotating Ring Spacer Installation
Figure 401/72-21-11-990-801-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Low Pressure Compressor (LPC) Rotating Ring Spacer Installation
Figure 401/72-21-11-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-21-11-400-801-G00

3. Low Pressure Compressor (LPC) Rotating Ring Spacer Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the Low Pressure Compressor (LPC) rotating ring spacer.

(2) You must do the tests that are listed in the power plant test reference table after you install the LPC rotating ring spacer.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
72-22-02-400-803-G00	Fan Blades (Set) Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-15154	Pin - Guide, Low Pressure Compressor Vane Part #: 956A8864G01 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00601 [CP2101]	Vaseline - Graphite Mineral	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Low Pressure Compressor (LPC) Rotating Ring Spacer Installation

SUBTASK 72-21-11-420-001-G00

WARNING USE THERMAL GLOVES WHEN YOU MOVE OR TOUCH HOT OR COLD PARTS. HOT OR COLD PARTS CAN CAUSE INJURY.

(1) Install the LPC rotating ring spacer [1] as follows:

(a) Install the three guide pins, SPL-15154, on the LPC spool [3].

(b) Make sure that there are 11 empty holes between each of the three guide pins, SPL-15154.

CAUTION DO NOT HEAT THE LPC ROTATING RING SPACER MORE THAN 248.0 DEGREES FAHRENHEIT (120.0 DEGREES CELSIUS). DAMAGE TO THE LPC ROTATING RING SPACER CAN OCCUR.

(c) Heat the LPC rotating ring spacer [1] to between 230.0°F (110.0°C) and 248.0°F (120.0°C) with the hot air blower.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Control the temperature of the inner surface of the LPC rotating ring spacer [1] with a thermocouple.

(d) Install the LPC rotating ring spacer [1] on the LPC spool [3].

1) Make sure that you install the LPC rotating ring spacer [1] in the correct position on the LPC spool [3].

(e) Lubricate the threads of the 36 bolts [2] with the graphite mineral vaseline, D00601 [CP2101].

(f) Install the 33 bolts [2] that attach the LPC rotating ring spacer [1] to the LPC spool [3].

(g) Remove the guide pins, SPL-15154, from the LPC spool [3].

(h) Install the last three bolts [2] that attach the LPC rotating ring spacer [1] to the LPC spool [3].

(i) Tighten the bolts [2] to 109.25 in-lb (12.34 N·m) - 120.75 in-lb (13.64 N·m), in a criss-cross pattern.

SUBTASK 72-21-11-420-002-G00

(2) Install the fan blades (TASK 72-22-02-400-803-G00).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-11-860-003-G00

(1) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 72-21-11-860-004-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

H. Low Pressure Compressor (LPC) Rotating Ring Spacer Installation Test

SUBTASK 72-21-11-710-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN OVERHEAT SENSORS - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Fan overheat sensors removal

(2) Fan overheat sensors installation.

TASK 72-21-12-000-801-G00

2. Fan Overheat Sensors Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the fan overheat sensors.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 72-21-12-860-005-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-21-12-860-006-G00

(2) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. Fan Overheat Sensors Removal

SUBTASK 72-21-12-020-002-G00

(1) Remove the upper overheat sensor [1] as follows:

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(a) Use the teflon-jawed pliers, STD-664 to disconnect the HJ6A harness electrical connector [3] and the HJ6B harness electrical connector [4] from the upper overheat sensor [1] as follows (TASK 70-00-01-910-803-G00):

1) Disconnect the HJ6A harness electrical connector [3] from the upper overheat sensor [1].
2) Disconnect the HJ6B harness electrical connector [4] from the upper overheat sensor [1].
3) Install the protective covers on the HJ6A harness electrical connector [3], the HJ6B harness electrical connector [4], and the two electrical receptacles on the upper overheat sensor [1].

(b) Remove the three bolts [5] that attach the upper overheat sensor [1] to the mounting bracket [6].

(c) Remove the upper overheat sensor [1] from the mounting bracket [6].

SUBTASK 72-21-12-020-003-G00

(2) Remove the lower overheat sensor [2] as follows:

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(a) Use the teflon-jawed pliers, STD-664 to disconnect the HJ5A harness electrical connector [7] and the HJ5B harness electrical connector [8] from the lower overheat sensor [2] as follows (TASK 70-00-01-910-803-G00):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Disconnect the HJ5A harness electrical connector [7] from the lower overheat sensor [2].

2) Disconnect the HJ5B harness electrical connector [8] from the lower overheat sensor [2].

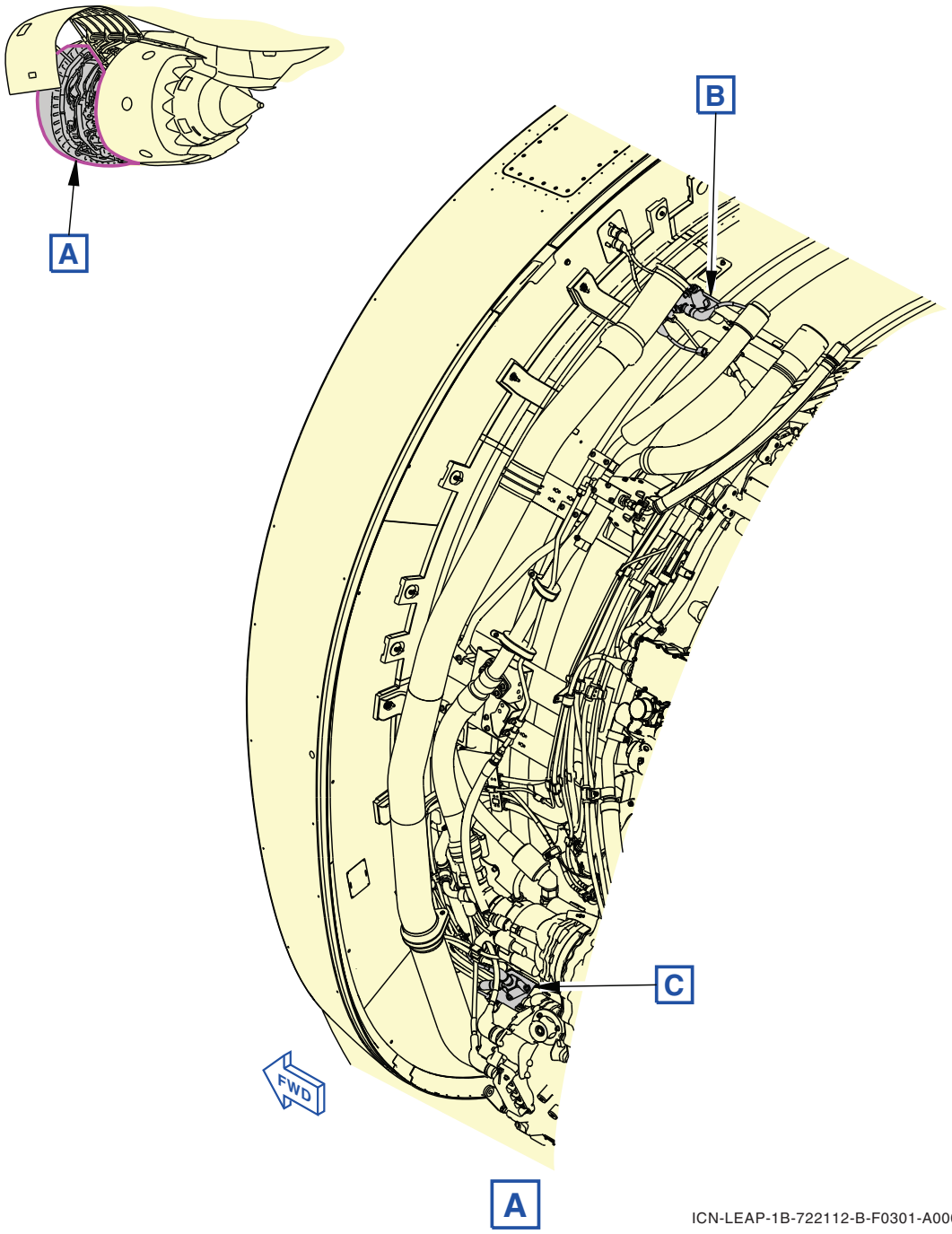
3) Install the protective covers on the HJ5A harness electrical connector [7], the HJ5B harness electrical connector [8], and the two electrical receptacles on the lower overheat sensor [2].

(b) Remove the three bolts [9] that attach the lower overheat sensor [2] to the mounting bracket [10].

(c) Remove the lower overheat sensor [2] from the mounting bracket [10].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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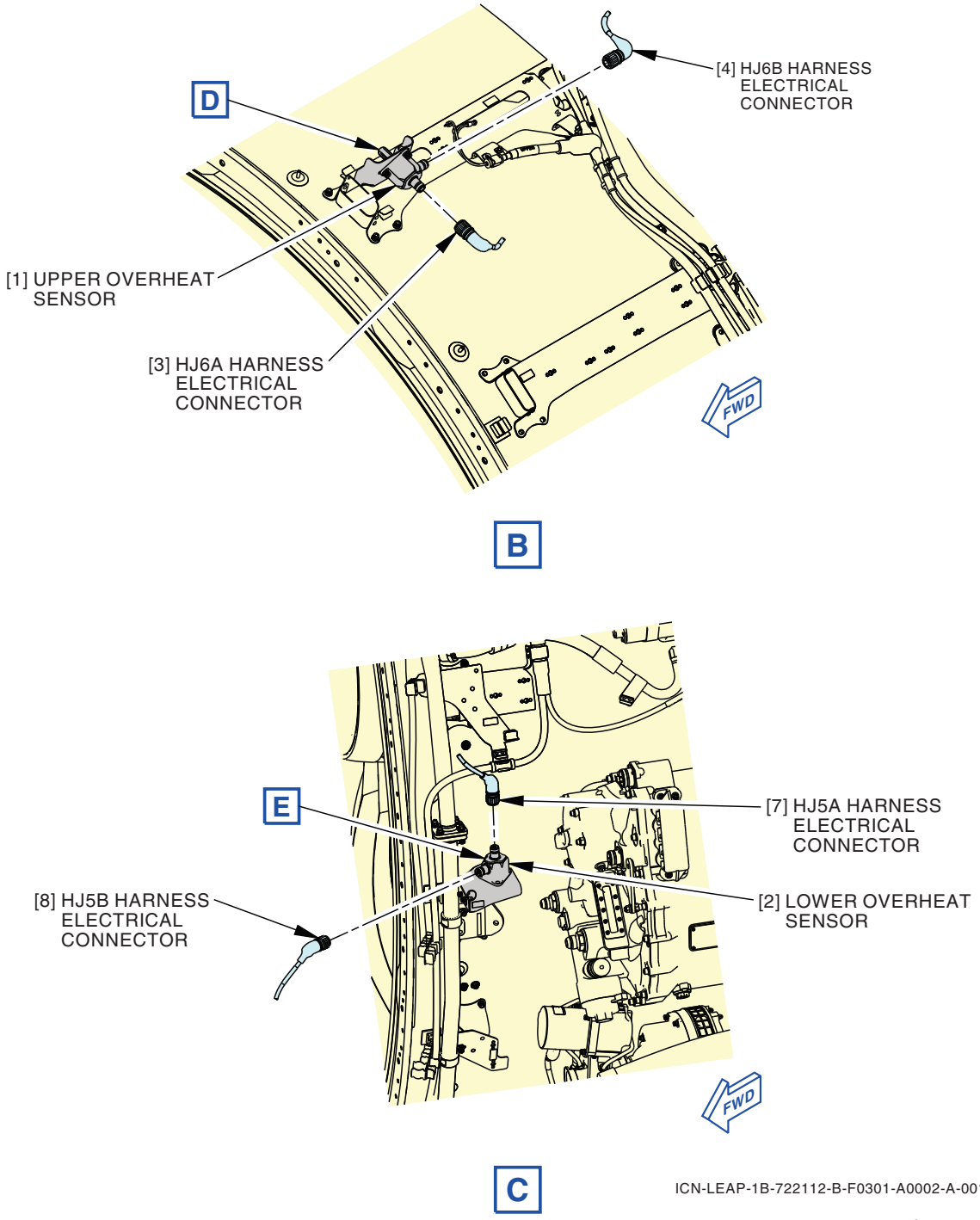
2570082 S0000616503_V1

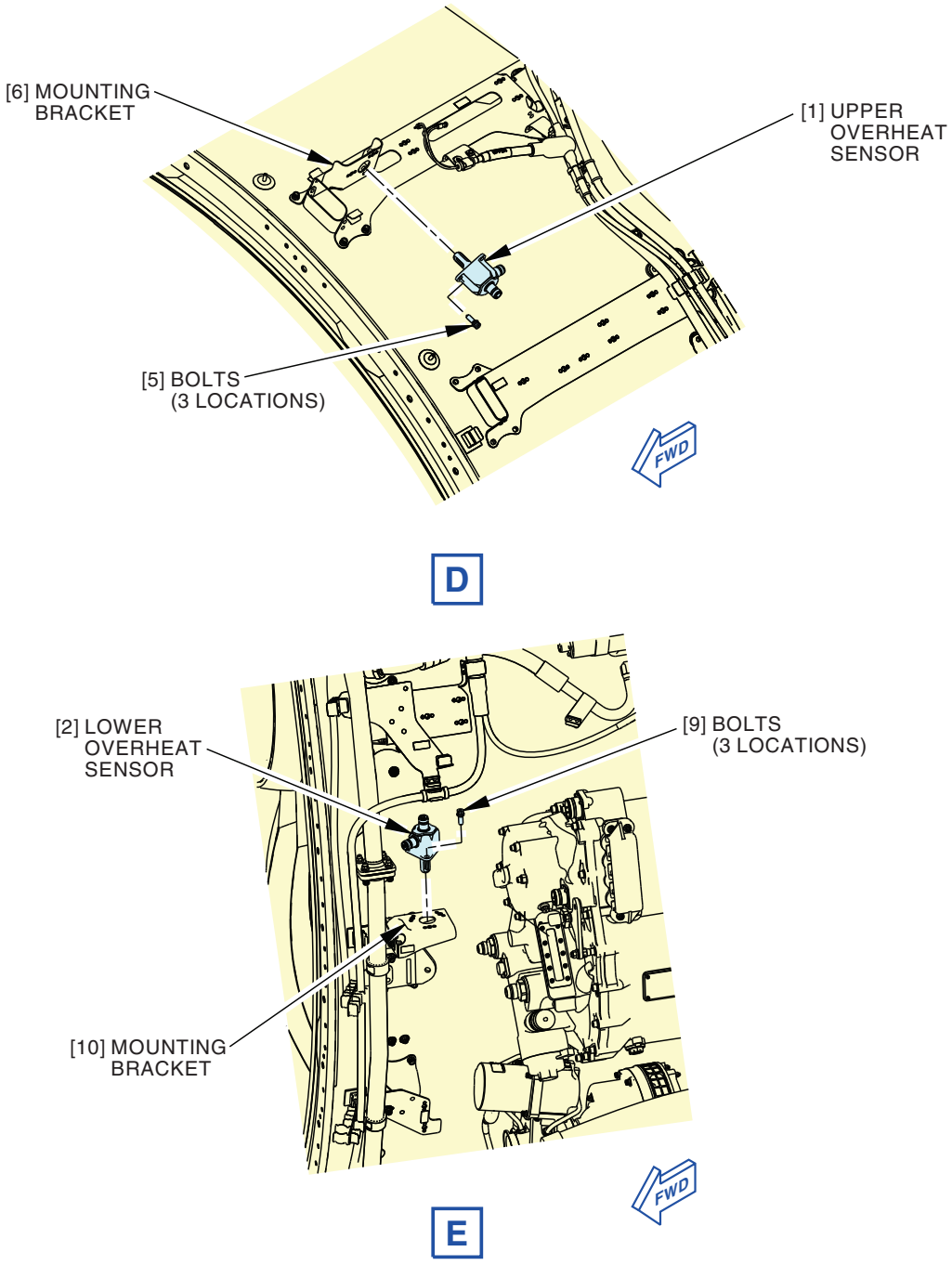
Fan Overheat Sensors Installation
Figure 401/72-21-12-990-802-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Overheat Sensors Installation
Figure 401/72-21-12-990-802-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

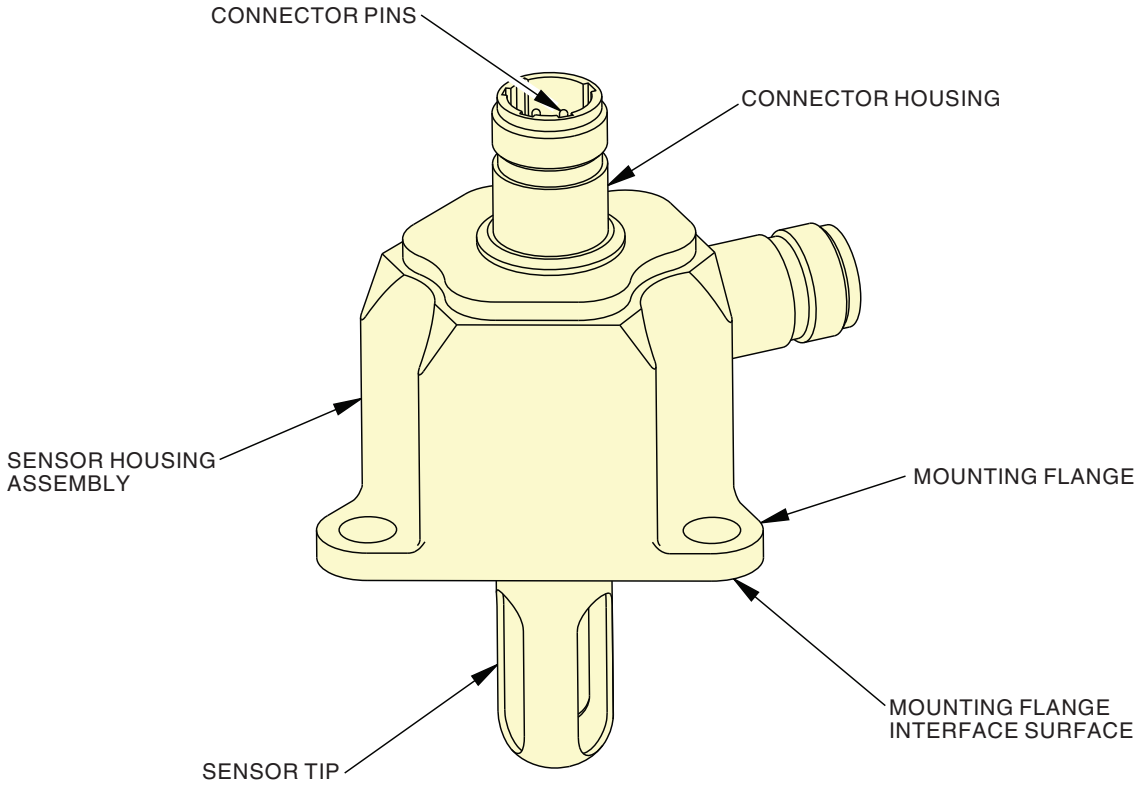


ICN-LEAP-1B-722112-B-F0301-A0003-A-001-01

2570088 S0000616505_V1

Fan Overheat Sensors Installation
Figure 401/72-21-12-990-802-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICN-LEAP-1B-722112-B-F0301-A0004-A-001-01

2570089 S0000616506_V1

Fan Overheat Sensors Installation
Figure 401/72-21-12-990-802-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-21-12-400-801-G00

3. Fan Overheat Sensors Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the fan overheat sensors.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. Fan Overheat Sensors Installation

SUBTASK 72-21-12-420-002-G00

(1) Install the upper overheat sensor [1] as follows:

(a) Install the upper overheat sensor [1] in the correct position on the mounting bracket [6].

(b) Lubricate the threads of the three bolts [5] with the engine oil, D00599 [CP2442].

(c) Install the three bolts [5] that attach the upper overheat sensor [1] to the mounting bracket [6].

(d) Tighten the three bolts [5] to 67.45 in-lb (7.6 N·m) - 74.55 in-lb (8.4 N·m).

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(e) Use the teflon-jawed pliers, STD-664 to connect the HJ6A harness electrical connector [3] and the HJ6B harness electrical connector [4] to the upper overheat sensor [1] as follows (TASK 70-00-01-910-803-G00):

1) Remove the protective covers from the HJ6A harness electrical connector [3], the HJ6B harness electrical connector [4], and the two electrical receptacles on the upper overheat sensor [1].

2) Connect the HJ6A harness electrical connector [3] to the upper overheat sensor [1].

3) Connect the HJ6B harness electrical connector [4] to the upper overheat sensor [1].

SUBTASK 72-21-12-420-003-G00

(2) Install the lower overheat sensor [2] as follows:

(a) Install the lower overheat sensor [2] in the correct position on the mounting bracket [10].

(b) Lubricate the threads of the three bolts [9] with the engine oil, D00599 [CP2442].

(c) Install the three bolts [9] that attach the lower overheat sensor [2] to the mounting bracket [10].

(d) Tighten the three bolts [9] to 67.45 in-lb (7.6 N·m) - 74.55 in-lb (8.4 N·m).

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(e) Use the teflon-jawed pliers, STD-664 to connect the HJ5A harness electrical connector [7] and the HJ5B harness electrical connector [8] to the lower overheat sensor [2] as follows (TASK 70-00-01-910-803-G00):

1) Remove the protective covers from the HJ5A harness electrical connector [7], the HJ5B harness electrical connector [8], and the two electrical receptacles on the lower overheat sensor [2].

2) Connect the HJ5A harness electrical connector [7] to the lower overheat sensor [2].

3) Connect the HJ5B harness electrical connector [8] to the lower overheat sensor [2].

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-12-940-001-G00

(1) Clean the work area and clear the tools and other items

SUBTASK 72-21-12-860-007-G00

(2) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 72-21-12-860-008-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN OVERHEAT SENSORS - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Fan overheat sensor inspection.

TASK 72-21-12-200-801-G00

2. Fan Overheat Sensor Inspection

(Figure 601)

A. General

(1) This task gives the instructions examine the fan overheat sensors.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-21-12-000-801-G00	Fan Overheat Sensors Removal (P/B 401)
72-21-12-400-801-G00	Fan Overheat Sensors Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

F. Prepare for the Inspection

SUBTASK 72-21-12-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-21-12-860-002-G00

(2) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
423	Left Fan Cowl, Engine 2

G. Fan Overheat Sensor Inspection

SUBTASK 72-21-12-220-001-G00

(1) Examine the fan overheat sensor [1] as follows:

(a) Sensor tip

1) Nick, scratches, and dents

a) Permitted, if it does not affect the sensor performance.

b) If it is necessary, replace the fan overheat sensor [1] (TASK 72-21-12-000-801-G00 and TASK 72-21-12-400-801-G00).

2) Raised metal

a) Not permitted.

b) Replace the fan overheat sensor [1] (TASK 72-21-12-000-801-G00 and TASK 72-21-12-400-801-G00).

(b) Sensor housing

1) Nick, scratches, and dents

a) Permitted up to a maximum depth of 0.005 in. (0.127 mm).

b) If it is necessary, replace the fan overheat sensor [1] (TASK 72-21-12-000-801-G00 and TASK 72-21-12-400-801-G00).

2) Cracks

a) Not permitted.

b) Replace the fan overheat sensor [1] (TASK 72-21-12-000-801-G00 and TASK 72-21-12-400-801-G00).

(c) Sensor mounting flange

1) Nick, scratches, and dents

a) Permitted up to a maximum depth of 0.005 in. (0.127 mm).

b) If it is necessary, replace the fan overheat sensor [1] (TASK 72-21-12-000-801-G00 and TASK 72-21-12-400-801-G00).

(d) Resistance fan detector

1) All finding (bent, etc)

a) Not permitted.

b) Replace the fan overheat sensor [1] (TASK 72-21-12-000-801-G00 and TASK 72-21-12-400-801-G00).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-12-860-003-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-21-12-860-004-G00

(2) Do these steps to remove the DO-NOT-OPERATE tags from the applicable ENGINE START switch and FUEL CONTROL switch:

(a) On the pilot's overhead panel, P5, remove the DO-NOT-OPERATE tag from the applicable ENGINE START switch.

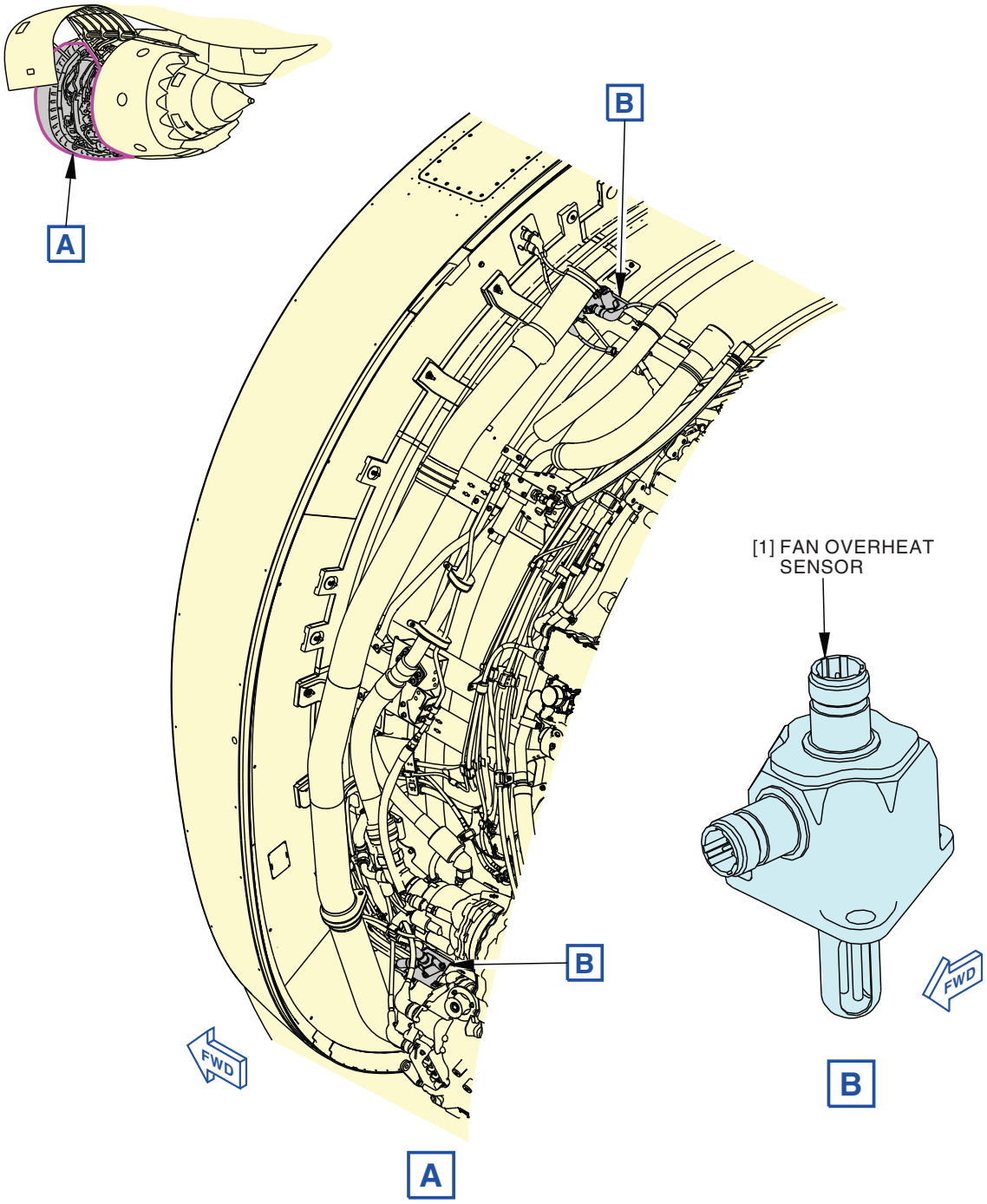
1) Put the applicable ENGINE START switch to the NORM (push in) position.

(b) On the pilot's aisle control stand, P8, remove the DO-NOT-OPERATE tag from the applicable FUEL CONTROL switch.

1) Put the applicable FUEL CONTROL switch to the CUTOFF position.

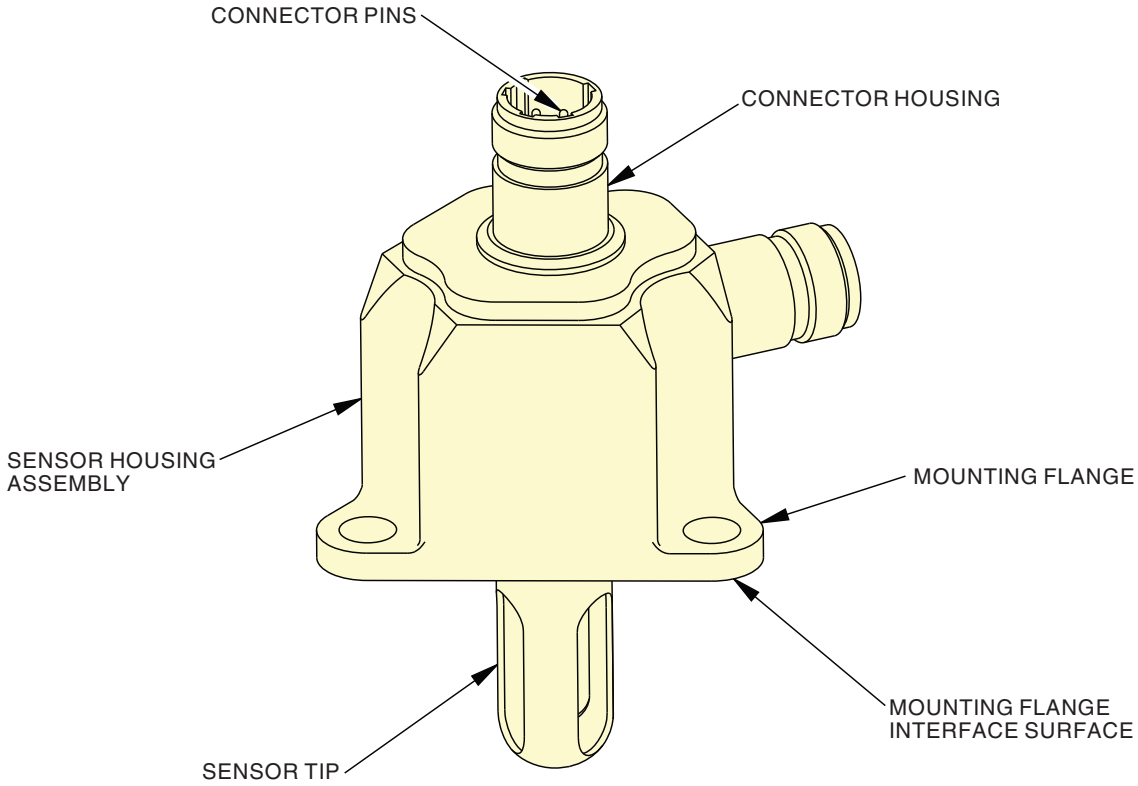
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Overheat Sensor Inspection
Figure 601/72-21-12-990-801-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Fan Overheat Sensor Inspection
Figure 601/72-21-12-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN BOOSTER ASSEMBLY - INSPECTION/CHECK

1. General

A. This procedure has these tasks:

(1) Flowpath Fairing Inspection

(2) Shroud Segment Inspection

(3) Fan and Booster Module inspection.

TASK 72-22-00-200-806-G00

2. Flowpath Fairing Inspection

A. General

(1) This task gives the instructions to examine the flowpath fairing.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-22-05-200-801-G00	Flowpath Fairing Inspection (P/B 601)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

F. Prepare for the Inspection

SUBTASK 72-22-00-600-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-22-00-860-003-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-22-00-860-004-G00

(3) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

G. Flowpath Fairing Inspection

SUBTASK 72-22-00-210-004-G00

(1) Examine the exposed areas of the flowpath fairing and flowpath fairing seal, do this task: Flowpath Fairing Inspection, TASK 72-22-05-200-801-G00.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-22-00-860-005-G00

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

(a) On the applicable engine, close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 72-22-00-860-006-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-22-00-860-007-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

TASK 72-22-00-200-807-G00

3. Shroud Segment Inspection

A. General

(1) This task gives the instructions to examine the shroud segment.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-22-05-200-802-G00	Shroud Segment Inspection (P/B 601)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

F. Prepare for the Inspection

SUBTASK 72-22-00-860-008-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-22-00-860-009-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-22-00-860-010-G00

(3) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

G. Shroud Segment Inspection

SUBTASK 72-22-00-220-021-G00

(1) Examine the exposed areas of the shroud segment, do this task: Shroud Segment Inspection, TASK 72-22-05-200-802-G00.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-22-00-860-011-G00

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

(a) On the applicable engine, close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 72-22-00-860-012-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-22-00-860-013-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-22-00-200-808-G00

4. Fan and Booster Module Inspection

(Figure 601, Figure 602, Figure 603, Figure 604, Figure 605, Figure 606, Figure 607, Figure 608, Figure 609, Figure 610, and Figure 611)

A. General

(1) This task gives the instructions for the inspection of the fan and booster module.

B. References

Reference	Title
72-22-00-300-801-G00	Platform Front Shroud Blend Repair (P/B 801)
72-22-00-900-801-G00	Self-Locking Shank Nuts on the Platform Front Shroud Replacement (P/B 801)
72-22-01-300-802-G00	Front Spinner Cone Attachment Holes Blending and Touch-Up of Sulfuric Anodizing (P/B 801)
72-22-40-300-801-G00	Low Pressure Compressor (LPC) Shaft Paint Touch-up Repair (P/B 801)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Inspection

SUBTASK 72-22-00-860-014-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-22-00-860-015-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

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F. Fan and Booster Module Inspection

SUBTASK 72-22-00-220-025-G00

(1) Examine the front spinner [1] as follows:

(a) Examine all areas for these conditions (Figure 601):

1) Cracks

a) Not permitted.

2) Dents

a) Permitted up to 0.019 in. (0.5 mm) in depth and 0.19 in. (5 mm) in length.

b) Continue in service limit - 100 cycles or 150 hours are permitted for a deviation of up to 0.055 in. (1.4 mm) in depth and 0.55 in. (14 mm) in length.

3) Nicks

a) Permitted up to 0.019 in. (0.5 mm) in depth and 0.19 in. (5 mm) in length.

b) Continue in service limit - 100 cycles or 150 hours are permitted for a deviation of up to 0.055 in. (1.4 mm) in depth and 0.55 in. (14 mm) in length.

4) Scratches

a) Permitted up to 0.019 in. (0.5 mm) in depth and 0.19 in. (5 mm) in length.

b) Continue in service limit - 100 cycles or 150 hours are permitted for a deviation of up to 0.055 in. (1.4 mm) in depth and 0.55 in. (14 mm) in length.

5) Missing black paint

a) Permitted if no corrosion mark is visible (white mushroom pattern).

b) Continue in service limit - 100 cycles or 150 hours are permitted if corrosion mark is visible.

6) Missing primer application

a) Permitted if no corrosion mark is visible (white mushroom pattern).

b) Continue in service limit - 100 cycles or 150 hours are permitted if corrosion mark is visible.

7) Missing sulfuric anodizing

a) Permitted if no corrosion mark is visible (white mushroom pattern).

b) Continue in service limit: 100 cycles or 150 hours are permitted if corrosion mark is visible.

(b) Examine Area A for these conditions (View B-B, Figure 601):

1) Dents

a) Permitted up to 0.011 in. (0.3 mm) in depth and 0.19 in. (5 mm) in length.

b) Continue in service limit - 100 cycles or 150 hours are permitted for a deviation of up to 0.03 in. (1 mm) in depth and 0.39 in. (10 mm) in length.

2) Nicks

a) Permitted up to 0.011 in. (0.3 mm) in depth and 0.19 in. (5 mm) in length.

b) Continue in service limit - 100 cycles or 150 hours are permitted for a deviation of up to 0.03 in. (1 mm) in depth and 0.39 in. (10 mm) in length.

3) Scratches

a) Permitted up to 0.011 in. (0.3 mm) in depth and 0.19 in. (5 mm) in length.

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b) Continue in service limit - 100 cycles or 150 hours are permitted for a deviation of up to 0.03 in. (1 mm) in depth and 0.39 in. (10 mm) in length.

(c) Examine Area B for erosion of parent material (Figure 601).

1) Permitted if DIM. A is between 0.1693 in. (4.30 mm) and 0.1850 in. (4.70 mm) and DIM. U is a minimum of 0.4382 in. (11.13 mm).

(d) Examine Area C except Area G for missing coating inner surfaces (Figure 601).

NOTE: Area C includes the complete internal and aft flange surfaces of the spinner. Area G includes the surfaces of the spinner attachment pads, and corresponding lateral surfaces of the aft flange.

1) Not permitted.

(e) Examine Area D for missing white spiral shaped mark on the front spinner [1] (View A, Figure 601).

1) Permitted.

(f) Examine Area E for these conditions (View C, Figure 601):

1) Dents

a) Permitted up to 0.0118 in. (0.30 mm) in depth, any length.

b) Continue in service limit: 100 cycles or 150 hours are permitted for a deviation of up to 0.039 in. (1.0 mm) in depth, any length.

c) Maximum repairable limit: In the limits of the applicable repair.

d) Repair method: Front Spinner Cone Attachment Holes Blending and Touch-Up of Sulfuric Anodizing, TASK 72-22-01-300-802-G00.

2) Nicks

a) Permitted up to 0.0118 in. (0.30 mm) in depth, any length.

b) Continue in service limit: 100 cycles or 150 hours are permitted for a deviation of up to 0.039 in. (1.0 mm) in depth, any length.

c) Maximum repairable limit: In the limits of the applicable repair.

d) Repair method: Front Spinner Cone Attachment Holes Blending and Touch-Up of Sulfuric Anodizing, TASK 72-22-01-300-802-G00.

3) Scratches

a) Permitted up to 0.0118 in. (0.30 mm) in depth, any length.

b) Continue in service limit: 100 cycles or 150 hours are permitted for a deviation of up to 0.039 in. (1.0 mm) in depth, any length.

c) Maximum repairable limit: In the limits of the applicable repair.

d) Repair method: Front Spinner Cone Attachment Holes Blending and Touch-Up of Sulfuric Anodizing, TASK 72-22-01-300-802-G00.

(g) Examine area F for these conditions (View D, Figure 601):

1) Dents

a) Permitted up to 0.0177 in. (0.45 mm) in depth, any length.

b) Maximum repairable limit: In the limits of the applicable repair.

c) Repair method: Front Spinner Cone Attachment Holes Blending and Touch-Up of Sulfuric Anodizing, TASK 72-22-01-300-802-G00.

2) Nicks

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Permitted up to 0.0177 in. (0.45 mm) in depth, any length.

b) Maximum repairable limit: In the limits of the applicable repair.

c) Repair method: Front Spinner Cone Attachment Holes Blending and Touch-Up of Sulfuric Anodizing, TASK 72-22-01-300-802-G00.

3) Scratches

a) Permitted up to 0.0177 in. (0.45 mm) in depth, any length.

b) Maximum repairable limit: In the limits of the applicable repair.

c) Repair method: Front Spinner Cone Attachment Holes Blending and Touch-Up of Sulfuric Anodizing, TASK 72-22-01-300-802-G00.

(h) Examine Area G for these conditions (Figure 601):

1) Missing primer application

a) Permitted if no corrosion mark is visible (white mushroom pattern).

b) Continue in service limit - 100 cycles or 150 hours are permitted if corrosion mark is visible.

2) Missing sulfuric anodizing

a) Permitted if no corrosion mark is visible (white mushroom pattern).

b) Continue in service limit: 100 cycles or 150 hours are permitted if corrosion mark is visible.

(i) Examine DIA B for contact marks of bolt threads (View B-B, Figure 601):

1) Permitted up to 0.003 in. (0.1 mm).

SUBTASK 72-22-00-220-026-G00

(2) Examine the front spinner cup [2] as follows (Figure 602):

(a) Examine all areas of the front spinner cup [2] for these conditions:

1) Dents

a) Permitted up to 0.003 in. (0.1 mm) in depth and 0.003 in. (0.1 mm) in length.

2) Nicks

a) Permitted up to 0.003 in. (0.1 mm) in depth and 0.003 in. (0.1 mm) in length.

3) Scratches

a) Permitted up to 0.003 in. (0.1 mm) in depth and 0.003 in. (0.1 mm) in length.

(b) Examine DIM. A of the front spinner cup [2] hole for peening.

1) Permitted if DIM. A is between 0.2717 in. (6.90 mm) and 0.2803 in. (7.12 mm).

(c) Examine DIM. B of the front spinner cup [2] for peening, fretting, and wear.

1) Permitted if DIM. B is between 0.0426 in. (1.08 mm) and 0.0519 in. (1.32 mm).

SUBTASK 72-22-00-220-027-G00

(3) Examine the platform front shroud [3] as follows:

(a) Examine all upper side areas except Area B and Area C for these conditions (Views C and B-B, Figure 603):

1) Dents

a) Permitted up to 0.003 in. (0.1 mm) in depth and 0.11 in. (3 mm) in length.

2) Nicks

a) Permitted up to 0.003 in. (0.1 mm) in depth and 0.11 in. (3 mm) in length.

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3) Scratches
   a) Permitted up to 0.003 in. (0.1 mm) in depth and 0.11 in. (3 mm) in length.
4) Cracks
   a) Not permitted.
5) Indication by rotating crimped self-locking nuts
   a) Not permitted

(b) Examine all lower side areas except Areas A1 and A2 for these conditions (View C-C, Figure 603):
1) Dents
   a) Permitted up to 0.003 in. (0.1 mm) in depth and 0.07 in. (2 mm) in length.
2) Nicks
   a) Permitted up to 0.003 in. (0.1 mm) in depth and0.07 in. (2 mm) in length.
3) Scratches
   a) Permitted up to 0.003 in. (0.1 mm) in depth and0.07 in. (2 mm) in length.
4) Cracks
   a) Not permitted.

(c) Examine the type A crimped self-locking nuts for these conditions (View B-B, Figure 603):
1) Loss of self-locking quality
   a) Maximum Serviceable Limit: Not serviceable if running torque is less than 3.5 in-lb (0.4 N·m).
   b) Maximum Repairable Limit: Any amount.
   c) Repair Method: Refer to Self-Locking Shank Nuts on the Platform Front Shroud Replacement, TASK 72-22-00-900-801-G00.
2) Looseness
   a) Maximum Serviceable Limit: Not serviceable.
   b) Maximum Repairable Limit: Any amount.
   c) Repair Method: Refer to Self-Locking Shank Nuts on the Platform Front Shroud Replacement, TASK 72-22-00-900-801-G00.
3) Missing
   a) Maximum Serviceable Limit: Not serviceable.
   b) Maximum Repairable Limit: Any amount.
   c) Repair Method: Refer to Self-Locking Shank Nuts on the Platform Front Shroud Replacement, TASK 72-22-00-900-801-G00.
4) Cracks
   a) Maximum Serviceable Limit: Not serviceable.
   b) Maximum Repairable Limit: Any amount.
   c) Repair Method: Refer to Self-Locking Shank Nuts on the Platform Front Shroud Replacement, TASK 72-22-00-900-801-G00.
5) Internal thread bent, torn, worn, missing fully or not
   a) Maximum Serviceable Limit: Not serviceable.
   b) Maximum Repairable Limit: Any amount.

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c) Repair Method: Refer to Self-Locking Shank Nuts on the Platform Front Shroud Replacement, TASK 72-22-00-900-801-G00.

(d) Examine DIM. L for peening, fretting, and wear (View B-B, Figure 603).

1) Permitted if DIM. L is between 0.0749 in. (1.90 mm) and 0.0826 in. (2.10 mm).

(e) Examine DIM. N for peening, fretting, and wear (View B-B, Figure 603).

1) Permitted if DIM. N is between 0.1733 in. (4.40 mm) and 0.1811 in. (4.60 mm).

(f) Examine DIA C for peening, fretting, and wear (View B-B, Figure 603).

1) Permitted if DIA C is between 17.8000 in. (452.12 mm) and 17.8039 in. (452.22 mm).

(g) Examine DIM. M for peening, fretting, and wear (View B-B, Figure 603).

1) Permitted if DIM. M is between 0.1162 in. (2.95 mm) and 0.1200 in. (3.05 mm).

(h) Examine DIA B for peening, fretting, and wear (View B-B, Figure 603).

1) Permitted if DIA B is between 15.9433 in. (404.96 mm) and 15.9456 in. (405.02 mm).

(i) Examine Area A1 for nicks, dents, and scratches (Views C-C and F, Figure 603).

1) Permitted up to 0.0019 in. (0.05 mm) in depth and any length, or up to 0.0039 in. (0.10 mm) in depth and 0.0393 in. (1.00 mm) in length.

2) Continue in service limit - 2000 cycles are permitted with the criteria up to 0.0295 in. (0.75 mm) in depth, or 800 cycles are permitted with the criteria up to 0.00551 in. (0.14 mm) in depth.

(j) Examine Area A2 for nicks, dents, and scratches (Views C-C and F, Figure 603).

1) Permitted up to 0.0007 in. (0.02 mm) in depth and any length, or up to 0.0015 in. (0.04 mm) in depth and 0.0157 in. (0.40 mm) in length.

2) Continue in service limit - 2000 cycles are permitted with the criteria up to 0.0098 in. (0.25 mm) in depth, or 800 cycles are permitted with the criteria up to 0.0157 in. (0.40 mm) in depth.

(k) Examine Area B for these conditions (View C, Figure 603):

1) Dents

a) Permitted up to 0.0031 in. (0.08 mm) in depth and 0.0031 in. (0.08 mm) in length.

b) Maximum Repairable Limit: Refer to the repair procedure.

c) Repair Method: Refer to Platform Front Shroud Blend Repair, TASK 72-22-00-300-801-G00.

2) Nicks

a) Permitted up to 0.0031 in. (0.08 mm) in depth and 0.0031 in. (0.08 mm) in length.

b) Maximum Repairable Limit: Refer to the repair procedure.

c) Repair Method: Refer to Platform Front Shroud Blend Repair, TASK 72-22-00-300-801-G00.

3) Scratches

a) Permitted up to 0.0031 in. (0.08 mm) in depth and 0.0031 in. (0.08 mm) in length.

b) Maximum Repairable Limit: Refer to the repair procedure.

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c) Repair Method: Refer to Platform Front Shroud Blend Repair, TASK 72-22-00-300-801-G00.

(l) Examine Area C for these conditions (View B-B, Figure 603):

1) Dents

a) Permitted up to 0.0023 in. (0.06 mm) in depth and 0.0027 in. (0.07 mm) in length.

2) Nicks

a) Permitted up to 0.0023 in. (0.06 mm) in depth and 0.0027 in. (0.07 mm) in length.

3) Scratches

a) Permitted up to 0.0031 in. (0.08 mm) in depth and 0.0031 in. (0.08 mm) in length.

(m) Examine Area D for stains (View C, Figure 603):

1) Stains with no depth are permitted.

(n) Examine Area P for these conditions (View B-B, Figure 603):

1) Dents

a) Permitted up to 0.0023 in. (0.06 mm) in depth and 0.0027 in. (0.07 mm) in length or 0.0007 in. (0.02 mm) in depth and all lengths.

2) Nicks

a) Permitted up to 0.0023 in. (0.06 mm) in depth and 0.0027 in. (0.07 mm) in length or 0.0007 in. (0.02 mm) in depth and all lengths.

3) Scratches

a) Permitted up to 0.0023 in. (0.06 mm) in depth and 0.0027 in. (0.07 mm) in length or 0.0007 in. (0.02 mm) in depth and all lengths.

4) Contact marks

a) Permitted up to 0.0027 in. (0.07 mm) in depth.

SUBTASK 72-22-00-220-028-G00

(4) Examine the fan disk [4] as follows:

(a) Examine the teeth top for these conditions:

1) Dents, Nicks, or Scratches

a) Any number up to 0.0019 in. (0.05 mm) in depth and any length.

b) A Continued-In-Service of 2000 cycles is permitted if indication is between 0.0019 in. (0.05 mm) and 0.0031 in. (0.08 mm) in depth.

<1> Within 2000 cycles after detection of the indication, the part must be removed.

2) Contact wear

a) Any number up to 0.0027 in. (0.07 mm) in depth and any length.

(b) Examine Area C for these conditions (View A-A, Figure 604):

1) Dents

a) Any number up to 0.0019 in. (0.05 mm) in depth and any length.

2) Nicks

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a) Any number up to 0.0019 in. (0.05 mm) in depth and any length.

3) Scratches

a) Any number up to 0.0019 in. (0.05 mm) in depth and any length.

(c) Examine Area E1 for these conditions (Views D, E, and I, Figure 604):

1) Dents

a) Any number up to 0.0039 in. (0.1 mm) in depth and any length.

2) Nicks

a) Any number up to 0.0039 in. (0.1 mm) in depth and any length.

3) Scratches

a) Any number up to 0.0039 in. (0.1 mm) in depth and any length.

(d) Examine Area E2 for these conditions (Views D, E, and I, Figure 604):

1) Dents

a) Any number up to 0.0019 in. (0.05 mm) in depth and any length.

2) Nicks

a) Any number up to 0.0019 in. (0.05 mm) in depth and any length.

3) Scratches

a) Any number up to 0.0019 in. (0.05 mm) in depth and any length.

(e) Examine Area F for these conditions (View A-A, Figure 604):

1) Dents

a) Permitted up to 0.00098 in. (0.025 mm) in depth and 0.0098 in. (0.25 mm) in length.

2) Nicks

a) Permitted up to 0.00098 in. (0.025 mm) in depth and 0.0098 in. (0.25 mm) in length.

3) Scratches

a) Permitted up to 0.00098 in. (0.025 mm) in depth and 0.0098 in. (0.25 mm) in length.

(f) Examine Area K for stains (View G, Figure 604).

1) Stains with no depth are permitted.

(g) Examine the shot-peened area N for these conditions (Views F, B-B, and C-C, Figure 604):

1) Dents

a) Permitted up to 0.0003 in. (0.01 mm) in depth and in all lengths.

2) Nicks

a) Permitted up to 0.0003 in. (0.01 mm) in depth and in all lengths.

3) Scratches

a) Permitted up to 0.0003 in. (0.01 mm) in depth and in all lengths.

(h) Examine the shot-peened area J for these conditions (View B-B, Figure 604):

1) Dents

a) Permitted up to 0.00098 in. (0.025 mm) in depth and any length.

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b) Continue in service limit - 2000 cycles are permitted with the criteria is between 0.00098 in. (0.025 mm) and 0.0023 in. (0.06 mm) in depth.

2) Nicks

a) Permitted up to 0.00098 in. (0.025 mm) in depth and any length.

b) Continue in service limit - 2000 cycles are permitted with the criteria is between 0.00098 in. (0.025 mm) and 0.0023 in. (0.06 mm) in depth.

3) Scratches

a) Permitted up to 0.00098 in. (0.025 mm) in depth and any length.

b) Continue in service limit - 2000 cycles are permitted with the criteria is between 0.00098 in. (0.025 mm) and 0.0023 in. (0.06 mm) in depth.

(i) Examine the shot-peened area L for these conditions (View B-B, Figure 604):

1) Dents

a) Permitted up to 0.0007 in. (0.02 mm) in depth and any length.

b) Continue in service limit - 2000 cycles are permitted with the criteria is between 0.0007 in. (0.02 mm) and 0.0023 in. (0.06 mm) in depth.

2) Nicks

a) Permitted up to 0.0007 in. (0.02 mm) in depth and any length.

b) Continue in service limit - 2000 cycles are permitted with the criteria is between 0.0007 in. (0.02 mm) and 0.0023 in. (0.06 mm) in depth.

3) Scratches

a) Permitted up to 0.0007 in. (0.02 mm) in depth and any lengths.

b) Continue in service limit - 2000 cycles are permitted with the criteria is between 0.0007 in. (0.02 mm) and 0.0023 in. (0.06 mm) in depth.

(j) Examine Area M for these conditions (Views B, E, and I, Figure 604):

1) Dents

a) Any number up to 0.0019 in. (0.05 mm) in depth and any length.

2) Nicks

a) Any number up to 0.0019 in. (0.05 mm) in depth and any length.

3) Scratches

a) Any number up to 0.0019 in. (0.05 mm) in depth and any length.

4) Contact marks

a) All contact marks with no depth are permitted.

(k) Examine Area Y for these conditions (View H, Figure 604):

1) Dents

a) Permitted up to 0.00098 in. (0.025 mm) in depth and 0.0098 in. (0.25 mm) in length.

2) Nicks

a) Permitted up to 0.00098 in. (0.025 mm) in depth and 0.0098 in. (0.25 mm) in length.

3) Scratches

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a) Permitted up to 0.00098 in. (0.025 mm) in depth and 0.0098 in. (0.25 mm) in length.

4) Fretting, wear, imprint mark.

a) Permitted up to 0.020 in. (0.51 mm) in depth.

SUBTASK 72-22-00-220-029-G00

(5) Examine the LPC shaft [5] as follows:

(a) Examine all areas that are not painted except Area K for these conditions (View C, Figure 605):

1) Dents

a) Permitted up to 0.003 in. (0.1 mm) in depth and 0.019 in. (0.5 mm) in length.

2) Nicks

a) Permitted up to 0.003 in. (0.1 mm) in depth and 0.019 in. (0.5 mm) in length.

3) Scratches

a) Permitted up to 0.003 in. (0.1 mm) in depth and 0.019 in. (0.5 mm) in length.

4) Corrosion or pitting

a) Not permitted.

5) Cracks

a) Not permitted.

6) Metal loss

a) Not permitted.

(b) Examine all painted areas for these conditions (Figure 605):

1) Nicks, dents, and scratches

a) Permitted if the parent material is not affected and if a minimum paint thickness of 0.00138 in. (0.035 mm) remains.

b) Maximum repairable limit: In the limits of the applicable repair. If necessary, do this task: Low Pressure Compressor (LPC) Shaft Paint Touch-up Repair, TASK 72-22-40-300-801-G00.

2) Corrosion or pitting

a) Not permitted.

3) Cracks

a) Not permitted.

4) Metal loss

a) Not permitted.

5) Missing paint except in Area P

a) Permitted if a minimum thickness of 0.00138 in. (0.035 mm) remains.

b) Maximum repairable limit: In the limits of the applicable repair. If necessary, do this task: Low Pressure Compressor (LPC) Shaft Paint Touch-up Repair, TASK 72-22-40-300-801-G00.

6) Peeling or Blistering

a) Not permitted.

(c) Examine Area K of the weep plug hole for these conditions (View C, Figure 605):

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1) Dents, nicks, and scratches
   a) Not permitted.
2) Fretting
   a) Not permitted.

(d) Examine the surface V of the forward anti-rotation lugs for fretting (View E, Figure 605).
1) Permitted up to 0.003 in. (0.1 mm) if DIM. M is between 0.0977 in. (2.48 mm) and 0.1082 in. (2.75 mm).

(e) Examine the surfaces W and X of the forward anti-rotation lugs for fretting (View E, Figure 605).
1) Permitted up to 0.003 in. (0.1 mm) if DIM. N is between 0.0792 in. (2.01 mm) and 0.0842 in. (2.14 mm).

(f) Examine the Area P of the forward anti-rotation lugs for missing paint (View D, Figure 605).
1) Permitted.

(g) Examine DIM. L as follows (View F, Figure 605):
1) Permitted if DIM. L is between 0.4075 in. (10.35 mm) and 0.4153 in. (10.55 mm).

SUBTASK 72-22-00-220-030-G00

(6) Examine the low pressure compressor cover [6] as follows:
(a) Examine all areas of the low pressure compressor cover [6] for these conditions (Figure 606):
1) Cracks
   a) Not permitted.
2) Dents
   a) Permitted up to 0.0098 in. (0.25 mm) in depth and 0.098 in. (2.5 mm) in length.
3) Nicks
   a) Permitted up to 0.0098 in. (0.25 mm) in depth and 0.098 in. (2.5 mm) in length.
4) Scratches
   a) Permitted up to 0.0098 in. (0.25 mm) in depth and 0.098 in. (2.5 mm) in length.

(b) Examine Area E of the seal groove for dents, nicks, and scratches (View B, Figure 606):
1) Not permitted.

(c) Examine DIA A for these conditions (View A-A, Figure 606):
1) Fretting or wear
   a) Permitted if DIA A is between 5.6000 in. (142.24 mm) and 5.6015 in. (142.28 mm).
2) Pitting
   a) Not permitted.

(d) Examine Area B of the anti-rotation tang/lug for these conditions (View A, Figure 606):
1) Peening, fretting, wear
   a) Permitted if DIM. C is between 0.3780 in. (9.60 mm) and 0.38582 in. (9.8 mm).
2) Breaking or distortion
   a) Not permitted.

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(e) Examine DIM. D of the anti-rotation tang/lug for fretting or wear (View C, Figure 606).

1) Permitted if DIM. B is between 0.1536 in. (3.90 mm) and 0.1614 in. (4.10 mm).

(f) Examine Area F of the thread for nicks, dents, scratches and indications that are not part of the thread (View D, Figure 606).

1) Not permitted.

SUBTASK 72-22-00-220-031-G00

(7) Examine the LPC slotted nut [7] as follows:

(a) Examine all areas of the LPC slotted nut [7] for these conditions (Figure 607):

1) Cracks

a) Not Permitted

2) Missing silver plating

a) Permitted.

(b) Examine all areas (except threads and slots) for these conditions (Figure 607):

1) Dents

a) Permitted up to 0.003 in. (0.1 mm) in depth

2) Nicks

a) Permitted up to 0.003 in. (0.1 mm) in depth.

3) Scratches

a) Permitted up to 0.003 in. (0.1 mm) in depth.

(c) Examine Area C of the slots for these conditions (Figure 607):

1) Dents

a) Permitted up to 0.003 in. (0.1 mm) in depth

2) Nicks

a) Permitted up to 0.003 in. (0.1 mm) in depth.

3) Scratches

a) Permitted up to 0.003 in. (0.1 mm) in depth.

SUBTASK 72-22-00-220-032-G00

(8) Examine the platform front shroud balance weight [8] as follows:

(a) Examine all areas of the balance weight [8] for these conditions (Figure 608):

1) Nicks, dents, and scratches

a) Permitted up to 0.0007 in. (0.02 mm) in depth.

2) Fretting or wear

a) Permitted up to 0.0007 in. (0.02 mm) in depth.

3) Contact marks

a) Permitted up to 0.0007 in. (0.02 mm) in depth.

4) Glazing

a) Permitted.

SUBTASK 72-22-00-220-033-G00

(9) Examine the silicone O-ring [10] between platform front shroud [3] and the fan platform [9] for these conditions (Figure 609):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Cracks

1) Permitted.

(b) Wear

1) Permitted.

(c) Missing adhesion

1) Permitted.

(d) Cut(s) on O-ring

1) Not permitted.

NOTE: If you find cut(s) on the O-ring, remove or replace it.

(e) Distortion

1) Permitted.

(f) Torn

1) Permitted.

(g) Nicks, dents, and scratches

1) Permitted.

(h) Partially exposed (after assembly)

1) Permitted, if the exposed area of the seal does not go beyond the flowpath profile of the platform front shroud.

(i) Missing silicone O-ring

1) Permitted.

SUBTASK 72-22-00-220-034-G00

(10) Examine surface A of the booster spool [11] for these conditions (Figure 610):

NOTE: With the fan disk [4] attached to surface A of the booster spool [11], not all of surface A is visible for inspection. This limits the inspection of surface A to the visible 18 areas corresponding to the 18 fan disks [4] counter dovetails.

(a) Cracks

1) Not permitted.

(b) Nicks, dents, and scratches

1) Permitted up to 0.00098 in. (0.025 mm) in depth.

(c) Fretting, wear

1) Permitted up to 0.00098 in. (0.025 mm) in depth.

SUBTASK 72-22-00-210-012-G00

(11) Examine the visible surface of the flowpath fairing and the shroud segment (Figure 611):

(a) Surface oxidation on screw head.

1) Permitted.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-22-00-860-016-G00

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-22-00-860-017-G00

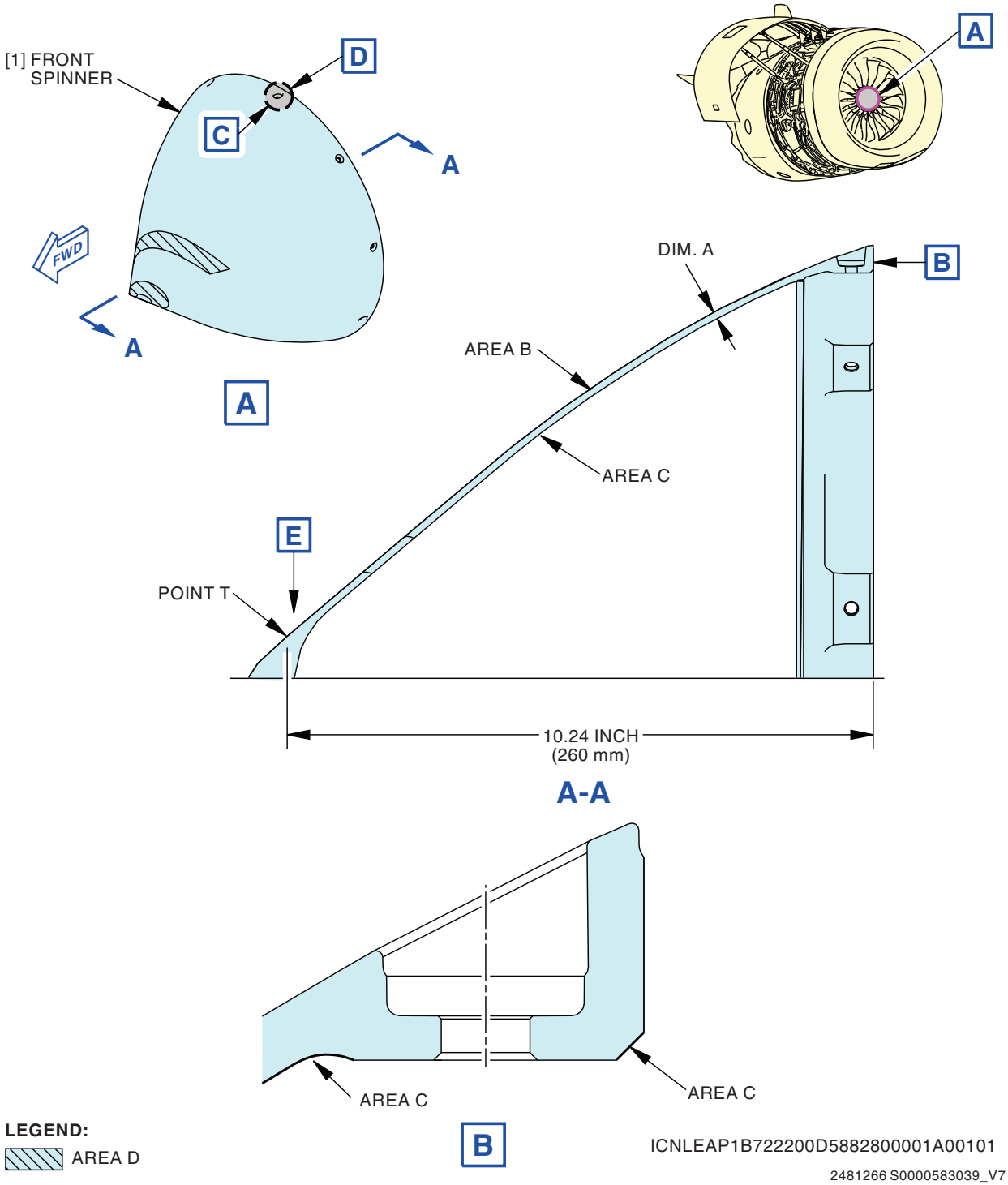
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

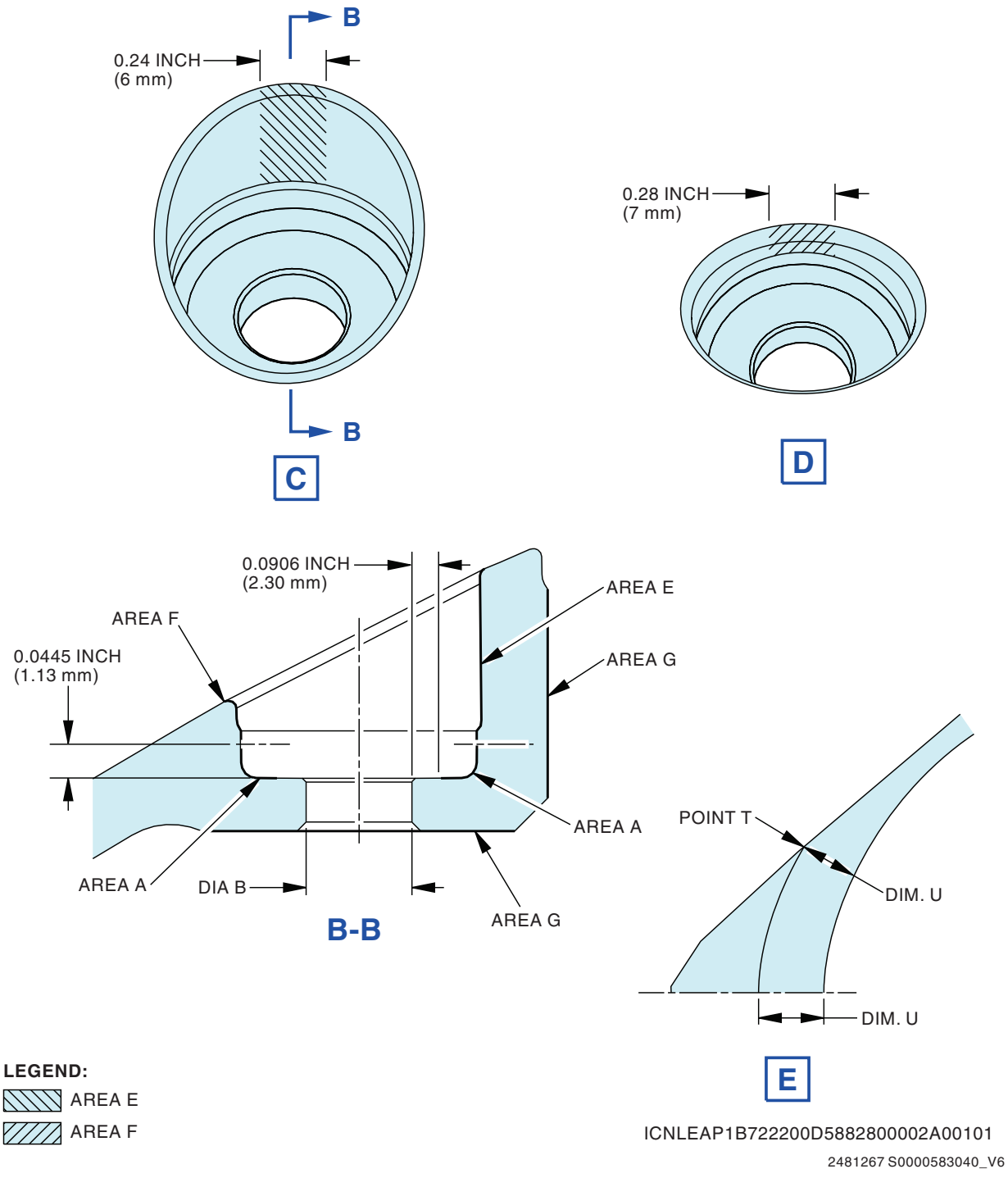
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



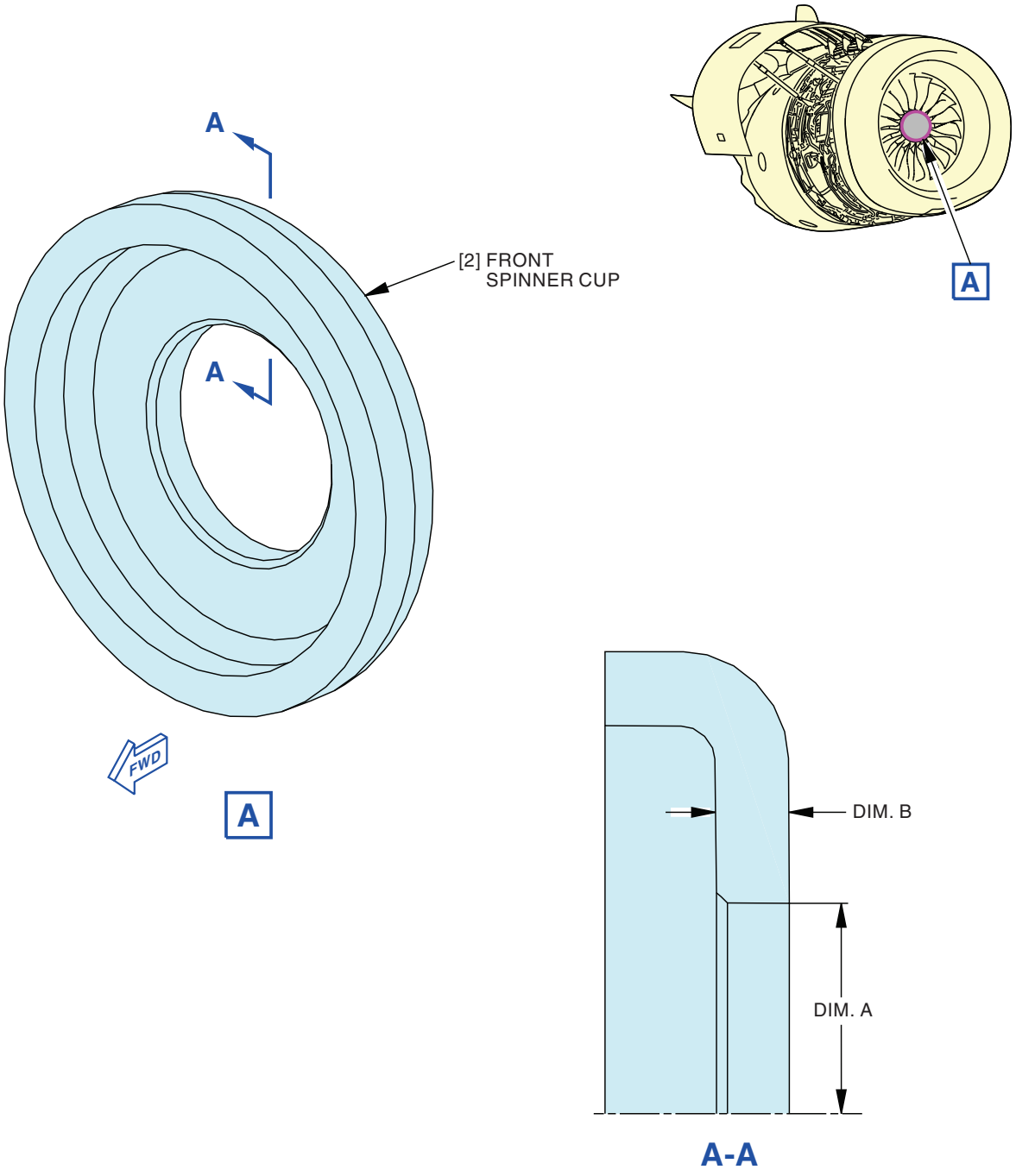
Front Spinner Inspection
Figure 601/72-22-00-990-811-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Front Spinner Inspection
Figure 601/72-22-00-990-811-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

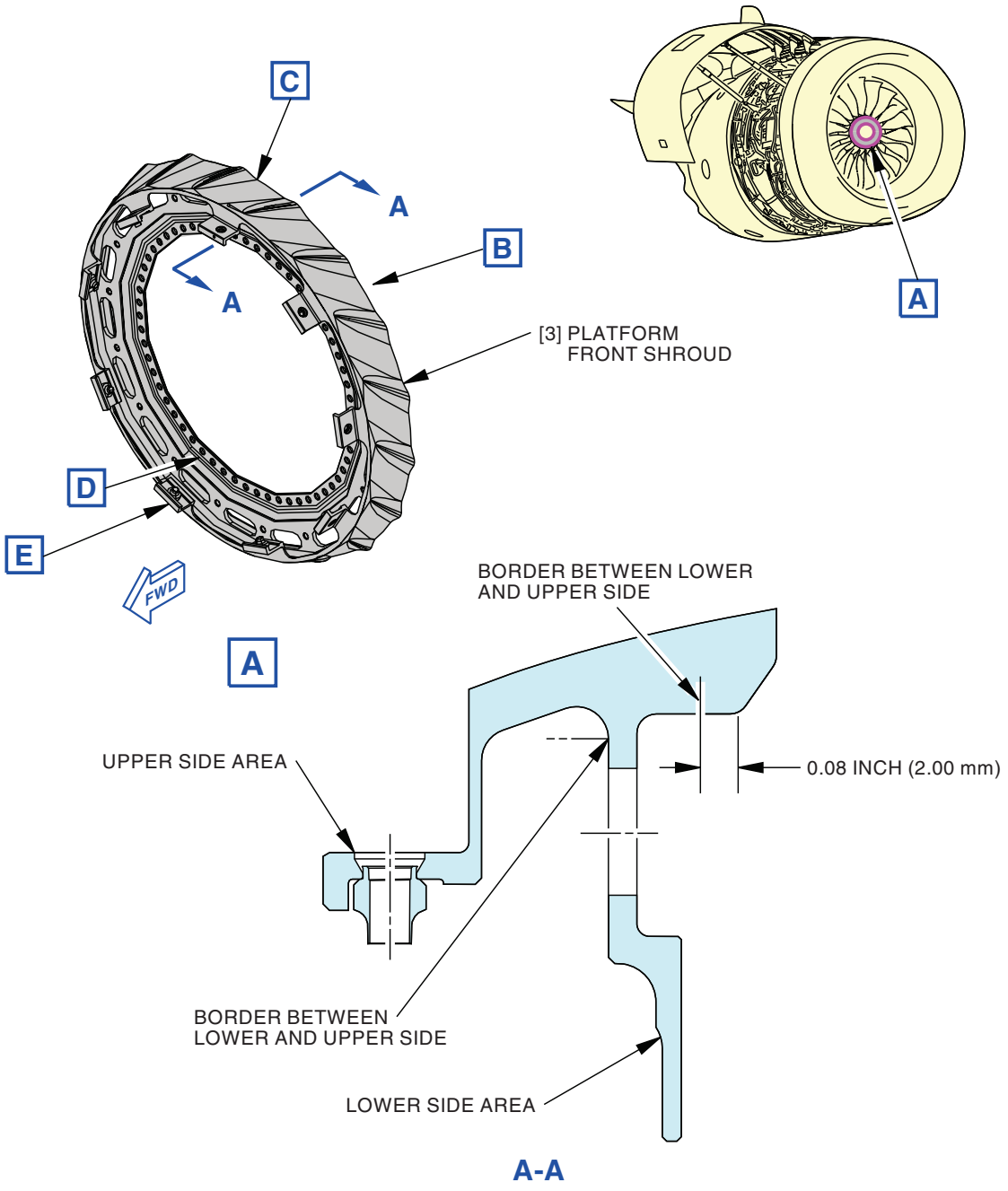


ICNLEAP1B722200D5882800003A00101

2481268 S0000583041_V4

Front Spinner Cup Inspection
Figure 602/72-22-00-990-814-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

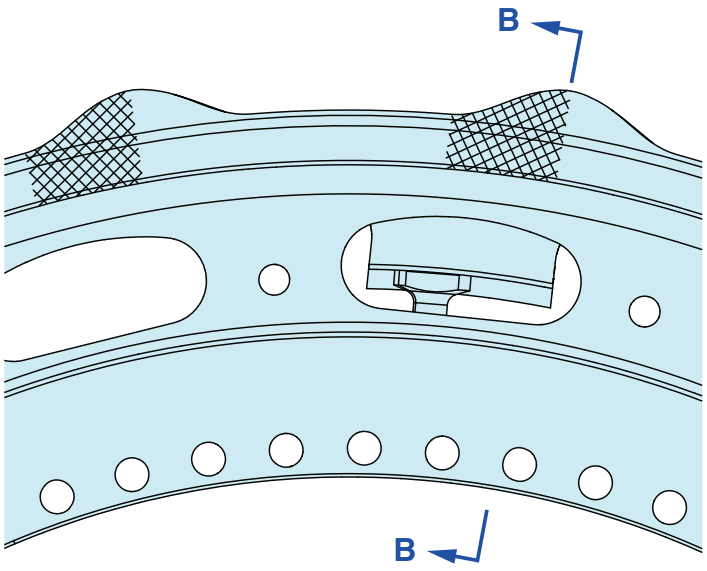


ICNLEAP1B722200D5882800004A00101

2481271 S0000583043_V6

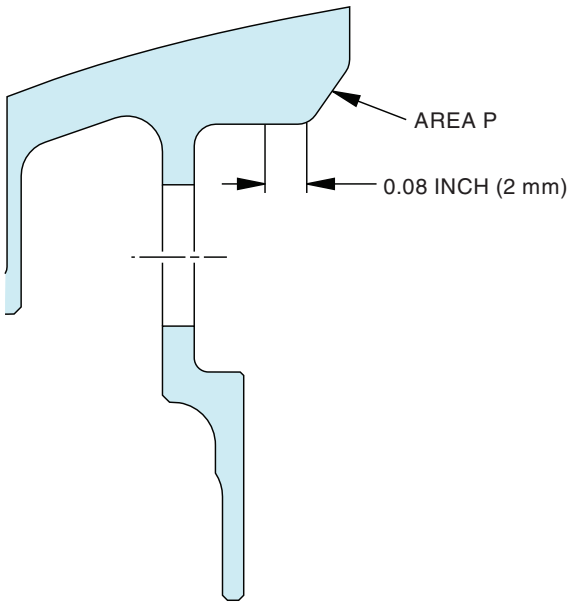
Platform Front Shroud Inspection
Figure 603/72-22-00-990-815-G00 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(VIEW IN THE FORWARD DIRECTION)

B



B-B

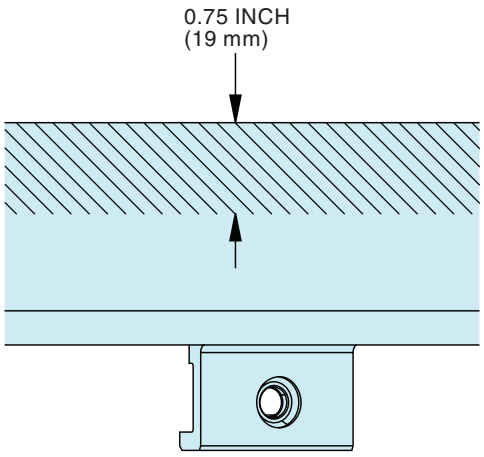
LEGEND:
AREA P

ICNLEAP1B722200D5882800005A00101

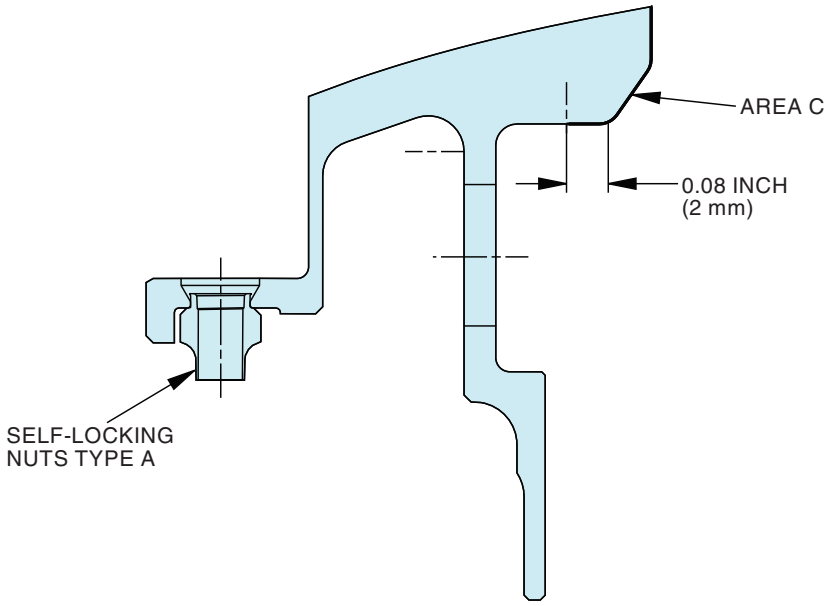
2567394 S0000614655_V3

Platform Front Shroud Inspection
Figure 603/72-22-00-990-815-G00 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



C



B-B

LEGEND:

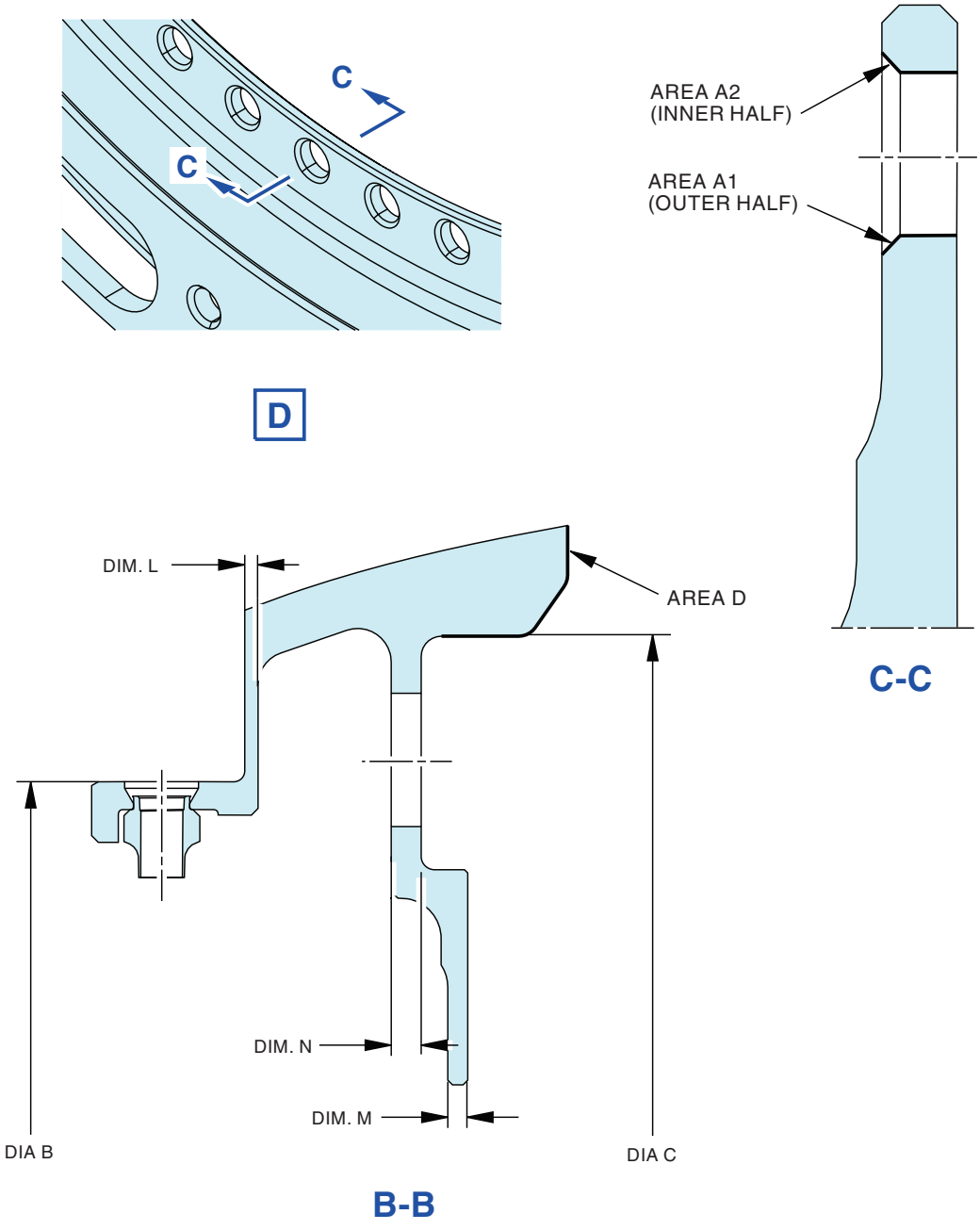
AREA B

ICNLEAP1B722200D5882800006A00101

2545719 S0000602144_V3

Platform Front Shroud Inspection
Figure 603/72-22-00-990-815-G00 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

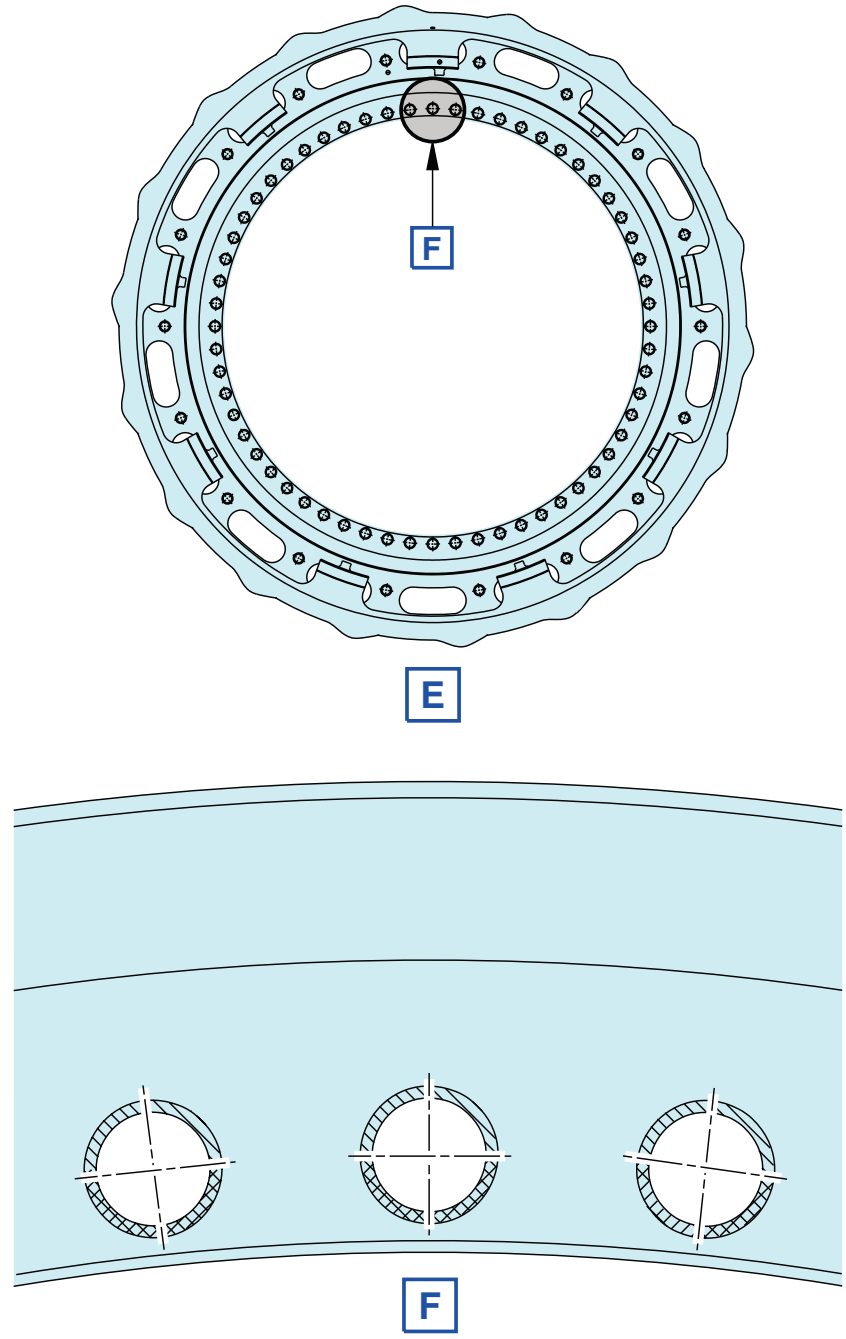


CNLEAP1B722200D5882800007A00101

2545726 S0000602147_V4

Platform Front Shroud Inspection
Figure 603/72-22-00-990-815-G00 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

AREA A1 (OUTER HALF)

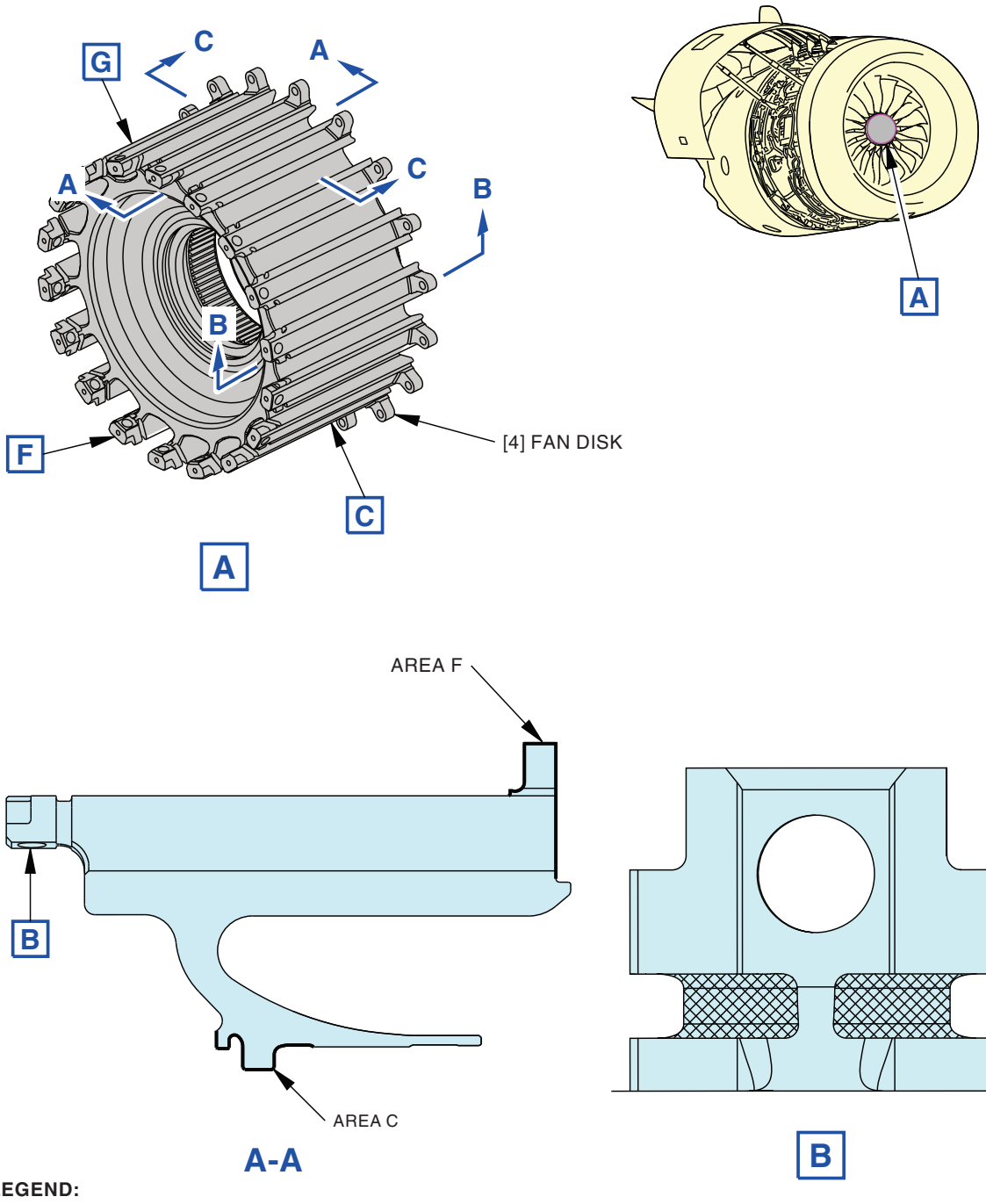
AREA A2 (INNER HALF)

ICNLEAP1B722200D5882800008A00101

2796430 S0000638088_V3

Platform Front Shroud Inspection
Figure 603/72-22-00-990-815-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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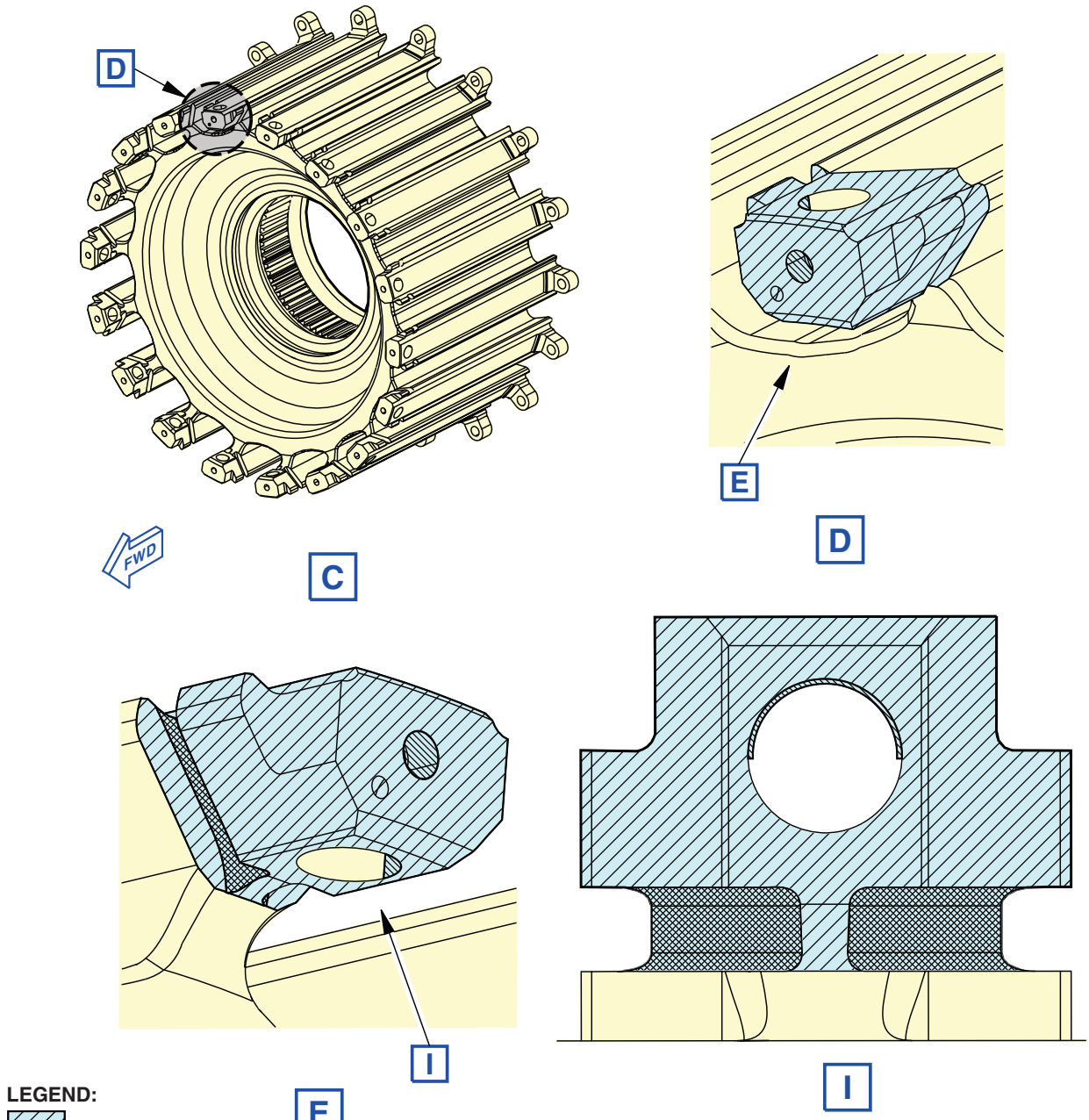
2545740 S0000602148_V7

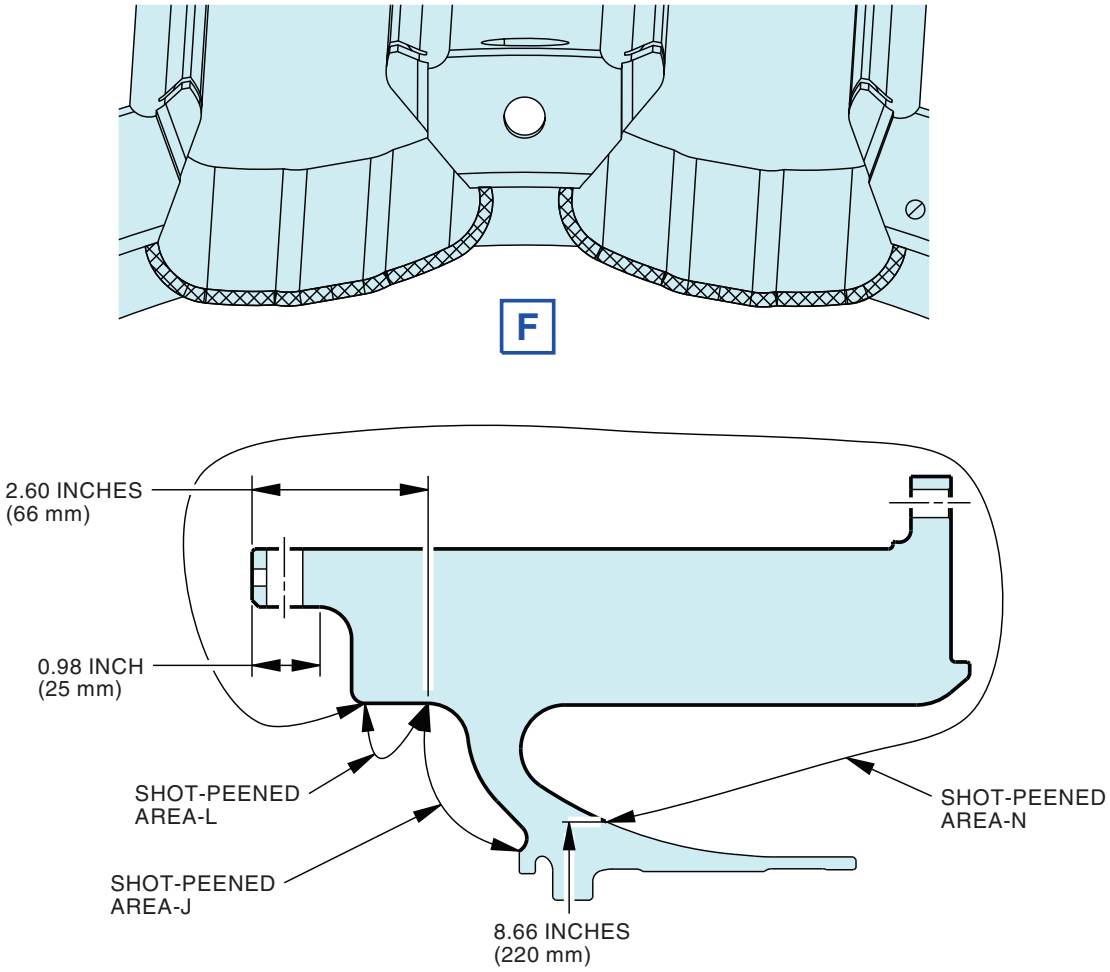
Fan Disk Inspection
Figure 604/72-22-00-990-816-G00 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



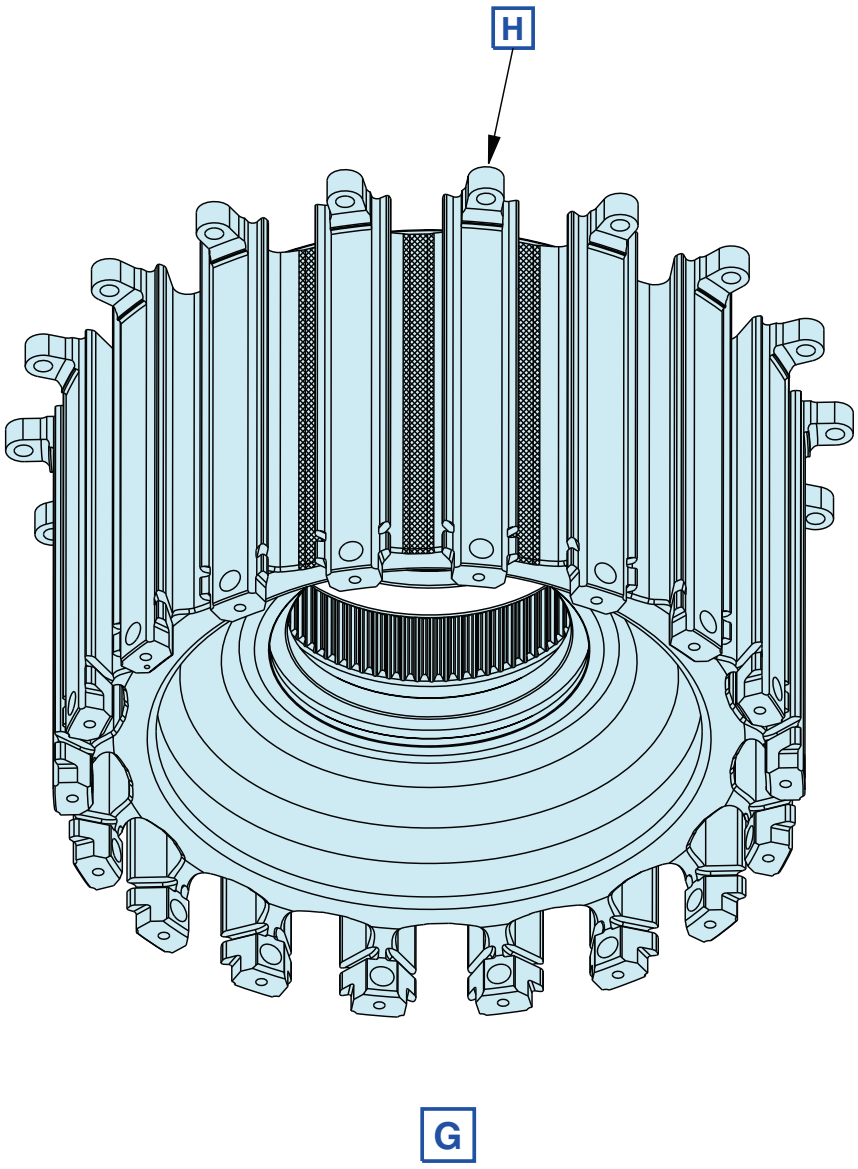
Fan Disk Inspection
Figure 604/72-22-00-990-816-G00 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Disk Inspection
Figure 604/72-22-00-990-816-G00 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

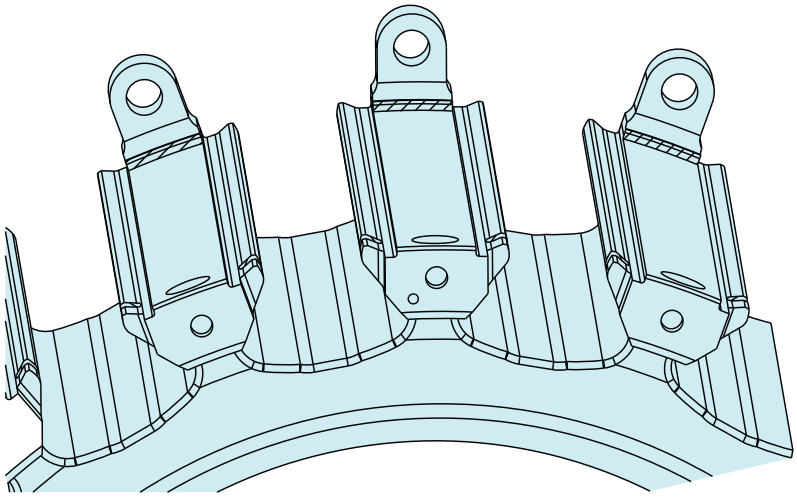
AREA K

ICNLEAP1B722200D58828000012A00101

2551524 S0000607480_V3

Fan Disk Inspection
Figure 604/72-22-00-990-816-G00 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



H

LEGEND:

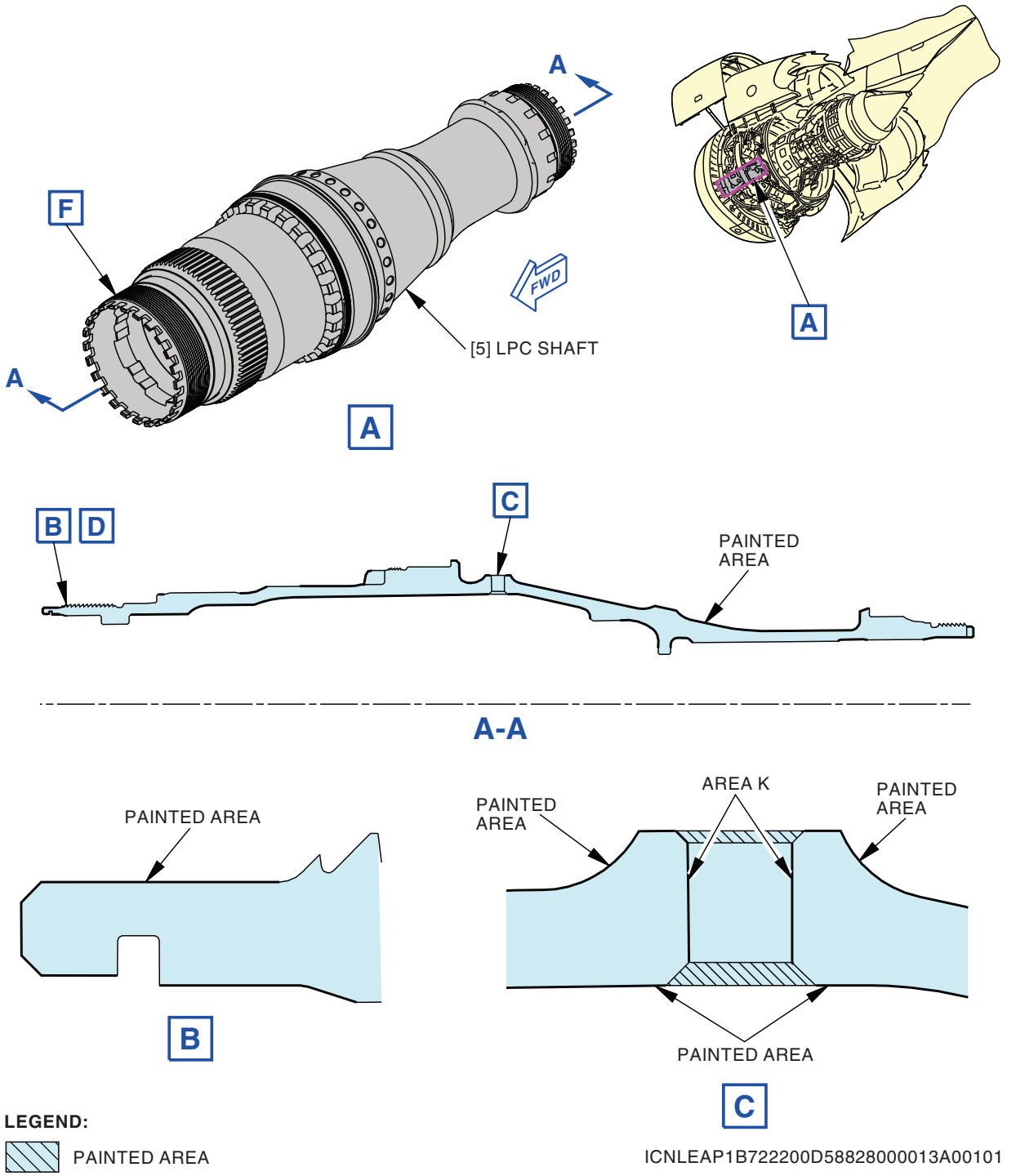
AREA Y

ICNLEAP1B722200D5882800021A00101

3001683 S0000777718_V1

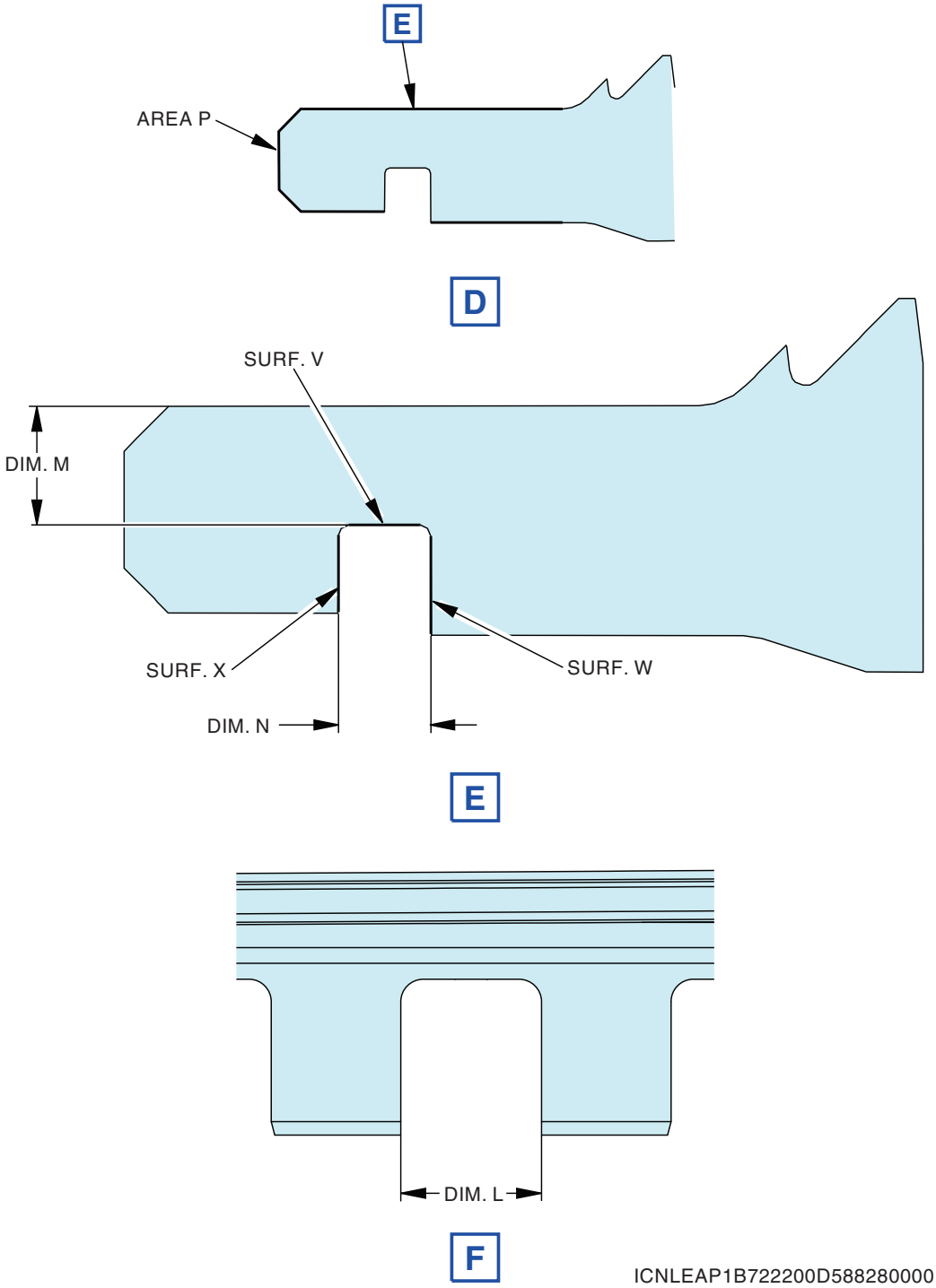
Fan Disk Inspection
Figure 604/72-22-00-990-816-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



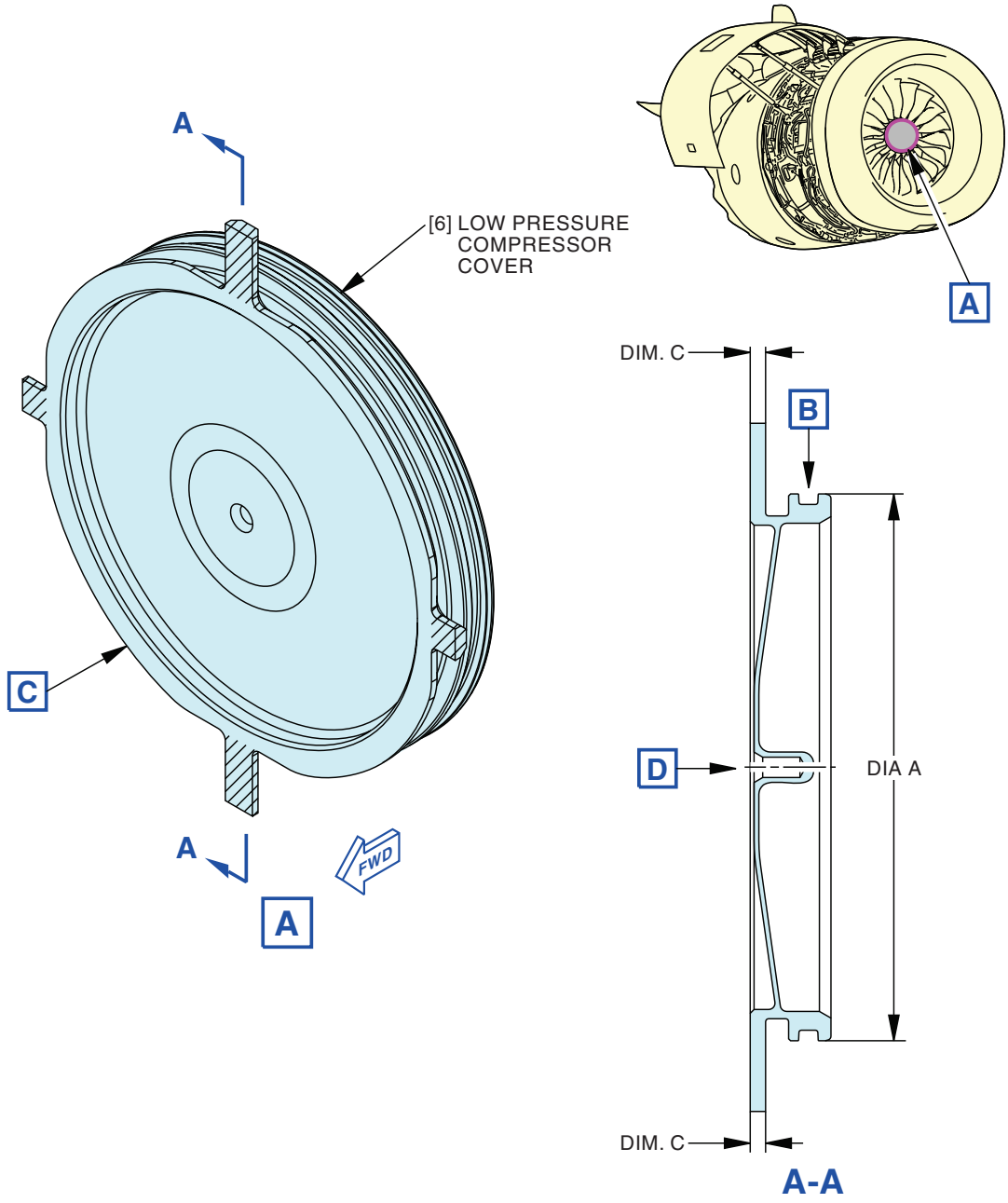
LPC Shaft Inspection
Figure 605/72-22-00-990-817-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LPC Shaft Inspection
Figure 605/72-22-00-990-817-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

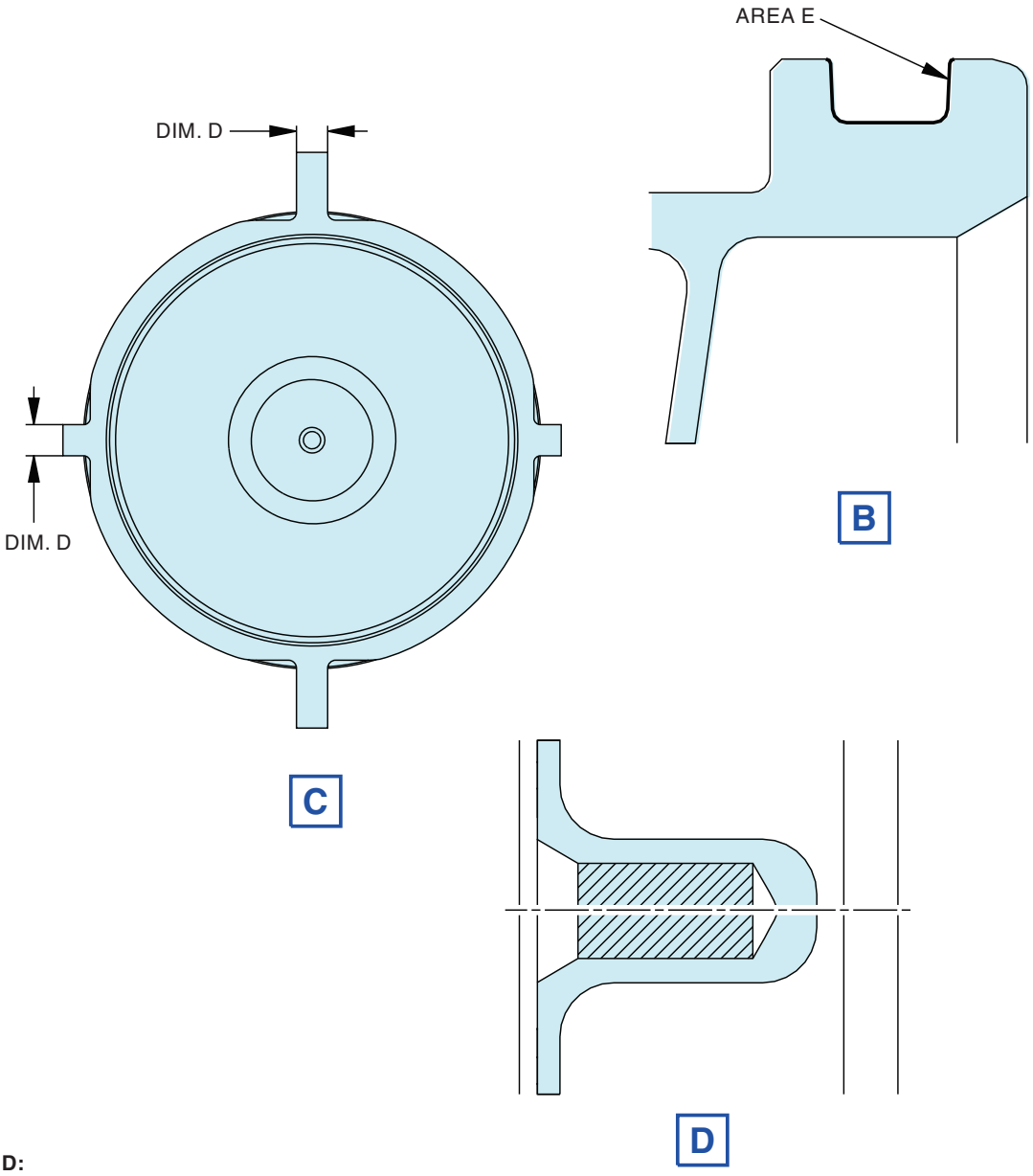
AREA B

ICNLEAP1B722200D588280000015A00101

2545893 S0000602152_V4

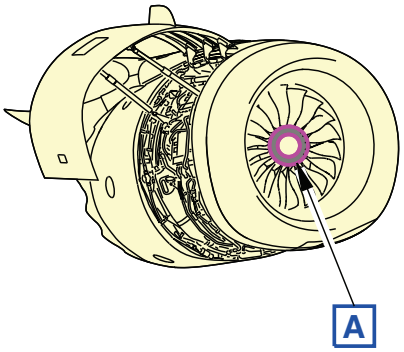
Low Pressure Compressor Cover Inspection
Figure 606/72-22-00-990-818-G00 (Sheet 1 of 2)

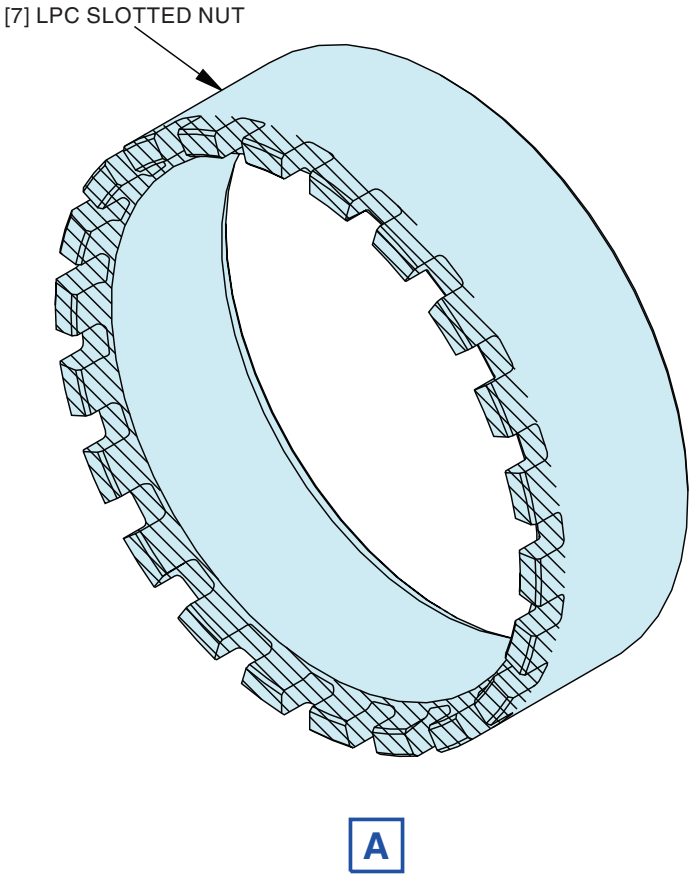
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Low Pressure Compressor Cover Inspection
Figure 606/72-22-00-990-818-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





LEGEND:

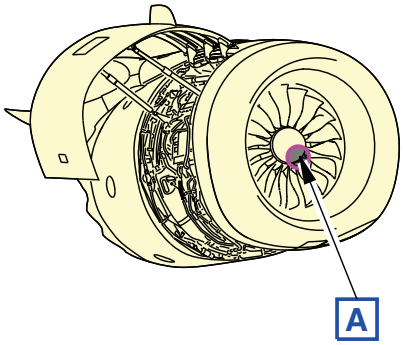
AREA C

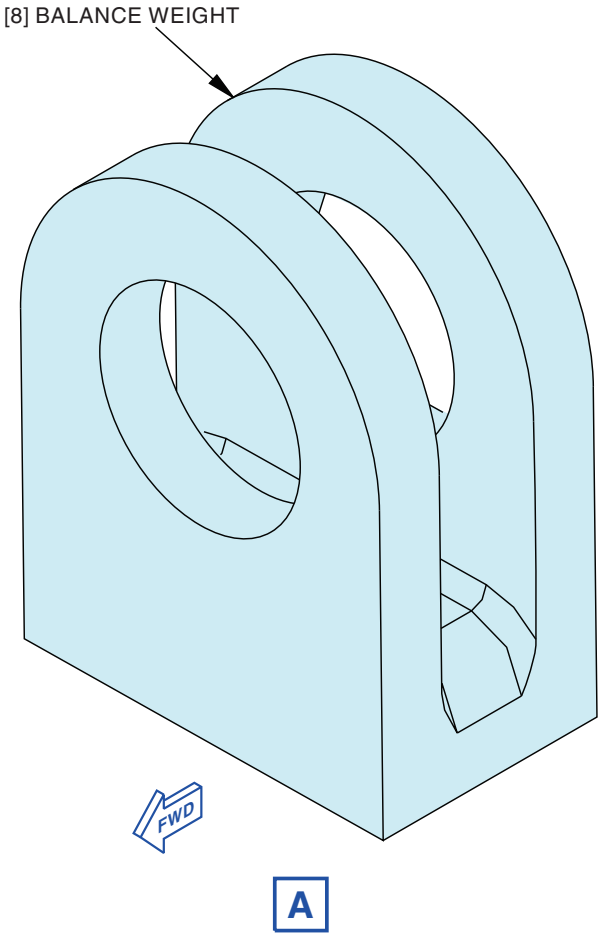
ICNLEAP1B722200D58828000017A00101

2545909 S0000602154_V2

LPC Slotted Nut Inspection
Figure 607/72-22-00-990-819-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



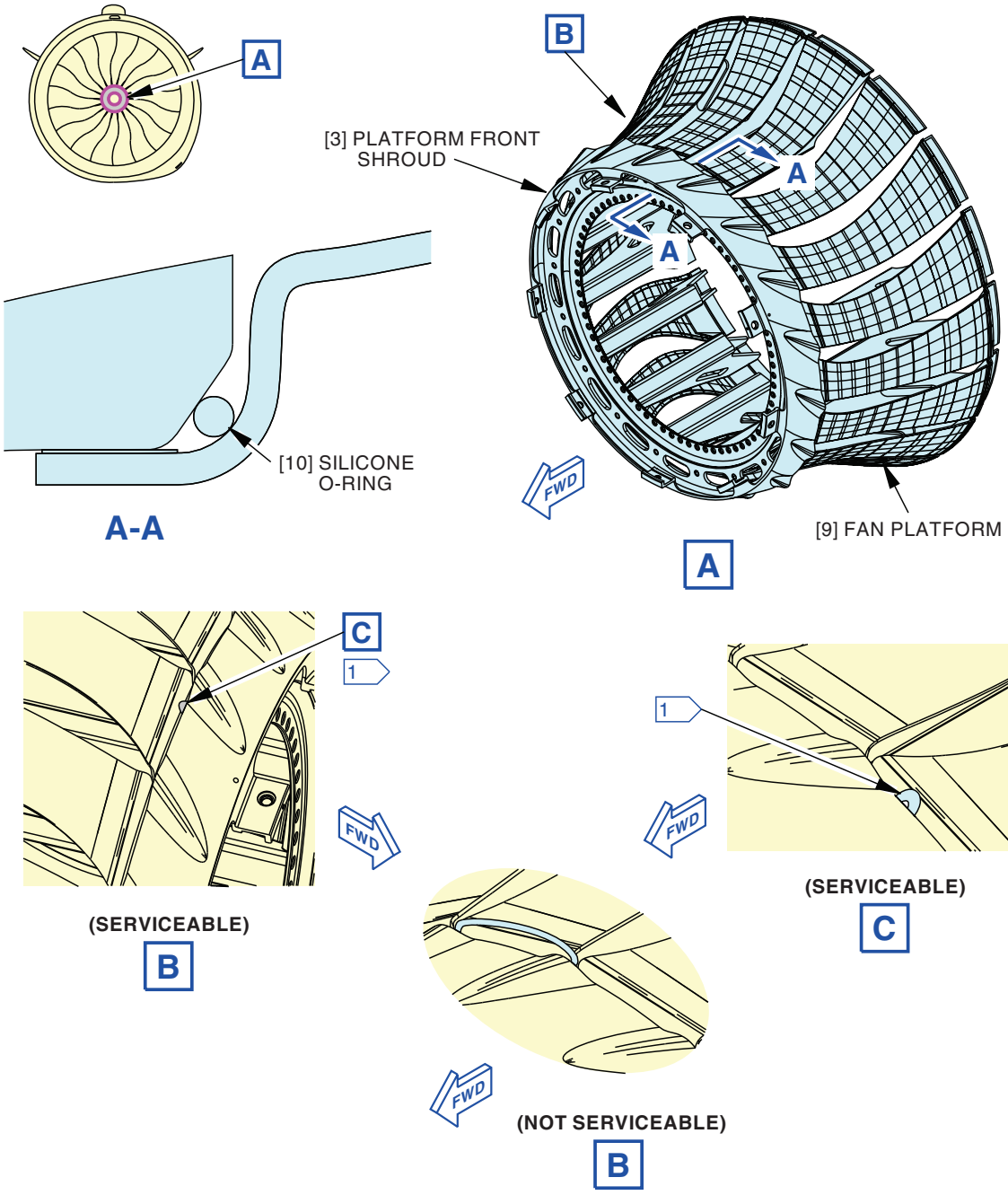


ICNLEAP1B722200D58828000018A00101

2545917 S0000602155_V2

Balance Weight Inspection
Figure 608/72-22-00-990-820-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 PARTIALLY EXPOSED O-RING IS PERMITTED IF THE AREA THAT SHOWS DO NOT GO BEYOND THE FLOWPATH PROFILE.

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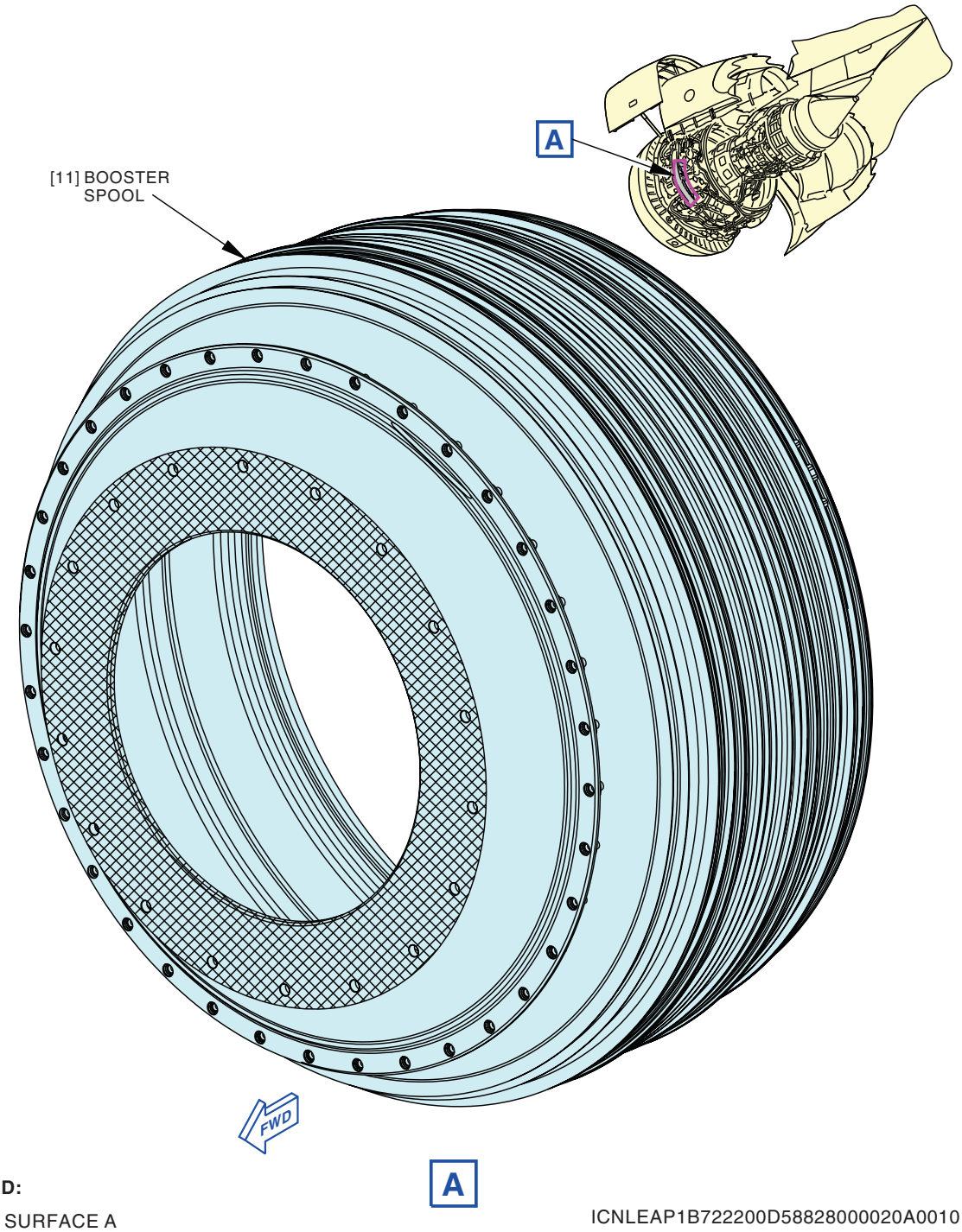
2574409 S0000617764_V3

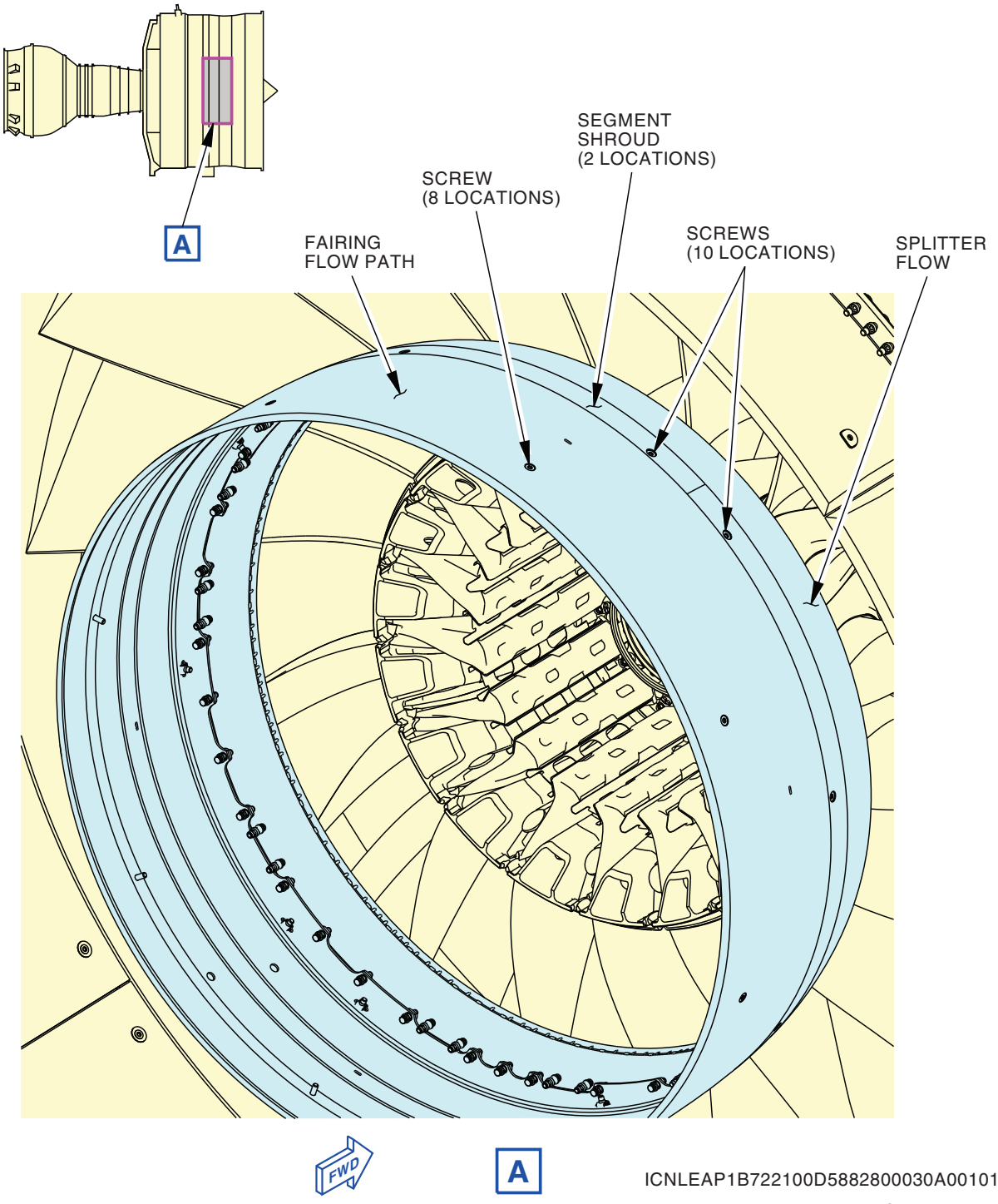
Silicone O-ring Inspection
Figure 609/72-22-00-990-821-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Booster Spool Inspection
Figure 610/72-22-00-990-822-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flowpath Fairing and Shroud Segment Surfaces Inspection
Figure 611/72-22-00-990-823-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN BOOSTER ASSEMBLY - REPAIRS

1. General

A. This procedure has these tasks:

(1) A blend repair of the platform front shroud

(2) A replacement of the self-locking shank nuts on the platform front shroud.

TASK 72-22-00-300-801-G00

2. Platform Front Shroud Blend Repair

(Figure 801)

A. General

(1) This task gives the instructions to do the blend repair of the platform front shroud.

B. References

Reference	Title
70-11-06-230-802-G00	Fluorescent Penetrant Inspection (P/B 201)
70-11-06-230-803-G00	Fluorescent Penetrant Inspection (General) (P/B 201)
70-21-23-100-801-G00	Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701)
70-43-24-100-802-G00	Swab Etching Procedure and Cleaning (P/B 201)
70-43-42-300-802-G00	Blending Procedures - Hand Blending (P/B 201)
70-43-42-300-803-G00	Blending Procedures - Power Blending (P/B 201)
72-22-01-000-801-G00	Front Spinner Cone Removal (P/B 401)
72-22-01-400-801-G00	Front Spinner Cone Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
B01058 [CP1039]	Solvent - Acetone, Reagent Grade	
G51412 [CP2721]	Cloth - Abrasive, Silicone Carbide - CS 333X	
G51789	Stone - Abrasive, Grade 150 or 180	

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Repair

SUBTASK 72-22-00-000-001-G00

(1) Remove the front spinner cone, do this task: Front Spinner Cone Removal, TASK 72-22-01-000-801-G00.

F. Platform Front Shroud Blend Repair

SUBTASK 72-22-00-220-035-G00

WARNING

REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

(1) Check the dimensional information as follows (Figure 801):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the minimum distance between two blend repairs is more than 1.97 in. (50 mm).

(b) Make sure that the number of blend repairs is less than or equal to 10.

(c) Do not blend on the bosses.

(d) Do not blend on DIM. X (View C, Figure 801).

NOTE: DIM. X is a strip of 0.08 in. (2 mm) from the upstream edge and of 0.08 in. (2 mm) from the downstream edge.

SUBTASK 72-22-00-870-001-G00

(2) Do the blending to remove the local indication (preferred method) as follows (Figure 801) (TASK 70-43-42-300-802-G00):

CAUTION

BLENDED AREAS MUST MAKE A SMOOTH CONTOUR INTO THE ADJACENT AREAS TO KEEP CORRECT AIRFLOW. IF YOU DO NOT OBEY, DAMAGE TO PARTS CAN OCCUR.

(a) Remove only a minimum quantity of metal in addition to the depth of the indication.

1) Make sure that the blended area transits smoothly into the adjacent area.

(b) Remove the defects with a fine file or abrasive stone, G51789 (TASK 70-43-42-300-802-G00).

1) Make sure that the DIM. L is equal to 2 x DIM. W (View B, Figure 801).

NOTE: DIM. W is equal to 0.47 in. (12 mm).

2) Make sure that DIM. D is less or equal to 0.0098 in. (0.25 mm) (View B-B, Figure 801).

3) Make sure that radius R is more than 0.79 in. (20 mm) (View B-B, Figure 801).

SUBTASK 72-22-00-870-002-G00

(3) Do the blending to remove the local indication (alternative method) as follows (Figure 801) (TASK 70-43-42-300-803-G00):

CAUTION

BLENDED AREAS MUST MAKE A SMOOTH CONTOUR INTO THE ADJACENT AREAS TO KEEP CORRECT AIRFLOW. IF YOU DO NOT OBEY, DAMAGE TO PARTS CAN OCCUR.

(a) Remove only a minimum quantity of metal in addition to the depth of the indication.

1) Make sure that the blended area transits smoothly into the adjacent area.

(b) If it is necessary, use the low-speed power tool, equipped with a grinding tool to shorten the time required to remove metal (TASK 70-43-42-300-803-G00).

(c) Do a hand finishing.

1) Make sure that the DIM. L is equal to 2 x DIM. W (View C, Figure 801).

NOTE: DIM. W is equal to 0.47 in. (12 mm).

2) Make sure that DIM. D is less or equal to 0.0098 in. (0.25 mm) (View B-B, Figure 801).

3) Make sure that radius R is more than 0.79 in. (20 mm) (View B-B, Figure 801).

SUBTASK 72-22-00-100-001-G00

(4) Polish with strips of abrasive cloth, G51412 [CP2721], to get a clean and smooth surface finish of 126 microinches (3.2 micrometers) (TASK 70-43-42-300-802-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that there are no sharp angles.

SUBTASK 72-22-00-220-036-G00

(5) Do a check of the surface finish of the part.

(a) Make sure that the surface finish (roughness) is not more than 126 microinches (3.2 micrometers) in the repair area.

(b) If the surface finish (roughness) is more than 126 microinches (3.2 micrometers), polish again the blend area.

SUBTASK 72-22-00-100-002-G00

(6) Clean the blended areas with a clean cloth soaked in acetone solvent, B01058 [CP1039], or alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

SUBTASK 72-22-00-100-003-G00

(7) Locally etch all the blended areas (TASK 70-43-24-100-802-G00).

SUBTASK 72-22-00-230-001-G00

(8) Do an Fluorescent Penetrant Inspection (FPI) of the repaired areas (TASK 70-11-06-230-803-G00 and TASK 70-11-06-230-802-G00).

(a) Do an FPI sensitivity level 3.

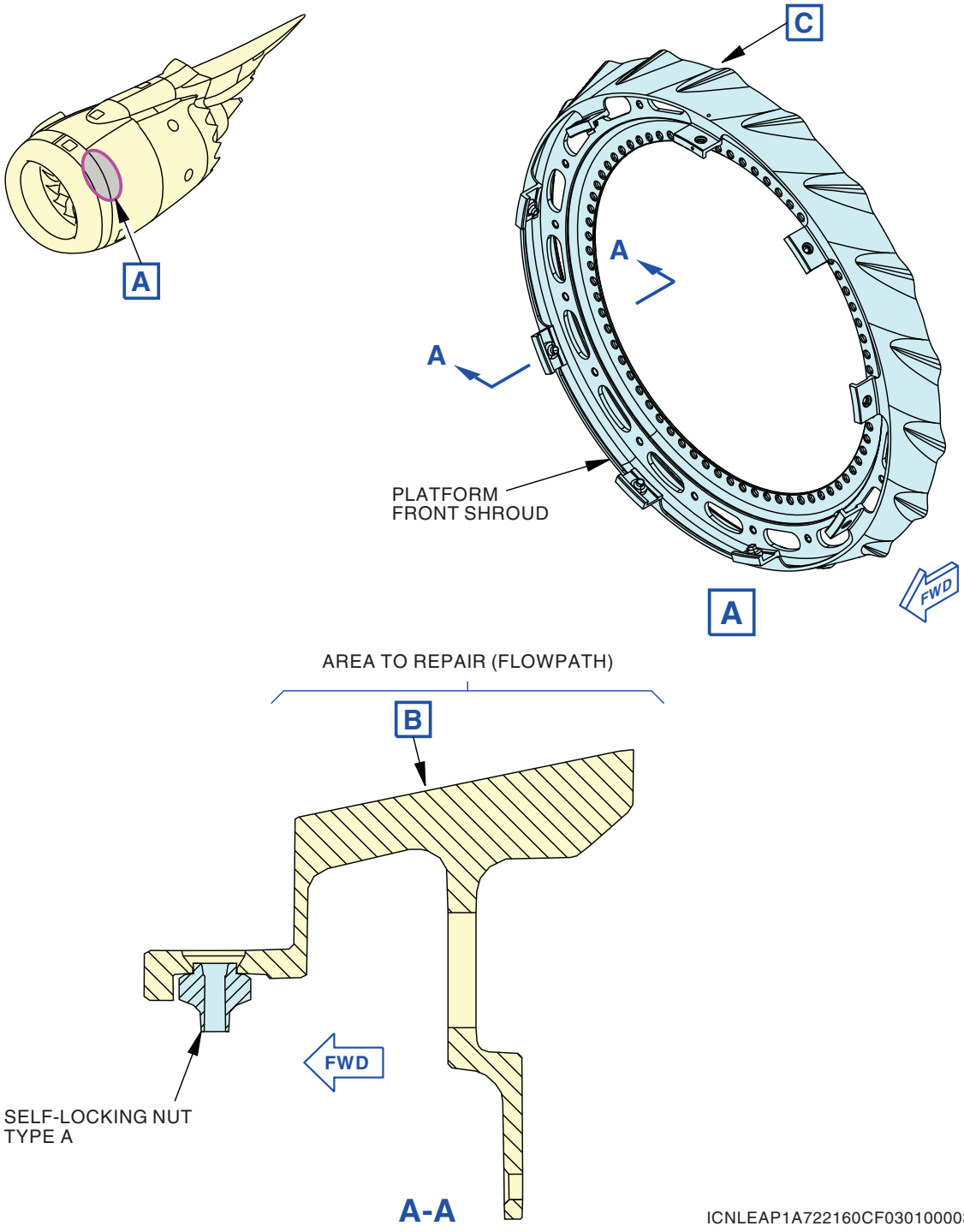
G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-22-00-400-001-G00

(1) Install the front spinner cone, do this task: Front Spinner Cone Installation, TASK 72-22-01-400-801-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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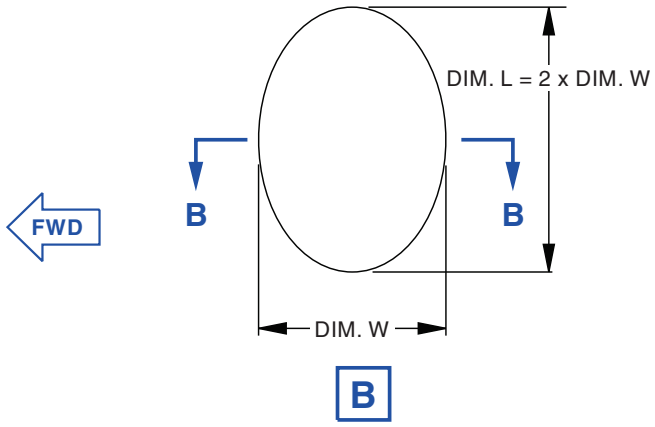
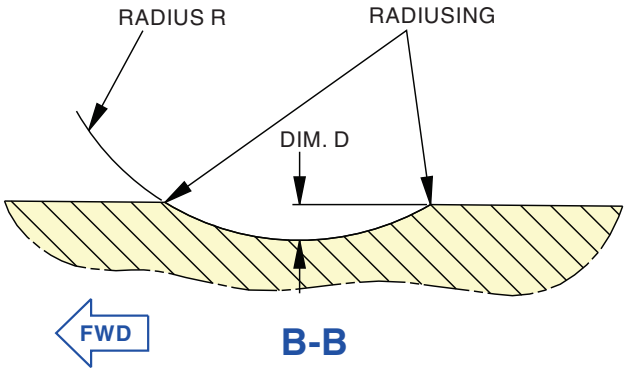
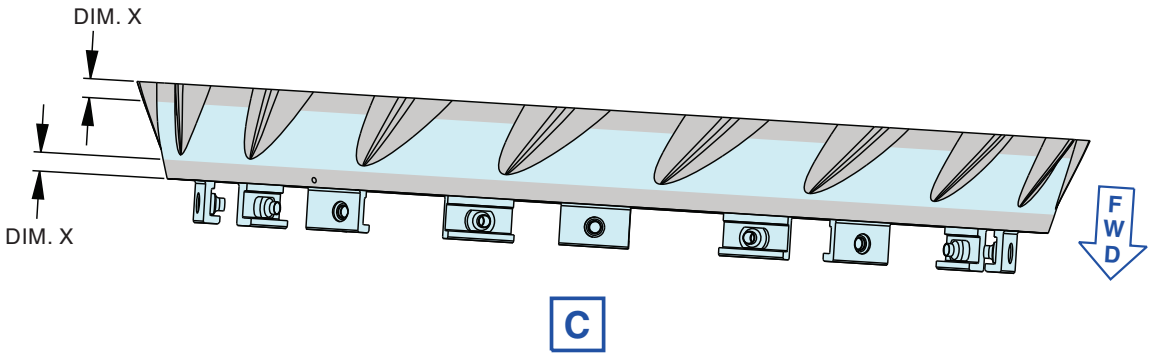
2999065 S0000775531_V1

Platform Front Shroud Blend Repair
Figure 801/72-22-00-990-812-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details







NOTE:

AREAS WHERE BLENDING IS FORBIDDEN

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Platform Front Shroud Blend Repair
Figure 801/72-22-00-990-812-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-22-00-900-801-G00

3. Self-Locking Shank Nuts on the Platform Front Shroud Replacement

(Figure 802)

A. General

(1) This task gives the instructions to replace the self-locking shank nuts on the platform front shroud.

B. References

Reference	Title
70-11-06-230-802-G00	Fluorescent Penetrant Inspection (P/B 201)
70-11-06-230-803-G00	Fluorescent Penetrant Inspection (General) (P/B 201)
70-21-23-100-801-G00	Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701)
72-22-01-000-801-G00	Front Spinner Cone Removal (P/B 401)
72-22-01-400-801-G00	Front Spinner Cone Installation (P/B 401)

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Prepare for the Replacement

SUBTASK 72-22-00-000-002-G00

(1) Remove the front spinner cone, do this task: Front Spinner Cone Removal, TASK 72-22-01-000-801-G00.

E. Self-Locking Shank Nuts on the Platform Front Shroud Replacement

SUBTASK 72-22-00-020-003-G00

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

CAUTION BE CAREFUL WHEN YOU REMOVE THE SELF-LOCKING SHANK NUT FROM THE PLATFORM FRONT SHROUD. IF YOU DO NOT OBEY, DAMAGE TO THE PLATFORM FRONT SHROUD CAN OCCUR.

(1) Remove the self-locking shank nut [1] to repair (View B, Figure 802).

(a) Make sure that the platform front shroud is not deformed.

SUBTASK 72-22-00-100-004-G00

(2) Do the crimping of the new self-locking shank nut [1], do these steps:

(a) Clean the hole and the adjacent areas (View B, Figure 802) (TASK 70-21-23-100-801-G00).

(b) Do a visual inspection of the hole and adjacent areas where self-locking shank nut [1] will be installed.

1) Make sure that there are no deviations to a correct condition.

(c) Do the dimensional inspection of the DIA B (View A-A, Figure 802).

1) Make sure that the DIA B is 0.3327 in. (8.45 mm) - 0.3366 in. (8.55 mm).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Do an Fluorescent Penetrant Inspection (FPI) of the hole and adjacent areas where self-locking shank nut [1] will be installed (TASK 70-11-06-230-803-G00 or TASK 70-11-06-230-802-G00).

1) Do an FPI sensitivity level 3.

2) Make sure that there are no cracks.

SUBTASK 72-22-00-420-003-G00

(3) Install the self-locking shank nut [1].

(a) Make sure that the self-locking shank nut [1] is installed correctly.

(b) Make sure that there are no cracks in the shank.

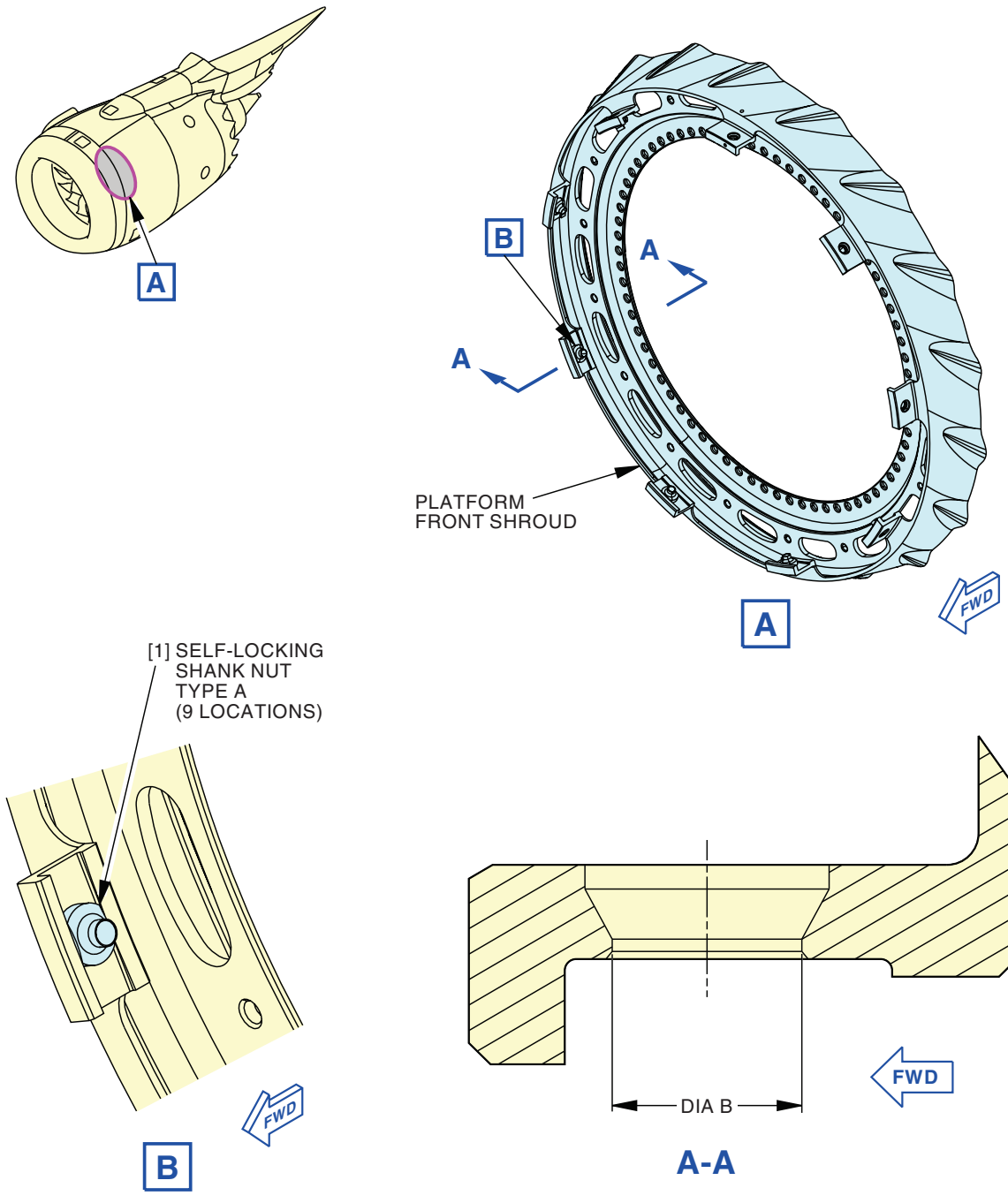
F. Put the Airplane Back to Its Usual Condition

SUBTASK 72-22-00-400-002-G00

(1) Install the front spinner cone, do this task: Front Spinner Cone Installation, TASK 72-22-01-400-801-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Self-Locking Shank Nuts Replacement
Figure 802/72-22-00-990-813-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FRONT SPINNER CONE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the front spinner cone

(2) An installation of the front spinner cone.

TASK 72-22-01-000-801-G00

2. Front Spinner Cone Removal

(Figure 401 and Figure 402)

A. General

(1) This task gives the instructions to remove the front spinner cone.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2586	Mat - Protective, Engine Inlet Part #: 956A7616G02 Supplier: 58828
SPL-14675	Tool - Removal, Spinner Cone Part #: 956A8670G02 Supplier: 58828 Opt Part #: 956A7617G01 Supplier: 58828 Opt Part #: 956A8670G01 Supplier: 58828
STD-585	Mat - Protective, Closed Cell Neoprene, 30-50 duro, Weather and Oil Resistant, 3/8 Inch Min Thickness, Minimum 42x60 Inches with Warning Streamers
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Prepare for the Removal

SUBTASK 72-22-01-210-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-22-01-860-001-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-22-01-490-001-G00

CAUTION

PUT THE PROTECTIVE MAT IN THE INLET COWL. THIS WILL PREVENT DAMAGE TO THE INLET COWL OR ENGINE BY TOOLS, PARTS, OR UNWANTED MATERIALS THAT FALL ON THE INLET COWL SURFACE.

(3) Put the protective mat, STD-585, or protective mat, SPL-2586, in the lower half of the inlet cowl to protect the inlet cowl and engine.

E. Front Spinner Cone Removal

SUBTASK 72-22-01-280-001-G00

(1) Locate the 12 o'clock index reference [5] on the platform front shroud [2].

SUBTASK 72-22-01-020-001-G00

(2) Remove the front spinner cone [1] as follows:

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0056

(a) Loosen the nine screws [3].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0056

(b) Loosen the nine bolts 6 lobes recess with cup [3].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0056

(c) Remove the three screws [3] and three washers [4] at the 1 (12:00 o'clock), 2 (4:00 o'clock), and 3 (8:00 o'clock) positions that attach the front spinner cone [1] to the platform front shroud [2].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0056

(d) Remove the three bolts 6 lobes recess with cup [3] at the 1 (12:00 o'clock), 2 (4:00 o'clock), and 3 (8:00 o'clock) positions that attach the front spinner cone [1] to the platform front shroud [2].

PTF ALL

(e) Install the spinner handling tool, SPL-14675, on the front spinner cone [1].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0056

CAUTION

MAKE SURE THAT THE SPINNER CONE DOES NOT FALL DOWN AFTER YOU REMOVE THE SCREWS. DAMAGE TO THE SPINNER CONE CAN OCCUR.

(f) Remove the six remaining screws [3] and six remaining washers [4] that attach the front spinner cone [1] to the platform front shroud [2].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0056

CAUTION MAKE SURE THAT THE SPINNER CONE DOES NOT FALL DOWN AFTER YOU REMOVE THE SCREWS. DAMAGE TO THE SPINNER CONE CAN OCCUR.

(g) Remove the six remaining bolts 6 lobes recess with cup [3] that attach the front spinner cone [1] to the platform front shroud [2].

PTF ALL

(h) Remove the front spinner cone [1] from the platform front shroud [2].

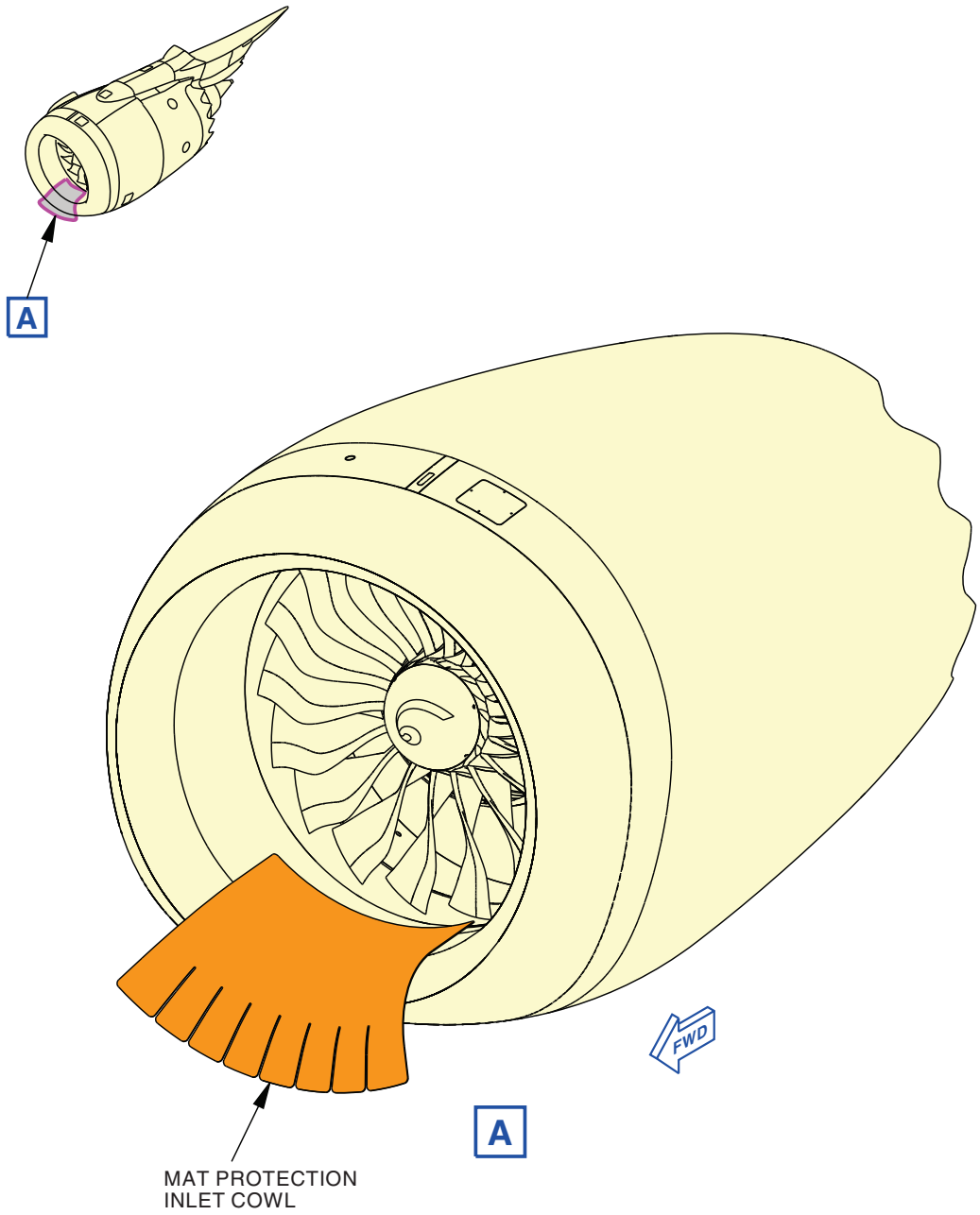
(i) Remove the front spinner cone [1] from the spinner handling tool, SPL-14675.

SUBTASK 72-22-01-480-001-G00

(3) Put a protective cover over the platform front shroud [2].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Mat Protection Inlet Cowl
Figure 401/72-22-01-990-807-G00

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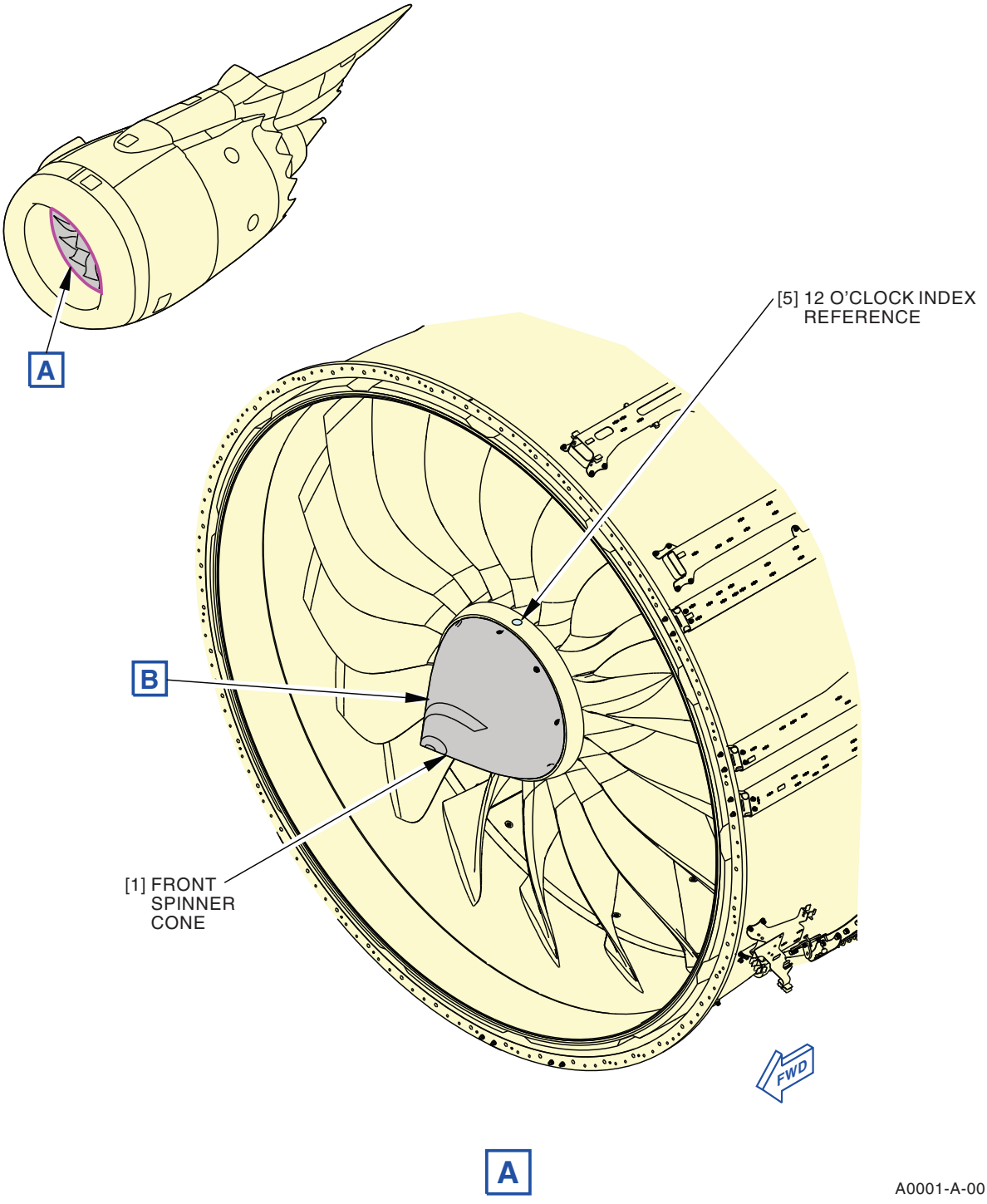
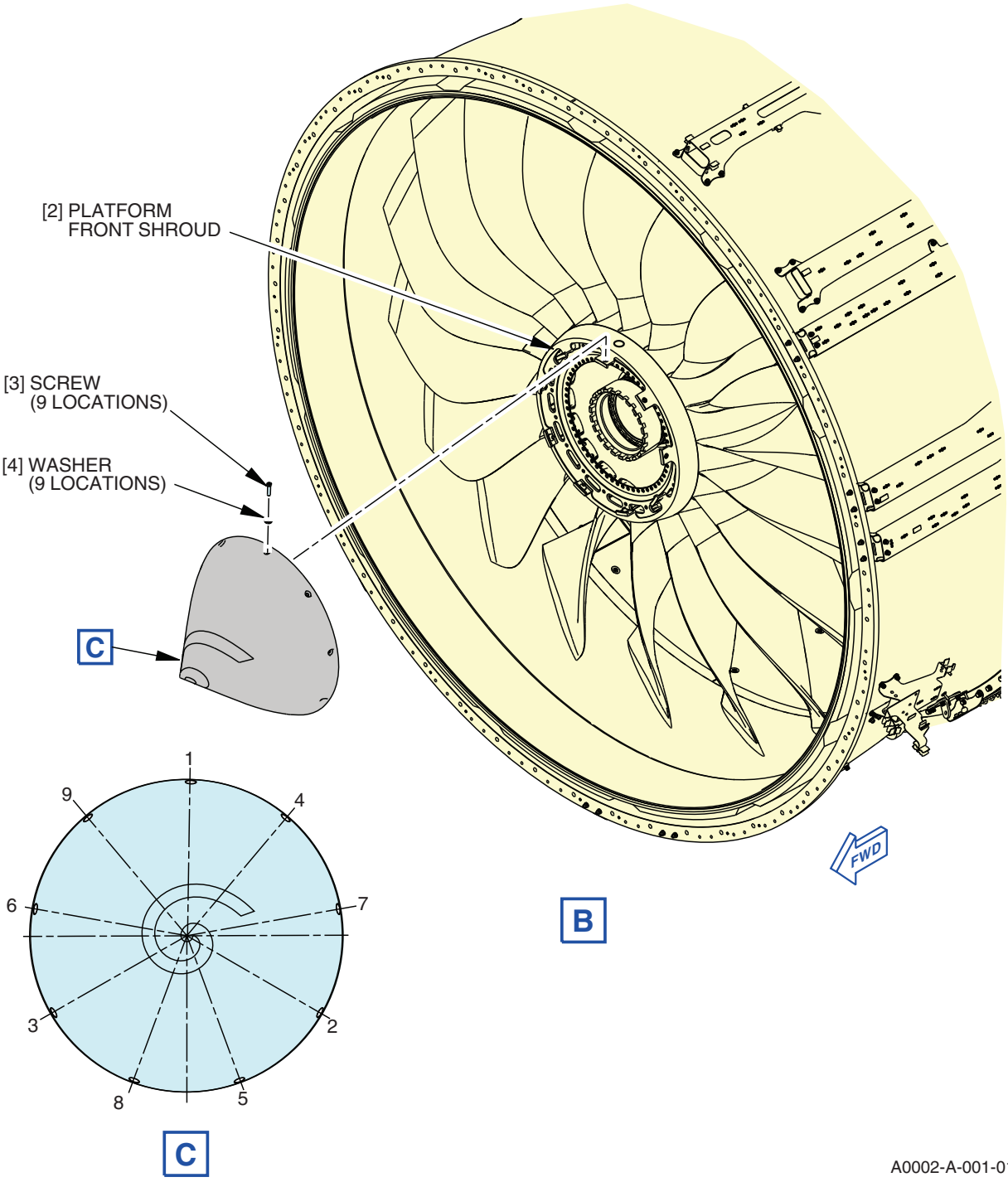


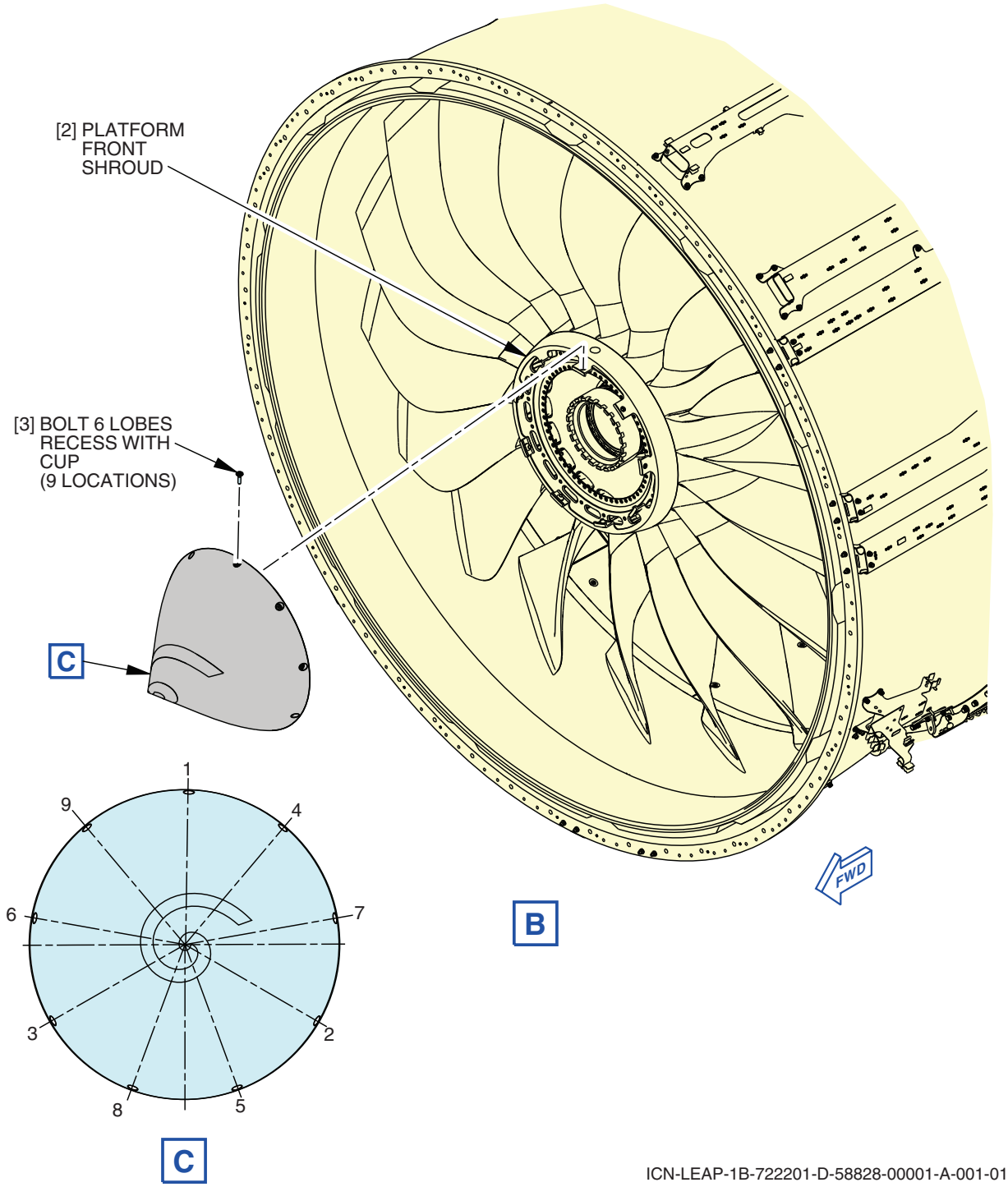
Front Spinner Cone Installation
Figure 402/72-22-01-990-806-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Front Spinner Cone Installation
Figure 402/72-22-01-990-806-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Front Spinner Cone Installation
Figure 402/72-22-01-990-806-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-22-01-400-801-G00

3. Front Spinner Cone Installation

(Figure 401 and Figure 402)

A. General

(1) This task gives the instructions to install the front spinner cone.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2586	Mat - Protective, Engine Inlet Part #: 956A7616G02 Supplier: 58828
SPL-14675	Tool - Removal, Spinner Cone Part #: 956A8670G02 Supplier: 58828 Opt Part #: 956A7617G01 Supplier: 58828 Opt Part #: 956A8670G01 Supplier: 58828
STD-585	Mat - Protective, Closed Cell Neoprene, 30-50 duro, Weather and Oil Resistant, 3/8 Inch Min Thickness, Minimum 42x60 Inches with Warning Streamers
STD-858	Tag - DO NOT OPERATE

C. Consumable Materials

Reference	Description	Specification
D00601 [CP2101]	Vaseline - Graphite Mineral	
G51335 [CP2674]	Lint-Free Cloth	

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Installation

SUBTASK 72-22-01-490-002-G00

(1) Make sure that the protective mat, STD-585, or protective mat, SPL-2586, is on the lower half of the inlet cowl.

F. Front Spinner Cone Installation

SUBTASK 72-22-01-540-001-G00

(1) Remove the protective cover from the platform front shroud [2].

SUBTASK 72-22-01-420-001-G00

(2) Install the front spinner cone [1] as follows:

(a) Make sure that the mating faces are clean and clear of unwanted materials.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0056

(b) Lubricate the threads and pressure faces of the nine screws [3] with graphite mineral vaseline, D00601 [CP2101].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0056

(c) Lubricate the threads and pressure faces of the nine bolts 6 lobes recess with cup [3] with graphite mineral vaseline, D00601 [CP2101].

PTF ALL

(d) Install the front spinner cone [1] on the spinner handling tool, SPL-14675.

(e) Install the front spinner cone [1] in a correct position on the platform front shroud [2].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0056

(f) Install the six screws [3] and six washers [4] in the 4, 5, 6, 7, 8, and 9 positions that attach the front spinner cone [1] to the platform front shroud [2].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0056

(g) Install the six bolts 6 lobes recess with cup [3] in the 4, 5, 6, 7, 8, and 9 positions that attach the front spinner cone [1] to the platform front shroud [2].

PTF ALL

(h) Remove the spinner handling tool, SPL-14675, from the front spinner cone [1].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0056

(i) Install the three screws [3] and three washers [4] at the 1 (12:00 o'clock), 2 (4:00 o'clock), and 3 (8:00 o'clock) positions that attach the front spinner cone [1] to the platform front shroud [2].

(j) Hold the front spinner cone [1] in the rear stop position on the platform front shroud [2] and tighten the three screws [3] in the 1 (12:00 o'clock), 2 (4:00 o'clock), and 3 (8:00 o'clock) positions to 67.45 in-lb (7.6 N·m) - 74.55 in-lb (8.4 N·m).

(k) Hold the front spinner cone [1] in the rear stop position on the platform front shroud [2] and tighten the six screws [3] in position 4, 5, 6, 7, 8, and 9 to 67.45 in-lb (7.6 N·m) - 74.55 in-lb (8.4 N·m).

(l) Remove unwanted graphite mineral vaseline, D00601 [CP2101], from the nine screws [3] with lint-free cloth, G51335 [CP2674].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0056

(m) Install the three bolts 6 lobes recess with cup [3] at the 1 (12:00 o'clock), 2 (4:00 o'clock), and 3 (8:00 o'clock) positions that attach the front spinner cone [1] to the platform front shroud [2].

(n) Hold the front spinner cone [1] in the rear stop position on the platform front shroud [2] and tighten the three bolts 6 lobes recess with cup [3] in the 1 (12:00 o'clock), 2 (4:00 o'clock), and 3 (8:00 o'clock) positions to 67.45 in-lb (7.6 N·m) - 74.55 in-lb (8.4 N·m).

(o) Hold the front spinner cone [1] in the rear stop position on the platform front shroud [2] and tighten the six bolts 6 lobes recess with cup [3] in position 4, 5, 6, 7, 8 and 9 to 67.45 in-lb (7.6 N·m) - 74.55 in-lb (8.4 N·m).

(p) Remove unwanted graphite mineral vaseline, D00601 [CP2101], from the nine bolts 6 lobes recess with cup [3] with lint-free cloth, G51335 [CP2674].

PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-22-01-100-001-G00

CAUTION

MAKE SURE THERE ARE NO TOOLS, PARTS OR OTHER UNWANTED MATERIAL IN THE INLET AREA. DAMAGE TO EQUIPMENT COULD OCCUR.

(1) Clean the work area and clear the tools and other items.

SUBTASK 72-22-01-090-001-G00

(2) Remove the protective mat, STD-585, or protective mat, SPL-2586, from the inlet cowl.

SUBTASK 72-22-01-860-002-G00

(3) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-22-01-860-003-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FRONT SPINNER CONE - INSPECTION/CHECK

1. General

A. This procedure contains one task:

(1) The front spinner cone inspection.

TASK 72-22-01-200-801-G00

2. Front Spinner Cone Inspection

A. General

(1) This task provides the instructions on how to examine the front spinner cone.

B. Tools/Equipment

Reference	Description
STD-585	Mat - Protective, Closed Cell Neoprene, 30-50 duro, Weather and Oil Resistant, 3/8 Inch Min Thickness, Minimum 42x60 Inches with Warning Streamers
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Prepare for the Inspection

SUBTASK 72-22-01-860-006-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-22-01-865-001-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-22-01-490-003-G00

CAUTION

PUT THE PROTECTIVE MAT IN THE INLET COWL. THIS WILL PREVENT DAMAGE TO THE INLET COWL OR ENGINE BY TOOLS, PARTS, OR UNWANTED MATERIALS THAT FALL ON THE INLET COWL SURFACE.

(3) Put the protective mat, STD-585 in the lower half of the inlet cowl to protect the inlet cowl and the engine.

E. Fan Rotor Spinner Inspection

SUBTASK 72-22-01-212-001-G00

(1) Visually examine the front spinner cone for damage:

(a) Cracks

1) Cracks are not permitted.

(b) Nicks and scratches

1) Nicks and scratches are permitted if they are not more than 0.030 inch (0.76 mm) in depth.

(c) Dents

1) Dents are permitted if they are not more than 0.125 inch (3.18 mm) in depth.

(d) Damaged surface protection

1) All amounts are permitted.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 72-22-01-100-003-G00

CAUTION

MAKE SURE THERE ARE NO TOOLS, PARTS OR OTHER UNWANTED MATERIAL IN THE INLET AREA. DAMAGE TO EQUIPMENT COULD OCCUR.

(1) Clean the work area and clear the tools and other items.

SUBTASK 72-22-01-090-002-G00

(2) Remove the protective mat, STD-585 from the inlet cowl.

SUBTASK 72-22-01-865-002-G00

(3) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-22-01-860-007-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FRONT SPINNER CONE - REPAIRS

1. General

A. This procedure has this task:

(1) A blending and touch-up of sulfuric anodizing on the front spinner cone attachment holes.

TASK 72-22-01-300-802-G00

2. Front Spinner Cone Attachment Holes Blending and Touch-Up of Sulfuric Anodizing

(Figure 801)

A. General

(1) This task gives instructions to blend and do a touch-up of sulfuric anodizing on the front spinner cone attachment holes.

(2) The table below gives a list of part numbers that are applicable to this repair.

NOTE: All part numbers are applicable to all paragraphs unless specified differently.

Table 801/72-22-01-993-801-G00 Applicable Part Numbers

Part Number	Designation
364-039-001-0	SPINNER, FRONT

(3) General limits for the repair are shown in the Figure 801.

(4) The table below gives dimensional information for the repair application.

NOTE: The dimensional information identified within this procedure was designed using millimeters as the base unit of measurement. The conversion to inches is rounded to two additional decimal digits compared to millimeters. For minimum dimensions, converted values are rounded up. For maximum dimensions, converted values are rounded down. To avoid conversion errors, do not re-convert inches to millimeters.

NOTE: If reduced number of decimal digits is required for the inch conversion, the minimum dimension must be rounded up and the maximum dimension must be rounded down.

NOTE: Reference dimensions do not require inspection.

Table 802/72-22-01-993-802-G00 Dimensional Information

Description	Minimum In-Process Dimension	Maximum In-Process Dimension	Minimum Finish Dimension	Maximum Finish Dimension	Reference Dimension
Dim. L	-	-	-	0.027 in. (0.7 mm)	-
Radius M	-	-	0.20 in. (5 mm)	-	-
Radius C	-	-	0.20 in. (5 mm)	-	-
Dia E	-	-	-	-	0.63 in. (16.0 mm) - 0.645 in. (16.4 mm)
Dim. H	-	-	-	-	0.115 in. (2.9 mm) - 0.129 in. (3.3 mm)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. References

Reference	Title
70-11-06-230-803-G00	Fluorescent Penetrant Inspection (General) (P/B 201)
70-43-07-300-801-G00	Specifications Chemical Touchup for Aluminum Standard Repair (P/B 201)
70-43-24-100-802-G00	Swab Etching Procedure and Cleaning (P/B 201)
70-43-42-300-802-G00	Blending Procedures - Hand Blending (P/B 201)
70-43-42-300-803-G00	Blending Procedures - Power Blending (P/B 201)
72-22-01-000-801-G00	Front Spinner Cone Removal (P/B 401)
72-22-01-400-801-G00	Front Spinner Cone Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
B50333 [CP2687]	Solvent - Degreasing, Diestone DLS	
G50143 [CP2208]	Tape - Glass Cloth with Silicone Adhesive, High Temperature, Abrasive Resistant - 3M 361	
G51478 [S1096]	Etchant - Aluminum Alloy	

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Repair

SUBTASK 72-22-01-020-005-G00

(1) Do this task: Front Spinner Cone Removal, TASK 72-22-01-000-801-G00.

F. Front Spinner Cone Attachment Holes Blending and Touch-Up of Sulfuric Anodizing

SUBTASK 72-22-01-300-006-G00

WARNING MAKE SURE THAT YOU READ THE MANUFACTURER'S SAFETY DATA SHEET (MSDS) FOR THE MATERIALS YOU WILL USE, AND WEAR THE PROTECTIVE CLOTHING AND EQUIPMENT IF IT IS SPECIFIED. IF YOU DO NOT OBEY, INJURY TO PERSONS CAN OCCUR.

(1) Do the visual inspection of the area J (Figure 801).

(a) If there are areas of missing anodizing that do not require blending, do these steps:

1) Do a spot touch-up of sulfuric anodizing (TASK 70-43-07-300-801-G00).

a) Mask the adjacent surfaces with 3M 361 tape, G50143 [CP2208].

b) Do a spot touch-up of sulfuric anodizing only on parent metal.

2) Remove the mask from the front spinner cone.

(b) If there are areas of missing anodizing that require blending within the limits, do these steps (Figure 801 and Table 802):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING DO NOT BREATHE THE PARTICLES FROM BLENDING OR LET THE PARTICLES TOUCH YOU. THE PARTICLES FROM BLENDING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PROTECTION EQUIPMENT. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR.

1) Remove local surface deviations, do one of these tasks: Blending Procedures - Hand Blending, TASK 70-43-42-300-802-G00 or Blending Procedures - Power Blending, TASK 70-43-42-300-803-G00.
   a) Do not blend the radius D.
   b) Do not remove more material than necessary.
   c) Make sure that the roughness of the blended areas is 126 microinches (3.2 micrometers).
2) Clean the blended areas with a clean cloth soaked in degreasing solvent, B50333 [CP2687], or alcohol, B00676 [CP1041].

CAUTION DO NOT LET THE ALUMINUM ALLOY ETCHANT (S1096) AND THE PENETRANT SOLUTION TOUCH THE ANTI-CORROSION PRIMER OR THE ADJACENT PAINT. IF YOU DO NOT OBEY, YOU CAN CAUSE DAMAGE TO THE FRONT SPINNER CONE.

3) Do the Fluorescent Penetrant Inspection (FPI) as follows:
   a) Do a local swab etching of the blended areas with aluminum alloy etchant, G51478 [S1096] (TASK 70-43-24-100-802-G00).
   b) Do the FPI of the blended areas (TASK 70-11-06-230-803-G00).
      <1> Use sensitivity level 3.
      <2> Indications more than 0.011 in. (0.3 mm) are not permitted.
      <3> Linear indications are not permitted.
      NOTE: Linear indications are indications that are at least four times longer than they are wide.

CAUTION TOUCH-UP THE SULFURIC ANODIZING IN LESS THAN 4 HOURS AFTER THE BLEND PROCEDURE. IF YOU DO NOT OBEY, YOU CAN CAUSE DAMAGE TO THE FRONT SPINNER CONE.

4) Do a spot touch-up of sulfuric anodizing on the blended areas (TASK 70-43-07-300-801-G00).
   a) Mask the adjacent surfaces with 3M 361 tape, G50143 [CP2208].
   b) Do a spot touch-up of sulfuric anodizing only on parent metal.
5) Remove the mask from the front spinner cone.

(c) If there are areas of missing anodizing where the required blending exceeds the limits, replace the front spinner cone (Figure 801 and Table 802).

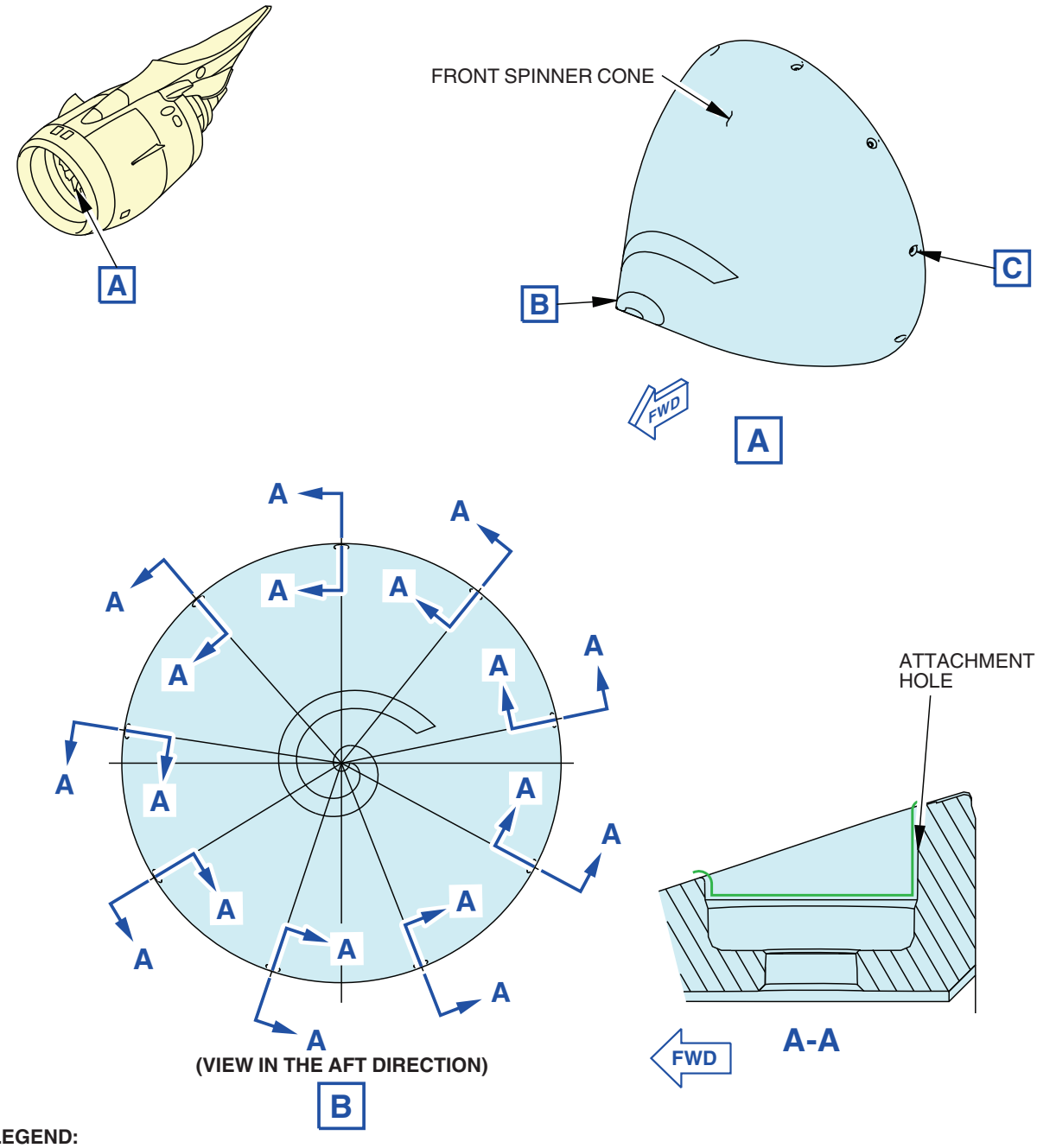
G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-22-01-420-004-G00

(1) Do this task: Front Spinner Cone Installation, TASK 72-22-01-400-801-G00.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

AREA J

MATERIAL: 2618A + SULFURIC ANODIZING + PRIMER + ANTI-EROSION PAINT

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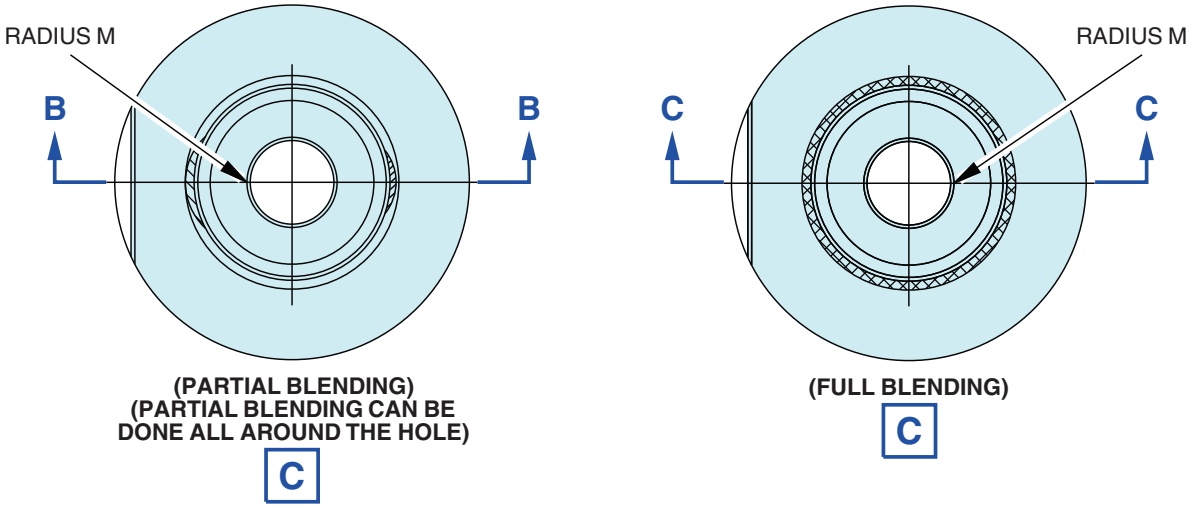
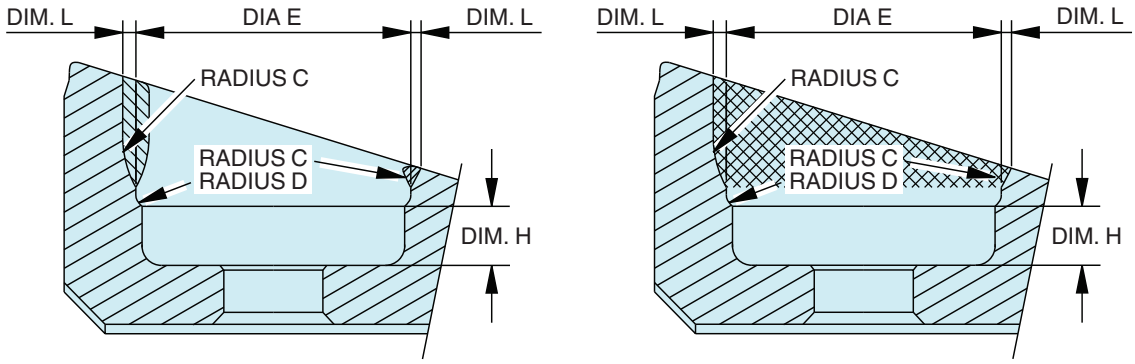
3132162 S0000861922_V1

Front Spinner Cone Attachment Holes Blending and Touch-Up of Sulfuric Anodizing
Figure 801/72-22-01-990-809-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





LEGEND:

- FRONT REPAIRED AREA
- REAR REPAIRED AREA
- FULL REPAIRED AREA

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3132163 S0000861923_V1

Front Spinner Cone Attachment Holes Blending and Touch-Up of Sulfuric Anodizing
Figure 801/72-22-01-990-809-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN BLADES - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the fan blades (selection)

(2) An installation of the fan blades (selection)

(3) A removal of the fan blades (pair)

(4) An installation of the fan blades (pair)

(5) A removal of the fan blades (set)

(6) An installation of the fan blades (set)

(7) A replacement of the fan blade.

TASK 72-22-02-000-804-G00

2. Fan Blades Removal (Selection)

A. Fan Blades Removal (Selection)

SUBTASK 72-22-02-020-007-G00

(1) Do one of these tasks to remove the fan blades:

(a) Do this task: Fan Blades (Pair) Removal, TASK 72-22-02-000-802-G00.

(b) Do this task: Fan Blades (Set) Removal, TASK 72-22-02-000-803-G00.

——— END OF TASK ———

TASK 72-22-02-400-804-G00

3. Fan Blades Installation (Selection)

A. Fan Blades Installation (Selection)

SUBTASK 72-22-02-420-007-G00

(1) Do one of these tasks to install the fan blades:

(a) Do this task: Fan Blades (Pair) Installation, TASK 72-22-02-400-802-G00.

(b) Do this task: Fan Blades (Set) Installation, TASK 72-22-02-400-803-G00.

——— END OF TASK ———

TASK 72-22-02-000-802-G00

4. Fan Blades (Pair) Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the fan blades (pair).

B. References

Reference	Title
72-22-01-000-801-G00	Front Spinner Cone Removal (P/B 401)
72-22-02-200-801-G00	Fan Blade Inspection (P/B 601)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-941	Tool - Removal, Spacer Fan Blade, Slide Hammer Part #: 956A7651G02 Supplier: 58828
SPL-14660	Positioner - Fan Disk Barrel Nut Part #: 956A6071G01 Supplier: 58828
SPL-14662	Puller - Fan Blade Spacer Part #: 956A6065G01 Supplier: 58828
SPL-14664	Tool - Platform Front Shroud Removal Part #: 956A8062G02 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
G50530 [CP5061]	Marker - Temporary	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Prepare for the Removal

SUBTASK 72-22-02-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-22-02-860-016-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-22-02-010-002-G00

(3) Remove the front spinner cone [2] (TASK 72-22-01-000-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Fan Blades (Pair) Removal

SUBTASK 72-22-02-010-003-G00

WARNING USE TWO PERSONS TO DO THIS PROCEDURE. ONE PERSON REMOVES THE FAN BLADES WHILE THE OTHER HOLDS THE FAN TO MAKE SURE THAT IT WILL NOT TURN. IF THE FAN TURNS, INJURY TO PERSONS AND DAMAGE TO THE ENGINE CAN OCCUR.

CAUTION USE THE STEPS AND THE SPECIFIED TOOLS TO REMOVE THE FAN BLADES. IF YOU DO NOT USE THE SPECIFIED TOOLS, DAMAGE TO THE FAN DISK OR ENGINE CAN OCCUR.

(1) Do these steps to remove the fan blade [1] (pair):

(a) Find the (No. 1) fan blade [1]:

1) Rotate the fan blade [1] and look for the number ”1” stamped on the forward face of the platform front shroud [3].

2) Put the (No. 1) fan blade [1] at the 12:00 o’clock position.

(b) Put marks on the external surface of the leading edge of the 18 fan blades [1]:

1) Use the temporary marker, G50530 [CP5061], to mark the (No. 1) fan blade [1].

2) Number the remaining fan blade [1] in counterclockwise direction from the (No. 2) fan blade [1] to (No. 18) fan blade [1].

NOTE: This is counterclockwise when you look on the front of the engine.

(c) Remove the platform front shroud [3]:

1) Loosen the 18 bolts [4] from the fan disk [10].

2) Remove the 16 bolts [4] that attach the platform front shroud [3] to the fan disk [10] except the two bolts [4] at the 3 o'clock to the 9 o'clock position.

CAUTION USE THE REMOVAL TOOL TO REMOVE THE PLATFORM FRONT SHROUD. IF YOU DO NOT USE THE TOOL, THE PLATFORM FRONT SHROUD CAN FALL AND CAUSE DAMAGE TO THE ENGINE.

3) Install the tool, SPL-14664, on the platform front shroud [3].

4) Remove the remaining two bolts [4] that attach the platform front shroud [3] to the fan disk [10].

5) Remove the platform front shroud [3] from the fan disk [10].

6) Remove the tool, SPL-14664, from the platform front shroud [3].

(d) Remove the fan blade platform [6] located around the fan blade [1]:

NOTE: The fan blade platform will come out easily if you turn the fan rotor to put the (No. 1) fan blade [1] at the 12 o'clock position.

1) Move the fan blade platform [6] along the platform axis.

2) Remove the fan blade platform [6] from the fan disk [10].

(e) Remove the fan blade platform [7] located around the fan blade [1]:

1) Move the fan blade platform [7] along the platform axis.

2) Remove the fan blade platform [7] from the fan disk [10].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Turn the fan rotor to put the (No. 1) fan blade [1] at the 6 o'clock position.

1) Put the fan blade [1] at the 6 o'clock position to loosen pressure on the fan blade spacer [9].

CAUTION

USE THE REMOVAL TOOL TO REMOVE THE FAN BLADE SPACER. IF YOU DO NOT USE THE TOOL, DAMAGE TO THE ENGINE CAN OCCUR.

(g) Remove the fan blade spacer [9] from the fan blade [1]:

1) Use the positioner, SPL-14660, to remove the barrel nut [5] from the fan disk [10] in front of the fan blade [1].
2) Install the puller, SPL-14662, on the fan blade spacer [9].
3) Put the pin of the puller, SPL-14662, through the hole of the fan blade spacer [9].
4) Move the fan blade spacer [9] along the spacer axis.
5) Remove the pin of the puller, SPL-14662, from the hole to release the fan blade spacer [9].
6) Remove the fan blade spacer [9] from the fan disk [10].

(h) Remove the fan blade spacer [9] from the fan blade [1] (alternative procedure):

1) Use the positioner, SPL-14660, to remove the barrel nut [5] from the fan disk [10] in front of the fan blade [1].
2) Install the hook of puller, SPL-14662, on the hole of the fan blade spacer [9].
3) Pull out the fan blade spacer [9] with two handles of puller, SPL-14662.
4) If the two handles of puller, SPL-14662, force is insufficient, use the slide hammer removal tool, SPL-941, to pull the fan blade spacer [9].

(i) Remove the fan blade lock [8] from the fan disk [10].

(j) Remove the fan blade [1] from the dovetail slot of the fan disk [10].

(k) Do an inspection of the fan blades [1] (Fan Blade Inspection, TASK 72-22-02-200-801-G00).

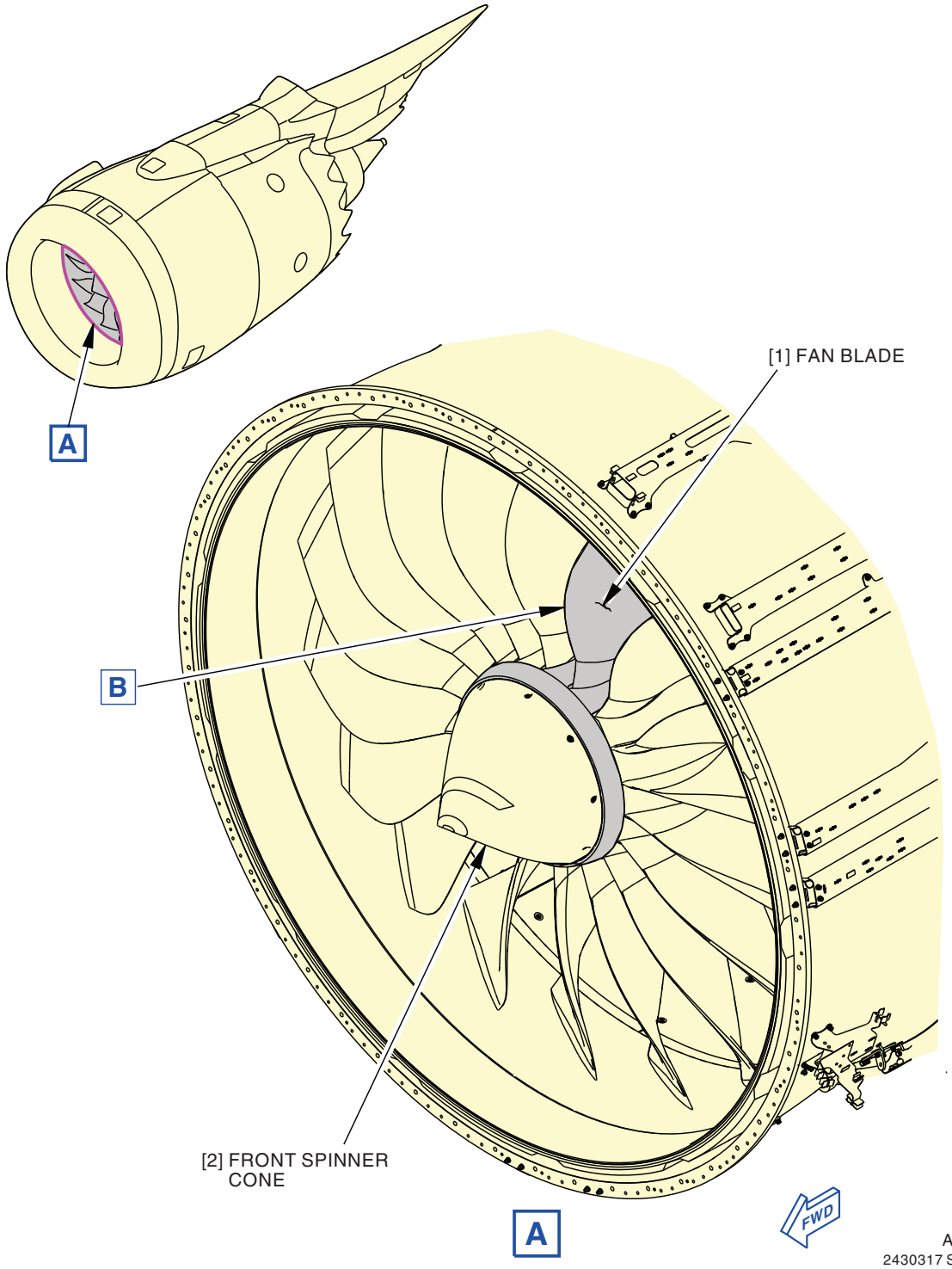
(l) Make sure that the two fan disk shims [11] are in position on the fan disk [10].

(m) Do the steps above again to remove the opposite blade.

NOTE: The opposite blade is at a position of 180 degrees apart.

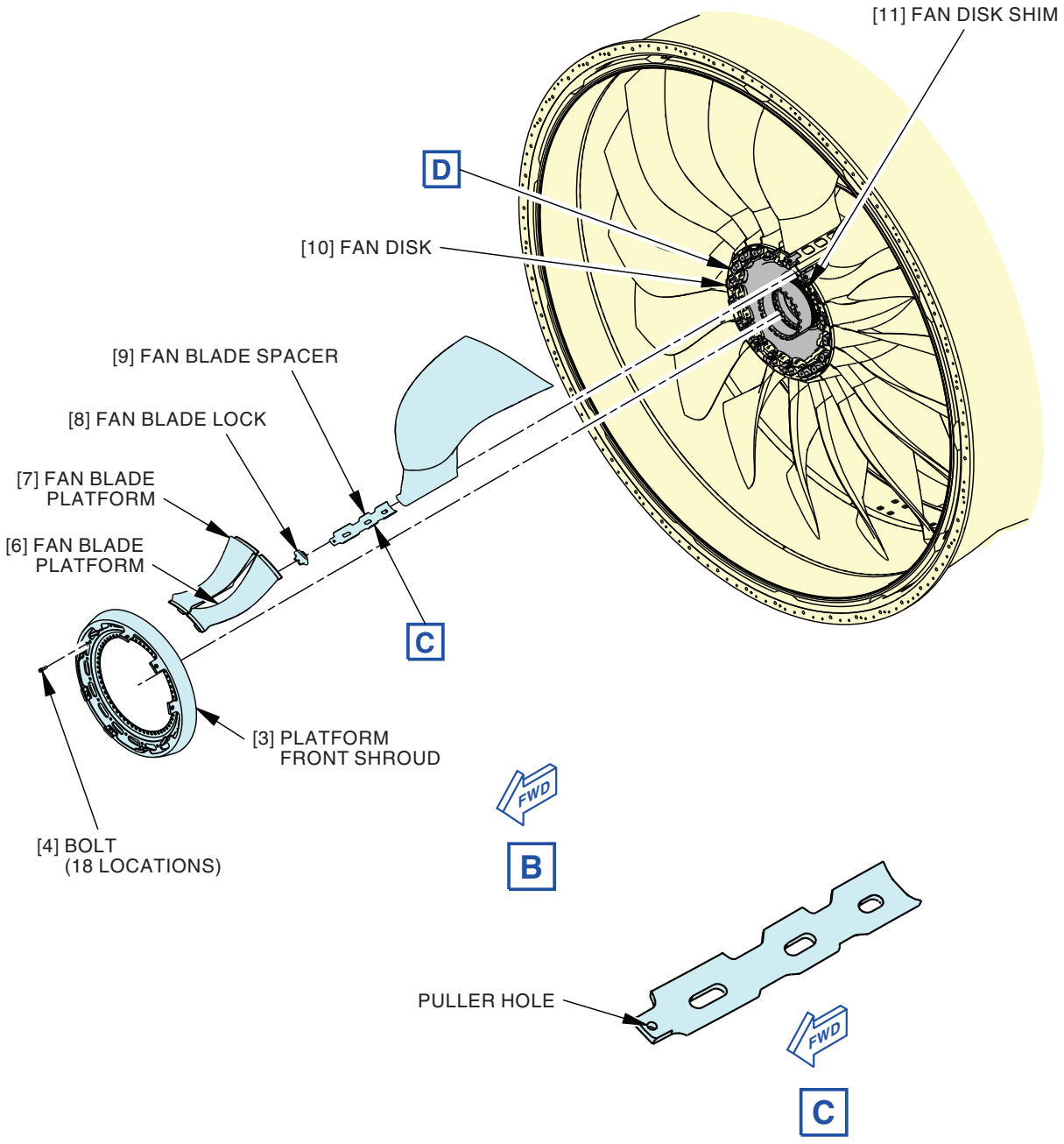
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Blades (Pair) Installation
Figure 401/72-22-02-990-824-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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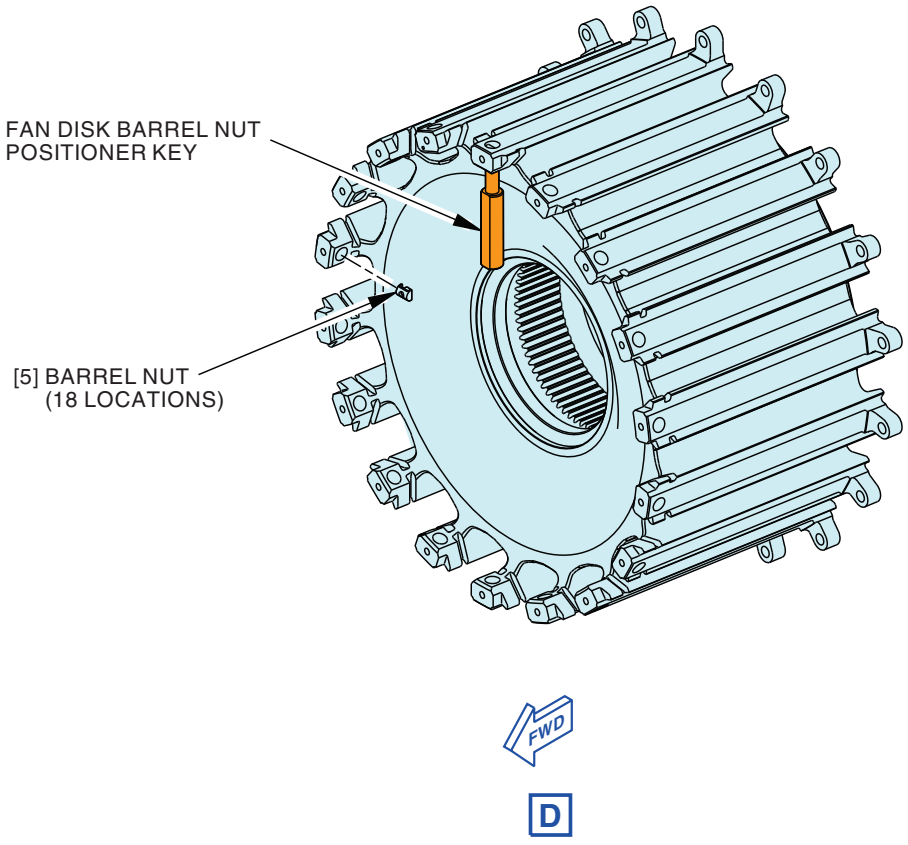
Fan Blades (Pair) Installation
Figure 401/72-22-02-990-824-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Fan Blades (Pair) Installation
Figure 401/72-22-02-990-824-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-22-02-400-802-G00

5. Fan Blades (Pair) Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the fan blades (pair).

(2) If the fan blade pair is replaced by a new pair, do this task: Fan Blade Replacement, TASK 72-22-02-960-802-G00.

(3) You must do the tests that are listed in the power plant test reference table after you install the fan blades (pair).

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
72-22-01-400-801-G00	Front Spinner Cone Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14660	Positioner - Fan Disk Barrel Nut Part #: 956A6071G01 Supplier: 58828
SPL-14661	Guide Pins - Platform Front Shroud Part #: 956A6067G01 Supplier: 58828
SPL-15044	Fixture - Install, Shroud Front Platform Part #: 956A8675G05 Supplier: 58828 Opt Part #: 956A8675G03 Supplier: 58828 Opt Part #: 956A8675G04 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00601 [CP2101]	Vaseline - Graphite Mineral	
G51335 [CP2674]	Lint-Free Cloth	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Fan Blades (Pair) Installation

SUBTASK 72-22-02-410-002-G00

WARNING USE TWO PERSONS TO DO THIS PROCEDURE. ONE PERSON INSTALLS THE FAN BLADES WHILE THE OTHER HOLDS THE FAN AND IT WILL NOT TURN. IF THE FAN TURNS, INJURY TO PERSONS AND DAMAGE TO THE ENGINE CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

CAUTION USE THE STEPS AND THE SPECIFIED TOOLS TO INSTALL THE FAN BLADES. IF YOU DO NOT USE THE SPECIFIED TOOLS, DAMAGE TO THE FAN DISK OR ENGINE CAN OCCUR.

(1) Install the fan blade [1] (pair).

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Turn the fan rotor to install the (No. 1) fan blade [1] at the 6 o'clock position.

1) Install the fan blade at the 6 o'clock position to loosen pressure on the fan blade spacer [9].

(c) Engage the dovetail of the fan blade [1] in the dovetail slot of the fan disk [10].

(d) Push the fan blade [1] rearwards.

NOTE: Push it until the fan lug on the rear of the fan blade engage with the recess on the fan disk.

(e) Engage the fan blade lock [8] on the fan disk [10].

CAUTION MAKE SURE THAT YOU INSTALL THE FAN BLADE SPACER CORRECTLY. THE PULLER HOLE MUST BE IN THE FORWARD DIRECTION. INCORRECT INSTALLATION CAN CAUSE DAMAGE TO THE FAN BLADE SPACER.

(f) Engage the fan blade spacer [9] in the dovetail slot between the fan blade [1] and fan disk [10].

(g) Install the fan blade spacer [9] of the fan blade [1] in a correct position on the fan disk [10].

(h) Install the fan blade platform [6].

1) Engage the fan blade platform [6] immediately at the right of the fan blade [1].

(i) Install the fan blade platform [7].

1) Engage the fan blade platform [7] immediately at the left of the fan blade [1].

(j) Do the steps above again to install the opposite blade.

NOTE: The opposite blade is at a position of 180 degrees apart.

(k) Install the platform front shroud [3].

1) Use the positioner, SPL-14660, to install the barrel nuts [5] in the correct position on the fan disk [10].

2) Check to make sure that the 18 barrel nuts [5] are in the correct positions on the fan disk [10].

a) If it is necessary, use the positioner, SPL-14660.

3) Lubricate the threads of the 18 bolts [4] with the graphite mineral vaseline, D00601 [CP2101].

4) Install the guide pins, SPL-14661, to align the platform front shroud [3] with the fan disk [10].

5) Install the platform front shroud [3] in the correct position on the fan disk [10] through the guide pins, SPL-14661.

6) Install the shroud front platform fixture, SPL-15044.

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7) Push the platform front shroud [3] onto the fan disk [10] with the shroud front platform fixture, SPL-15044.

8) Remove the guide pins, SPL-14661, and the shroud front platform fixture, SPL-15044, from the platform front shroud [3].

9) Install the 18 bolts [4] that attach the platform front shroud [3] to the fan disk [10].

10) Make sure that the 18 bolts [4] align correctly with the 18 barrel nuts [5].

11) Tighten the bolts [4] to 133 in-lb (15 N·m) - 186 in-lb (21 N·m).

12) Remove the excess of graphite mineral vaseline, D00601 [CP2101], with lint-free cloth, G51335 [CP2674].

(l) Clean the work area and clear the tools and other items.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-22-02-410-003-G00

(1) Install the front spinner cone [2] (TASK 72-22-01-400-801-G00).

SUBTASK 72-22-02-860-019-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-22-02-860-002-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

H. Fan Blades (Pair) Installation Test

SUBTASK 72-22-02-710-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

TASK 72-22-02-000-803-G00

6. Fan Blades (Set) Removal

(Figure 402)

A. General

(1) This task gives the instructions to remove the fan blades (set).

B. References

Reference	Title
72-22-01-000-801-G00	Front Spinner Cone Removal (P/B 401)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
72-22-02-200-801-G00	Fan Blade Inspection (P/B 601)
72-22-03-000-801-G00	Fan Blade Platforms - Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-941	Tool - Removal, Spacer Fan Blade, Slide Hammer Part #: 956A7651G02 Supplier: 58828
SPL-14660	Positioner - Fan Disk Barrel Nut Part #: 956A6071G01 Supplier: 58828
SPL-14662	Puller - Fan Blade Spacer Part #: 956A6065G01 Supplier: 58828
SPL-14664	Tool - Platform Front Shroud Removal Part #: 956A8062G02 Supplier: 58828
SPL-14677	Rack - Fan Blade Storage Part #: 956A6634G01 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
G50530 [CP5061]	Marker - Temporary	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Prepare for the Removal

SUBTASK 72-22-02-860-003-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-22-02-860-017-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-22-02-020-002-G00

(3) Do this task: Front Spinner Cone Removal, TASK 72-22-01-000-801-G00.

G. Fan Blades (Set) Removal

SUBTASK 72-22-02-020-003-G00

WARNING USE TWO PERSONS TO DO THIS PROCEDURE. ONE PERSON REMOVES THE FAN BLADES WHILE THE OTHER HOLDS THE FAN TO MAKE SURE THAT IT WILL NOT TURN. IF THE FAN TURNS, INJURY TO PERSONS AND DAMAGE TO THE ENGINE CAN OCCUR.

CAUTION USE THE STEPS AND THE SPECIFIED TOOLS TO REMOVE THE FAN BLADES. IF YOU DO NOT USE THE SPECIFIED TOOLS, DAMAGE TO THE FAN DISK OR ENGINE CAN OCCUR.

(1) Do these steps to remove the fan blade [1] (set):

(a) Find the (No. 1) fan blade [1]:

1) Rotate the fan blade [1] and look for the number “1” stamped on the forward face of the platform front shroud [3].

2) Put the (No. 1) fan blade [1] at the 12:00 o’clock position.

(b) Put marks on the external surface of the leading edge of the 18 fan blades [1]:

1) Use the temporary marker, G50530 [CP5061], to mark the (No. 1) fan blade [1].

2) Number the remaining fan blade [1] in counterclockwise direction from (No. 2) fan blade [1] to (No. 18) fan blade [1].

NOTE: This is counterclockwise when you look on the front of the engine.

(c) Remove the platform front shroud [3]:

1) Loosen the 18 bolts [4] from the fan disk [10].

2) Remove the 16 bolts [4] that attach the platform front shroud [3] to the fan disk [10] except the two bolts [4] at the 3 o'clock to the 9 o'clock position.

CAUTION USE THE REMOVAL TOOL TO REMOVE THE PLATFORM FRONT SHROUD. IF YOU DO NOT USE THE TOOL, THE PLATFORM FRONT SHROUD CAN FALL AND CAUSE DAMAGE TO THE ENGINE.

3) Install the tool, SPL-14664 on the platform front shroud [3].

4) Remove the remaining two bolts [4] that attach the platform front shroud [3] to the fan disk [10].

5) Remove the platform front shroud [3] from the fan disk [10].

6) Remove the tool, SPL-14664, from the platform front shroud [3].

(d) Remove the 18 fan blade platforms [6] (TASK 72-22-03-000-801-G00).

(e) Turn the fan rotor to put the (No. 1) fan blade [1] at 6:00 o'clock position.

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(f) Put the fan blade at the 6 o'clock position to loosen the pressure on the fan blade spacer [9].

CAUTION

USE THE REMOVAL TOOL TO REMOVE THE FAN BLADE SPACER. IF YOU DO NOT USE THE TOOL, DAMAGE TO THE ENGINE CAN OCCUR.

(g) Remove the (No. 1) fan blade spacer [9] from the (No. 1) fan blade [1]:

1) Use the positioner, SPL-14660, to remove the barrel nut [5] from the fan disk [10] in front of the located fan blade [1].
2) Install the puller, SPL-14662, on the (No. 1) fan blade spacer [9].
3) Move the (No. 1) fan blade spacer [9] along the spacer axis and not along the engine center line.
4) Remove the (No. 1) fan blade spacer [9] from the groove of the fan disk [10].
5) Remove the (No. 1) fan blade spacer [9] from the puller, SPL-14662.
6) Remove the fan blade lock [8] from the fan disk [10].

(h) Remove the (No. 1) fan blade spacer [9] from the (No. 1) fan blade [1] (alternative procedure):

1) Use the positioner, SPL-14660, to remove the barrel nut [5] from the fan disk [10] in front of the fan blade [1].
2) Install the hook of puller, SPL-14662, on the hole of the fan blade spacer [9].
3) Pull out the fan blade spacer [9] with two handles of puller, SPL-14662.
4) If the two handles of puller, SPL-14662, force is insufficient, use the slide hammer removal tool, SPL-941, to pull the fan blade spacer [9].

(i) Remove the (No. 1) fan blade lock [8] from the fan disk [10].

(j) Remove the (No. 1) fan blade [1] from the fan disk [10].

(k) Do an inspection of the fan blades [1] (Fan Blade Inspection, TASK 72-22-02-200-801-G00).

(l) Install the (No. 1) fan blade [1] on the rack, SPL-14677.

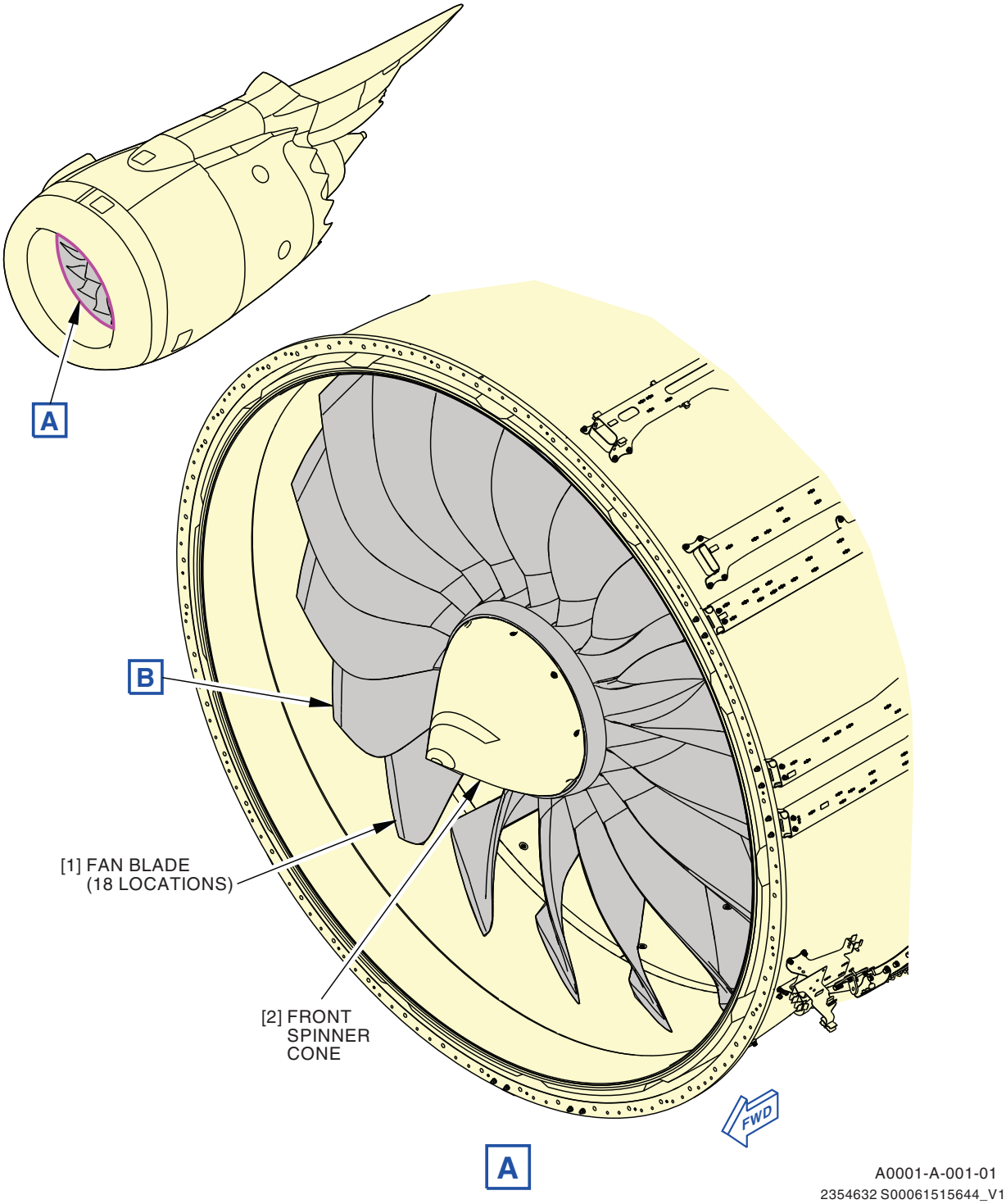
(m) Remove the remaining 17 fan blades [1]:

1) Turn the fan rotor to put the fan blade [1] at the 6 o'clock position.
2) Put the fan blade at the 6 o'clock position to loose the pressure on the spacers.
3) Remove the remaining 17 barrel nuts [5] from the fan disk [10].
4) Remove the remaining 17 fan blade spacers [9] from the remaining 17 fan blades [1].
5) Remove the remaining 17 fan blade locks [8] from the fan disk [10].
6) Remove the remaining 17 fan blades [1] from the fan disk [10].
7) Install the remaining 17 fan blades [1] on the rack, SPL-14677.

(n) Make sure that the 18 fan disk shims [11] are in position on the fan disk [10].

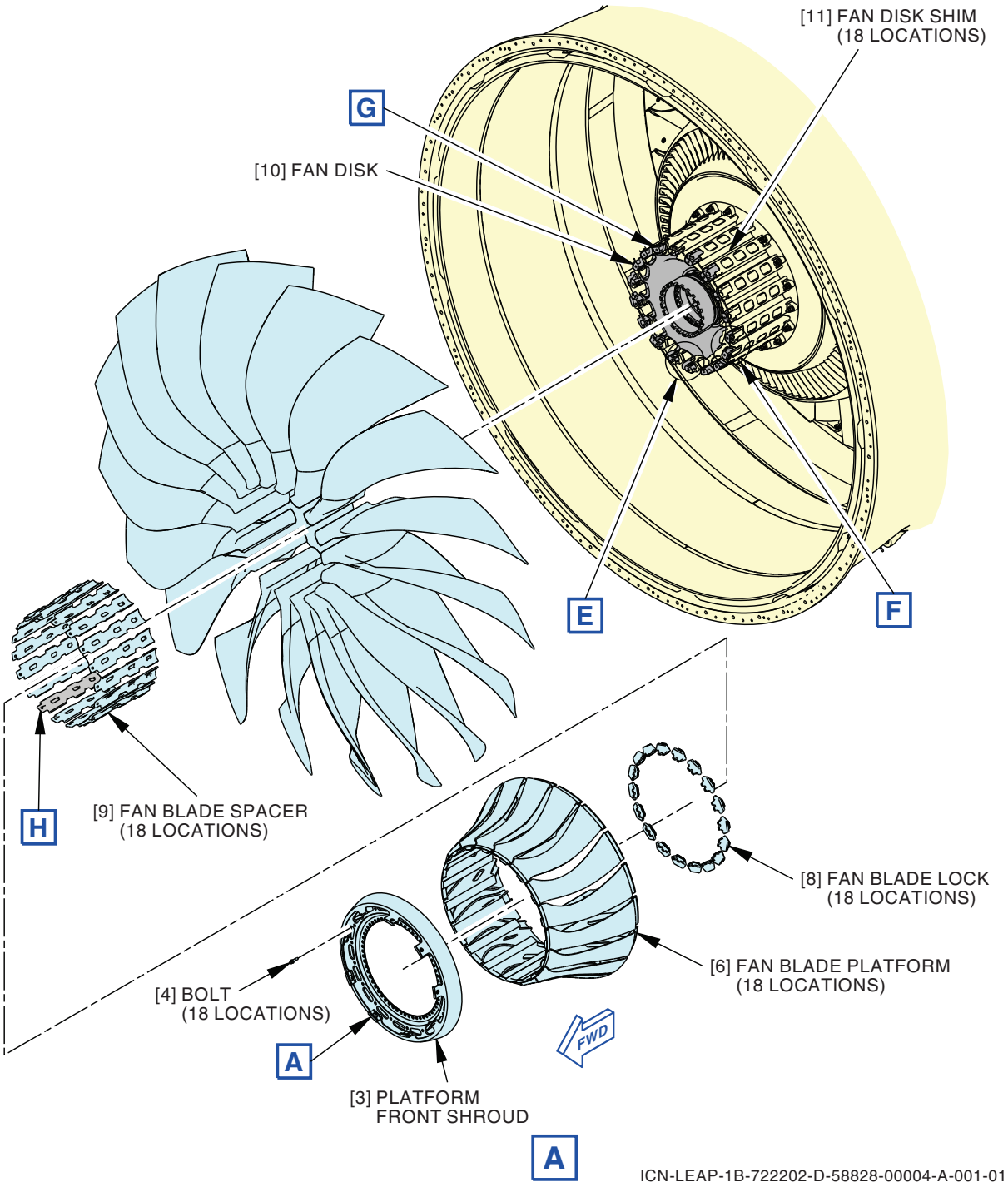
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



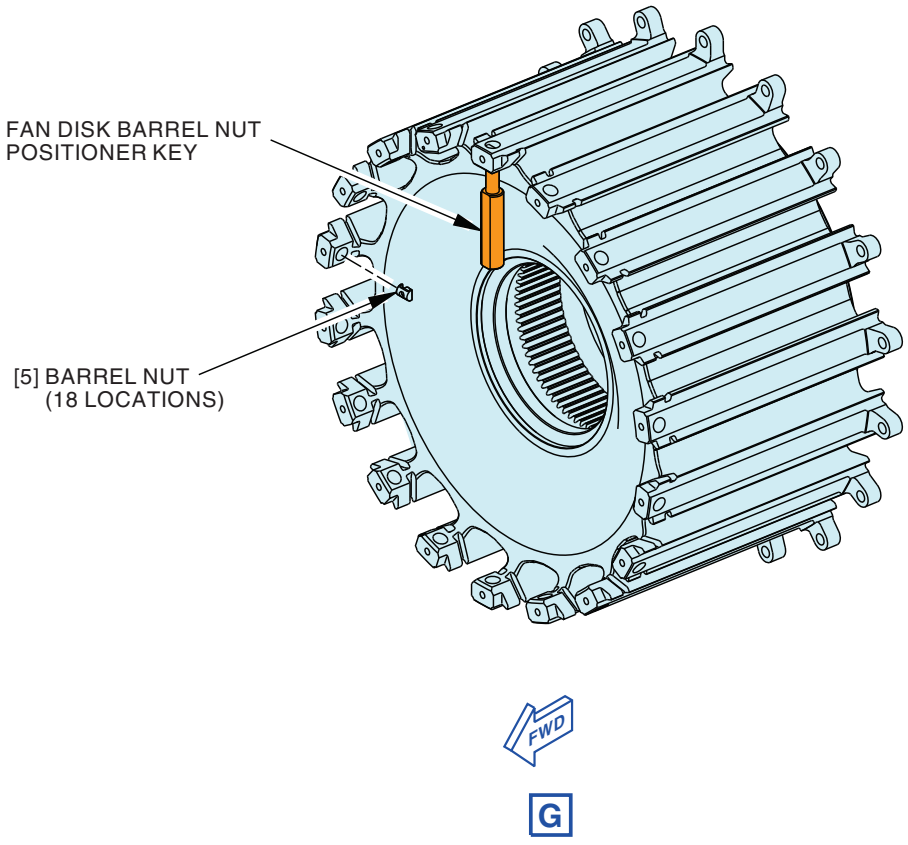
Fan Blades (Set) Installation
Figure 402/72-22-02-990-805-G00 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Blades (Set) Installation
Figure 402/72-22-02-990-805-G00 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

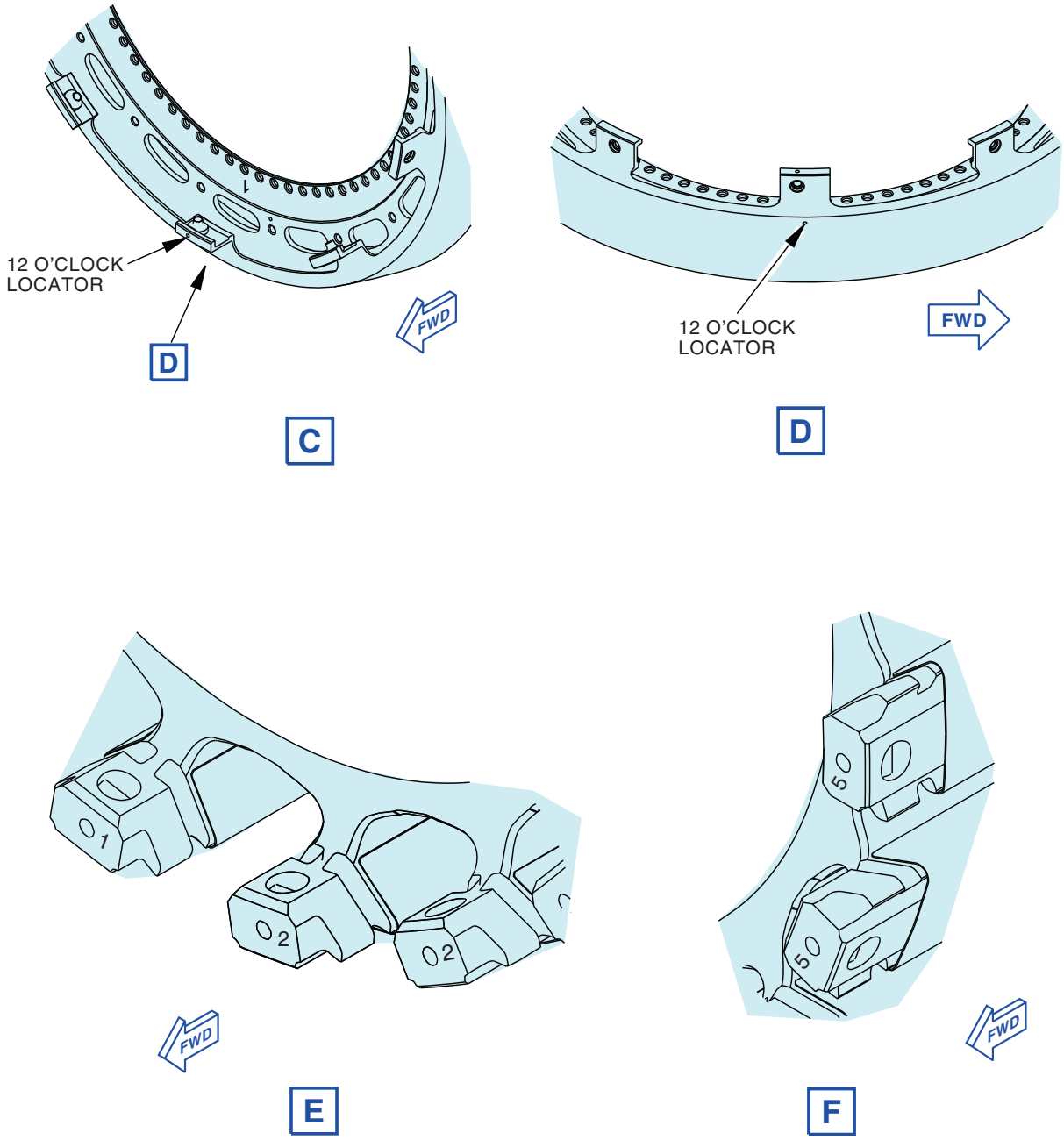


ICN-LEAP-1B-722202-D-58828-00005-A-001-01

2814289 S0000648520_V1

Fan Blades (Set) Installation
Figure 402/72-22-02-990-805-G00 (Sheet 3 of 5)

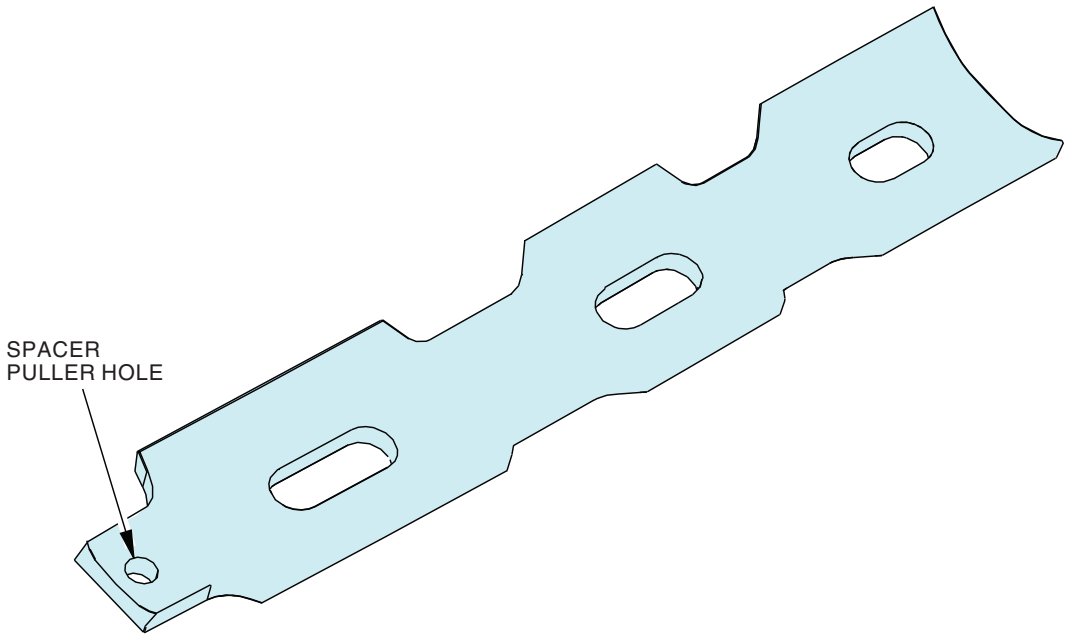
ECCN 9E991 BOEING PROPRIETARY - See title page for details



A0003-A-001-01
2431338 S0000562738_V2

Fan Blades (Set) Installation
Figure 402/72-22-02-990-805-G00 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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2869652 S0000677593_V1

Fan Blades (Set) Installation
Figure 402/72-22-02-990-805-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-22-02-400-803-G00

7. Fan Blades (Set) Installation

(Figure 402)

A. General

(1) This task gives the instructions to install the fan blades (set).

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
72-22-01-400-801-G00	Front Spinner Cone Installation (P/B 401)
72-22-03-400-801-G00	Fan Blade Platforms - Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14660	Positioner - Fan Disk Barrel Nut Part #: 956A6071G01 Supplier: 58828
SPL-14661	Guide Pins - Platform Front Shroud Part #: 956A6067G01 Supplier: 58828
SPL-15044	Fixture - Install, Shroud Front Platform Part #: 956A8675G05 Supplier: 58828 Opt Part #: 956A8675G03 Supplier: 58828 Opt Part #: 956A8675G04 Supplier: 58828
STD-583	Mallet - Non-metallic
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00601 [CP2101]	Vaseline - Graphite Mineral	
G51335 [CP2674]	Lint-Free Cloth	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Fan Blades (Set) Installation

SUBTASK 72-22-02-930-001-G00

(1) Make sure that the 18 fan disk shims [11] are in position on the fan disk [10].

SUBTASK 72-22-02-930-002-G00

WARNING USE TWO PERSONS TO DO THIS PROCEDURE. ONE PERSON INSTALLS THE FAN BLADES WHILE THE OTHER HOLDS THE FAN AND IT WILL NOT TURN. IF THE FAN TURNS, INJURY TO PERSONS AND DAMAGE TO THE ENGINE CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

CAUTION USE THE STEPS AND THE SPECIFIED TOOLS TO INSTALL THE FAN BLADES. IF YOU DO NOT USE THE SPECIFIED TOOLS, DAMAGE TO THE FAN DISK OR ENGINE CAN OCCUR.

(2) Install the fan blade [1] (set).

(a) Make sure that the mating faces are clean and clear of unwanted materials.

NOTE: The fan blades are installed in the clockwise direction. You start with the (No. 18) fan blade and continue down to the (No. 1) fan blade.

(b) Engage the dovetail of the (No. 18) fan blade [1] with the dovetail slot No. 18 of the fan disk [10].

(c) Push the (No. 18) fan blade [1] rearwards.

NOTE: Push it until the fan lug on the rear of the fan blade engages with the recess on the fan disk.

(d) Engage the (No. 18) fan blade lock [8] on the fan disk [10].

CAUTION MAKE SURE THAT YOU INSTALL THE FAN BLADE SPACER CORRECTLY. THE PULLER HOLE MUST BE IN THE FORWARD DIRECTION. INCORRECT INSTALLATION CAN CAUSE DAMAGE TO THE FAN BLADE SPACER.

(e) Engage the (No. 18) fan blade spacer [9] in the dovetail slot between the (No. 18) fan blade [1] and fan disk [10].

1) Use a non-metallic mallet, STD-583, to gently tap the fan blade spacer [9] into position.

(f) Do the steps above again to install the remaining 17 fan blades [1] and fan blade spacer [9].

(g) Install the 18 fan blade platforms [6] (TASK 72-22-03-400-801-G00).

(h) Make sure that the fan blades [1], fan blade platforms [6], and fan blade spacer [9] are in the correct positions.

(i) Install the platform front shroud [3].

1) Use the positioner, SPL-14660, to install the barrel nuts [5] in the correct position on the fan disk [10].

2) Lubricate the threads of the 18 bolts [4] with the graphite mineral vaseline, D00601 [CP2101].

3) Install the guide pins, SPL-14661, at the 3 o'clock and 9 o'clock positions to align the platform front shroud [3] with the fan disk [10].

4) Install the platform front shroud [3] in the correct position on the fan disk [10] through the guide pins, SPL-14661.

5) Install the shroud front platform fixture, SPL-15044.

6) Push the platform front shroud [3] on the fan disk [10] with the shroud front platform fixture, SPL-15044.

7) Remove the guide pins, SPL-14661.

8) Install six bolts [4] to attach the platform front shroud [3] to the fan disk [10].

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9) Remove the shroud front platform fixture, SPL-15044, from the platform front shroud [3]

10) Install the remaining 12 bolts [4] that attach the platform front shroud [3] to the fan disk [10].

11) Make sure that the 18 bolts [4] align correctly with the 18 barrel nuts [5].

12) Tighten the bolts [4] to 133 in-lb (15 N·m) - 186 in-lb (21 N·m).

13) Remove the excess of graphite mineral vaseline, D00601 [CP2101], with lint-free cloth, G51335 [CP2674].

(j) Clean the work area and clear the tools and other items.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-22-02-410-004-G00

(1) Do this task: Front Spinner Cone Installation, TASK 72-22-01-400-801-G00.

SUBTASK 72-22-02-080-003-G00

(2) Make sure that you remove the tape that holds the fan blade platforms to the fan blades.

SUBTASK 72-22-02-860-018-G00

(3) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-22-02-860-004-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

H. Fan Blades (Set) Installation Test

SUBTASK 72-22-02-710-002-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

TASK 72-22-02-960-802-G00

8. Fan Blade Replacement

(Figure 404 or Figure 403)

A. General

(1) This task gives the instructions to replace the fan blade.

(2) You can get access to the fan blades through the inlet cowl after you remove the front spinner cone and platform front shroud.

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(3) Each engine has eighteen fan blades that are installed in the fan disk.

(4) Each fan blade is attached with a spacer and lock and held in its position with a platform.

(5) Replace the fan blades in single pair or in multiple pair sets. In case a pair is not available, there is a procedure to replace a individual fan blade.

NOTE: Three moment weights named as axial, tangential, and radial are engraved on the fan blade root. These moment weights are used when a fan blade mapping is necessary. For the fan blade replacement of less than or equal to three pairs, only the radial moment weight is to be considered.

NOTE: Spare blades are delivered in pairs so that the difference between the radial moment weights of the two fan blades is less than or equal to 0.52 in-lb (600 cm-g), as for all opposite fan blades constituting a set.

NOTE: If you replace more than three pairs of fan blades you must do a fan blade mapping and a vibration data check.

B. References

Reference	Title
71-00-00-720-803-G00	Test No. 9 - Low Pressure System (Fan/LPT) Vibration Survey (P/B 501)
71-00-00-720-805-G00	Test No. 11 - Low Pressure System (Fan) Trim Balance (With ONS) (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-15594	Software, Mapping - Fan Blade Part #: 956A7671G01 Supplier: 58828 Part #: 956A7671G02 Supplier: 58828
STD-585	Mat - Protective, Closed Cell Neoprene, 30-50 duro, Weather and Oil Resistant, 3/8 Inch Min Thickness, Minimum 42x60 Inches with Warning Streamers
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Replacement

SUBTASK 72-22-02-860-026-G00

(1) Do these steps to make sure that the ENGINE START switch and FUEL CONTROL switch are not operated:

(a) On the P5 pilot's overhead panel, make sure that the applicable ENGINE START switch is in the NORM (push in) position.

1) Put the DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) On the P8 pilot's aisle control stand, make sure that the applicable FUEL CONTROL switch is in the CUTOFF position.

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1) Put the DO NOT OPERATE tag, STD-858, on the applicable FUEL CONTROL switch.

SUBTASK 72-22-02-480-004-G00

CAUTION

PUT THE PROTECTIVE MAT IN THE INLET COWL. THIS WILL PREVENT DAMAGE TO THE INLET COWL OR ENGINE BY TOOLS, PARTS, OR UNWANTED MATERIALS THAT FALL ON THE INLET COWL SURFACE.

(2) Put the protective mat, STD-585, in the lower half of the inlet cowl to prevent damage to the inlet cowl and the engine.

F. Fan Blade Pairs Replacement

SUBTASK 72-22-02-960-009-G00

(1) Two conditions can exist as given below:

(a) Up to three pairs replaced.

(b) More than three pairs replaced.

1) If you replace more than three pairs of fan blades, you must do the following steps:

a) Remove bolt nuts and balance weights from the platform front shroud.

b) Do a blade mapping with the mapping software, SPL-15594.

c) Reinstall the fan blades, do this task: Fan Blades (Pair) Installation, TASK 72-22-02-400-802-G00.

d) Do a vibration survey test, do this task: Test No. 9 - Low Pressure System (Fan/LPT) Vibration Survey, TASK 71-00-00-720-803-G00.

e) If the vibration survey test is not satisfactory, do a fan trim balance test, do this task: Test No. 11 - Low Pressure System (Fan) Trim Balance (With ONS), TASK 71-00-00-720-805-G00.

SUBTASK 72-22-02-020-011-G00

(2) Do this task: Fan Blades (Pair) Removal, TASK 72-22-02-000-802-G00.

(a) Record the S/N, moment weights, and position of each fan blade (position of the heaviest and lightest blade for each pair replaced).

SUBTASK 72-22-02-960-010-G00

(3) If the quantity of pairs replaced is less than or equal to three and the radial moment weight for each pair is less than or equal to 0.52 in-lb (600 cm-g), do these steps:

(a) Do not remove the weight balance on the platform front shroud.

(b) Install the new fan blades with the same orientation (heaviest and lightest) (TASK 72-22-02-400-802-G00).

(c) Return the engine to service.

NOTE: A Vibration Survey test (Test No. 9 - Low Pressure System (Fan/LPT) Vibration Survey, TASK 71-00-00-720-803-G00) is at customer convenience.

SUBTASK 72-22-02-960-011-G00

(4) If the quantity of pairs is more than three, or the radial moment weight of at least one pair is more than 0.52 in-lb (600 cm-g), do these steps:

(a) Remove the bolt, nuts, and balance weight from the platform front shroud.

(b) Do a blade mapping with the mapping software, SPL-15594.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Reinstall the fan blades, do this task: Fan Blades (Pair) Installation, TASK 72-22-02-400-802-G00.

(d) Do a vibration survey test, do this task: Test No. 9 - Low Pressure System (Fan/LPT) Vibration Survey, TASK 71-00-00-720-803-G00.

(e) If the vibration survey test is not satisfactory, do a fan trim balance test, do this task: Test No. 11 - Low Pressure System (Fan) Trim Balance (With ONS), TASK 71-00-00-720-805-G00.

G. Individual Fan Blade Replacement

SUBTASK 72-22-02-020-012-G00

(1) Remove the fan blade to be replaced and the opposite fan blade (TASK 72-22-02-000-802-G00).

(a) Record the S/N, moment weights, and position of each fan blade (position of the heaviest and lightest blade for each pair replaced).

SUBTASK 72-22-02-960-012-G00

(2) If the number of blades to be replaced is less than or equal to three, do the applicable step:

(a) If the radial moment weight of the blades to be replaced is less than or equal to 0.52 in-lb (600 cm-g) versus the opposite remaining blade, do the fan blade pairs replacement procedure above.

(b) If the number of blades to be replaced is less than or equal to three, and the radial moment weight of the blades to be replaced is more than or equal to 0.52 in-lb (600 cm-g) versus the opposite remaining blade, do these steps:

1) Do a static correction with the mapping software, SPL-15594.

a) If the static correction solution is possible, position bolts and weights as recommended by the software (TASK 71-00-00-720-805-G00).

NOTE: There are three configurations of balancing device. They are as follows:

- Hole-free
- Hole with bolt and nut
- Hole with bolt, nut, and weight.

<1> Do a vibration survey test, do this task: Test No. 9 - Low Pressure System (Fan/LPT) Vibration Survey, TASK 71-00-00-720-803-G00.

<2> If the vibration survey test is not satisfactory, do a fan trim balance test, do this task: Test No. 11 - Low Pressure System (Fan) Trim Balance (With ONS), TASK 71-00-00-720-805-G00.

b) Do these steps if the static correction solution is not given by the software due to the static correction more than the maximum correction capacity:

<1> Remove the bolt nuts and balance weight from the platform front shroud.

<2> Do a blade mapping with the mapping software, SPL-15594.

<3> Reinstall the fan blades, do this task: Fan Blades (Pair) Installation, TASK 72-22-02-400-802-G00.

<4> Do a vibration survey test, do this task: Test No. 9 - Low Pressure System (Fan/LPT) Vibration Survey, TASK 71-00-00-720-803-G00.

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<5> If the vibration survey test is not satisfactory, do a fan trim balance test, do this task: Test No. 11 - Low Pressure System (Fan) Trim Balance (With ONS), TASK 71-00-00-720-805-G00.

SUBTASK 72-22-02-960-008-G00

(3) Do these steps if the number of blades to be replaced is more than three:

(a) Remove bolt nuts and balance weights from the platform front shroud.

(b) Do a blade mapping with the mapping software, SPL-15594.

(c) Reinstall the fan blades, do this task: Fan Blades (Pair) Installation, TASK 72-22-02-400-802-G00.

(d) Do a vibration survey test, do this task: Test No. 9 - Low Pressure System (Fan/LPT) Vibration Survey, TASK 71-00-00-720-803-G00.

(e) If the vibration survey test is not satisfactory, do a fan trim balance test, do this task: Test No. 11 - Low Pressure System (Fan) Trim Balance (With ONS), TASK 71-00-00-720-805-G00.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-22-02-960-007-G00

CAUTION

MAKE SURE THERE ARE NO TOOLS, PARTS OR OTHER UNWANTED MATERIAL IN THE INLET AREA. DAMAGE TO EQUIPMENT COULD OCCUR.

CAUTION

MAKE SURE THERE ARE NO TOOLS, PARTS, OR OTHER UNWANTED MATERIAL IN THE INLET COWL. IF YOU DO NOT OBEY THESE INSTRUCTIONS, YOU CAN CAUSE DAMAGE TO THE ENGINE.

(1) Remove the protective mat, STD-585, from the lower half of the inlet cowl.

SUBTASK 72-22-02-840-009-G00

(2) Do these steps to remove the DO NOT OPERATE tags from the applicable ENGINE START switch and FUEL CONTROL switch:

(a) On the P5 pilot's overhead panel, remove the DO NOT OPERATE tag from the applicable ENGINE START switch.

1) Put the applicable ENGINE START switch to the NORM (push in) position.

(b) On the P8 pilot's aisle control stand, remove the DO NOT OPERATE tag from the applicable FUEL CONTROL switch.

1) Put the applicable FUEL CONTROL switch to the CUTOFF position.

END OF TASK

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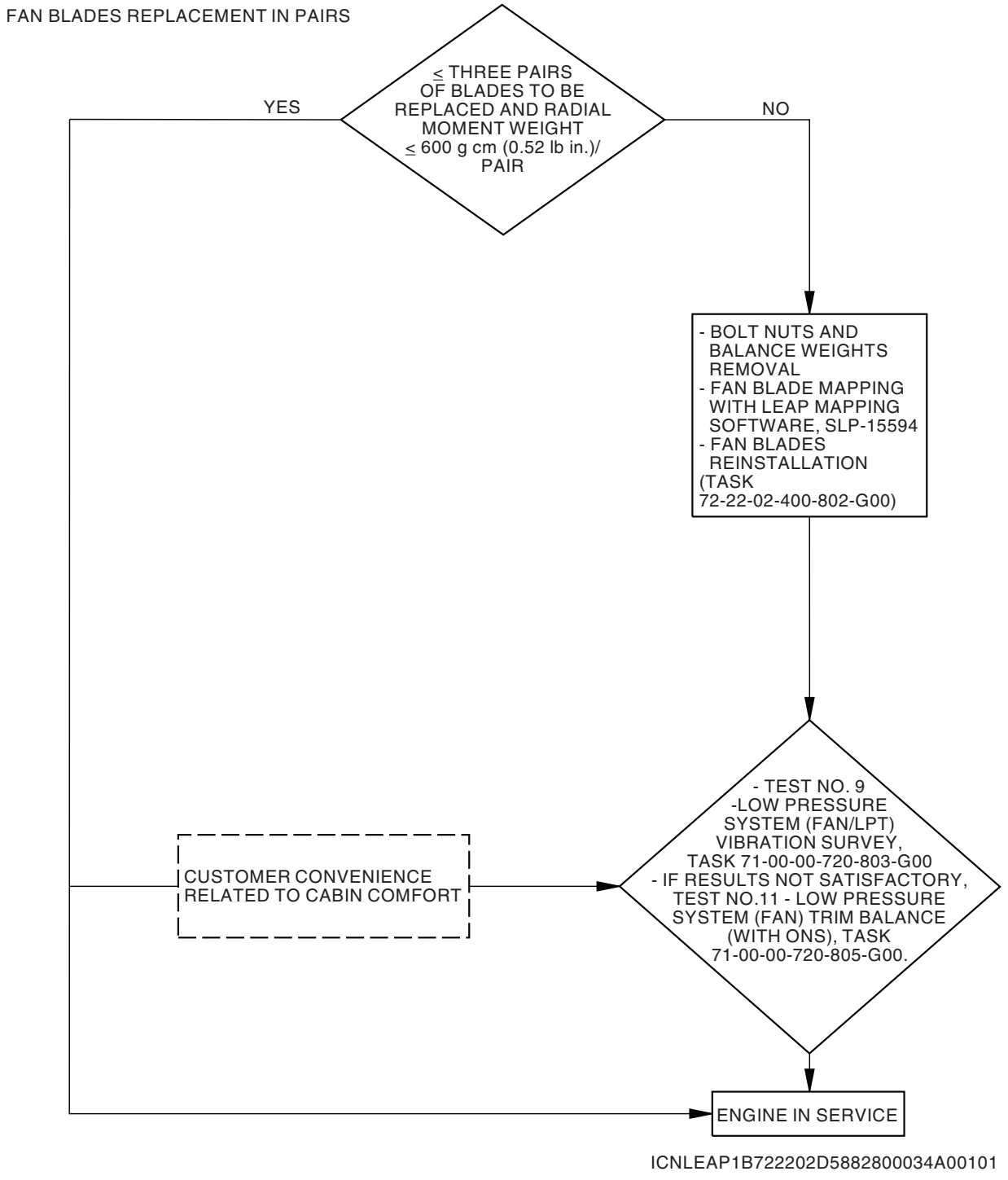
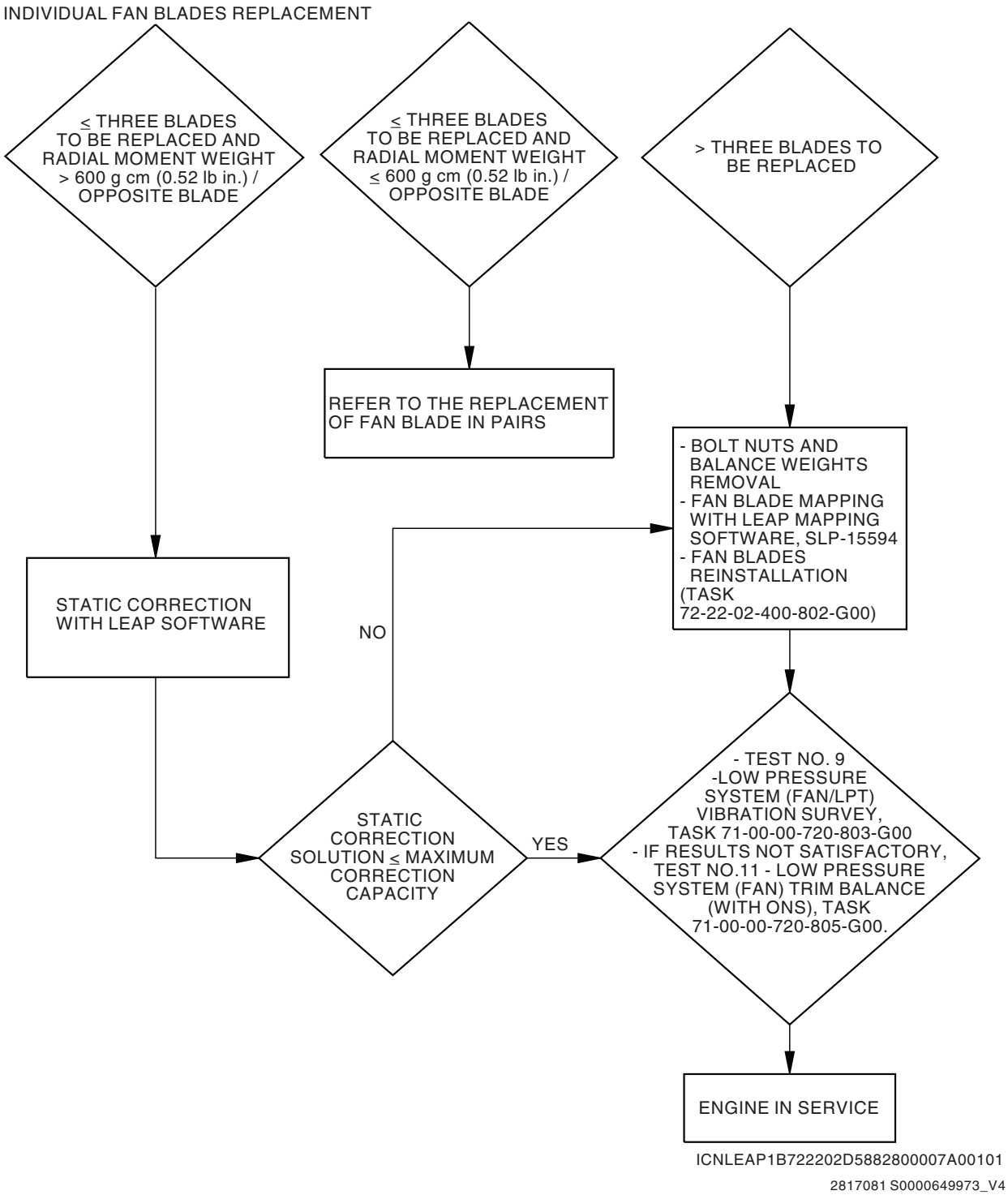


Fan Blade Pairs Replacement
Figure 403/72-22-02-990-837-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Individual Fan Blade Replacement
Figure 404/72-22-02-990-836-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN BLADES - INSPECTION/CHECK

1. General

A. This procedure has this task:

(1) Fan Blade Inspection.

TASK 72-22-02-200-801-G00

2. Fan Blade Inspection

(Figure 601, Figure 602, Figure 603, Figure 604, and Figure 605)

A. General

(1) This task gives the instructions to examine the fan blades.

B. References

Reference	Title
72-22-02-300-801-G00	Fan Blade Repair of the Missing Polyurethane (PU) Film with PU Paste (P/B 801)
72-22-02-900-802-G00	Fan Blade Coating Replacement (Local Repair of Anti-Erosion Paint) (P/B 801)
72-22-02-900-803-G00	Fan Blade Coating Replacement (Local Repair of Platform Wear-Strips) (P/B 801)
72-22-02-900-804-G00	Fan Blade Label Replacement (P/B 801)
72-22-02-900-805-G00	Fan Blade Metal Leading Edge Blending (P/B 801)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Inspection

SUBTASK 72-22-02-860-005-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-22-02-860-006-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

F. Fan Blade Inspection

NOTE: In case of unserviceable condition of the fan blade due to bird/Foreign Object Damage (FOD) ingestion, do the inspection of the dovetail and shank (root section of the fan blade up to the inner annulus flow path line). Any findings in the composite (carbon fiber) more than manual limits that show of a possible crack or missing material in the dovetail and shank must be reported to CFM.

SUBTASK 72-22-02-211-003-G00

(1) Do these steps to examine the fan blade [1]:

(a) Examine the Metallic Leading Edge (MLE) sidewall in area A for these conditions (Figure 601 (Sheet 1)):

1) Disbonding on the concave (pressure) or convex (suction) side

a) One area disbonded with an area of 0.223 in^2 (144 mm^2) is permitted.

b) Do a tap test of the MLE with an aluminum mallet that weighs between 1.59 oz (45.08 g) and 3.53 oz (100.07 g) as follows:

NOTE: If it is necessary, make a mallet to the weight specified in the step and as shown in Figure 601 (Sheet 23).

NOTE: If a solid click is heard, the area is bonded. If a hollow sound is heard, the area is disbonded.

<1> Tap on each side (convex and concave) of the fan blade.

<2> Tap from the end of the MLE nose to the MLE sidewalls.

<3> Tap from the bottom of the MLE to the tip.

2) Nicks, dents and scratches at concave (pressure) side

a) Permitted, up to 0.0007 in. (0.02 mm) in depth.

3) Wear and erosion

a) Not permitted.

4) Crease

a) Not permitted.

5) Cracks and tears

a) Permitted in the concave (pressure) side.

b) Not permitted on the convex (suction) side.

6) Glazing

a) Any amount if less than 0.003 in. (0.1 mm) of depth on the concave side.

b) Any amount without depth on the convex side.

(b) Examine the MLE nose of the area B for these conditions (Figure 601 (Sheet 1)):

1) Nicks, dents and scratches

a) Permitted, up to 0.0059 in. (0.15 mm) in depth.

b) Maximum Repairable Limit: In the limits of applicable repair.

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c) Repair Method: Fan Blade Metal Leading Edge Blending, TASK 72-22-02-900-805-G00.

2) Cracks and tears

a) Not permitted.

(c) Examine the MLE sidewall in area B for these conditions (Figure 601 (Sheet 1)):

1) Disbonding on the concave (pressure) or convex (suction) side

a) One area disbonded with an area of 0.223 in^2 (144 mm^2) is permitted.

b) Do a tap test of the MLE with an aluminum mallet that weighs between 1.59 oz (45.08 g) and 3.53 oz (100.07 g) as follows:

NOTE: If it is necessary, make a mallet to the weight specified in the step and as shown in Figure 601 (Sheet 23).

NOTE: If a solid click is heard, the area is bonded. If a hollow sound is heard, the area is disbonded.

<1> Tap on each side (convex and concave) of the fan blade.

<2> Tap from the end of the MLE nose to the MLE sidewalls.

<3> Tap from the bottom of the MLE to the tip.

2) Nicks and dents at concave (pressure) side

a) Maximum Serviceable Limit: If the distance from the tip is 22.17 in. (563 mm) - 23.03 in. (585 mm).

<1> If the distance from the leading edge is ≤ 0.71 in. (18 mm).

<a> Any number up to 0.0035 in. (0.09 mm) in depth.

<2> If the distance from the leading edge is 0.71 in. (18 mm) - 1.77 in. (45 mm).

<a> Any number up to 0.0027 in. (0.07 mm) in depth.

<3> If the distance from the leading edge is > 1.77 in. (45 mm).

<a> Any number up to 0.0019 in. (0.05 mm) in depth.

b) Maximum Serviceable Limit: If the distance from the tip is 23.03 in. (585 mm) - 24.37 in. (619 mm).

<1> If the distance from the leading edge is ≤ 0.63 in. (16 mm).

<a> Any number up to 0.0027 in. (0.07 mm) in depth.

<2> If the distance from the leading edge is > 0.63 in. (16 mm).

<a> Any number up to 0.0019 in. (0.05 mm) in depth.

c) Maximum Serviceable Limit: If the distance from the tip is 24.37 in. (619 mm) - 25.23 in. (641 mm).

<1> Any number up to 0.0011 in. (0.03 mm) in depth.

3) Scratches at concave (pressure) side

a) Permitted, up to 0.0007 in. (0.02 mm) in depth.

4) Wear and erosion

a) Not permitted.

5) Cracks and tears

a) Not permitted.

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6) Crease

a) Not permitted.

(d) Examine the MLE Nose of the area C (Figure 601 (Sheet 2)):

1) Nicks, dents and scratches.

a) Permitted, up to 0.0078 in. (0.20 mm) in depth.

b) Maximum Repairable Limit: In the limits of applicable repair.

c) Repair Method: Fan Blade Metal Leading Edge Blending, TASK 72-22-02-900-805-G00.

2) Cracks and tears

a) Not permitted.

(e) Examine the MLE sidewall in area C for these conditions (Figure 601 (Sheet 2)):

1) Disbonding on the concave (pressure) or convex (suction) side

a) One area disbonded with an area of 0.223 in^2 (144 mm^2) is permitted.

b) Do a tap test of the MLE with an aluminum mallet that weighs between 1.59 oz (45.08 g) and 3.53 oz (100.07 g) as follows:

NOTE: If it is necessary, make a mallet to the weight specified in the step and as shown in Figure 601 (Sheet 23).

NOTE: If a solid click is heard, the area is bonded. If a hollow sound is heard, the area is disbonded.

<1> Tap on each side (convex and concave) of the fan blade.

<2> Tap from the end of the MLE nose to the MLE sidewalls.

<3> Tap from the bottom of the MLE to the tip.

2) Nicks and dents at concave (pressure) side

a) Maximum Serviceable Limit: If the distance from the tip is 19.10 in. (485 mm) - 21.02 in. (534 mm).

<1> If the distance from the leading edge is ≤ 1.15 in. (29 mm).

<a> Any number up to 0.0059 in. (0.15 mm) in depth.

<2> If the distance from the leading edge is 1.15 in. (29 mm) - 2.08 in. (53 mm).

<a> Any number up to 0.0047 in. (0.12 mm) in depth.

<3> If the distance from the leading edge is > 2.08 in. (53 mm).

<a> Any number up to 0.0027 in. (0.07 mm) in depth.

b) Maximum Serviceable Limit: If the distance from the tip is 21.02 in. (534 mm) - 22.16 in. (563 mm).

<1> If the distance from the leading edge is ≤ 0.60 in. (15 mm).

<a> Any number up to 0.0059 in. (0.15 mm) in depth.

<2> If the distance from the leading edge is 0.60 in. (15 mm) - 2.16 in. (55 mm).

<a> Any number up to 0.0027 in. (0.07 mm) in depth.

<3> If the distance from the leading edge is > 2.16 in. (55 mm).

<a> Any number up to 0.0019 in. (0.05 mm) in depth.

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3) Scratches at concave (pressure) side
   a) Permitted, up to 0.0007 in. (0.02 mm) in depth.
4) Nicks, dents, and scratches at convex (suction) side
   a) Permitted, up to 0.0007 in. (0.02 mm) in depth.
5) Wear and erosion
   a) Not permitted.
6) Cracks and tears
   a) Not permitted.
7) Crease
   a) Not permitted.

(f) Examine the MLE nose of the area D for these conditions (Figure 601 (Sheet 2)):
1) Nicks, dents and scratches
   a) Permitted, up to 0.0098 in. (0.25 mm) in depth.
   b) Continue-In-Service limit: 50 cycles is permitted with the criteria Permitted, up to 0.011 in. (0.3 mm) in depth.
      NOTE: This over serviceable limit extension is for a one-time continue-in-service application.
   c) Maximum Repairable Limit: In the limits of applicable repair.
   d) Repair Method: Fan Blade Metal Leading Edge Blending, TASK 72-22-02-900-805-G00.
2) Cracks and tears
   a) Not permitted.

(g) Examine the MLE sidewall in area D for these conditions (Figure 601 (Sheet 2)):
1) Disbonding on the concave (pressure) or convex (suction) side
   a) One area disbonded with an area of 0.223 in^2 (144 mm^2) is permitted.
   b) Do a tap test of the MLE with an aluminum mallet that weighs between 1.59 oz (45.08 g) and 3.53 oz (100.07 g) as follows:
      NOTE: If it is necessary, make a mallet to the weight specified in the step and as shown in Figure 601 (Sheet 23).
      NOTE: If a solid click is heard, the area is bonded. If a hollow sound is heard, the area is disbonded.
      <1> Tap on each side (convex and concave) of the fan blade.
      <2> Tap from the end of the MLE nose to the MLE sidewalls.
      <3> Tap from the bottom of the MLE to the tip.
2) Nicks and dents at concave (pressure) side
   a) Maximum Serviceable Limit: If the distance from the tip is 15.75 in. (400 mm) - 17.32 in. (440 mm).
      <1> If the distance from the leading edge is ≤ 2.33 in. (59 mm).
         <a> Any number up to 0.0059 in. (0.15 mm) in depth.
      <2> If the distance from the leading edge is > 2.33 in. (59 mm).
         <a> Any number up to 0.0047 in. (0.12 mm) in depth.

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b) Maximum Serviceable Limit: If the distance from the tip is 17.32 in. (440 mm) - 19.09 in. (485 mm).

<1> If the distance from the leading edge is ≤ 1.78 in. (45 mm).

<a> Any number up to 0.0059 in. (0.15 mm) in depth.

<2> If the distance from the leading edge is 1.78 in. (45 mm) - 2.95 in. (75 mm).

<a> Any number up to 0.0047 in. (0.12 mm) in depth.

<3> If the distance from the leading edge is > 2.95 in. (75 mm).

<a> Any number up to 0.0039 in. (0.10 mm) in depth.

3) Scratches at concave (pressure) side

a) Permitted, up to 0.0039 in. (0.10 mm) in depth.

4) Nicks, dents and scratches at convex (suction) side

a) Permitted, up to 0.0059 in. (0.15 mm) in depth.

5) Wear and erosion

a) Not permitted.

6) Cracks and tears

a) Not permitted.

7) Crease

a) Not permitted.

(h) Examine the MLE nose of the area E for these conditions (Figure 601 (Sheet 3)):

1) Nicks, dents and scratches

a) Permitted, up to 0.0137 in. (0.35 mm) in depth.

b) Continue-In-Service limit: 50 cycles is permitted with all amounts criteria, up to 0.0157 in. (0.40 mm) in depth.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

c) Maximum Repairable Limit: In the limits of applicable repair.

d) Repair Method: Fan Blade Metal Leading Edge Blending, TASK 72-22-02-900-805-G00.

2) Cracks and tears.

a) Not permitted.

(i) Examine the MLE sidewall in area E for these conditions (Figure 601 (Sheet 3)):

1) Nicks and dents at concave (pressure) side (Figure 601 (Sheet 12))

a) Maximum Serviceable Limit: If the axial distance between the MLE paint transition line and the indication is ≤ 0.28 in. (7 mm).

<1> Any number up to 0.0007 in. (0.02 mm) in depth.

b) Maximum Serviceable Limit: If the axial distance between the MLE paint transition line and the indication is > 0.28 in. (7 mm).

<1> If the distance from the tip is 4.57 in. (116 mm) - 8.93 in. (227 mm).

<a> If the distance from the leading edge is ≤ 2.17 in. (55 mm), any number up to 0.0059 in. (0.15 mm) in depth.

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<b> If the distance from the leading edge is > 2.17 in. (55 mm), any number up to 0.0047 in. (0.12 mm) in depth.

<2> If the distance from the tip is 8.93 in. (227 mm) - 12.24 in. (311 mm).

<a> If the distance from the leading edge is ≤ 2.96 in. (75 mm), any number up to 0.0059 in. (0.15 mm) in depth.

<b> If the distance from the leading edge is 2.96 in. (75 mm) - 4.13 in. (105 mm), any number up to 0.0047 in. (0.12 mm) in depth.

<c> If the distance from the leading edge is > 4.13 in. (105 mm), any number up to 0.0035 in. (0.09 mm) in depth.

<3> If the distance from the tip is 12.24 in. (311 mm) - 15.74 in. (400 mm).

<a> If the distance from the leading edge is ≤ 2.96 in. (75 mm), any number up to 0.0059 in. (0.15 mm) in depth.

<b> If the distance from the leading edge is > 2.96 in. (75 mm), any number up to 0.0047 in. (0.12 mm) in depth.

2) Scratches at concave (pressure) side.

a) Permitted, up to 0.0007 in. (0.02 mm) in depth.

3) Nicks, dents and scratches at convex (suction) side

a) Permitted, up to 0.0019 in. (0.05 mm) in depth.

4) Wear and erosion

a) Not permitted.

5) Cracks and tears

a) Not permitted.

6) Crease

a) Not permitted.

7) Disbonding at concave (pressure) and convex (suction) side

a) Not permitted.

(j) Examine the MLE nose of the area F for these conditions (Figure 601 (Sheet 3)):

1) Nicks , dents and scratches

a) Permitted, up to 0.039 in. (1.0 mm) in depth.

b) Continue-In-Service limit: 50 cycles is permitted with the up to 0.0531 in. (1.35 mm) in depth criteria.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

c) Maximum Repairable Limit: In the limits of applicable repair.

d) Repair Method: Fan Blade Metal Leading Edge Blending, TASK 72-22-02-900-805-G00.

2) Cracks and tears

a) Not permitted.

3) Coloration up to 0.197 in. (5.0 mm) from tip at concave side (Figure 603).

a) Permitted.

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(k) Examine the MLE sidewall in area F for these conditions (Figure 601 (Sheet 3) and Figure 601 (Sheet 5)):

1) Nicks, dents and scratches

a) Permitted, up to 0.0059 in. (0.15 mm) in depth.

2) Wear and erosion

a) Not permitted.

3) Cracks and tears

a) Not permitted.

4) Crease.

a) Not permitted.

5) Coloration up to 0.197 in. (5.0 mm) from tip at concave side (Figure 603).

a) Permitted.

6) Disbonding at concave (pressure) side in area F except F1 area

a) Not permitted.

7) Disbonding at concave (pressure) side in area F1

a) Permitted.

b) If disbonding is detected, get an aluminum mallet that weighs between 1.59 oz (45.08 g) and 3.53 oz (100.07 g).

NOTE: If it is necessary, make a mallet to the weight specified in the step and as shown in Figure 601 (Sheet 23).

c) Do a tap test to determine the disbonded surface on the MLE sidewall on the areas C, D, E, F, F1, and F2 on concave (pressure) side as follows:

NOTE: If a solid click is heard, the area is bonded. If a hollow sound is heard, the area is disbonded,

<1> Tap on each concave side of the fan blade.

<2> Tap from the end of the MLE nose to the MLE sidewalls.

<3> Tap from the bottom of the MLE to the tip.

<4> Visually check the convex (suction) side for disbonding, per corresponding paragraphs of the task.

8) Disbonding at concave (pressure) side in area F2

a) Not permitted.

b) Continue-In-Service limit: 100 cycles is permitted for one of the two engines and for three fan blades, with all amounts permitted.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

c) If disbonding is detected, get an aluminum mallet that weighs between 1.59 oz (45.08 g) and 3.53 oz (100.07 g).

NOTE: If it is necessary, make a mallet to the weight specified in the step and as shown in Figure 601 (Sheet 23).

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d) Do a tap test to determine the disbonded surface on the MLE sidewall on the areas C, D, E, F, F1, and F2 on concave (pressure) side as follows:

NOTE: If a solid click is heard, the area is bonded. If a hollow sound is heard, the area is disbonded.

<1> Tap on each concave side of the fan blade.

<2> Tap from the end of the MLE nose to the MLE sidewalls.

<3> Tap from the bottom of the MLE to the tip.

<4> Visually check the convex (suction) side for disbonding, per corresponding paragraphs of the task.

9) Step between sidewall and blade profile at concave (pressure) side in area F1 and F2

a) 0.059 in. (1.5 mm) at the tip.

b) If disbonding is detected, get an aluminum mallet that weighs between 1.59 oz (45.08 g) and 3.53 oz (100.07 g).

NOTE: If it is necessary, make a mallet to the weight specified in the step and as shown in Figure 601 (Sheet 23).

c) Do a tap test to determine the disbonded surface on the MLE sidewall on the areas C, D, E, F, F1 and F2 on concave (pressure) side as follows:

NOTE: If a solid click is heard, the area is bonded. If a hollow sound is heard, the area is disbonded.

<1> Tap on each concave side of the fan blade.

<2> Tap from the end of the MLE nose to the MLE sidewalls.

<3> Tap from the bottom of the MLE to the tip.

<4> Visually check the convex (suction) side for disbonding, per corresponding paragraphs of the task.

d) Maximum number of 6 blades on each engine.

e) Continue-In-Service limit: 100 cycles is permitted with the criteria:

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

<1> 0.31 in. (7.9 mm) at the tip.

<2> Maximum one fan blade for one of the two engines.

10) Disbonding at convex (suction) side

a) Up to a total of 0.47 in. (12 mm) in width, up to 3.93 in. (100 mm) cumulative length.

b) Continue-In-Service limit: 100 cycles is permitted for one of the two engines and for three fan blades, with the criteria: Up to a total of 0.78 in. (20 mm) in width, up to 4.33 in. (110 mm) cumulative length.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

c) If disbonding is detected, get an aluminum mallet that weighs between 1.59 oz (45.08 g) and 3.53 oz (100.07 g).

NOTE: If it is necessary, make a mallet to the weight specified in the step and as shown in Figure 601 (Sheet 23).

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d) Do a tap test to determine the disbonded surface on the MLE sidewall on the areas C, D, E, F, F1 and F2 on concave (pressure) side as follows:

NOTE: If a solid click is heard, the area is bonded. If a hollow sound is heard, the area is disbonded.

<1> Tap on each concave side of the fan blade.

<2> Tap from the end of the MLE nose to the MLE sidewalls.

<3> Tap from the bottom of the MLE to the tip.

<4> Visually check the convex (suction) side for disbonding, per corresponding paragraphs of the task.

e) Only one engine is approved to have fan blades over the serviceable limit and the total number of fan blades over the serviceable limit is limited to 3 for this engine.

(l) Examine the MLE glue/adhesive transition areas (Figure 604):

1) Porosities

a) Permitted any number, any amount, if restricted within the glue (not in contact with the composite).

2) Micro-cracking

a) Permitted any number, any amount.

3) Gap between the MLE glue and the platform wearstrip on the concave and convex sides (Figure 604 (Sheet 2)).

a) Permitted up to 0.019 in. (0.5 mm) in axial width.

(m) Examine the blade tip in area G for these conditions (Figure 601 (Sheet 3)):

1) Tip contact, tip rub: melt resin, frayed carbon fiber

a) Permitted.

2) Material deposit on the surface

a) Permitted.

3) Decohesions on convex side

a) One permitted, up to 1.96 in. (50 mm) in length in radial direction.

b) If an indication is detected, do the visual inspection of MLE sidewall on the areas C, D, E, F and F1 on concave (pressure) and convex (suction) sides, per corresponding paragraphs of the task.

4) Tears, cracks, fiber breakage on convex (suction) side

a) Not permitted.

5) Decohesions on concave (pressure) side

a) One permitted, up to 0.78 in. (20 mm) in length in radial direction.

b) If an indication is detected, do the visual inspection of MLE sidewall on the areas C, D, E, F and F1 on concave (pressure) and convex (suction) sides, per corresponding paragraphs of the task.

6) Tears, cracks, fiber breakage on concave (pressure) side

a) Not permitted.

(n) Examine the blade tip in area H for these conditions (Figure 601 (Sheet 3)):

1) Tip contact, tip rub: melt resin, frayed carbon fiber

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a) Permitted.

2) Material deposit on the surface

a) Permitted.

3) Decohesions, tears, cracks, fiber breakage

a) Permitted, tangential length up to 0.11 in. (3 mm) (Figure 601 (Sheet 4)).

(o) Examine the blade tip in area U for these conditions (Figure 601 (Sheet 3)):

1) Tip contact, tip rub: melt resin, frayed carbon fiber

a) Permitted.

2) Material deposit on the surface

a) Permitted.

3) Decohesions, tears, cracks, fiber breakage in 0.19 in. (5 mm) from tip

a) Permitted, tangential length up to 0.11 in. (3 mm) (Figure 601 (Sheet 4)).

4) Decohesions line if not starting from trailing edge more than 0.19 in. (5 mm) to 4.33 in. (110 mm) from tip

a) Permitted.

b) If an indication is detected, do the visual inspection of MLE sidewall on the areas C, D, E, F and F1 on concave (pressure) and convex (suction) sides, per corresponding paragraphs of the task.

5) Missing material on the concave (pressure) and the convex (suction) sides going through the fan blade thickness (visible on both sides):

a) Up to 0.1969 in. (5 mm) in cumulated axial (chord direction) length.

b) Up to 0.1260 in. (3.2 mm) from fan blade tip.

c) Up to 3 fan blades per engine.

(p) Examine the blade in areas G, H and U for these conditions (Figure 601 (Sheet 3)):

1) Missing material on the concave (pressure) side or on the convex (suction) side not going through the fan blade thickness.

a) Up to 1.969 in. (50 mm) in cumulated axial (chord direction) length.

b) Up to 0.1260 in. (3.2 mm) from fan blade tip.

c) Any depth if not going through the fan blade thickness.

2) Missing fiber on tip surface in contact with the abradable strip (if the fan blade is removed from the fan disk).

a) Up to 1.969 in. (50 mm) in cumulated axial (chord direction) length.

b) Up to 0.1260 in. (3.2 mm) in depth.

(q) Examine the blade in area J for these conditions (Figure 601 (Sheet 6)):

1) Decohesions line if not starting from trailing edge

a) Permitted.

b) If an indication is detected, do the visual inspection of MLE sidewall on the areas C, D, E, F and F1 on concave (pressure) and convex (suction) sides, per corresponding paragraphs of the task.

(r) Examine the blade trailing edge areas U, J, K, and L for these conditions (Figure 601 (Sheet 3) and Figure 601 (Sheet 6)):

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1) Decohesions, tears, cracks, fiber breakage in 0.19 in. (5 mm) from trailing edge
   a) Permitted.
2) Decohesions, tears, cracks, fiber breakage for a distance more than 0.19 in. (5 mm) to 1.18 in. (30 mm) from trailing edge for areas U, J, K and L
   a) Up to 2 indications in area L.
   b) Up to 5 indications cumulated permitted over areas U, J, K.
   c) Cumulative axial length from trailing edge: up to 5.11 in. (130 mm).
   d) If an indication is detected, do the visual inspection of MLE sidewall on the areas C, D, E, F and F1 on concave (pressure) and convex (suction) sides, per corresponding paragraphs of the task.
3) Decohesions, tears, cracks, fiber breakage in trailing edge area more than 1.18 in. (30 mm) to 1.57 in. (40 mm) from trailing edge
   a) Not permitted.
   b) Continue-In-Service limit: 50 Cycles permitted with the criteria:

      NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

      <1> Up to 2 indications permitted in area L.
      <2> Up to 5 indications cumulated permitted.
      <3> Cumulative axial length from trailing edge: up to 7.08 in. (180 mm).
      <4> If an indication is detected, do the visual inspection of MLE sidewall on the areas C, D, E, F and F1 on concave (pressure) and convex (suction) sides, per corresponding paragraphs of the task.
4) Step (s) in trailing edge area in areas U, J, and K
   a) Up to 0.039 in. (1.0 mm).
   b) If an indication is detected, do the visual inspection of MLE sidewall on the areas C, D, E, F and F1 on concave (pressure) and convex (suction) sides, per corresponding paragraphs of the task.
5) Step (s) in trailing edge area in area L
   a) Not permitted.

(s) Examine the blade airfoil in area M for these conditions (Figure 601 (Sheet 6)):
   1) Decohesions, tears, cracks, fiber breakage.
      a) Not permitted.

(t) Examine the blade airfoil for the cosmetic paint indications in areas G, H, U, J, K, L, and M (Figure 605).
   1) Linear indication.
      a) Axial indented lines in the adhesive visible on the paint coat on concave (pressure) side.
         <1> 10 indented permitted.

(u) Examine the blade airfoil in areas G, H, U, J, K, L, and M for these conditions (Figure 601 (Sheet 3), Figure 601 (Sheet 6) and Figure 601 (Sheet 7)):
   1) Micro cracking
      a) Permitted.

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2) Black paint coat on the concave (pressure) and convex (suction) side for missing material, disbondings, wear, indentations
   a) Permitted.
   b) Maximum repairable limit: any amount.
   c) To repair the coating, do this task: Fan Blade Coating Replacement (Local Repair of Anti-Erosion Paint), TASK 72-22-02-900-802-G00.
3) Primer on the convex (suction) side for missing material
   a) Up to 0.0387 in^2 (25 mm^2) if the loose coating is cut away.
   b) Continue-In-Service limit: 100 cycles is permitted on all amounts criteria.
      NOTE: This over serviceable limit extension is for a one-time continue-in-service application.
   c) Maximum repairable limit: any amount.
   d) To repair the coating, do this task: Fan Blade Coating Replacement (Local Repair of Anti-Erosion Paint), TASK 72-22-02-900-802-G00.
4) Primer on the convex (suction) side for erosion
   a) Permitted if composite fiber is not visible.
   b) Maximum repairable limit: any amount.
   c) To repair the coating, do this task: Fan Blade Coating Replacement (Local Repair of Anti-Erosion Paint), TASK 72-22-02-900-802-G00.
5) Primer on the concave (pressure) side for erosion, loose or missing material
   a) Permitted.
   b) Maximum repairable limit: any amount.
   c) To repair the coating, do this task: Fan Blade Coating Replacement (Local Repair of Anti-Erosion Paint), TASK 72-22-02-900-802-G00.
6) Black rubber polyurethane coating on the concave (pressure) side for erosion or disbonding
   a) Permitted if composite fiber is not visible.
   b) Continue-In-Service limit: 100 cycles is permitted on all amounts criteria.
      NOTE: This over serviceable limit extension is for a one-time continue-in-service application.
   c) Maximum repairable limit: in the limit of the applicable repair.
   d) Repair Method: Fan Blade Repair of the Missing Polyurethane (PU) Film with PU Paste, TASK 72-22-02-300-801-G00.
7) Black rubber polyurethane coating on the concave (pressure) side for loose or missing material from 0 in. (0 mm) to 0.39 in. (10 mm) from tip.
   a) Up to 0.6975 in^2 (450 mm^2) if the loose coating is cut away.
   b) Maximum cumulated surface for the 18 blades: 3.4875 in^2 (2250 mm^2).
   c) Continue-In-Service limit: 100 cycles are permitted with the criteria of any amount.
      NOTE: This over serviceable limit extension is for a one-time Continue-In-Service application.
   d) Maximum repairable limit: in the limit of the applicable repair.

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e) Repair Method: Fan Blade Repair of the Missing Polyurethane (PU) Film with PU Paste, TASK 72-22-02-300-801-G00.

8) Black rubber polyurethane coating on the concave (pressure) side for loose or missing material beyond 0.39 in. (10 mm) from tip.

a) Up to 0.0387 in^2 (25 mm^2) if the loose coating is cut away

b) Continue-In-Service limit: 100 cycles are permitted with the criteria of any amount.

NOTE: This over serviceable limit extension is for a one-time Continue-In-Service application.

9) Nicks or dents in composite on the concave (pressure) and convex (suction) side. Depth ≤ 0.0078 in. (0.20 mm)

NOTE: If nicks or dents depth measurement is performed from the intact black-painted surface around, it is necessary to consider a coating thickness of 0.01783 in. (0.453 mm) on the concave (pressure) side and 0.00216 in. (0.055 mm) on the convex (suction) side.

a) Permitted

b) Maximum repairable limit: in the limit of the applicable repair.

c) Repair Method: Fan Blade Repair of the Missing Polyurethane (PU) Film with PU Paste, TASK 72-22-02-300-801-G00.

10) Nicks or dents in composite on the concave (pressure) and convex (suction) side. 0.0078 in. (0.20 mm) < Depth ≤ 0.019 in. (0.5 mm)

NOTE: If nicks or dents depth measurement is performed from the intact black-painted surface around, it is necessary to consider a coating thickness of 0.01783 in. (0.453 mm) on the concave (pressure) side and 0.00216 in. (0.055 mm) on the convex (suction) side.

a) Permitted

<1> Up to 0.015 in^2 (10 mm^2) in surface for each nick or dent.

11) Nicks or dents in composite on the concave (pressure) and convex (suction) side. 0.019 in. (0.5 mm) < Depth ≤ 0.039 in. (1.0 mm)

NOTE: If nicks or dents depth measurement is performed from the intact black-painted surface around, it is necessary to consider a coating thickness of 0.01783 in. (0.453 mm) on the concave (pressure) side and 0.00216 in. (0.055 mm) on the convex (suction) side.

a) Not permitted.

b) Continue-In-Service limit: 100 cycles is permitted with all amounts criteria.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

<1> Up to 0.02 in^2 (13 mm^2) in surface for each nick or dent.

<2> Minimum distance of 0.78 in. (20 mm) between 2 nicks or dents for areas H, U, J, K, L, and M.

<3> Minimum distance of 1.58 in. (40 mm) between 2 nicks or dents for area G.

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12) Scratches in composite on the concave (pressure) and convex (suction) side. Depth ≤ 0.0078 in. (0.20 mm)

NOTE: If nicks or dents depth measurement is performed from the intact black-painted surface around, it is necessary to consider a coating thickness of 0.01783 in. (0.453 mm) on the concave (pressure) side and 0.00216 in. (0.055 mm) on the convex (suction) side.

a) Permitted.

13) Scratches in composite on the concave (pressure) and convex (suction) side. 0.0078 in. (0.20 mm) < Depth ≤ 0.019 in. (0.5 mm)

NOTE: If nicks or dents depth measurement is performed from the intact black-painted surface around, it is necessary to consider a coating thickness of 0.01783 in. (0.453 mm) on the concave (pressure) side and 0.00216 in. (0.055 mm) on the convex (suction) side.

a) Not permitted.

b) Continue-In-Service limit: 100 cycles is permitted with all amounts criteria.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

14) Indentation line of the paint at the end of metallic leading edge side wall on the convex (suction) and concave (pressure) side

a) Permitted.

15) Rectangular indentation, paint cracking or missing paint in the area located at the TIP of the inner curve of the fan blade side near leading edge (area G and area H) (Figure 601 (Sheet 25)).

a) Permitted up to 3.93 in. (100 mm) in axial length and 1.96 in. (50 mm) in radial length.

16) Rectangular indentation, black rubber polyurethane coating on the concave (pressure) side visible.

a) Permitted up to 3.93 in. (100 mm) in axial length and 1.96 in. (50 mm) in radial length.

NOTE: If black rubber polyurethane coating missing, refer to black rubber polyurethane coating on the concave (pressure) side for loose or missing material from 0 in. (0 mm) to 0.39 in. (10 mm) from tip.

17) Paint porosities, foreign material covered by paint or fish eyes (Figure 601 (Sheet 24)).

a) Permitted up to 32 cumulated indications:

<1> Up to 32 indications with width less than 0.039 in. (1.0 mm).

<2> Up to 12 indications with width more than 0.039 in. (1.0 mm) and less than 0.079 in. (2.0 mm).

<3> Up to 2 indications with width more than 0.079 in. (2.0 mm) and less than 0.197 in. (5.0 mm).

(v) Examine the blade dovetail and shank if the fan blade removed from the fan disk in area N for these conditions (Figure 601 (Sheet 1) and Figure 601 (Sheet 8)):

1) Cracks, tears, and fiber breakage

a) Not permitted.

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2) Nicks or dents for concave (pressure) side
   a) Up to 0.01575 in. (0.40 mm) in depth, 0.15748 in. (4.0 mm) in width.
      <1> Cumulated length: Up to 1.97 in. (50 mm) in length.
   b) Continue-In-Service limit: 100 cycles is permitted with the criteria of up to 0.04 in. (1 mm) in depth, 0.2 in. (5 mm) in width.
      NOTE: This over serviceable limit extension is for a one-time continue-in-service application.
      <1> Cumulated length: Up to 1.97 in. (50 mm) in length.
3) Nicks or dents for convex (suction) side
   a) Up to 0.01575 in. (0.40 mm) in depth, 0.15748 in. (4.0 mm) in width.
      <1> Cumulated length: Up to 1.97 in. (50 mm) in length.
   b) Continue-In-Service limit: 100 cycles is permitted with the criteria up to 0.04 in. (1 mm) in depth, 0.2 in. (5 mm) in width.
      NOTE: This over serviceable limit extension is for a one-time continue-in-service application.
      <1> Cumulated length: Up to 1.97 in. (50 mm) in length.
4) Scratches for concave (pressure) side
   a) Permitted up to 0.004 in. (0.1 mm) in depth, any length.
5) Scratches for concave (pressure) side
   a) Permitted if depth more than 0.004 in. (0.1 mm) and up to 0.008 in. (0.20 mm), 1.18 in. (30 mm) in length.
6) Scratches for concave (pressure) side
   a) Not permitted if depth more than 0.008 in. (0.20 mm) and up to 0.04 in. (1 mm).
   b) Continue-In-Service limit: 100 cycles with the criteria one scratch permitted. Up to 3.54 in. (90 mm) in length.
      NOTE: This over serviceable limit extension is for a one-time continue-in-service application.
7) Scratches for convex (suction) side
   a) Permitted up to 0.004 in. (0.1 mm) in depth, any length.
8) Scratches for convex (suction) side
   a) Permitted if depth more than 0.004 in. (0.1 mm) and up to 0.008 in. (0.20 mm), 1.18 in. (30 mm) in length.
9) Scratches for convex (suction) side
   a) Not permitted if depth more than 0.008 in. (0.20 mm) and up to 0.04 in. (1 mm).
   b) Continue-In-Service limit: 100 cycles with the criteria of one scratch permitted. Up to 3.54 in. (90 mm) in length.
      NOTE: This over serviceable limit extension is for a one-time continue-in-service application.
10) Disbonding, missing material of the label
   a) Not permitted.

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b) Continue-In-Service limit: 1000 cycles with the criteria of missing or disbonded if P/N and S/N of MLE is fully legible (Figure 602).

NOTE: This over serviceable limit extension is for a reduced interval. Repeat the inspection for legible MLE to allow a new Continue-In-Service limit of 1000 cycles.

c) Maximum repairable limit: in the limits of the applicable repair.

d) To replace the label, do this task: Fan Blade Label Replacement, TASK 72-22-02-900-804-G00.

11) Micro-cracking in the resin on concave (pressure) surface and convex (suction) surface

a) Maximum Serviceable Limit: Any amount.

(w) Examine the blade dovetail and shank if the fan blade removed from the fan disk in area O for these conditions (Figure 601 (Sheet 1) and Figure 601 (Sheet 8)):

1) Cracks, tears, and fiber breakage

a) Not permitted.

2) Nicks or dents for concave (pressure) side

a) In an area below 1.555 in. (39.5 mm) from the blade root.

b) Up to 0.008 in. (0.20 mm) in depth, 0.08 in. (2 mm) in width.

<1> Cumulated length: Up to 0.39 in. (10 mm) in length.

c) Continue-In-Service limit: 100 cycles is permitted with the criteria of up to 0.02 in. (0.5 mm) in depth, 0.08 in. (2 mm) in width.

<1> Cumulated length: Up to 0.39 in. (10 mm) in length.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

<2> More than 1.555 in. (39.5 mm) from the blade root.

d) Up to 0.01575 in. (0.40 mm) in depth, 0.15748 in. (4.0 mm) in width.

<1> Cumulated length: Up to 0.39 in. (10 mm) in length.

e) Continue-In-Service limit: 100 cycles is permitted with the criteria of up to 0.02 in. (0.51 mm) in depth, 0.08 in. (2.0 mm) in width.

<1> Cumulated length: Up to 0.39 in. (10 mm) in length.

NOTE: This over serviceable limit extension is for a one-time-continue-in-service application.

3) Nicks or dents for convex (suction) side

a) In an area below 1.555 in. (39.5 mm) from the blade root.

b) Up to 0.008 in. (0.20 mm) in depth, 0.08 in. (2 mm) in width.

<1> Cumulated length: Up to 0.39 in. (10 mm) in length.

c) Continue-In-Service limit: 100 cycles is permitted with the criteria of up to 0.02 in. (0.5 mm) in depth, 0.08 in. (2 mm) in width.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

<1> Cumulated length: Up to 0.39 in. (10 mm) in length.

<2> More than 1.555 in. (39.5 mm) from the blade root.

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d) Up to 0.01575 in. (0.40 mm) in depth, 0.15748 in. (4.0 mm) in width.

<1> Cumulated length: Up to 0.39 in. (10 mm) in length.

e) Continue-In-Service limit: 100 cycles is permitted with the criteria of up to 0.02 in. (0.51 mm) in depth, 0.08 in. (2.0 mm) in width.

NOTE: This over serviceable limit extension is for a one-time-continue-in-service application.

<1> Cumulated length: Up to 0.39 in. (10 mm) in length.

4) Scratches for concave (pressure) side

a) Permitted up to 0.004 in. (0.1 mm) in depth, any amount.

<1> Any length.

5) Scratches for concave (pressure) side

a) Permitted if depth more than 0.004 in. (0.1 mm) and up to 0.008 in. (0.2 mm), 1.18 in. (30 mm) in length and a minimum distance between two scratches of 1.18 in. (30 mm).

<1> Cumulated length: Up to 4.72 in. (120 mm).

b) Continue-In-Service limit: 100 cycles with the criteria one scratch permitted up to 0.02 in. (0.5 mm) in depth and 2.36 in. (60 mm) in length.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

6) Scratches for convex (suction) side

a) Permitted up to 0.004 in. (0.1 mm) in depth, any amount.

<1> Any length.

7) Scratches for convex (suction) side

a) Permitted if depth more than 0.004 in. (0.1 mm) and up to 0.008 in. (0.2 mm), 1.18 in. (30 mm) in length and a minimum distance between two scratches of 1.18 in. (30 mm).

<1> Cumulated length: Up to 4.72 in. (120 mm).

b) Continue-In-Service limit: 100 cycles with the criteria one scratch permitted of up to 0.02 in. (0.5 mm) in depth and 2.36 in. (60 mm) in length.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

8) Micro-cracking in the resin on concave (pressure) surface and convex (suction) surface

a) Maximum Serviceable Limit:

<1> Any amount in radial direction only if the distance from the root is ≤ 0.965 in. (24.5 mm).

<2> Any amount in all directions if the distance from the root is > 0.965 in. (24.5 mm).

(x) Examine the blade dovetail and shank if the fan blade removed from the fan disk in areas N and O for these conditions (Figure 601 (Sheet 1), Figure 601 (Sheet 26), Figure 601 (Sheet 8), and Figure 601 (Sheet 11)):

1) Nicks or dents on leading edge and trailing edge radius over 1.18 in. (30 mm) from bottom root

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a) Any number 0.029 in. (0.74 mm) in depth, up to 0.00155 in^2 (1 mm^2) in damaged section, any length.

b) Continue-In-Service limit: 100 cycles is permitted with the criteria up to 1 permitted per radius 0.039 in. (1.0 mm) in depth, up to 0.0054 in^2 (3.5 mm^2) in damaged section, any length.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

2) Nicks or dents on leading edge and trailing edge radius up to 1.18 in. (30 mm) from bottom root

a) Any number 0.02 in. (0.51 mm) in depth, up to 0.00116 in^2 (0.75 mm^2) in damaged section, any length.

b) Continue-In-Service limit: 100 cycles is permitted with the criteria up to 0.0295 in. (0.75 mm) in depth, up to 0.00155 in^2 (1 mm^2) in damaged section, any length.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

<1> Cumulated surface: Up to 0.007 in^2 (5 mm^2) per radius.

3) Nicks or dents on leading edge and trailing edge radius bottom end 0 in. (0 mm) to 0.059 in. (1.5 mm) from root

a) 1 permitted per radius 0.0098 in. (0.25 mm) in depth, 0.0492 in. (1.25 mm) in length and 0.039 in. (1.0 mm) in width.

b) Continue-In-Service limit: 100 cycles is permitted with the criteria up to 1 permitted per radius 0.019 in. (0.5 mm) in depth, 0.059 in. (1.5 mm) in length and 0.0492 in. (1.25 mm) in width.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

4) Scratches on leading edge and trailing edge radius

a) Up to 1 permitted per radius 0.0078 in. (0.20 mm) in depth and 0.78 in. (20 mm) in length.

b) Continue-In-Service limit: 100 cycles is permitted with the criteria up to 1 permitted per radius 0.019 in. (0.5 mm) in depth and 1.96 in. (50 mm) in length.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

5) Nicks or dents on leading edge surface

a) Up to 0.0078 in. (0.20 mm) in depth with a unitary surface 0.01 in^2 (6 mm^2).

<1> Cumulated surface Up to 0.03 in^2 (19 mm^2).

b) Continue-In-Service limit: 100 cycles is permitted with the criteria up to 0.019 in. (0.5 mm) in depth with a unitary surface 0.03 in^2 (19 mm^2).

<1> Cumulated surface Up to 0.07 in^2 (45 mm^2).

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

6) Scratches on leading edge surface

a) Up to 2 permitted. 0.0078 in. (0.20 mm) in depth, 0.78 in. (20 mm) in length.

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b) Continue-In-Service limit: 100 cycles is permitted with the criteria up to 1 permitted 0.019 in. (0.5 mm) in depth and 0.78 in. (20 mm) in length.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

7) Nicks or dents on trailing edge surface

a) Up to 0.019 in. (0.5 mm) in depth with a unitary surface 0.01 in^2 (6 mm^2).

<1> Cumulated surface Up to 0.06 in^2 (39 mm^2).

b) Continue-In-Service limit: 100 cycles is permitted with the criteria up to 0.019 in. (0.5 mm) in depth with a unitary surface 0.03 in^2 (19 mm^2).

<1> Cumulated surface Up to 0.12 in^2 (77 mm^2)

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

8) Scratches on trailing edge surface

a) Up to 0.0078 in. (0.20 mm) in depth, 1.18 in. (30 mm) in length.

<1> Cumulated length: Up to 3.93 in. (100 mm).

b) Continue-In-Service limit: 100 cycles is permitted with the criteria up to 0.019 in. (0.5 mm) in depth, 1.57 in. (40 mm) in length.

<1> Cumulated length: Up to 3.93 in. (100 mm).

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

9) Decohesion on leading edge and trailing edge radius.

a) Any amount.

(y) Examine the blade dovetail and shank if the fan blade removed from the fan disk in Area P for these conditions (Figure 601 (Sheet 1) and Figure 601 (Sheet 8)):

1) Cracks, tears, and fiber breakage

a) Not permitted.

2) Nicks or dents

a) Up to 0.0098 in. (0.25 mm) in depth, unitary surface: 0.01 in^2 (6 mm^2) and a minimum distance between two indications of 0.39 in. (10 mm).

b) Continue-In-Service limit: 100 cycles is permitted with the criteria up to 0.019 in. (0.5 mm) in depth, unitary surface of 0.03 in^2 (19 mm^2) and a minimum distance between two indications of 0.019 in. (0.5 mm).

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

3) Scratches

a) Up to 0.004 in. (0.1 mm) in depth, any length, any number.

4) Scratches

a) More than 0.004 in. (0.1 mm) and up to 0.008 in. (0.2 mm) in depth, 1.97 in. (50 mm) in length.

<1> Cumulated length: Up to 9.84 in. (250 mm).

5) Scratches

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a) Not permitted if depth more than 0.008 in. (0.20 mm) and up to 0.02 in. (0.51 mm).

b) Continue-In-Service limit: 100 cycles is permitted with the criteria up to 3.15 in. (80 mm) in length.

<1> Cumulated length: Up to 9.84 in. (250 mm).

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

6) Micro-cracking in the resin on concave (pressure) surface and convex (suction) surface

a) Maximum Serviceable Limit: Any amount in radial direction only.

(z) Examine the blade dovetail and shank if the fan blade removed from the fan disk in area Q for these conditions (Figure 601 (Sheet 8) and Figure 601 (Sheet 9)):

1) Decohesion on leading edge face concave (pressure) side

a) Up to 1 permitted, all lengths, 0.11 in. (3 mm) in width.

2) Decohesion on leading edge face convex (suction) side

a) Up to 1 permitted, all lengths, 0.11 in. (3 mm) in width.

(aa) Examine the blade dovetail and shank if the fan blade removed from the fan disk in area R for these conditions (Figure 601 (Sheet 8) and Figure 601 (Sheet 10)):

1) Micro-cracking in the resin on concave (pressure) side and convex (suction) side

a) Permitted.

(ab) Examine the blade dovetail and shank if the fan blade removed from the fan disk in Area S for these conditions (Figure 601 (Sheet 8) and Figure 601 (Sheet 10)):

1) Decohesion on concave (pressure) side and convex (suction) side

a) Permitted.

(ac) Examine the dovetail wear strips if the fan blade removed from the fan disk for these conditions (Figure 601 (Sheet 7) and Figure 601 (Sheet 8)):

1) Loose or missing

a) Permitted if the missing material is up to 0.039 in. (1.0 mm) from one of the edges

b) Continue-In-Service limit: 100 cycles is permitted on all amounts criteria.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

2) Wear

a) Permitted, if not worn through to the composite blade surface, and the worn area is not frayed.

b) Continue-In-Service limit: 100 cycles is permitted on all amounts criteria.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

3) Nicks, dents and scratches

a) Permitted if the composite is not visible and the wear strips surface not frayed.

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b) Continue-In-Service limit: 100 cycles is permitted on all amounts criteria.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

4) Frayed and disbonded corners

a) Permitted if the frayed or disbonded area up to 0.11 in. (3 mm) in length across the diagonal corner of the wear strip.

b) Continue-In-Service limit: 100 cycles is permitted on all amounts criteria.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

5) Frayed and disbonded edges

a) Permitted up to 0.039 in. (1.0 mm) in length from one of the edges.

b) Continue-In-Service limit: 100 cycles is permitted on all amounts criteria.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

6) Discoloration.

a) Permitted.

NOTE: Platform wear strips can be brown or gray.

(ad) Examine the platform wear strips for these conditions (Figure 601 (Sheet 7) and Figure 601 (Sheet 8)):

1) Visual examination of visible platform wear strips if fan blade is not removed.

a) Loose or missing

<1> Permitted if the missing material is up to 0.039 in. (1.0 mm) in length from one of the edges

<2> Continue-In-Service limit: 100 cycles is permitted on all amounts criteria.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

b) Wear on visible pressure side platform wear strip area.

<1> Permitted, up to 5.2 in. (132 mm) wear strip length.

<2> Continue-In-Service limit: 100 cycles is permitted with the criteria: Up to 7.1 in. (180 mm) of wear strip length.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

<3> Maximum repairable limit: in the limits of the applicable repair.

<4> To repair the coating, do this task: Fan Blade Coating Replacement (Local Repair of Platform Wear-Strips), TASK 72-22-02-900-803-G00.

c) Wear on visible suction side platform wear strip area.

<1> Any amount, if not worn through to the composite blade surface, and the worn area is not frayed.

<2> Continue-In-Service limit: 100 cycles is permitted on all amounts criteria.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

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d) Nicks, dents and scratches

<1> Any amount, if not worn through to the composite blade surface, and the worn area is not frayed.

<2> Continue-In-Service limit: 100 cycles is permitted on all amounts criteria.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

e) Frayed and disbonded corners

<1> Permitted if the frayed or disbonded area up to 0.11 in. (3 mm) in length across the diagonal corner of the wear strip.

<2> Continue-In-Service limit: 100 cycles is permitted on all amounts criteria.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

f) Frayed and disbonded edges

<1> Permitted up to 0.039 in. (1.0 mm) in length from one of the edges.

<2> Continue-In-Service limit: 100 cycles is permitted on all amounts criteria.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

g) Discoloration.

<1> Permitted.

NOTE: Platform wear strips can be brown or gray.

2) Visual examination of visible platform wear strips if fan blade is removed (Figure 601 (Sheet 7) and Figure 601 (Sheet 8)).

a) Loose or missing

<1> Permitted if the missing material is up to 0.039 in. (1.0 mm) in length from one of the edges.

<2> Continue-In-Service limit: 100 cycles is permitted on all amounts criteria.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

b) Wear on upper pressure side platform wear strip area (Figure 601 (Sheet 7)).

<1> Permitted, up to 5.23 in. (133 mm) wear strip length.

<2> Continue-In-Service limit: 100 cycles is permitted with the criteria: Up to 7.12 in. (181 mm) of wear strip length.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

<3> Maximum repairable limit: in the limits of the applicable repair.

<4> To repair the coating, do this task: Fan Blade Coating Replacement (Local Repair of Platform Wear-Strips), TASK 72-22-02-900-803-G00.

c) Wear on suction and pressure side platform wear strip area.

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<1> Any amount, if not worn through to the composite blade surface, and the worn area is not frayed.

<2> Continue-In-Service limit: 100 cycles is permitted on all amounts criteria.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

d) Nicks, dents and scratches

<1> Any amount, if not worn through to the composite blade surface, and the worn area is not frayed.

<2> Continue-In-Service limit: 100 cycles is permitted on all amounts criteria.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

e) Frayed and disbonded corners

<1> Permitted if the frayed or disbonded area up to 0.11 in. (3 mm) in length across the diagonal corner of the wear strip.

<2> Continue-In-Service limit: 100 cycles is permitted on all amounts criteria.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

f) Frayed and disbonded edges

<1> Permitted if the frayed or disbonded area up to 0.039 in. (1.0 mm) in length from one of the edges.

<2> Continue-In-Service limit: 100 cycles is permitted on all amounts criteria.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

g) Discoloration.

<1> Permitted.

NOTE: Platform wear strips can be brown or gray.

(ae) Examine the leading edge and trailing edge wear strips if the fan blade removed from the fan disk for these conditions (Figure 601 (Sheet 7) and Figure 601 (Sheet 8)):

1) Loose or missing

a) Permitted if the missing material is up to 0.039 in. (1.0 mm) in length from one of the edges

b) Continue-In-Service limit: 100 cycles is permitted on all amounts criteria.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

2) Wear

a) Permitted, if not worn through to the composite blade surface, and the worn area is not frayed.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

b) Continue-In-Service limit: 100 cycles is permitted on all amounts criteria.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

3) Nicks, dents and scratches

a) Permitted if the composite is not visible and the wear strips surface not frayed.

b) Continue-In-Service limit: 100 cycles is permitted on all amounts criteria.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

4) Frayed and disbonded corners

a) Permitted if the frayed or disbonded area up to 0.11 in. (3 mm) in length across the diagonal corner of the wear strip.

b) Continue-In-Service limit: 100 cycles is permitted on all amounts criteria.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

5) Frayed and disbonded edges

a) Permitted up to 0.039 in. (1.0 mm) in length from one of the edges.

b) Continue-In-Service limit: 100 cycles is permitted on all amounts criteria.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

6) Discoloration.

a) Permitted.

NOTE: Platform wear strips can be brown or gray.

SUBTASK 72-22-02-211-004-G00

(2) Do these steps to examine the fan blade lock [2]:

(a) Examine all the titanium areas for these conditions (Figure 601 (Sheet 15)):

1) Cracks

a) Not permitted.

2) Dents

a) Permitted up to 0.0039 in. (0.10 mm) in depth.

3) Nicks

a) Permitted up to 0.0039 in. (0.10 mm) in depth.

4) Scratches.

a) Permitted up to 0.0039 in. (0.10 mm) in depth.

(b) Examine the contact area with the fan disk for these conditions (Figure 601 (Sheet 15)):

1) Fretting / Wear

a) Permitted, if dimension A is between 0.2815 in. (7.15 mm) and 0.2854 in. (7.25 mm).

(c) Examine the dimension B for these conditions (Figure 601 (Sheet 15)):

1) Fretting / Wear

a) Permitted, if dimension B is between 0.3957 in. (10.05 mm) and 0.4074 in. (10.35 mm).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Examine the dimension C for these conditions (Figure 601 (Sheet 15)):

1) Fretting / Wear

a) Permitted, if dimension C is between 1.1477 in. (29.15 mm) and 1.1515 in. (29.25 mm).

(e) Examine the area A for these conditions (Figure 601 (Sheet 15)):

1) Fretting / Wear

a) Permitted if less than 0.0011 in. (0.03 mm) in depth.

(f) Examine all honeycomb and sheet metal cover areas for these conditions (Figure 601 (Sheet 14) and Figure 601 (Sheet 15)):

1) Honeycomb crushed or collapsed cells out of fan blade contact area

a) Not more than three locations, up to 0.02311 in. (0.587 mm) in depth, 0.17074 in. (4.337 mm) in length.

2) Honeycomb crushed or collapsed cells on fan blade contact area

a) Not more than three cells, up to 0.02499 in. (0.635 mm) in depth.

3) Honeycomb disbonding of the adhesive

a) Not permitted.

4) Honeycomb cracks in the adhesive

a) Not permitted.

5) Honeycomb sheet metal cover nicks, dents and scratches out of fan blade contact area

a) Not more than three locations, up to 0.02311 in. (0.587 mm) in depth, 0.17074 in. (4.337 mm) in length.

6) Honeycomb sheet metal cover nicks, dents and scratches on fan blade contact area

a) Not more than three cells, up to 0.01181 in. (0.300 mm) in depth, 0.06830 in. (1.735 mm) in length.

(g) Examine the dimension G for these conditions (Figure 601 (Sheet 15)):

1) Fretting / Wear

a) Permitted if dimension G is between 0.2067 in. (5.25 mm) and 0.2185 in. (5.55 mm).

(h) Examine the dimension H for these conditions (Figure 601 (Sheet 15)):

1) Fretting / Wear

a) Permitted if dimension H is between 0.0142 in. (0.36 mm) and 0.0173 in. (0.44 mm).

(i) Examine the honeycomb cells for these conditions (Figure 601 (Sheet 16)):

1) Local twist from manufacturing

a) Permitted.

2) Local holes from manufacturing

a) Permitted.

3) Local cut in honeycomb cell without crushing from manufacturing

a) Permitted.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0331

SUBTASK 72-22-02-211-005-G00

(3) Examine the wear strips in the flat area of the fan blade spacer [3] for these conditions (Figure 601 (Sheet 20)):

(a) Missing material, frayed, disbonded, wear, dents.

1) Any amount.

SUBTASK 72-22-02-211-017-G00

(4) Examine the composite material in the area B (Figure 601 (Sheet 20)):

(a) Wear.

1) Maximum Serviceable Limit: Any amount up to 0.004 in. (0.1 mm) in depth.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0331

SUBTASK 72-22-02-211-014-G00

(5) Do these steps to examine the fan blade spacer [3]:

(a) Examine the composite material outside head area for these conditions (Figure 601 (Sheet 17) and Figure 601 (Sheet 18)):

1) Wear

a) Make sure that the surface D (in contact with fan blade) has a thickness not less than 0.1831 in. (4.65 mm).

b) Surface E is permitted.

c) Make sure that the surface F (in contact with fan disk) has a thickness not less than 0.178 in. (4.52 mm).

d) Make sure that the surface G (in contact with fan disk) has a thickness not less than 0.1831 in. (4.65 mm).

2) Nicks, dents, missing material areas

a) Make sure that the maximum depth is no greater than 0.019 in. (0.48 mm).

Make sure the maximum surface area is no greater than 0.0155 in^2 (10.00 mm^2).

b) Make sure there is a minimum of 0.79 in. (20.07 mm) between two indications.

3) Scratches

a) There can be any number of scratches as long as the depth is less than 0.003 in. (0.08 mm).

b) There can be any number of scratches as long as the depth is less than 0.019 in. (0.48 mm) and if total length is not more than 8.85 in. (224.79 mm).

4) Micro-cracks, decohesion

a) There can be two indications if not more than 0.19 in. (4.83 mm) in length.

5) Tear, fiber breakage.

a) Not permitted.

(b) Examine the composite material in the head area for these conditions (Figure 601 (Sheet 18), Figure 601 (Sheet 19), and Figure 601 (Sheet 21)):

1) Tears, fiber breakage in extraction hole area

a) Any amount.

2) Missing material in extraction hole area

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0331 (Continued)

a) There can be any amount if thickness H is not less than 0.138 in. (3.51 mm)

3) Tears, fiber breakage outside extraction hole area

a) Not permitted.

4) Micro-cracks, decohesion in extraction hole area

a) Not permitted.

5) Micro-cracks, decohesion outside extraction hole area

a) There can be up to two indications if not more than 0.19 in. (4.83 mm) in length.

6) Ovalisation of extraction hole

a) Permitted.

7) Nicks, dents, missing material areas

a) Make sure that the maximum depth is no greater than 0.019 in. (0.48 mm).

b) Make sure the maximum surface area is no greater than 0.0155 in^2 (10.00 mm^2).

8) Scratches

a) There can be any number as long as the depth is less than 0.019 in. (0.48 mm).

9) Wear in the head area (Figure 601 (Sheet 21)):

a) There can be any amount up to 0.004 in. (0.1 mm) in depth.

PTF ALL

SUBTASK 72-22-02-211-006-G00

(6) Do these steps to examine the fan disk shim [4] (Figure 601 (Sheet 22)):

(a) Examine all areas for these conditions:

1) Nicks / Dents / Scratches and Fretting on all surfaces (does not include the pressure face)

a) Not permitted.

2) Cracks, tears and punctures

a) Not permitted.

(b) Examination the pressure faces for these conditions:

1) Nicks / Dents / Scratches and Fretting.

a) Not permitted.

SUBTASK 72-22-02-211-007-G00

(7) Use an aluminum mallet between 1.59 oz (45.08 g) and 3.53 oz (100.07 g), to do the tap-test of the MLE (Figure 601 (Sheet 23)).

NOTE: You can make or use an equivalent tool with similar material and shape.

NOTE: If a solid click is heard, the area is bonded. If a hollow sound is heard, the area is disbonded

(a) Tap on each side (convex and concave) of the fan blade.

(b) Tap from the end of the MLE nose to the MLE sidewalls.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Tap from the bottom of the MLE to the tip.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-22-02-860-007-G00

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-22-02-860-008-G00

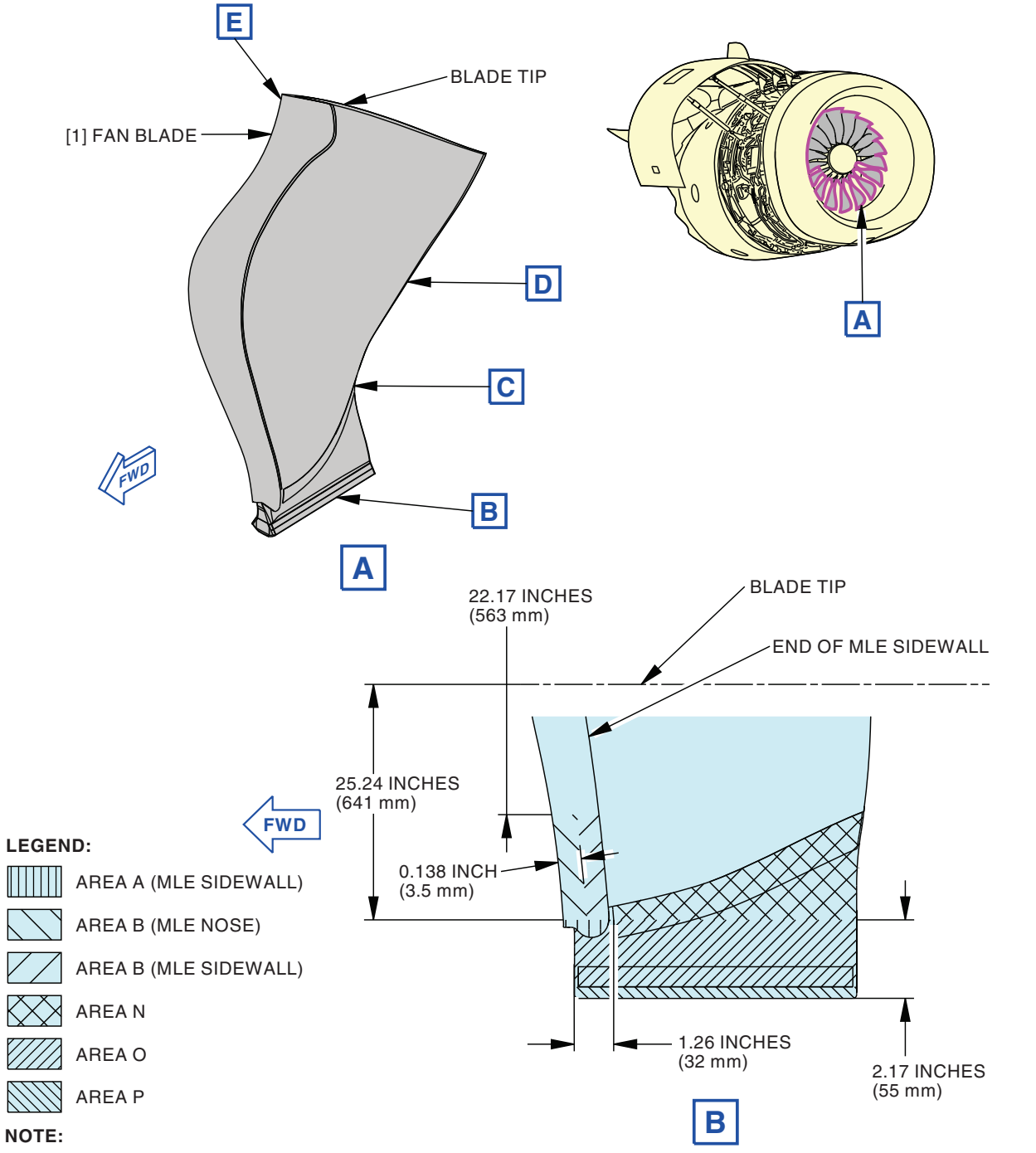
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

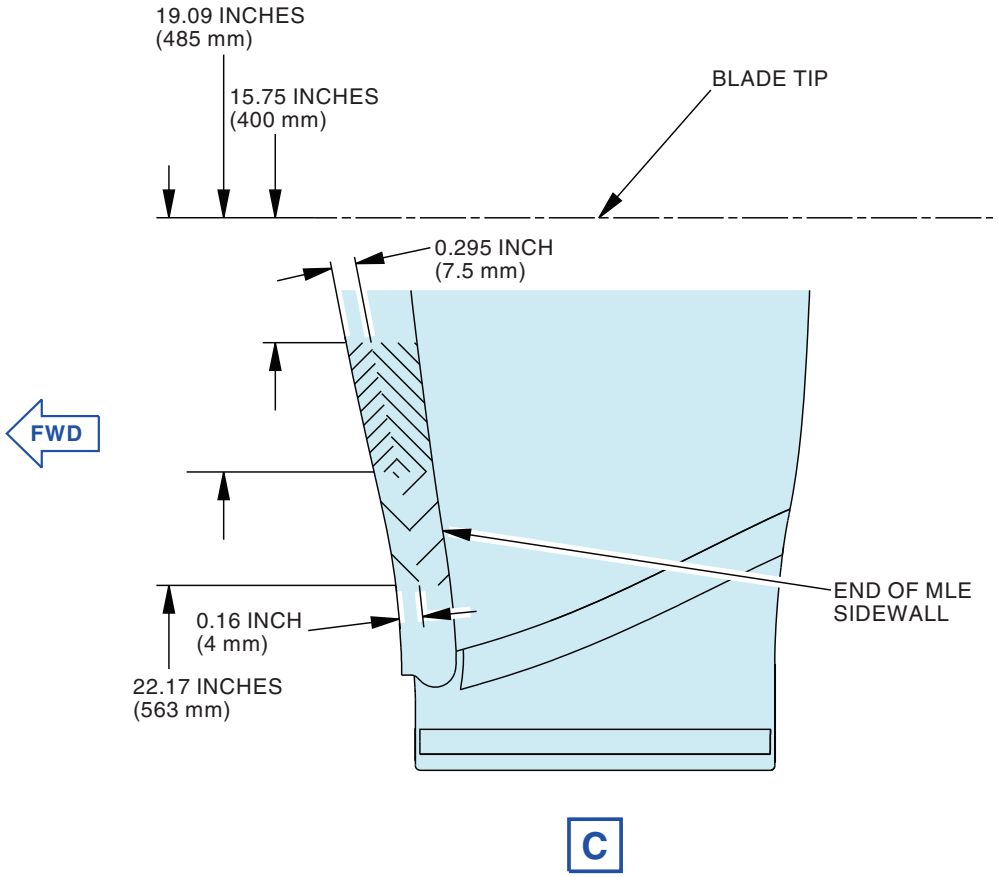
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Blade Inspection
Figure 601/72-22-02-990-810-G00 (Sheet 1 of 26)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

- AREA C (MLE NOSE)
- AREA C (MLE SIDEWALL)
- AREA D (MLE NOSE)
- AREA D (MLE SIDEWALL)

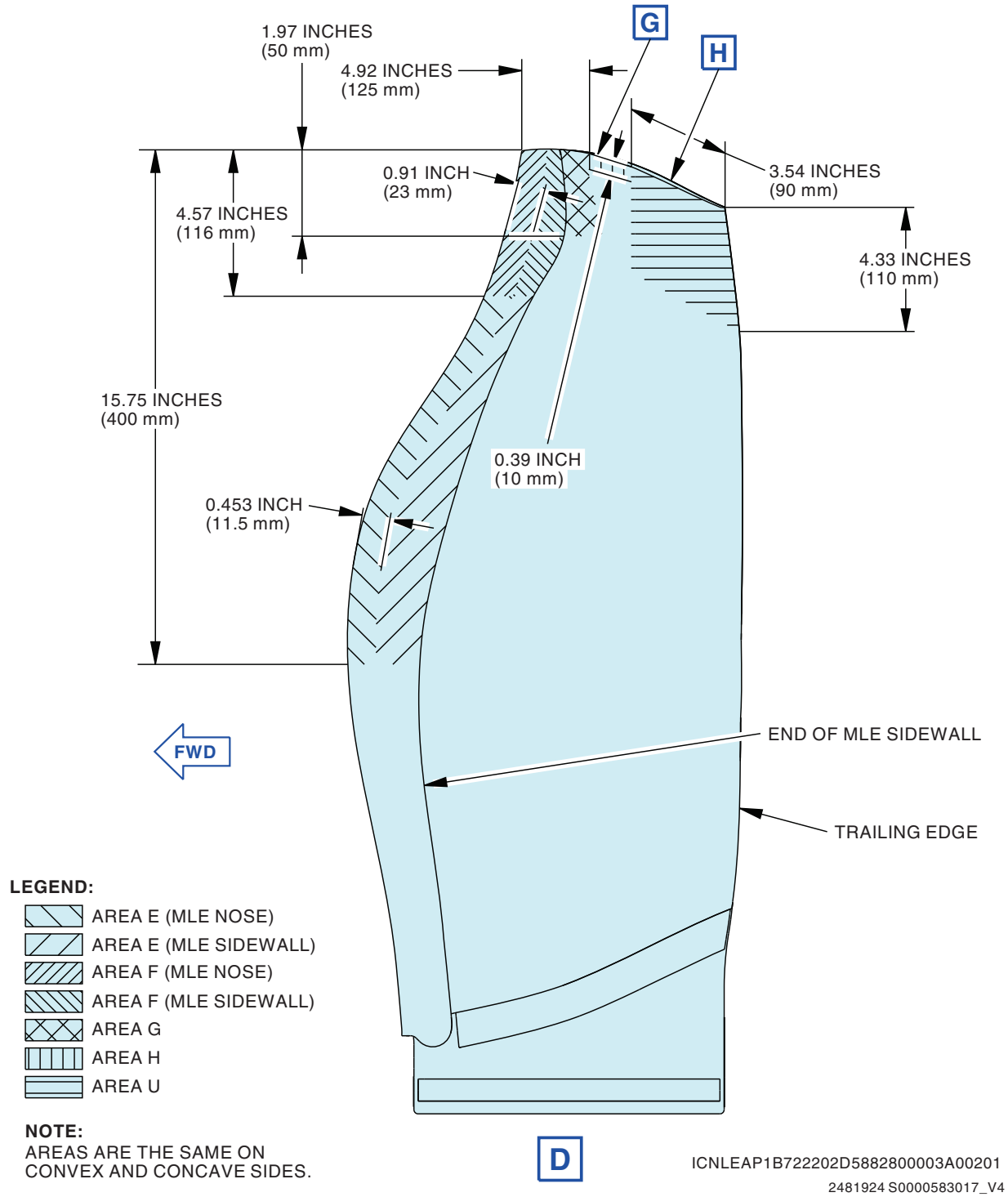
NOTE:

AREAS ARE THE SAME ON
CONVEX AND CONCAVE SIDES.

2481922 S0000583016_V2

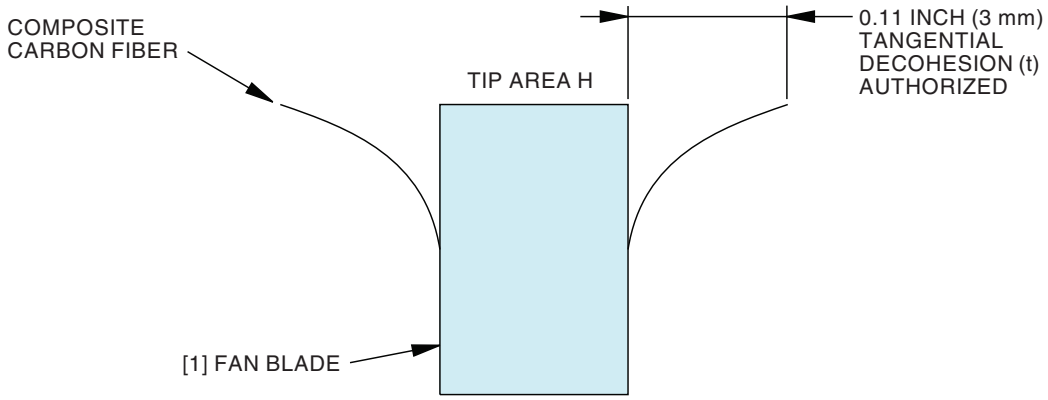
Fan Blade Inspection
Figure 601/72-22-02-990-810-G00 (Sheet 2 of 26)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



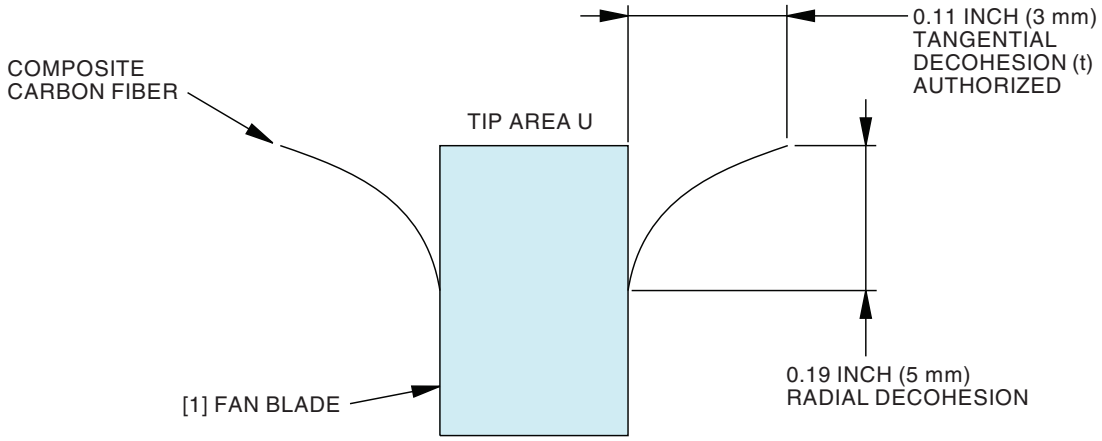
Fan Blade Inspection
Figure 601/72-22-02-990-810-G00 (Sheet 3 of 26)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TIP DECOHESION IN AREA H

G



TIP DECOHESION IN AREA U

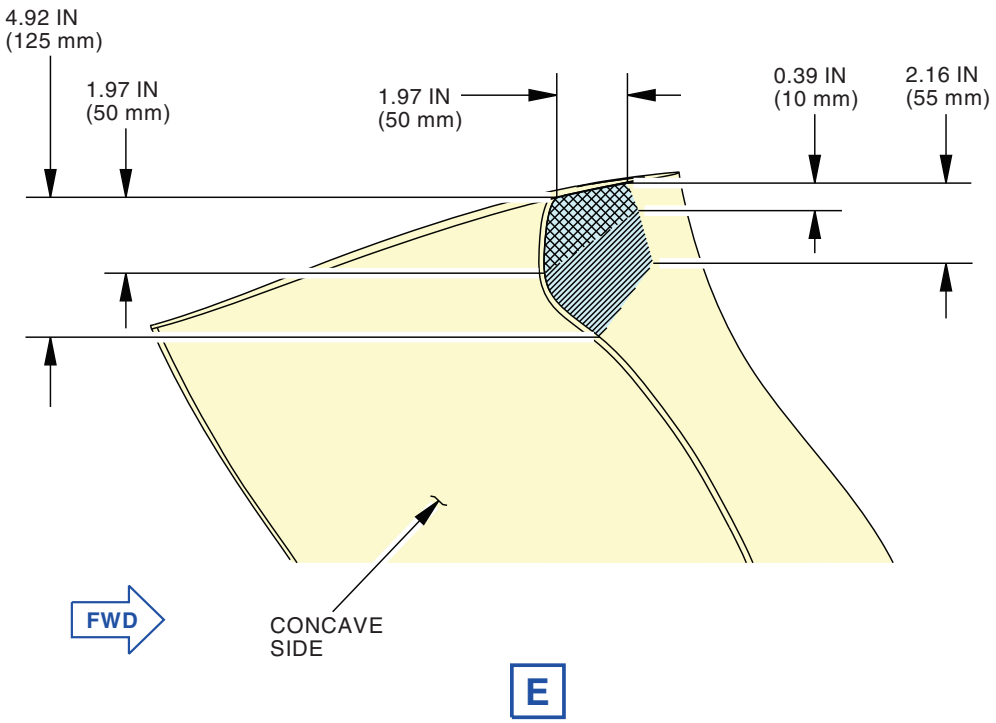
H

ICNLEAP1B722202D5882800010A00201

2853436 S0000669160_V1

Fan Blade Inspection
Figure 601/72-22-02-990-810-G00 (Sheet 4 of 26)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

AREA F1

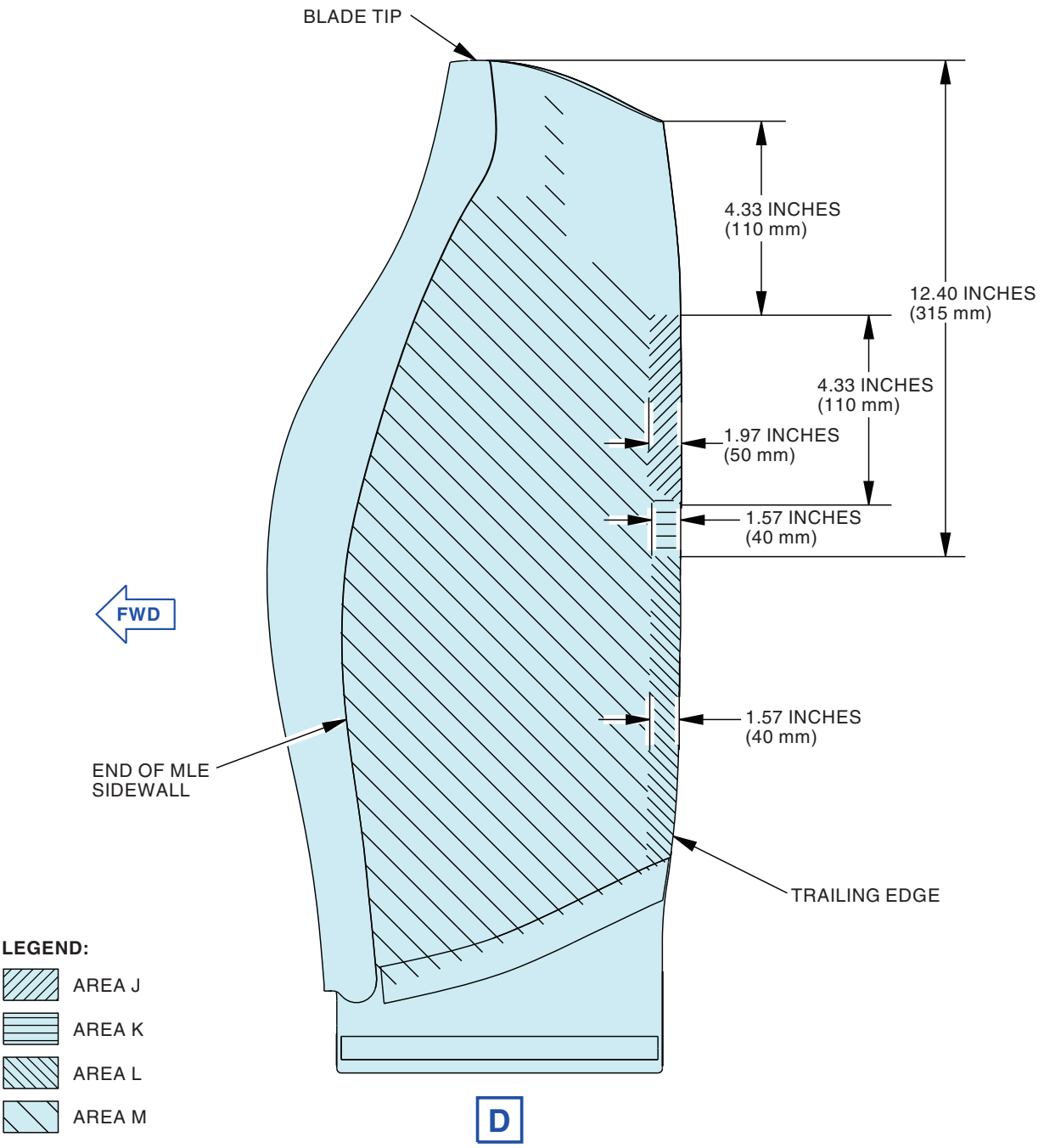
AREA F2

ICNLEAP1B722202D5882800004A00101

2481927 S0000583018_V3

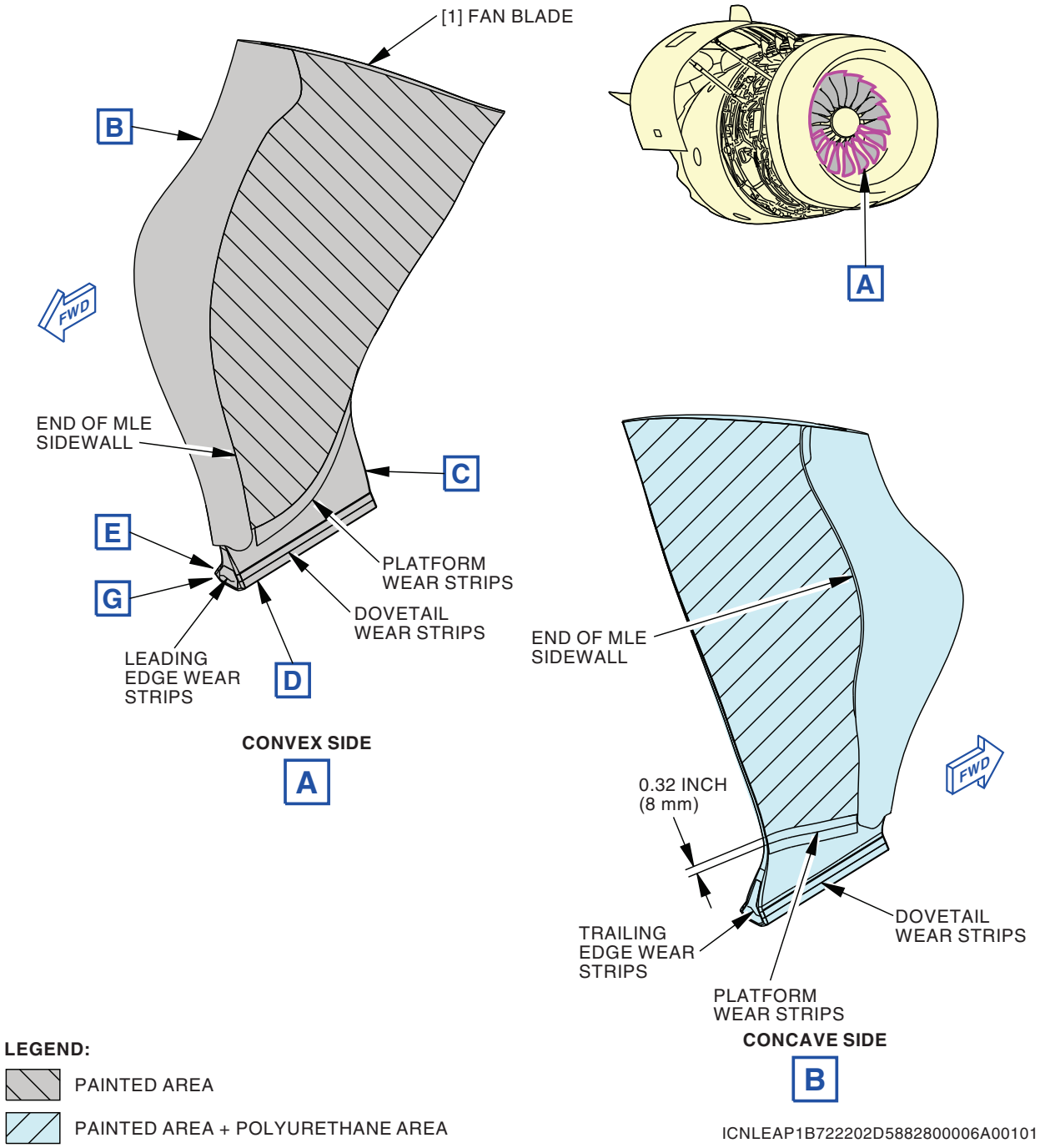
Fan Blade Inspection
Figure 601/72-22-02-990-810-G00 (Sheet 5 of 26)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Blade Inspection
Figure 601/72-22-02-990-810-G00 (Sheet 6 of 26)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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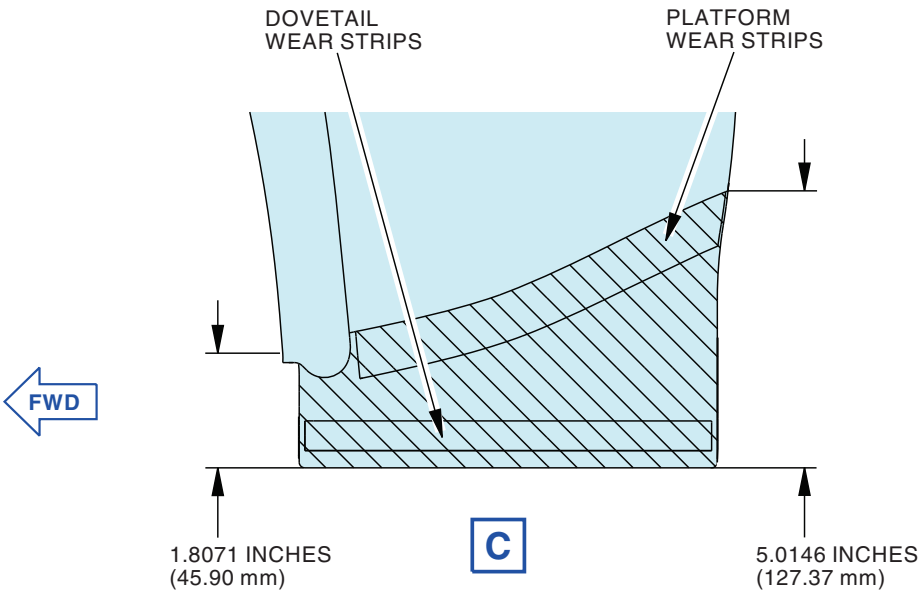
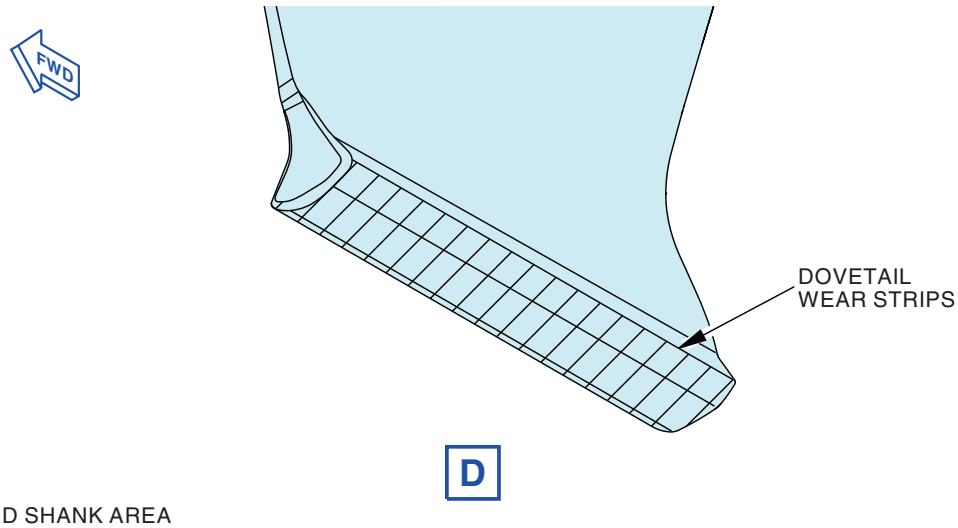
Fan Blade Inspection
Figure 601/72-22-02-990-810-G00 (Sheet 7 of 26)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





LEGEND:

ROOT AND SHANK AREA

AREA P

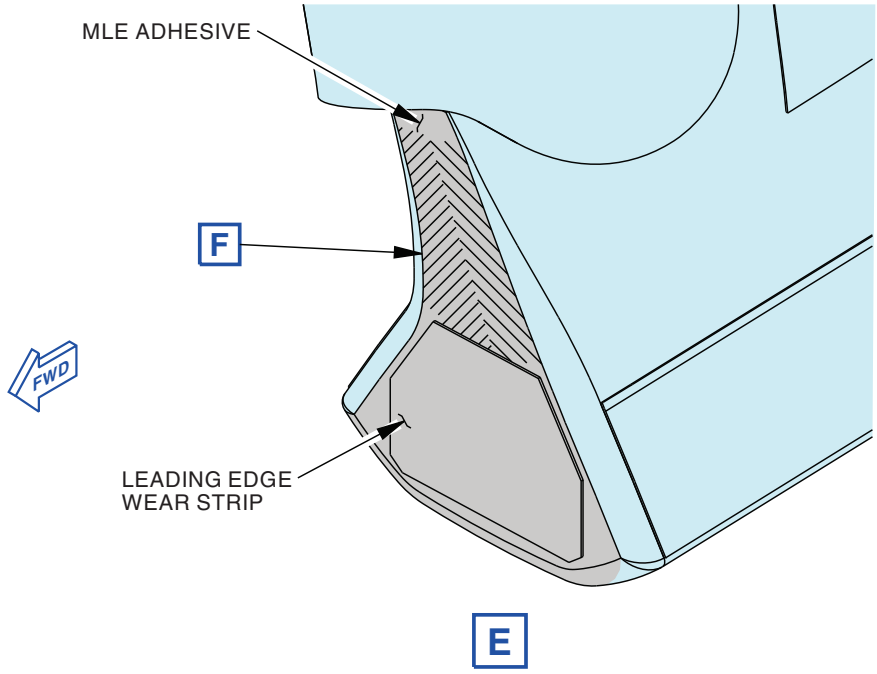
NOTE:

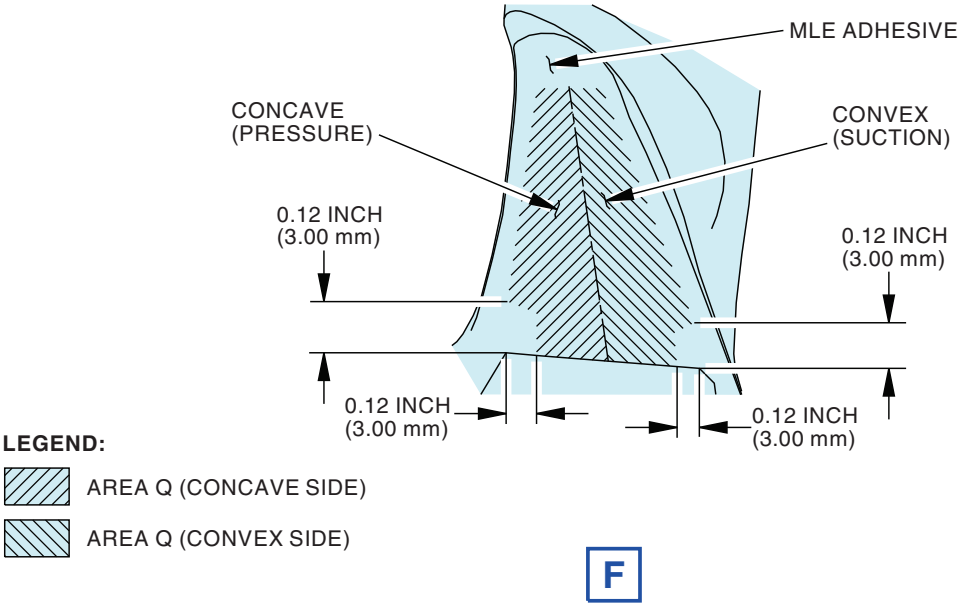
AREAS ARE THE SAME ON
CONVEX AND CONCAVE SIDES.

2545936 S0000605150_V1

Fan Blade Inspection
Figure 601/72-22-02-990-810-G00 (Sheet 8 of 26)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

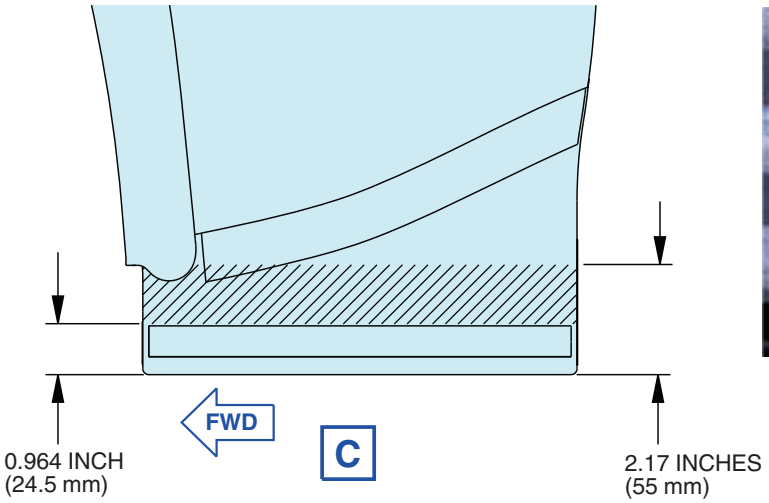


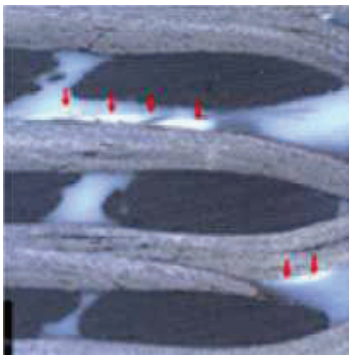


2545942 S0000605151_V1

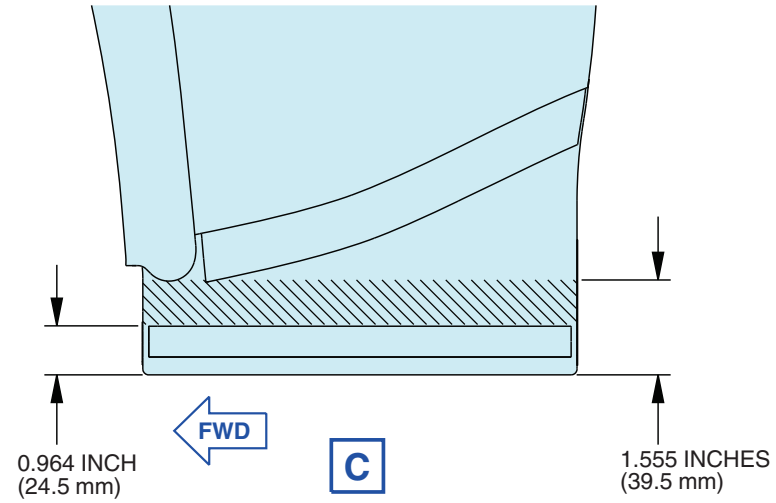
Fan Blade Inspection
Figure 601/72-22-02-990-810-G00 (Sheet 9 of 26)

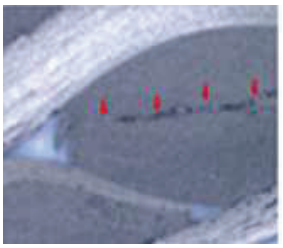
ECCN 9E991 BOEING PROPRIETARY - See title page for details





MICRO-CRACKING





DECOHESION

LEGEND:

AREA R

AREA S

NOTE:

AREAS ARE THE SAME ON CONVEX AND CONCAVE SIDE.

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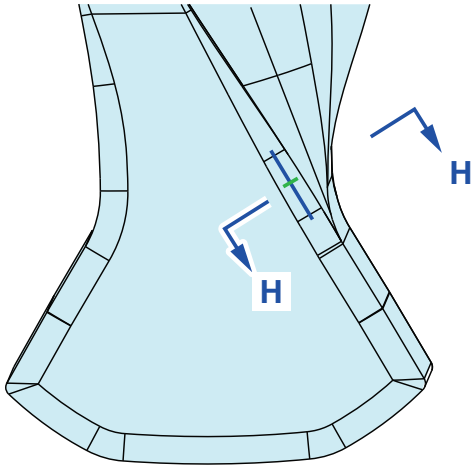
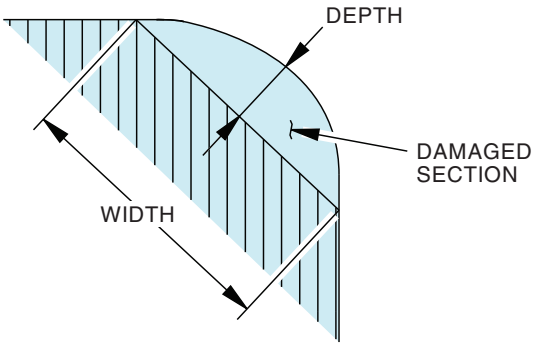
2545948 S0000605153_V2

Fan Blade Inspection
Figure 601/72-22-02-990-810-G00 (Sheet 10 of 26)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



G



H-H

LEGEND:

— DAMAGED SECTION PERPENDICULAR TO THE RADIUS (WIDTH)

— DIRECTION PARALLEL TO THE RADIUS (LENGTH)

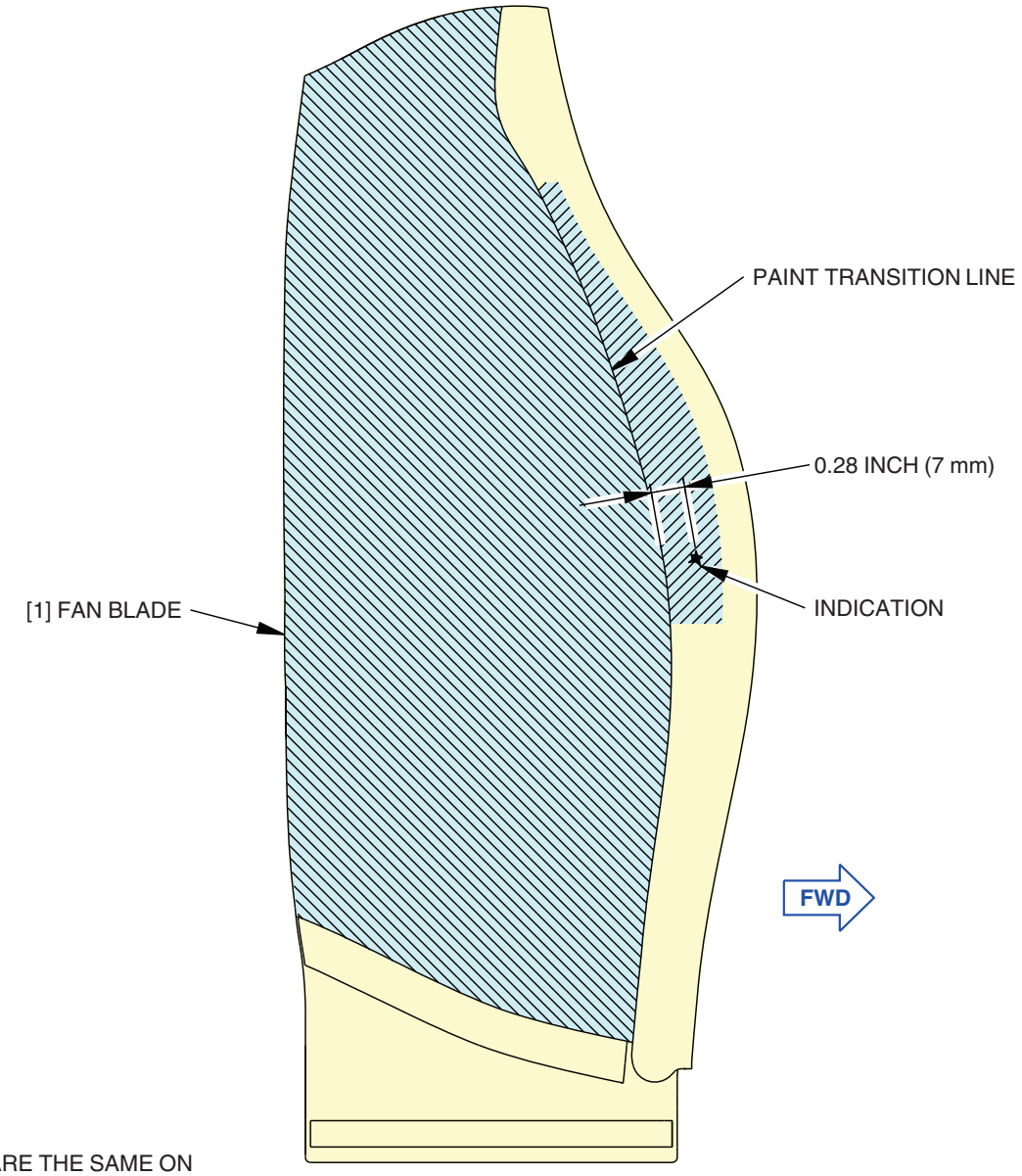
DAMAGED SECTION:

AREA OBTAINED BY MULTIPLYING THE GREATEST DEPTH OF THE ANOMALY BY ITS GREATEST WIDTH

ICNLEAP1B722202D5882800031A00101
2966764 S0000745956_V1

Fan Blade Inspection
Figure 601/72-22-02-990-810-G00 (Sheet 11 of 26)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:
AREAS ARE THE SAME ON CONVEX AND CONCAVE SIDES.

LEGEND:
AREA E (MLE SIDEWALL)
PAINTED AREA

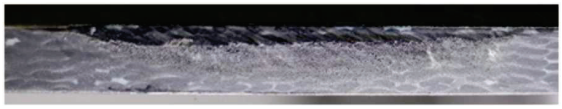
ICNLEAP1B722202D5882800061A00101
3120734 S0000855605_V1

Fan Blade Inspection
Figure 601/72-22-02-990-810-G00 (Sheet 12 of 26)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TIP RUB

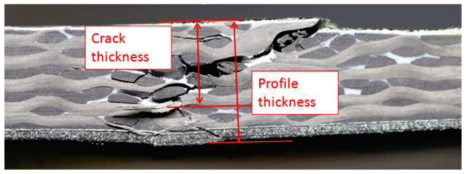
TIP RUB



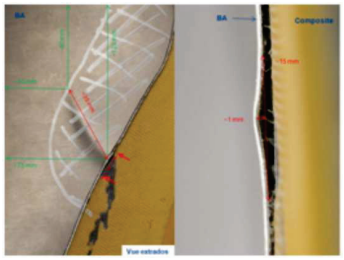
TIP RUB

GREY PLATFORM WEAR-STRIP

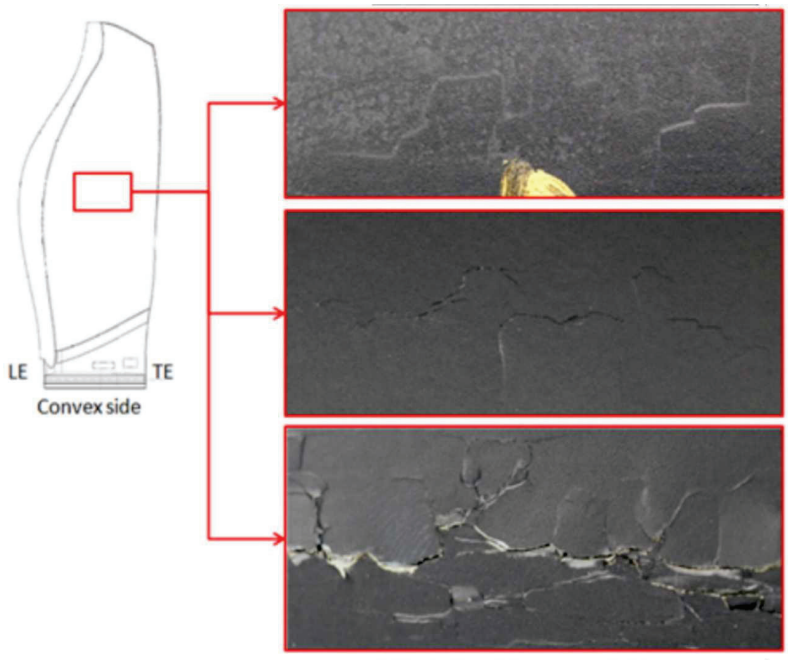
BROWN PLATFORM WEAR-STRIP



CRACK THICKNESS



MLE CREASE



PAINTING CREASING DUE TO UNDERLYING RESIN MICRO-CRACKING

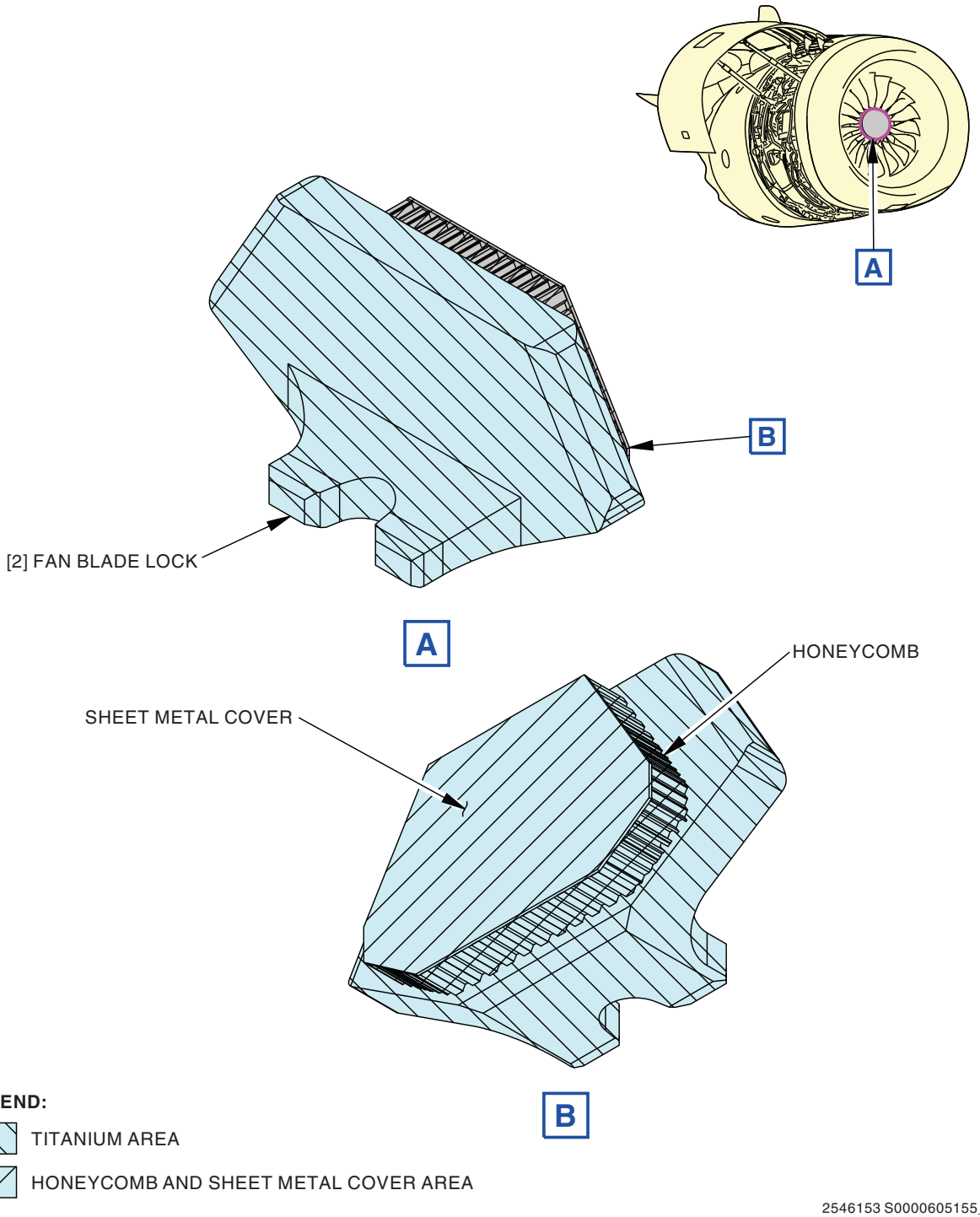
PAINTING FLAKING DUE TO UNDERLYING DECOHESION

CRACKS (FIBER BREAKAGE)

2545962 S0000605154_V2

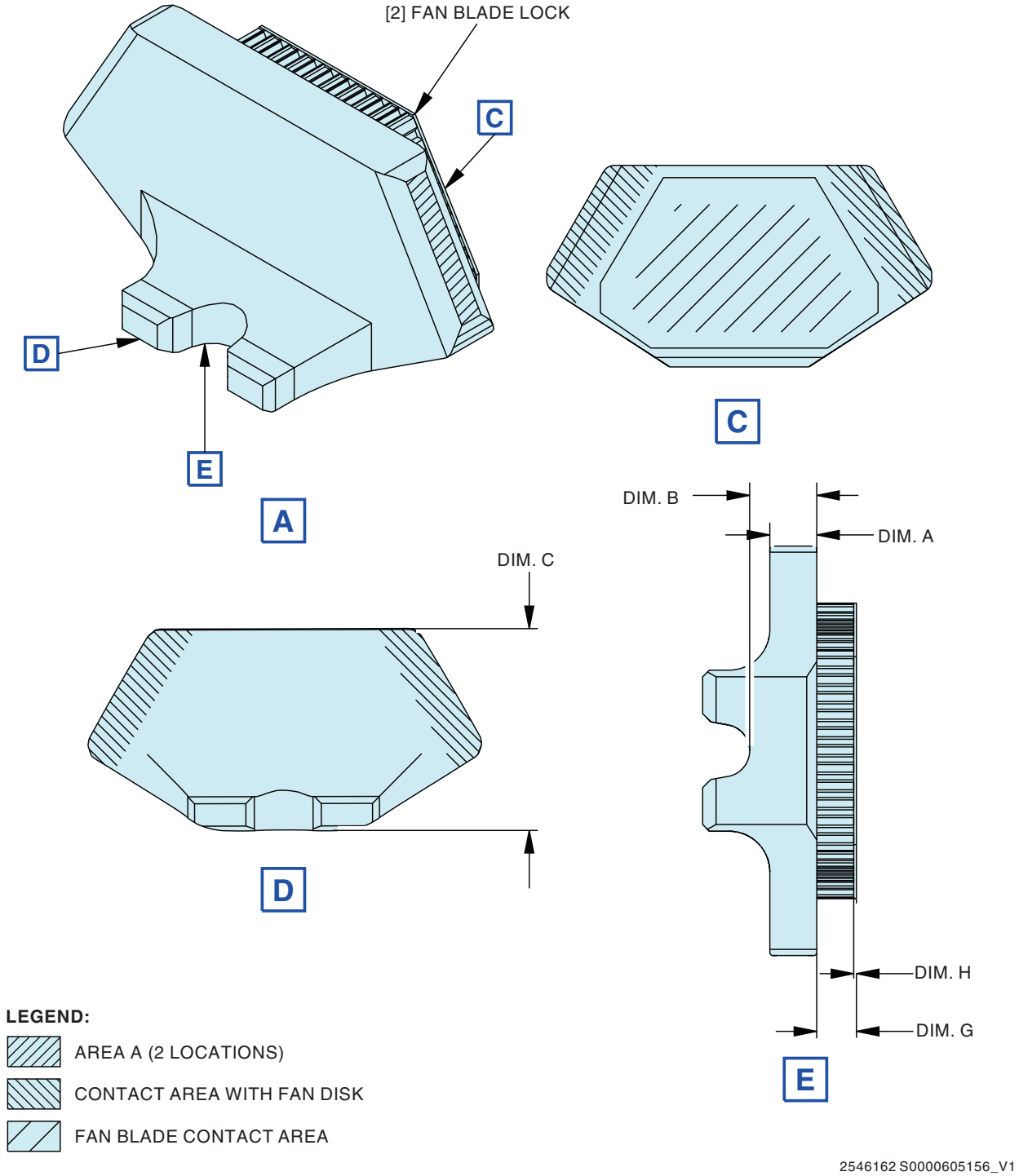
Fan Blade Inspection
Figure 601/72-22-02-990-810-G00 (Sheet 13 of 26)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Blade Inspection
Figure 601/72-22-02-990-810-G00 (Sheet 14 of 26)

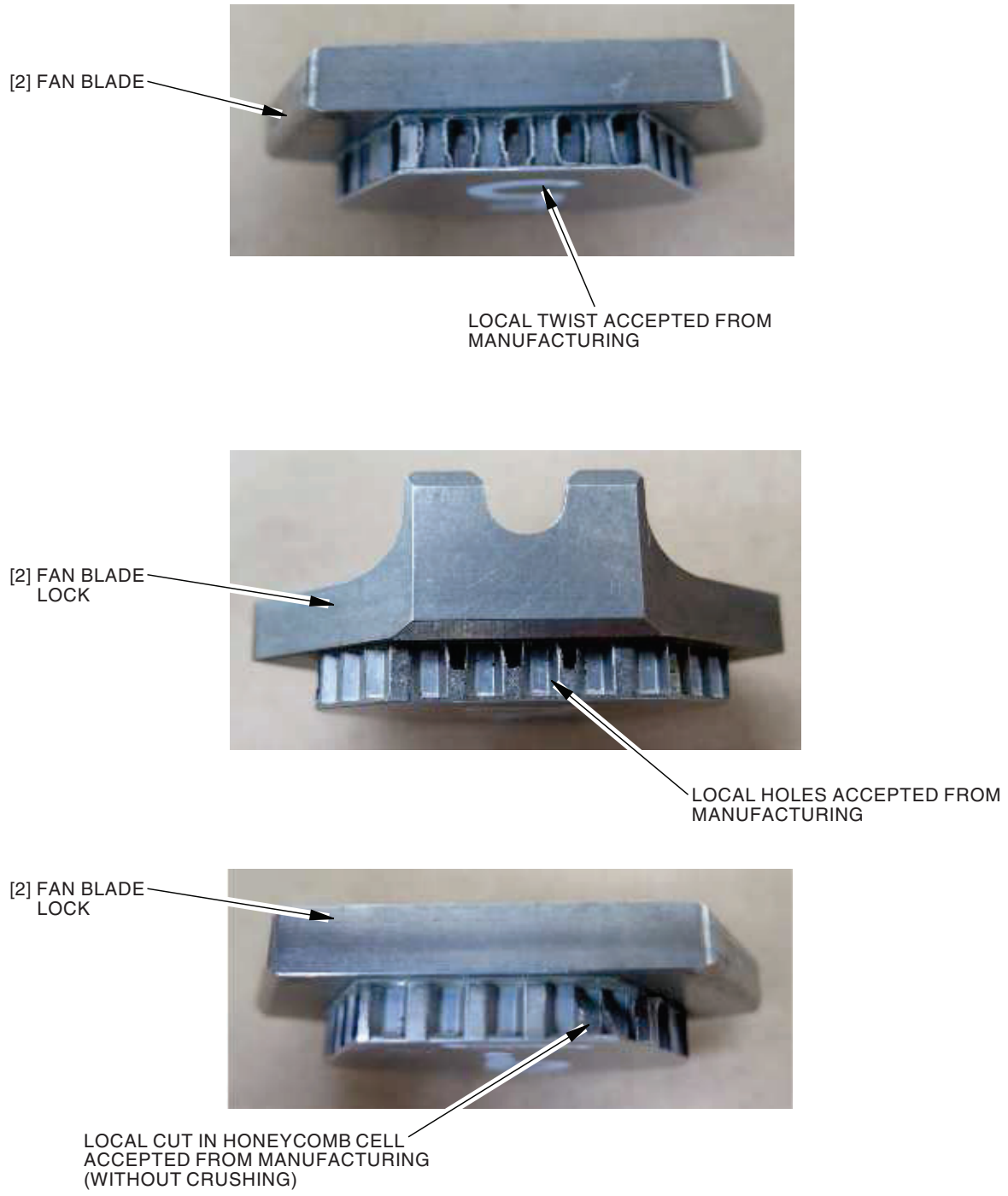
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2546162 S0000605156_V1

Fan Blade Inspection
Figure 601/72-22-02-990-810-G00 (Sheet 15 of 26)

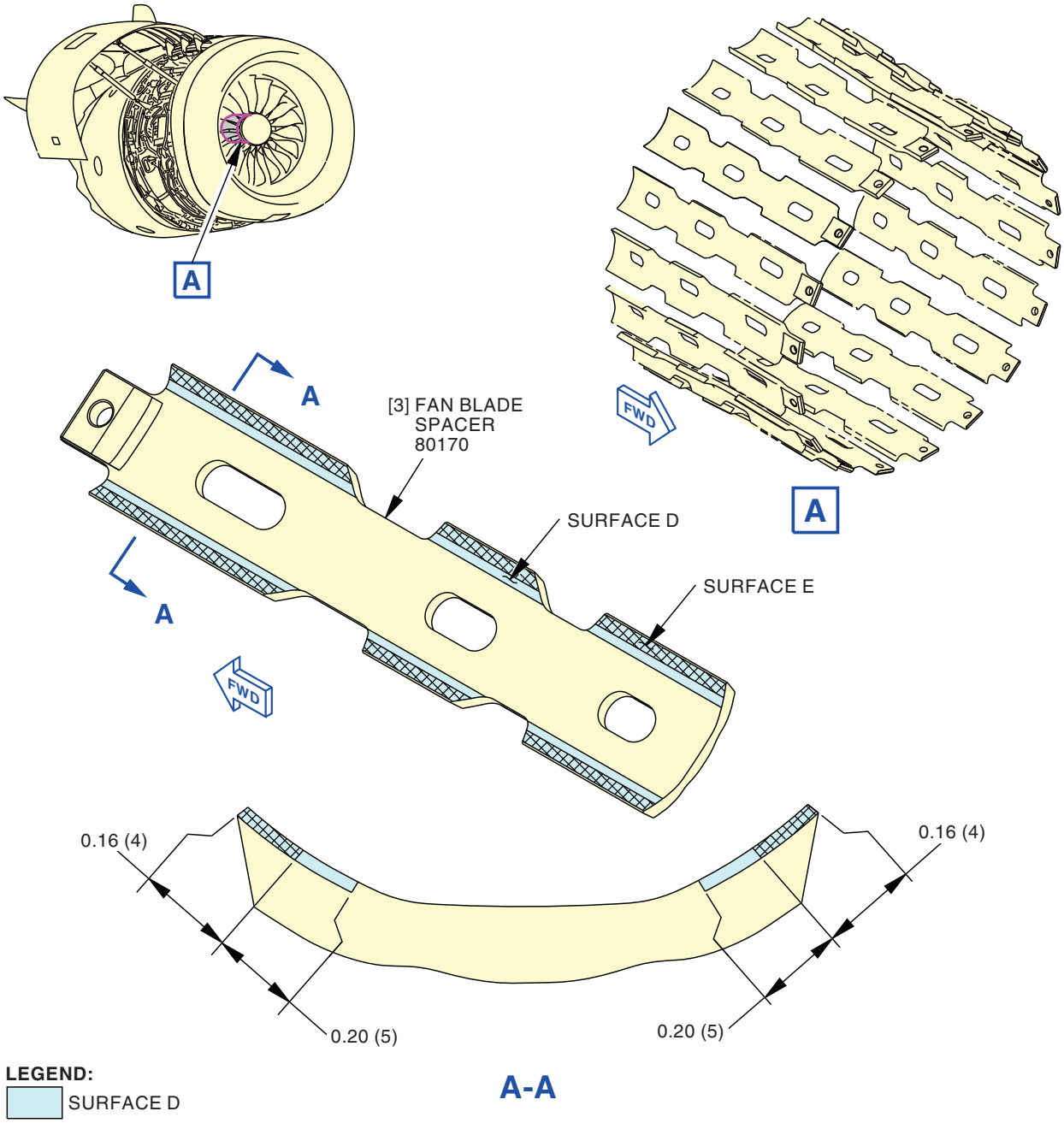
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2546435 S0000605157_V1

Fan Blade Inspection
Figure 601/72-22-02-990-810-G00 (Sheet 16 of 26)

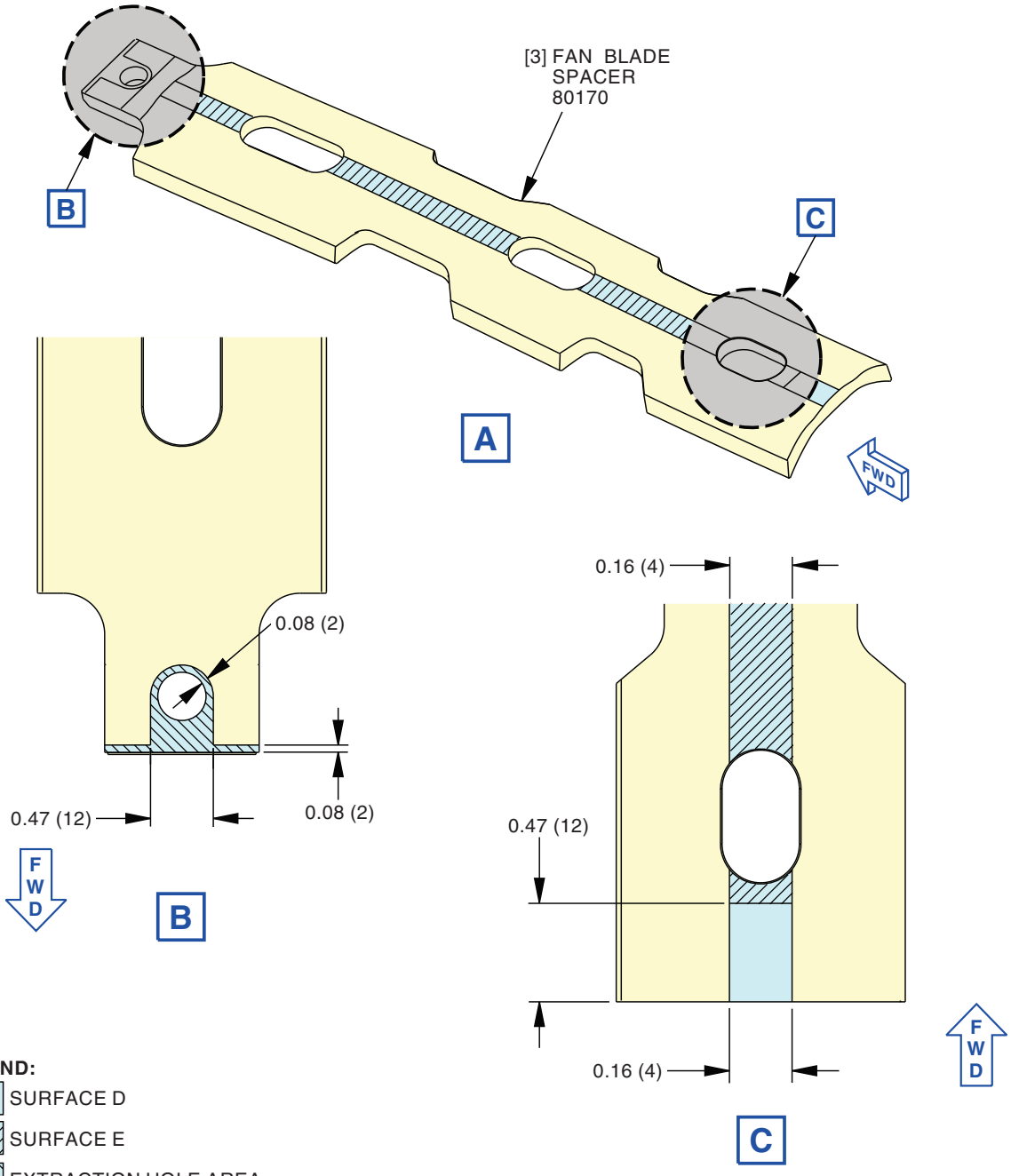
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2954516 S0000731802_V1

Fan Blade Inspection
Figure 601/72-22-02-990-810-G00 (Sheet 17 of 26)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

SURFACE D

SURFACE E

EXTRACTION HOLE AREA

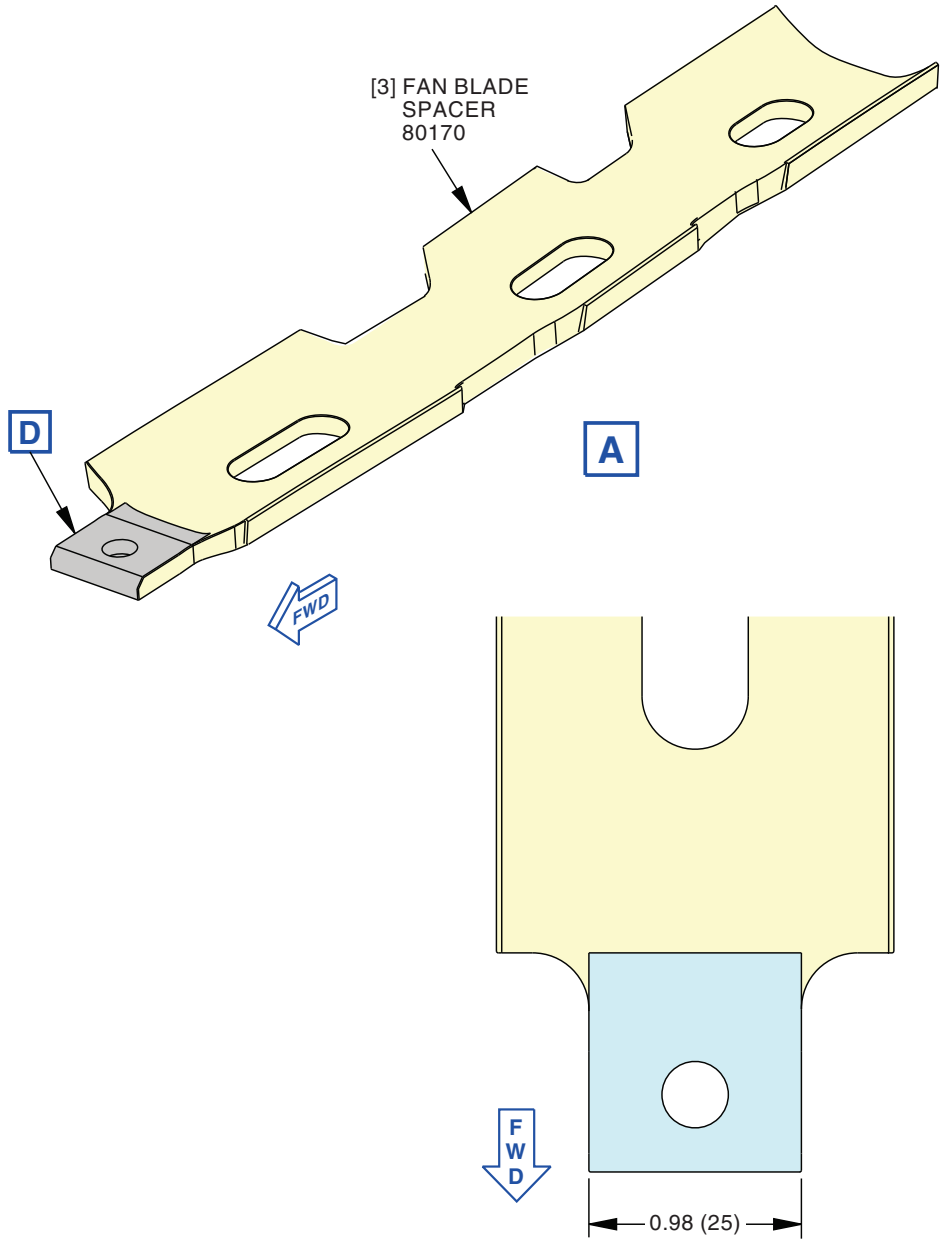
NOTE:

ALL DIMENSIONS ARE IN INCHES
(MILLIMETERS ARE IN PARENTHESES).

ICNLEAP1B722202D5882800022A00101
2954517 S0000731804_V1

Fan Blade Inspection
Figure 601/72-22-02-990-810-G00 (Sheet 18 of 26)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

HEAD AREA

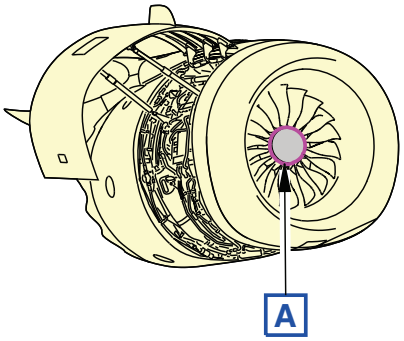
NOTE:

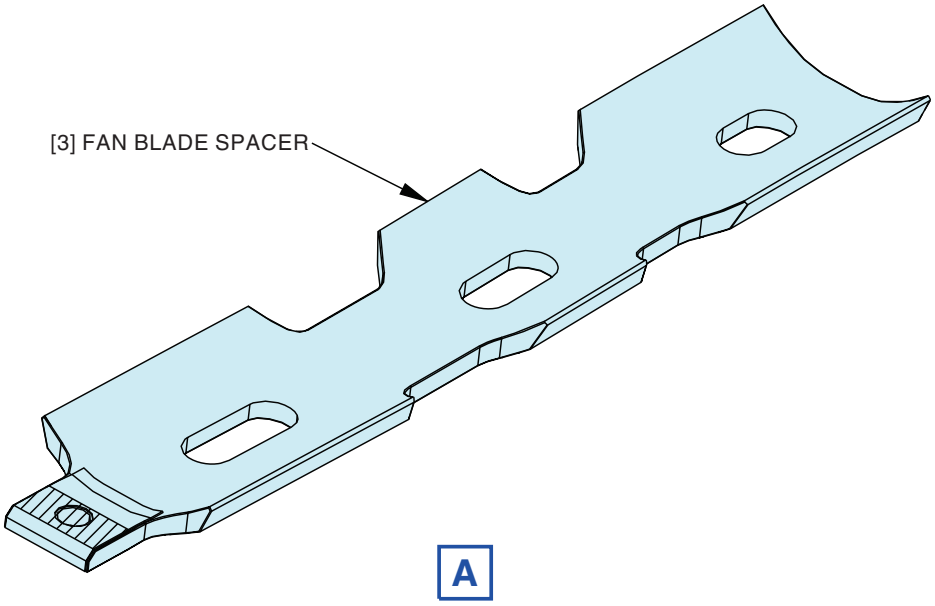
ALL DIMENSIONS ARE IN INCHES
(MILLIMETERS ARE IN PARENTHESES).

ICNLEAP1B722202D5882800023A00101
2954518 S0000731805_V1

Fan Blade Inspection
Figure 601/72-22-02-990-810-G00 (Sheet 19 of 26)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





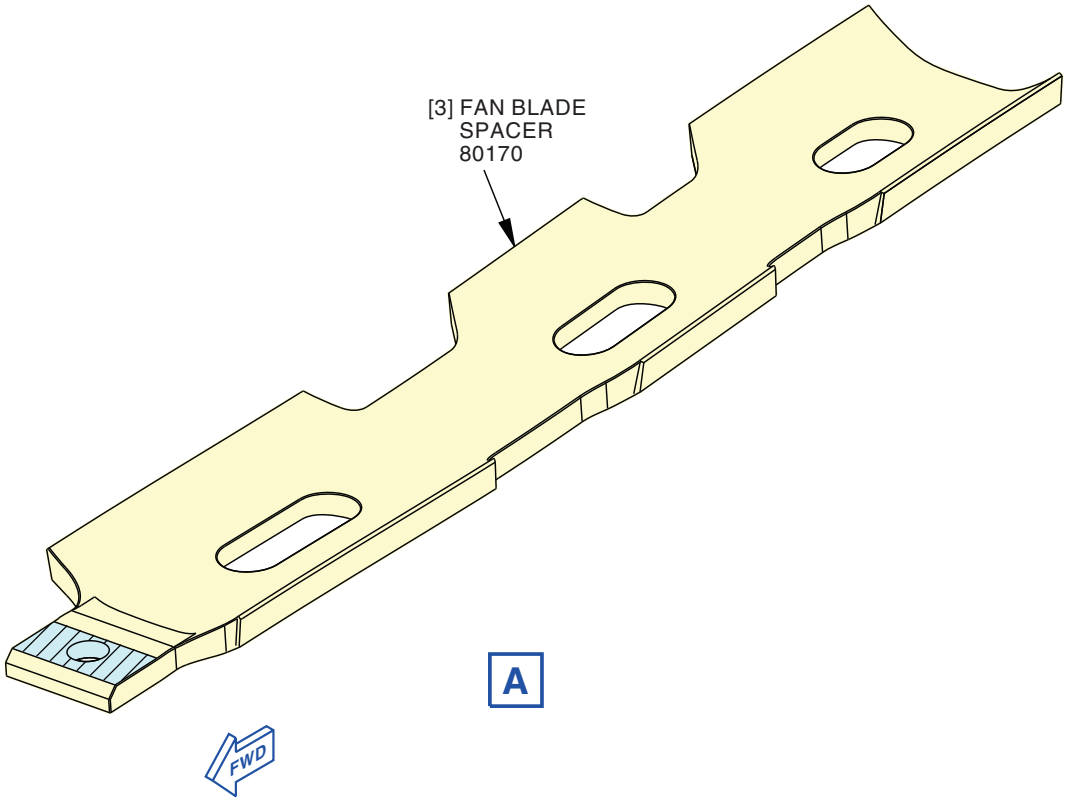
LEGEND:

AREA B (FLAT AREA)

2546438 S0000605158_V1

Fan Blade Inspection
Figure 601/72-22-02-990-810-G00 (Sheet 20 of 26)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

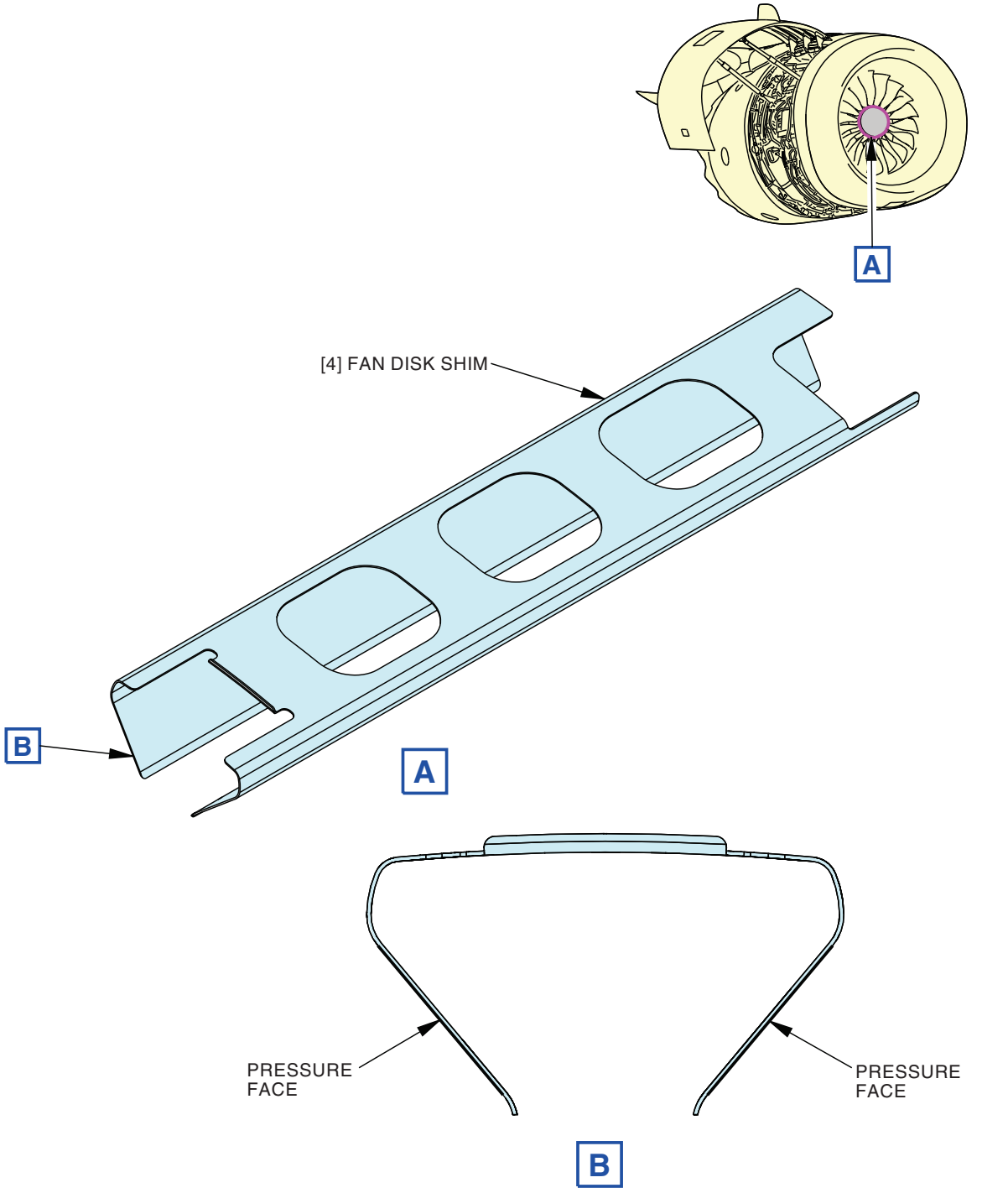


LEGEND:
HEAD AREA

ICNLEAP1B722202D5882800024A00101
2954519 S0000731807_V1

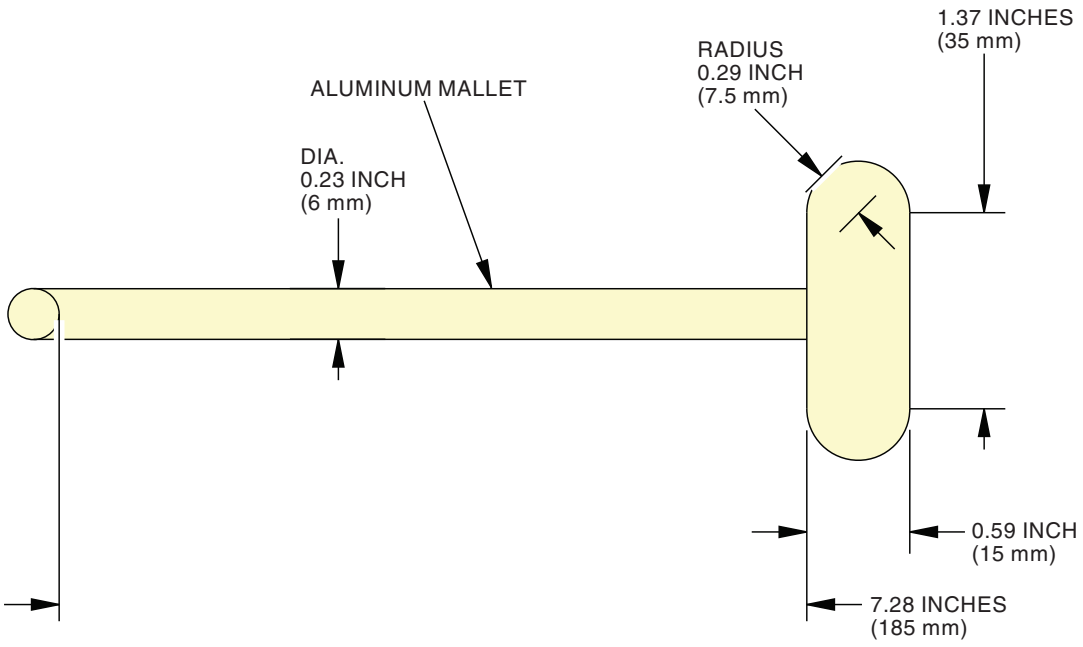
Fan Blade Inspection
Figure 601/72-22-02-990-810-G00 (Sheet 21 of 26)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Blade Inspection
Figure 601/72-22-02-990-810-G00 (Sheet 22 of 26)

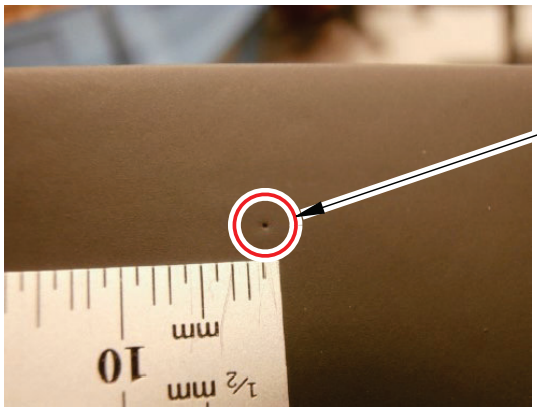
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2546447 S0000605160_V2

Fan Blade Inspection
Figure 601/72-22-02-990-810-G00 (Sheet 23 of 26)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



POROSITY

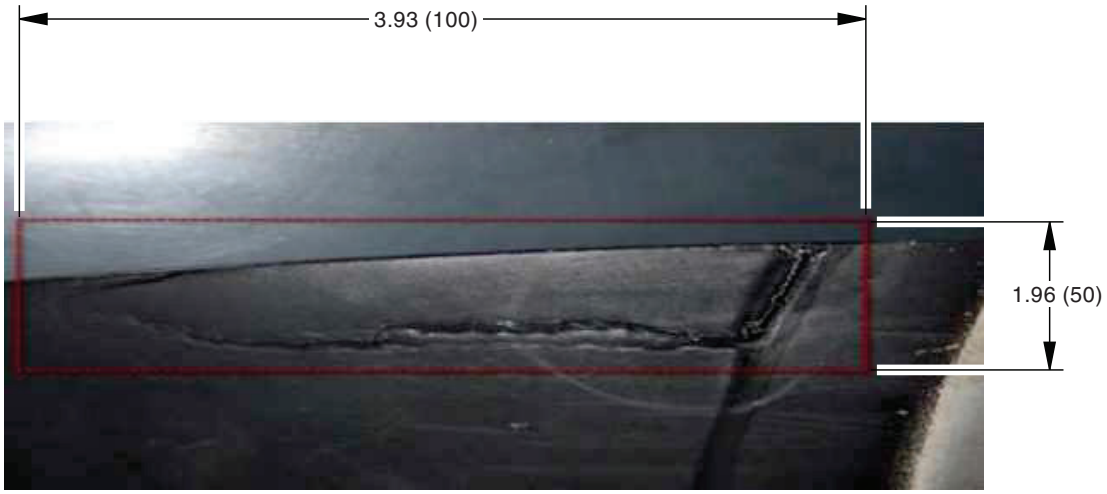
FOREIGN MATERIAL

FISH EYES

3002821 S0000778584_V1

Fan Blade Inspection
Figure 601/72-22-02-990-810-G00 (Sheet 24 of 26)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:
ALL DIMENSIONS ARE IN INCHES
(MILLIMETERS ARE IN PARENTHESES).

ICNLEAP1B722202D5882800033A00101
3041455 S0000809095_V1

Fan Blade Inspection
Figure 601/72-22-02-990-810-G00 (Sheet 25 of 26)

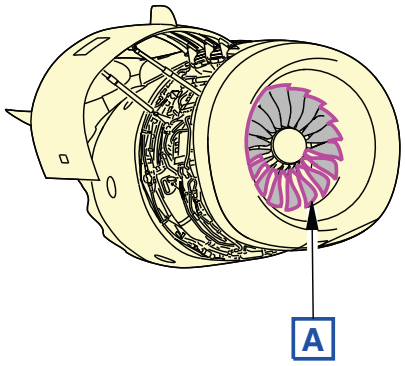
ECCN 9E991 BOEING PROPRIETARY - See title page for details

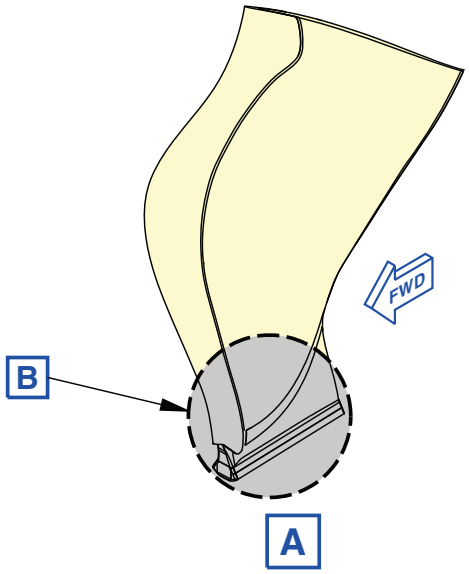
EXAMPLE OF DECOHESION

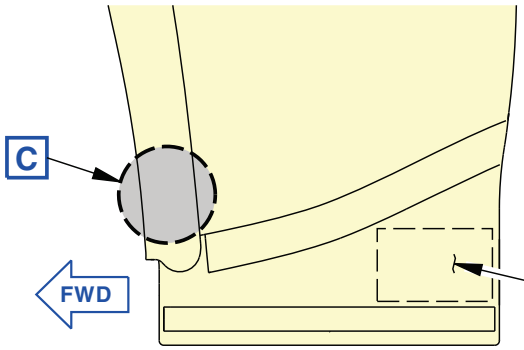
ICNLEAP1B722120D5882800001A00101
3083290 S0000838375_V1

Fan Blade Inspection
Figure 601/72-22-02-990-810-G00 (Sheet 26 of 26)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

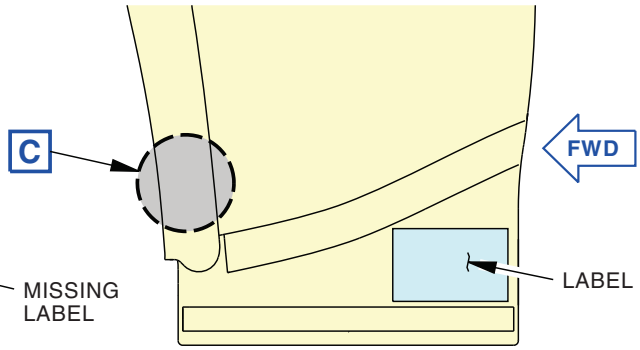






(SERVICEABLE WITHOUT LABEL)

B 1



(SERVICEABLE WITH LABEL)

B

PART NO. AND SERIAL NO. OF METALLIC LEADING EDGE (MLE)

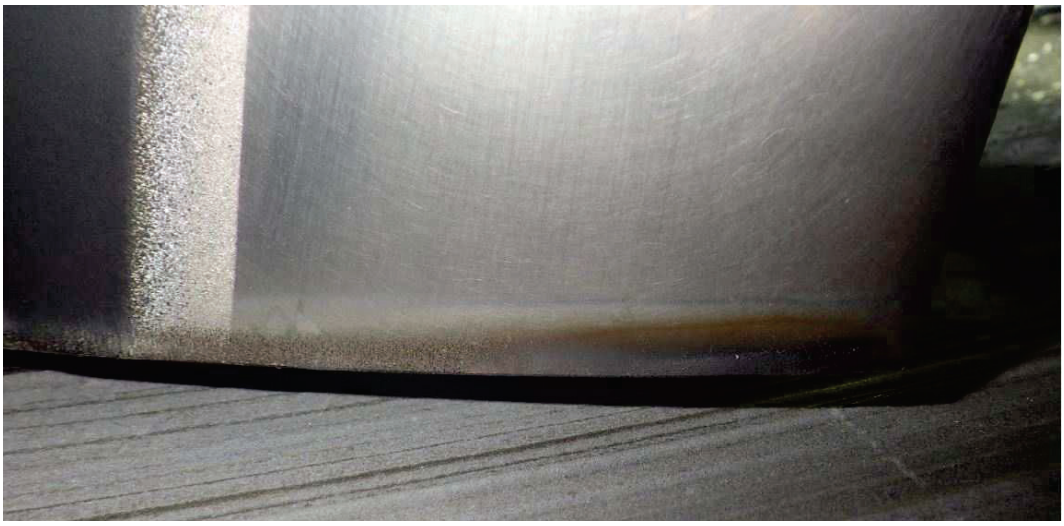
C

1 CONTINUE-IN-SERVICE LIMIT: 1000 CYCLES WITH THE CRITERIA MISSING OR DISBONDED IF PART NO. AND SERIAL NO. OF METALLIC LEADING EDGE (MLE) IS FULLY LEGIBLE.

ICNLEAP1B722202D5882800018A00101
2897453 S0000691009_V1

CIS 1000 Cycles Serviceable Without Label
Figure 602/72-22-02-990-827-G00

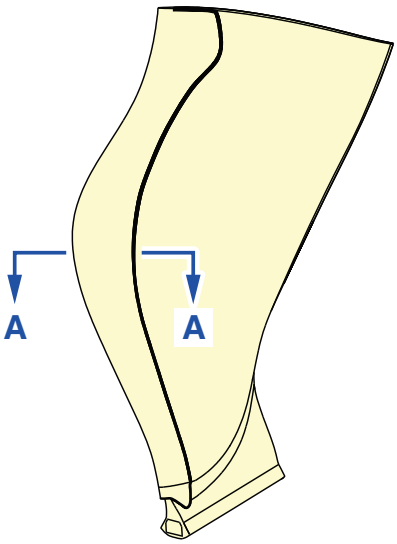
ECCN 9E991 BOEING PROPRIETARY - See title page for details

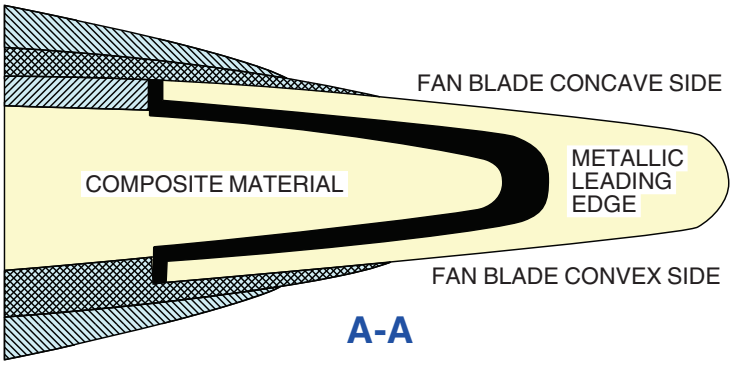


2931413 S0000709260_V1

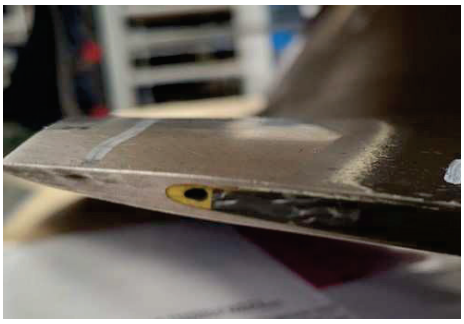
Fan Blade Inspection - Coloration on Metallic Leading Edge
Figure 603/72-22-02-990-828-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

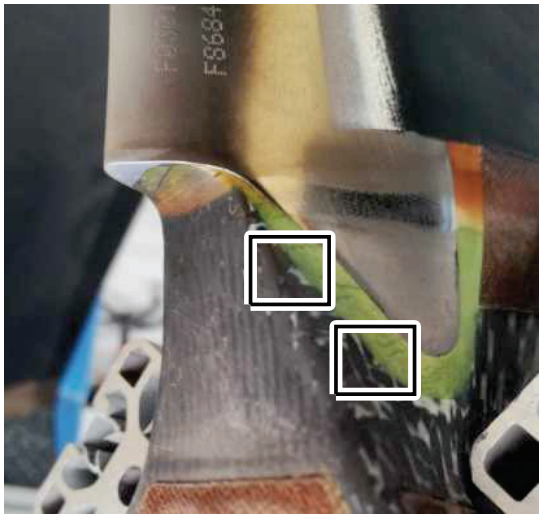




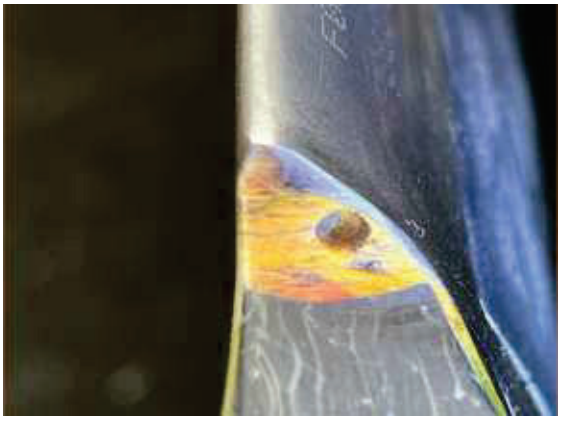
A-A



POROSITY ON TOP OF THE MLE NOSE



MICRO-CRACKS
(EXAMPLE)



POROSITY ON BOTTOM OF THE MLE NOSE

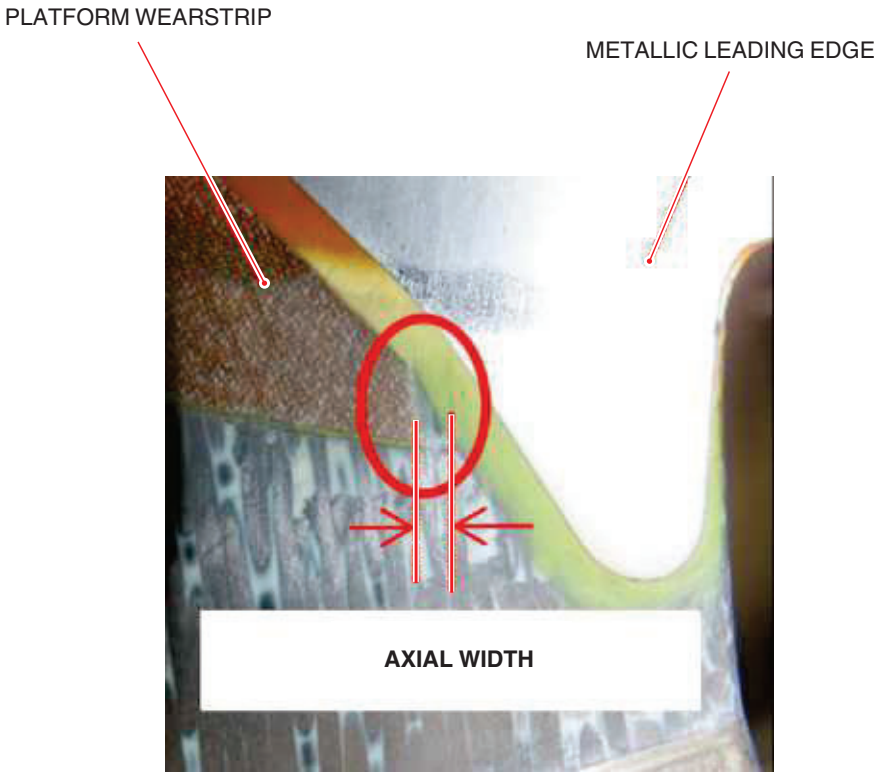
LEGEND:

- MLE GLUE
- PRIMER
- POLYURETHANE (PU)
- BLACK PAINT

ICNLEAP1B722202D5882800039A00101
3095523 S0000843806_V1

Fan Blade Inspection - Metallic Leading Edge Glue/Adhesive Transition Areas
Figure 604/72-22-02-990-848-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

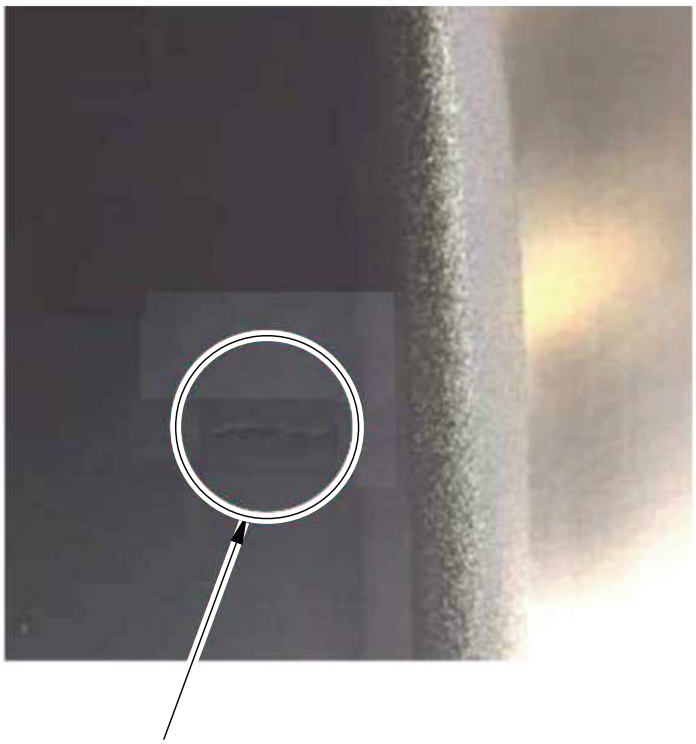


LOCALIZATION OF AXIAL WIDTH MEASUREMENT

ICNLEAP1B722202D5882800060A00101
3112514 S0000851116_V1

Fan Blade Inspection - Metallic Leading Edge Glue/Adhesive Transition Areas
Figure 604/72-22-02-990-848-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B722202D5882800057A00101

3130406 S0000861472_V1

Cosmetic Paint, Linear Indication on the Paint Coat on Concave (Pressure) Side
Figure 605/72-22-02-990-849-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN BLADES - CLEANING/PAINTING

1. General

A. This procedure has one task:

(1) Fan blade cleaning.

TASK 72-22-02-100-801-G00

2. Fan Blade Cleaning

A. General

(1) This task has instructions on how to clean the fan blades.

B. Tools/Equipment

Reference	Description
STD-585	Mat - Protective, Closed Cell Neoprene, 30-50 duro, Weather and Oil Resistant, 3/8 Inch Min Thickness, Minimum 42x60 Inches with Warning Streamers
STD-858	Tag - DO NOT OPERATE

C. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50139	Water - Distilled	

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Cleaning

SUBTASK 72-22-02-930-007-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-22-02-490-002-G00

CAUTION PUT THE PROTECTIVE MAT IN THE INLET COWL. THIS WILL PREVENT DAMAGE TO THE INLET COWL OR ENGINE BY TOOLS, PARTS, OR UNWANTED MATERIALS THAT FALL ON THE INLET COWL SURFACE.

(2) Put the protective mat, STD-585 in the lower half of the inlet cowl.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Fan Blade Cleaning

SUBTASK 72-22-02-941-003-G00

WARNING

DO NOT GET CLEANING SOLUTION IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM CLEANING SOLUTION. PUT ON GOGGLES, AND GLOVES WHEN YOU USE CLEANING SOLUTION. CLEANING SOLUTION IS POISONOUS, AND CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

WARNING

KEEP THE CLEANING SOLUTIONS AWAY FROM SPARKS, FLAMES, AND HEAT. CLEANING SOLUTIONS ARE FLAMMABLE, AND CAN CAUSE INJURIES TO PERSONNEL.

(1) Wipe the fan blades with a cloth moist with alcohol, B00676 [CP1041].

SUBTASK 72-22-02-110-001-G00

(2) If the fan blades are not clean, do the steps that follow:

(a) Remove residue on the surfaces of the blades with a sponge or nylon net moist with distilled water, G50139.

(b) Wipe the fan blades with a clean cotton wiper, G00034.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-22-02-090-002-G00

CAUTION

MAKE SURE THERE ARE NO TOOLS, PARTS OR OTHER UNWANTED MATERIAL IN THE INLET AREA. DAMAGE TO EQUIPMENT COULD OCCUR.

(1) Remove the protective mat, STD-585 from the inlet cowl.

SUBTASK 72-22-02-930-008-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN BLADES - REPAIRS

1. General

A. This procedure has these tasks:

(1) A replacement of the fan blade coating (local repair of anti-erosion paint)

(2) A replacement of the fan blade coating (local repair of platform wear-strips)

(3) A bonding of the fan blade

(4) A blending of the fan blade metal leading edge

(5) A fan blade repair of the missing Polyurethane (PU) film with PU paste.

TASK 72-22-02-900-802-G00

2. Fan Blade Coating Replacement (Local Repair of Anti-Erosion Paint)

(Figure 801, Figure 802, and Figure 803)

A. General

(1) This task gives instructions to locally repair the fan blade paint.

(2) The following equipment is required for the application of this coating:

(a) The paint area must be enclosed and equipped with a ventilation system do not let contamination likely to adversely affect the final characteristics of the film.

(b) The paint booth must be equipped with a temperature and hygrometry recording system and must enable the following conditions during paint application.

(c) Temperature: 59°F (15°C) - 95°F (35°C).

(d) Hygrometry: maximum of 80%.

(e) The paint station air must be filtered, dry, and oil-free.

(f) Protective safety masks.

B. Tools/Equipment

Reference	Description
STD-585	Mat - Protective, Closed Cell Neoprene, 30-50 duro, Weather and Oil Resistant, 3/8 Inch Min Thickness, Minimum 42x60 Inches with Warning Streamers
STD-858	Tag - DO NOT OPERATE

C. Consumable Materials

Reference	Description	Specification
A50488 [CP2724]	Primer - Epoxy, Polyamide - PCEH 100	
A50563 [CP2950]	Hardener - LBYH 200	
B00676 [CP1041]	Alcohol - Isopropyl	
B50383 [CP2949]	Thinner - DL 1511	
C50360 [CP2948]	Paint - Finishing, Top Coat, Black - LBYH 203	
G51261	Abrasive - Non-Silicone Based, 240 Grit (Non-Stearated)	
G51335 [CP2674]	Lint-Free Cloth	
G51700 [CP2968]	Tape - Masking, High-Temperature - 3M 501E	

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Fan Blade Coating Replacement

SUBTASK 72-22-02-860-022-G00

(1) Do these steps to make sure that the ENGINE START switch and FUEL CONTROL switch are not operated:

(a) On the P5 pilot's overhead panel, make sure that the applicable ENGINE START switch is in the NORM (push in) position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) On the P8 pilot's aisle control stand, make sure that the applicable FUEL CONTROL switch is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable FUEL CONTROL switch.

SUBTASK 72-22-02-480-003-G00

CAUTION PUT THE PROTECTIVE MAT IN THE INLET COWL. THIS WILL PREVENT DAMAGE TO THE INLET COWL OR ENGINE BY TOOLS, PARTS, OR UNWANTED MATERIALS THAT FALL ON THE INLET COWL SURFACE.

(2) Put the protective mat, STD-585, in the lower half of the inlet cowl to prevent damage to the inlet cowl and engine.

F. Fan Blade Coating Replacement

SUBTASK 72-22-02-110-002-G00

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

WARNING USE PROTECTIVE GLOVES TO PREVENT INJURY FROM SHARP EDGES AND ROUGH SURFACES. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

(1) If it is necessary, clean the fan blade [1] with lint-free cloth, G51335 [CP2674], moist with alcohol, B00676 [CP1041] (Figure 801).

NOTE: The tip and platform wear strips must not be painted.

SUBTASK 72-22-02-350-001-G00

WARNING DO NOT BREATHE THE PARTICLES WHEN YOU SAND. THE PARTICLES CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PROTECTION EQUIPMENT. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR.

CAUTION DO NOT SAND INTO THE COMPOSITE MATERIAL ON THE ENGINE FAN BLADES. IF YOU DO NOT OBEY, DAMAGE TO THE FAN BLADES CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

DO NOT SAND INTO THE LEADING EDGE ADHESIVE ON THE ENGINE FAN BLADES. IF YOU DO NOT OBEY, DAMAGE TO THE FAN BLADES CAN OCCUR.

CAUTION

BE CAREFUL WHEN YOU REMOVE THE PAINT FROM THE LEADING EDGE SECTION ON THE ENGINE FAN BLADE. IF YOU CHANGE MECHANICAL PROPERTIES OF THE FAN BLADE, DAMAGE TO THE FAN BLADE CAN OCCUR.

(2) If the total surface of paint to repair is less than 0.62 in^2 (400.00 mm^2) and primer is not visible, do the following steps:

(a) Lightly sand the finish paint on the area to repair with 240 grit abrasive paper, G51261.

NOTE: You must not see the primer.

(b) Clean the area to repair with lint-free cloth, G51335 [CP2674], moist with alcohol, B00676 [CP1041].

1) Do not heat part over 248°F (120°C).

2) Do not change the leading edge surface roughness.

(c) Prepare the finish paint as follows:

1) Mix the finish paint in the proportions given below:

NOTE: The pot life of the finish paint is 2 hours maximum.

a) 3.53 oz (100 g) of base of LBYH 203 paint, C50360 [CP2948]

b) 0.35 oz (10 g) of hardener, A50563 [CP2950]

c) thinner, B50383 [CP2949], or deionized water: max 5% by weight of the total mixture.

(d) Paint the area to repair with a brush (Figure 801 and Figure 802).

(e) Dry the finish paint for 30 minutes minimum at ambient temperature.

(f) Cure to 167°F (75°C) - 185°F (85°C) during 30 minutes minimum.

(g) Do a visual inspection as follows:

1) The color and appearance of the repaired area must be homogenous, mat.

2) The repaired area must be smooth and free of run, pinhole, bump, crater or blister, and without inclusions of foreign object.

3) The interface between the repaired area and remaining paint must be smooth.

SUBTASK 72-22-02-350-002-G00

WARNING

USE EYE PROTECTION WHEN YOU CUT MATERIAL OR PARTS. THE PARTICLES CAN CAUSE DAMAGE TO YOUR EYES.

WARNING

DO NOT BREATHE THE PARTICLES WHEN SANDING. THE PARTICLES CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

CAUTION DO NOT SAND INTO THE COMPOSITE MATERIAL ON THE ENGINE FAN BLADES. IF YOU DO NOT OBEY, DAMAGE TO THE FAN BLADES CAN OCCUR.

CAUTION DO NOT SAND INTO THE LEADING EDGE ADHESIVE ON THE ENGINE FAN BLADES. IF YOU DO NOT OBEY, DAMAGE TO THE FAN BLADES CAN OCCUR.

CAUTION BE CAREFUL WHEN YOU REMOVE THE PAINT FROM THE LEADING EDGE SECTION ON THE ENGINE FAN BLADE. IF YOU CHANGE MECHANICAL PROPERTIES OF THE FAN BLADE, DAMAGE TO THE FAN BLADE CAN OCCUR.

(3) If the total surface of paint to repair is not less than 0.62 in^2 (400.00 mm^2) and there is remaining paint around the repair area, do the following steps:

(a) Apply 3M 501E tape, G51700 [CP2968], around the area to repair.

(b) Remove the paint and primer to repair as follows:

1) Use minimum 240 grit abrasive paper, G51261, and 3M 501E tape, G51700 [CP2968], to remove the finish paint and primer as follows:

NOTE: A veil of primer can remain.

a) Make sure that you do not sand the adhesive of the leading edge.

NOTE: The sanding dust must not be black.

b) Do not remove parent material on the leading edge.

c) Do not remove composite material or polyurethane film.

d) Do not heat the part over 248°F (120°C).

e) Do not change the leading edge surface roughness.

f) Do not use materials that can changed the mechanical properties of the leading edge.

2) Remove the mask.

WARNING USE EYE PROTECTION WHEN YOU CUT MATERIAL OR PARTS. THE PARTICLES CAN CAUSE DAMAGE TO YOUR EYES.

3) Blow the sanded area with oil-free and dry air.

4) Clean the area to repair with lint-free cloth, G51335 [CP2674], moist with alcohol, B00676 [CP1041].

5) Do a visual inspection as follows:

a) Non-through scratches on polyurethane film are permitted.

b) Disbonding, foreign objects, step, lack of material or cut on polyurethane film are not permitted.

c) Removal of composite material is not permitted.

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(c) Make sure that the time between the application of the primer and finish paint is not more than 72 hours.

(d) Apply the primer as follows (Figure 801, Figure 802, and Figure 803):

1) Prepare Priam PCEH 100 primer, A50488 [CP2724], as follows:

a) Mix Priam PCEH 100 primer, A50488 [CP2724], in the proportions given below:

NOTE: The pot life of the primer paint is 3 hours maximum.

<1> 3.53 oz (100 g) of base of Priam PCEH 100 primer, A50488 [CP2724]

<2> 3.88 oz (110 g) of hardening agent of Priam PCEH 100 primer, A50488 [CP2724]

<3> thinner, B50383 [CP2949], or deionized water: max 10% by weight of the total mixture.

2) Paint the area with a spray gun as follows:

a) The thickness of the primer in area A must be 0.00039 in. (0.010 mm) -0.00118 in. (0.030 mm) (Figure 802).

b) The thickness of primer must decrease up to 0 in. (0 mm) in the transition paint area (Figure 802).

c) A thin layer of primer can be seen more that the finish paint on the leading edge.

3) Dry the primer for 30 minutes minimum at ambient temperature.

(e) Cure to 167°F (75°C) - 185°F (85°C) during one hour minimum.

(f) Apply the finish paint as follows (Figure 801, Figure 802, and Figure 803):

1) Prepare the finish paint as follows:

a) Mix the finish paint in the proportions given below:

NOTE: The pot life of the finish paint is 2 hours maximum.

<1> 3.53 oz (100 g) of base of LBYH 203 paint, C50360 [CP2948]

<2> 0.35 oz (10 g) of hardener, A50563 [CP2950]

<3> thinner, B50383 [CP2949], or deionized water: max 5% by weight of the total mixture.

2) Paint the area with a spray gun as follows:

a) The thickness of the finishing paint in area A must be less than 0.00177 in. (0.045 mm) (Figure 802).

b) The thickness of finish paint must decrease up to 0 in. (0 mm) in the transition paint area (Figure 802).

c) The overspray areas must not be larger than DIM. F and the surface must be rough and clean in the overspray area before paint overspray (Figure 803).

NOTE: Paint overspray on adjacent areas is permitted.

<1> Dimensional Information.

NOTE: All dimensions are in inches (mm).

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Table 801/72-22-02-993-805-G00 Dimensional Information

Description	Minimum In-Process Dimension	Maximum In-Process Dimension	Minimum Finish Dimension	Maximum Finish Dimension	Reference Dimension
DIM. A	-	-	-	0.59 in. (15 mm)	-
DIM. B	-	-	-	0.59 in. (15 mm)	-
DIM. C	-	-	-	-	0.65295 in. (16.585 mm)
DIM. D	-	-	-	-	2.9528 in. (75.0 mm)
DIM. E	-	-	-	-	1.49606 in. (38.0 mm)
DIM. F	-	-	-	1.18 in. (30 mm)	-

3) Dry the finish paint for 30 minutes minimum at ambient temperature.

(g) Cure to 167°F (75°C) - 185°F (85°C) during 30 minutes minimum.

(h) Do a visual inspection as follows:

1) A thin layer of primer can be seen more that the finish paint on the leading edge.

2) The color and appearance of the repaired area must be homogenous, mat.

3) The repaired area must be smooth and free of any run, pinhole, bump, crater or blister and without inclusions of foreign object.

4) The interface between the repaired area and the remaining paint must be smooth.

5) Paint step is not permitted unless specified differently.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-22-02-080-004-G00

CAUTION MAKE SURE THERE ARE NO TOOLS, PARTS OR OTHER UNWANTED MATERIAL IN THE INLET AREA. DAMAGE TO EQUIPMENT COULD OCCUR.

CAUTION MAKE SURE THERE ARE NO TOOLS, PARTS, OR OTHER UNWANTED MATERIAL IN THE INLET COWL. IF YOU DO NOT OBEY THESE INSTRUCTIONS, YOU CAN CAUSE DAMAGE TO THE ENGINE.

(1) Remove the protective mat, STD-585, from the lower half of the inlet cowl.

SUBTASK 72-22-02-860-023-G00

(2) Do these steps to remove the DO NOT OPERATE tags from the applicable ENGINE START switch and FUEL CONTROL switch:

(a) On the P5 pilot's overhead panel, remove the DO NOT OPERATE tag from the applicable ENGINE START switch.

1) Put the applicable ENGINE START switch to the NORM (push in) position.

(b) On the P8 pilot's aisle control stand, remove the DO NOT OPERATE tag from the applicable FUEL CONTROL switch.

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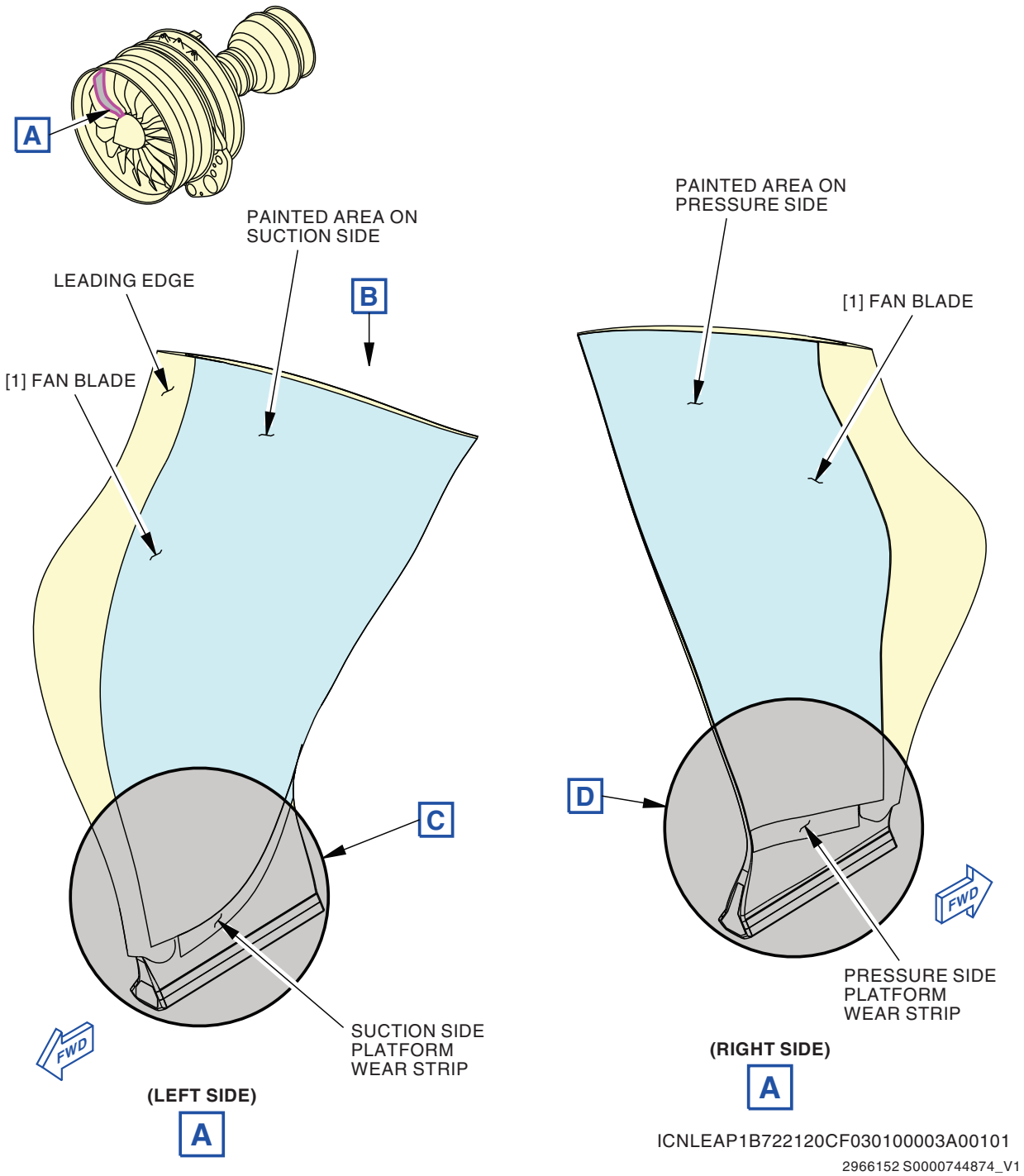
1) Put the applicable FUEL CONTROL switch to the CUTOFF position.

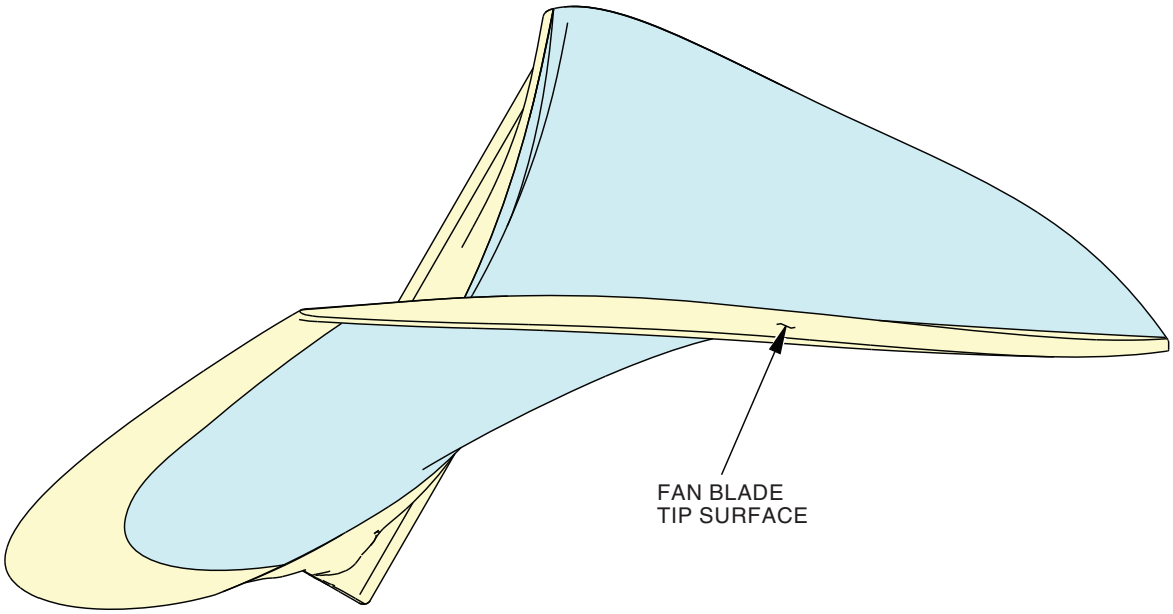
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Blade Painted Areas
Figure 801/72-22-02-990-832-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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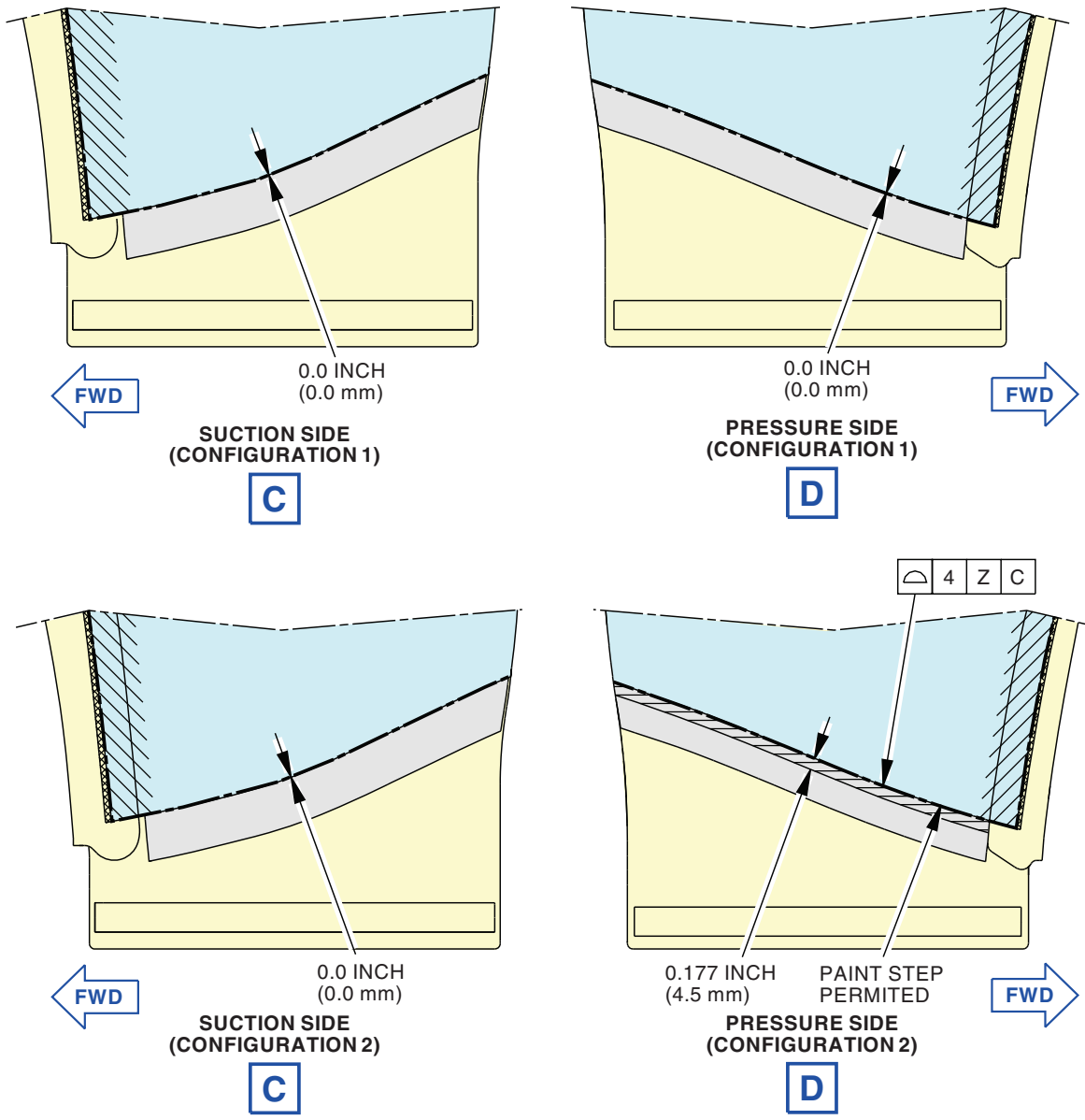
Fan Blade Painted Areas
Figure 801/72-22-02-990-832-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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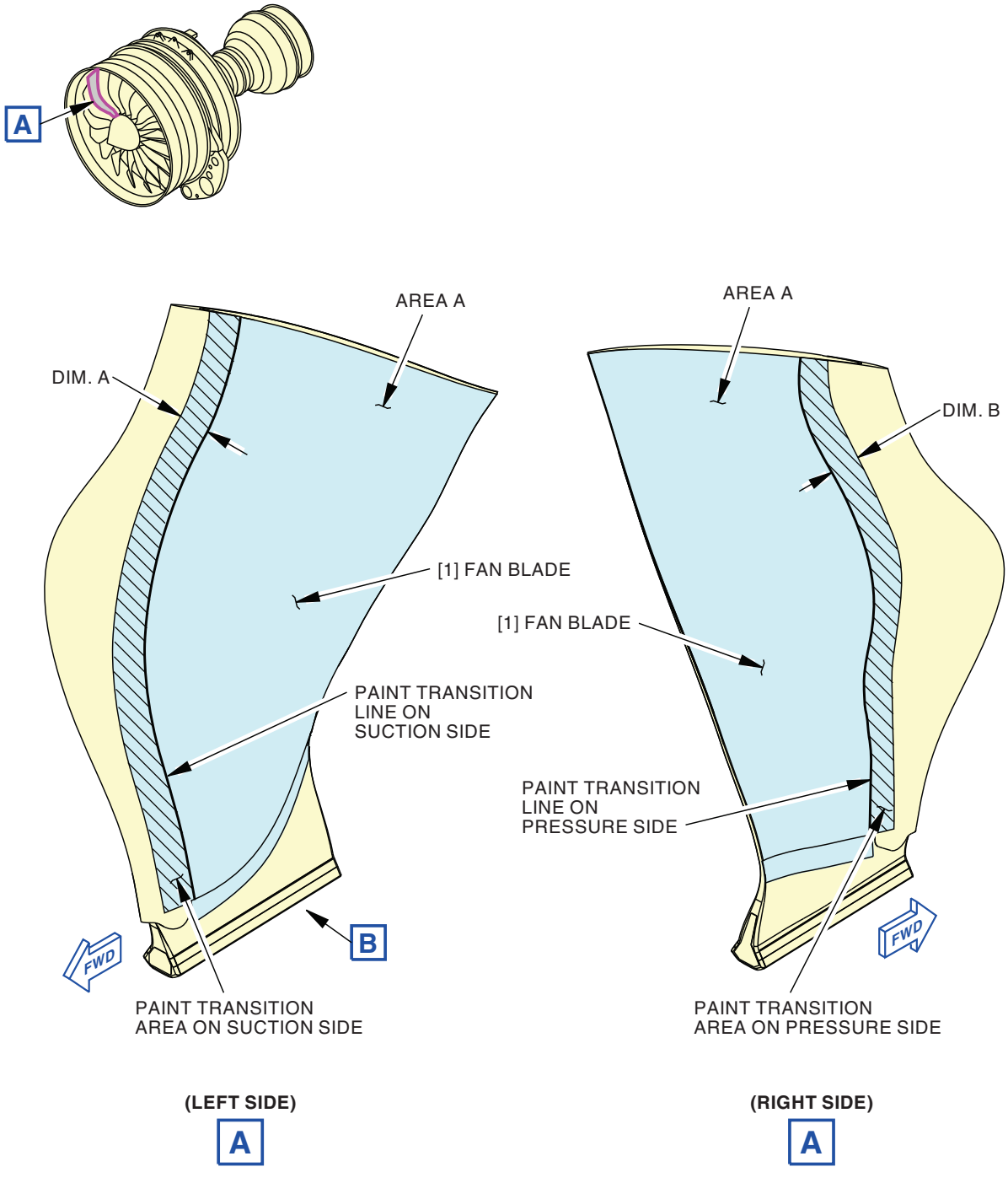
Fan Blade Painted Areas
Figure 801/72-22-02-990-832-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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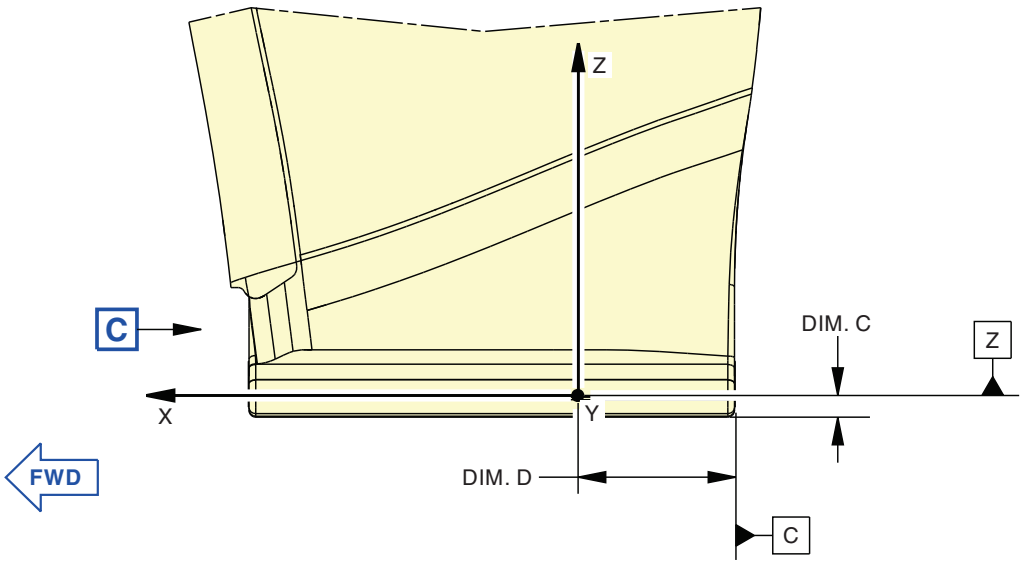
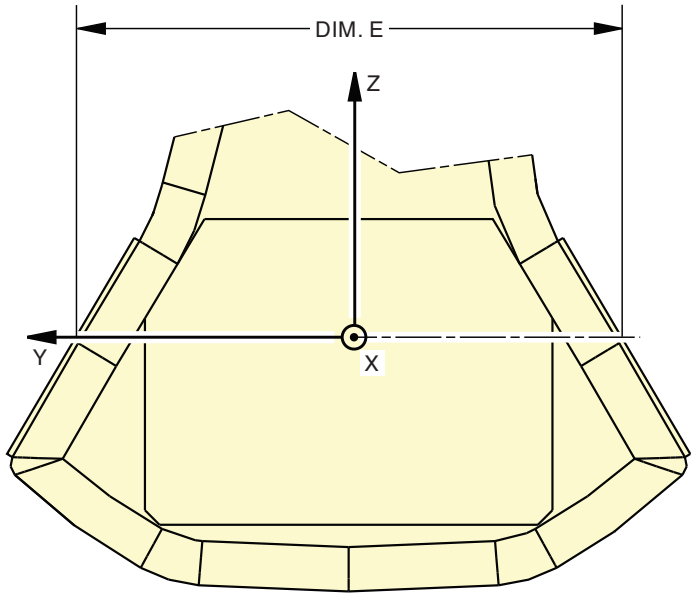
Fan Blade Transition Paint Areas
Figure 802/72-22-02-990-833-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





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Fan Blade Transition Paint Areas
Figure 802/72-22-02-990-833-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EXT RADOS	X/PTQ INCH	X/PTQ mm	Y/PTQ INCH	Y/PTQ mm	Z/PTQ INCH	Z/PTQ mm
PT 1	0.8925	22.67	-0.9630	-24.46	26.6873	677.86
PT 2	0.8323	21.14	-1.0425	-26.48	25.9653	659.52
PT 3	0.8209	20.85	-1.0425	-26.48	25.2334	640.93
PT 4	0.8571	21.77	-0.9453	-24.01	24.5063	622.46
PT 5	0.9583	24.34	-0.8256	-20.97	23.7909	604.29
PT 6	1.1114	28.23	-0.6815	-17.31	23.0893	586.47
PT 7	1.3154	33.41	-0.5315	-13.50	22.4015	569.00
PT 8	1.5508	39.39	-0.3799	-9.65	21.7248	551.81
PT 9	1.7976	45.66	-0.2287	-5.81	21.0515	534.71
PT 10	2.0386	51.78	-0.0807	-2.05	20.3755	517.54
PT 11	2.2638	57.50	0.0657	1.67	19.6941	500.23
PT 12	2.4736	62.83	0.2083	5.29	19.0067	482.77
PT 13	2.6732	67.90	0.3346	8.50	18.3130	465.15
PT 14	2.8583	72.60	0.4406	11.19	17.6122	447.35
PT 15	3.0236	76.80	0.5256	13.35	16.9035	429.35
PT 16	3.1716	80.56	0.5831	14.81	16.1882	411.18
PT 17	3.3012	83.85	0.6087	15.46	15.4677	392.88
PT 18	3.4071	86.54	0.6059	15.39	14.7429	374.47
PT 19	3.4772	88.32	0.5803	14.74	14.0141	355.96
PT 20	3.5008	88.92	0.5362	13.62	13.2830	337.39
PT 21	3.4724	88.20	0.4768	12.11	12.5535	318.86
PT 22	3.3988	86.33	0.4098	10.41	11.8279	300.43
PT 23	3.3118	84.12	0.3500	8.89	11.1027	282.01
PT 24	3.2539	82.65	0.3102	7.88	10.3736	263.49
PT 25	3.2472	82.48	0.2976	7.56	9.6413	244.89
PT 26	3.2705	83.07	0.3012	7.65	8.9090	226.29
PT 27	3.3071	84.00	0.3083	7.83	8.1783	207.73
PT 28	3.3858	86.00	0.3209	8.15	7.4504	189.24
PT 29	3.4925	88.71	0.3287	8.35	6.7256	170.83
PT 30	3.5638	90.52	0.3177	8.07	5.9964	152.31
PT 31	3.5929	91.26	0.2925	7.43	5.2650	133.73
PT 32	3.6027	91.51	0.2720	6.91	4.5327	115.13
PT 33	3.6051	91.57	0.2579	6.55	3.8000	96.52
PT 34	3.6047	91.56	0.2468	6.27	3.0673	77.91
PT 35	3.5957	91.33	0.2327	5.91	2.3350	59.31

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Fan Blade Transition Paint Areas
Figure 802/72-22-02-990-833-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EXT RADOS	X/PTQ INCH	X/PTQ mm	Y/PTQ INCH	Y/PTQ mm	Z/PTQ INCH	Z/PTQ mm
PT 1	1.4237	36.16	-0.6076	-15.43	26.7441	679.30
PT 2	1.3354	33.92	-0.7938	-20.16	26.0149	660.78
PT 3	1.2667	32.17	-0.9285	-23.59	25.2727	641.93
PT 4	1.2656	32.15	-0.9537	-24.22	24.5172	622.74
PT 5	1.4043	35.67	-0.7921	-20.12	23.7942	604.37
PT 6	1.6870	42.85	-0.4619	-11.73	23.1762	588.68
PT 7	2.0379	51.76	-0.0837	-2.13	22.6212	574.58
PT 8	2.3816	60.49	0.2135	5.42	22.0162	559.21
PT 9	2.6942	68.43	0.4086	10.38	21.3543	542.40
PT 10	2.9901	75.95	0.5625	14.29	20.6739	525.12
PT 11	3.2712	83.09	0.7065	17.94	19.9851	507.62
PT 12	3.5274	89.60	0.8352	21.21	19.2836	489.81
PT 13	3.7670	95.68	0.9414	23.91	18.5726	471.74
PT 14	3.9996	101.59	1.0284	26.12	17.8566	453.56
PT 15	4.2189	107.16	1.0923	27.74	17.1341	435.21
PT 16	4.4262	112.43	1.1292	28.68	16.4062	416.72
PT 17	4.6256	117.49	1.1402	28.96	15.6752	398.15
PT 18	4.8103	122.18	1.1252	28.58	14.9404	379.49
PT 19	4.9742	126.34	1.0912	27.72	14.2013	360.71
PT 20	5.1063	129.70	1.0391	26.39	13.4571	341.81
PT 21	5.1889	131.80	0.9629	24.46	12.7078	322.78
PT 22	5.2279	132.79	0.8703	22.11	11.9567	303.70
PT 23	5.2361	133.00	0.7674	19.49	11.2060	284.63
PT 24	5.2141	132.44	0.6500	16.51	10.4577	265.63
PT 25	5.1634	131.15	0.5201	13.21	9.7129	246.71
PT 26	5.0871	129.21	0.3827	9.72	8.9715	227.88
PT 27	4.9890	126.72	0.2423	6.15	8.2333	209.13
PT 28	4.8763	123.86	0.1034	2.63	7.4969	190.42
PT 29	4.7598	120.90	-0.0293	-0.74	6.7600	171.70
PT 30	4.6530	118.19	-0.1535	-3.90	6.0201	152.91
PT 31	4.5627	115.89	-0.2650	-6.73	5.2760	134.01
PT 32	4.4819	113.84	-0.3639	-9.24	4.5290	115.04
PT 33	4.4014	111.80	-0.4536	-11.52	3.7808	96.03
PT 34	4.3238	109.82	-0.5386	-13.68	3.0318	77.01
PT 35	4.2517	107.99	-0.6230	-15.82	2.2821	57.97

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Fan Blade Transition Paint Areas
Figure 802/72-22-02-990-833-G00 (Sheet 4 of 4)

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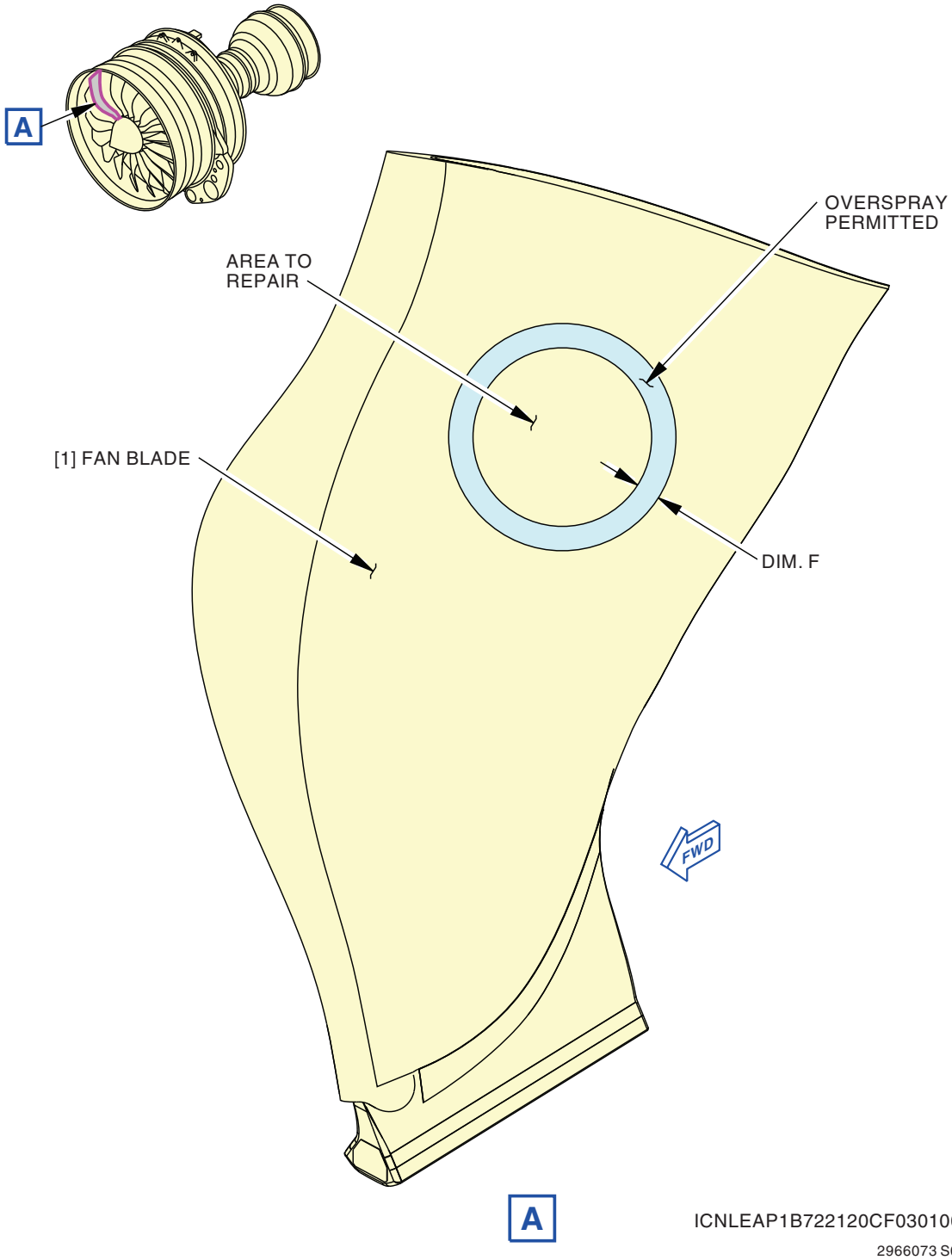

Fan Blade Overspray Dimension
Figure 803/72-22-02-990-834-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-22-02-900-803-G00

3. Fan Blade Coating Replacement (Local Repair of Platform Wear-Strips)

(Figure 804 and Figure 805)

A. General

(1) This task gives instructions to repair the platform wear-strip area of the fan blade on the concave (pressure) side with Scotch-Weld Repair Paste 2110, A50562 [CP2946].

B. References

Reference	Title
72-22-02-200-801-G00	Fan Blade Inspection (P/B 601)

C. Tools/Equipment

Reference	Description
STD-585	Mat - Protective, Closed Cell Neoprene, 30-50 duro, Weather and Oil Resistant, 3/8 Inch Min Thickness, Minimum 42x60 Inches with Warning Streamers
STD-858	Tag - DO NOT OPERATE
STD-14873	Spatula - Teflon

D. Consumable Materials

Reference	Description	Specification
A50562 [CP2946]	Paste - Repair, Scotch-Weld 2110	
B00676 [CP1041]	Alcohol - Isopropyl	
G50767 [CP2279]	Sheet - Abrasive Sheet For Cleaning Up And Finishing, Scotch-Brite Type S CFS	
G51335 [CP2674]	Lint-Free Cloth	
G51693 [CP2967]	Tape - Teflease MG2	
G51694 [CP2348]	Film - Separator - VAC PACK 3760 P1	
G51697 [CP2929]	Tape - Flashbreaker 1	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Fan blade	72-22-02-02-390	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Prepare for the Fan Blade Coating Replacement

SUBTASK 72-22-02-840-001-G00

(1) Do these steps to make sure that the ENGINE START switch and FUEL CONTROL switch are not operated:

(a) On the P5 pilot's overhead panel, make sure that the applicable ENGINE START switch is in the NORM (push in) position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) On the P8 pilot's aisle control stand, make sure that the applicable FUEL CONTROL switch is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable FUEL CONTROL switch.

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SUBTASK 72-22-02-840-002-G00

CAUTION PUT THE PROTECTIVE MAT IN THE INLET COWL. THIS WILL PREVENT DAMAGE TO THE INLET COWL OR ENGINE BY TOOLS, PARTS, OR UNWANTED MATERIALS THAT FALL ON THE INLET COWL SURFACE.

(2) Put the protective mat, STD-585, in the lower half of the inlet cowl to prevent damage to the inlet cowl and the engine.

H. Fan Blade Coating Replacement

SUBTASK 72-22-02-220-002-G00

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

WARNING DO NOT BREATHE THE PARTICLES FROM THE REPAIR OF COMPOSITE MATERIALS OR LET THE PARTICLES TOUCH YOU. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR. THE PARTICLES CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT.

(1) Do a visual and dimensional inspection of the area that you will repair on the concave side of the fan blade [1] as follows:

(a) The wear strip worn dimension A is minimum 0.276 in. (7.0 mm) and maximum 0.315 in. (8.0 mm).

NOTE: If the wear strip is worn more than dimension A, you cannot repair the fan blade with this task.

(b) Wear on the composite is not permitted.

NOTE: If there is wear on the composite, you cannot repair the fan blade with this task.

SUBTASK 72-22-02-950-001-G00

WARNING USE EYE PROTECTION WHEN YOU CUT MATERIAL OR PARTS. THE PARTICLES CAN CAUSE DAMAGE TO YOUR EYES.

WARNING BE CAREFUL WHEN USE SHARP OR POINTED TOOLS. THESE TOOLS CAN CAUSE AN INJURY TO YOU.

CAUTION MAKE SURE THAT YOU DO NOT CUT INTO THE LOWER PRESSURE-SIDE PLATFORM WEAR-STRIP PAINTED AREA OF THE FAN BLADE OR METAL LEADING EDGE. IF YOU DO NOT OBEY, DAMAGE TO THE FAN BLADE CAN OCCUR.

(2) Remove the upper pressure-side platform wear-strip of the fan blade [1] as follows (Figure 804):

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(a) Apply mask on the adjacent areas (painted area, Metal Leading Edge (MLE) area, lower pressure-side platform wear-strip, MLE adhesive) with Teflease MG2 tape, G51693 [CP2967].

(b) Apply mask again on the first layer of masking with Flashbreaker 1 tape, G51697 [CP2929].

(c) Use a sharp tool to cut the pressure-side platform wear-strip along line A.

1) Do not cut into the composite material.

(d) Remove the wear-strip with a sharp or pointed tool.

(e) Make sure that you fully removed the worn platform wear-strip in the area that you repair.

NOTE: You can use a metal shim to help the removal of the upper pressure-side platform wear-strip.

SUBTASK 72-22-02-120-001-G00

WARNING

DO NOT BREATHE THE PARTICLES FROM THE REPAIR OF COMPOSITE MATERIALS OR LET THE PARTICLES TOUCH YOU. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR. THE PARTICLES CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT.

CAUTION

MAKE SURE THAT YOU DO NOT CUT INTO THE LOWER PRESSURE-SIDE PLATFORM WEAR-STRIP PAINTED AREA OF THE FAN BLADE OR METAL LEADING EDGE. IF YOU DO NOT OBEY, DAMAGE TO THE FAN BLADE CAN OCCUR.

(3) Sand the areas that you repair as follows (Figure 804 and Figure 805):

(a) Apply mask on the adjacent areas (painted area, MLE area, lower pressure-side platform wear-strip, MLE adhesive) with Teflease MG2 tape, G51693 [CP2967].

(b) Apply mask again on the first layer of masking with Flashbreaker 1 tape, G51697 [CP2929].

(c) Do a lightly hand sanding with Scotch-Brite Type S CFS Sheet, G50767 [CP2279] (minimum grade P240), to remove all the remaining layer of wear-strip adhesive and to make the surface rough.

NOTE: The sanding dust must not be black.

(d) Remove the masking.

SUBTASK 72-22-02-100-001-G00

(4) Clean the area that you repair with lint-free cloth, G51335 [CP2674], moist with alcohol, B00676 [CP1041].

SUBTASK 72-22-02-210-004-G00

(5) Do a visual inspection of the area that you repair (Figure 804).

(a) Make sure that the lower pressure-side platform wear-strip is correct (Fan Blade Inspection, TASK 72-22-02-200-801-G00).

(b) Wear on the composite is not permitted.

(c) Make sure that there is no remaining wear-strip fiber on the upper area.

(d) Make sure that the surface roughness is homogeneous.

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SUBTASK 72-22-02-950-002-G00

(6) Apply mask on the adjacent areas (painted area, MLE area, lower pressure-side platform wear-strip, MLE adhesive) with Teflease MG2 tape, G51693 [CP2967] (Figure 804 and Figure 805).

SUBTASK 72-22-02-100-002-G00

CAUTION DO NOT TOUCH THE CLEANED FAN BLADE SURFACES WITH YOUR BARE HANDS. USE WHITE COTTON GLOVES WHEN YOU TOUCH THE FAN BLADE OR YOU CAN COMPROMISE THE BONDLINE.

(7) Clean the area that you repair with lint-free cloth, G51335 [CP2674], moist with alcohol, B00676 [CP1041].

(a) Do not touch the cleaned surface with your bare hands, if you do it clean the area again.

SUBTASK 72-22-02-300-002-G00

(8) Apply Scotch-Weld Repair Paste 2110, A50562 [CP2946], to the fan blade [1] area that you repair as follows:

(a) Apply a small quantity of Scotch-Weld Repair Paste 2110, A50562 [CP2946], to the fan blade [1] area.

(b) Make sure that Scotch-Weld Repair Paste 2110, A50562 [CP2946], smooth to make it flush with the lower pressure-side platform wear-strip and the painted airfoil fan blade [1].

1) If it is necessary, use the teflon spatula, STD-14873, to make Scotch-Weld Repair Paste 2110, A50562 [CP2946], smooth.

2) If it is necessary, apply a piece of VAC-PACK 3760 P1 separator film, G51694 [CP2348], on the top of Scotch-Weld Repair Paste 2110, A50562 [CP2946].

a) Use a roller on the release film VAC-PACK 3760 P1 separator film, G51694 [CP2348], to make Scotch-Weld Repair Paste 2110, A50562 [CP2946], smooth.

(c) Make sure that you do not put Scotch-Weld Repair Paste 2110, A50562 [CP2946], smooth on the MLE.

SUBTASK 72-22-02-300-003-G00

WARNING USE THERMAL GLOVES WHEN YOU MOVE OR TOUCH HOT OR COLD PARTS. HOT OR COLD PARTS CAN CAUSE INJURY.

(9) Let Scotch-Weld Repair Paste 2110, A50562 [CP2946], cure as follows:

NOTE: A relative humidity more than 40 percent can prevent full curing of Scotch-Weld Repair Paste 2110, A50562 [CP2946]. Use a heating device if the relative humidity is more than 40 percent.

(a) If the temperature is 64°F (18°C) - 84°F (29°C), let Scotch-Weld Repair Paste 2110, A50562 [CP2946], dry for 24 hours.

1) Apply a piece of release film VAC-PACK 3760 P1 separator film, G51694 [CP2348], on top of Scotch-Weld Repair Paste 2110, A50562 [CP2946], and attach the film with tape.

(b) If the temperature is 131°F (55°C) - 171°F (77°C), let Scotch-Weld Repair Paste 2110, A50562 [CP2946], dry for 120 - 150 minutes.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) If the temperature is 171°F (77°C) - 190°F (88°C), let Scotch-Weld Repair Paste 2110, A50562 [CP2946], dry for 60 - 90 minutes.

SUBTASK 72-22-02-020-010-G00

(10) If you applied VAC-PACK 3760 P1 separator film, G51694 [CP2348], remove it and remove masking from the fan blade [1].

(a) Remove unwanted Scotch-Weld Repair Paste 2110, A50562 [CP2946].

SUBTASK 72-22-02-211-015-G00

(11) Do a visual inspection of the repaired area as follows:

(a) Make sure that Scotch-Weld Repair Paste 2110, A50562 [CP2946], is flush with the lower pressure-side platform wear-strip and the painted airfoil of the fan blade [1].

(b) Blisters are not permitted.

(c) Contamination more than 0.0039 in. (0.10 mm) in depth is not permitted.

(d) Scotch-Weld Repair Paste 2110, A50562 [CP2946], on the MLE adhesive is permitted.

(e) Make sure that the fan blade [1] repaired area is fully covered with Scotch-Weld Repair Paste 2110, A50562 [CP2946].

1) If the fan blade [1] repaired area is not fully covered, do the repair again on the missing Scotch-Weld Repair Paste 2110, A50562 [CP2946], area.

SUBTASK 72-22-02-700-001-G00

(12) Make sure that Scotch-Weld Repair Paste 2110, A50562 [CP2946], is cured.

(a) Do an inspection of surface shape memory as follows:

1) Apply force with the teflon spatula, STD-14873, to the cured surface to cause a movement.

2) Remove the teflon spatula, STD-14873.

3) Make sure that the surface of Scotch-Weld Repair Paste 2110, A50562 [CP2946], keeps the initial shape.

(b) Make sure that the cured surface dry and not wet.

(c) If Scotch-Weld Repair Paste 2110, A50562 [CP2946], is not cured, remove the non-cured Scotch-Weld Repair Paste 2110, A50562 [CP2946], from the repaired area and do the repair again.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-22-02-840-003-G00

CAUTION

MAKE SURE THERE ARE NO TOOLS, PARTS OR OTHER UNWANTED MATERIAL IN THE INLET AREA. DAMAGE TO EQUIPMENT COULD OCCUR.

CAUTION

MAKE SURE THERE ARE NO TOOLS, PARTS, OR OTHER UNWANTED MATERIAL IN THE INLET COWL. IF YOU DO NOT OBEY THESE INSTRUCTIONS, YOU CAN CAUSE DAMAGE TO THE ENGINE.

(1) Remove the protective mat, STD-585, from the lower half of the inlet cowl.

SUBTASK 72-22-02-840-004-G00

(2) Do these steps to remove the DO NOT OPERATE tags from the applicable ENGINE START switch and FUEL CONTROL switch:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) On the P5 pilot's overhead panel, remove the DO NOT OPERATE tag from the applicable ENGINE START switch.

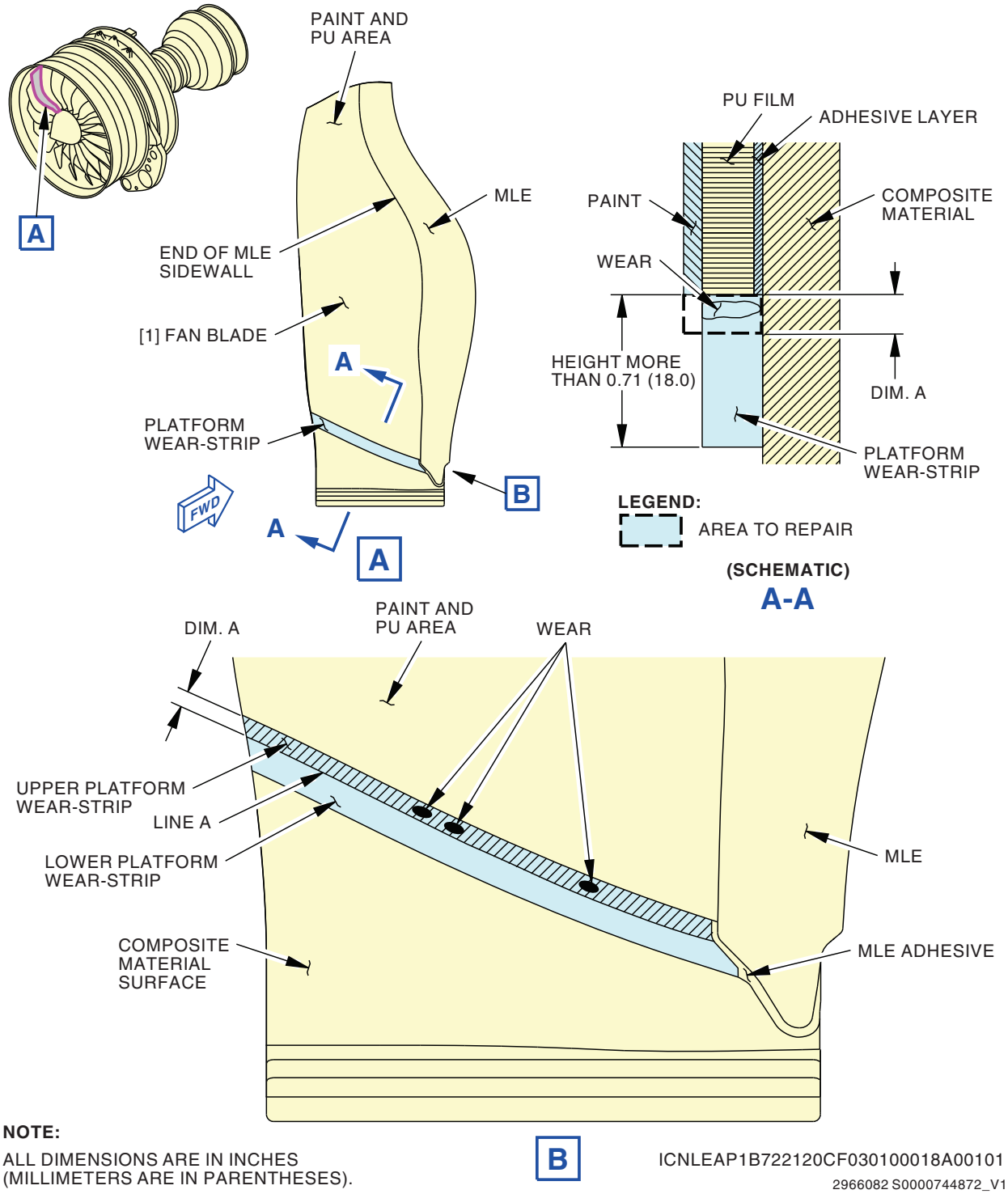
1) Put the applicable ENGINE START switch to the NORM (push in) position.

(b) On the P8 pilot's aisle control stand, remove the DO NOT OPERATE tag from the applicable FUEL CONTROL switch.

1) Put the applicable FUEL CONTROL switch to the CUTOFF position.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

ALL DIMENSIONS ARE IN INCHES
(MILLIMETERS ARE IN PARENTHESES).

B

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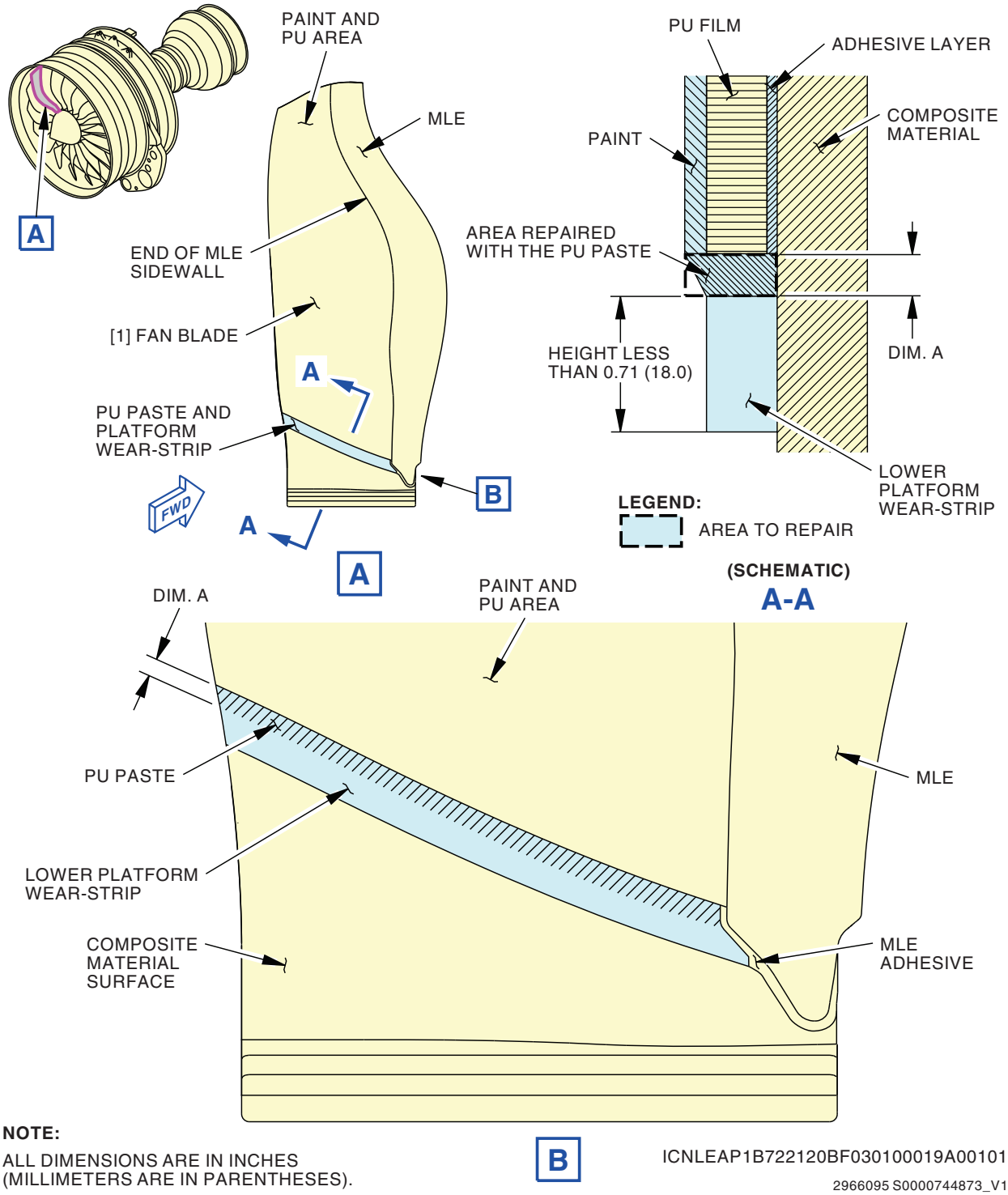
Fan Blade with All Platform Wear-Strip on Concave Side
Figure 804/72-22-02-990-830-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

ALL DIMENSIONS ARE IN INCHES
(MILLIMETERS ARE IN PARENTHESES).

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Fan Blade with PU Paste and Platform Wear-Strip on Concave Side
Figure 805/72-22-02-990-831-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-22-02-900-804-G00

4. Fan Blade Label Replacement

(Figure 806)

A. General

(1) This task gives instructions to replace the identification and/or moments-weight labels.

B. References

Reference	Title
72-22-02-000-804-G00	Fan Blades Removal (Selection) (P/B 401)
72-22-02-400-804-G00	Fan Blades Installation (Selection) (P/B 401)

C. Tools/Equipment

Reference	Description
STD-585	Mat - Protective, Closed Cell Neoprene, 30-50 duro, Weather and Oil Resistant, 3/8 Inch Min Thickness, Minimum 42x60 Inches with Warning Streamers
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
C50435 [CP8019]	Varnish Base - A1500-UVR	
G51261	Abrasive - Non-Silicone Based, 240 Grit (Non-Stearated)	
G51697 [CP2929]	Tape - Flashbreaker 1	
G51698 [CP8020]	Veil - Glass (Hollingsworth Vose P/N 8000110)	
G51699 [CP8024]	Label - Paper, Cotton	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Fan blade	72-22-02-02-390	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Prepare for the Fan Blade Label Replacement

SUBTASK 72-22-02-840-005-G00

(1) Do these steps to make sure that the ENGINE START switch and FUEL CONTROL switch are not operated:

(a) On the P5 pilot's overhead panel, make sure that the applicable ENGINE START switch is in the NORM (push in) position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) On the P8 pilot's aisle control stand, make sure that the applicable FUEL CONTROL switch is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable FUEL CONTROL switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-22-02-840-006-G00

CAUTION

PUT THE PROTECTIVE MAT IN THE INLET COWL. THIS WILL PREVENT DAMAGE TO THE INLET COWL OR ENGINE BY TOOLS, PARTS, OR UNWANTED MATERIALS THAT FALL ON THE INLET COWL SURFACE.

(2) Put the protective mat, STD-585, in the lower half of the inlet cowl to prevent damage to the inlet cowl and the engine.

SUBTASK 72-22-02-820-001-G00

(3) If it is necessary, measure the axial, tangential and radial moments-weight of the fan blade [1].

(a) Make sure that the scale is calibrated.

(b) Install the fan blade [1].

(c) Weigh the blade and record the moments-weight.

NOTE: Refer to the scale manufacturer's procedures for weighing.

SUBTASK 72-22-02-800-001-G00

(4) For the moments-weight label, write the information on cotton paper label, G51699 [CP8024], with an Optical Character Recognition (OCR) and scan legible print font as follows:

(a) Character height must be 0.098 in. (2.5 mm).

(b) The space between two characters must be between 0.2 and 0.4 times the height of the characters.

(c) The space between two markings on the same line must be at least equivalent to three characters.

(d) The minimum line spacing must be at 0.0469 in. (1.19 mm).

(e) The margin between the information written and the sides of the label must be at least 0.5 times the character height.

(f) Write the information correctly (View B, Figure 806).

(g) The location of the data matrix can be on the right side or the left side of the label.

(h) Make sure that the ink is dry.

(i) Make sure that you can read information from the label.

SUBTASK 72-22-02-800-002-G00

(5) For the identification label, write the information on cotton paper label, G51699 [CP8024], with an OCR and scan legible print font as follows:

(a) Character height must be 0.098 in. (2.5 mm).

(b) The space between two characters must be between 0.2 and 0.4 times the height of the characters.

(c) The space between two markings on the same line must be at least equivalent to three characters.

(d) The minimum line spacing must be at 0.0469 in. (1.19 mm).

(e) The margin between the information written and the sides of the label must be at least 0.5 times the character height.

(f) Write the information correctly (View B, Figure 806).

(g) The location of the data matrix can be on the right side or the left side of the label.

(h) Make sure that the ink is dry.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(i) Make sure that you can read information from the label.

H. Fan Blade Label Replacement

SUBTASK 72-22-02-860-029-G00

(1) Do this task to remove the fan blade: Fan Blades Removal (Selection), TASK 72-22-02-000-804-G00.

SUBTASK 72-22-02-950-003-G00

(2) Apply masking Flashbreaker 1 tape, G51697 [CP2929], around the varnished area of the label that you replace.

SUBTASK 72-22-02-970-008-G00

(3) If it is necessary, identify and record the information marked on the labels and included in the data matrix.

SUBTASK 72-22-02-000-001-G00

CAUTION MAKE SURE THAT YOU DO NOT CUT INTO THE CARBON-FIBER COMPOSITE AREA OF THE FAN BLADE. DAMAGE TO THE FAN BLADE CAN OCCUR.

(4) Remove the label and varnish from the fan blade [1] with a scalpel or any other sharp tool.

(a) Remove the remaining varnish with 240 grit abrasive paper, G51261.

1) Do not remove the composite material.

(b) Clean the area where you removed the label with alcohol, B00676 [CP1041].

SUBTASK 72-22-02-210-005-G00

(5) Do a visual inspection of the area where you remove the label.

(a) Make sure that the surface is mat.

1) A bright area is not permitted.

(b) Make sure that here is no residual varnish.

(c) Make sure that there is no worn fibers or cracks.

SUBTASK 72-22-02-410-006-G00

(6) Bond the cotton labels as follows (Figure 806):

(a) Apply A1500-UVR varnish base, C50435 [CP8019], on the varnished area.

(b) Apply cotton paper label, G51699 [CP8024], on the varnished area.

1) Make sure that you can read the text when the fan blade is tip down.

(c) Apply A1500-UVR varnish base, C50435 [CP8019], on the label.

(d) Apply glass veil, G51698 [CP8020], on the varnished label.

1) The dimensions of the glass veil must be larger than the label (View C, Figure 806).

(e) Apply A1500-UVR varnish base, C50435 [CP8019], on the label.

1) The dimensions of the varnished area must be larger than the glass veil (View C, Figure 806).

2) Do not apply A1500-UVR varnish base, C50435 [CP8019], on the wear-strips.

(f) Let A1500-UVR varnish base, C50435 [CP8019], dry at ambient temperature during 7 - 9 hours or at a temperature 140°F (60°C) - 149°F (65°C) during 3 hours.

SUBTASK 72-22-02-210-006-G00

(7) Do a visual and dimensional examination as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Foreign objects more than 0.04 in. (1 mm) in diameter are not permitted.

(b) Air bubbles more than 0.04 in. (1 mm) in diameter are not permitted.

(c) Non-soak areas more than 0.0062 in^2 (4 mm^2) are not permitted.

(d) Make sure that you can read information from the label.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-22-02-860-028-G00

(1) Do this task to install the fan blade: Fan Blades Installation (Selection), TASK 72-22-02-400-804-G00.

SUBTASK 72-22-02-840-007-G00

CAUTION

MAKE SURE THERE ARE NO TOOLS, PARTS OR OTHER UNWANTED MATERIAL IN THE INLET AREA. DAMAGE TO EQUIPMENT COULD OCCUR.

CAUTION

MAKE SURE THERE ARE NO TOOLS, PARTS, OR OTHER UNWANTED MATERIAL IN THE INLET COWL. IF YOU DO NOT OBEY THESE INSTRUCTIONS, YOU CAN CAUSE DAMAGE TO THE ENGINE.

(2) Remove the protective mat, STD-585, from the lower half of the inlet cowl.

SUBTASK 72-22-02-840-008-G00

(3) Do these steps to remove the DO NOT OPERATE tags from the applicable ENGINE START switch and FUEL CONTROL switch:

(a) On the P5 pilot's overhead panel, remove the DO NOT OPERATE tag from the applicable ENGINE START switch.

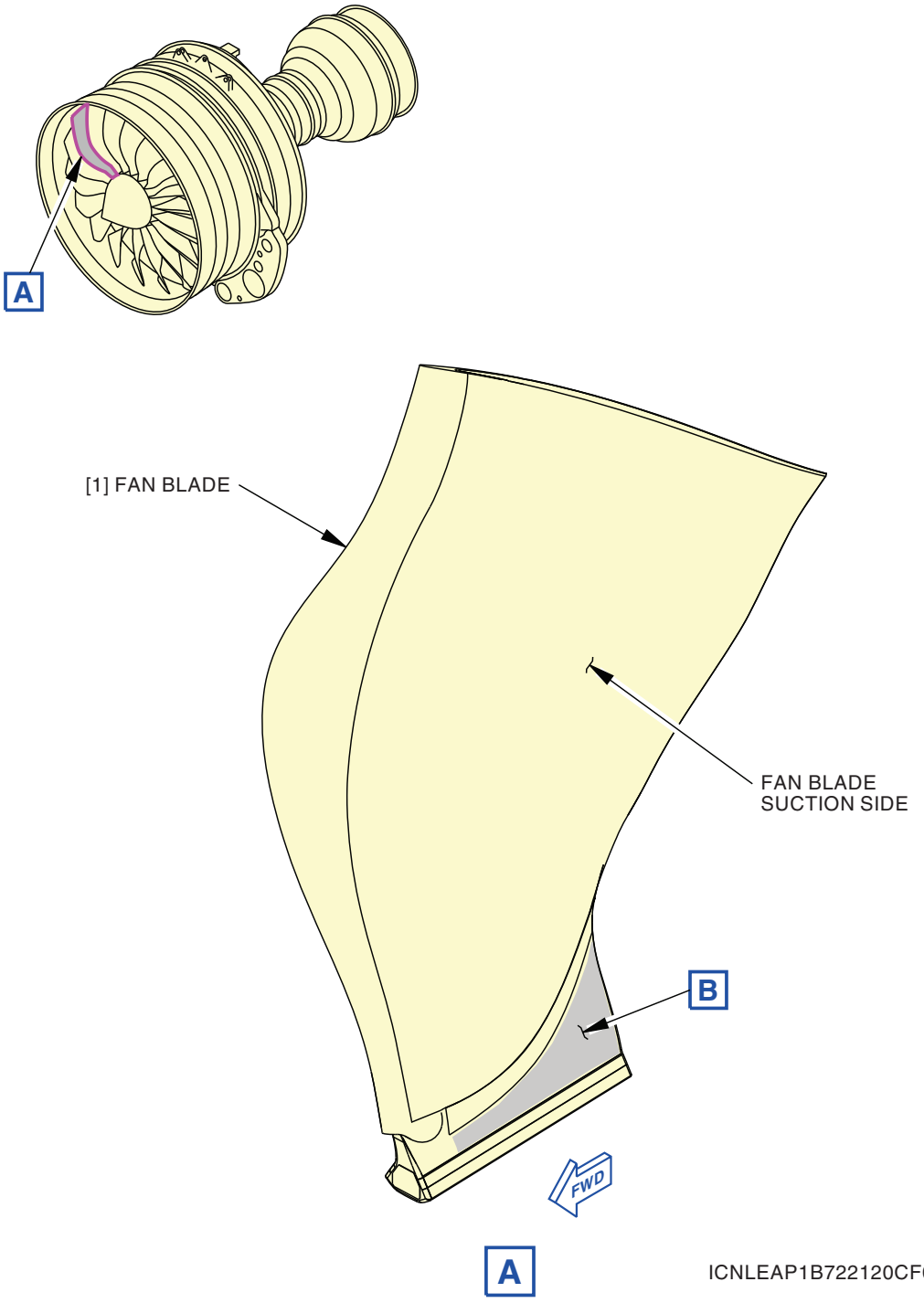
1) Put the applicable ENGINE START switch to the NORM (push in) position.

(b) On the P8 pilot's aisle control stand, remove the DO NOT OPERATE tag from the applicable FUEL CONTROL switch.

1) Put the applicable FUEL CONTROL switch to the CUTOFF position.

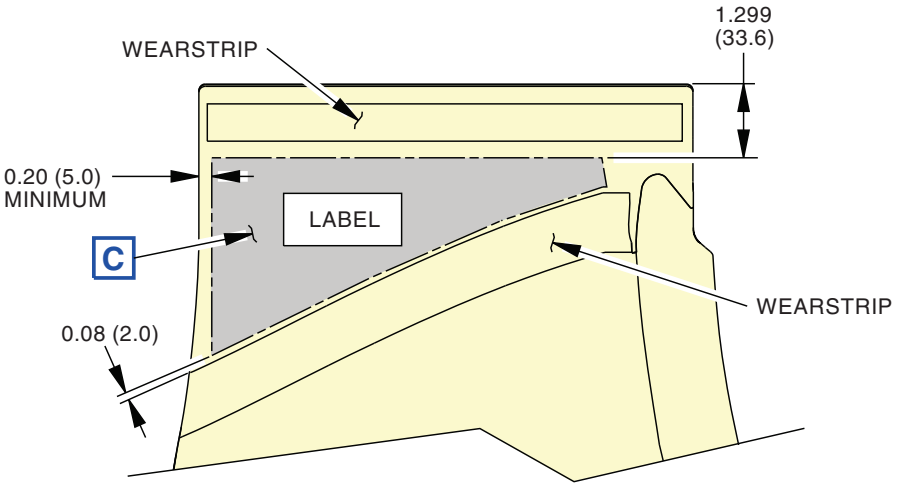
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Blade Label Bonding
Figure 806/72-22-02-990-835-G00 (Sheet 1 of 3)

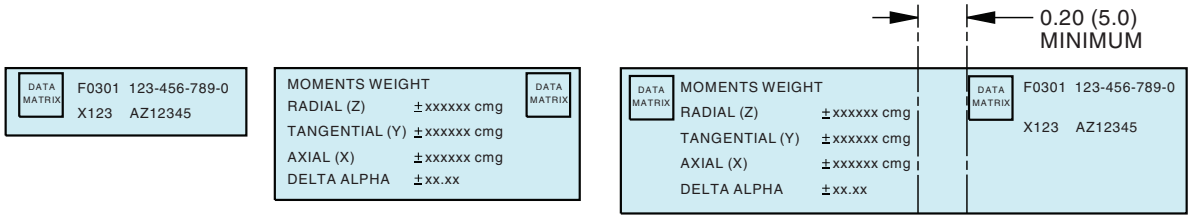
ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

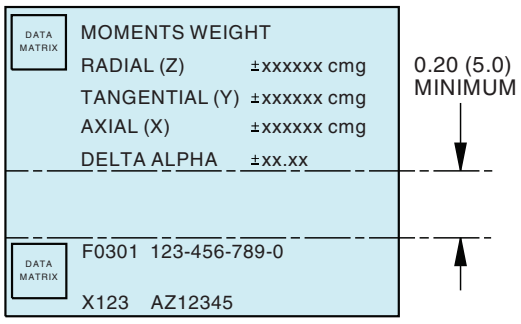
PERMITTED AREA TO BOND THE LABEL(S)

B



OPTION 1: TWO LABELS

OPTION 2: ONE LABEL HORIZONTAL



OPTION 3: ONE LABEL VERTICAL

NOTE:

ALL DIMENSIONS ARE IN INCHES
(MILLIMETERS ARE IN PARENTHESES).

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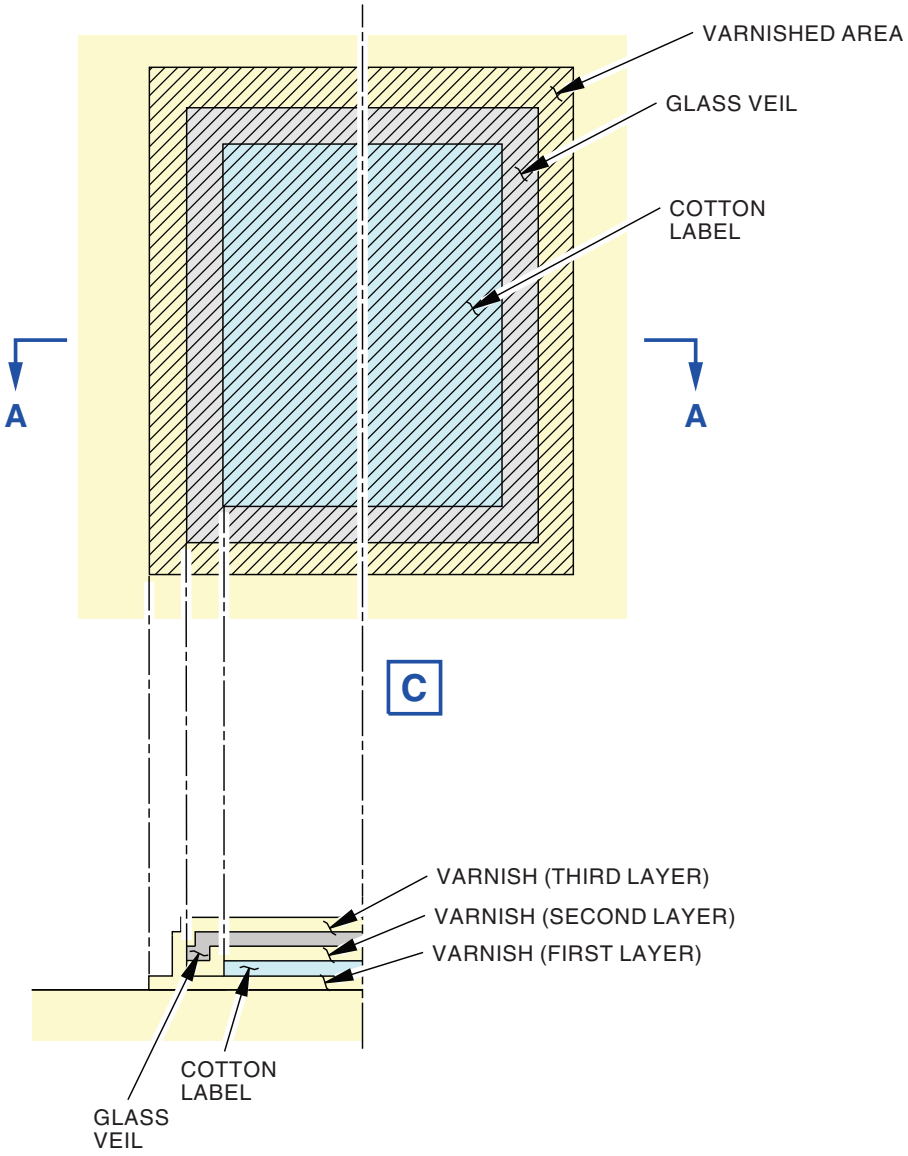
Fan Blade Label Bonding
Figure 806/72-22-02-990-835-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Fan Blade Label Bonding
Figure 806/72-22-02-990-835-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-22-02-900-805-G00

5. Fan Blade Metal Leading Edge Blending

(Figure 807, Figure 808, Figure 809, Figure 810, Figure 811, and Figure 812)

A. General

(1) This task gives instructions to blend the nose of the fan blade Metal Leading Edge (MLE).

B. References

Reference	Title
70-11-06-230-802-G00	Fluorescent Penetrant Inspection (P/B 201)
70-11-06-230-803-G00	Fluorescent Penetrant Inspection (General) (P/B 201)
70-21-23-100-801-G00	Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701)
70-43-24-100-802-G00	Swab Etching Procedure and Cleaning (P/B 201)
70-43-42-300-802-G00	Blending Procedures - Hand Blending (P/B 201)
70-43-42-300-803-G00	Blending Procedures - Power Blending (P/B 201)
72-22-02-200-801-G00	Fan Blade Inspection (P/B 601)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-21303	Bit, Burr Milling - Moto Tool Part #: VWB800D Supplier: 55719 Part #: ZYA 0413 6S 3 Plus Supplier: $1355
STD-585	Mat - Protective, Closed Cell Neoprene, 30-50 duro, Weather and Oil Resistant, 3/8 Inch Min Thickness, Minimum 42x60 Inches with Warning Streamers
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G51754	Paper - Abrasive, Silicon Carbide, 220 Grit	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Prepare for the Fan Blade Metal Leading Edge Blending

SUBTASK 72-22-02-840-010-G00

(1) Do these steps to make sure that the ENGINE START switch and FUEL CONTROL switch are not operated:

(a) On the P5 pilot's overhead panel, make sure that the applicable ENGINE START switch is in the NORM (push in) position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) On the P8 pilot's aisle control stand, make sure that the applicable FUEL CONTROL switch is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable FUEL CONTROL switch.

SUBTASK 72-22-02-840-011-G00

(2) Put the protective mat, STD-585, in the lower half of the inlet cowl to prevent damage to the inlet cowl and engine.

G. Fan Blade Metal Leading Edge Blending

SUBTASK 72-22-02-110-003-G00

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

(1) If it is necessary, clean the area with clean cotton wiper, G00034, moist with alcohol, B00676 [CP1041], or distilled water (TASK 70-21-23-100-801-G00).

SUBTASK 72-22-02-110-004-G00

WARNING DO NOT LET TITANIUM DUST COLLECT. TITANIUM DUST IS VERY FLAMMABLE. DISCARD ALL TITANIUM WASTE IN AN APPROVED CONTAINER. TITANIUM DUST CAN CAUSE INJURY TO PERSONNEL.

WARNING DO NOT USE WATER TO EXTINGUISH A TITANIUM FIRE. EXTINGUISH WITH AN APPROVED CHEMICAL FOR METAL FIRES. IF YOU DO NOT OBEY, OPERATION OF A WATER EXTINGUISHER COULD CAUSE INJURY TO PERSONNEL.

WARNING DO NOT USE CHLORINATED HYDROCARBONS AND HALOGEN MATERIALS. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

CAUTION DO NOT USE TOOLS ON TITANIUM THAT WERE USED ON OTHER MATERIALS. IT WILL CAUSE CONTAMINATION OF THE TITANIUM. KEEP THE BLENDING ABRASIVE TOOL USED FOR THE TITANIUM ISOLATED FROM TOOLS USED FOR OTHER METALS. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT THE METAL LEADING EDGE IS NOT HOT. IF YOU DO NOT OBEY, DAMAGE TO THE BOND JOINT ADHESIVE AND COMPOSITE MATERIAL CAN OCCUR.

(2) Make sure that the repair can be done with cumulated limits below (Figure 808):

NOTE: If one of the measured cumulated lengths is not in the limits, the blade repair cannot be done.

(a) For each blade, the maximum cumulative length is 6.69 in. (170 mm) with the maximum lengths for the areas E and F:

1) 2.91 in. (74 mm) in area E

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) 1.96 in. (50 mm) in area F.

(b) For each engine, the maximum cumulative length is 32.26 in. (819 mm) with the maximum lengths for the areas E and F:

1) 8.74 in. (222 mm) in area E

2) 5.91 in. (150 mm) in area F.

SUBTASK 72-22-02-320-003-G00

(3) Do one of the fan blade MLE blending as follows:

(a) Do a hand blending of the MLE nose (Figure 808) (TASK 70-43-42-300-802-G00).

1) Refer to the blend repair dimensions below.

(b) Alternative procedure. Do a power blending of the MLE nose as follows:

1) Start with carbide milling tool (milling bit, COM-21303) (Figure 808) (TASK 70-43-42-300-803-G00).

a) Refer to the blend repair dimensions below.

2) Finish with 220-grit abrasive paper, G51754, for a minimum of 0.004 in. (0.1 mm) depth (Figure 808) (TASK 70-43-42-300-803-G00).

a) Refer to the blend repair dimensions below.

(c) Blend repair dimensions:

1) Maximum depth DIM. P:

a) Area B: 0.031 in. (0.8 mm)

b) Area C: 0.04 in. (1 mm)

c) Area D: 0.07 in. (2 mm)

d) Area E: 0.12 in. (3 mm)

e) Area F: 0.16 in. (4 mm).

2) If two blendings overlap, blend out the bottom of the two repairs tangentially.

3) In the Area F, if the DIM. N is less than two times the depth of DIM. P, blend as shown in Figure 808.

4) The radius R1 is 20 (twenty) times the minimum of the depth DIM. P.

5) The radius R2 is 0.20 in. (5 mm) minimum and 0.31 in. (8 mm) maximum.

6) For each blade, the maximum cumulative length is 6.69 in. (170 mm) with the maximum lengths for the Areas E and F:

a) 2.91 in. (74 mm) in area E

b) 1.96 in. (50 mm) in area F.

7) For each engine, the maximum cumulative length is 32.26 in. (819 mm) with the maximum lengths for the areas E and F:

a) 8.74 in. (222 mm) in area E

b) 5.91 in. (150 mm) in area F.

SUBTASK 72-22-02-220-003-G00

(4) Make sure that the contour of the MLE nose after blending is correct (Figure 812).

SUBTASK 72-22-02-220-004-G00

(5) Make sure that the blending is on the limits defined above.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-22-02-110-005-G00

(6) Clean the repaired areas with clean cotton wiper, G00034, moist with alcohol, B00676 [CP1041], or distilled water (TASK 70-21-23-100-801-G00).

SUBTASK 72-22-02-230-001-G00

CAUTION DO NOT SPRAY FLUORESCENT PENETRANT INSPECTION PRODUCTS ON THE PART MADE FROM THE COMPOUND MATERIAL. IF YOU DO NOT OBEY, DAMAGE TO THE PART CAN OCCUR.

(7) Do the Fluorescent Penetrant Inspection (FPI).

(a) For the blend repairs in the areas C, D, E, or F:

1) Alternative procedure available. Do the step G.(3) to do the FPI after blending.

a) Alternative procedure. Do the step G.(3) to do the FPI within 10 flight cycles.

(b) For the blend repairs in area B, the FPI must be performed immediately after blending with the step G.(3).

NOTE: No FPI deferral is permitted.

(c) Do the FPI as follows:

1) If it is necessary, apply mask on the composite area.

2) Do one of the following:

a) If the hand blending is done before, do these steps:

<1> Do the local swab etching of the repaired areas (TASK 70-43-24-100-802-G00).

<2> Do the FPI of the repaired areas (TASK 70-11-06-230-803-G00 and TASK 70-11-06-230-802-G00).

<a> Do an FPI sensitivity level 3.

<3> The FPI acceptance criterias are the following:

<a> Cracks are not permitted.

<b> Indications more than 0.012 in. (0.3 mm) are not permitted.

<c> Indications less than or equal to 0.012 in. (0.3 mm) are permitted.

b) Alternative procedure. If the hand blending is done before, do these steps:

<1> Do the FPI of the repaired areas (TASK 70-11-06-230-803-G00 and TASK 70-11-06-230-802-G00).

<a> Do an FPI sensitivity level 4.

<2> The FPI acceptance criterias are the following:

<a> Cracks are not permitted.

<b> Indications more than 0.012 in. (0.3 mm) are not permitted.

<c> Indications less than or equal to 0.012 in. (0.3 mm) are permitted.

c) If the power blending is done before, do these steps:

<1> Do the local swab etching of the repaired areas (TASK 70-43-24-100-802-G00)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

<2> Do the FPI of the repaired areas (TASK 70-11-06-230-803-G00 and TASK 70-11-06-230-802-G00).

<a> Do an FPI sensitivity level 3.

<3> The FPI acceptance criterias are the following:

<a> Cracks are not permitted.

<b> Indications more than 0.012 in. (0.3 mm) are not permitted.

<c> Indications less than or equal to 0.012 in. (0.3 mm) are permitted.

SUBTASK 72-22-02-110-006-G00

(8) Clean the repaired areas with clean cotton wiper, G00034, moist with alcohol, B00676 [CP1041], or distilled water (TASK 70-21-23-100-801-G00).

SUBTASK 72-22-02-280-001-G00

(9) If the blending depth is equal or more than 0.002 in. (0.05 mm), do a tap inspection of the bond joints between the MLE sidewalls and composite part of the fan blade as follows:

(a) If it is necessary, make the aluminum mallet (Figure 809).

(b) Do a tap test of the surface of the MLE sidewalls (Figure 810, Figure 811) (TASK 72-22-02-200-801-G00).

1) Tap for a distance of 5.12 in. (130 mm) above and below the repaired area.

2) For the acceptance criteria, refer to Fan Blade Inspection, TASK 72-22-02-200-801-G00.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-22-02-080-005-G00

CAUTION

MAKE SURE THERE ARE NO TOOLS, PARTS OR OTHER UNWANTED MATERIAL IN THE INLET AREA. DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION

MAKE SURE THERE ARE NO TOOLS, PARTS OR OTHER UNWANTED MATERIAL IN THE INLET COWL. IF YOU DO NOT OBEY THIS INSTRUCTION YOU CAN CAUSE DAMAGE TO THE ENGINE.

(1) Remove the protective mat, STD-585, from the lower half of the inlet cowl.

SUBTASK 72-22-02-860-027-G00

(2) Do these steps to remove the DO NOT OPERATE tags from the applicable ENGINE START switch and FUEL CONTROL switch:

(a) On the P5 pilot's overhead panel, remove the DO NOT OPERATE tag from the applicable ENGINE START switch.

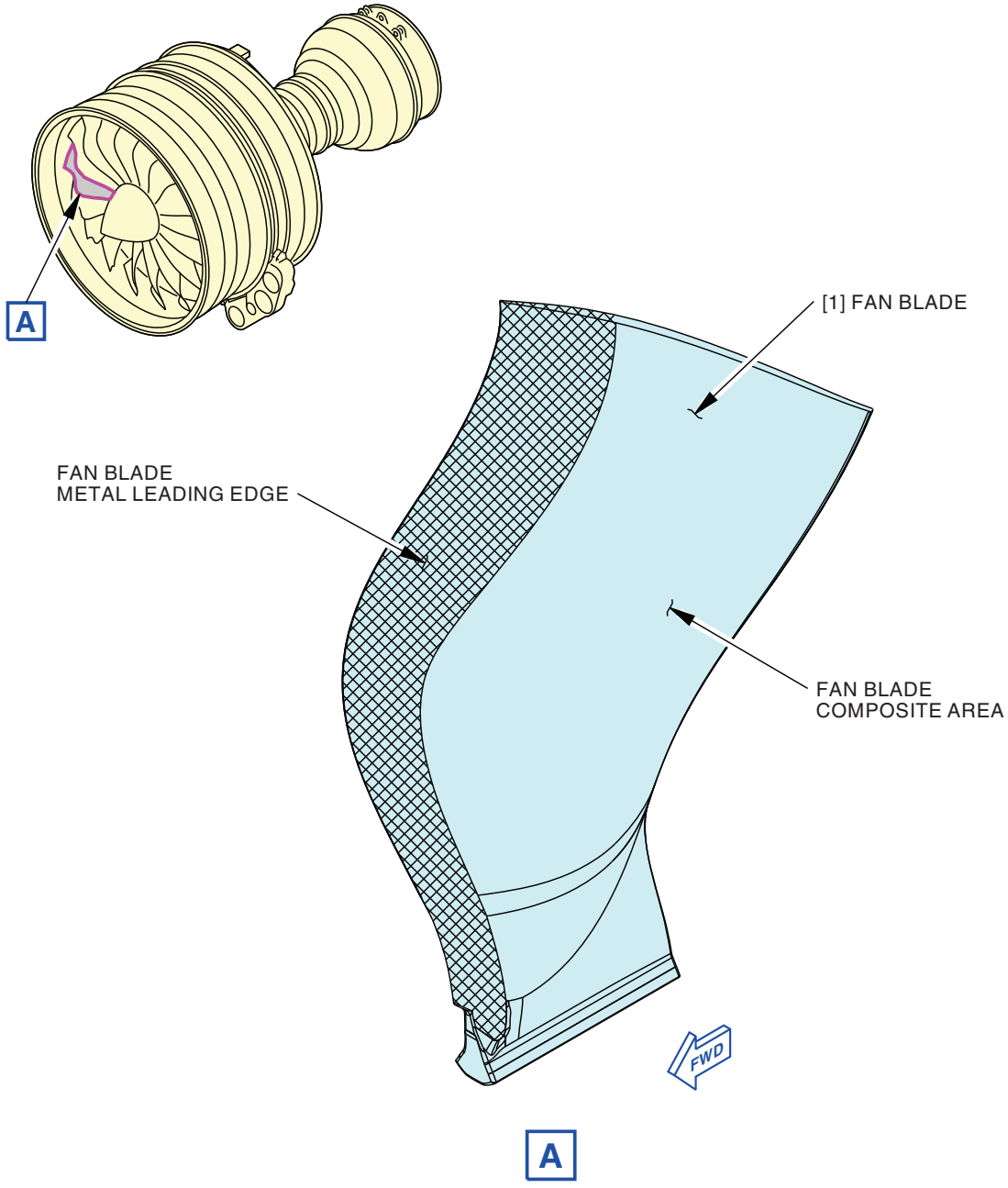
1) Put the applicable ENGINE START switch to the NORM (push in) position.

(b) On the P8 pilot's aisle control stand, remove the DO NOT OPERATE tag from the applicable FUEL CONTROL switch.

1) Put the applicable FUEL CONTROL switch to the CUTOFF position.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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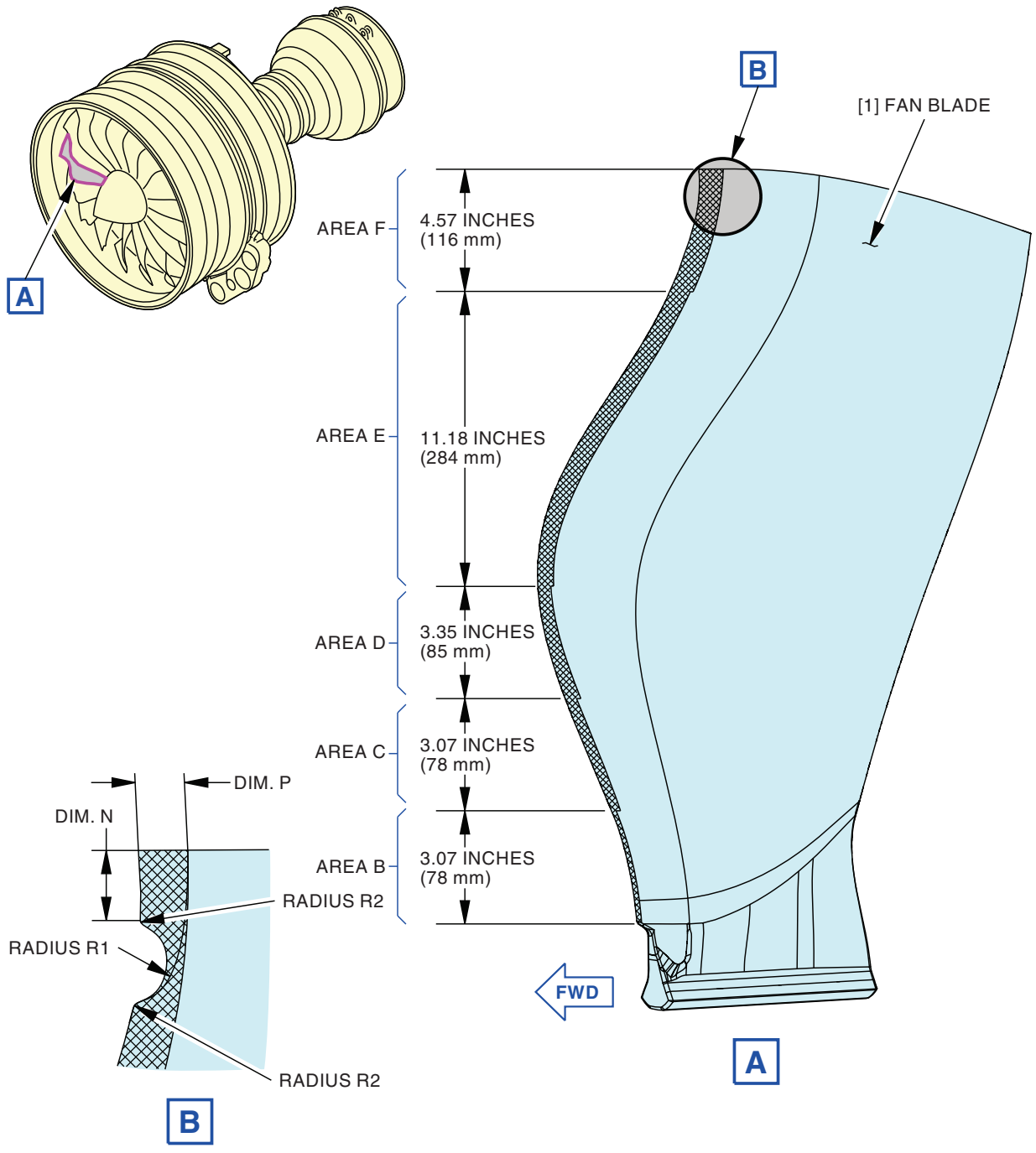
Fan Blade Metal Leading Edge Blending
Figure 807/72-22-02-990-838-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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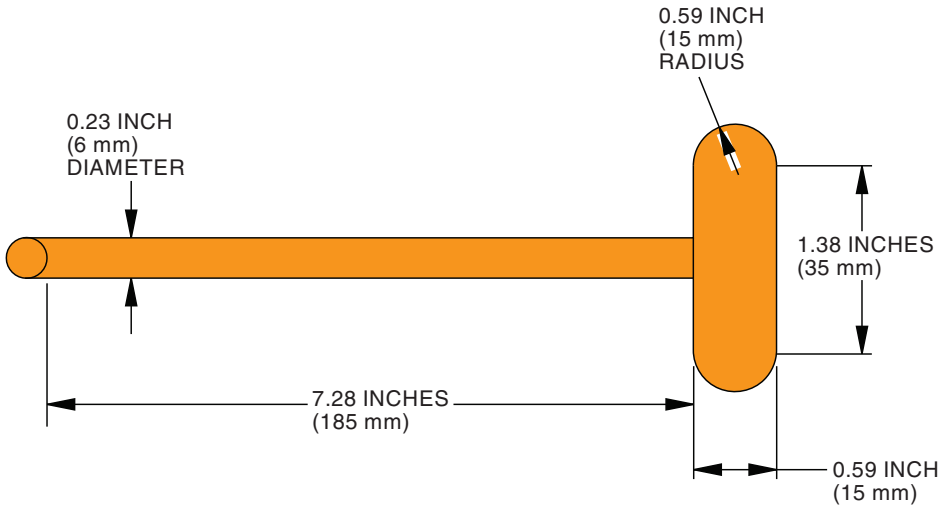
Fan Blade Blending Repair Limits
Figure 808/72-22-02-990-839-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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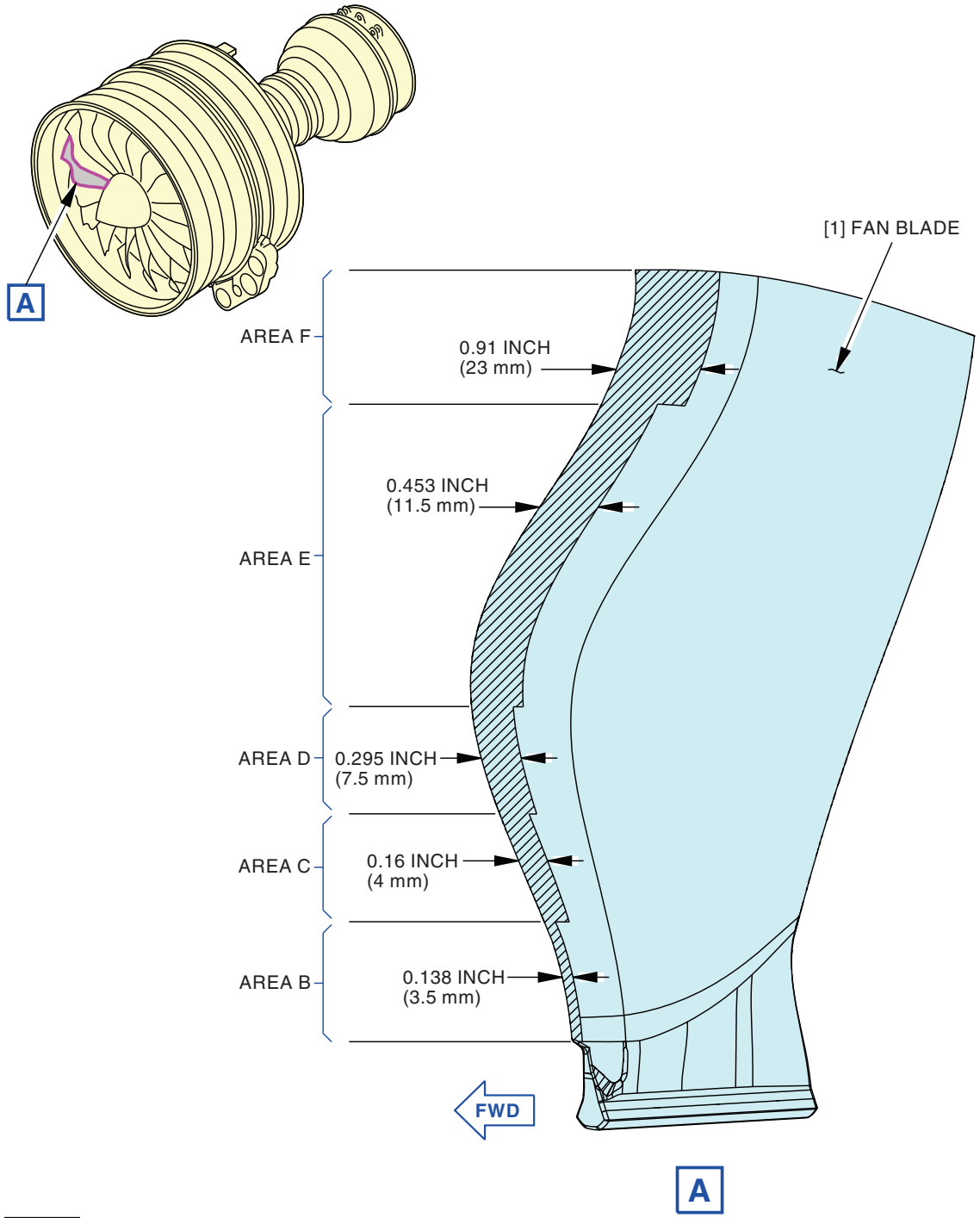
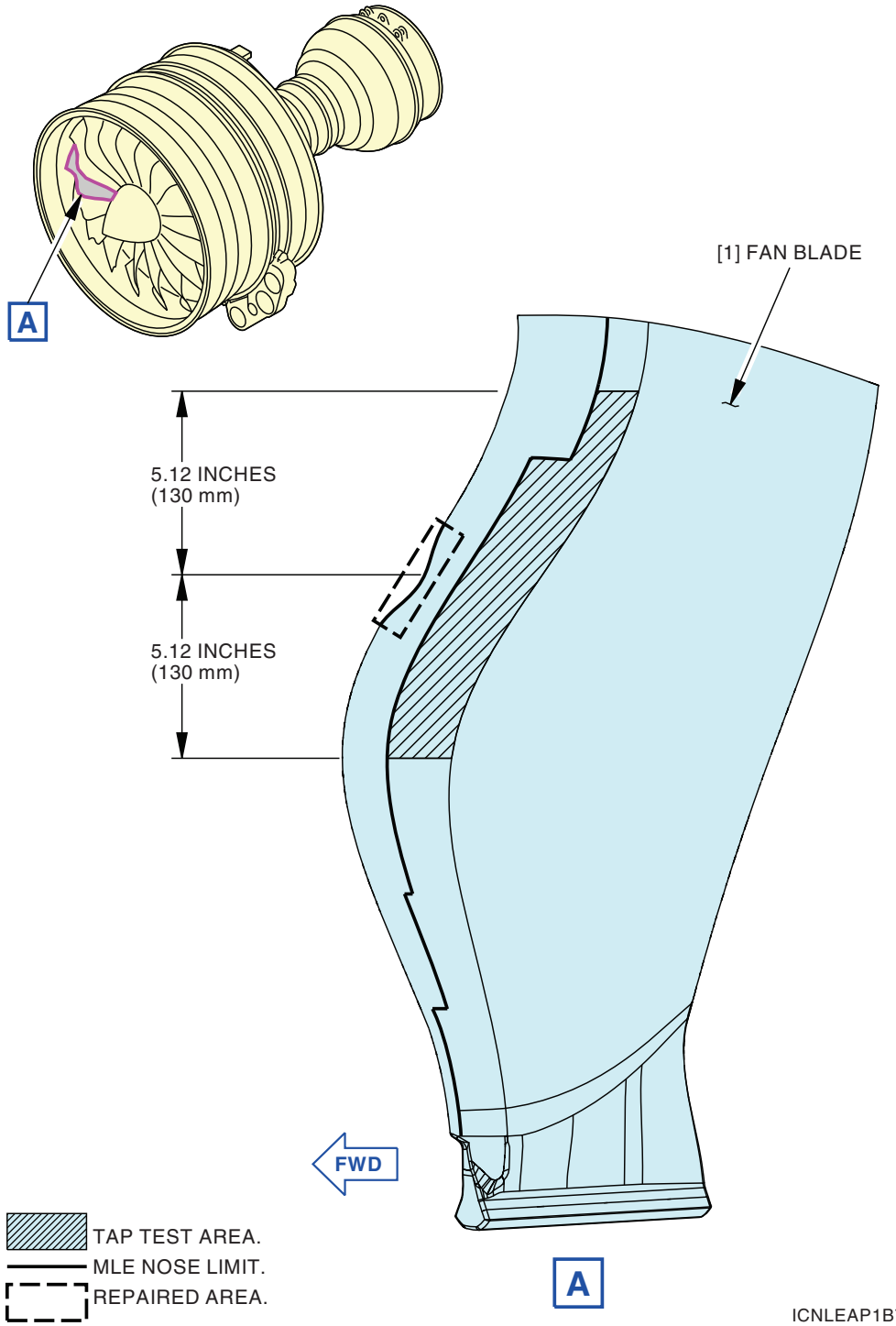
Aluminum Mallet
Figure 809/72-22-02-990-840-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



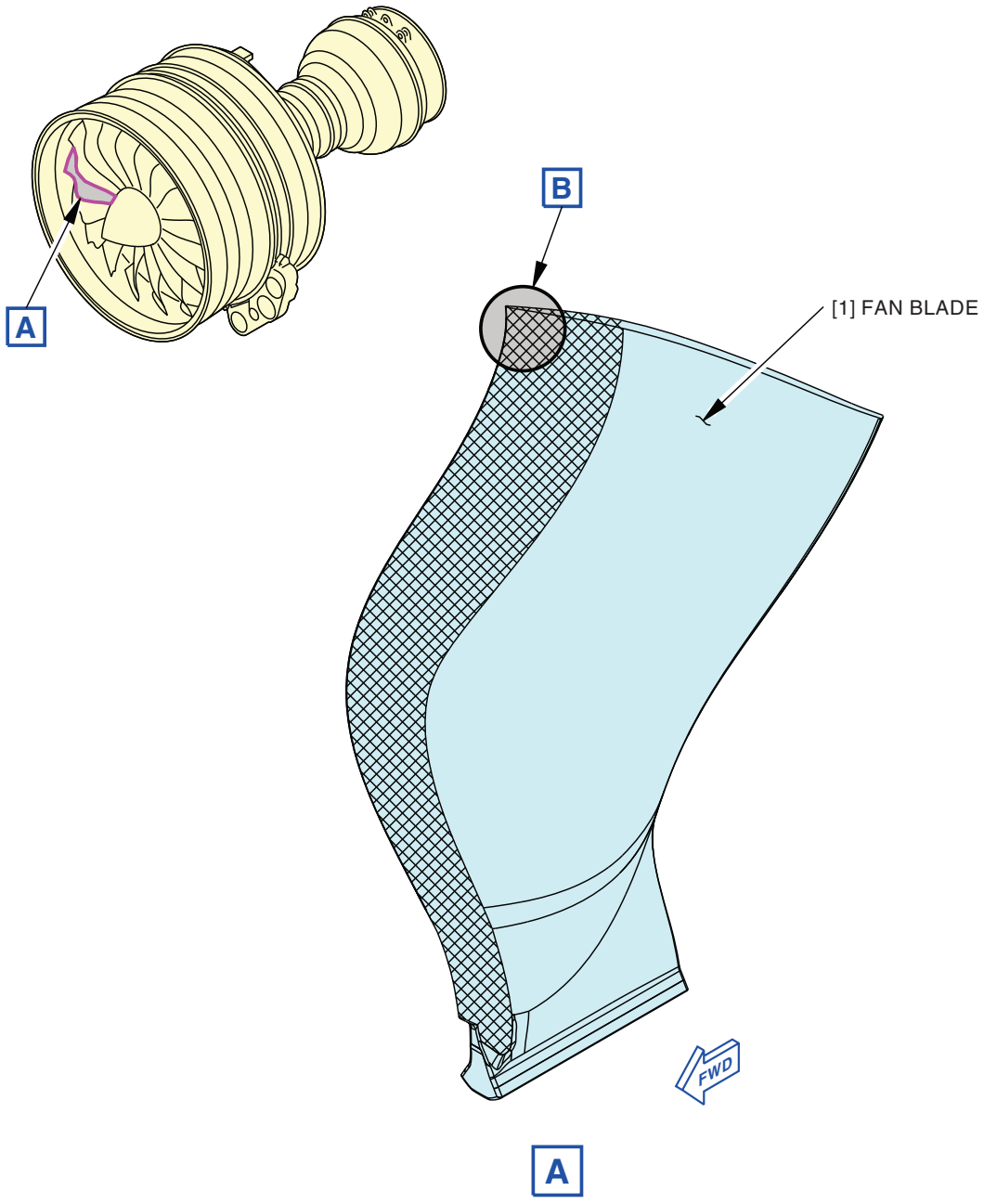
Metal Leading Edge Nose Limits
Figure 810/72-22-02-990-841-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Tap Test Area
Figure 811/72-22-02-990-842-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AREA TO BE REPAIRED.

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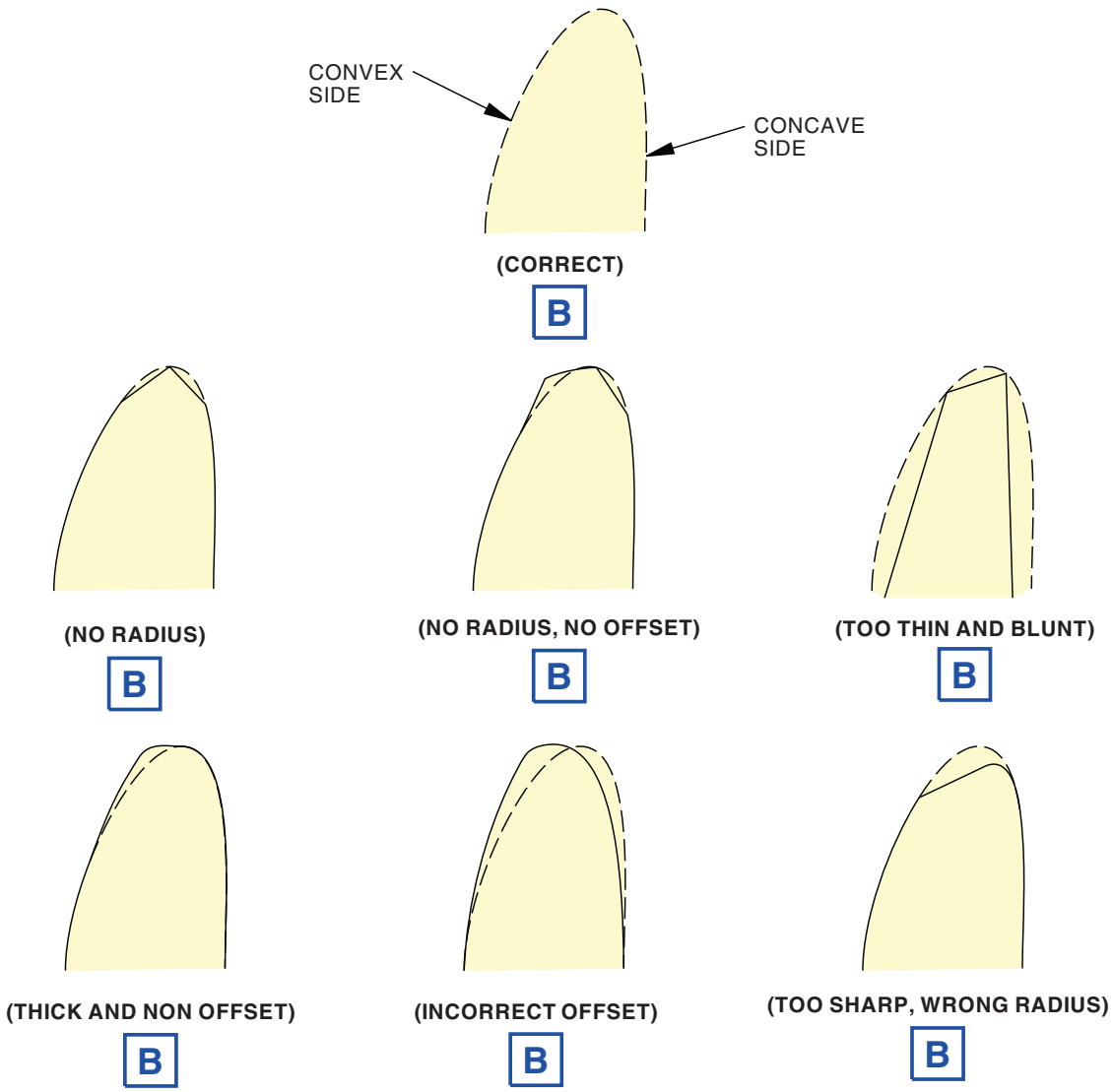
Typical Examples of the Incorrect Leading Edge Contours
Figure 812/72-22-02-990-843-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Typical Examples of the Incorrect Leading Edge Contours
Figure 812/72-22-02-990-843-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-22-02-300-801-G00

6. Fan Blade Repair of the Missing Polyurethane (PU) Film with PU Paste

(Figure 813, Figure 814, Figure 815, and Figure 816)

A. General

(1) This task gives instructions to repair the missing Polyurethane (PU) film and composite of the fan blade with PU paste.

B. Tools/Equipment

Reference	Description
STD-585	Mat - Protective, Closed Cell Neoprene, 30-50 duro, Weather and Oil Resistant, 3/8 Inch Min Thickness, Minimum 42x60 Inches with Warning Streamers
STD-858	Tag - DO NOT OPERATE
STD-14873	Spatula - Teflon

C. Consumable Materials

Reference	Description	Specification
A50562 [CP2946]	Paste - Repair, Scotch-Weld 2110	
B00676 [CP1041]	Alcohol - Isopropyl	
G51099 [CP2505]	Film - Airtech A4000 R Release Film	
G51693 [CP2967]	Tape - Teflease MG2	
G51694 [CP2348]	Film - Separator - VAC PACK 3760 P1	
G51824	Abrasive - Scotch-Brite Roloc Disc, Medium	
G51826 [CP2508]	Tape - Flashbreaker No. 5	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Fan blade	72-22-02-02-390	PTF ALL

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Prepare for the Repair

SUBTASK 72-22-02-840-012-G00

(1) Do these steps to make sure that the ENGINE START switch and FUEL CONTROL switch are not operated:

(a) On the P5 pilot's overhead panel, make sure that the applicable ENGINE START switch is in the NORM (push in) position.

1) Attach a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) On the P8 pilot's aisle control stand, make sure that the applicable FUEL CONTROL switch is in the CUTOFF position.

1) Attach a DO NOT OPERATE tag, STD-858, on the applicable FUEL CONTROL switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-22-02-840-013-G00

CAUTION PUT THE PROTECTIVE MAT IN THE INLET COWL. THIS WILL PREVENT DAMAGE TO THE INLET COWL OR ENGINE BY TOOLS, PARTS, OR UNWANTED MATERIALS THAT FALL ON THE INLET COWL SURFACE.

(2) Put the protective mat, STD-585, in the lower half of the inlet cowl to prevent damage to the inlet cowl and engine.

G. Repair of the Missing Polyurethane (PU) Film with PU Paste

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

SUBTASK 72-22-02-220-005-G00

(1) Do a visual and dimensional inspection of the indentation to be repaired (Figure 813).

(a) The maximum surface repairable with this task is 0.038 in^2 (25 mm^2) of each missing PU film.

(b) The maximum indentation depth repairable with this procedure is DIM. A.

Table 802/72-22-02-993-806-G00 Dimensional Information

Description	Minimum In-Process Dimension	Maximum In-Process Dimension	Minimum Finish Dimension	Maximum Finish Dimension	Reference Dimension
DIM. A	-	-	-	0.007 in. (0.2 mm)	-

NOTE: The dimensional data identified within this procedure was designed using millimeters as the base unit of measurement for dimensional values. The conversion to inches is rounded to two additional decimal digits than millimeters. For minimum dimensions, converted values are rounded up. For maximum dimensions, converted values are rounded down. To avoid conversion errors, do not re-convert from inches to millimeters.

NOTE: Reference dimensions do not require inspection.

(c) If the surface or depth to repair are over, the fan blade [1] is not repairable with this repair task.

SUBTASK 72-22-02-020-013-G00

CAUTION DURING THE REMOVAL OF THE EXCEEDING POLYURETHANE (PU) FILM FROM THE FAN BLADE, DO NOT REMOVE THE COMPOSITE MATERIAL. IF YOU DO NOT OBEY, DAMAGE TO THE FAN BLADE CAN OCCUR.

(2) Prepare the area to be repaired as follows (Figure 814):

(a) If it is necessary, remove the exceeding PU film with a sharp tool around the area to repair.

1) Keep the area to repair of missing PU 0.038 in^2 (25 mm^2).

(b) Clean the area to repair with a lint-free tissue moist with alcohol, B00676 [CP1041].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-22-02-300-004-G00

(3) Apply mask on the adjacent repaired areas (painted area, Metal Leading Edge (MLE) area, lower pressure-side platform wear-strip, MLE adhesive) with masking Teflease MG2 tape, G51693 [CP2967] (Figure 815).

(a) If there is no finishing paint around the area to repair, put the masking Teflease MG2 tape, G51693 [CP2967], on the primer.

(b) If there is no primer around the area to repair, put the masking Teflease MG2 tape, G51693 [CP2967], on the PU film.

SUBTASK 72-22-02-100-003-G00

CAUTION

DO NOT TOUCH THE CLEANED FAN BLADE SURFACES WITH YOUR BARE HANDS. IF YOU DO NOT OBEY, FINGER OILS CAN CAUSE INSUFFICIENT BONDING.

(4) Clean the area to be repaired with alcohol, B00676 [CP1041], as follows:

(a) Use a lint-free tissue moist and piped with IPA to be sure that you can clean the indentation.

(b) After cleaning, dry the area to be repaired with pressurized dry air.

(c) Let at least 10 minutes at ambient air.

(d) Do not touch the cleaned surfaces with your bare hands, if you do it clean again the area to repair.

SUBTASK 72-22-02-840-014-G00

CAUTION

YOU MUST LET THE ISOPROPYL ALCOHOL DRY FOR 10 MINUTES. IF YOU DO NOT OBEY, THE PU PASTE WILL NOT ATTACH CORRECTLY.

(5) Prepare the PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]), according to the manufacturer's recommendations.

SUBTASK 72-22-02-300-005-G00

CAUTION

DO NOT PUT PU PASTE ON THE METAL LEADING EDGE (MLE). IF YOU DO NOT OBEY, DAMAGE TO THE MLE CAN OCCUR.

(6) Apply the PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]), to the fan blade [1] area to repair as follows (Figure 815):

(a) Apply a small quantity of the PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]), to the fan blade [1] area to repair.

(b) Make the PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]), smooth to make it flush with the masking tape around with one of the following methods:

1) Use the teflon spatula, STD-14873, to make PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]), smooth.

a) Smooth the PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]), at once going on one side to the other.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Alternative procedure. Apply a piece of release film (VAC-PACK 3760 P1 separator film, G51694 [CP2348], or Airtech A4000 R release film, G51099 [CP2505]), on the top of the PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]).

a) Use a roller on the release film (VAC-PACK 3760 P1 separator film, G51694 [CP2348], or Airtech A4000 R release film, G51099 [CP2505]), to make the PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]), smooth.

b) Smooth the PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]), at once going on one side to the other.

(c) Do not put the PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]), on the MLE.

SUBTASK 72-22-02-020-014-G00

(7) When the PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]), is frozen, remove the masking tape Teflease MG2 tape, G51693 [CP2967].

SUBTASK 72-22-02-350-003-G00

WARNING

USE THERMAL GLOVES WHEN YOU MOVE OR TOUCH HOT OR COLD PARTS. HOT OR COLD PARTS CAN CAUSE INJURY.

(8) Cure the PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]), with one of the following methods:

NOTE: A relative humidity more than 40 percent can prevent full curing of the PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]). It is recommended to use a heating device if the relative humidity more than 40 percent.

NOTE: It is recommended to apply the release film (VAC-PACK 3760 P1 separator film, G51694 [CP2348], or Airtech A4000 R release film, G51099 [CP2505]), if the fan blade is not cured in a low moisture environment (relative humidity is less than 40 percent). This is to get a satisfactory cure.

(a) Method 1. Cure the PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]), at room temperature 64°F (18°C) - 84°F (29°C) for 24 hours.

(b) Method 2. Cure the PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]), at a temperature range 131°F (55°C) - 171°F (77°C) for 120 minutes to 150 minutes.

(c) Method 3. Cure the PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]), at a temperature range 171°F (77°C) - 190°F (88°C) for 60 minutes to 90 minutes.

SUBTASK 72-22-02-020-015-G00

(9) If it is necessary, remove the release film (VAC-PACK 3760 P1 separator film, G51694 [CP2348], or Airtech A4000 R release film, G51099 [CP2505]).

SUBTASK 72-22-02-020-016-G00

(10) If it is necessary, remove the unwanted PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]).

SUBTASK 72-22-02-211-016-G00

(11) Do a visual inspection of the repaired area as follows:

(a) Make sure that there is PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]), on all the repaired area.

1) If there is not PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]), on all the repaired area, do the repair again on the missing PU paste area.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-22-02-210-007-G00

(12) Do an inspection of the PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]), to make sure that it is cured as follows:

(a) Do an inspection of surface shape memory as follows:

1) Apply force with the teflon spatula, STD-14873, to the cured surface to cause a movement.

2) Remove the teflon spatula, STD-14873.

3) Make sure that the surface of PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]), keeps the initial shape.

(b) Make sure that the cured surface is dry and not wet.

1) If the PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]), is not cured, remove the non-cured PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]), from the repaired area and do the repair again.

SUBTASK 72-22-02-120-002-G00

WARNING DO NOT BREATHE THE PARTICLES WHEN SANDING. THE PARTICLES CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR.

(13) Sand the PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]), as follows (Figure 816):

(a) Mask the coatings around the area to be repaired with Flashbreaker 5 tape, G51826 [CP2508].

(b) Sand the PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]), to make it flush with the coatings with an abrasive paper 240 grit minimum, or Scotch-Brite disc, G51824, or fin.

1) If there is no finishing paint around the area to repair, make the PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]), flush with the primer.

2) If there is no primer around the area to repair, make the PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]), flush with the PU film.

(c) Remove Flashbreaker 5 tape, G51826 [CP2508].

(d) Clean the area to repair with a lint-free tissue moist with alcohol, B00676 [CP1041].

SUBTASK 72-22-02-210-008-G00

(14) Do a visual inspection of the repaired area as follows:

(a) Make sure that the PU paste (Scotch-Weld Repair Paste 2110, A50562 [CP2946]), is flush with the coating.

(b) The repaired area must be smooth and free of run, bump, pinhole, crater or blister, and without inclusion of foreign object.

(c) Bubble with dimension more than 0.0039 in. (0.10 mm) is not permitted.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-22-02-840-015-G00

CAUTION

MAKE SURE THERE ARE NO TOOLS, PARTS OR OTHER UNWANTED MATERIAL IN THE INLET AREA. DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION

MAKE SURE THERE ARE NO TOOLS, PARTS OR OTHER UNWANTED MATERIAL IN THE INLET COWL. IF YOU DO NOT OBEY THIS INSTRUCTION YOU CAN CAUSE DAMAGE TO THE ENGINE.

(1) Remove the protective mat, STD-585, from the lower half of the inlet cowl.

SUBTASK 72-22-02-840-016-G00

(2) Do these steps to remove the DO NOT OPERATE tags from the applicable ENGINE START switch and FUEL CONTROL switch:

(a) On the P5 pilot's overhead panel, remove the DO NOT OPERATE tag from the applicable ENGINE START switch.

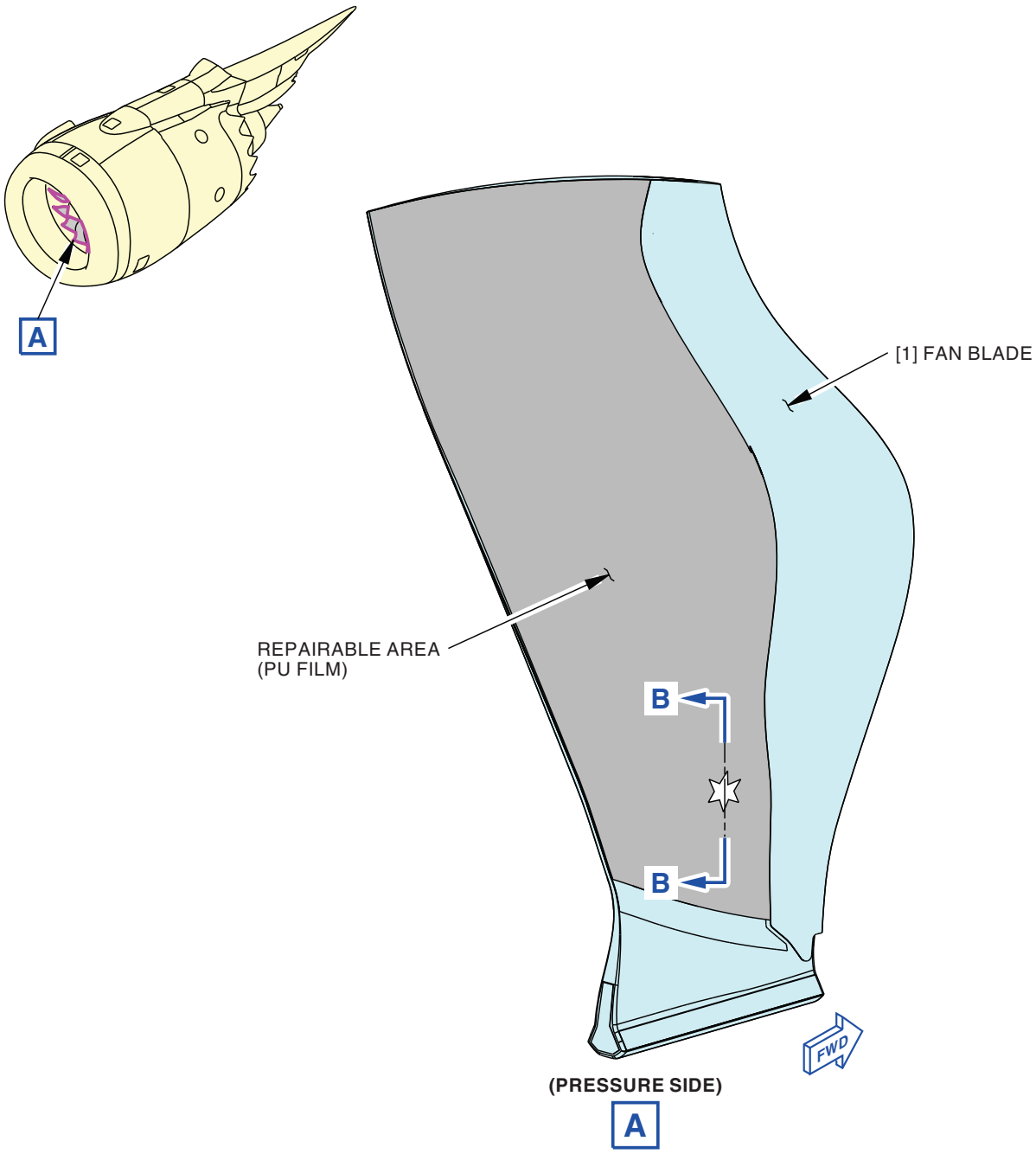
1) Put the applicable ENGINE START switch to the NORM (push in) position.

(b) On the P8 pilot's aisle control stand, remove the DO NOT OPERATE tag from the applicable FUEL CONTROL switch.

1) Put the applicable FUEL CONTROL switch to the CUTOFF position.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



MATERIAL: INTERLOCK WOVEN COMPOSITE INJECTED WITH EPOXY RESIN.
COATING: POLYURETHAN FILM PRIMER PCEH100 PAINT LBYH203.

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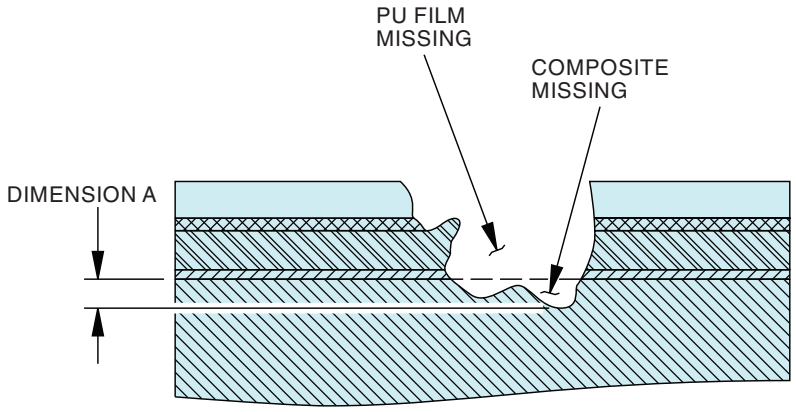
Fan Blade Pressure Side - Repairable Area
Figure 813/72-22-02-990-844-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(TYPICAL INDENTATION TO BE REPAIRED)

B-B

PAINT

PRIMER (SOME P/N DOES NOT HAVE PRIMER)

PU FILM

PU FILM ADHESIVE

COMPOSITE

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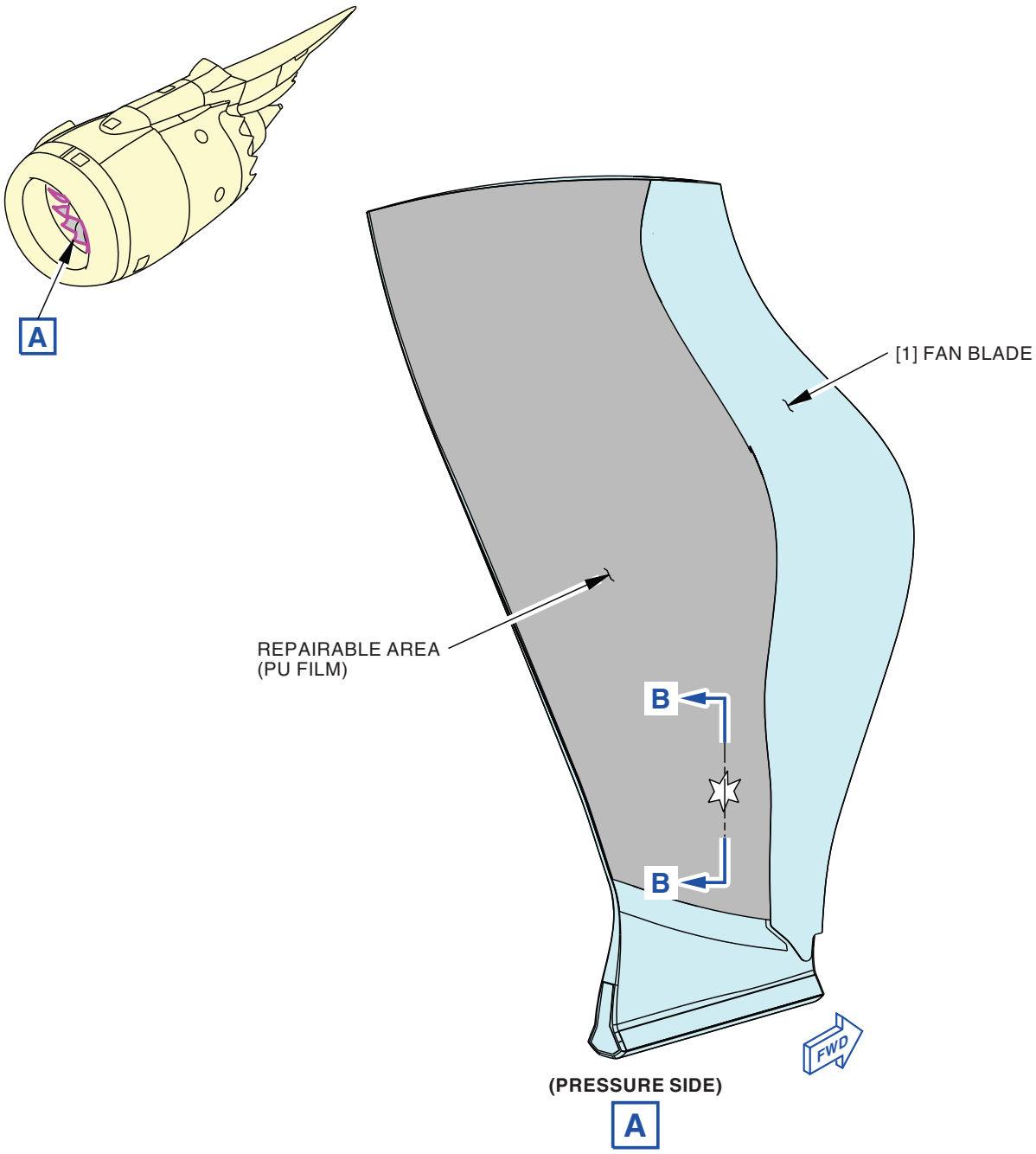
Fan Blade Pressure Side - Repairable Area
Figure 813/72-22-02-990-844-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



MATERIAL: INTERLOCK WOVEN COMPOSITE INJECTED WITH EPOXY RESIN.
COATING: POLYURETHAN FILM PRIMER PCEH100 PAINT LBYH203.

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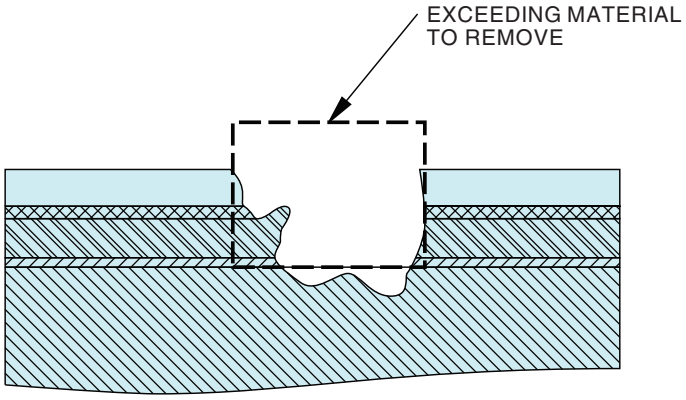
Indentation To Be Repaired - Exceeding Material to Remove
Figure 814/72-22-02-990-845-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**(TYPICAL INDENTATION TO BE REPAIRED.
EXCEEDING MATERIAL TO REMOVE)**

B-B

PAINT

PRIMER (SOME P/N DOES NOT HAVE PRIMER)

PU FILM

PU FILM ADHESIVE

COMPOSITE

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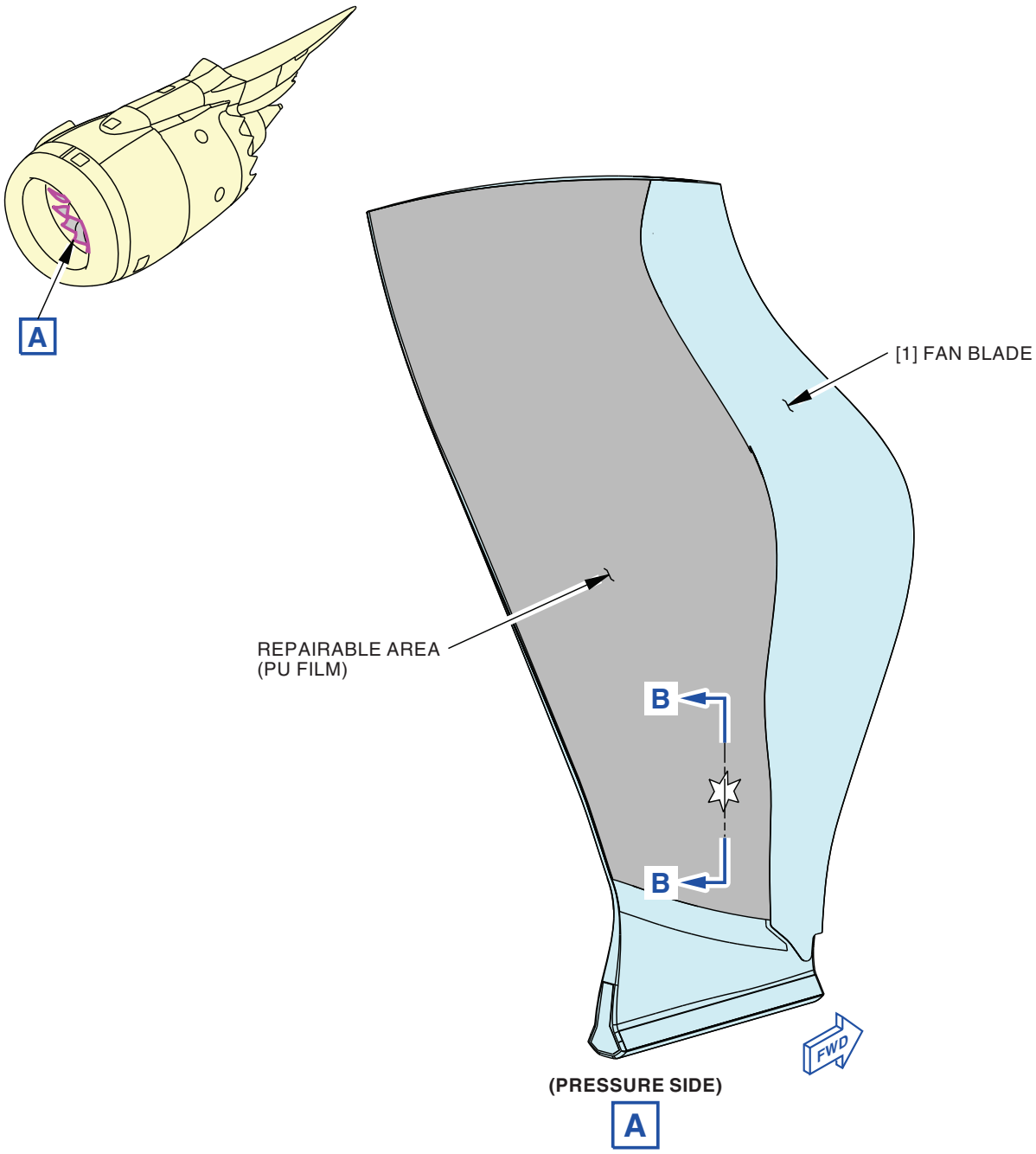
**Indentation To Be Repaired - Exceeding Material to Remove
Figure 814/72-22-02-990-845-G00 (Sheet 2 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



MATERIAL: INTERLOCK WOVEN COMPOSITE INJECTED WITH EPOXY RESIN.
COATING: POLYURETHAN FILM PRIMER PCEH100 PAINT LBYH203.

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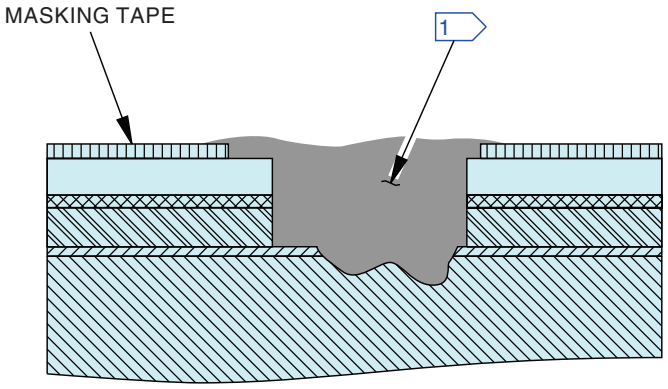
Indentation To Be Repaired - Mask and PU Paste to Apply
Figure 815/72-22-02-990-846-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**(TYPICAL INDENTATION TO BE REPAIRED.
MASK AND PU PASTE APPLICATION)**

B-B

MASKING TAPE (CP2967)

PAINT

PRIMER (SOME P/N DOES NOT HAVE PRIMER)

PU FILM

PU FILM ADHESIVE

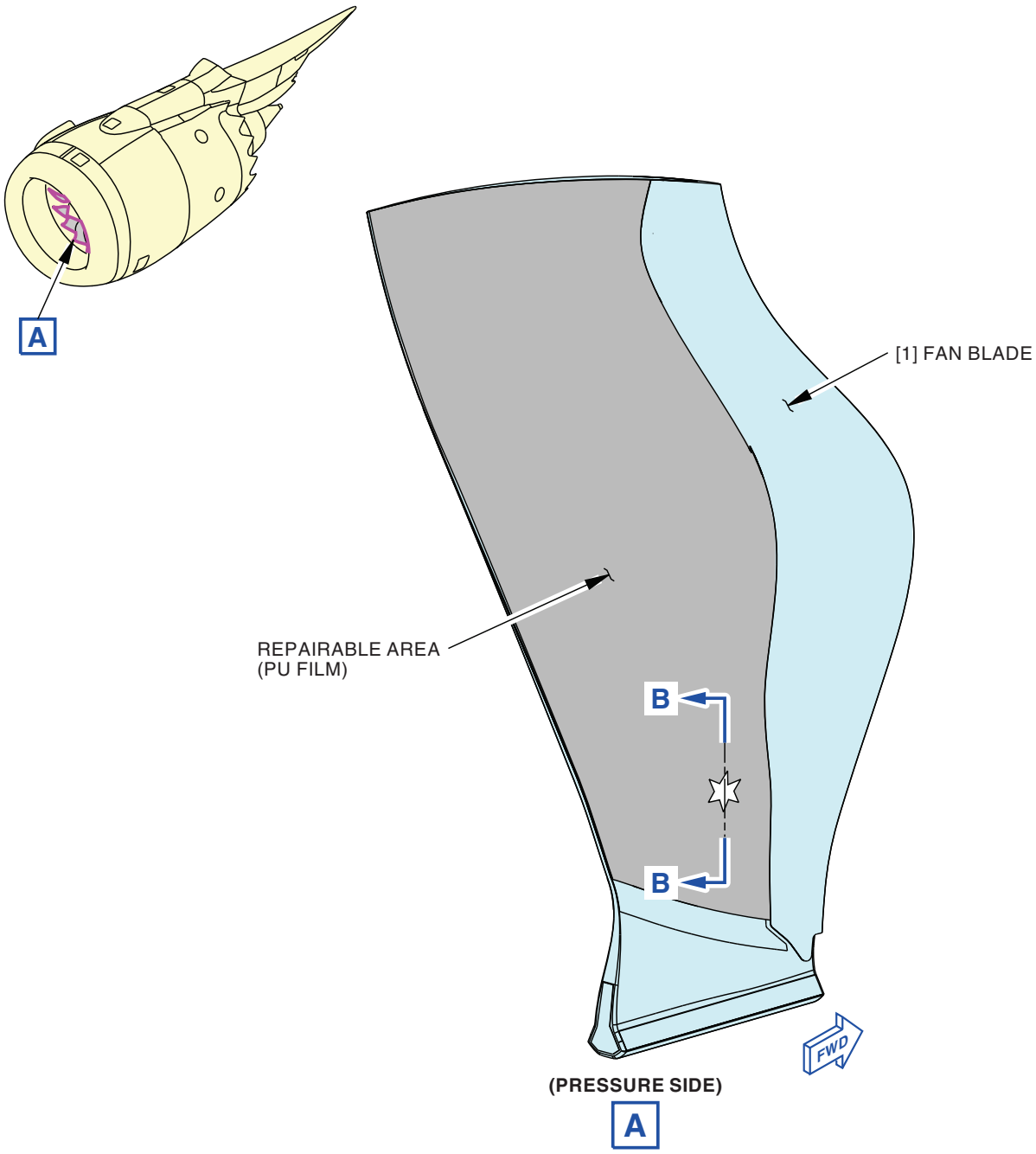
COMPOSITE

1 PU PASTE (CP2946) TO APPLY FLUSH WITH THE COATING

ICNLEAP1B722120CF030100001A00104
3012385 S0000785962_V1

**Indentation To Be Repaired - Mask and PU Paste to Apply
Figure 815/72-22-02-990-846-G00 (Sheet 2 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



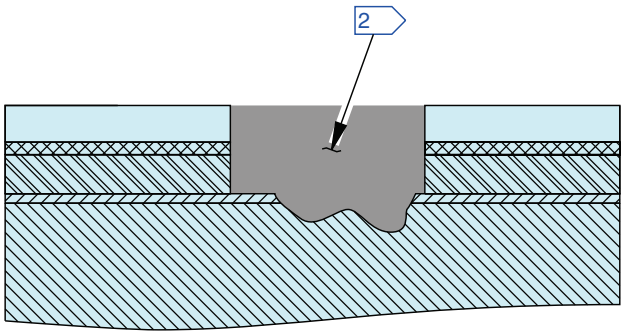
MATERIAL: INTERLOCK WOVEN COMPOSITE INJECTED WITH EPOXY RESIN.
COATING: POLYURETHAN FILM PRIMER PCEH100 PAINT LBYH203.

ICNLEAP1B722120CF030100001A00101

3012376 S0000785959_V1

Indentation To Be Repaired - PU Paste After Sanding
Figure 816/72-22-02-990-847-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**(TYPICAL INDENTATION TO BE REPAIRED.
PU PASTE AFTER SANDING)**

B-B

PAINT

PRIMER (SOME P/N DOES NOT HAVE PRIMER)

PU FILM

PU FILM ADHESIVE

COMPOSITE

2 PU PASTE FLUSH WITH THE COATING

ICNLEAP1B722120CF030100001A00105
3012388 S0000785963_V1

**Indentation To Be Repaired - PU Paste After Sanding
Figure 816/72-22-02-990-847-G00 (Sheet 2 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN BLADE PLATFORMS - REMOVAL/INSTALLATION

1. General

A. This procedure contains the following tasks:

(1) A removal Fan Blade Platforms

(2) An installation Fan Blade Platforms.

TASK 72-22-03-000-801-G00

2. Fan Blade Platforms - Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the fan blade platform.

B. References

Reference	Title
72-22-01-000-801-G00	Front Spinner Cone Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14664	Tool - Platform Front Shroud Removal Part #: 956A8062G02 Supplier: 58828

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Removal

SUBTASK 72-22-03-010-001-G00

(1) Remove the front spinner cone [2] (TASK 72-22-01-000-801-G00).

F. Fan Blade Platform Removal

SUBTASK 72-22-03-930-001-G00

(1) Find the (No. 1) fan blade.

(a) Rotate the fan blade rotor and look for the number “1” on the forward face of the platform front shroud [3].

(b) Put the (No. 1) fan blade at the 12:00 o'clock position.

SUBTASK 72-22-03-020-001-G00

(2) Remove the fan blade platform [1].

(a) Remove the platform front shroud [3].

1) Loosen the 18 bolts [4] from the fan disk [6].

2) Remove 16 bolts [4] that attach the platform front shroud [3] to the fan disk [6] except the two bolts [4] at the 3 o'clock to the 9 o'clock position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION USE THE REMOVAL TOOL TO REMOVE THE PLATFORM FRONT SHROUD. IF YOU DO NOT USE THE TOOL, THE PLATFORM FRONT SHROUD CAN FALL AND CAUSE DAMAGE TO THE ENGINE.

3) Install the tool, SPL-14664, on the platform front shroud [3].

4) Remove the remaining two bolts [4] that attach the platform front shroud [3] to the fan disk [6].

5) Remove the platform front shroud [3] from the fan disk [6].

6) Remove the tool, SPL-14664, from the platform front shroud [3].

(b) Use the temporary marking to make a mark of the position on each of the 18 fan blade platforms [1].

NOTE: Numbered location of the fan blade platforms is to keep engine balanced after installation.

(c) Remove the fan blade platform [1] from the fan disk [6]

SUBTASK 72-22-03-020-002-G00

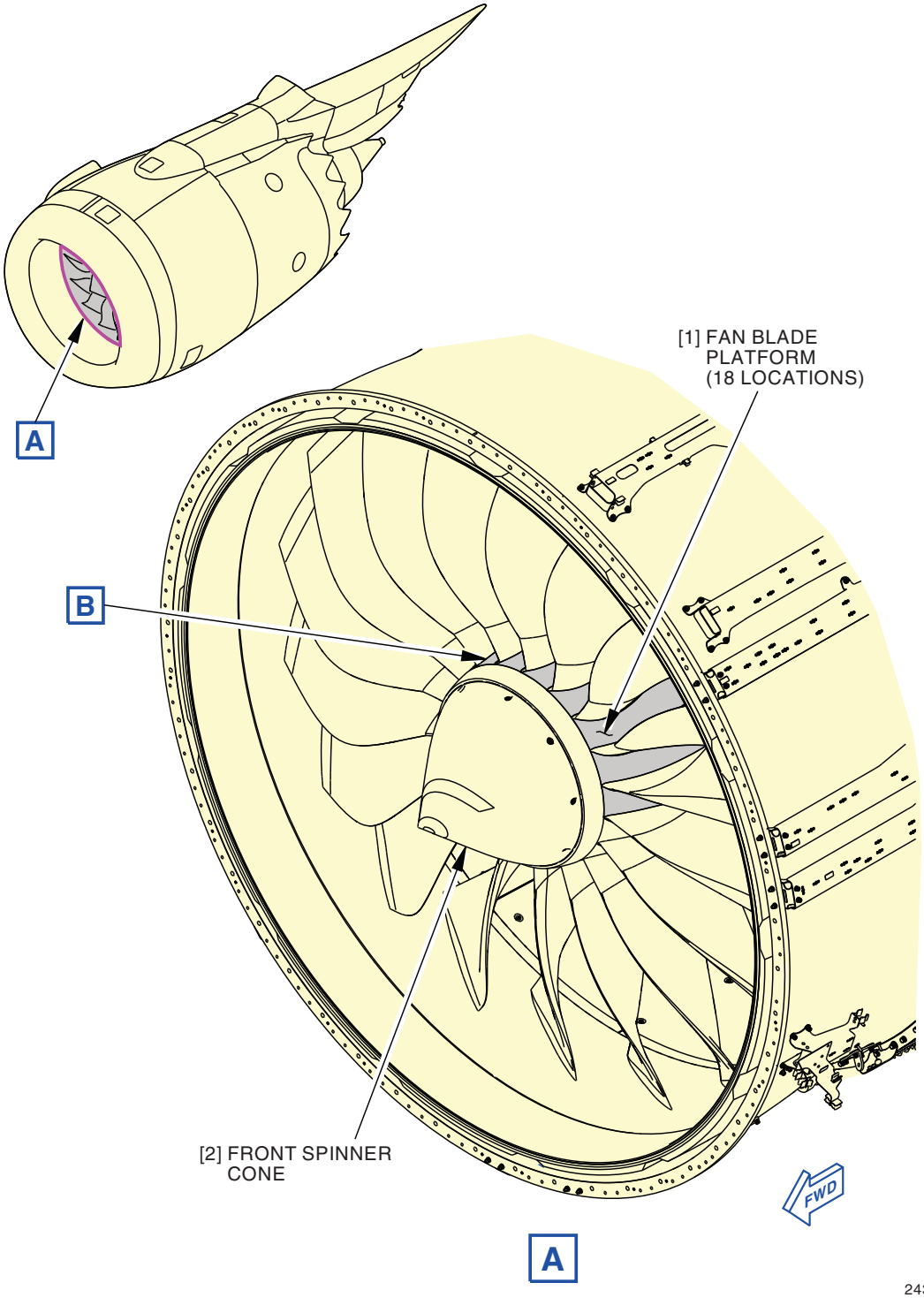
(3) If it is necessary, remove the remaining fan blade platforms [1] from the fan disk [6].

SUBTASK 72-22-03-480-001-G00

(4) Install the protective covers on the openings.

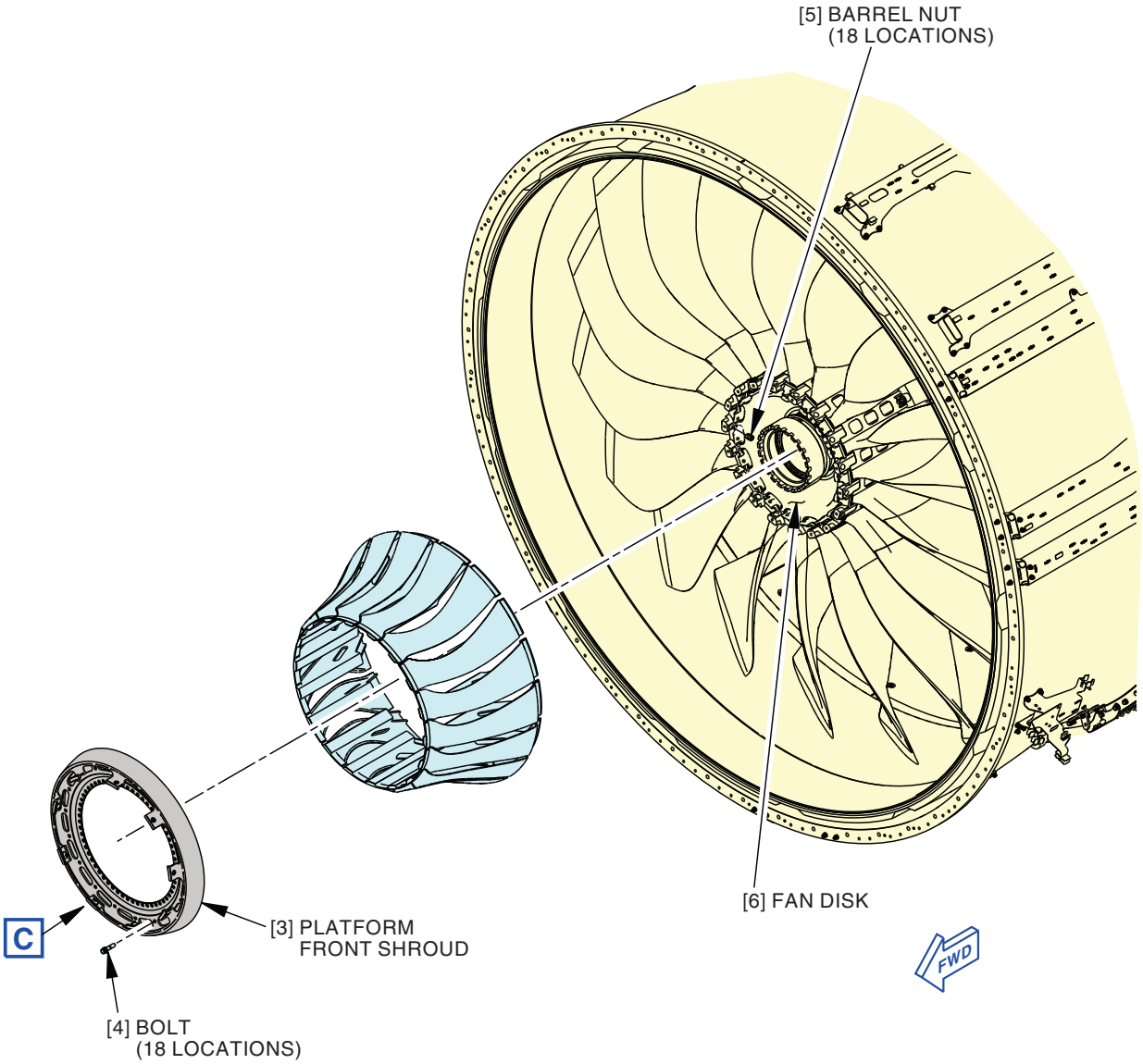
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Blade Platforms Installation
Figure 401/72-22-03-990-809-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



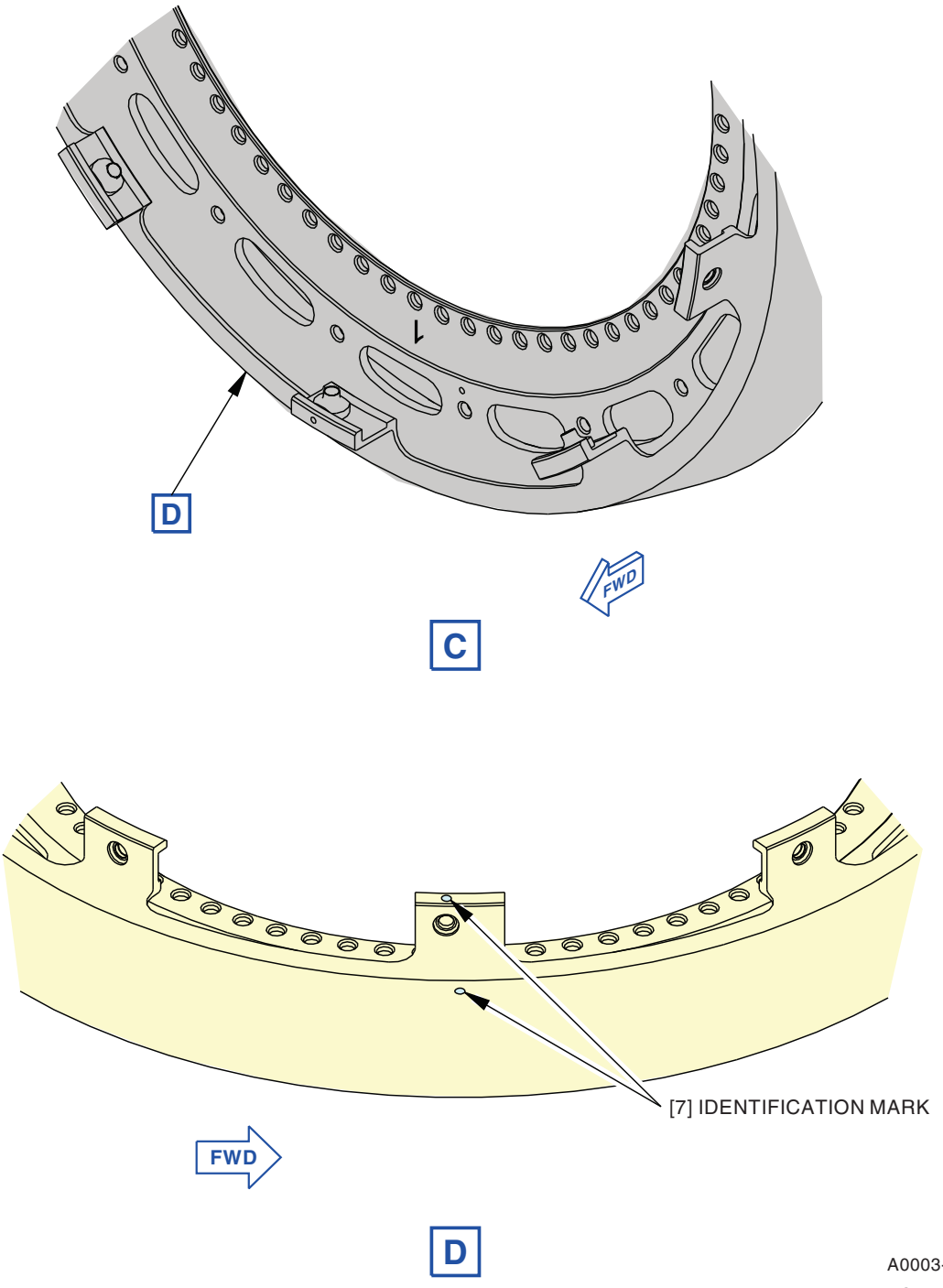
B

ICN-LEAP-1B-722203-B-F0301-A0002-A-001-01

2434325 S0000564159_V3

Fan Blade Platforms Installation
Figure 401/72-22-03-990-809-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Blade Platforms Installation
Figure 401/72-22-03-990-809-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-22-03-400-801-G00

3. Fan Blade Platforms - Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the fan blade platform.

(2) You must do the tests that are listed in the power plant test reference table after you install the fan blade platform(s).

B. References

Reference	Title
72-22-01-400-801-G00	Front Spinner Cone Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14660	Positioner - Fan Disk Barrel Nut Part #: 956A6071G01 Supplier: 58828
SPL-14661	Guide Pins - Platform Front Shroud Part #: 956A6067G01 Supplier: 58828
SPL-14664	Tool - Platform Front Shroud Removal Part #: 956A8062G02 Supplier: 58828

D. Consumable Materials

Reference	Description	Specification
D00601 [CP2101]	Vaseline - Graphite Mineral	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Fan Blade Platform Installation

SUBTASK 72-22-03-080-001-G00

(1) Remove the protective covers from the openings.

SUBTASK 72-22-03-420-001-G00

(2) Install the fan blade platforms [1].

(a) Put one fan blade platform [1] between fan blades in the correct recorded position on the fan disk [6].

(b) If you will install all fan blade platforms, do this step:

1) Put the remaining 17 fan blade platforms [1] in their recorded positions on the fan disk [6].

(c) Lubricate the threads of the 18 bolts [4] with graphite mineral vaseline, D00601 [CP2101].

(d) Make sure that the 18 barrel nuts [5] are in the correct position and alignment on the fan disk [6].

1) If it is necessary, use the positioner, SPL-14660.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Install the platform front shroud guide pins, SPL-14661, to align the platform front shroud [3] with the fan disk [6].

(f) Install the platform front shroud [3] in the correct position on the fan disk [6], using the identification mark [7] to locate the 12 o'clock position, through the platform front shroud guide pins, SPL-14661.

1) Install the tool, SPL-14664, on the platform front shroud [3].

2) Push the platform front shroud [3] onto the fan disk [6].

3) Make sure that the platform front shroud [3] is in contact with the fan disk [6].

(g) Install two bolts [4] at the 3 o'clock and 9 o'clock positions in the platform front shroud [3].

(h) Remove the platform front shroud guide pins, SPL-14661, from the platform front shroud [3].

(i) Remove the tool, SPL-14664, from the platform front shroud [3].

(j) Install the remaining 16 bolts [4] that attach the platform front shroud [3] to the fan disk [6].

(k) Tighten the bolts [4] to 133 in-lb (15 N·m) - 186 in-lb (21 N·m).

SUBTASK 72-22-03-160-001-G00

(3) Clean the area and remove the tools from where you did the work.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-22-03-420-002-G00

(1) Install the front spinner cone [2] (TASK 72-22-01-400-801-G00).

H. Fan Blade Platform Installation Test

SUBTASK 72-22-03-840-001-G00

(1) No Test Required

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN BLADE PLATFORMS - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Fan blade platforms inspection.

TASK 72-22-03-200-801-G00

2. Fan Blade Platforms Inspection

(Figure 601)

A. General

(1) This task gives the instructions on how to examine the installed fan blade platforms.

(2) This inspection is performed with the fan blades installed on the engine.

B. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

C. Fan Blade Platforms Inspection

NOTE: There are three layers of the fan blade platforms:

A black polyurethane topskin film

A translucent gray glass fibre layer

A black dark composite layer.

SUBTASK 72-22-03-212-001-G00

(1) Examine the exposed fan blade platform flowpath surfaces for these conditions:

(a) Scratches, tears, nicks, chips, or erosion in the black polyurethane topskin film

1) Permitted if you cannot see the fiberglass ply layer below the polyurethane.

2) If the fiberglass ply layer is visible but not damaged and black dark composite layer fibres are not visible, the Continue-In-Service (C-I-S) limit is 50 cycles.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

NOTE: The polyurethane film top coat can extend past the edges of the platform's top surface and partially cover the platform seals. Detachment of the film from the seals is permitted.

(b) Cracks, tears, and broken fibers of the gray/translucent fiberglass ply layer

1) Not permitted.

2) The Continue-In-Service (C-I-S) limit for cracks, tears and broken fibers is 25 cycles, with these conditions:

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

a) All loose materials are removed.

b) Black dark composite layer is not visible.

c) The cracks, tears, and broken fibers have a maximum length of 1.0 in. (25 mm).

(c) Erosion in the gray/translucent fiberglass ply layer

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Not permitted.

2) The Continue-In-Service (C-I-S) limit for erosion is 25 cycles, if the black dark composite layer is not visible.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

(d) Delamination of the black dark composite layer

1) Not permitted.

(e) Damaged edges (crushed fiber or laminate) along the perimeter of the platform

1) Permitted a maximum depth of 0.02 in. (0.5 mm), with these conditions:

a) All loose materials are removed

b) No delamination is observed

c) Adjacent fan blades are examined for damage.

(f) External surface resin crazing at the leading and trailing edge radii:

NOTE: The resin at the leading and trailing edge radii will only be visible if the polyurethane surface coating is missing in these areas.

1) External surface resin crazing at the leading and trailing edge radii are permitted if there are no black dark composite layer fibres seen below the resin.

NOTE: Black dark composite layer fibers that can be seen are not sealed by the resin.

2) The Continue-In-Service (C-I-S) limit for surface resin crazing at the leading and trailing edge radii where you can see the black dark composite layer fiber plies below the resin is 50 cycles.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

3) The Continue-In-Service (C-I-S) limit for surface resin crazing at the leading and trailing edge radii where you can see the black dark composite layer fiber plies below the resin is 50 cycles.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

(g) Chipped external surface resin at the leading and trailing edge radii:

1) Permitted if there is no carbon fiber plies seen below the resin.

2) The Continue-In-Service (C-I-S) limit for chipped external surface resin at the leading and trailing edge radii where you can see the black dark composite layer fiber plies below the resin is 10 cycles.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

SUBTASK 72-22-03-212-002-G00

(2) Examine the seals of the fan blade platforms for these conditions:

(a) Seal erosion

NOTE: The top of the gray seal lip must be visible between the fan platform edge and the fan blade wear strip interface along the two sides of the platform.

1) Erosion of seal lip is permitted to a maximum of 2.0 in. (51 mm) along each side.

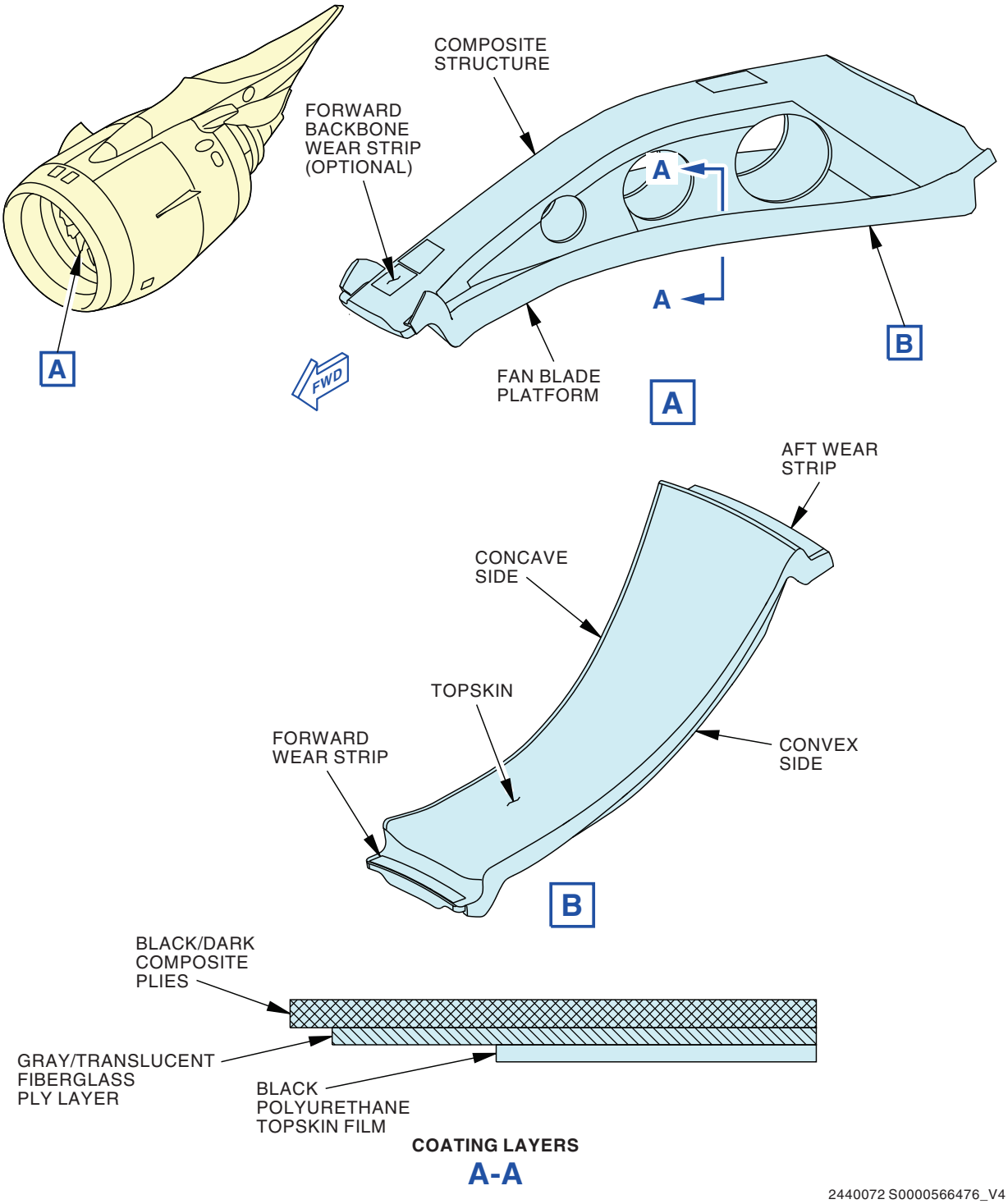
ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) The Continue-In-Service (C-I-S) limit for the missing seal lip is 50 cycles.

NOTE: This over serviceable limit extension is for a one-time continue-in-service application.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Blade Platform Inspection
Figure 601/72-22-03-990-810-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLOWPATH FAIRING - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Flowpath Fairing Removal

(2) Flowpath Fairing Installation.

TASK 72-22-05-000-801-G00

2. Flowpath Fairing Removal

(Figure 401, Figure 402, and Figure 403)

A. General

(1) This task gives the instructions to remove the flowpath fairing.

B. References

Reference	Title
72-22-02-000-803-G00	Fan Blades (Set) Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-15093	Fixture, Remove/Install - Panel Flow Path Part #: 956A8074G01 Supplier: 58828
SPL-25854	Fixture - Heater, Fairing Flow Path Part #: 956A6035G01 Supplier: 58828
SPL-25855	Protector - Fan Disk Part #: 956A8067P01 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Removal

SUBTASK 72-22-05-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

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SUBTASK 72-22-05-860-006-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

F. Flowpath Fairing Removal

SUBTASK 72-22-05-020-003-G00

(1) Do this task: Fan Blades (Set) Removal, TASK 72-22-02-000-803-G00.

SUBTASK 72-22-05-480-003-G00

(2) Make sure that the fan disk protector, SPL-25855, is installed (Figure 401).

SUBTASK 72-22-05-020-004-G00

(3) Remove the five screws [4] that attach the flowpath (ring half) fairing [2] (top or bottom) to the splitter fairing [3] (Figure 401).

SUBTASK 72-22-05-010-001-G00

(4) Remove the flowpath (ring half) fairing [2] from the engine.

SUBTASK 72-22-05-480-001-G00

(5) Install the panel flow path remove/install fixture, SPL-15093, as follows (Figure 402):

(a) Loosen the indexing plunger [9].

(b) Put the sliding axis [10] in the most outward position.

(c) Hand tighten the indexing plunger [9] to maintain the sliding axis [10] in position.

(d) Install the tool body [8] on the flowpath fairing [1] with TOP mark aligned at the 12 o'clock location.

(e) Loosen the indexing plunger [9].

(f) Put the sliding axis [10] in the most inward position into the slots of the flowpath fairing [1].

(g) Hand tighten the indexing plunger [9] to maintain the sliding axis [10] in position.

SUBTASK 72-22-05-020-005-G00

(6) Remove the eight screws [5] that attach the flowpath fairing [1] to the bracket [6] (Figure 401).

SUBTASK 72-22-05-480-004-G00

WARNING

USE THERMAL GLOVES WHEN YOU MOVE OR TOUCH HOT OR COLD PARTS. HOT OR COLD PARTS CAN CAUSE INJURY.

(7) If it is necessary, install the heater fixture, SPL-25854, on the flowpath fairing [1] (Figure 402).

(a) Use the velcros hooks to attach the heating blanket [12], heating blanket [14], and heating blanket [13] around the flowpath fairing [1] (Figure 403).

(b) Connect the heating blanket [12], heating blanket [14], and heating blanket [13] electrically to each other once they are all secured together.

NOTE: The connection of the sensor is not necessary.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Use the power extension [15] and platinum probe extension [16] to plug the main heating blanket [12] into the heater control panel [17] inside of the control box container [18].

(d) Turn on the heater control panel [17].

(e) Continue to heat up.

SUBTASK 72-22-05-020-006-G00

(8) Remove the flowpath fairing [1] from the engine (Figure 401).

NOTE: If it is necessary, use three slide hammers [11] on the panel flow path remove/install fixture, SPL-15093, to remove the flowpath fairing [1] (Figure 402).

SUBTASK 72-22-05-020-007-G00

(9) Remove and discard the seal [7] from the groove on the flowpath fairing [1] (Figure 401).

SUBTASK 72-22-05-080-005-G00

WARNING

USE THERMAL GLOVES WHEN YOU MOVE OR TOUCH HOT OR COLD PARTS. HOT OR COLD PARTS CAN CAUSE INJURY.

(10) If it is necessary, remove the heater fixture, SPL-25854, from the flowpath fairing [1] as follows (Figure 402):

(a) Turn off the heater control panel [17] (Figure 403).

(b) Unplug the heater control panel [17] from the power outlet.

(c) Disconnect the heating blanket [12], heating blanket [14], and heating blanket [13].

(d) Remove the heating blanket [12], heating blanket [14], and heating blanket [13] from the flowpath fairing [1].

SUBTASK 72-22-05-080-001-G00

(11) Remove the panel flow path remove/install fixture, SPL-15093, from the flowpath fairing [1] as follows (Figure 402):

(a) Loosen the indexing plunger [9].

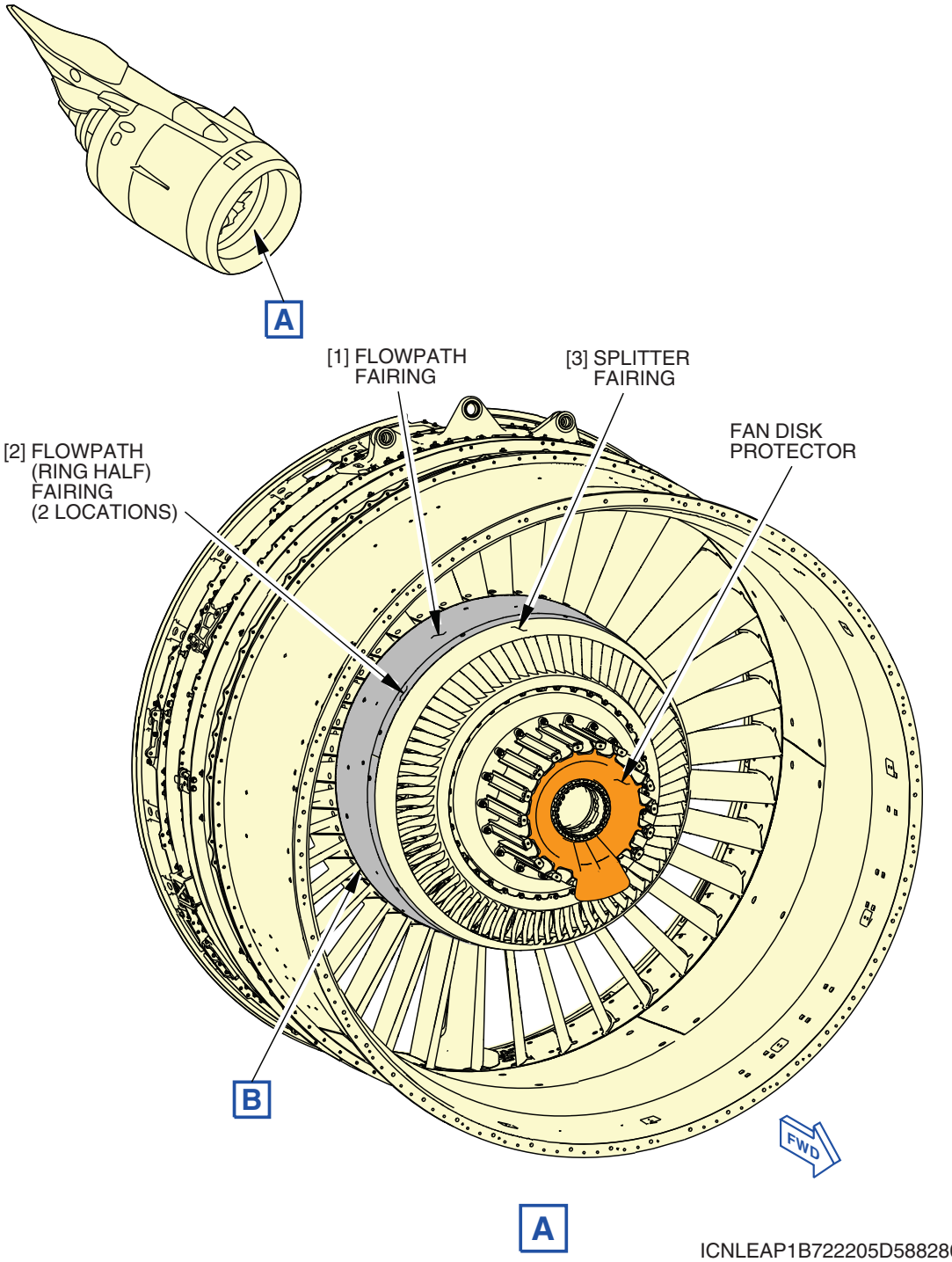
(b) Put the sliding axis [10] in the outward position to disengage from the slots of the flowpath fairing [1].

(c) Hand tighten the indexing plunger [9] to maintain the sliding axis [10] in position.

(d) Remove the tool body [8] from the flowpath fairing [1].

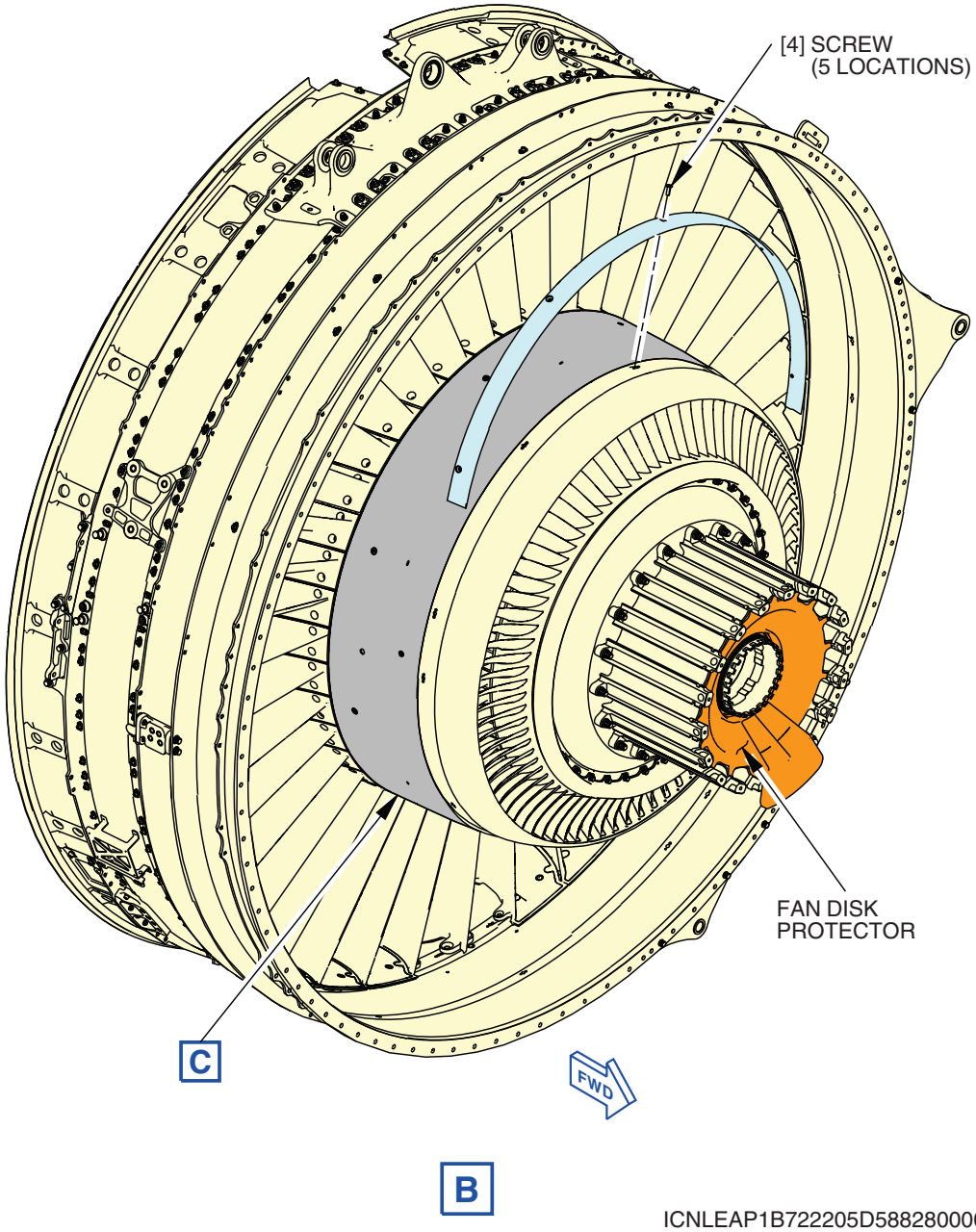
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



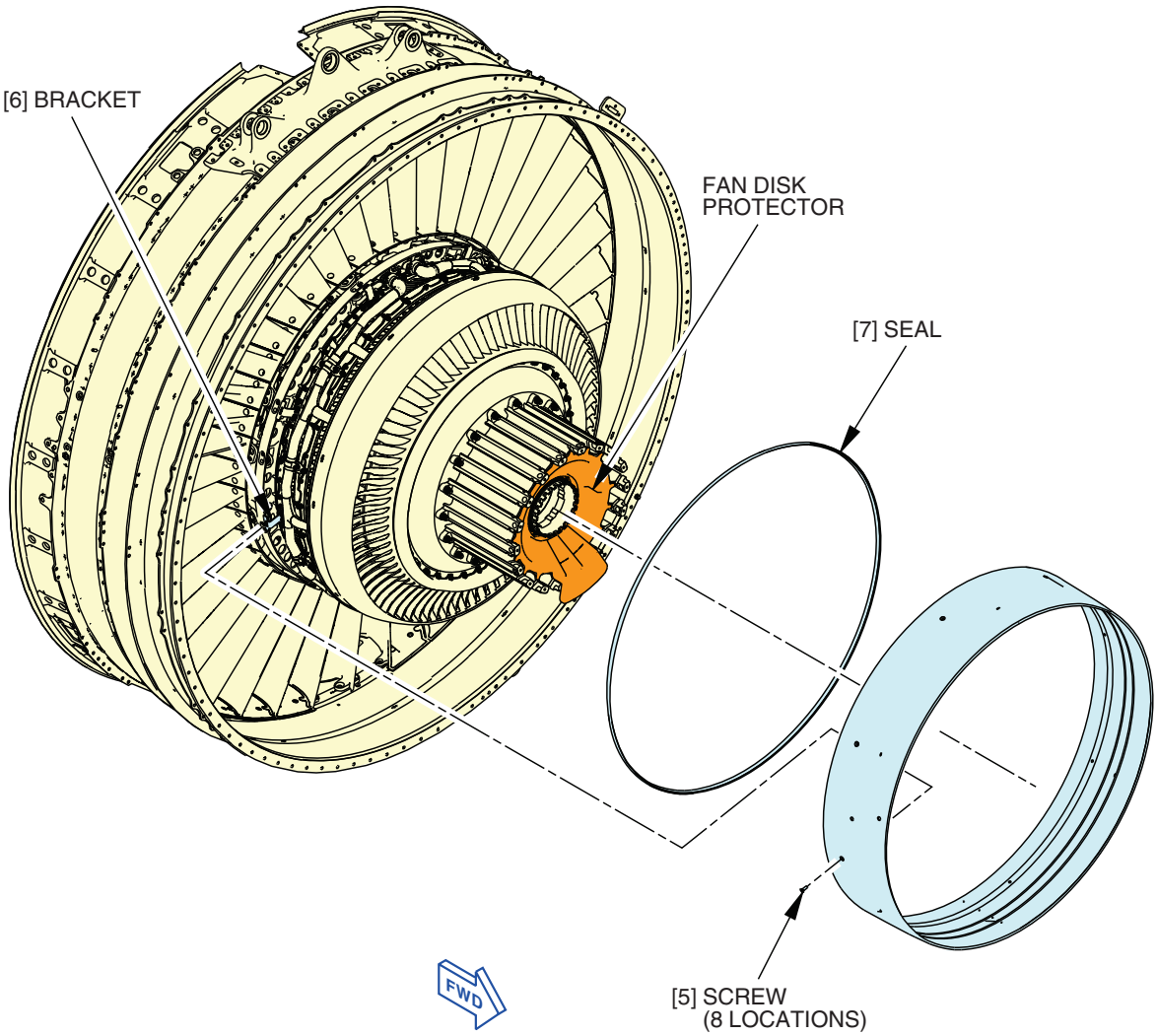
Flowpath Fairing Installation
Figure 401/72-22-05-990-802-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flowpath Fairing Installation
Figure 401/72-22-05-990-802-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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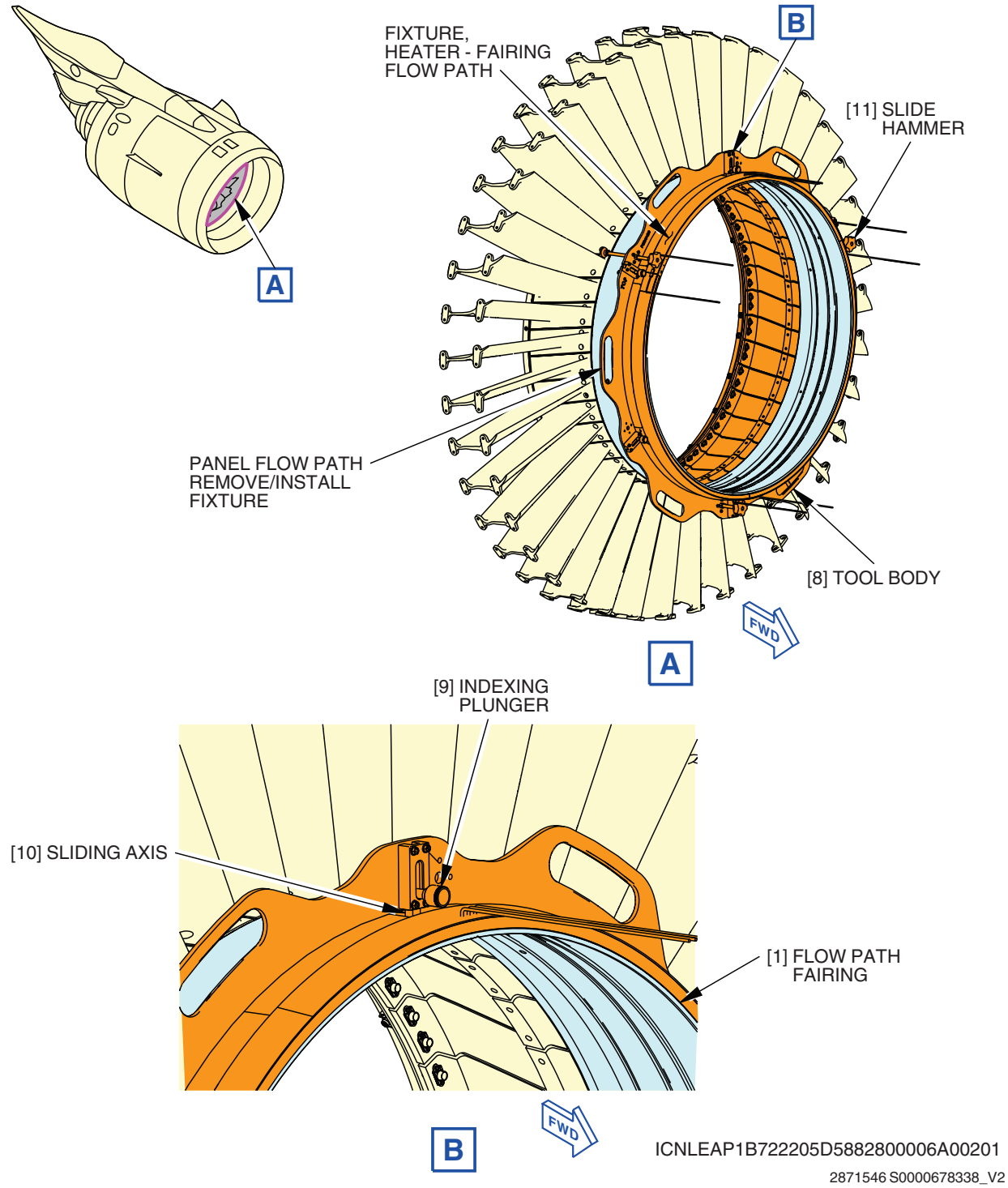
Flowpath Fairing Installation
Figure 401/72-22-05-990-802-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flowpath Fairing Removal/Installation Fixture
Figure 402/72-22-05-990-805-G00

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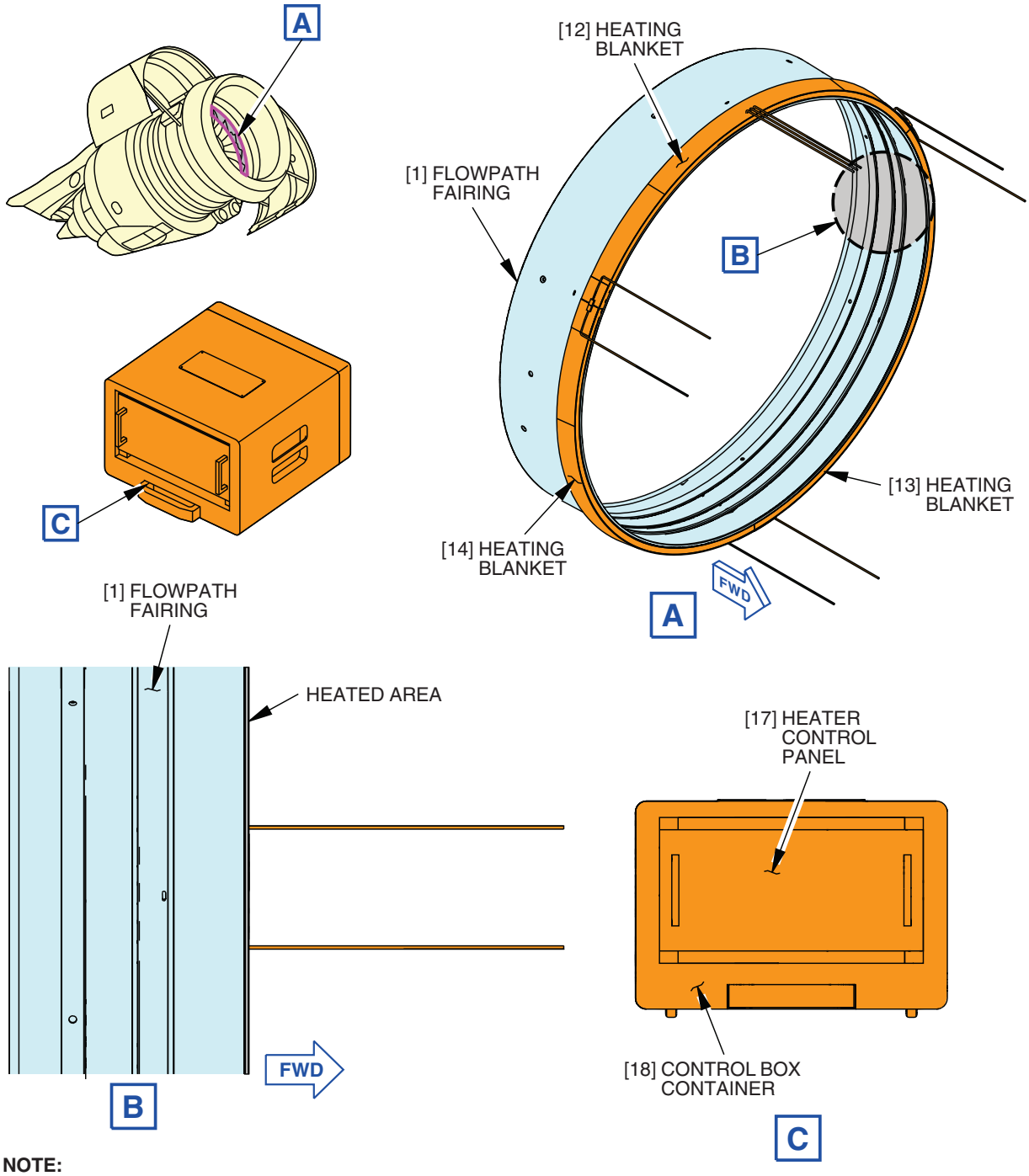

NOTE:

[15] POWER EXTENSION 4M : ITEM NOT REPRESENTED
[16] PLATINIUM PROBE EXTENSION 4M : ITEM NOT REPRESENTED

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Flowpath Fairing Heating
Figure 403/72-22-05-990-807-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-22-05-400-801-G00

3. Flowpath Fairing Installation

(Figure 401, Figure 402, and Figure 403)

A. General

(1) This task gives the instructions to install the flowpath fairing.

(2) You must do the tests that are listed in the power plant test reference table after you install the flowpath fairing.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
72-22-02-400-803-G00	Fan Blades (Set) Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-15093	Fixture, Remove/Install - Panel Flow Path Part #: 956A8074G01 Supplier: 58828
SPL-25854	Fixture - Heater, Fairing Flow Path Part #: 956A6035G01 Supplier: 58828
SPL-25855	Protector - Fan Disk Part #: 956A8067P01 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A50447	Compound - Jointing [CP2773], Corrosion Inhibiting, Chromate Free, Mastinox CA1010	
G00834	Cloth - Lint-free Cotton	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
7	Seal	72-21-00-02-060	PTF ALL
		72-21-00-02A-060	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Flowpath Fairing Installation

SUBTASK 72-22-05-100-001-G00

(1) Make sure that the mating faces are clean and clear of unwanted materials.

SUBTASK 72-22-05-480-006-G00

(2) Make sure that the fan disk protector, SPL-25855, is installed (Figure 401).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-22-05-080-002-G00

(3) Install the panel flow path remove/install fixture, SPL-15093, on the flowpath fairing [1] as follows (Figure 402):

(a) Loosen the indexing plunger [9].

(b) Put the sliding axis [10] in the most outward position to disengage from the slots of the flowpath fairing [1].

(c) Hand tighten the indexing plunger [9] to maintain the sliding axis [10] in position.

(d) Install the tool body [8] on the flowpath fairing [1] with TOP mark aligned at the 12 o'clock position.

SUBTASK 72-22-05-640-001-G00

(4) Apply mastinox chromatefree jointing compound, A50447, on the pressure face and the threads of the eight screws [5] (Figure 401).

SUBTASK 72-22-05-420-004-G00

(5) Install the new seal [7] on the groove of the flowpath fairing [1].

SUBTASK 72-22-05-480-005-G00

WARNING USE THERMAL GLOVES WHEN YOU MOVE OR TOUCH HOT OR COLD PARTS. HOT OR COLD PARTS CAN CAUSE INJURY.

(6) Do these steps to increase the temperature of the forward flange of the flowpath fairing [1] (preferred method):

(a) Install the heater fixture, SPL-25854, on the flowpath fairing [1] as follows (Figure 402):

1) Use the velcros hooks to attach the heating blanket [12], heating blanket [14], and heating blanket [13] around the flowpath fairing [1] (Figure 403).

2) Connect the heating blanket [12], heating blanket [14], and heating blanket [13] electrically to each other once they are all secured together.

NOTE: The connection of the sensor is not necessary.

3) Use the power extension [15] and platinum probe extension [16] to plug the heating blanket [12] into the heater control panel [17] inside of the control box container [18].

4) Turn on the heater control panel [17].

5) Continue to heat up.

SUBTASK 72-22-05-880-001-G00

CAUTION OBEY THE INSTRUCTIONS IN THE PROCEDURE TO INCREASE THE TEMPERATURE OF THE FLOWPATH FAIRING. IF YOU DO NOT OBEY THE INSTRUCTIONS, DAMAGE TO THE FLOWPATH FAIRING CAN OCCUR.

CAUTION DO NOT USE A BLOWTORCH TO HEAT THE FLOWPATH FAIRING. IF YOU DO NOT OBEY, DAMAGE TO THE FLOWPATH FAIRING CAN OCCUR.

(7) Do this step to increase the temperature of the forward flange of the flowpath fairing [1] (alternative method):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Increase the temperature of the forward flange of the flowpath fairing [1] gradually and equally to 176°F (80°C) maximum with a heat gun.

SUBTASK 72-22-05-420-005-G00

(8) Install the flowpath fairing [1] in a correct position on the engine (Figure 401).

SUBTASK 72-22-05-420-006-G00

(9) Install the eight screws [5] that attach the flowpath fairing [1] to the bracket [6].

SUBTASK 72-22-05-420-007-G00

(10) Tighten in a criss-cross pattern the eight screws [5] to 128 in-lb (14.5 N·m) - 155 in-lb (17.5 N·m).

SUBTASK 72-22-05-020-009-G00

(11) Remove the unwanted mastinox chromatefree jointing compound, A50447, with lint-free cloth, G00834.

SUBTASK 72-22-05-080-006-G00

WARNING

USE THERMAL GLOVES WHEN YOU MOVE OR TOUCH HOT OR COLD PARTS. HOT OR COLD PARTS CAN CAUSE INJURY.

(12) If it is installed, remove the heater fixture, SPL-25854, from the flowpath fairing [1] as follows (Figure 402):

(a) Turn off the heater control panel [17] (Figure 403).

(b) Unplug the heater control panel [17] from the power outlet.

(c) Disconnect the heating blanket [12], heating blanket [14], and heating blanket [13].

(d) Remove the heating blanket [12], heating blanket [14], and heating blanket [13] from the flowpath fairing [1].

SUBTASK 72-22-05-080-004-G00

(13) Remove the panel flow path remove/install fixture, SPL-15093, from the flowpath fairing [1] as follows (Figure 402):

(a) Loosen the indexing plunger [9].

(b) Put the sliding axis [10] in the outward position to disengage from the slots of the flowpath fairing [1].

(c) Hand tighten the indexing plunger [9] to maintain the sliding axis [10] in position.

(d) Remove the tool body [8] from the flowpath fairing [1].

SUBTASK 72-22-05-410-001-G00

(14) Install the two flowpath (ring half) fairings [2] as follows (Figure 401):

(a) Install the flowpath (ring half) fairing [2] (top or bottom) in the correct position on the splitter fairing [3].

(b) Apply mastinox chromatefree jointing compound, A50447, on the pressure face and the threads of the five screws [4].

(c) Install the five screws [4] that attach the flowpath (ring half) fairing [2] to the splitter fairing [3].

(d) Tighten in a criss-cross pattern the five screws [4] to 128 in-lb (14.5 N·m) - 155 in-lb (17.5 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Remove the unwanted mastinox chromatefree jointing compound, A50447, with lint-free cloth, G00834.

SUBTASK 72-22-05-410-003-G00

(15) Do this task: Fan Blades (Set) Installation, TASK 72-22-02-400-803-G00.

SUBTASK 72-22-05-100-002-G00

(16) Clean the work area and remove the tools from where you did the work.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-22-05-860-007-G00

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-22-05-860-004-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

I. Flowpath Fairing Installation Test

SUBTASK 72-22-05-710-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLOWPATH FAIRING - INSPECTION/CHECK

1. General

A. This procedure has these task:

(1) Flowpath Fairing Inspection

(2) Shroud Segment Inspection.

TASK 72-22-05-200-801-G00

2. Flowpath Fairing Inspection

A. General

(1) This task gives the instructions to examine the flowpath fairing.

B. References

Reference	Title
72-22-05-000-801-G00	Flowpath Fairing Removal (P/B 401)
72-22-05-300-801-G00	Flowpath Fairing Spot Touch-Up of Anodization (P/B 801)

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Prepare for the Inspection

SUBTASK 72-22-05-010-003-G00

(1) Do this task: Flowpath Fairing Removal, TASK 72-22-05-000-801-G00.

E. Flowpath Fairing Inspection

SUBTASK 72-22-05-220-001-G00

(1) Examine the forward centering surface A, the surfaces B and C, and rear surface E for these conditions:

(a) Cracks and tears

1) Not permitted.

(b) Nicks and dents

1) Permitted up to 0.011 in. (0.3 mm) in depth with 0.500 in. (12.7 mm) minimum separation between two nicks or dents.

(c) Scratches

1) Permitted up to 0.011 in. (0.3 mm) in depth.

(d) Contact marks

1) Permitted up to 0.0007 in. (0.02 mm) in depth.

(e) Missing of surface coating (sulfuric anodizing)

1) Permitted.

(f) Pitting and corrosion

1) Not permitted.

(g) Wear, fretting on surface C

1) Permitted up to 0.0059 in. (0.15 mm) in depth.

(h) Wear, fretting on forward centering surface A

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Permitted up to 0.003 in. (0.1 mm) deep or if DIA A is between 38.150 in. (969.0 mm) and 38.157 in. (969.2 mm).

(i) Wear, fretting on surfaces B

1) Permitted up to 0.003 in. (0.1 mm) in depth.

SUBTASK 72-22-05-220-002-G00

(2) Examine the extracting holes, the borescope pass through holes and drain holes for these conditions:

(a) Cracks and tears

1) Not permitted.

(b) Nicks and dents

1) Permitted up to 0.011 in. (0.3 mm) in depth with 0.500 in. (12.7 mm) minimum separation between two nicks or dents.

(c) Scratches

1) Permitted up to 0.0051 in. (0.13 mm) in depth.

(d) Contact marks

1) Permitted up to 0.0007 in. (0.02 mm) in depth.

(e) Missing of surface coating (sulfuric anodizing)

1) Permitted.

(f) Pitting and corrosion

1) Not permitted.

(g) Thread marks

1) Permitted up to 0.003 in. (0.1 mm) in depth.

(h) Erosion of parent metal

1) Not permitted.

SUBTASK 72-22-05-220-003-G00

(3) Examine the attachment areas for these conditions:

(a) Cracks and tears

1) Not permitted.

(b) Nicks and dents

1) Permitted up to 0.011 in. (0.3 mm) in depth.

(c) Scratches

1) Permitted up to 0.011 in. (0.3 mm) in depth.

(d) Missing of surface coating (sulfuric anodizing)

1) Permitted.

(e) Pitting and corrosion

1) Not permitted.

(f) Thread marks or hole ovalization in cylindrical area

1) Permitted up to 0.007 in. (0.2 mm) in depth.

(g) Erosion of parent metal

1) Not permitted.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-22-05-220-004-G00

(4) Examine all other areas of the flowpath fairing except the seal for these conditions:

NOTE: These are areas outside of the forward centering surface A, the surfaces B and C, the rear surface E, the extracting holes, the borescope pass through holes, the drain holes and the attachment areas.

(a) Cracks and tears

1) Not permitted.

(b) Nicks and dents

1) Permitted up to 0.011 in. (0.3 mm) in depth with 0.500 in. (12.7 mm) minimum separation between two nicks or dents.

(c) Scratches

1) Permitted up to 0.011 in. (0.3 mm) in depth.

(d) Contact marks of rivet around attachment areas on internal surface D

1) Permitted up to 0.0007 in. (0.02 mm) in depth.

(e) Missing of surface coating (sulfuric anodizing)

1) Permitted up to 10% of the total area.

2) Maximum Repairable Limit: Any amount.

3) Repair Method: Flowpath Fairing Spot Touch-Up of Anodization, TASK 72-22-05-300-801-G00.

(f) Erosion of parent metal

1) Not permitted.

(g) Pitting and corrosion

1) Not permitted.

SUBTASK 72-22-05-210-001-G00

(5) Examine the flowpath fairing seal [3] for these conditions:

(a) Missing material, cutting

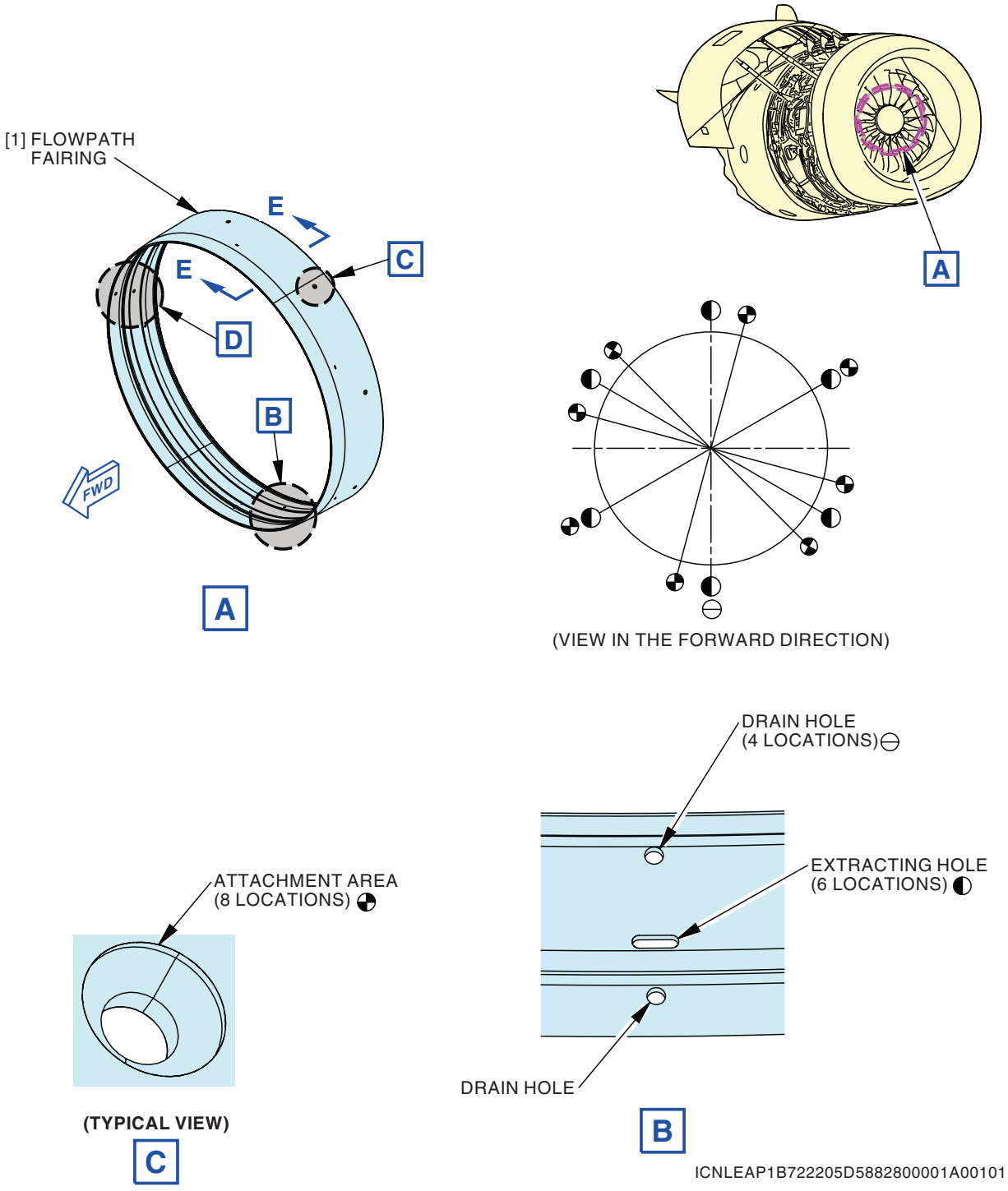
1) Not permitted.

(b) Seal separation

1) Not permitted.

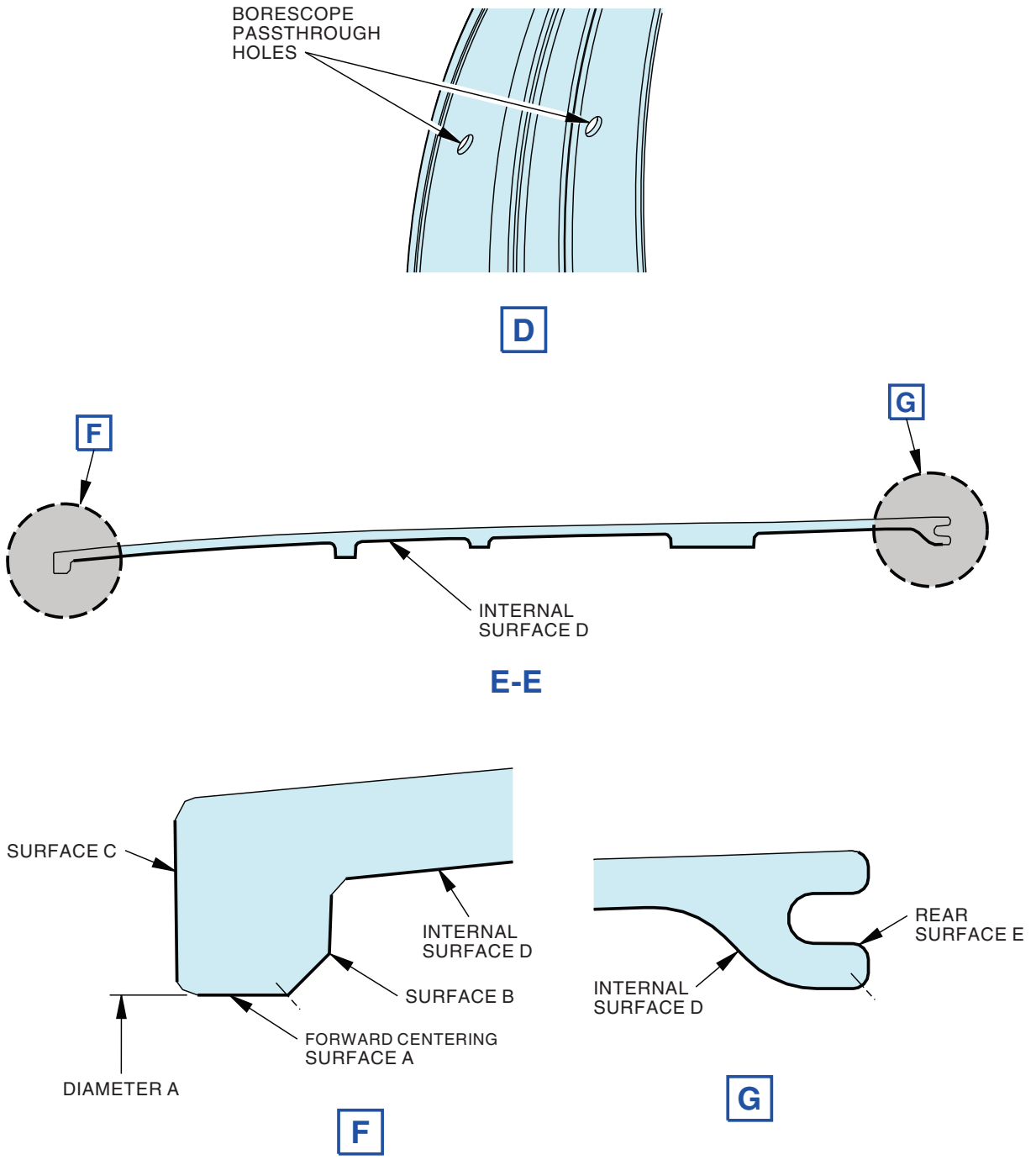
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



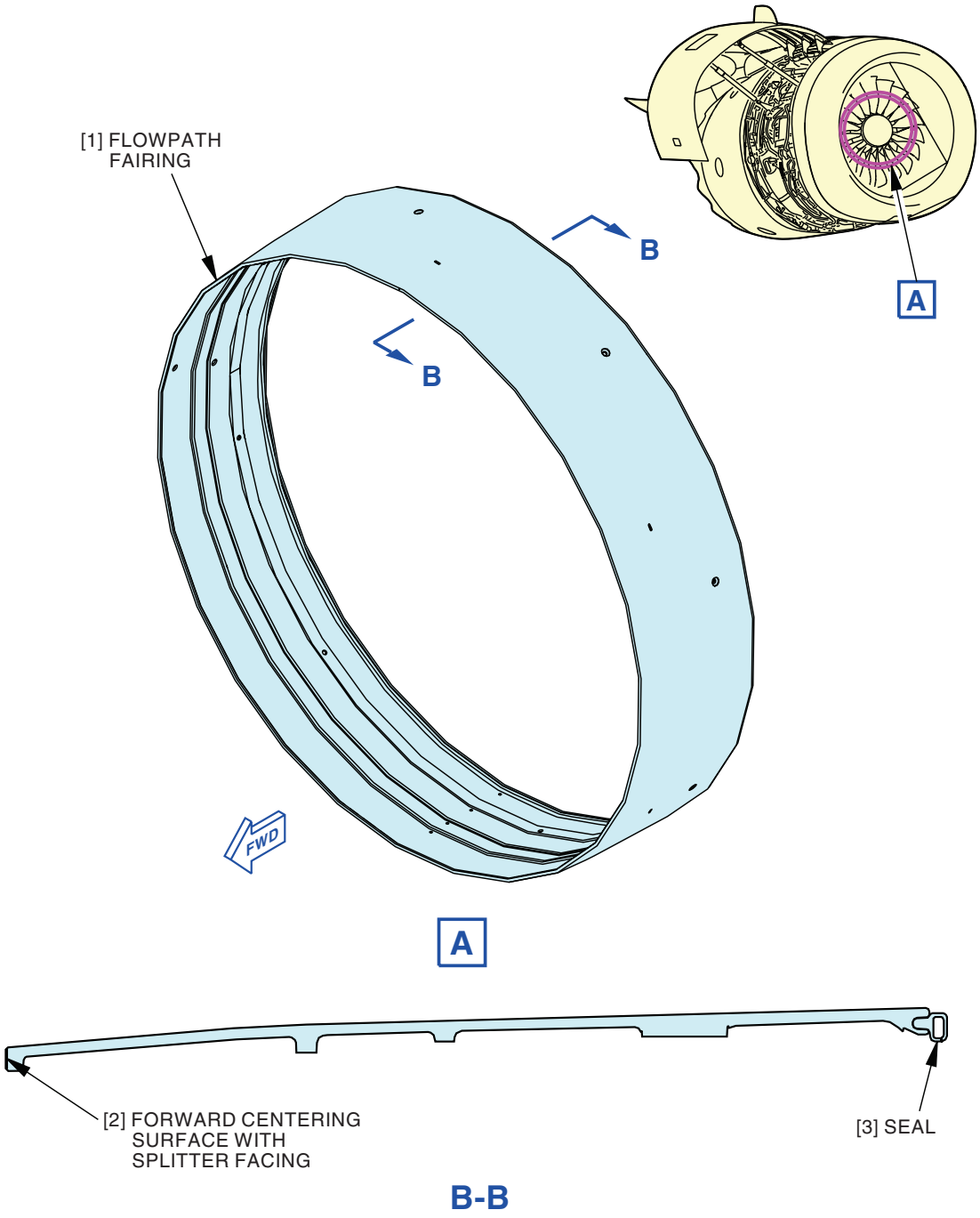
Flowpath Fairing Inspection
Figure 601/72-22-05-990-803-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flowpath Fairing Inspection
Figure 601/72-22-05-990-803-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flowpath Fairing Inspection
Figure 601/72-22-05-990-803-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-22-05-200-802-G00

3. Shroud Segment Inspection

(Figure 602)

A. General

(1) This task gives the instructions to examine the shroud segment.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

F. Prepare for the Inspection

SUBTASK 72-22-05-860-008-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-22-05-860-009-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-22-05-860-010-G00

(3) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

G. Flowpath Fairing Half Ring Inspection

SUBTASK 72-22-05-220-005-G00

(1) Examine all surfaces of the segment shroud [1], except the attachment areas and the surfaces F, G, H and I, for these conditions:

(a) Cracks and tears

1) Not permitted.

(b) Nicks and dents

1) Permitted up to 0.003 in. (0.1 mm) in depth with a minimum of 0.500 in. (12.7 mm) between two indications.

(c) Scratches

1) Permitted up to 0.0019 in. (0.05 mm) in depth and 0.03 in. (1 mm) in length with a minimum of 0.500 in. (12.7 mm) between two indications.

(d) Missing surface coating (sulfuric anodization)

1) Permitted up to a maximum of 10% of the total area.

(e) Erosion of parent metal

1) Not permitted.

(f) Pitting and corrosion

1) Not permitted.

SUBTASK 72-22-05-220-006-G00

(2) Examine the attachment areas for these conditions:

(a) Cracks and tears

1) Not permitted.

(b) Nicks and dents

1) Not permitted.

(c) Scratches

1) Not permitted.

(d) Contact marks

1) Permitted up to 0.0007 in. (0.02 mm) in depth.

(e) Missing surface coating (sulfuric anodization)

1) Permitted.

(f) Pitting and corrosion

1) Not permitted.

(g) Thread marks in the cylindrical area

1) Permitted up to 0.007 in. (0.18 mm) in depth.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-22-05-220-007-G00

(3) Examine the surfaces F, G and H for these conditions:

(a) Cracks and tears

1) Not permitted.

(b) Nicks and dents

1) Permitted up to 0.003 in. (0.1 mm) in depth with a minimum of 0.500 in. (12.7 mm) between two indications.

(c) Scratches

1) Permitted up to 0.0011 in. (0.03 mm) in depth and 0.023 in. (0.6 mm) in length with a minimum of 0.500 in. (12.7 mm) between two indications.

(d) Contact marks

1) Permitted up to 0.0007 in. (0.02 mm) in depth.

(e) Missing surface coating (sulfuric anodization)

1) Permitted.

(f) Pitting and corrosion

1) Not permitted.

(g) Wear or fretting

1) Permitted up to 0.003 in. (0.1 mm) in depth.

SUBTASK 72-22-05-220-008-G00

(4) Examine the surface I for these conditions:

(a) Cracks and tears

1) Not permitted.

(b) Nicks and dents

1) Permitted up to 0.003 in. (0.1 mm) in depth with a minimum of 0.500 in. (12.7 mm) between two indications.

(c) Scratches

1) Permitted up to 0.0019 in. (0.05 mm) in depth and 0.03 in. (1 mm) in length with a minimum of 0.500 in. (12.7 mm) between two indications.

(d) Contact marks

1) Permitted up to 0.0007 in. (0.02 mm) in depth.

(e) Missing surface coating (sulfuric anodization)

1) Permitted.

(f) Pitting and corrosion

1) Not permitted.

(g) Wear or fretting

1) Permitted up to 0.0059 in. (0.15 mm) in depth.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-22-05-860-011-G00

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) On the applicable engine, close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 72-22-05-860-012-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-22-05-860-013-G00

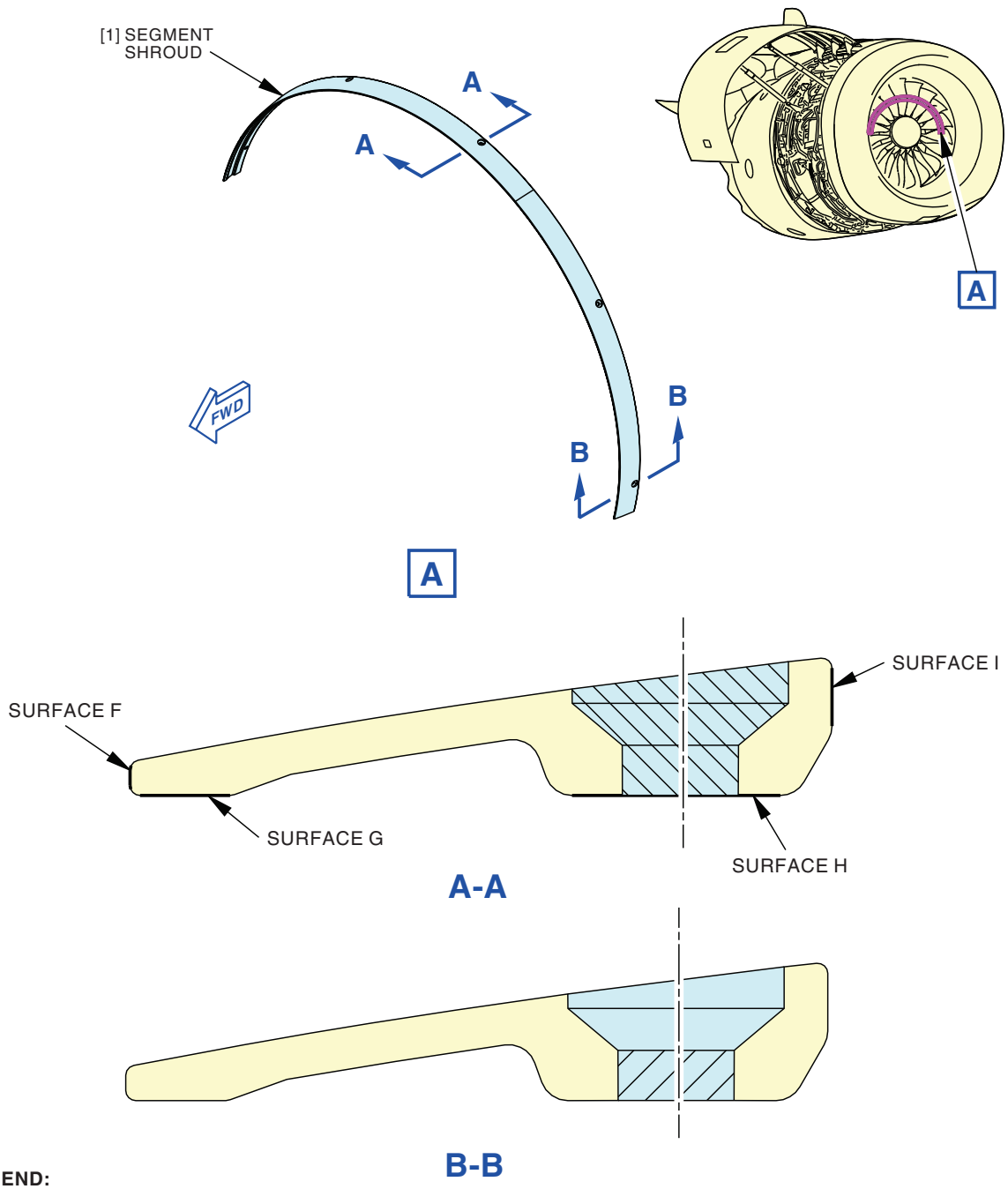
(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Segment Shroud Inspection
Figure 602/72-22-05-990-804-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FLOWPATH FAIRING - REPAIR

1. General

A. This procedure has this task:

(1) A spot touch-up of anodization of the flow path fairing.

TASK 72-22-05-300-801-G00

2. Flowpath Fairing Spot Touch-Up of Anodization

(Figure 801)

A. General

(1) This task gives the instructions to do the spot touch-up of the sulfuric anodizing of the flowpath fairing.

(2) The Table 801 gives a list of the part numbers that are applicable to this repair. All part numbers are applicable to all paragraphs unless specified differently.

Table 801/72-22-05-993-801-G00

Part Number	Service Bulletin	Designation
364-043-410-0	72-0001	Flowpath Fairing
364-043-411-0	72-0005	

B. References

Reference	Title
70-21-23-100-801-G00	Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701)
70-43-07-300-801-G00	Specifications Chemical Touchup for Aluminum Standard Repair (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-22-05-000-801-G00	Flowpath Fairing Removal (P/B 401)
72-22-05-400-801-G00	Flowpath Fairing Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
B01058 [CP1039]	Solvent - Acetone, Reagent Grade	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

G. Prepare for the Touch-Up

SUBTASK 72-22-05-860-014-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-22-05-860-015-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-22-05-860-016-G00

(3) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 72-22-05-010-004-G00

(4) If it is necessary, remove the flowpath fairing (TASK 72-22-05-000-801-G00).

H. Flowpath Fairing Spot Touch-Up of Anodization

SUBTASK 72-22-05-110-001-G00

WARNING SOLVENTS ARE POISONOUS AND FLAMMABLE. DO NOT LET SOLVENTS TOUCH SKIN OR BREATHE FUMES. YOU MUST HAVE PRECAUTIONS AGAINST FIRE. IF YOU DO NOT OBEY INSTRUCTIONS, INJURIES TO PERSONNEL CAN OCCUR.

(1) If it is necessary, clean the part with a cloth moistened with acetone solvent, B01058 [CP1039], or alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-22-05-370-001-G00

(2) Do the spot touch-up of the sulfuric anodizing, do this task: Specifications Chemical Touchup for Aluminum Standard Repair, TASK 70-43-07-300-801-G00.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-22-05-410-004-G00

(1) If it was removed, install the flowpath fairing (TASK 72-22-05-400-801-G00).

SUBTASK 72-22-05-860-017-G00

(2) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

(a) On the applicable engine, close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 72-22-05-860-018-G00

(3) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-22-05-860-019-G00

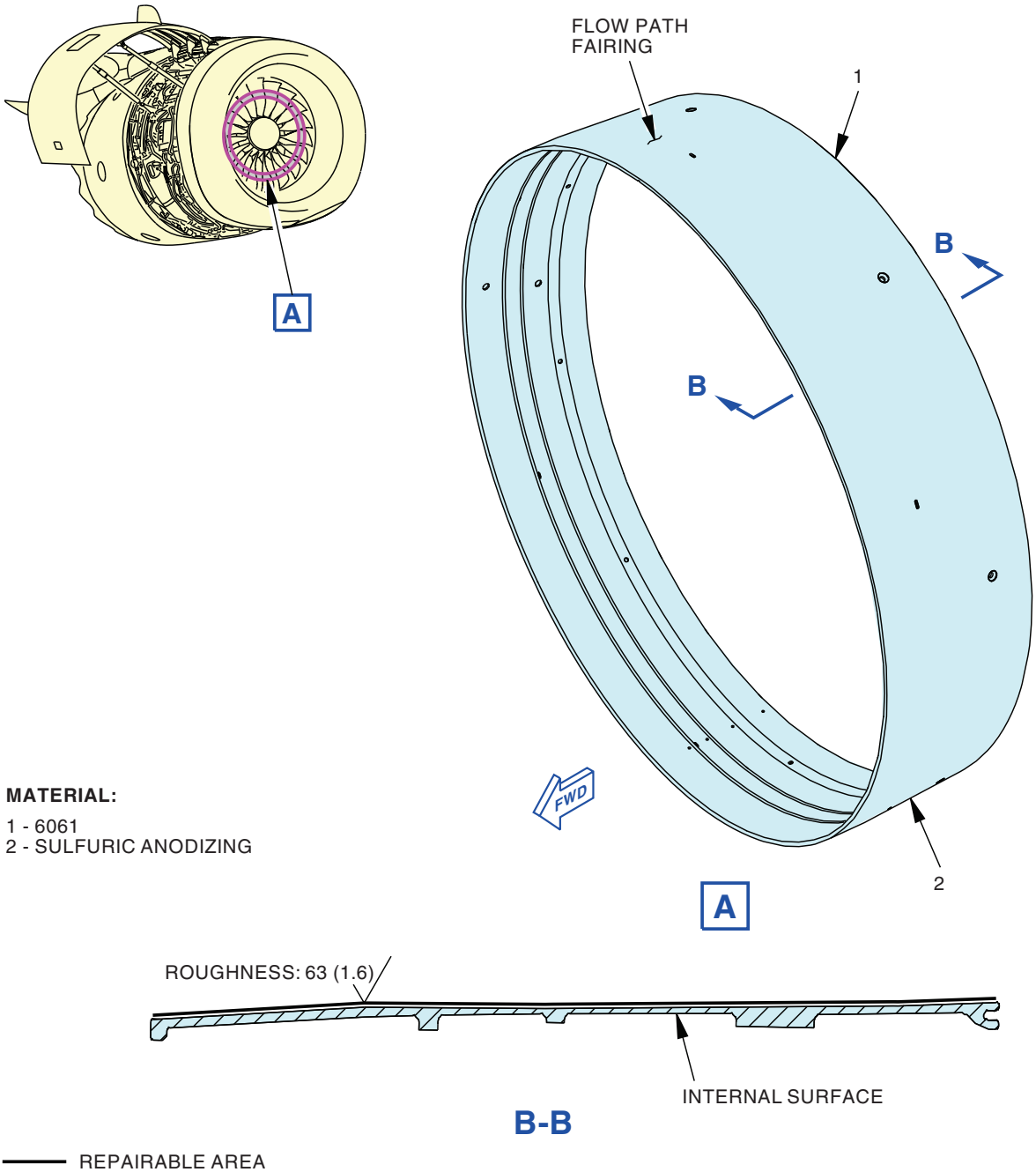
(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



MATERIAL:

1 - 6061
2 - SULFURIC ANODIZING

REPAIRABLE AREA

NOTE:

ROUGHNESS IS IN MICROINCHES
WITH MICROMETERS IN PARENTHESES.

2990642 S0000767883_V1

Flowpath Fairing Repair
Figure 801/72-22-05-990-806-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SPLITTER FAIRING - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Splitter fairing removal

(2) Splitter fairing installation.

TASK 72-22-06-000-801-G00

2. Splitter Fairing Removal

(Figure 401 and Figure 402)

A. General

(1) This task provides the instructions on how to remove the splitter fairing.

B. References

Reference	Title
72-22-05-000-801-G00	Flowpath Fairing Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-15091	Fixture, Remove/Install - Splitter Fairing Part #: 956A7015G01 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Removal

SUBTASK 72-22-06-860-001-G00

(1) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 72-22-06-860-002-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-22-06-010-001-G00

(3) Do this task: Flowpath Fairing Removal, TASK 72-22-05-000-801-G00.

F. Splitter Fairing Removal

SUBTASK 72-22-06-020-001-G00

(1) Remove the splitter fairing [1] as follows:

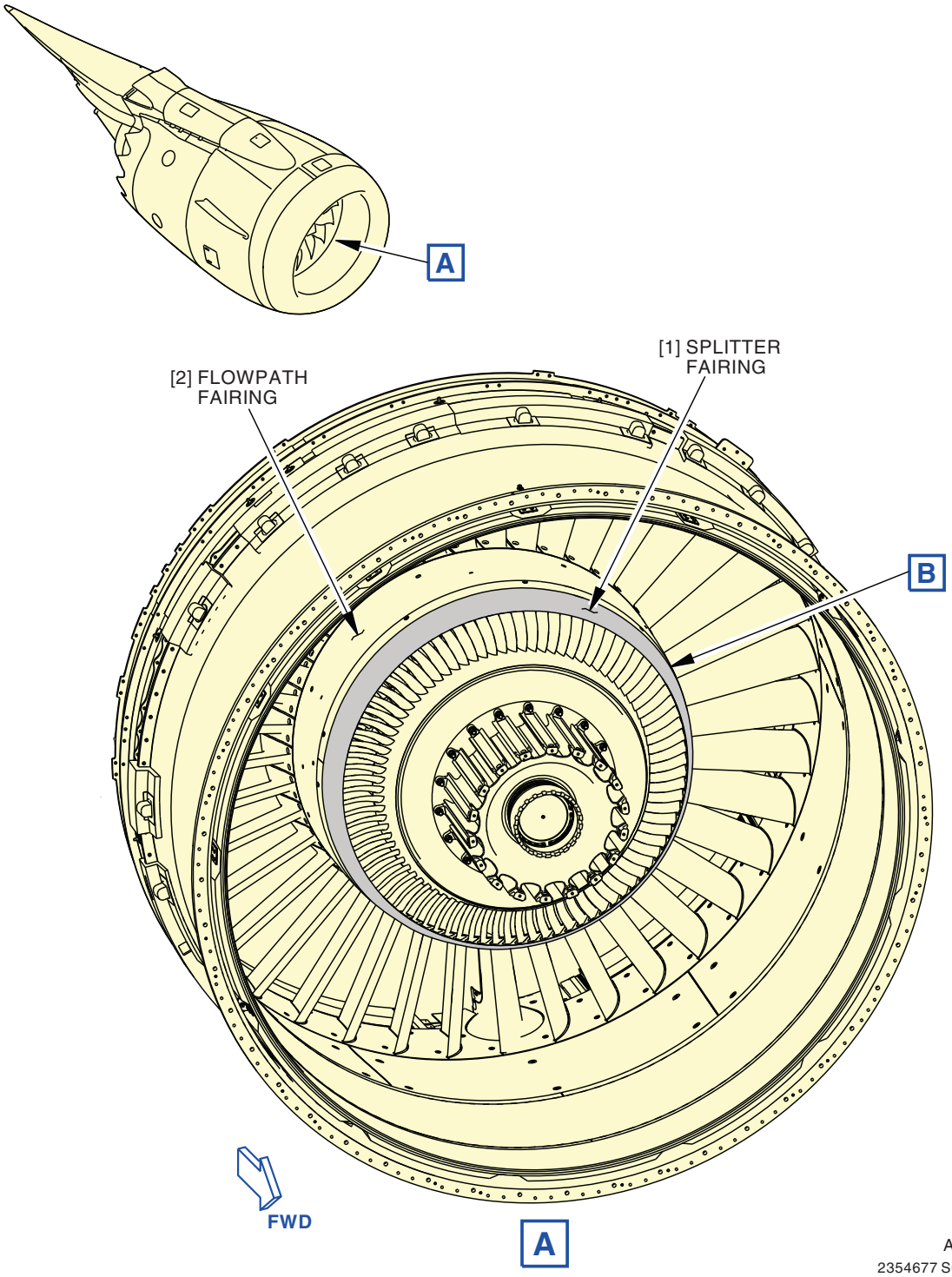
(a) Install the splitter fairing remove/install fixture, SPL-15091, around the splitter fairing [1].

(b) Remove the 36 bolts [3] that attach the splitter fairing [1] to the LPC stage 1 IGV [4].

(c) Remove the splitter fairing [1] from the engine.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Splitter Fairing Installation
Figure 401/72-22-06-990-801-G00 (Sheet 1 of 2)

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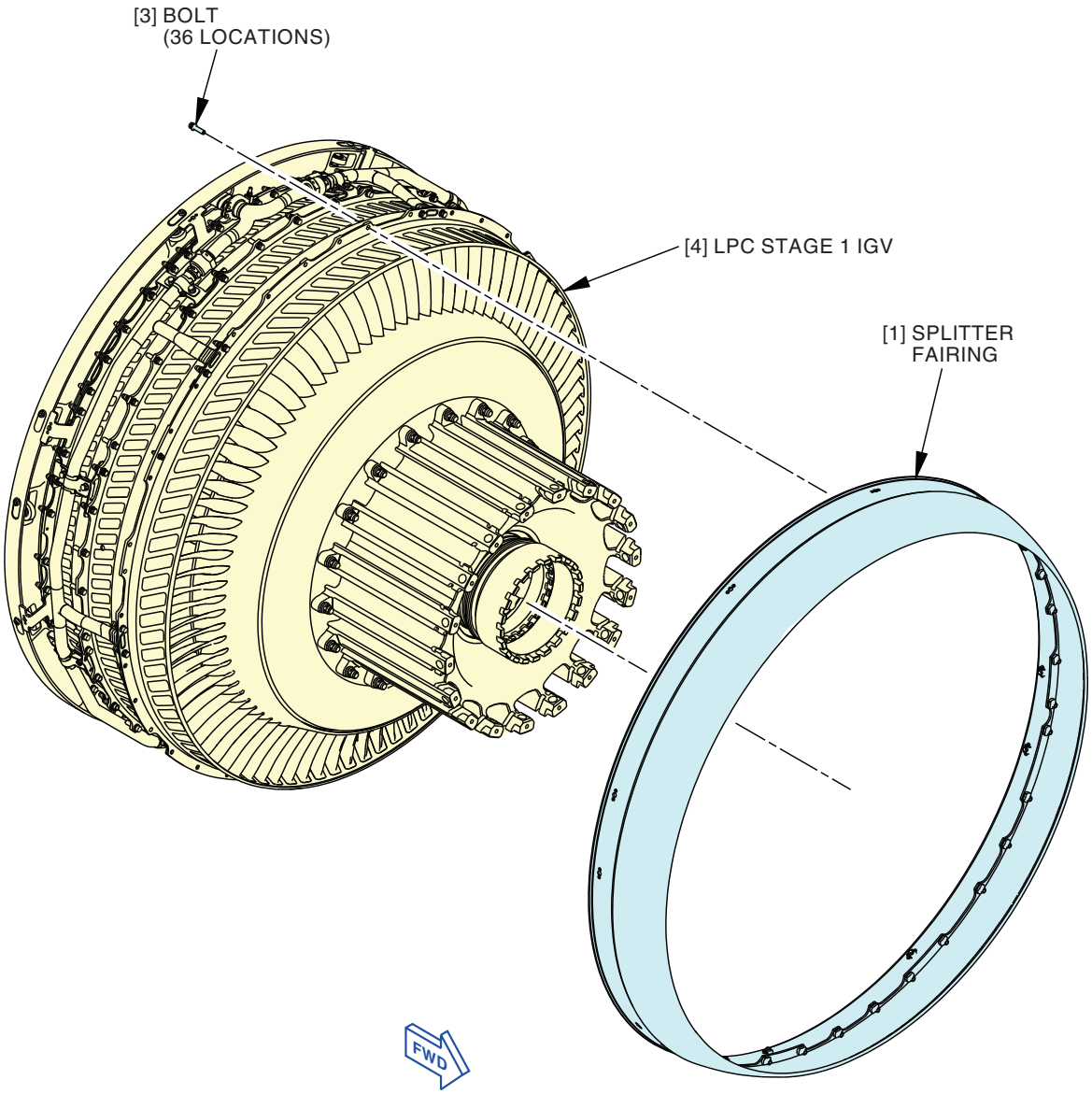

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Splitter Fairing Installation
Figure 401/72-22-06-990-801-G00 (Sheet 2 of 2)

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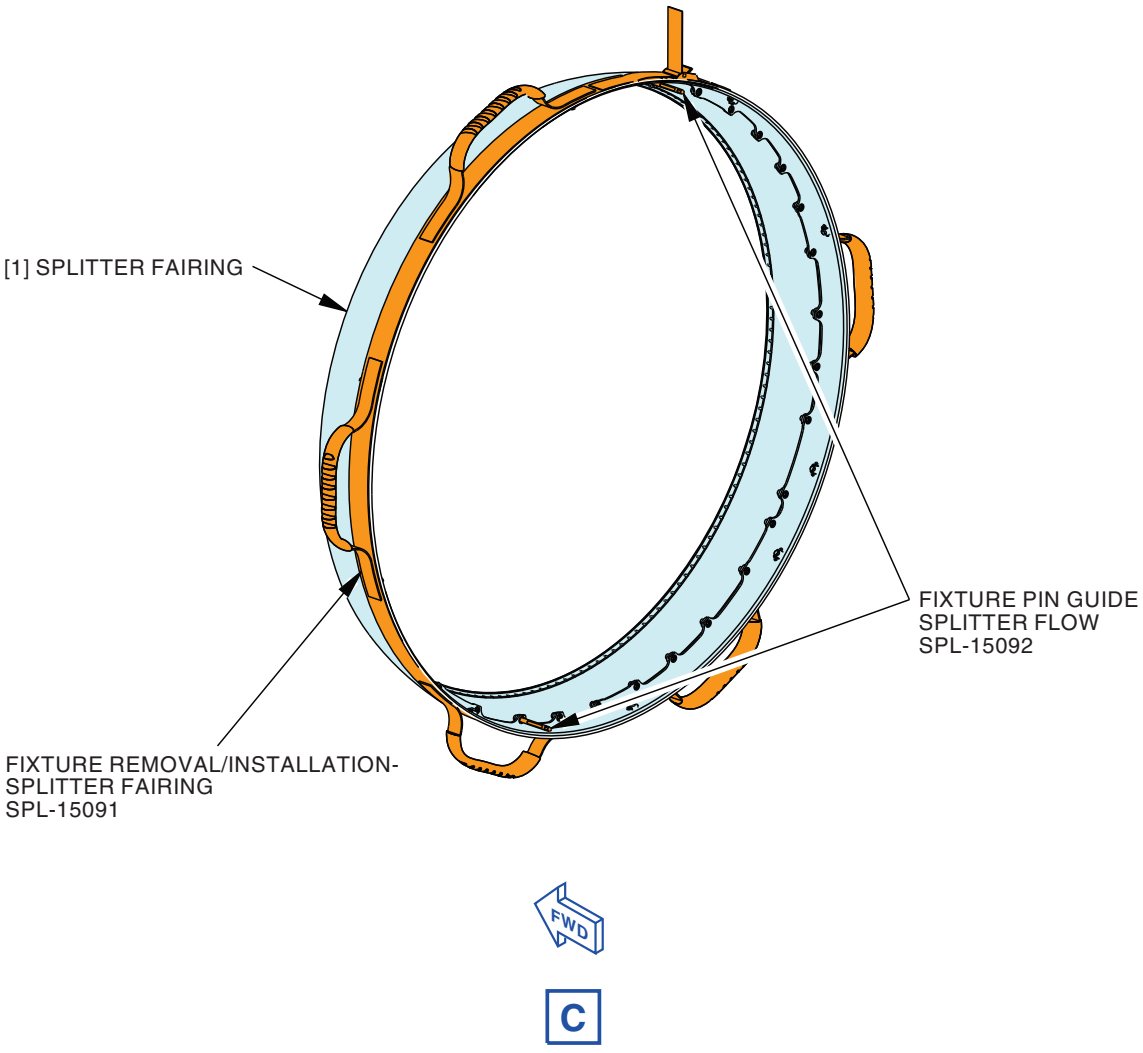

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Splitter Fairing Removal and Installation Tools
Figure 402/72-22-06-990-804-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-22-06-400-801-G00

3. Splitter Fairing Installation

(Figure 401 and Figure 402)

A. General

(1) This task provides the instructions on how to install the splitter fairing.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
72-22-05-400-801-G00	Flowpath Fairing Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-15091	Fixture, Remove/Install - Splitter Fairing Part #: 956A7015G01 Supplier: 58828
SPL-15092	Pins, Guide - Splitter Fairing Part #: 956A8040G01 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00601 [CP2101]	Vaseline - Graphite Mineral	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Splitter Fairing Installation

SUBTASK 72-22-06-390-001-G00

(1) Install the splitter fairing [1] as follows:

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Install the splitter fairing remove/install fixture, SPL-15091, around the splitter fairing [1].

(c) Install two splitter fairing guide pin, SPL-15092, 180 degrees apart in the splitter faring flange

(d) Install the splitter fairing [1] in the correct position on the engine.

(e) Lubricate the threads of the 36 bolts [3] with the graphite mineral vaseline, D00601 [CP2101].

(f) Install 34 of the bolts [3] that attach the splitter fairing [1] to the LPC stage 1 IGV [4].

(g) Remove the two splitter fairing guide pin, SPL-15092.

(h) Install the last two bolts [3] that attach the splitter fairing [1], to the LPC stage 1 IGV [4].

(i) Tighten the bolts [3] to 109.25 in-lb (12.34 N·m) - 120.75 in-lb (13.64 N·m).

(j) Remove the splitter fairing remove/install fixture, SPL-15091.

(k) Clean the work area and clear the tools and other items.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-22-06-410-001-G00

(1) Do this task: Flowpath Fairing Installation, TASK 72-22-05-400-801-G00.

SUBTASK 72-22-06-860-003-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-22-06-860-004-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

H. Splitter Fairing Installation Test

SUBTASK 72-22-06-710-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SPLITTER FAIRING INSPECTION/CHECK

1. General

A. This procedure has one task.

(1) Splitter fairing inspection.

TASK 72-22-06-200-801-G00

2. Splitter Fairing Inspection

(Figure 601)

A. General

(1) This task gives the instructions to examine the splitter fairing.

B. References

Reference	Title
72-22-06-000-801-G00	Splitter Fairing Removal (P/B 401)
72-22-06-400-801-G00	Splitter Fairing Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-585	Mat - Protective, Closed Cell Neoprene, 30-50 duro, Weather and Oil Resistant, 3/8 Inch Min Thickness, Minimum 42x60 Inches with Warning Streamers
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Inspection

SUBTASK 72-22-06-860-009-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-22-06-860-007-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-22-06-410-002-G00

CAUTION PUT THE PROTECTIVE MAT IN THE INLET COWL. THIS WILL PREVENT DAMAGE TO THE INLET COWL OR ENGINE BY TOOLS, PARTS, OR UNWANTED MATERIALS THAT FALL ON THE INLET COWL SURFACE.

(3) Put the protective mat, STD-585 on the lower half of the inlet cowl.

F. Splitter Fairing Inspection

SUBTASK 72-22-06-211-001-G00

(1) Examine the splitter fairing [1] for these conditions:

NOTE: This is a visual inspection of the splitter fairing through the fan blades. If it is necessary to do a full inspection, it will be necessary to remove the splitter fairing (TASK 72-22-06-000-801-G00).

(a) Cracks and tears

1) Not permitted.

(b) Nicks in Area H

1) Any number up to 0.0059 in. (0.15 mm) in depth.

(c) Dents in Area H

1) Any number up to 0.0059 in. (0.15 mm) in depth.

(d) Scratches in Area H

1) Any number up to 0.0039 in. (0.10 mm) in depth.

(e) Nicks in Area G

1) Any number up to 0.0059 in. (0.15 mm) in depth.

(f) Dents in Area G

1) Any number up to 0.0059 in. (0.15 mm) in depth.

(g) Scratches in Area G

1) Any number up to 0.0039 in. (0.10 mm) in depth.

(h) Erosion

1) Any number up to 0.0059 in. (0.15 mm) in depth.

(i) Pitting

1) Not permitted.

(j) Deposits in gap (dimension A) between the IGV and the splitter fairing [1]

1) Permitted with these conditions:

a) The deposits are not more than 0.512 in. (13.0 mm) long.

b) The distance between two deposits is more than 7.480 in. (190.0 mm).

c) A maximum of 5% of the circumference can be affected.

SUBTASK 72-22-06-211-002-G00

(2) Do a dimensional examination of the splitter fairing [1], dimension A.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) The minimum serviceable limit is 0.01 in. (0.25 mm).

(b) The maximum serviceable limit is 0.0335 in. (0.85 mm).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-22-06-410-004-G00

(1) If it is necessary, do this task: TASK 72-22-06-400-801-G00.

SUBTASK 72-22-06-010-002-G00

CAUTION

MAKE SURE THERE ARE NO TOOLS, PARTS OR OTHER UNWANTED MATERIAL IN THE INLET AREA. DAMAGE TO EQUIPMENT COULD OCCUR.

(2) Remove the protective mat, STD-585 from the engine inlet cowl.

SUBTASK 72-22-06-860-008-G00

(3) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-22-06-860-010-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

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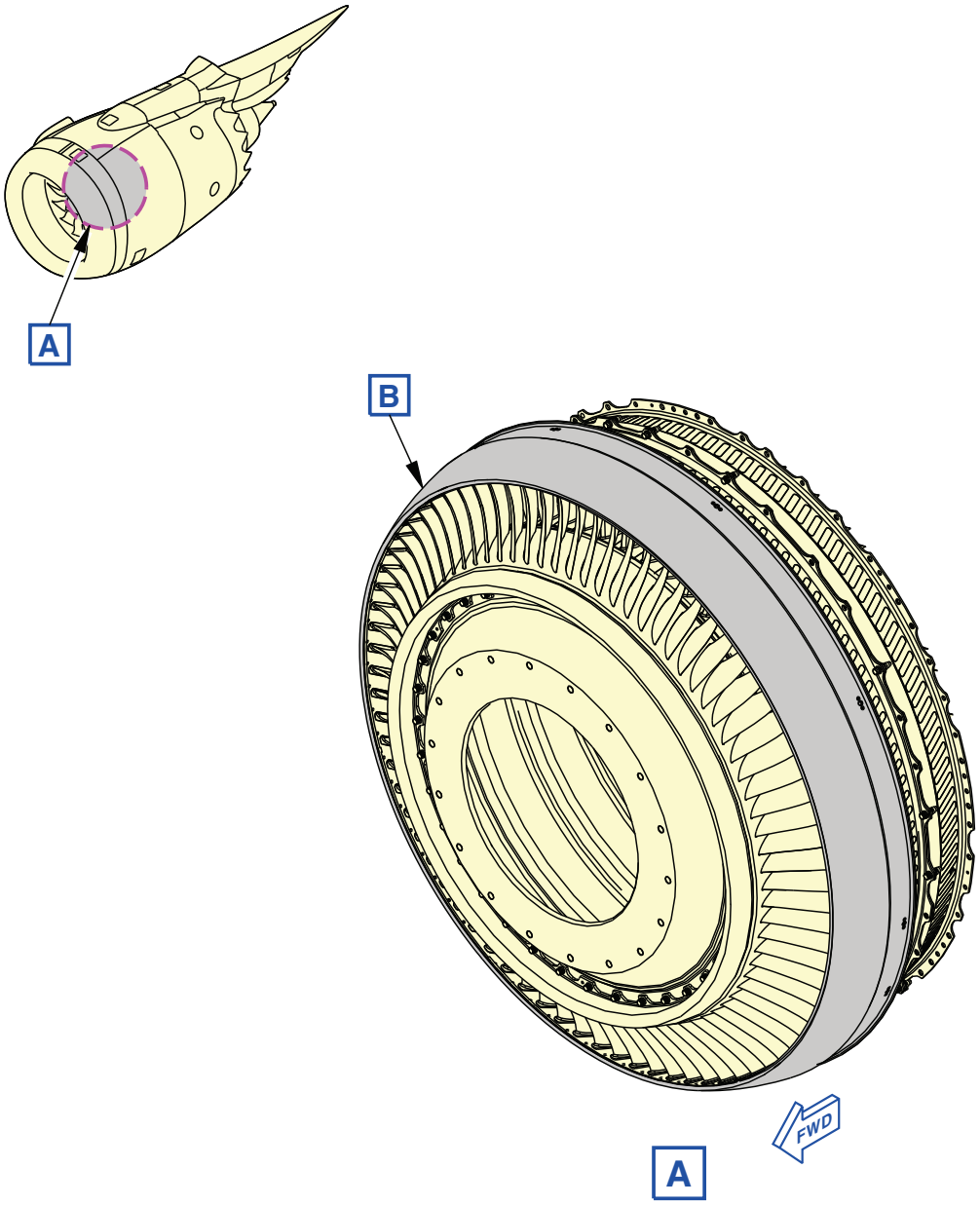

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Splitter Fairing Inspection
Figure 601/72-22-06-990-803-G00 (Sheet 1 of 3)

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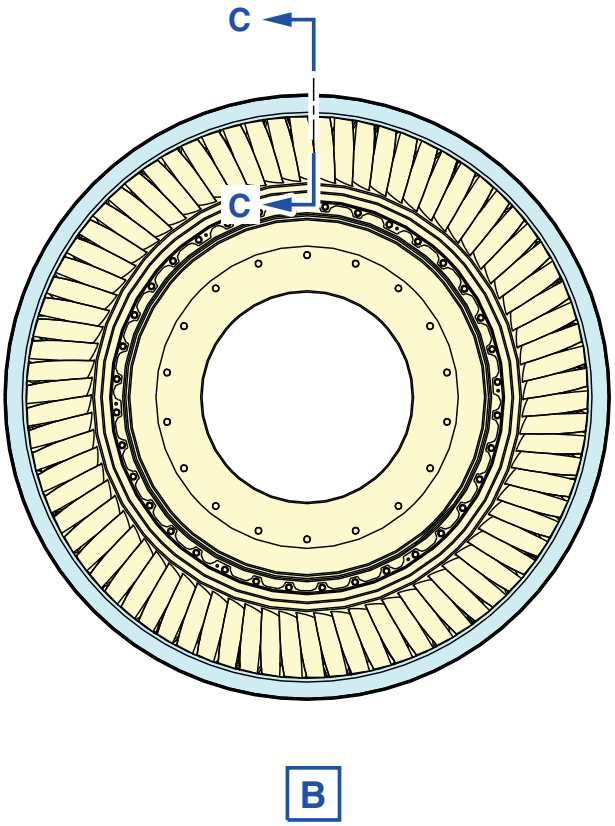

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Splitter Fairing Inspection
Figure 601/72-22-06-990-803-G00 (Sheet 2 of 3)

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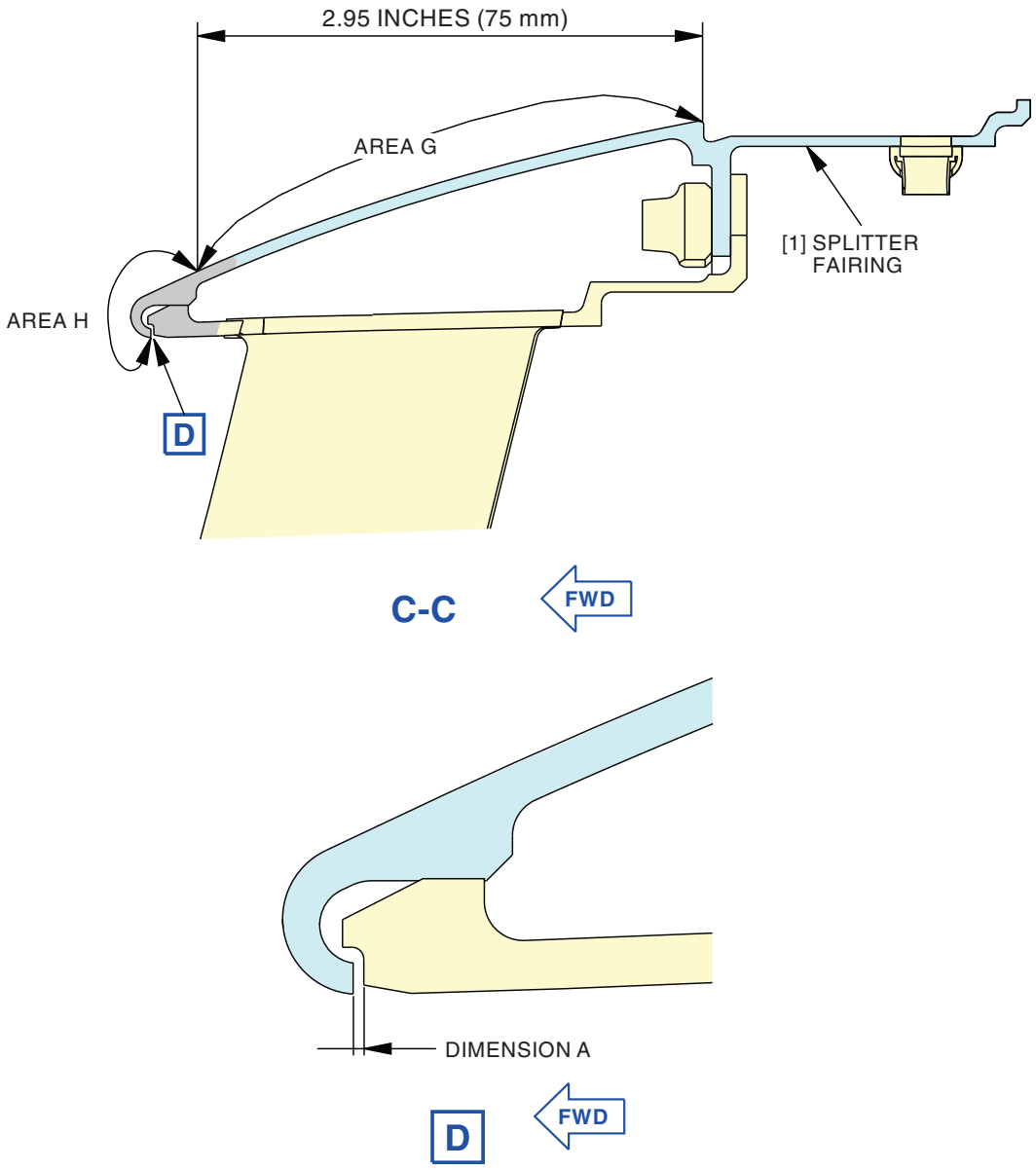

NOTE:
ALL DIMENSIONS ARE IN INCHES (mm).

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Splitter Fairing Inspection
Figure 601/72-22-06-990-803-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

BOOSTER ANTI-ICE DUCT - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Booster anti-ice inspection duct.

TASK 72-22-08-200-801-G00

2. Booster Anti-Ice Duct Inspection

(Figure 601)

A. General

(1) This task gives the instructions to examine the booster anti-ice duct.

B. References

Reference	Title
72-22-05-000-801-G00	Flowpath Fairing Removal (P/B 401)
72-22-05-400-801-G00	Flowpath Fairing Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Inspection

SUBTASK 72-22-08-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-22-08-860-002-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-22-08-020-001-G00

(3) Do this task: Flowpath Fairing Removal, TASK 72-22-05-000-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Booster Anti-Ice Duct Inspection

SUBTASK 72-22-08-210-001-G00

(1) Examine the booster anti-ice duct [1] for this condition:

(a) Cracks

1) Not permitted.

2) If you find cracks, contact CFM.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-22-08-420-001-G00

(1) Do this task: Flowpath Fairing Installation, TASK 72-22-05-400-801-G00.

SUBTASK 72-22-08-860-003-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-22-08-860-004-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

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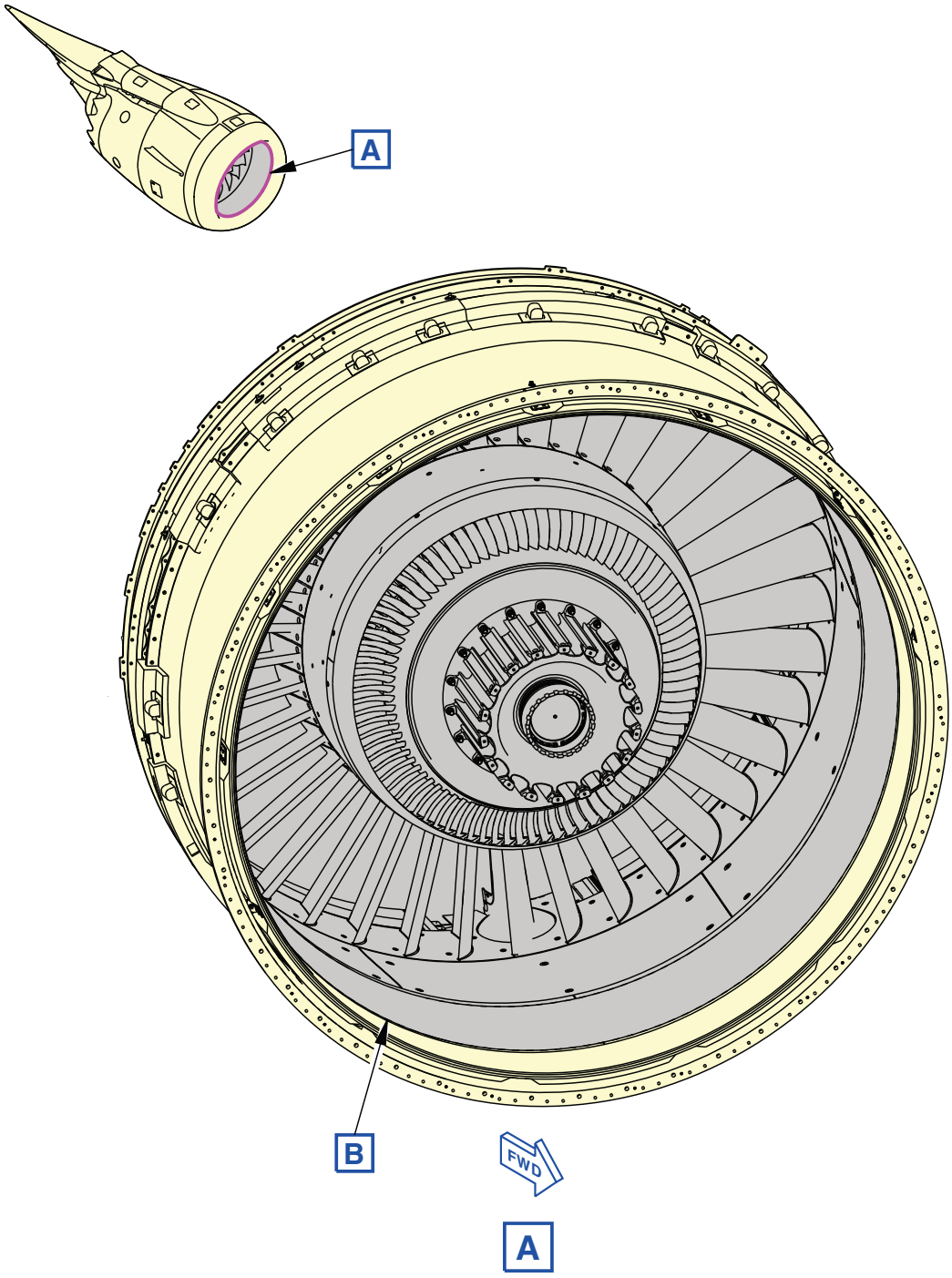

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Booster Anti-Ice Duct Inspection
Figure 601/72-22-08-990-801-G00 (Sheet 1 of 3)

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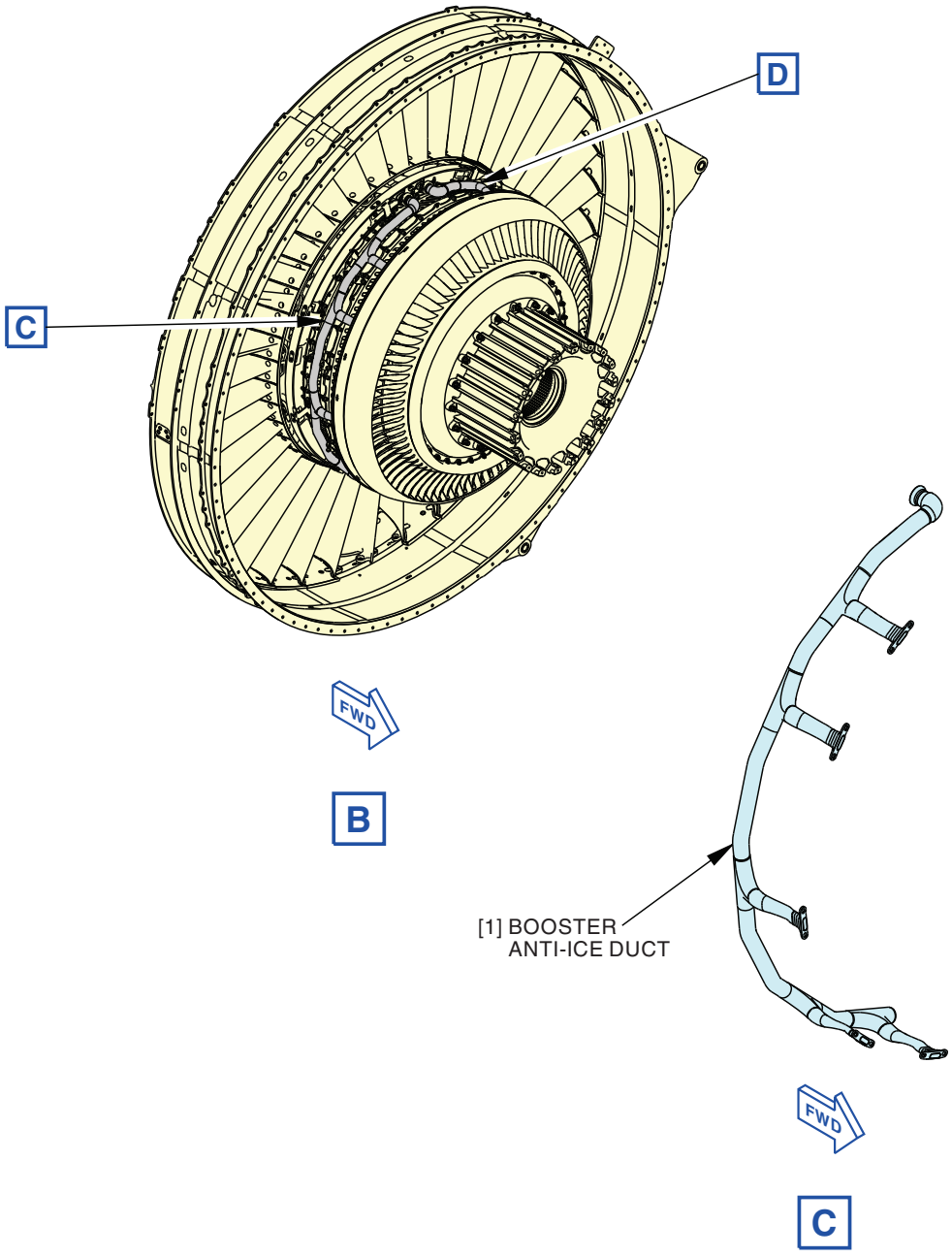

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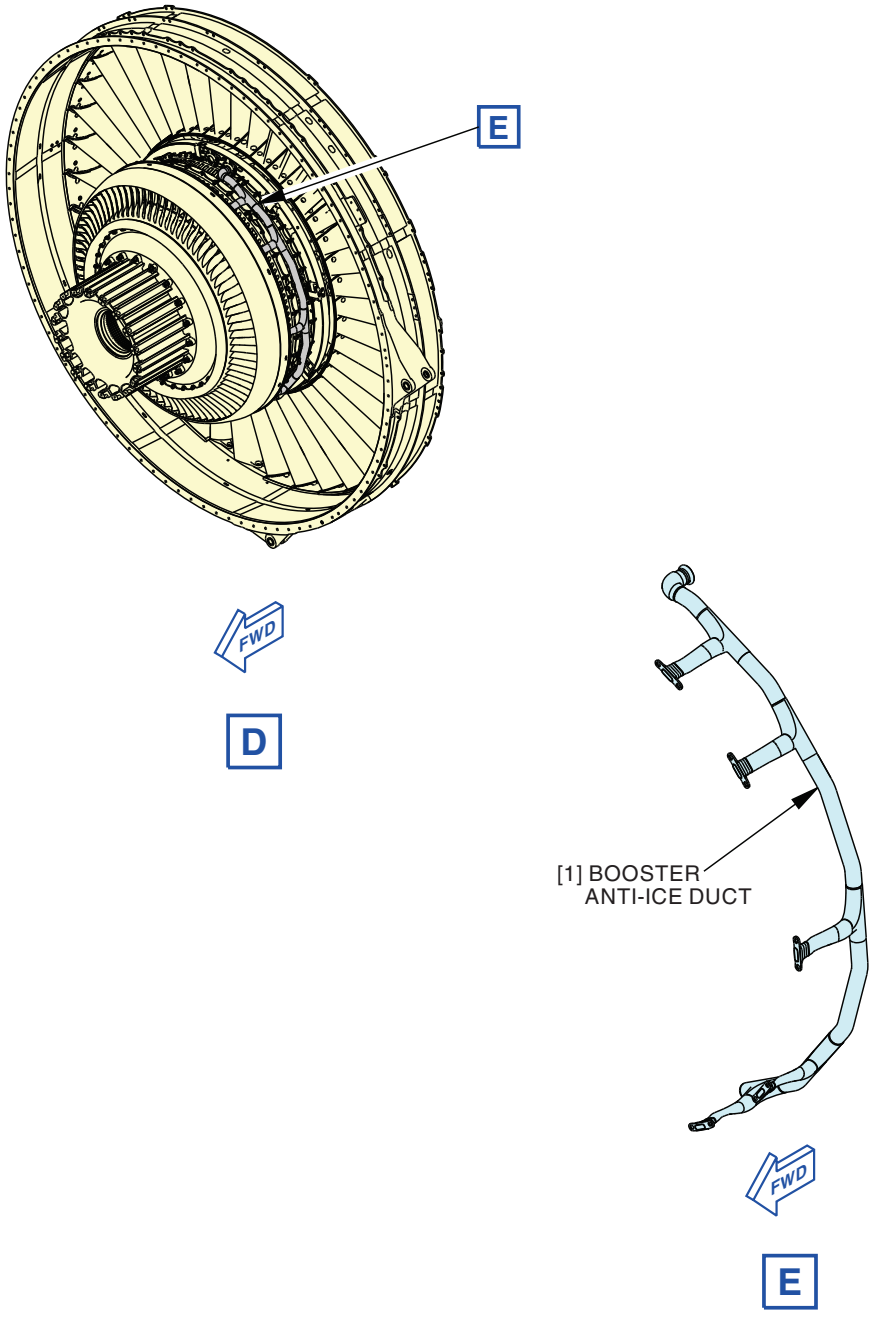
Booster Anti-Ice Duct Inspection
Figure 601/72-22-08-990-801-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Booster Anti-Ice Duct Inspection
Figure 601/72-22-08-990-801-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LOW PRESSURE COMPRESSOR SHAFT - REPAIR

1. General

A. This procedure has one task.

(1) Low Pressure Compressor (LPC) Shaft Paint Touch-up Repair.

TASK 72-22-40-300-801-G00

2. Low Pressure Compressor (LPC) Shaft Paint Touch-up Repair

(Figure 801)

A. General

(1) This task gives the instructions to do a paint touch-up repair of the Low Pressure Compressor (LPC) shaft.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
B01058 [CP1039]	Solvent - Acetone, Reagent Grade	
G51335 [CP2674]	Lint-Free Cloth	
G51412 [CP2721]	Cloth - Abrasive, Silicone Carbide - CS 333X	
G51479 [CP2696]	Tape - Masking - 3M 2525	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
426	Right Thrust Reverser, Engine 2

G. Prepare for the Repair

SUBTASK 72-22-40-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-22-40-860-002-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Low Pressure Compressor (LPC) Shaft Paint Touch-up Repair

SUBTASK 72-22-40-100-001-G00

WARNING MAKE SURE THAT YOU READ THE MANUFACTURER'S SAFETY DATA SHEET (MSDS) FOR THE MATERIALS YOU WILL USE, AND WEAR THE PROTECTIVE CLOTHING AND EQUIPMENT IF IT IS SPECIFIED. IF YOU DO NOT OBEY, INJURY TO PERSONS CAN OCCUR.

WARNING PUT ON GLOVES FOR PROTECTION BEFORE YOU TOUCH THE SHARP EDGES. THE SHARP EDGES CAN CUT YOU.

(1) If it is necessary, remove grease from the part.

(a) Clean the part with a moist lint-free cloth, G51335 [CP2674] and acetone solvent, B01058 [CP1039] or alcohol, B00676 [CP1041].

SUBTASK 72-22-40-220-001-G00

(2) Examine the outer painted areas of the LPC shaft for these conditions.

(a) Missing paint

1) Permitted if less than 1.00 in^2 (6.5 cm^2) for each zone and less than 28.2 in^2 (181.9 cm^2) total for all zone.

(b) If you cannot see the parent material in areas A and B, do a paint touch-up.

(c) Missing paint in area E

1) Not repairable with this procedure.

SUBTASK 72-22-40-370-001-G00

(3) Do a paint touch-up repair as follows:

(a) Use an abrasive cloth, G51412 [CP2721] to clean the area to be repaired.

NOTE: Do not clean down to the parent metal in areas A or B. You can clean down to the parent metal in areas C or D.

(b) Remove grease with a moist lint-free cloth, G51335 [CP2674] and acetone solvent, B01058 [CP1039] or alcohol, B00676 [CP1041].

(c) Do these steps to apply the aluminum paint:

1) Paint the part and the test specimen at the same time.

2) Use local purchase eddy current measurement tool in the touch-up area with the standard tool.

a) In areas A, C, and D, make sure that the paint thickness is between 0.00138 in. (0.035 mm) and 0.00295 in. (0.075 mm).

b) In area B, there is no thickness requirement.

3) Do a peeling-off test for paint stripping with 3M 2525 tape, G51479 [CP2696] as follows:

NOTE: Do this test only if the touch-up area is in areas A, B, or D. Peeling-off test is not required in the area C.

a) Use a peeling strength of 6N per cm of width.

b) To bond the tape, burnish and apply a pressure which does not cause damage to the paint.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-22-40-860-003-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-22-40-860-004-G00

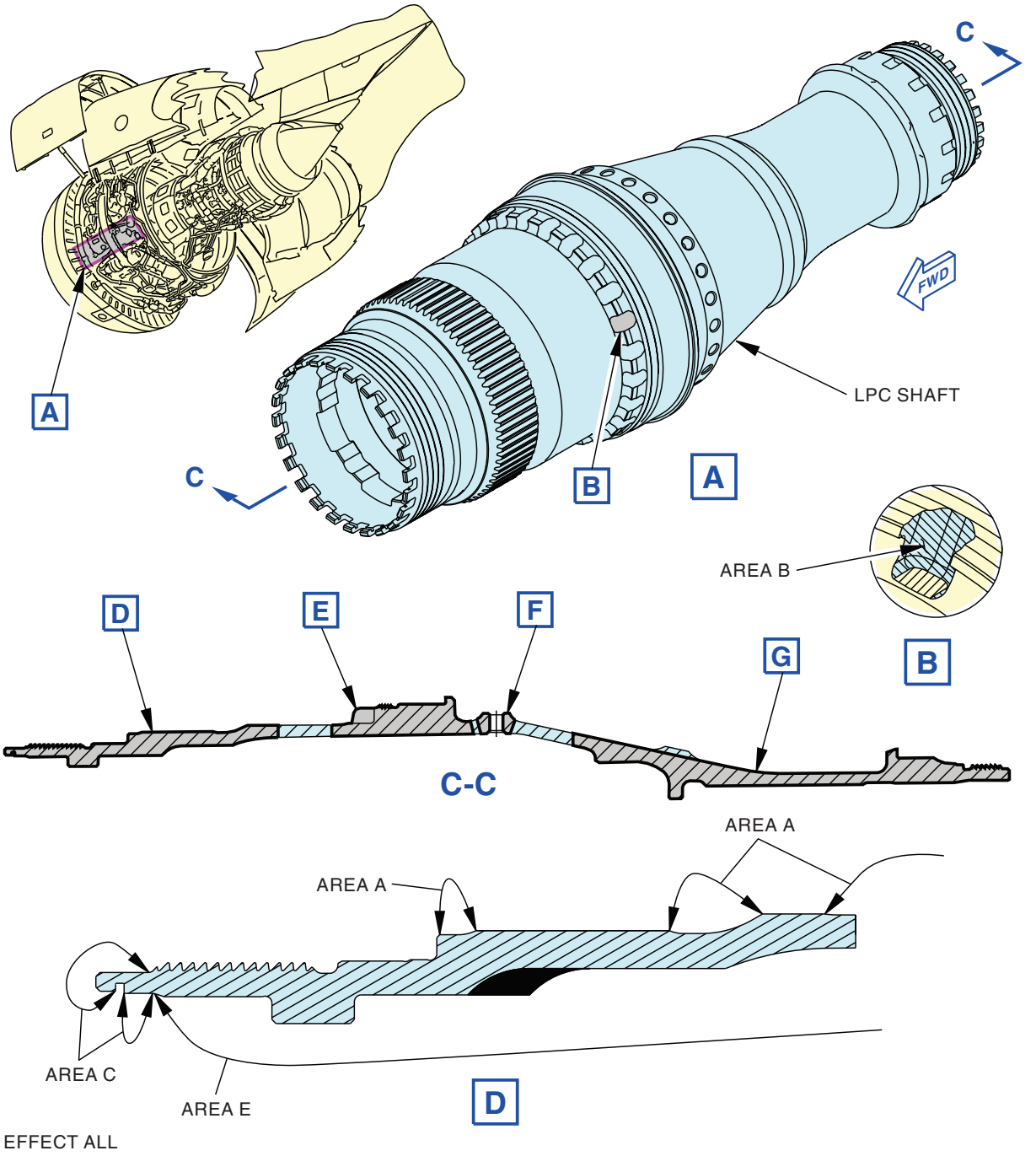
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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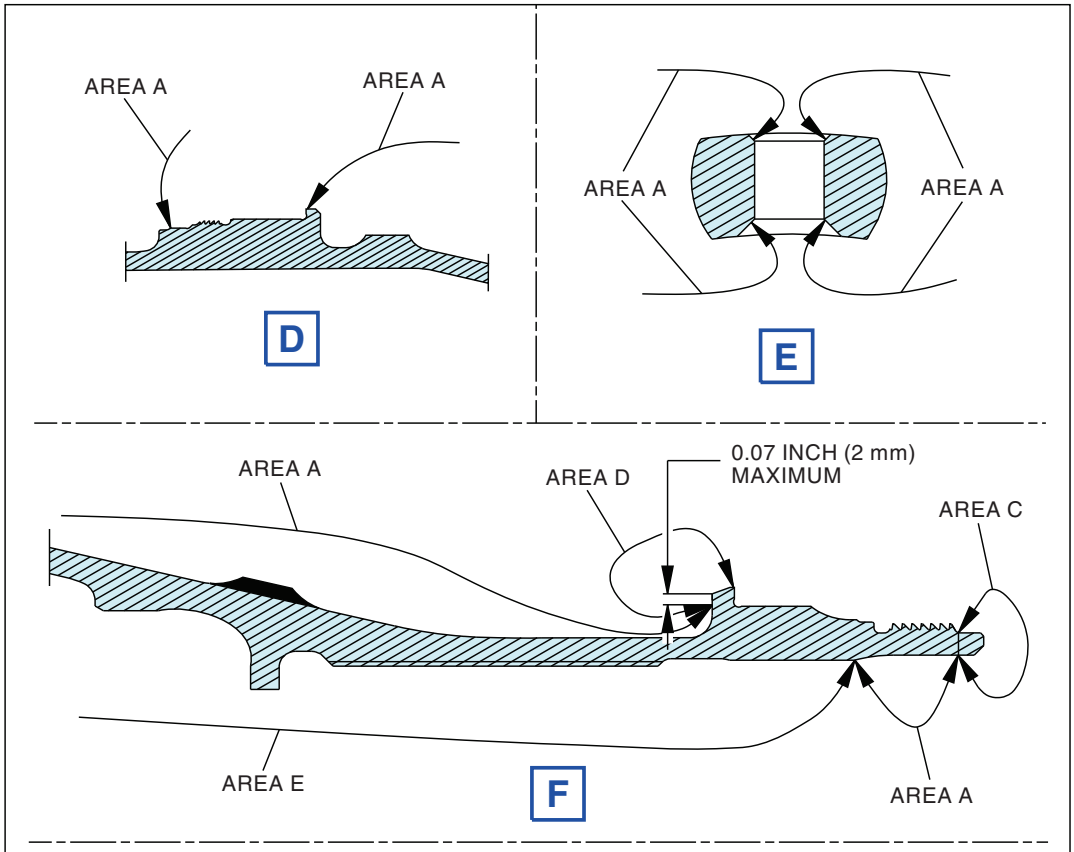
2799414 S0000640536_V1

Low Pressure Compressor (LPC) Shaft Paint Touch-up Repair
Figure 801/72-22-40-990-801-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND :

AREA A: PAINTED AREAS (THICKNESS BETWEEN 0.00137 INCH (0.035 MM) AND 0.00295 INCH (0.075 MM) NOT REPAIRABLE IF YOU SEE PARENT METAL.
PEELING OFF TEST REQUIRED.

AREA B: PAINTED AREAS (WITHOUT THICKNESS REQUIREMENT)
NOT REPAIRABLE IF YOU SEE PARENT METAL.
PEELING OFF TEST REQUIRED.

AREA C: PAINTED AREAS (THICKNESS BETWEEN 0.00137 INCH (0.035 MM) AND 0.00295 INCH (0.075 MM) REPAIRABLE IF YOU SEE PARENT METAL.
PEELING OFF TEST REQUIRED.

AREA D: PAINTED AREAS (THICKNESS BETWEEN 0.00137 INCH (0.035 MM) AND 0.00295 INCH (0.075 MM) REPAIRABLE IF YOU SEE PARENT METAL.
PEELING OFF TEST REQUIRED.

AREA E: AREA NOT REPAIRABLE WITH THIS PROCEDURE.

EFFECT ALL

ICNLEAP1B722240BF03100002A0101
2799609 S0000640537_V1

Low Pressure Compressor (LPC) Shaft Paint Touch-up Repair
Figure 801/72-22-40-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN FRAME MODULE - INSPECTION/CHECK

1. General

A. This procedure has these tasks:

(1) Fan Frame Module Inspection

(2) Fan Frame AGB Fittings and Axial Mount Inspection.

TASK 72-23-00-211-804-G00

2. Fan Frame Module Inspection

(Figure 601, Figure 602, Figure 603, Figure 604, Figure 605, Figure 606, Figure 607, Figure 608, Figure 609, Figure 610, Figure 611, Figure 612, Figure 613, Figure 614, Figure 615, Figure 616, Figure 617, Figure 618, Figure 619, Figure 620, Figure 621, Figure 622, Figure 623, Figure 624, Figure 625, Figure 626, Figure 627, Figure 628, Figure 629, Figure 630, Figure 631, Figure 632, Figure 633, Figure 634, Figure 635, Figure 636, Figure 637, Figure 638, Figure 639, Figure 640, and Figure 641)

A. General

(1) This task gives the instructions to do the inspection of the fan frame module.

B. References

Reference	Title
70-43-07-300-801-G00	Specifications Chemical Touchup for Aluminum Standard Repair (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-21-03-320-801-G00	Aluminum Outlet Guide Vane Leading and Trailing Edge Blending Repair (P/B 801)
72-21-03-320-802-G00	Outlet Guide Vane Struts 1 and 5 Leading Edge Blending Repair (P/B 801)
72-21-03-320-803-G00	Titanium Outlet Guide Vane Leading Edge Blending Repair (P/B 801)
72-21-03-370-801-G00	Outlet Guide Vanes Coating Local Repair (P/B 801)
72-21-03-370-802-G00	Outlet Guide Vanes Paint Primer Local Repair (P/B 801)
72-21-03-960-801-G00	Outlet Guide Vane Two-Lug Nut Plate/Assembly Gang Channel Nuts Replacement (P/B 801)
72-21-09-960-802-G00	Outer Outlet Guide Vane Platform - Damper Replacement (P/B 801)
72-23-00-300-801-G00	Strut 1 Outer Fitting - Blending (P/B 801)
72-23-00-300-802-G00	Strut 5 Outer Fitting - Blending (P/B 801)
72-23-00-900-801-G00	Fan Frame Hub Inserts Replacement (P/B 801)
72-23-04-300-801-G00	Fan Frame Shroud - Key-locked Inserts Replacement (P/B 801)
72-23-04-300-802-G00	Fan Frame Shroud Repair (PTFE Lubricant Varnish in the V-groove) (P/B 801)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A50472 [CP2734]	Product - Silicone Elastomer Sealing	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Inspection

SUBTASK 72-23-00-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-00-860-002-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Fan Frame Module Inspection

SUBTASK 72-23-00-211-005-G00

(1) Examine the strut 1 outer fitting (RH) [1] (Figure 601 (Sheet 1), Figure 601 (Sheet 2), Figure 601 (Sheet 3)).

(a) Do a visual examination of all areas.

1) Dents

a) Permitted up to 0.019 in. (0.5 mm) in depth.

b) Maximum repairable limit: in the limits of the applicable repair.

<1> If the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm) (TASK 72-23-00-300-801-G00).

2) Cracks

a) Not permitted.

3) Nicks and scratches

a) Not permitted.

b) Maximum repairable limit: in the limits of the applicable repair.

<1> If the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm) (TASK 72-23-00-300-801-G00).

(b) Do a visual examination of area 3.

1) Nicks and scratches

a) Permitted up to 0.023 in. (0.6 mm) in depth.

(c) Do a visual examination of area 4.

1) Nicks, dents, and scratches

a) Not permitted.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

b) Maximum repairable limit: in the limits of the applicable repair.

<1> If the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm) (TASK 72-23-00-300-801-G00).

(d) Do a visual examination of area 5.

1) Fretting, erosion or wear

a) Permitted as follows:

<1> Dim A is between 0.091 in. (2.3 mm) and 0.106 in. (2.7 mm).

<2> Dim B is between 0.1161 in. (2.95 mm) and 0.1358 in. (3.45 mm).

<3> Dim C is between 0.060 in. (1.5 mm) and 0.145 in. (3.7 mm).

<4> Dim D is between 0.071 in. (1.8 mm) and 0.165 in. (4.2 mm).

(e) Do a visual examination of area 6.

1) Wear

a) Permitted up to 0.0019 in. (0.05 mm) in depth.

(f) Do a visual examination of replaceable nuts.

1) Missing

a) Not permitted.

2) Looseness

a) Not permitted.

(g) Do a visual examination of the black silicone sealing elastomer RTV 103 adhesive, A50472 [CP2734].

1) Missing

a) Permitted.

2) Looseness

a) Permitted.

3) Wear

a) Permitted.

SUBTASK 72-23-00-211-006-G00

(2) Examine the strut 1 outer fitting (LH) [2] (Figure 602 (Sheet 1), Figure 602 (Sheet 2), Figure 602 (Sheet 3) and Figure 602 (Sheet 4)).

(a) Do a visual examination of all areas.

1) Cracks

a) Not permitted.

(b) Do a visual examination of area 1.

1) Dents

a) Permitted up to 0.023 in. (0.6 mm) in depth.

b) Maximum repairable limit: in the limits of the applicable repair.

<1> If the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm) (TASK 72-23-00-300-801-G00).

2) Nicks and scratches

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Not permitted.

b) Maximum repairable limit: in the limits of the applicable repair.

<1> If the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm) (TASK 72-23-00-300-801-G00).

(c) Do a visual examination of area 2.

1) Dents, nicks, scratches

a) Not permitted.

b) Maximum repairable limit: in the limits of the applicable repair.

<1> If the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm) (TASK 72-23-00-300-801-G00).

(d) Do a visual examination of area 3.

1) Nicks and scratches

a) Permitted up to 0.023 in. (0.6 mm) in depth.

(e) Do a visual examination of area 4.

1) Nicks, dents, and scratches

a) Not permitted.

b) Maximum repairable limit: in the limits of the applicable repair.

<1> If the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm) (TASK 72-23-00-300-801-G00).

(f) Do a visual examination of area 5.

1) Fretting, erosion, or wear

a) Permitted as follows:

<1> Dim E is between 0.091 in. (2.3 mm) and 0.106 in. (2.7 mm).

<2> Dim F is between 0.1161 in. (2.95 mm) and 0.1358 in. (3.45 mm).

<3> Dim G is between 0.060 in. (1.5 mm) and 0.145 in. (3.7 mm).

<4> Dim H is between 0.071 in. (1.8 mm) and 0.165 in. (4.2 mm).

(g) Do a visual examination of area 6.

1) Wear

a) Permitted up to 0.0019 in. (0.05 mm) in depth.

(h) Do a visual examination of replaceable nuts.

1) Missing

a) Not permitted.

2) Looseness

a) Not permitted.

(i) Do a visual examination of the black silicone sealing elastomer RTV 103 adhesive, A50472 [CP2734].

1) Missing

a) Permitted.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Looseness

a) Permitted.

3) Wear

a) Permitted.

SUBTASK 72-23-00-211-007-G00

(3) Examine the strut 5 outer fitting [3] (Figure 603 (Sheet 1), Figure 603 (Sheet 2), Figure 603 (Sheet 3) and Figure 603 (Sheet 4)).

(a) Do a visual examination of all areas.

1) Cracks

a) Not permitted.

(b) Do a visual examination of area 1.

1) Dents

a) Permitted up to 0.019 in. (0.5 mm) in depth.

b) Maximum repairable limit: in the limits of the applicable repair.

<1> If the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm) (TASK 72-23-00-300-802-G00).

(c) Do a visual examination of area 2.

1) Dents

a) Not permitted.

b) Maximum repairable limit: in the limits of the applicable repair.

<1> If the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm) (TASK 72-23-00-300-802-G00).

(d) Do a visual examination of area 3.

1) Nicks and scratches

a) Permitted up to 0.007 in. (0.2 mm) in depth.

(e) Do a visual examination of area 4.

1) Nicks and scratches

a) Not permitted.

b) Maximum repairable limit: in the limits of the applicable repair.

<1> If the depth is less than 0.023 in. (0.6 mm) or if it can be blended to a finished blend depth of 0.023 in. (0.6 mm) (TASK 72-23-00-300-802-G00).

(f) Do a visual examination of area 5.

1) Fretting, erosion, or wear

a) Permitted as follows:

<1> Dim A is between 0.048 in. (1.2 mm) and 0.149 in. (3.8 mm).

<2> Dim B is between 0.111 in. (2.8 mm) and 0.204 in. (5.2 mm).

<3> Dim C is between 0.032 in. (0.8 mm) and 0.125 in. (3.2 mm).

(g) Do a visual examination of area 6.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Wear
   a) Permitted up to 0.0019 in. (0.05 mm) in depth.

(h) Do a visual examination of replaceable nuts.
1) Missing
   a) Not permitted.
2) Looseness
   a) Not permitted.

(i) Do a visual examination of the black silicone sealing elastomer RTV 103 adhesive, A50472 [CP2734].
1) Missing
   a) Permitted.
2) Looseness
   a) Permitted.
3) Wear
   a) Permitted.

(j) Examine the screws of the strut 5 outer fitting (Figure 606):
1) Surface oxidation on screw head
   a) Permitted.

SUBTASK 72-23-00-211-008-G00

(4) Examine the strut 1 fairing support [4] (Figure 604).
(a) Do a visual examination of all areas.
1) Cracks
   a) Not permitted.
2) Dents
   a) Not permitted.
3) Nicks and scratches
   a) Not permitted.

(b) Do a visual examination of area 3.
1) Fretting, erosion and wear
   a) Permitted if Dim A is up to equal to or more than 0.0666 in. (1.69 mm).

(c) Do a visual examination of replaceable nuts.
1) Missing
   a) Not permitted.
2) Looseness
   a) Not permitted.

(d) Do a visual examination of the black silicone sealing elastomer RTV 103 adhesive, A50472 [CP2734].
1) Missing
   a) Permitted.
2) Looseness

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Permitted.

3) Wear

a) Permitted.

SUBTASK 72-23-00-211-009-G00

(5) Examine the strut 1 fireproof plate [5] (Figure 605).

(a) Do a visual examination of all areas.

1) Cracks

a) Not permitted.

2) Nicks and scratches

a) Not permitted.

(b) Do a visual examination of area 1.

1) Dents

a) Not permitted.

(c) Do a visual examination of area 2.

1) Dents

a) Permitted up to 0.023 in. (0.6 mm) in depth.

(d) Do a visual examination of area 3.

1) Fretting, erosion and wear

a) Permitted if Dim A is up to equal to or more than 0.036 in. (0.9 mm).

(e) Do a visual examination of replaceable nuts.

1) Missing

a) Not permitted.

2) Looseness

a) Not permitted.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181

SUBTASK 72-23-00-211-010-G00

(6) Examine the fan frame shroud [6] (Figure 607).

(a) Do a visual examination of the shroud fan frame flanges and Outlet Guide Vane (OGV) flanges (Figure 607 (Sheet 1), Figure 607 (Sheet 2), and Figure 607 (Sheet 3)).

1) Dents, nicks, and scratches

a) Permitted up to 0.011 in. (0.3 mm) in depth.

2) Fretting/wear on mating surfaces around holes 50, 51, 55, 56, 57, 59, 69, 70, 77 and 78 of B2 flange

a) Permitted up to 0.0011 in. (0.03 mm) in depth.

3) Fretting/wear on mating surfaces around holes 41 and 44 of B11 flange

a) Permitted up to 0.0011 in. (0.03 mm) in depth.

4) Fretting/wear on mating surfaces around hole A4

a) Permitted up to 0.0007 in. (0.02 mm) in depth.

5) Fretting/worn areas on other mating surfaces (but not B2, B11, and B16 flanges)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Permitted up to 0.0011 in. (0.03 mm) in depth and up to 20% of the area around each hole up to 0.0098 in. (0.25 mm) in depth.

6) Fretting/worn areas on other mating surfaces (but not B12, B13, and OGV flanges)

a) Permitted up to 0.0007 in. (0.02 mm) in depth and up to 20% of the area around each hole up to 0.0098 in. (0.25 mm) in depth.

7) Thread marks in holes 50, 51, 55, 56, 57, 59, 69, 70, 77 and 78 of B2 flange

a) Not permitted.

8) Thread marks in holes 41 and 44 of B11 flange

a) Not permitted

9) Thread marks in hole A4

a) Not permitted.

10) Thread marks in other holes

a) Permitted up to 0.0019 in. (0.05 mm) in depth.

11) Tears and cracks

a) Not permitted.

(b) Do a visual examination of the V-groove (Figure 607 (Sheet 5)).

1) Fretting on the sealing surface

a) Permitted up to 0.0003 in. (0.01 mm) in depth.

2) Dents, nicks, scratches on the sealing surface

a) Permitted up to 0.0003 in. (0.01 mm) in depth.

3) Dents, nicks, scratches on the other surfaces

a) Permitted up to 0.011 in. (0.3 mm) in depth.

4) Fretting on base metal of the varnish surface (Figure 641).

NOTE: Fretting area starting at 6 and 12 o'clock, 0.3937 in. (10 mm) from the edge of the flange.

a) V-Groove bottom (area B) (Figure 607 (Sheet 6)).

<1> Permitted up to 0.0098 in. (0.25 mm) in depth.

<2> Permitted if dim. X1 is greater than 0.1405 in. (3.57 mm).

b) Forward flange (area A).

<1> Permitted up to 0.0039 in. (0.10 mm) in depth.

c) Afterward flange (area C).

<1> Permitted up to 0.0118 in. (0.3 mm) in depth.

<2> Permitted if dim. X2 is upper than 0.1882 in. (4.78 mm).

5) Missing varnish PTFE (Polytetraflouroethylene (Teflon)) on varnish surface

a) Permitted.

b) Maximum repairable limit: any amount (TASK 72-23-04-300-802-G00).

6) Distortion of the forward flange

a) Not permitted.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181 (Continued)

7) Tears and cracks
   a) Not permitted.

(c) Do a visual examination of DIM. CT (Figure 607 (Sheet 7)).
1) Dents, nicks, scratches on DIM. CT
   a) Permitted up to 0.011 in. (0.3 mm) in depth.
2) Fretting on DIM. CT
   a) Permitted if DIM. CT is between 2.55905 in. (65.000 mm) and 2.56023 in. (65.030 mm).
3) Tears/cracks on DIM. CT
   a) Not permitted.
4) Fretting on base metal of the varnish surface
   a) Permitted up to 0.0007 in. (0.02 mm) in depth.
5) Missing varnish PTFE on varnish surface
   a) Permitted.
6) Distortion of the forward flange
   a) Not permitted.
7) Tears and cracks
   a) Not permitted.

(d) Do a visual examination of the handling mount surface (Figure 607 (Sheet 7)).
1) Dents, nicks, scratches
   a) Permitted up to 0.0393 in. (1.0 mm) in depth.

(e) Do a visual examination of the handling mount flange (Figure 607 (Sheet 7)).
1) Dents, nicks, scratches
   a) Permitted up to 0.011 in. (0.3 mm) in depth.
2) Fretting/wear on mating surfaces
   a) Permitted up to 0.003 in. (0.1 mm) in depth or any amount if Dim CS is between 0.8445 in. (21.45 mm) and 0.8602 in. (21.85 mm).
3) Tears/cracks
   a) Not permitted.

(f) Do a visual examination of the handling mount lugs (Figure 607 (Sheet 8)).
1) Dents, nicks, scratches
   a) Permitted up to 0.011 in. (0.3 mm) in depth.
2) Tears/cracks
   a) Not permitted.
3) Fretting on mating surface DH
   a) Permitted if Dim DH is between 0.0749 in. (1.90 mm) and 0.0826 in. (2.10 mm).
4) Fretting on mating surface DR

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a) Permitted if Dim DR is between 0.0749 in. (1.90 mm) and 0.0826 in. (2.10 mm).

5) Fretting/wear on diameter CQ

a) Permitted if Dia CQ is between 0.78390 in. (19.911 mm) and 0.79177 in. (20.111 mm).

(g) Do a visual examination of the Accessory Gear Box (AGB) fittings (TASK 72-23-00-211-805-G00).

(h) Do a visual examination of the AGB Axial Mount (TASK 72-23-00-211-805-G00).

(i) Do a visual examination of the Fittings (Figure 607 (Sheet 9)).

1) Dents, nicks, scratches

a) Permitted up to 0.011 in. (0.3 mm) in depth.

2) Tears/cracks

a) Not permitted.

3) Fretting/wear on mating surface DM

a) Permitted up to 0.0007 in. (0.02 mm) in depth or if Dim DM is between 0.08808 in. (2.237 mm) and 0.09303 in. (2.363 mm).

4) Fretting/wear on mating surface DG

a) Permitted up to 0.0007 in. (0.02 mm) in depth or if Dim DG is between 0.4686 in. (11.90 mm) and 0.4763 in. (12.10 mm).

5) Fretting/wear on diameter Q

a) Permitted if Dia Q is between 1.18736 in. (30.159 mm) and 1.18814 in. (30.179 mm).

6) Fretting/wear on diameter U

a) Permitted if Dia U is between 0.87607 in. (22.252 mm) and 0.87684 in. (22.272 mm).

(j) Do a visual examination of the Fail Safe Fitting (Figure 607 (Sheet 10)).

1) Dents, nicks, scratches

a) Permitted up to 0.011 in. (0.3 mm) in depth.

2) Contact marks in the bushing inner diameter

a) Permitted up to 0.0007 in. (0.02 mm) in depth.

3) Fretting in the bushing inner diameter

a) Permitted if Dia J is between 1.40351 in. (35.649 mm) and 1.41137 in. (35.849 mm).

4) Tears/cracks

a) Not permitted.

(k) Do a visual examination of Dim CO and Dia CM (Figure 607 (Sheet 10)).

1) Dents, nicks, scratches

a) Permitted up to 0.003 in. (0.1 mm) in depth.

2) Fretting

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181 (Continued)

a) Permitted if Dim CO is between 0.31300 in. (7.950 mm) and 0.31692 in. (8.050 mm) and if Dia CM is between 0.31292 in. (7.948 mm) and 0.31354 in. (7.964 mm).

3) Tears/cracks

a) Not permitted.

(l) Do a visual examination of the mating faces of the Transfer Gearbox (TGB) Mounts (Figure 607 (Sheet 10)).

1) Dents, nicks, scratches

a) Permitted up to 0.003 in. (0.1 mm) in depth.

2) Fretting/wear on mating surfaces

a) Permitted up to 0.003 in. (0.1 mm) in depth on 10% of the surface and 20% of the area around each hole up to 0.0098 in. (0.25 mm) in depth.

3) Tears/cracks

a) Not permitted.

(m) Do a visual examination of the mating faces of the Oil Tank Mounts (Figure 607 (Sheet 11)).

1) Dents, nicks, scratches

a) Permitted up to 0.003 in. (0.1 mm) in depth.

2) Fretting/wear on mating surfaces

a) Permitted up to 0.003 in. (0.1 mm) in depth and up to 20% of the area around each hole up to 0.0098 in. (0.25 mm) in depth.

3) Tears/cracks

a) Not permitted.

(n) Do a visual examination of the mating face of the Spigot Mount (Figure 607 (Sheet 12)).

1) Dents, nicks, scratches

a) Permitted up to 0.003 in. (0.1 mm) in depth.

2) Fretting

a) Permitted up to 0.003 in. (0.1 mm) in depth and up to 20% of the area around each hole up to 0.0098 in. (0.25 mm) in depth.

3) Tears/cracks

a) Not permitted.

(o) Do a visual examination of the mating face of the Acoustic Panel Flange and the mating faces of the Acoustic Panel Bosses (Figure 607 (Sheet 13)).

1) Dents, nicks, scratches

a) Permitted up to 0.003 in. (0.1 mm) in depth.

2) Fretting

a) Permitted up to 0.0007 in. (0.02 mm) in depth and up to 20% of the area around each hole up to 0.0098 in. (0.25 mm) in depth.

3) Tears/cracks

a) Not permitted.

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